

Lean Formalization Summary for the Explicit RandNLA CACM Algorithms

LeanFpAnalysis repository

Abstract

This note summarizes the Lean formalization attached to the three explicit meta-algorithms in Drineas and Mahoney’s CACM survey, [RandNLA: Randomized Numerical Linear Algebra](#). The formalized results focus on stability and on the paper-level approximation guarantees that are currently present in the repository: Algorithm 1 entrywise sampling stability, a hard-thresholded element-wise spectral theorem used as a source-aligned support variant, and a faithful nontruncated Algorithm 1 Frobenius/Markov floating-point corollary with an explicit scalar radius, plus a literal Algorithm 1 source-rate specialization under a no-small-entry condition and a nonconditional literal-law support-radius theorem with all reciprocal-entry and floating-point terms expanded, Algorithm 2 row-sampling Gram approximation and its leverage-score specialization, the Algorithm 2 leverage-score least-squares consequence behind equation (8), and Algorithm 3 matrix-preconditioning stability. It also records the remaining accuracy backlog: the source-uniform sharp literal Algorithm 1 equation (2) matrix-Bernstein theorem for $p_{ij} = A_{ij}^2 / \|A\|_F^2$, with no reciprocal-entry or no-small-entry dependence, its fully rectangular CACM-prose variant, the implementation-backed rounded least-squares sketch and solver pipeline for equation (8), and the later paper-level randomized approximation claims. Each mathematical statement below is paired with its Lean theorem names.

1 Common Floating-Point Model

The repository uses an axiomatic floating-point model with unit roundoff u . Each basic arithmetic operation satisfies the standard relative error model

$$\text{fl}(x \circ y) = (x \circ y)(1 + \delta), \quad |\delta| \leq u.$$

Accumulated dot-product and matrix-product errors use Higham’s

$$\gamma_k = \frac{ku}{1 - ku},$$

with Lean side condition `gammaValid fp k`.

2 Algorithm 1: Element-Wise Sampling

Algorithm 1 samples entries (i, j) independently with probabilities

$$p_{ij} = \frac{A_{ij}^2}{\sum_{k,\ell} A_{k\ell}^2} = \frac{A_{ij}^2}{\|A\|_F^2}.$$

The current formalization studies exact unbiasedness, the one-step mean-zero residual increments, the floating-point stability of the sampled entry update, and the random hit counter of a fixed entry.

By current convention, the probabilities p_{ij} are exact mathematical sampling laws. The implementation-facing floating-point statements therefore charge the arithmetic performed after sampling: the sampled-entry rescaling division, repeated accumulation into the sketch, and residual formation. They do not charge floating-point error for constructing, normalizing, repairing, or sampling from the probability table. The ‘WithProb’ Lean functions are exact supplied-law adapters, useful for deterministic transfers, not certificates for probability construction.

Lean formalization.

```
sqMagProb
sqMagProb_sum_eq_one
elementwiseSampleUpdateWithProb
fl_elementwiseSampleUpdateWithProb_error_bound
fl_elementwiseTraceSketchWithProb_error_bound
```

Theorem 1 (Exact unbiasedness). *Assume the trace length equals the Algorithm 1 sample count, $\text{steps} = s$, and $s \neq 0$. Under the independent squared-magnitude trace law, the exact zero-initialized estimator is unbiased entrywise:*

$$\mathbb{E}[\tilde{A}_{ij}] = A_{ij}.$$

Equivalently, as a matrix-valued statement,

$$(\mathbb{E}[\tilde{A}_{ij}])_{i,j} = A.$$

Lean formalization.

```
sqMagTraceProbMass_marginal_one
sqMagTraceProbability_eventProb_sampleHits
sqMagTraceProbability_expectationReal_hitCount_eq
elementwiseTraceSketch_zero_init_of_entry_eq_zero
sqMagTraceProbability_expectationReal_elementwiseTraceSketch_nonzero_entry
sqMagTraceProbability_expectationReal_elementwiseTraceSketch_zero_entry
sqMagTraceProbability_expectationReal_elementwiseTraceSketch_entry
sqMagTraceProbability_expectationReal_elementwiseTraceSketch_matrix
```

Lemma 2 (Zero-mean residual increments). *For one squared-magnitude sample X , define the residual increment*

$$Z_X(i, j) = \frac{A_{ij}}{s} - \mathbf{1}_{\{X=(i,j)\}} \frac{A_{ij}}{sp_{ij}}.$$

If $s \neq 0$ and $\|A\|_F^2 > 0$, then

$$\mathbb{E}[Z_X(i, j)] = 0.$$

For an s -sample trace $X_1, \dots, X_s$, the exact residual decomposes as

$$A - \tilde{A} = \sum_{t=1}^s Z_{X_t}.$$

Consequently, for every fixed vector x ,

$$\mathbb{E}[Z_X x] = 0, \quad (A - \tilde{A})x = \sum_{t=1}^s Z_{X_t} x.$$

Lean formalization.

```

elementwiseSampleContribution
elementwiseTraceContribution_eq_sampleContribution
elementwiseSampleResidualIncrement
elementwiseTraceResidual_eq_sum_sampleResidualIncrement
rectMatMulVec_elementwiseTraceResidual_eq_sum_sampleResidualIncrement
sqMagProb_sum_elementwiseSampleContribution_entry
sqMagProb_sum_elementwiseSampleResidualIncrement_entry_eq_zero
sqMagProb_sum_rectMatMulVec_elementwiseSampleResidualIncrement_eq_zero
sqMagTraceProbability_expectationReal_elementwiseTraceResidual_entry_eq_zero
sqMagTraceProbability_expectationReal_rectMatMulVec_elementwiseTraceResidual_eq_zero

```

Theorem 3 (Deterministic entrywise stability). *Fix an entry (i, j) , and let q_{ij} be the number of times this entry is sampled in a deterministic Algorithm 1 trace. For nonzero A_{ij} and valid $\gamma_{q_{ij}}$, the computed entry satisfies*

$$\left| \widehat{A}_{ij} - \left(\widetilde{A}_{ij}^{(0)} + q_{ij} \frac{\|A\|_F^2}{sA_{ij}} \right) \right| \leq \left| \widetilde{A}_{ij}^{(0)} \right| \gamma_{q_{ij}} + q_{ij} \left| \frac{\|A\|_F^2}{sA_{ij}} \right| \gamma_{q_{ij}+1}.$$

Lean formalization.

```

fl_elementwiseTraceSketch_sqMag_error_bound
fl_elementwiseTraceSketch_sqMag_error_bound_exact
fl_elementwiseSketch_zero_init_sqMag_error_bound

```

Corollary 4 (High-probability stability from hit-count concentration). *If a proved concentration theorem gives*

$$\mathbb{P}[q_{ij} \leq Q] \geq 1 - \delta,$$

then the same event implies the deterministic stability bound with the budget

$$B(Q) = |\widetilde{A}_{ij}^{(0)}| \gamma_Q + Q \left| \frac{\|A\|_F^2}{sA_{ij}} \right| \gamma_{Q+1}.$$

The repository instantiates this transfer with Markov, pairwise-Chebyshev, and independent Chernoff concentration for the Algorithm 1 product trace law.

Lean formalization.

```

highProbability_sqMagTraceStability_of_markov_budget
highProbability_sqMagTraceStability_of_pairwise_chebyshev_budget
highProbability_sqMagTraceStability_of_independent_chernoff_budget
highProbability_sqMagTraceStability_of_independent_chernoff_optimized_tail_budget

```

3 Algorithm 1: Spectral Transfer for Equation (2)

The CACM survey’s equation (2) is a high-probability spectral-norm concentration theorem for the exact sampled residual $A - \tilde{A}$. The repository contains both the deterministic floating-point transfer layer and a closed source-aligned square theorem for the truncated squared-magnitude route used in Drineas–Zouzias. The operator norm is stated in rectangular vector-action form:

$$\|M\|_{2,\text{act}} \leq c \iff \|Mx\|_2 \leq c \|x\|_2 \quad \text{for all } x.$$

The first few statements in this section are support adapters: they expose the generic perturbation matrix used internally by the Lean library. They are not the advertised equation (2) result. The end-of-section corollary eliminates that matrix and gives explicit γ_{s+1} -radii: first for the hard-thresholded source variant, and then for a faithful nontruncated Frobenius/Markov Algorithm 1 corollary. It also records a literal source-rate corollary under the explicit condition that the source threshold $\varepsilon/(2n)$ would not remove any nonzero entry. The fully unconditional sharp literal CACM equation (2) matrix-Bernstein theorem remains open.

Theorem 5 (Support adapter: floating-point transfer for an exact spectral event). *Let $R = A - \tilde{A}$ be the exact Algorithm 1 residual for a fixed trace, and let $\hat{R} = A - \hat{\tilde{A}}$ be the floating-point residual for the same trace. Suppose*

$$\|Rx\|_2 \leq \varepsilon \|x\|_2 \quad \text{for all } x, \quad |\hat{\tilde{A}}_{ij} - \tilde{A}_{ij}| \leq B_{ij}, \quad B_{ij} \geq 0.$$

Then

$$\|\hat{R}x\|_2 \leq (\varepsilon + \|B\|_F) \|x\|_2 \quad \text{for all } x.$$

Lean formalization.

```
frobNormRect
rectOpNorm2Le
rectOpNorm2Le_add_of_rectOpNorm2Le_of_frobNormRect_le
fl_elementwiseTraceResidual_rectOpNorm2Le_of_exact
```

Corollary 6 (Support adapter: hit-count-budget transfer). *Assume the sampled-entry denominators are nonzero for every entry considered (the Lean corollary uses the sufficient hypothesis $A_{ij} \neq 0$ for all (i, j)), and assume the required γ_Q and γ_{Q+1} validity conditions. If every entry hit count in a fixed trace is at most Q , then the entrywise Algorithm 1 stability budget $B(Q)$ can be used as the matrix B above. Consequently an exact spectral residual bound transfers to the floating-point residual with additive perturbation $\|B(Q)\|_F$.*

Lean formalization.

```
fl_elementwiseTraceResidual_rectOpNorm2Le_of_exact_and_hitCount_le
```

Corollary 7 (Support adapter: probability transfer). *On a finite probability space, if the exact spectral event has probability at least ρ , and the entrywise floating-point budget holds for every outcome, then the floating-point spectral event has probability at least ρ .*

Lean formalization.

```
algorithm1ExactSpectralEvent_subset_flSpectralEvent
probability_algorithm1_fl_spectral_of_exact_spectral
```

Theorem 8 (Self-adjoint dilation support). *Let $R = A - \tilde{A}$ be the exact rectangular residual, and define its self-adjoint dilation by*

$$\mathcal{D}(R) = \begin{bmatrix} 0 & R \\ R^T & 0 \end{bmatrix}.$$

If the square dilation satisfies the vector-action bound

$$\|\mathcal{D}(R)z\|_2 \leq \varepsilon \|z\|_2 \quad \text{for all block vectors } z,$$

then the rectangular residual satisfies

$$\|Rx\|_2 \leq \varepsilon \|x\|_2 \quad \text{for all } x.$$

Consequently, any proved high-probability event for the dilation transfers to the exact rectangular residual event. The generic floating-point handoff is kept in Lean as a support adapter; the displayed equation (2) endpoint below uses the explicit γ_{s+1} -radius instead.

Lemma 9 (Dilation increment foundations). *For an s -sample Algorithm 1 trace,*

$$\mathcal{D}(A - \tilde{A}) = \sum_{t=1}^s \mathcal{D}(Z_{X_t}),$$

where Z_X is the one-sample residual increment. Under the one-step squared-magnitude law, every entry of $\mathcal{D}(Z_X)$ has mean zero, and

$$\mathbb{E} \|\mathcal{D}(Z_X)\|_F^2 \leq 2mn \frac{\|A\|_F^2}{s^2}.$$

The finite-probability library also now contains the real-valued exponential Markov kernel

$$\mathbb{P}[T \leq X] \leq \exp(-\lambda T) \mathbb{E} \exp(\lambda X), \quad \lambda > 0.$$

The square finite matrix layer contains product and trace operations, trace additivity, trace homogeneity, cyclicity $\text{tr}(MN) = \text{tr}(NM)$, finite identity and transpose operations, quadratic forms, PSD and Loewner-order predicates, quadratic-form add/neg/sub algebra, two-sided Loewner-to-absolute quadratic-form control, symmetric-square PSD facts, the deterministic implication

$$\|Mx\|_2 \leq L \|x\|_2 \quad \forall x \quad \implies \quad M^2 \preceq L^2 I$$

for symmetric finite matrices, and the dilation trace-of-square identity

$$\text{tr}(\mathcal{D}(M)^2) = 2 \|M\|_F^2.$$

Conversely, for a symmetric finite matrix and $L \geq 0$,

$$M^2 \preceq L^2 I \quad \implies \quad \|Mx\|_2 \leq L \|x\|_2 \quad \text{for every } x.$$

Therefore, for an Algorithm 1 residual R , the squared dilation event

$$\mathcal{D}(R)^2 \preceq \varepsilon^2 I$$

implies both the exact dilation operator event and the rectangular event

$$\|Rx\|_2 \leq \varepsilon \|x\|_2 \quad \text{for every } x.$$

The same event transfers to the floating-point residual after adding the proved entrywise perturbation budget. It also proves the quadratic-form variance proxy

$$\mathbb{E} z^T \mathcal{D}(Z_X)^2 z \leq 2mn \frac{\|A\|_F^2}{s^2} \|z\|_2^2$$

for every block vector z . Equivalently, the one-step variance matrix

$$V_1 = \mathbb{E} \mathcal{D}(Z_X)^2$$

is positive semidefinite and satisfies the Loewner bound

$$0 \preceq V_1 \preceq 2mn \frac{\|A\|_F^2}{s^2} I.$$

After summing the s independent one-step proxies, the variance matrix

$$V_s = \sum_{t=1}^s \mathbb{E} \mathcal{D}(Z_{X_t})^2$$

is positive semidefinite and satisfies

$$\sum_{t=1}^s \mathbb{E} z^T \mathcal{D}(Z_{X_t})^2 z \leq 2mn \frac{\|A\|_F^2}{s} \|z\|_2^2.$$

Equivalently, $0 \preceq V_s \preceq 2mn \|A\|_F^2 I/s$. The trace form is also formalized:

$$\text{tr}(V_1) \leq 2mn \frac{\|A\|_F^2}{s^2} (m+n), \quad \text{tr}(V_s) \leq 2mn \frac{\|A\|_F^2}{s} (m+n).$$

The product-law bridge is now formalized explicitly as well. If $X_1, \dots, X_s$ are the independent Algorithm 1 samples and $R = \sum_t Z_{X_t}$, then every entry of the full dilation residual is mean-zero:

$$\mathbb{E} \mathcal{D}(R)_{ab} = 0.$$

Moreover the trace-law expected variance matrix

$$\sum_{t=1}^s \mathbb{E} \mathcal{D}(Z_{X_t})^2$$

is the same object bounded above by $2mn \|A\|_F^2 I/s$ in Loewner order. The scalar product-law MGF bridge is also formalized. For any one-step statistic $f: (i, j) \mapsto \mathbb{R}$,

$$\mathbb{E} \exp \left(\lambda \sum_{t=1}^s f(X_t) \right) = \left(\sum_{i,j} p_{ij} \exp(\lambda f(i, j)) \right)^s.$$

Consequently the repository has exponential-Markov upper-tail and complement forms for $\sum_t f(X_t)$, including variants that consume an explicit one-step bound $\sum_{i,j} p_{ij} \exp(\lambda f(i, j)) \leq \exp(\psi)$ and return the tail $\exp(s\psi - \lambda T)$. This is useful scalar concentration infrastructure for fixed-projection or covering arguments. These results are not by themselves the noncommutative trace-exponential theorem needed for matrix Bernstein, but the later trace-MGF and Bernstein sections compose the source-aligned route into a closed high-probability spectral event and floating-point transfer.

Corollary 10 (Finite test-vector quadratic-form MGF support). *Let $\mathcal{N}$ be a finite index set of deterministic test vectors $z_a \in \mathbb{R}^{m+n}$. For each $a \in \mathcal{N}$, define the one-step scalar statistic*

$$f_a(i, j) = z_a^T D(\Delta_{ij}) z_a,$$

where Δ_{ij} is the one-sample residual increment of Algorithm 1 and $D(\cdot)$ is the self-adjoint dilation. The formalization proves the identity

$$z_a^T D(R) z_a = \sum_{t=1}^s f_a(X_t).$$

Consequently, if for every $a \in \mathcal{N}$

$$\sum_{i,j} p_{ij} \exp(\lambda_a f_a(i, j)) \leq \exp(\psi_a), \quad \lambda_a > 0,$$

then

$$\mathbb{P}[z_a^T D(R) z_a \leq T_a \text{ for all } a \in \mathcal{N}] \geq 1 - \sum_{a \in \mathcal{N}} \exp(s\psi_a - \lambda_a T_a).$$

The one-step MGF hypothesis can itself be discharged from a pointwise bound: if $f_a(i, j) \leq B_a$ for every sampled entry, then the same conclusion holds with $\psi_a = \lambda_a B_a$. There is also a support-aware form: it is enough to prove $f_a(i, j) \leq B_a$ only when $p_{ij} > 0$, since the zero-probability entries contribute zero mass to the one-step MGF. This support-aware statement is the version used by the truncated source-aligned route, where retained-entry side conditions are available exactly on positive-probability samples. This is a finite-family scalar-MGF theorem for supplied test vectors. It does not construct a net, prove sharp Bernstein/Hoeffding-type one-step MGF bounds, or prove a largest-eigenvalue or matrix-Bernstein theorem.

Corollary 11 (One-sided dilation event transfer). *Let $R = A - \tilde{A}$ be the exact Algorithm 1 residual and let*

$$D(R) = \begin{bmatrix} 0 & R \\ R^T & 0 \end{bmatrix}.$$

If $\varepsilon \geq 0$ and

$$D(R) \preceq \varepsilon I,$$

then

$$\|R\|_2 \leq \varepsilon.$$

Consequently, any high-probability largest-eigenvalue theorem proving the one-sided dilation event transfers to the repository's exact rectangular operator event, and then to the floating-point event after adding the proved entrywise FP perturbation budget. This is an event-transfer theorem only: it does not prove that $D(R) \preceq \varepsilon I$ occurs with high probability.

Lean formalization.

```
finiteVecNorm2
finiteMatVec
finiteFrobNormSq
finiteMatMul
finiteIdMatrix
```

finiteBasisVec
 finiteTranspose
 finiteTrace
 finiteTrace_add
 finiteTrace_smul
 finiteTrace_neg
 finiteTrace_sub
 finiteTrace_finiteIdMatrix
 finiteTrace_smul_finiteIdMatrix
 finiteTrace_fintype_sum
 finiteTrace_fintype_sum_smul
 finiteTrace_nonneg_of_finitePSD
 finiteTrace_mono_of_finiteLoewnerLe
 finiteTrace_le_of_finiteLoewnerLe_smul_id
 finiteTrace_finiteMatMul_comm
 finiteMatVec_finiteMatMul
 finiteMatVec_finiteIdMatrix
 finiteMatVec_finiteBasisVec
 finiteVecNorm2Sq_finiteMatVec_le_finiteFrobNormSq_mul
 finiteOpNorm2Le
 finiteQuadraticForm
 finitePSD
 finiteLoewnerLe
 finitePSD_iff_zero_loewnerLe
 finiteQuadraticForm_finiteIdMatrix
 finiteQuadraticForm_finiteBasisVec
 finiteQuadraticForm_smul_finiteIdMatrix
 finiteQuadraticForm_add
 finiteQuadraticForm_neg
 finiteQuadraticForm_sub
 finiteLoewnerLe_iff_sub_finitePSD
 finiteLoewnerLe_add
 finiteLoewnerLe_smul_of_nonneg
 finiteQuadraticForm_finset_sum
 finiteQuadraticForm_finset_sum_smul
 finiteQuadraticForm_fintype_sum
 finiteQuadraticForm_fintype_sum_smul
 finiteLoewnerLe_fintype_sum
 finiteLoewnerLe_fintype_sum_smul_of_nonneg
 finitePSD_fintype_sum_of_finitePSD

finitePSD_fintype_sum_smul_of_nonneg
 abs_finiteQuadraticForm_le_of_loewnerLe_neg
 finiteVecInnerProduct_sq_le
 abs_finiteVecInnerProduct_le_finiteVecNorm2_mul
 abs_finiteVecInnerProduct_finiteMatVec_le_of_finiteOpNorm2Le
 abs_finiteQuadraticForm_le_of_finiteOpNorm2Le
 finiteQuadraticForm_le_of_finiteOpNorm2Le
 finiteLoewnerLe_smul_id_of_finiteOpNorm2Le
 finiteLoewnerLe_neg_smul_id_of_finiteOpNorm2Le
 finiteVecInnerProduct_finiteMatVec_eq_transpose
 finiteVecInnerProduct_finiteMatVec_left_eq_right_of_symmetric
 finiteQuadraticForm_finiteMatMul_self_of_symmetric
 finitePSD_finiteMatMul_self_of_symmetric
 finiteQuadraticForm_finiteMatMul_self_le_finiteFrobNormSq_mul_of_symmetric
 finiteMatMul_self_loewnerLe_scalar_id_of_finiteOpNorm2Le
 finiteTrace_finiteMatMul_self_le_card_mul_sq_of_finiteOpNorm2Le
 finiteOpNorm2Le_of_finiteMatMul_self_loewnerLe_scalar_id
 IsSymmetricFiniteMatrix
 IsSymmetricFiniteMatrix.to_matrix_isSymm
 Matrix_isSymm.to_IsSymmetricFiniteMatrix
 IsSymmetricFiniteMatrix.to_matrix_isHermitian
 Matrix_isHermitian.to_IsSymmetricFiniteMatrix
 finitePSD.to_matrix_posSemidef
 Matrix_posSemidef.to_finitePSD
 finitePSD_iff_matrix_posSemidef_of_symmetric
 finiteLoewnerLe.to_matrix_posSemidef_sub
 Matrix_posSemidef_sub.to_finiteLoewnerLe
 finiteLoewnerLe_iff_matrix_posSemidef_sub_of_symmetric
 finiteHermitianEigenvalues
 finiteHermitianEigenvalues_mem_spectrum_real
 finiteTrace_eq_sum_finiteHermitianEigenvalues
 finiteDiagonal
 finiteMatrixExp
 finiteMatrixExp_smul_finiteIdMatrix
 finiteTrace_finiteMatrixExp_smul_finiteIdMatrix
 finiteMatrixExp_symmetric
 finiteMatrixExp_finiteDiagonal
 finiteTrace_finiteMatrixExp_finiteDiagonal
 finiteHermitianCfcExp
 finiteTrace_finiteHermitianCfcExp_eq_sum_exp_finiteHermitianEigenvalues

finiteTrace_finiteMatrixExp_eq_sum_exp_finiteHermitianEigenvalues
 finiteTrace_finiteMatrixExp_neg_eq_sum_exp_neg_finiteHermitianEigenvalues
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 finiteTrace_finiteMatrixExp_nonneg
 exp_neg_le_finiteTrace_finiteMatrixExp_neg_of_finiteHermitianEigenvalue_le
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 FiniteProbability.eventProb_exists_finiteHermitianEigenvalue_le_le_exp_mul_expected_trace_exp_neg
 FiniteProbability.eventProb_forall_abs_finiteHermitianEigenvalue_lt_ge_one_sub_exp_neg_mul_
 expected_trace_exp_add
 finitePSD_iff_finiteHermitianEigenvalues_nonneg
 finiteLoewnerLe_iff_sub_finiteHermitianEigenvalues_nonneg
 smulFiniteIdMatrix_symmetric
 finiteScalarUpperDiff_symmetric
 finiteScalarUpperDiffEigenvalues
 finiteLoewnerLe_smul_id_iff_finiteScalarUpperDiffEigenvalues_nonneg
 finiteLoewnerLe_smul_id_iff_sub_finiteHermitianEigenvalues_nonneg
 rectSelfAdjointDilation
 rectSelfAdjointDilation_symmetric
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 rectSelfAdjointDilation_square_loewnerLe_scalar_id_of_finiteOpNorm2Le
 rectSelfAdjointDilation_opNorm2Le_of_square_loewnerLe_scalar_id
 sumBothVec
 finiteVecNorm2Sq_sumBothVec
 finiteMatVec_rectSelfAdjointDilation_sumBothVec
 finiteQuadraticForm_rectSelfAdjointDilation_sumBothVec
 rectOpNorm2Le_of_selfAdjointDilation_loewnerLe_scalar_id
 rectOpNorm2Le_of_selfAdjointDilation_square_loewnerLe_scalar_id
 finiteFrobNormSq_rectSelfAdjointDilation
 finiteTrace_finiteMatMul_rectSelfAdjointDilation_self
 finiteMatVec_rectSelfAdjointDilation_sumInr
 rectOpNorm2Le_of_selfAdjointDilation
 algorithm1ExactDilationEvent
 algorithm1ExactDilationUpperEvent
 algorithm1ExactDilationEigenUpperEvent
 algorithm1ExactDilationEigenUpperIndexEvent
 algorithm1ExactDilationSquareEvent
 algorithm1ExactDilationEvent_subset_exactSpectralEvent
 algorithm1ExactDilationUpperEvent_subset_exactSpectralEvent
 algorithm1ExactDilationEigenUpperEvent_subset_exactDilationUpperEvent

algorithm1ExactDilationEigenUpperEvent_subset_exactSpectralEvent
 algorithm1ExactDilationSquareEvent_subset_exactDilationEvent
 algorithm1ExactDilationSquareEvent_subset_exactSpectralEvent
 probability_algorithm1_exact_spectral_of_dilation
 probability_algorithm1_exact_spectral_of_dilation_upper
 probability_algorithm1_exact_spectral_of_dilation_eigen_upper
 probability_algorithm1_exact_dilation_eigen_upper_of_index_bounds
 probability_algorithm1_exact_spectral_of_dilation_eigen_upper_index_bounds
 probability_algorithm1_exact_spectral_of_dilation_square
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 probability_algorithm1_fl_spectral_of_exact_dilation_eigen_upper
 probability_algorithm1_fl_spectral_of_exact_dilation_eigen_upper_index_bounds
 probability_algorithm1_fl_spectral_of_exact_dilation_square
 rectSelfAdjointDilation_elementwiseTraceResidual_eq_sum_sampleResidualIncrement
 sqMagProb_sum_rectSelfAdjointDilation_elementwiseSampleResidualIncrement_eq_zero
 sqMagProb_sum_finiteFrobNormSq_rectSelfAdjointDilation_elementwiseSampleResidualIncrement_le
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 sqMagProb_sum_steps_finiteQuadraticForm_rectSelfAdjointDilation_square_le
 sqMagProb_sum_rectSelfAdjointDilation_square_loewnerLe_scalar_id
 sqMagProb_sum_rectSelfAdjointDilation_square_psd
 sqMagProb_sum_steps_rectSelfAdjointDilation_square_loewnerLe_scalar_id
 sqMagProb_sum_steps_rectSelfAdjointDilation_square_psd
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 sqMagProb_sum_steps_finiteTrace_rectSelfAdjointDilation_square_le
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 exp_sum_stepFunction_eq_prod
 sqMagTraceProbMass_exp_sum_stepFunction_eq
 sqMagTraceProbability_expectationReal_exp_sum_stepFunction_eq
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 sqMagTraceProbability_eventProb_sum_stepFunction_ge_le_exp_of_one_step_mgf_bound
 sqMagTraceProbability_eventProb_sum_stepFunction_le_ge_one_sub_exp_of_one_step_mgf_bound
 sqMagTraceProbability_eventProb_forall_sum_stepFunction_le_ge_one_sub_sum_exp_of_one_step_mgf_bound
 sqMagProb_sum_exp_stepFunction_le_exp_of_forall_le
 sqMagProb_sum_exp_stepFunction_le_exp_of_support_forall_le
 sqMagTraceProbability_eventProb_forall_sum_stepFunction_le_ge_one_sub_sum_exp_of_pointwise_bound
 sqMagTraceProbability_eventProb_forall_sum_stepFunction_le_ge_one_sub_sum_exp_of_support_pointwise_bound

finiteQuadraticForm_rectSelfAdjointDilation_elementwiseTraceResidual_eq_sum_
 sampleResidualIncrement
 sqMagTraceProbability_eventProb_forall_finiteQuadraticForm_rectSelfAdjointDilation_
 elementwiseTraceResidual_le_ge_one_sub_sum_exp_of_one_step_mgf_bound
 sqMagTraceProbability_eventProb_forall_finiteQuadraticForm_rectSelfAdjointDilation_
 elementwiseTraceResidual_le_ge_one_sub_sum_exp_of_pointwise_bound
 sqMagTraceProbability_eventProb_forall_finiteQuadraticForm_rectSelfAdjointDilation_
 elementwiseTraceResidual_le_ge_one_sub_sum_exp_of_support_pointwise_bound
 sqMagTraceProbability_expectationReal_rectSelfAdjointDilation_sampleResidualIncrement_eq_zero
 sqMagTraceProbability_expectationReal_rectSelfAdjointDilation_elementwiseTraceResidual_eq_zero
 sqMagTraceProbability_expectationReal_finiteQuadraticForm_rectSelfAdjointDilation_
 sampleResidualIncrement_square_le
 sqMagTraceProbability_expectationReal_sum_finiteQuadraticForm_rectSelfAdjointDilation_
 sampleResidualIncrement_square_le
 sqMagTraceProbability_sum_expectationReal_rectSelfAdjointDilation_square_loewnerLe_scalar_id
 sqMagTraceProbability_sum_expectationReal_rectSelfAdjointDilation_square_psd
 FiniteProbability.eventProb_real_ge_le_exp_mul_mgf
 FiniteProbability.eventProb_real_le_ge_one_sub_exp_mul_mgf
 FiniteProbability.eventProb_real_le_ge_one_sub_exp_of_mgf_bound
 FiniteProbability.eventProb_real_le_mean_add_ge_one_sub_exp_sq_of_subgaussian_mgf
 FiniteProbability.log_mgf_differential_le_of_entropyReal_exp_mul_le
 FiniteProbability.log_mgf_div_sub_quadratic_antitoneOn_of_differential_le
 FiniteProbability.tendsto_log_mgf_div_nhdsGT_zero
 FiniteProbability.log_mgf_le_mean_add_quadratic_of_differential_le
 FiniteProbability.log_mgf_le_mean_add_quadratic_of_entropyReal_exp_mul_le
 FiniteProbability.expectationReal_exp_centered_le_exp_of_log_mgf_le
 FiniteProbability.eventProb_real_le_mean_add_ge_one_sub_exp_sq_of_log_mgf_bound
 FiniteProbability.eventProb_real_le_mean_add_ge_one_sub_exp_sq_of_entropyReal_exp_mul_le

Lemma 12 (Finite MGF, entropy, and Laplace calculus). *Let X be a real random variable on a finite probability space. The finite moment-generating function*

$$M(\lambda) = \mathbb{E} \exp\{\lambda X\}$$

is strictly positive, and

$$M'(\lambda) = \mathbb{E}[X \exp\{\lambda X\}], \quad (\log M)'(\lambda) = \frac{\mathbb{E}[X \exp\{\lambda X\}]}{\mathbb{E}[\exp\{\lambda X\}]}.$$

For the real entropy functional

$$\text{Ent}(Z) = \mathbb{E}[Z \log Z] - \mathbb{E}[Z] \log(\mathbb{E}[Z]),$$

the exponential tilt satisfies

$$\text{Ent}(e^{\lambda X}) = \lambda \mathbb{E}[X e^{\lambda X}] - \mathbb{E}[e^{\lambda X}] \log \mathbb{E}[e^{\lambda X}].$$

Consequently, any explicit entropy bound

$$\text{Ent}(e^{\lambda X}) \leq c\lambda^2 M(\lambda)$$

implies the pointwise Herbst differential inequality

$$\lambda \frac{M'(\lambda)}{M(\lambda)} - \log M(\lambda) \leq c\lambda^2.$$

If this differential inequality holds for every $\lambda > 0$, then the corrected quotient

$$\lambda \mapsto \frac{\log M(\lambda)}{\lambda} - c\lambda$$

is antitone on $(0, \infty)$. The endpoint value is formalized as

$$\frac{\log M(\lambda)}{\lambda} \longrightarrow \mathbb{E}X \quad (\lambda \downarrow 0),$$

and therefore the same differential inequality implies the finite Herbst log-Laplace estimate

$$\log M(\lambda) \leq \lambda \mathbb{E}X + c\lambda^2 \quad (\lambda > 0).$$

Equivalently, the formalization packages the direct entropy-to-log-Laplace wrapper: the entropy bound

$$\text{Ent}(e^{\lambda X}) \leq c\lambda^2 M(\lambda) \quad (\lambda > 0)$$

itself implies the same log-Laplace estimate. Also, any visible log-Laplace estimate

$$\log M(\lambda) \leq \lambda\mu + R$$

implies the centered MGF estimate

$$\mathbb{E} \exp\{\lambda(X - \mu)\} \leq e^R.$$

In particular, if $\sigma > 0$, $t > 0$, and

$$\log M(\lambda) \leq \lambda\mu + \lambda^2\sigma^2/2 \quad (\lambda > 0),$$

then

$$\mathbb{P}\{X \leq \mu + t\} \geq 1 - \exp\left(-\frac{t^2}{2\sigma^2}\right).$$

In the special case where the entropy bound has the Ledoux/Herbst normalization

$$\text{Ent}(e^{\lambda X}) \leq \frac{\sigma^2}{2} \lambda^2 M(\lambda) \quad (\lambda > 0),$$

the same tail bound follows directly.

Lean formalization.

```
Analysis/FiniteProbability.lean FiniteProbability.expectationReal_exp_pos
FiniteProbability.hasDerivAt_expectationReal_exp_mul
FiniteProbability.hasDerivAt_log_expectationReal_exp_mul
```

```

FiniteProbability.entropyReal
FiniteProbability.entropyReal_exp_mul_eq
FiniteProbability.log_mgf_differential_le_of_entropyReal_exp_mul_le
FiniteProbability.log_mgf_div_sub_quadratic_antitoneOn_of_differential_le
FiniteProbability.tendsto_log_mgf_div_nhdsGT_zero
FiniteProbability.log_mgf_le_mean_add_quadratic_of_differential_le
FiniteProbability.log_mgf_le_mean_add_quadratic_of_entropyReal_exp_mul_le
FiniteProbability.expectationReal_exp_centered_le_exp_of_log_mgf_le
FiniteProbability.eventProb_real_le_mean_add_ge_one_sub_exp_sq_of_log_mgf_bound
FiniteProbability.eventProb_real_le_mean_add_ge_one_sub_exp_sq_of_entropyReal_exp_mul_le

```

Lemma 13 (Finite product entropy chain rule). *Let P and Q be finite probability laws on finite sets Ω and Λ . For any real function $Z : \Omega \times \Lambda \rightarrow \mathbb{R}$, define*

$$\bar{Z}(\omega) := \mathbb{E}_Q Z(\omega, \cdot).$$

Then the repository-native finite entropy functional satisfies the exact product decomposition

$$\text{Ent}_{P \otimes Q}(Z) = \mathbb{E}_P [\text{Ent}_Q(Z(\omega, \cdot))] + \text{Ent}_P(\bar{Z}).$$

Lean formalization.

```

Analysis/FiniteProbability.lean FiniteProbability.prod_expectationReal_eq
FiniteProbability.prod_expectationReal fst_eq
FiniteProbability.prod_expectationReal snd_eq
FiniteProbability.entropyReal_prod_eq_expectation_entropyReal_add_entropyReal_expectation

```

Lemma 14 (Scalar and fixed-vector independent-copy symmetrization). *Let P be a finite probability law on a finite set Ω . Let (ω, ω') be distributed according to the product law $P \otimes P$. For a scalar random variable $X : \Omega \rightarrow \mathbb{R}$,*

$$\mathbb{E}_P |X - \mathbb{E}_P X| \leq \mathbb{E}_{P \otimes P} |X(\omega) - X(\omega')|.$$

Consequently, if $\mathbb{E}_P X = 0$, then

$$\mathbb{E}_P |X| \leq \mathbb{E}_{P \otimes P} |X(\omega) - X(\omega')|.$$

Let $d \in \mathbb{N}$, and let $Y : \Omega \rightarrow \mathbb{R}^d$. Write

$$\mu_i = \mathbb{E}_P Y_i \quad (1 \leq i \leq d), \quad \|z\|_2 = \left(\sum_{i=1}^d z_i^2 \right)^{1/2}.$$

Then the Euclidean vector form is

$$\mathbb{E}_P \|Y - \mu\|_2 \leq \mathbb{E}_{P \otimes P} \|Y(\omega) - Y(\omega')\|_2.$$

In particular, if $\mathbb{E}_P Y_i = 0$ for every coordinate i , then

$$\mathbb{E}_P \|Y\|_2 \leq \mathbb{E}_{P \otimes P} \|Y(\omega) - Y(\omega')\|_2.$$

For Algorithm 1, let $A \in \mathbb{R}^{m \times n}$, let $\tilde{A}(\omega)$ be the exact trace sketch produced from the exact squared-magnitude product law, and fix $x \in \mathbb{R}^n$. With

$$Y(\omega) = (A - \tilde{A}(\omega))x \in \mathbb{R}^m,$$

the exact mean-zero residual identity gives

$$\mathbb{E} \left\| (A - \tilde{A})x \right\|_2 \leq \mathbb{E} \left\| (A - \tilde{A}(\omega))x - (A - \tilde{A}(\omega'))x \right\|_2.$$

The law P is exact by convention. All quantities in this lemma are exact analysis objects; no floating-point operation or computed non-probability quantity is introduced at this layer. The result is fixed-vector Khintchine-route infrastructure and is not a uniform spectral-norm tail for equation (2).

Lean formalization.

```
FiniteProbability.expectationReal_abs_sub_mean_le_prod_expectationReal_abs_sub
FiniteProbability.expectationReal_abs_le_prod_expectationReal_abs_sub_of_expectation_eq_zero
sqMagTraceProbability.expectationReal_abs_rectSelfAdjointDilation_elementwiseTraceResidual_le_
prod_abs_sub
FiniteProbability.expectationReal_vecNorm2_mean_le_expectationReal_vecNorm2
FiniteProbability.expectationReal_vecNorm2_sub_mean_le_prod_expectationReal_vecNorm2_sub
FiniteProbability.expectationReal_vecNorm2_le_prod_expectationReal_vecNorm2_sub_of_expectation_
eq_zero
sqMagTraceProbability.expectationReal_vecNorm2_rectMatMulVec_elementwiseTraceResidual_le_prod_
vecNorm2_sub
```

Lemma 15 (Operator copy-difference symmetrization bridge). *Let P be a finite probability law on a finite set Ω . Let $m, n \in \mathbb{N}$, and let $X : \Omega \rightarrow \mathbb{R}^{m \times n}$ be a matrix-valued random variable satisfying*

$$\mathbb{E}_P X_{ij} = 0 \quad (1 \leq i \leq m, 1 \leq j \leq n).$$

Fix an outcome $\omega \in \Omega$ and a radius $L \in \mathbb{R}$. Suppose that for every independent-copy outcome $\eta \in \Omega$,

$$\|(X(\omega) - X(\eta))z\|_2 \leq L\|z\|_2 \quad \text{for every } z \in \mathbb{R}^n.$$

Then

$$\|X(\omega)z\|_2 \leq L\|z\|_2 \quad \text{for every } z \in \mathbb{R}^n.$$

For Algorithm 1, let P be the exact squared-magnitude product law and write

$$R(\omega) = A - \tilde{A}(\omega).$$

Define the exact all-copy-differences event

$$\mathcal{E}_{\text{diff}}(L) = \{\omega : \|(R(\omega) - R(\eta))z\|_2 \leq L\|z\|_2 \text{ for every } \eta \text{ and every } z \in \mathbb{R}^n\}.$$

If $P(\mathcal{E}_{\text{diff}}(L)) \geq \rho$, then the exact Algorithm 1 spectral event

$$\{\omega : \|R(\omega)z\|_2 \leq L\|z\|_2 \text{ for every } z \in \mathbb{R}^n\}$$

also has probability at least ρ . This is exact-probability and exact-arithmetic infrastructure. No floating-point computation is modeled in this bridge, and the probability tail for $\mathcal{E}_{\text{diff}}(L)$ is a separate Khintchine-route target.

Lean formalization.

```
FiniteProbability.rectOpNorm2Le_of_entrywise_mean_zero_of_copy_diff_rectOpNorm2Le
sqMagTraceProbability_rectOpNorm2Le_elementwiseTraceResidual_of_all_copy_diffs
algorithm1ExactAllCopyDiffSpectralEvent
algorithm1ExactAllCopyDiffSpectralEvent_subset_exactSpectralEvent
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_ge_of_all_copy_diff
```

Lemma 16 (Finite-test Rademacher signed-matrix quadratic forms). *Let $m, n \in \mathbb{N}$. Let*

$$\Omega_m = \{-1, 1\}^m$$

with the exact uniform Rademacher law, and for $\omega \in \Omega_m$ write $\varepsilon_k(\omega) \in \{-1, 1\}$ for the k -th exact sign. Let

$$M_k \in \mathbb{R}^{n \times n} \quad (1 \leq k \leq m)$$

be a fixed exact matrix family. For $z \in \mathbb{R}^n$, define the exact quadratic form

$$q(M, z) = z^\top M z.$$

Then, for every $z \in \mathbb{R}^n$ and every exact sign outcome ω ,

$$q\left(\sum_{k=1}^m \varepsilon_k(\omega) M_k, z\right) = \sum_{k=1}^m q(M_k, z) \varepsilon_k(\omega).$$

Let $\mathcal{A}$ be a finite index set. For each $a \in \mathcal{A}$, let $z_a \in \mathbb{R}^n$, $T_a > 0$, and $\sigma_a > 0$. Assume the exact variance proxy

$$\sum_{k=1}^m q(M_k, z_a)^2 \leq \sigma_a^2 \quad (a \in \mathcal{A}).$$

Then the exact Rademacher law satisfies

$$\mathbb{P}\left\{\forall a \in \mathcal{A}, \quad \left|q\left(\sum_{k=1}^m \varepsilon_k M_k, z_a\right)\right| \leq T_a\right\} \geq 1 - \sum_{a \in \mathcal{A}} 2 \exp\left(-\frac{T_a^2}{2\sigma_a^2}\right).$$

This is the finite-test scalar-Hoeffding layer for a later matrix Khintchine/Bernstein route. It is exact-probability and exact-arithmetic only: the signs are drawn from the exact law, the matrices and test vectors are exact analysis objects, and no floating-point computation or computed non-probability quantity appears in this lemma. It does not by itself prove an operator-norm or spectral event for Algorithm 1.

Lean formalization.

```
Algorithms/RandNLA/Preconditioning.lean    finiteQuadraticForm_rademacher_signed_matrix_
sum_eq_sum
rademacherTraceProbability_eventProb_forall_abs_finiteQuadraticForm_signed_matrix_sum_le_ge_one_
sub_sum_two_mul_exp_neg_sq_div
```


Corollary 17 (Algorithm 1 Rademacherized stepwise copy-difference tests). *Let $A \in \mathbb{R}^{m \times n}$, let $s \in \mathbb{N}$, and let*

$$S = (S_1, \dots, S_s), \quad S' = (S'_1, \dots, S'_s)$$

be two fixed exact Algorithm 1 traces. For a one-step sample $S_t = (i_t, j_t)$, let $Z_t(S_t) \in \mathbb{R}^{m \times n}$ denote the exact one-sample residual increment

$$Z_t(S_t) = s^{-1}A - \frac{A_{i_t j_t}}{s p_{i_t j_t}} e_{i_t} e_{j_t}^\top,$$

with the formalized zero-probability branch used when the exact sampling probability $p_{i_t j_t}$ is zero. Define the exact self-adjoint dilation difference

$$\Delta_t = \begin{bmatrix} 0 & Z_t(S_t) - Z_t(S'_t) \\ (Z_t(S_t) - Z_t(S'_t))^\top & 0 \end{bmatrix} \in \mathbb{R}^{(m+n) \times (m+n)}.$$

For $w \in \mathbb{R}^{m+n}$, define

$$q(\Delta, w) = w^\top \Delta w.$$

Let $\mathcal{A}$ be a finite test set. For each $a \in \mathcal{A}$, let $w_a \in \mathbb{R}^{m+n}$, $T_a > 0$, and $\sigma_a > 0$. Assume

$$\sum_{t=1}^s q(\Delta_t, w_a)^2 \leq \sigma_a^2 \quad (a \in \mathcal{A}).$$

Let $\varepsilon_1, \dots, \varepsilon_s$ be exact independent Rademacher signs. No additional computed quantity is randomized or charged in this statement. Then, under the exact Rademacher law,

$$\mathbb{P} \left\{ \forall a \in \mathcal{A}, \quad \left| q \left(\sum_{t=1}^s \varepsilon_t \Delta_t, w_a \right) \right| \leq T_a \right\} \geq 1 - \sum_{a \in \mathcal{A}} 2 \exp \left(-\frac{T_a^2}{2\sigma_a^2} \right).$$

This corollary is exact-probability and exact-arithmetic only. The traces S, S' , the matrices Δ_t , and the test vectors w_a are fixed analysis objects; the only randomness in the statement is the exact Rademacher sign law. No floating-point computation or computed non-probability quantity is introduced. The corollary is the Algorithm 1 finite-test Rademacherization adapter for future copy-difference Khintchine/Bernstein work; it is not yet the spectral-norm probability tail for the all-copy-differences event.

Lean formalization.

```
Algorithms/RandNLA/Preconditioning.lean    finiteQuadraticForm_rademacher_signed_matrix_
sum_eq_sum_fintype

rademacherTraceProbability_eventProb_forall_abs_finiteQuadraticForm_signed_matrix_sum_fintype_le_
ge_one_sub_sum_two_mul_exp_neg_sq_div

Algorithms/RandNLA/ElementwiseSpectral.lean sqMagTraceProbability_eventProb_forall_abs_
finiteQuadraticForm_rademacher_signed_rectSelfAdjointDilation_sampleResidualIncrement_diff_le_ge_
one_sub_sum_two_mul_exp_neg_sq_div
```

Corollary 18 (Finite-cover Rademacher operator event for fixed Algorithm 1 copy differences). *Use the notation of the preceding corollary. Let $\mathcal{K} = \{1, \dots, m\} \sqcup \{1, \dots, n\}$, and view each Δ_t as an exact self-adjoint matrix on $\mathbb{R}^{\mathcal{K}}$. Let $\mathcal{A}$ be a finite index set and let $w_a \in \mathbb{R}^{\mathcal{K}}$ ($a \in \mathcal{A}$) be an exact*

finite cover of the Euclidean unit ball at radius $\rho \geq 0$: for every $x \in \mathbb{R}^{\mathcal{K}}$ with $\|x\|_2 \leq 1$, some $a \in \mathcal{A}$ satisfies $\|x - w_a\|_2 \leq \rho$. Let $\eta > 0$, let $L \geq 0$, and let $\sigma_a > 0$ for $a \in \mathcal{A}$. Define

$$D(\varepsilon) = \sum_{t=1}^s \varepsilon_t \Delta_t, \quad R(\varepsilon) = \sum_{t=1}^s \varepsilon_t (Z_t(S_t) - Z_t(S'_t)).$$

Thus $D(\varepsilon)$ is the exact self-adjoint dilation of $R(\varepsilon)$. Assume the exact deterministic coarse radius

$$\|D(\varepsilon)\|_{2 \rightarrow 2} \leq L \quad \text{for every exact sign outcome } \varepsilon,$$

and the exact finite-test variance proxies

$$\sum_{t=1}^s q(\Delta_t, w_a)^2 \leq \sigma_a^2 \quad (a \in \mathcal{A}).$$

Then, under the exact Rademacher sign law,

$$\mathbb{P}\{\|R(\varepsilon)\|_{2 \rightarrow 2} \leq \eta + L(2\rho + \rho^2)\} \geq 1 - \sum_{a \in \mathcal{A}} 2 \exp\left(-\frac{\eta^2}{2\sigma_a^2}\right).$$

This is an exact-law and exact-arithmetic cover upgrade from finite quadratic-form tests to a rectangular operator event for fixed signed copy-difference sums. No floating-point computation, stored matrix, computed basis, or probability-construction error appears. The coarse radius L is a visible deterministic exact hypothesis, not a computed certificate and not a probability event. Consequently this corollary is intermediate infrastructure; it is not the final source-uniform Algorithm 1 spectral tail or the CACM equation (2) theorem.

Lean formalization.

```
Analysis/MatrixAlgebra.lean finiteUnitBallCover
finiteLoewnerLe_of_finite_unit_ball_cover_quadraticForm
Algorithms/RandNLA/ElementwiseSpectral.lean      rademacherTraceProbability_eventProb_
rectOpNorm2Le_signed_sampleResidualIncrement_diff_ge_one_sub_sum_two_mul_exp_neg_sq_div_of_
finiteUnitBallCover
```

Theorem 19 (Small-entry obstruction for the literal all-copy-differences event). *Let $L \in \mathbb{R}$ be any proposed rectangular operator radius and define the exact literal Algorithm 1 input*

$$A_L = \begin{bmatrix} 1 & \alpha \end{bmatrix} \in \mathbb{R}^{1 \times 2}, \quad \alpha = (|L| + 2)^{-1}.$$

Let p_{01} denote the exact squared-magnitude sampling probability of the tiny entry $A_{0,1} = \alpha$. Then $p_{01} > 0$. Let S_{tiny} be the exact one-step trace that samples entry $(0, 1)$, and let S_{unit} be the exact one-step trace that samples entry $(0, 0)$. If $R(S) = A_L - \tilde{A}(S)$ denotes the exact one-step residual, then

$$\neg(\|R(S_{\text{tiny}}) - R(S_{\text{unit}})\|_{2 \rightarrow 2} \leq L).$$

Equivalently, S_{tiny} is not in the all-copy-differences event

$$\mathcal{E}_{\text{allcopy}}(A_L, L) = \{S : \forall S', \|R(S) - R(S')\|_{2 \rightarrow 2} \leq L\}.$$

Consequently, under the exact one-step squared-magnitude law for A_L ,

$$p_{01} \leq \mathbb{P}\{S : S \notin \mathcal{E}_{\text{allcopy}}(A_L, L)\}.$$

Equivalently, for every $\delta \in \mathbb{R}$, any lower bound

$$\mathbb{P}\{S : S \in \mathcal{E}_{\text{allcopy}}(A_L, L)\} \geq 1 - \delta$$

forces

$$p_{01} \leq \delta.$$

In particular, the all-copy-differences support event has positive failure probability for this literal untruncated input.

This is an exact-law and exact-arithmetic obstruction result. It introduces no floating-point computation and no computed non-probability quantity. It does not say that every possible high-probability spectral tail is false; it says that the literal untruncated all-copy support event cannot be treated as a probability-one prerequisite, nor can the missing Algorithm 1 equation (2) tail be closed by hiding a uniform deterministic copy-difference radius.

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSpectral.lean    algorithm1SmallEntrySupportMatrix_trace_
residual_small_unit_diff_not_rectOpNorm2Le

sqMagTraceProbability_eventProb_not_algorithm1ExactAllCopyDiffSpectralEvent_smallEntry_ge
sqMagTraceProbability_eventProb_not_algorithm1ExactAllCopyDiffSpectralEvent_smallEntry_pos
sqMagTraceProbability_algorithm1ExactAllCopyDiffSpectralEvent_smallEntry_delta_ge
```

Theorem 20 (Sample-count obstruction for the literal exact spectral event). *Let $s \geq 1$ be the Algorithm 1 sample count and also the update denominator used in the exact rescaling. Let $L \in \mathbb{R}$ be any proposed rectangular operator radius. Define*

$$A_L = \begin{bmatrix} 1 & \alpha \end{bmatrix} \in \mathbb{R}^{1 \times 2}, \quad \alpha = (|L| + 2)^{-1},$$

and let

$$p_{01} = \frac{\alpha^2}{1 + \alpha^2}$$

be the exact squared-magnitude probability of the tiny entry $A_{0,1}$. Let $S_{\text{tiny}}^{(s)}$ be the exact length- s trace that samples $(0, 1)$ at every step. For any exact length- s trace S , write

$$R_s(S) = A_L - \tilde{A}_s(S)$$

for the exact Algorithm 1 residual formed with denominator s . Then

$$\neg(\|R_s(S_{\text{tiny}}^{(s)})\|_{2 \rightarrow 2} \leq L).$$

Consequently, under the exact independent length- s squared-magnitude law,

$$p_{01}^s \leq \mathbb{P}\{S : \|R_s(S)\|_{2 \rightarrow 2} > L\}.$$

Equivalently, the exact success probability satisfies the upper bound

$$\mathbb{P}\{S : \|R_s(S)\|_{2 \rightarrow 2} \leq L\} \leq 1 - p_{01}^s < 1.$$

Equivalently, if the exact spectral event

$$\mathcal{E}_{\text{spec}}(A_L, L, s) = \{S : \|R_s(S)\|_{2 \rightarrow 2} \leq L\}$$

is claimed to satisfy

$$\mathbb{P}\{S : S \in \mathcal{E}_{\text{spec}}(A_L, L, s)\} \geq 1 - \delta,$$

then the failure budget must obey

$$p_{01}^s \leq \delta.$$

For $\delta > 0$, this is equivalently bounded in logarithmic sample-count form by the proved inequality

$$\log(1/\delta) \leq s \log(1/p_{01}) = s \log\left(\frac{1 + \alpha^2}{\alpha^2}\right).$$

Thus any such claimed literal exact spectral theorem must, at minimum, have

$$s \geq \frac{\log(1/\delta)}{\log((1 + \alpha^2)/\alpha^2)} = \Omega_L(\log(1/\delta)),$$

where the subscript means L , hence $\alpha = (|L| + 2)^{-1}$, is fixed. The divided inequality is formalized directly: $0 < p_{01} < 1$, so the logarithmic denominator is positive and no additional analytic assumption is introduced.

Equivalently, the same Lean development proves the contrapositive forms: if

$$\delta < p_{01}^s,$$

then the displayed $1 - \delta$ lower bound for $\mathcal{E}_{\text{spec}}(A_L, L, s)$ cannot hold. If $\delta > 0$ and

$$s < \frac{\log(1/\delta)}{\log((1 + \alpha^2)/\alpha^2)},$$

then the same $1 - \delta$ lower bound also cannot hold. These are exact arithmetic contradictions derived from the exact product-law mass of the all-tiny trace; no floating-point operation or probability-construction error appears in the statement.

The same exact product-mass argument also gives a fixed-sample existential form: for every positive s and every L , there is a positive δ , for instance $\delta = p_{01}^s/2$, for which the claimed $1 - \delta$ lower bound cannot hold.

The formalization also includes a concrete source-budget witness. For

$$A = \begin{bmatrix} 1 & 1/102 \end{bmatrix}, \quad s = 1, \quad L = 100, \quad \delta = 1/30000,$$

with $m = 1$, $n = 2$, and

$$C = 2 \max\{1, 2\} + \frac{4\sqrt{2}}{3}\sqrt{2},$$

the rectangular source-style budget

$$4C\|A\|_F^2 \log\left(\frac{2(1+2)}{1/30000}\right) \leq 100^2$$

is true in exact arithmetic, while the exact lower bound

$$\mathbb{P}\{S : \|R_1(S)\|_{2 \rightarrow 2} \leq 100\} \geq 1 - \frac{1}{30000}$$

is formally impossible for the literal squared-magnitude law. Writing $\mathcal{B}(A)$ for the displayed source-style budget at these fixed parameters with A in place of the concrete witness, the formalization also proves

$$\neg \left[\forall A, \text{sqMagProbDen}(A) > 0 \wedge \mathcal{B}(A) \implies \mathbb{P}\{S : \|R_1(S)\|_{2 \rightarrow 2} \leq 100\} \geq 1 - \frac{1}{30000} \right].$$

Thus the failure is a formal refutation of the source-budget-only implication at these fixed parameters, not just a warning about the displayed witness.

This is still an exact-law and exact-arithmetic obstruction, not a floating-point theorem. It is stronger than the previous all-copy support obstruction because it applies directly to the exact spectral event itself. It does not rule out a different matrix-tail proof, but it shows that any literal untruncated source-uniform theorem must pay at least this all-tiny trace mass, or else expose reciprocal-entry, truncation, or no-small-entry dependence.

Lean formalization.

```

Analysis/FiniteProbability.lean FiniteProbability.prob_le_eventProb_of_mem
Algorithms/RandNLA/ElementwiseSpectral.lean sqMagTraceProbMass_
algorithm1SmallEntrySupportMatrix_all_tiny
algorithm1SmallEntrySupportMatrix_all_tiny_trace_residual_not_rectOpNorm2Le
sqMagTraceProbability_eventProb_not_algorithm1ExactSpectralEvent_all_tiny_smallEntry_ge
sqMagTraceProbability_algorithm1ExactSpectralEvent_all_tiny_smallEntry_delta_ge
sqMagProb_algorithm1SmallEntrySupportMatrix_small_lt_one
log_inv_delta_le_nat_mul_log_inv_of_pow_le
log_inv_delta_div_log_inv_le_nat_of_pow_le
sqMagTraceProbability_algorithm1ExactSpectralEvent_all_tiny_smallEntry_log_delta_le
sqMagTraceProbability_algorithm1ExactSpectralEvent_all_tiny_smallEntry_sample_count_ge
sqMagTraceProbability_not_algorithm1ExactSpectralEvent_all_tiny_smallEntry_of_delta_lt_pow
sqMagTraceProbability_not_algorithm1ExactSpectralEvent_all_tiny_smallEntry_of_sample_count_lt
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_all_tiny_smallEntry_le_one_sub_pow
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_all_tiny_smallEntry_lt_one
exists_delta_not_sqMagTraceProbability_algorithm1ExactSpectralEvent_all_tiny_smallEntry
real_log_180000_le_18
algorithm1RectSourceBudget_1_2_100_one_div_30000
algorithm1SmallEntrySupportMatrix_100_small_prob_gt_one_div_30000
algorithm1SmallEntrySupportMatrix_100_rect_source_budget_one_div_30000
sqMagTraceProbability_not_algorithm1ExactSpectralEvent_rect_source_budget_witness
not_forall_algorithm1ExactSpectralEvent_of_rect_source_budget_one_div_30000

```

Lemma 21 (Unbiased Bernoulli coordinate law). *Let $B = \{0, 1\}$ and let ν be the unbiased probability law on B : $\nu(\{0\}) = \nu(\{1\}) = 1/2$. For every $X : B \rightarrow \mathbb{R}$,*

$$\mathbb{E}_\nu X = \frac{X(0) + X(1)}{2}.$$

For every $Z : B \rightarrow \mathbb{R}$, the finite entropy functional satisfies

$$\text{Ent}_\nu(Z) = \frac{Z(0) \log Z(0) + Z(1) \log Z(1)}{2} - \frac{Z(0) + Z(1)}{2} \log \left(\frac{Z(0) + Z(1)}{2} \right).$$

Lean formalization.

```
Analysis/FiniteProbability.lean FiniteProbability.boolUniformProbability
FiniteProbability.boolUniformProbability_prob
FiniteProbability.boolUniformProbability_expectationReal
FiniteProbability.entropyReal_boolUniformProbability_eq
```

Lemma 22 (Two-point Bernoulli log-Sobolev inequality). *For $a, b > 0$, the two-point entropy satisfies*

$$\frac{a \log a + b \log b}{2} - \frac{a+b}{2} \log \left(\frac{a+b}{2} \right) \leq \frac{(a-b)^2}{2(a+b)}.$$

Consequently, if $g : B \rightarrow (0, \infty)$, then

$$\text{Ent}_\nu(g^2) \leq (g(1) - g(0))^2.$$

Lean formalization.

```
Analysis/FiniteProbability.lean FiniteProbability.twoPointEntropy_le_sq_sub_div_of_pos
FiniteProbability.entropyReal_boolUniformProbability_sq_le_sq_sub_of_pos
```

Lemma 23 (One-coordinate Bernoulli tensorization peel-off). *Let P be any finite probability law on Ω , let ν be the unbiased Bernoulli law on $B = \{0, 1\}$, and let $g : \Omega \times B \rightarrow (0, \infty)$. Then*

$$\text{Ent}_{P \otimes \nu}(g^2) \leq \mathbb{E}_P \left[(g(\omega, 1) - g(\omega, 0))^2 \right] + \text{Ent}_P(\mathbb{E}_\nu g(\omega, \cdot)^2).$$

Lean formalization.

```
Analysis/FiniteProbability.lean FiniteProbability.entropyReal_prod_boolUniformProbability_sq_le_coordinate_add_entropy
```

Lemma 24 (Finite L^2 section-norm bridge). *Let P be a finite probability law on Ω . For real-valued functions $X, Y : \Omega \rightarrow \mathbb{R}$, the finite $L^2(P)$ seminorm satisfies the reverse triangle inequality*

$$\left| \sqrt{\mathbb{E}_P X^2} - \sqrt{\mathbb{E}_P Y^2} \right| \leq \sqrt{\mathbb{E}_P (X - Y)^2}.$$

In particular, for the unbiased Bernoulli coordinate law ν ,

$$\left| \sqrt{\mathbb{E}_\nu u^2} - \sqrt{\mathbb{E}_\nu v^2} \right| \leq \sqrt{\mathbb{E}_\nu (u - v)^2}.$$

Lean formalization.

```
Analysis/FiniteProbability.lean FiniteProbability.expectationReal_sq_nonneg
FiniteProbability.abs_expectationReal_mul_le_sqrt_mul_sqrt
FiniteProbability.sqrt_expectationReal_sq_add_le
FiniteProbability.abs_sqrt_expectationReal_sq_sub_le_sqrt_expectationReal_sub_sq
FiniteProbability.boolUniformProbability_abs_sqrt_expectationReal_sq_sub_le_sqrt_expectationReal_sub_sq
```

Lemma 25 (Abstract Bernoulli-product induction lift). *Let P be a finite probability law on Ω , let ν be the unbiased Bernoulli law, and let $\{T_i : \Omega \rightarrow \Omega\}_{i \in I}$ be a finite family of coordinate moves. Assume that P satisfies the entropy-gradient bound*

$$\text{Ent}_P(h^2) \leq \sum_{i \in I} \mathbb{E}_P[(h(\omega) - h(T_i \omega))^2]$$

for every positive $h : \Omega \rightarrow (0, \infty)$. Then every positive $g : \Omega \times \{0, 1\} \rightarrow (0, \infty)$ satisfies

$$\begin{aligned} \text{Ent}_{P \otimes \nu}(g^2) &\leq \mathbb{E}_P[(g(\omega, 1) - g(\omega, 0))^2] \\ &\quad + \sum_{i \in I} \mathbb{E}_{P \otimes \nu}[(g(\omega, b) - g(T_i \omega, b))^2]. \end{aligned}$$

Lean formalization.

```
Analysis/FiniteProbability.lean FiniteProbability.entropyReal_prod_boolUniformProbability_
sq_le_lifted_diff_sum_add
```

Theorem 26 (Concrete Rademacher-cube entropy-gradient inequality). *Let $\Omega_m = \{0, 1\}^m$ with the uniform Rademacher law μ_m . For $\omega \in \Omega_m$, let $\omega^{(i)}$ denote the trace obtained by flipping coordinate i . If $g : \Omega_m \rightarrow (0, \infty)$, then*

$$\text{Ent}_{\mu_m}(g^2) \leq \sum_{i=1}^m \mathbb{E}_{\mu_m}[(g(\omega) - g(\omega^{(i)}))^2].$$

Lean formalization.

```
Algorithms/RandNLA/Preconditioning.lean rademacherTraceFlip
rademacherTraceProbMass_snoc
rademacherTraceProbability_expectationReal_succ_last_eq
rademacherTraceProbability_expectationReal_succ_eq_prod
rademacherTraceProbability_entropyReal_succ_eq_prod
rademacherTraceFlip_castSucc_snoc
rademacherTraceFlip_last_snoc
rademacherTraceProbability_entropyReal_sq_le_sum_flip
```

Corollary 27 (Conditional exponential-tilt reduction on the Rademacher cube). *Let $X : \Omega_m \rightarrow \mathbb{R}$, let $\lambda \in \mathbb{R}$, and let $c \in \mathbb{R}$. Assume the deterministic tilt flip-gradient estimate*

$$\sum_{i=1}^m \mathbb{E}_{\mu_m} \left[\left(\exp\{\lambda X(\omega)/2\} - \exp\{\lambda X(\omega^{(i)})/2\} \right)^2 \right] \leq c\lambda^2 \mathbb{E}_{\mu_m} \exp\{\lambda X(\omega)\}.$$

Then

$$\text{Ent}_{\mu_m}(\exp\{\lambda X\}) \leq c\lambda^2 \mathbb{E}_{\mu_m} \exp\{\lambda X\}.$$

Consequently, if $\sigma > 0$ and, for every $\lambda > 0$, the displayed flip-gradient estimate holds with $c = \sigma^2/2$, then for every $t > 0$,

$$\mu_m\{X \leq \mathbb{E}_{\mu_m} X + t\} \geq 1 - \exp\left\{-\frac{t^2}{2\sigma^2}\right\}.$$

Lean formalization.

```
Algorithms/RandNLA/Preconditioning.lean rademacherTraceProbability_entropyReal_exp_mul_
le_of_flip_tilt_sq_sum_bound
rademacherTraceProbability_eventProb_real_le_mean_add_ge_one_sub_exp_sq_of_flip_tilt_sq_sum_bound
```

Lemma 28 (Scalar and finite-cube tilt-gradient symmetrization). *For real a, b ,*

$$\left(e^{a/2} - e^{b/2}\right)^2 \leq 2 \left(\frac{a}{2} - \frac{b}{2}\right)^2 (e^a + e^b).$$

Consequently, on the finite Rademacher cube, if coefficients $L_i \geq 0$ satisfy

$$|X(\omega) - X(\omega^{(i)})| \leq L_i$$

for every coordinate flip, and if

$$2 \sum_i \frac{\lambda^2}{2} L_i^2 \leq c \lambda^2,$$

then the flip-gradient hypothesis in Corollary 27 holds with that c .

Lean formalization.

```
Analysis/FiniteProbability.lean real_exp_sub_one_le_mul_exp
real_abs_exp_sub_exp_le_abs_sub_mul_exp_add_exp
real_exp_half_sub_sq_le_two_mul_half_diff_sq
Algorithms/RandNLA/Preconditioning.lean rademacherTraceProbability_expectationReal_flip
rademacherTraceProbability_flip_tilt_sq_sum_bound_of_pointwise_halfdiff_sq_le
rademacherTraceProbability_flip_tilt_sq_sum_bound_of_pointwise_absdiff_le
```

Lemma 29 (Finite subgaussian Chernoff tail). *Let X be a real random variable on a finite probability space. Suppose that for every $\lambda > 0$,*

$$\mathbb{E} \exp\{\lambda(X - \mu)\} \leq \exp\{\lambda^2 \sigma^2 / 2\}, \quad \sigma > 0.$$

Then for every $t > 0$,

$$\mathbb{P}\{X \leq \mu + t\} \geq 1 - \exp\left(-\frac{t^2}{2\sigma^2}\right).$$

Lean formalization.

```
Analysis/FiniteProbability.lean FiniteProbability.eventProb_real_le_ge_one_sub_exp_of_mgf_
bound
FiniteProbability.eventProb_real_le_mean_add_ge_one_sub_exp_sq_of_subgaussian_mgf
```

Lemma 30 (Finite spectral Loewner bridge). *Let $M \in \mathbb{R}^{d \times d}$ be symmetric, and let $\lambda_1(M), \dots, \lambda_d(M)$ denote the Hermitian eigenvalues supplied by mathlib. The formalization proves*

$$M \succeq 0 \iff \lambda_i(M) \geq 0 \text{ for all } i.$$

More generally, for symmetric M, N ,

$$M \preceq N \iff \lambda_i(N - M) \geq 0 \text{ for all } i.$$

In particular,

$$M \preceq LI \iff \lambda_i(LI - M) \geq 0 \text{ for all } i.$$

This is the deterministic endpoint needed by a future largest-eigenvalue tail bound. It does not give a probability estimate for the event $M \preceq LI$.

Lean formalization.

```
finitePSD_iff_finiteHermitianEigenvalues_nonneg
finiteLoewnerLe_iff_sub_finiteHermitianEigenvalues_nonneg
finiteScalarUpperDiffEigenvalues
finiteLoewnerLe_smul_id_iff_finiteScalarUpperDiffEigenvalues_nonneg
finiteLoewnerLe_smul_id_iff_sub_finiteHermitianEigenvalues_nonneg
```

Lemma 31 (Scalar matrix-exponential normalization). *Let I_d be the $d \times d$ identity matrix and let $L \in \mathbb{R}$. The formalization defines a repository-native matrix exponential $\text{Exp}(M)$ by routing finite matrices through mathlib's matrix exponential, and proves*

$$\text{Exp}(LI_d) = e^L I_d, \quad \text{tr}(\text{Exp}(LI_d)) = de^L.$$

For a diagonal matrix $\text{diag}(v)$, it also proves

$$\text{Exp}(\text{diag}(v)) = \text{diag}(i \mapsto e^{v_i}), \quad \text{tr}(\text{Exp}(\text{diag}(v))) = \sum_i e^{v_i}.$$

The same formal layer proves that $\text{Exp}(M)$ is symmetric when M is symmetric. This is the scalar normalization term that occurs in trace-MGF and matrix-Bernstein proofs. It is not a trace-exponential domination theorem and does not by itself imply a largest-eigenvalue tail bound.

Lean formalization.

```
finiteDiagonal
finiteMatrixExp
finiteMatrixExp_smul_finiteIdMatrix
finiteTrace_finiteMatrixExp_smul_finiteIdMatrix
finiteMatrixExp_symmetric
finiteMatrixExp_finiteDiagonal
finiteTrace_finiteMatrixExp_finiteDiagonal
```

Lemma 32 (Hermitian spectral-calculus trace diagonalization). *Let $M \in \mathbb{R}^{d \times d}$ be symmetric, and let $\lambda_1(M), \dots, \lambda_d(M)$ be the Hermitian eigenvalues used by the repository. The formalization defines the Hermitian continuous-functional-calculus exponential $E_{\text{cfc}}(M)$ and proves*

$$\text{tr}(E_{\text{cfc}}(M)) = \sum_{i=1}^d e^{\lambda_i(M)}.$$

This is the diagonalized trace identity normally used inside trace-MGF arguments. It is still not a trace-exponential domination theorem for random matrix sums, and it does not close equation (2).

Lean formalization.

```
finiteHermitianCfcExp
finiteTrace_finiteHermitianCfcExp_eq_sum_exp_finiteHermitianEigenvalues
```

Lemma 33 (Power-series trace diagonalization). *Let $M \in \mathbb{R}^{d \times d}$ be symmetric, and let $\text{Exp}(M)$ denote the repository-native matrix exponential defined by `mathlib`'s power-series matrix exponential. Then*

$$\text{tr}(\text{Exp}(M)) = \sum_{i=1}^d e^{\lambda_i(M)}.$$

It also proves the lower-tail companion

$$\text{tr}(\text{Exp}(-M)) = \sum_{i=1}^d e^{-\lambda_i(M)}.$$

The proof diagonalizes M by the Hermitian spectral theorem, applies invariance of the matrix exponential under unitary conjugation, and then uses trace cyclicity and the diagonal exponential formula. This closes the power-series trace-diagonalization dependency. It is still not a trace-exponential domination theorem for random matrix sums, and it does not close equation (2).

Lean formalization.

```
finiteTrace_finiteMatrixExp_eq_sum_exp_finiteHermitianEigenvalues
finiteTrace_finiteMatrixExp_neg_eq_sum_exp_neg_finiteHermitianEigenvalues
```

Lemma 34 (Scalar Loewner bound to trace-exponential bound). *If $M \in \mathbb{R}^{d \times d}$ is symmetric and*

$$M \preceq cI,$$

then every Hermitian eigenvalue satisfies $\lambda_i(M) \leq c$, and hence

$$\text{tr}(\text{Exp}(M)) \leq de^c.$$

The lower-tail companion is stated as

$$-M \preceq cI \implies \text{tr}(\text{Exp}(-M)) \leq de^c.$$

This is the deterministic scalarization step that a future matrix-CGF Loewner estimate can feed into the trace-MGF interface.

Lean formalization.

```
finiteQuadraticForm_finiteHermitianEigenvector_eq_eigenvalue_mul_norm_sq
finiteVecNorm2Sq_finiteHermitianEigenvector_eq_one
finiteHermitianEigenvalues_le_of_finiteLoewnerLe_smul_id
finiteTrace_finiteMatrixExp_le_card_mul_exp_of_finiteLoewnerLe_smul_id
finiteTrace_finiteMatrixExp_neg_le_card_mul_exp_of_neg_finiteLoewnerLe_smul_id
```

Lemma 35 (Trace-exponential eigenvalue tail interface). *Let $M(\omega) \in \mathbb{R}^{d \times d}$ be a finite random symmetric matrix family and let $T \in \mathbb{R}$. The formalization proves*

$$\mathbb{P}\{\exists i : \lambda_i(M(\omega)) \geq T\} \leq e^{-T} \mathbb{E}[\text{tr}(\text{Exp}(M(\omega)))].$$

Consequently, if

$$\mathbb{E}[\text{tr}(\text{Exp}(M(\omega)))] \leq B,$$

then

$$\mathbb{P}\{\exists i : \lambda_i(M(\omega)) \geq T\} \leq e^{-T} B.$$

The same formal layer also proves the high-probability complement

$$\mathbb{P}\{\forall i : \lambda_i(M(\omega)) < T\} \geq 1 - e^{-T} \mathbb{E}[\text{tr}(\text{Exp}(M(\omega)))],$$

and its scalar-bound version with B . These are useful final-mile forms for any future trace-MGF domination theorem. The lower-tail companion is also formalized:

$$\mathbb{P}\{\exists i : \lambda_i(M(\omega)) \leq T\} \leq e^T \mathbb{E}[\text{tr}(\text{Exp}(-M(\omega)))],$$

and therefore

$$\mathbb{P}\{\forall i : T < \lambda_i(M(\omega))\} \geq 1 - e^T \mathbb{E}[\text{tr}(\text{Exp}(-M(\omega)))],$$

again with scalar-bound versions. Combining the upper and lower complements gives the two-sided event

$$\mathbb{P}\{\forall i : |\lambda_i(M(\omega))| < T\} \geq 1 - e^{-T} \mathbb{E}[\text{tr}(\text{Exp}(-M(\omega)))] - e^{-T} \mathbb{E}[\text{tr}(\text{Exp}(M(\omega)))],$$

with a scalar-bound version obtained by replacing the two expectations by proved bounds B_- and B_+ . This is scalar Markov applied to the nonnegative random variable $\text{tr}(\text{Exp}(\pm M(\omega)))$, using the deterministic witnesses

$$\lambda_i(M) \geq T \implies e^T \leq \text{tr}(\text{Exp}(M)).$$

and

$$\lambda_i(M) \leq T \implies e^{-T} \leq \text{tr}(\text{Exp}(-M)).$$

It is a genuine MGF-to-eigenvalue bridge, but it still requires a separate trace-MGF domination theorem to bound the expectation on the right.

Lean formalization.

```
exp_le_finiteTrace_finiteMatrixExp_of_finiteHermitianEigenvalue_ge
finiteTrace_finiteMatrixExp_nonneg
exp_neg_le_finiteTrace_finiteMatrixExp_neg_of_finiteHermitianEigenvalue_le
finiteTrace_finiteMatrixExp_neg_nonneg
FiniteProbability.eventProb_exists_finiteHermitianEigenvalue_ge_le_exp_neg_mul_expected_trace_exp
FiniteProbability.eventProb_exists_finiteHermitianEigenvalue_ge_le_exp_neg_mul_trace_bound
FiniteProbability.eventProb_forall_finiteHermitianEigenvalue_lt_ge_one_sub_exp_neg_mul_expected_trace_exp
FiniteProbability.eventProb_forall_finiteHermitianEigenvalue_lt_ge_one_sub_exp_neg_mul_trace_bound
FiniteProbability.eventProb_exists_finiteHermitianEigenvalue_le_le_exp_mul_expected_trace_exp_neg
```

```

FiniteProbability.eventProb_exists_finiteHermitianEigenvalue_le_le_exp_mul_trace_bound
FiniteProbability.eventProb_forall_finiteHermitianEigenvalue_gt_ge_one_sub_exp_mul_expected_trace_exp_neg
FiniteProbability.eventProb_forall_finiteHermitianEigenvalue_gt_ge_one_sub_exp_mul_trace_bound
FiniteProbability.eventProb_forall_abs_finiteHermitianEigenvalue_lt_ge_one_sub_exp_neg_mul_expected_trace_exp_add
FiniteProbability.eventProb_forall_abs_finiteHermitianEigenvalue_lt_ge_one_sub_exp_neg_mul_trace_bound_add

```

Corollary 36 (Eigenvalue form of the dilation upper event). *Let $R = A - \tilde{A}$ be the exact Algorithm 1 residual and let*

$$D(R) = \begin{bmatrix} 0 & R \\ R^T & 0 \end{bmatrix}.$$

If $\varepsilon \geq 0$ and every Hermitian eigenvalue of $\varepsilon I - D(R)$ is nonnegative, then

$$D(R) \preceq \varepsilon I \quad \implies \quad \|R\|_2 \leq \varepsilon.$$

Consequently, any probability lower bound for this eigenvalue event transfers to the exact Algorithm 1 spectral residual event and, after the entrywise floating-point perturbation budget is supplied, to the floating-point spectral event. The formalization also includes a finite union-bound form: if each single eigenvalue inequality has failure budget δ_a , then the full eigenvalue event has failure budget $\sum_a \delta_a$. This corollary is only an event adapter: it does not prove the high-probability eigenvalue event or the single-eigenvalue scalar tail bounds.

Lean formalization.

```

algorithm1ExactDilationEigenUpperEvent
algorithm1ExactDilationEigenUpperIndexEvent
algorithm1ExactDilationEigenUpperEvent_subset_exactDilationUpperEvent
algorithm1ExactDilationEigenUpperEvent_subset_exactSpectralEvent
probability_algorithm1_exact_spectral_of_dilation_eigen_upper
probability_algorithm1_exact_dilation_eigen_upper_of_index_bounds
probability_algorithm1_exact_spectral_of_dilation_eigen_upper_index_bounds
probability_algorithm1_fl_spectral_of_exact_dilation_eigen_upper
probability_algorithm1_fl_spectral_of_exact_dilation_eigen_upper_index_bounds

```

Remark (External concentration source chain). The next matrix-concentration target is guided by Tropp, *User-friendly tail bounds for sums of random matrices*, Foundations of Computational Mathematics 12 (2012), 389–434, <https://arxiv.org/abs/1004.4389>. The relevant source locations are Theorem 3.2 (Lieb trace concavity), Corollary 3.3 (the one-step trace-expectation inequality), Theorem 3.6 (master tail bound for independent self-adjoint sums), Corollary 3.7 (trace-exponential tail bound), Theorem 1.4 (matrix Bernstein), and Theorem 1.6 (rectangular matrix Bernstein via self-adjoint dilation). Tropp’s monograph *An Introduction to Matrix Concentration Inequalities*, <https://tropp.caltech.edu/books/Tro14-Introduction-Matrix-preprint.pdf>, Theorem 8.1.1, records the same Lieb concavity theorem and expands the relative-entropy proof route. These references guided theorem design only: no Lean theorem in this repository cites them as a hypothesis. The source-aligned square theorem below formalizes the needed Tropp/Lieb-style

trace-MGF, Bernstein, and truncation steps locally. An older Ahlswede–Winter operator Chernoff route remains recorded in the ledger as an advisory alternative, but it is no longer the active route for the closed hard-thresholded source-variant theorem. The local symmetry, PSD, and Loewner predicates for finite square matrices are now connected to mathlib’s `Matrix.IsSymm`, `Matrix.IsHermitian`, and `Matrix.PosSemidef` APIs, so the next formal trace-exponential or largest-eigenvalue proof can use mathlib’s spectral and matrix-order vocabulary without changing the RandNLA algorithm statements. A small spectral bridge also names the Hermitian eigenvalues of a local symmetric finite matrix, proves trace equals their sum, and identifies local PSD and scalar-identity Loewner upper bounds with nonnegativity of the Hermitian eigenvalues of the corresponding difference matrices. The same bridge now includes the scalar matrix-exponential normalization $\text{tr}(\exp(LI)) = de^L$ and preservation of symmetry by matrix exponentiation, plus the diagonal identity $\text{tr}(\exp(\text{diag}(v))) = \sum_i e^{v_i}$. It also includes the Hermitian spectral-calculus identity $\text{tr}(E_{\text{cfc}}(M)) = \sum_i e^{\lambda_i(M)}$, and the repository-native matrix-exponential trace identity $\text{tr}(\text{Exp}(M)) = \sum_i e^{\lambda_i(M)}$. These bridges are foundational only; they are not themselves concentration theorems. For source alignment, Drineas–Zouzias, *A Note on Element-wise Matrix Sparsification via a Matrix-valued Bernstein Inequality*, arXiv:1006.0407, Algorithm 1 and Theorem 1, uses a truncated matrix $\hat{A}$: small entries are zeroed out before sampling from $\hat{A}_{ij}^2 / \|\hat{A}\|_F^2$. The deterministic truncation transfer layer, the matrix-Bernstein half-budget event for sampling $\hat{A}$, and the support-aware floating-point transfer are formalized below. The untruncated user-requested law $A_{ij}^2 / \|A\|_F^2$ remains a separate theorem target.

Lemma 37 (Operator-log monotonicity bridge). *For complex $d \times d$ C^* -matrices X, Y , if $0 < X \preceq Y$, then*

$$\log X \preceq \log Y.$$

Lean formalization.

```
Analysis/OperatorLog.lean cstarMatrix_log_le_log
```

Lemma 38 (Self-adjoint logarithm–exponential inverse). *Let X be a self-adjoint complex $d \times d$ C^* -matrix. Then*

$$\log(\exp X) = X.$$

The same identity is formalized both for the normed-algebra exponential and for the real and complex continuous-functional-calculus exponentials:

$$\log(\exp_{\text{cfc}, \mathbb{C}} X) = X, \quad \log(\exp_{\text{cfc}, \mathbb{R}} X) = X.$$

Lean formalization.

```
Analysis/OperatorLog.lean cstarMatrix_normedSpaceExp_isTopologicalRing
cstarMatrix_normedRingExp_isTopologicalRing
cstarMatrix_realContinuousFunctionalCalculus
cstarMatrix_log_normedSpace_exp_of_isSelfAdjoint
cstarMatrix_log_cfc_complex_exp_of_isSelfAdjoint
cstarMatrix_log_cfc_real_exp_of_isSelfAdjoint
```

Lemma 39 (Self-adjoint matrix exponentials are strictly positive). *Let X be a self-adjoint complex $d \times d$ C^* -matrix. Then*

$$\exp X \succ 0.$$

The same strict-positivity statement is formalized for the normed-algebra exponential and for the real and complex continuous-functional-calculus exponentials:

$$\exp_{\text{cfc},\mathbb{C}}(X) \succ 0, \quad \exp_{\text{cfc},\mathbb{R}}(X) \succ 0.$$

Consequently, for self-adjoint H , the matrix

$$\exp_{\text{cfc},\mathbb{C}}(H + \log A)$$

appearing in the local Lieb trace functional is strictly positive.

Lean formalization.

```
Analysis/OperatorLog.lean cstarMatrix_normedSpace_exp_isStrictlyPositive_of_isSelfAdjoint
cstarMatrix_cfc_complex_exp_isStrictlyPositive_of_isSelfAdjoint
cstarMatrix_cfc_real_exp_isStrictlyPositive_of_isSelfAdjoint
Analysis/LiebTrace.lean liebTraceCfcExp_isStrictlyPositive
```

Theorem 40 (Scalar Bernstein parabola). *Let $a \geq 0$. For every $x \leq 1$,*

$$e^{ax} \leq 1 + ax + (e^a - a - 1)x^2.$$

Equivalently, if $\theta \geq 0$, $R > 0$, and $x \leq R$, then

$$e^{\theta x} \leq 1 + \theta x + \frac{e^{\theta R} - \theta R - 1}{R^2} x^2.$$

Lean formalization.

```
Analysis/LiebTrace.lean real_exp_quadratic_remainder_monotone
real_exp_tail_two_hasSum
real_exp_mul_le_quadratic_of_nonneg_of_le_one
real_exp_mul_le_quadratic_scaled_of_nonneg_of_pos_of_le
```

Lemma 41 (CFC Bernstein parabola lift). *Let X be a self-adjoint finite complex C^* -matrix and let $\theta, \beta \in \mathbb{R}$. Suppose the scalar inequality*

$$\exp(\theta x) \leq 1 + \theta x + \beta x^2 \quad \text{for every } x \in \sigma_{\mathbb{R}}(X)$$

holds on the real spectrum of X . Then the same inequality lifts to Loewner order:

$$\exp_{\text{cfc},\mathbb{R}}(\theta X) \preceq I + \theta X + \beta X^2.$$

The formalization also proves the normalization identity

$$\text{cfc}(x \mapsto 1 + \theta x + \beta x^2, X) = I + \theta X + \beta X^2.$$

Lean formalization.

```
Analysis/LiebTrace.lean cstarMatrix_cfc_quadratic_eq
cstarMatrix_cfc_real_exp_mul_le_quadratic_of_spectrum
cstarMatrix_cfc_real_exp_mul_le_bernstein_quadratic_of_spectrum_le
```

Theorem 42 (One-sample Bernstein log-CGF variance proxy). *Let X be a self-adjoint finite complex C^* -matrix-valued random variable on a finite probability space. Assume*

$$\mathbb{E}X = 0, \quad x \leq R \quad \text{for all } x \in \sigma_{\mathbb{R}}(X(\omega)), \quad \theta \geq 0, \quad R > 0.$$

Define

$$g(\theta, R) = \frac{e^{\theta R} - \theta R - 1}{R^2}, \quad V = \mathbb{E}[X^2].$$

Then

$$\log \mathbb{E} \exp(\theta X) \preceq g(\theta, R) V.$$

Equivalently, before taking logarithms,

$$\mathbb{E} \exp_{\text{cfc}}(\theta X) \preceq I + g(\theta, R) \mathbb{E}[X^2].$$

Lean formalization.

```
Analysis/CStarMatrixExpectation.lean FiniteProbability.expectationCStarMatrix_real_smul
Analysis/LiebTrace.lean cstarMatrix_real_smul_isSelfAdjoint
cstarMatrix_cfc_real_exp_mul_eq_normedSpace_exp_real_smul
cstarMatrix_selfAdjoint_mul_self_nonneg
cstarMatrix_one_add_le_normedSpace_exp_of_nonneg
cstarMatrix_log_one_add_le_self_of_nonneg
FiniteProbability.expectationCStarMatrix_nonneg_of_prob_pos
FiniteProbability.expectationCStarMatrix_mono_of_prob_pos
FiniteProbability.expectationCStarMatrix_cfc_real_exp_mul_le_bernstein_variance_proxy
FiniteProbability.cstarMatrix_log_expectationCStarMatrix_cfc_real_exp_mul_le_bernstein_variance_proxy
FiniteProbability.cstarMatrix_log_expectationCStarMatrix_normed_exp_real_smul_le_bernstein_variance_proxy
FiniteProbability.expectationCStarMatrix_cfc_real_exp_mul_le_bernstein_variance_proxy_of_prob_pos
FiniteProbability.cstarMatrix_log_expectationCStarMatrix_cfc_real_exp_mul_le_bernstein_variance_proxy_of_prob_pos
FiniteProbability.cstarMatrix_log_expectationCStarMatrix_normed_exp_real_smul_le_bernstein_variance_proxy_of_prob_pos
```

Corollary 43 (Algorithm 1 truncated one-sample log-CGF). *Let $\hat{A}$ be the hard-thresholded matrix used in the Drineas–Zouzias truncated element-wise sampler, let S be one squared-magnitude sample from $\hat{A}$, and define the one-sample residual increment*

$$Z(S) = s^{-1} \hat{A} - C(S).$$

Let $D(Z(S))$ be its self-adjoint dilation and assume $s > 0$, $\tau > 0$, and $\|\hat{A}\|_F^2 > 0$. Put

$$R_{\tau,s}(A) = \sqrt{2} \left(s^{-1} \|\hat{A}\|_F + \frac{\|\hat{A}\|_F^2}{s\tau} \right).$$

Then, for every $\theta \geq 0$,

$$\log \mathbb{E} \exp(\theta D(Z(S))) \preceq \frac{e^{\theta R_{\tau,s}(A)} - \theta R_{\tau,s}(A) - 1}{R_{\tau,s}(A)^2} \mathbb{E}[D(Z(S))^2].$$

Lean formalization.

```
Analysis/OperatorLog.lean   cstarMatrix_spectrum_le_of_le_real_smul_one
Algorithms/RandNLA/ElementwiseSpectral.lean   sqMagSampleProbability_
expectationCStarMatrix_rectSelfAdjointDilation_sampleResidualIncrement_eq_zero

sqMagSampleProbability_finiteComplex_rectSelfAdjointDilation_sampleResidualIncrement_truncated_
spectrum_le

sqMagSampleProbability_cstarMatrix_log_expectationCStarMatrix_normed_exp_rectSelfAdjointDilation_
truncated_sampleResidualIncrement_le
```

Lemma 44 (Algorithm 1 one-sample dilation variance proxy). *For the same one-sample residual increment $Z(S)$,*

$$\mathbb{E} D(Z(S))^2 \preceq 2mn \frac{\|\hat{A}\|_F^2}{s^2} I_{m+n}.$$

Here the expectation is under the squared-magnitude one-sample law for $\hat{A}$, and the order is the complex C^ -matrix Loewner order after embedding the real self-adjoint dilation.*

Lean formalization.

```
Analysis/CStarMatrixBridge.lean   finiteComplexCStarMatrix_mul
Analysis/MatrixAlgebra.lean   finiteMatMul_self_symmetric_of_symmetric
IsSymmetricFiniteMatrix.sum_smul

Algorithms/RandNLA/ElementwiseSpectral.lean   sqMagSampleProbability_
expectationCStarMatrix_rectSelfAdjointDilation_sampleResidualIncrement_square_le
```

Lemma 45 (Source-sharp square dilation variance proxy). *For square $n \times n$ input matrices, the Drineas–Zouzias route now has the sharper one-step proxy*

$$\mathbb{E} D(Z(S))^2 \preceq n \frac{\|\hat{A}\|_F^2}{s^2} I_{2n}.$$

This theorem avoids both the earlier Frobenius-detour factor $2n^2$ and the extra $\sqrt{2}$ support-radius loss.

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSpectral.lean   sqMagProb_sum_vecNorm2Sq_rectMatMulVec_
elementwiseSampleResidualIncrement_le_sharp

sqMagProb_sum_vecNorm2Sq_transposeRectMatMulVec_elementwiseSampleResidualIncrement_le_sharp

sqMagProb_sum_finiteQuadraticForm_rectSelfAdjointDilation_square_le_sharp_square

sqMagProb_sum_rectSelfAdjointDilation_square_loewnerLe_scalar_id_sharp_square

sqMagSampleProbability_expectationCStarMatrix_rectSelfAdjointDilation_sampleResidualIncrement_
square_le_sharp_square
```

Lemma 46 (Source-aligned rectangular dilation variance proxy). *Let $A \in \mathbb{R}^{m \times n}$, let $s > 0$, and let one Algorithm 1 sample be drawn from the exact squared-magnitude law*

$$p_{ij} = \frac{A_{ij}^2}{\|A\|_F^2}.$$

For a sample $S = (i, j)$, define the exact residual increment

$$Z(S) = s^{-1}A - \frac{A_{ij}}{s p_{ij}} e_i e_j^\top,$$

with the usual zero-probability branch used by the formalized sampler. Let

$$D(Z) = \begin{bmatrix} 0 & Z \\ Z^\top & 0 \end{bmatrix}$$

be the self-adjoint dilation. Then, in exact arithmetic,

$$\mathbb{E} D(Z(S))^2 \preceq \max\{m, n\} \frac{\|A\|_F^2}{s^2} I_{m+n}.$$

Equivalently, for every $x \in \mathbb{R}^{m+n}$,

$$\mathbb{E} x^\top D(Z(S))^2 x \leq \max\{m, n\} \frac{\|A\|_F^2}{s^2} \|x\|_2^2.$$

The same result is also formalized in the C-star matrix order used by the trace-MGF/log-CGF infrastructure. This closes the unconditional rectangular variance-proxy dependency for the literal Algorithm 1 law; it is not yet the final matrix-Bernstein/Khintchine tail theorem for CACM equation (2).

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSpectral.lean      sqMagProb_sum_finiteQuadraticForm_
rectSelfAdjointDilation_square_le_sharp_rect
sqMagProb_sum_rectSelfAdjointDilation_square_loewnerLe_scalar_id_sharp_rect
sqMagSampleProbability_expectationCStarMatrix_rectSelfAdjointDilation_sampleResidualIncrement_
square_le_sharp_rect
```

Lemma 47 (Source-aligned rectangular summed variance proxy). *Let $Z_1, \dots, Z_s$ be the exact independent Algorithm 1 residual increments under the squared-magnitude trace law above, and let $D(Z_t)$ be their self-adjoint dilations. In exact arithmetic,*

$$\mathbb{E} \sum_{t=1}^s D(Z_t)^2 \preceq \max\{m, n\} \frac{\|A\|_F^2}{s} I_{m+n}.$$

Equivalently, for every $x \in \mathbb{R}^{m+n}$,

$$\mathbb{E} \sum_{t=1}^s x^\top D(Z_t)^2 x \leq \max\{m, n\} \frac{\|A\|_F^2}{s} \|x\|_2^2.$$

The trace form is also proved:

$$\text{tr} \left(\mathbb{E} \sum_{t=1}^s D(Z_t)^2 \right) \leq \max\{m, n\} \frac{\|A\|_F^2}{s} (m + n).$$

This is the exact-law variance parameter expected by a rectangular self-adjoint-dilation Bernstein route. It still does not supply the missing unconditional largest-eigenvalue tail for CACM equation (2).

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSpectral.lean    sqMagProb_sum_steps_finiteQuadraticForm_
rectSelfAdjointDilation_square_le_sharp_rect

sqMagProb_sum_steps_rectSelfAdjointDilation_square_loewnerLe_scalar_id_sharp_rect

sqMagProb_sum_finiteTrace_rectSelfAdjointDilation_square_le_sharp_rect

sqMagProb_sum_steps_finiteTrace_rectSelfAdjointDilation_square_le_sharp_rect

sqMagTraceProbability_expectationReal_finiteQuadraticForm_rectSelfAdjointDilation_
sampleResidualIncrement_square_le_sharp_rect

sqMagTraceProbability_expectationReal_sum_finiteQuadraticForm_rectSelfAdjointDilation_
sampleResidualIncrement_square_le_sharp_rect

sqMagTraceProbability_sum_expectationReal_rectSelfAdjointDilation_square_loewnerLe_scalar_id_
sharp_rect
```

Lemma 48 (Literal untruncated bounded-increment obstruction). *The exact literal squared-magnitude law does not give a uniform bounded one-step support radius for all untruncated inputs. For every proposed scalar radius L , define the 1×2 exact matrix*

$$A_L = \begin{bmatrix} 1 & (|L| + 2)^{-1} \end{bmatrix}.$$

Its squared-magnitude denominator is positive, and the small entry is sampled with positive probability. If S is that small-entry sample, then the exact one-step residual increment with $s = 1$ satisfies

$$|Z(S)_{1,2}| = |L| + 2 > L,$$

and consequently the repository's rectangular operator support predicate

$$\text{rectOpNorm2Le}(Z(S), L)$$

is false. Moreover, under the exact one-step squared-magnitude product law for A_L , the event

$$\{\omega : \neg \text{rectOpNorm2Le}(A_L - \tilde{A}_L(\omega), L)\}$$

has strictly positive probability.

Remark. This is an exact-arithmetic route-elimination theorem, not a floating-point theorem and not the final CACM equation (2) tail bound. Sampling laws remain exact mathematical inputs. The result shows that the currently formalized support-aware Bernstein API cannot close the source-uniform literal untruncated Algorithm 1 theorem by inserting a uniform deterministic bounded-increment radius. The closed source-rate path must therefore keep its truncation/no-small-entry hypothesis, or the literal theorem must use a different concentration route. The nonconditional support-radius theorem below follows the latter principle by replacing the impossible uniform radius with an explicit reciprocal-entry radius.

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSpectral.lean algorithm1SmallEntrySupportMatrix

sqMagProbDen_algorithm1SmallEntrySupportMatrix_pos

sqMagProb_algorithm1SmallEntrySupportMatrix_small_pos

algorithm1SmallEntrySupportMatrix_residual_increment_abs_eq
```

```

algorithm1SmallEntrySupportMatrix_residual_increment_not_rect0pNorm2Le
exists_sqMagPositive_sampleResidualIncrement_entry_abs_gt
exists_sqMagPositive_sampleResidualIncrement_not_rect0pNorm2Le
sqMagTraceProbability_eventProb_not_algorithm1ExactSpectralEvent_smallEntry_pos

```

Corollary 49 (Truncated trace-MGF scalarization). *Let*

$$L_{\tau,s}(A) = \sqrt{2} \left(s^{-1} \|\hat{A}\|_F + \frac{\|\hat{A}\|_F^2}{s\tau} \right), \quad V_{\tau,s}(A) = 2mn \frac{\|\hat{A}\|_F^2}{s^2}.$$

For every $\theta \geq 0$, define

$$g(\theta, L) = \frac{\exp(\theta L) - \theta L - 1}{L^2}.$$

Then the iid trace-MGF expression for the positive dilation increments $\theta D(Z_t)$ satisfies

$$\text{tr} \exp \left(\sum_{t=1}^s K_t^+ \right) \leq (m+n) \exp(s g(\theta, L_{\tau,s}(A)) V_{\tau,s}(A)),$$

where $K_t^+ = \log \mathbb{E} \exp(\theta D(Z_t))$. The same bound holds for the negative increments $-\theta D(Z_t)$, with $K_t^- = \log \mathbb{E} \exp(-\theta D(Z_t))$.

Lean formalization.

```

Analysis/CStarMatrixTrace.lean      cstarMatrix_normedSpace_exp_le_real_smul_one_of_le_real_
smul_one

cstarMatrixTrace_normedSpace_exp_re_le_card_mul_exp_of_le_real_smul_one

Algorithms/RandNLA/ElementwiseTraceMGF.lean sqMagTraceProbabilityFiniteRealTraceMGFLogBound_
zero_le_card_mul_exp_of_log_le_real_smul_one

Algorithms/RandNLA/ElementwiseSpectral.lean sqMagSampleProbability_neg_finiteComplex_
rectSelfAdjointDilation_sampleResidualIncrement_truncated_spectrum_le

sqMagTraceProbabilityFiniteRealTraceMGFLogBound_rectSelfAdjointDilation_truncated_
sampleResidualIncrement_le

sqMagTraceProbabilityFiniteRealTraceMGFLogBound_neg_rectSelfAdjointDilation_truncated_
sampleResidualIncrement_le

```

Corollary 50 (Source-sharp rectangular truncated trace-MGF skeleton). *For rectangular $m \times n$ matrices, the positive and negative truncated trace-MGF scalarizations are also formalized with the sharper rectangular variance proxy*

$$V_{\tau,s}^{\text{rect}}(A) = \max\{m, n\} \frac{\|\hat{A}\|_F^2}{s^2}, \quad \hat{A} = \text{trunc}_{\tau}(A).$$

The support radius remains the generic truncated one

$$L_{\tau,s}^{\text{rect}}(A) = \sqrt{2} \left(s^{-1} \|\hat{A}\|_F + \frac{\|\hat{A}\|_F^2}{s\tau} \right).$$

Thus the two-sided scaled dilation eigenvalue event has an explicit $1 - \delta$ form with

$$B_{\text{rect}} = (m+n) \exp(s g(\theta, L_{\tau,s}^{\text{rect}}(A)) V_{\tau,s}^{\text{rect}}(A)), \quad T = \log(2B_{\text{rect}}/\delta).$$

This is exact-arithmetic, exact-probability matrix-concentration infrastructure. It is consumed below by the rectangular source-budget and floating-point gamma-transfer corollaries; it is still distinct from the literal untruncated CACM law when tiny nonzero entries are not hard-thresholded.

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSpectral.lean sqMagTraceProbabilityFiniteRealTraceMGFLogBound_
rectSelfAdjointDilation_truncated_sampleResidualIncrement_le_sharp_rect

sqMagTraceProbabilityFiniteRealTraceMGFLogBound_neg_rectSelfAdjointDilation_truncated_
sampleResidualIncrement_le_sharp_rect

sqMagTraceProbability_eventProb_forall_abs_finiteHermitianEigenvalue_scaled_
rectSelfAdjointDilation_truncatedTraceResidual_lt_ge_exp_sharp_rect

sqMagTraceProbability_eventProb_forall_abs_finiteHermitianEigenvalue_scaled_
rectSelfAdjointDilation_truncatedTraceResidual_lt_ge_one_sub_delta_sharp_rect
```

Corollary 51 (Source-sharp square trace-MGF scalarization). *For square $n \times n$ matrices, the same positive and negative trace-MGF scalarizations are now formalized with*

$$L_{\tau,s}^{\sharp}(A) = s^{-1} \|\widehat{A}\|_F + \frac{\|\widehat{A}\|_F^2}{s\tau}, \quad V_{\tau,s}^{\sharp}(A) = n \frac{\|\widehat{A}\|_F^2}{s^2}.$$

Consequently the two-sided truncated dilation eigenvalue event has an explicit $1 - \delta$ form with

$$B_{\sharp} = 2n \exp\left(s g(\theta, L_{\tau,s}^{\sharp}(A)) V_{\tau,s}^{\sharp}(A)\right), \quad T = \log(2B_{\sharp}/\delta).$$

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSpectral.lean sqMagTraceProbabilityFiniteRealTraceMGFLogBound_
rectSelfAdjointDilation_truncated_sampleResidualIncrement_le_sharp_square

sqMagTraceProbabilityFiniteRealTraceMGFLogBound_neg_rectSelfAdjointDilation_truncated_
sampleResidualIncrement_le_sharp_square

sqMagTraceProbability_eventProb_forall_abs_finiteHermitianEigenvalue_scaled_
rectSelfAdjointDilation_truncatedTraceResidual_lt_ge_exp_sharp_square

sqMagTraceProbability_eventProb_forall_abs_finiteHermitianEigenvalue_scaled_
rectSelfAdjointDilation_truncatedTraceResidual_lt_ge_one_sub_delta_sharp_square
```

Corollary 52 (Parameterized two-sided Bernstein tail skeleton). *For $T \in \mathbb{R}$ and $\theta \geq 0$,*

$$\mathbb{P}\left[\lambda_{\max}(\theta D(A - \tilde{A})) \geq T\right] \leq e^{-T} (m + n) \exp(s g(\theta, L_{\tau,s}(A)) V_{\tau,s}(A)).$$

Moreover,

$$\mathbb{P}\left[\forall r, \left|\lambda_r(\theta D(A - \tilde{A}))\right| < T\right] \geq 1 - 2e^{-T} (m + n) \exp(s g(\theta, L_{\tau,s}(A)) V_{\tau,s}(A)).$$

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSpectral.lean sqMagTraceProbability_eventProb_exists_
finiteHermitianEigenvalue_scaled_rectSelfAdjointDilation_truncatedTraceResidual_ge_le_exp

sqMagTraceProbability_eventProb_forall_abs_finiteHermitianEigenvalue_scaled_
rectSelfAdjointDilation_truncatedTraceResidual_lt_ge_exp
```

Corollary 53 (Explicit $1 - \delta$ scaled dilation tail). *Let*

$$B_{\theta,\tau,s}(A) = (m + n) \exp(s g(\theta, L_{\tau,s}(A)) V_{\tau,s}(A)).$$

Assume $B_{\theta,\tau,s}(A) > 0$ and $\delta > 0$. With the choice

$$T = \log\left(\frac{2B_{\theta,\tau,s}(A)}{\delta}\right),$$

the formalization proves

$$\mathbb{P}\left[\forall r, \left|\lambda_r(\theta D(A - \tilde{A}))\right| < \log\left(\frac{2B_{\theta,\tau,s}(A)}{\delta}\right)\right] \geq 1 - \delta.$$

This is still a scaled dilation-eigenvalue statement. It removes the free tail parameter T , but it does not yet optimize θ or convert the result to the final CACM equation (2) rectangular spectral-norm constants.

Lean formalization.

```
Analysis/MatrixConcentration.lean real_exp_neg_log_two_mul_div_mul_self_add
Algorithms/RandNLA/ElementwiseSpectral.lean sqMagTraceProbability_eventProb_forall_abs_
finiteHermitianEigenvalue_scaled_rectSelfAdjointDilation_truncatedTraceResidual_lt_ge_one_sub_
delta
```

Corollary 54 (Explicit $1 - \delta$ spectral event with unoptimized radius). *Assume additionally that $\theta > 0$ and $0 < \delta \leq 1$. Then the formalization proves the rectangular spectral event*

$$\mathbb{P}\left[\left\|\hat{A} - \tilde{A}\right\|_2 \leq \frac{\log(2B_{\theta,\tau,s}(A)/\delta)}{\theta}\right] \geq 1 - \delta.$$

Here $\hat{A}$ is the hard-thresholded matrix used by the truncated Algorithm 1 route, and $\tilde{A}$ is the exact sampled sketch from that truncated distribution. This is now a genuine spectral-norm event, not only a scaled eigenvalue event. Its radius is still unoptimized; the CACM equation (2) constant form requires choosing θ and simplifying the resulting expression.

Lean formalization.

```
Analysis/MatrixAlgebra.lean finiteLoewnerLe_of_smul_left_le_smul_id
Analysis/MatrixSpectral.lean finiteLoewnerLe_smul_id_of_finiteHermitianEigenvalues_le
Algorithms/RandNLA/ElementwiseSpectral.lean algorithm1ScaledDilationAbsEigenvalueEvent
algorithm1ScaledDilationAbsEigenvalueEvent_subset_exactSpectralEvent
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_truncatedTraceResidual_ge_one_sub_
delta_scaled_radius
```

Corollary 55 (Bennett-optimized $1 - \delta$ spectral event). *Let*

$$W_{\tau,s}(A) = s V_{\tau,s}(A), \quad q = \log\left(\frac{2(m+n)}{\delta}\right),$$

and fix a desired radius $r > 0$. Assume $L_{\tau,s}(A) > 0$, $W_{\tau,s}(A) > 0$, and the Bennett budget

$$q \leq \frac{W_{\tau,s}(A)}{L_{\tau,s}(A)^2} \left[\left(1 + \frac{L_{\tau,s}(A)r}{W_{\tau,s}(A)}\right) \log\left(1 + \frac{L_{\tau,s}(A)r}{W_{\tau,s}(A)}\right) - \frac{L_{\tau,s}(A)r}{W_{\tau,s}(A)} \right].$$

With

$$\theta = \frac{\log(1 + L_{\tau,s}(A)r/W_{\tau,s}(A))}{L_{\tau,s}(A)},$$

the formalization proves

$$\mathbb{P}\left[\left\|\hat{A} - \tilde{A}\right\|_2 \leq r\right] \geq 1 - \delta.$$

In Lean, the positivity of $L_{\tau,s}(A)$ and $W_{\tau,s}(A)$ is derived from $s > 0$, $\tau > 0$, a positive squared-magnitude denominator, and the domain assumption $mn > 0$. The theorem is still the truncated exact Algorithm 1 route; it does not yet simplify the Bennett budget to the final Drineas–Zouzias/CACM sample-complexity constants or compose with floating-point perturbation.

Lean formalization.

```
Analysis/MatrixConcentration.lean real_bernstein_exact_radius_le_of_log_le
Algorithms/RandNLA/ElementwiseSpectral.lean algorithm1ExactSpectralEvent_mono
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_truncatedTraceResidual_ge_one_sub_
delta_bennett_radius
```

Corollary 56 (Source-sharp Bennett spectral event). *In the square $n \times n$ source-sharp route, set*

$$L_{\tau,s}^\sharp(A) = s^{-1} \|\hat{A}\|_F + \frac{\|\hat{A}\|_F^2}{s\tau}, \quad V_{\tau,s}^\sharp(A) = n \frac{\|\hat{A}\|_F^2}{s^2}, \quad W_{\tau,s}^\sharp(A) = sV_{\tau,s}^\sharp(A).$$

Let

$$q_\sharp = \log\left(\frac{2(2n)}{\delta}\right),$$

and fix $r > 0$. If

$$q_\sharp \leq \frac{W_{\tau,s}^\sharp(A)}{(L_{\tau,s}^\sharp(A))^2} \left[\left(1 + \frac{L_{\tau,s}^\sharp(A)r}{W_{\tau,s}^\sharp(A)}\right) \log\left(1 + \frac{L_{\tau,s}^\sharp(A)r}{W_{\tau,s}^\sharp(A)}\right) - \frac{L_{\tau,s}^\sharp(A)r}{W_{\tau,s}^\sharp(A)} \right],$$

then the exact truncated Algorithm 1 sketch satisfies

$$\mathbb{P}\left[\left\|\hat{A} - \tilde{A}\right\|_2 \leq r\right] \geq 1 - \delta.$$

This is now the source-aligned Bennett step: it uses $V_{\tau,s}^\sharp(A)$ and the no- $\sqrt{2}$ support radius. It still leaves open the algebraic simplification of this budget to the final Drineas–Zouzias/CACM equation (2) sample-complexity constants, followed by truncation and floating-point transfer at those constants.

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSpectral.lean sqMagTraceProbability_eventProb_
algorithm1ExactSpectralEvent_truncatedTraceResidual_ge_one_sub_delta_scaled_radius_sharp_square
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_truncatedTraceResidual_ge_one_sub_
delta_bennett_radius_sharp_square
```

Corollary 57 (Conservative source-sharp denominator form). *With the notation of the previous corollary, suppose instead of the Bennett transform budget that*

$$q_\sharp \leq \frac{r^2}{2W_{\tau,s}^\sharp(A) + L_{\tau,s}^\sharp(A)r}.$$

Then

$$\mathbb{P}\left[\left\|\hat{A} - \tilde{A}\right\|_2 \leq r\right] \geq 1 - \delta.$$

This statement is intentionally marked as a conservative fallback: the denominator $2W + Lr$ is weaker than the sharper Bernstein denominator $2W + \frac{2}{3}Lr$ used in the Drineas–Zouzias final constant simplification. It is nevertheless a closed local theorem, because the scalar inequality

$$\frac{x^2}{2+x} \leq (1+x) \log(1+x) - x, \quad x \geq 0,$$

is formalized and then composed with the source-sharp Bennett spectral event.

Lean formalization.

```
Analysis/MatrixConcentration.lean real_bennett_transform_lower_bound_two_add
real_bennett_budget_of_quadratic_denominator_two_add
Algorithms/RandNLA/ElementwiseSpectral.lean sqMagTraceProbability_eventProb_
algorithm1ExactSpectralEvent_truncatedTraceResidual_ge_one_sub_delta_bernstein_denominator_
sharp_square
```

Corollary 58 (Source sample budget and truncation transfer). *Let $A \in \mathbb{R}^{n \times n}$, $n > 0$, $s > 0$, $\varepsilon > 0$, $0 < \delta \leq 1$, and set*

$$\tau = \frac{\varepsilon}{2n}, \quad \hat{A} = \text{trunc}_\tau(A).$$

Assume $\|\hat{A}\|_F > 0$ and

$$14n \|A\|_F^2 \log\left(\frac{2(2n)}{\delta}\right) \leq s\varepsilon^2.$$

Then the exact truncated Algorithm 1 sampler satisfies

$$\mathbb{P}\left[\left\|\hat{A} - \tilde{A}\right\|_2 \leq \frac{\varepsilon}{2}\right] \geq 1 - \delta.$$

Moreover, after adding the deterministic truncation error,

$$\mathbb{P}\left[\left\|A - \tilde{A}\right\|_2 \leq \varepsilon\right] \geq 1 - \delta.$$

Lean formalization.

```
Analysis/MatrixConcentration.lean real_bennett_transform_lower_bound_two_add_two_thirds
real_bennett_budget_of_quadratic_denominator_two_add_two_thirds
Algorithms/RandNLA/ElementwiseSpectral.lean elementwiseTruncate_tau_le_frobNormRect_of_
sqMagProbDen_pos
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_truncatedTraceResidual_ge_one_sub_
delta_source_sample_budget_sharp_square
sqMagTraceProbability_eventProb_algorithm1ExactTruncatedSpectralEvent_ge_one_sub_delta_source_
sample_budget_sharp_square
```

Corollary 59 (Floating-point transfer with an explicit perturbation budget). *Under the hypotheses of the previous corollary, let $B \in \mathbb{R}^{n \times n}$ be an entrywise nonnegative perturbation budget. If for every trace sample ω and every entry (i, j) ,*

$$\left| \mathfrak{fl}(\tilde{A}_\tau)(\omega)_{ij} - \tilde{A}_\tau(\omega)_{ij} \right| \leq B_{ij},$$

then the rounded truncated Algorithm 1 sketch satisfies

$$\mathbb{P}\left[\left\|A - \mathfrak{fl}(\tilde{A}_\tau)\right\|_2 \leq \varepsilon + \|B\|_F\right] \geq 1 - \delta.$$

This is a genuine FP transfer theorem, but it still records the entrywise perturbation budget as an explicit hypothesis. The next corollary discharges that budget for the canonical squared-magnitude trace law by using the probability-one positive-support event.

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSpectral.lean      sqMagTraceProbability_eventProb_
algorithm1FlTruncatedSpectralEvent_ge_one_sub_delta_source_sample_budget_sharp_square
```

Corollary 60 (Floating-point source budget with derived gamma budget). *Under the source sample-budget hypotheses above, assume additionally γ_s and γ_{s+1} are valid for the floating-point model. Let*

$$B_{ij} = \text{budget}_{ij} = \gamma_s |0| + s \left\| \frac{\widehat{A}}{s \widehat{A}_{ij}} \right\|_F^2 \gamma_{s+1},$$

where Lean writes this as `sqMagTraceErrorBudget(fp, s, s,  $\widehat{A}$ , 0)ij`. Then

$$\mathbb{P}\left[\left\|A - \mathfrak{fl}(\tilde{A}_\tau)\right\|_2 \leq \varepsilon + \|B\|_F\right] \geq 1 - \delta.$$

The proof does not assume the entrywise perturbation event. It intersects the exact source-budget event with the event that every sampled entry of $\widehat{A}$ has positive squared-magnitude probability. That support event has probability one under the canonical product law. On the support event, all actually used denominators are nonzero, each hit count is at most s , and the local Algorithm 1 gamma stability theorem supplies the displayed budget. Traces that hit zero-probability entries are irrelevant because they have zero mass, and the floating-point model intentionally makes no claim about division by a zero denominator on those impossible traces.

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSampling.lean hitCount_le_steps
hitCount_eq_zero_of_forall_not_hit
fl_elementwiseTraceSketch_zero_init_eq_zero_of_forall_not_hit
sqMagTraceErrorBudget_nonneg
Algorithms/RandNLA/ElementwiseSpectral.lean fl_elementwiseTraceSketch_zero_init_sqMag_
error_bound_of_positiveProb
sqMagTraceProbability_eventProb_algorithm1FlTruncatedSpectralEvent_ge_one_sub_delta_source_
sample_budget_gamma_square
```

Corollary 61 (Intermediate Algorithm 1 floating-point radius expansion). *Assume the source sample-budget hypotheses of the previous corollaries:*

$$0 < n, \quad 0 < s, \quad 0 < \varepsilon, \quad 0 < \delta \leq 1,$$

$$14n \|A\|_F^2 \log\left(\frac{2(2n)}{\delta}\right) \leq s\varepsilon^2.$$

Let

$$\tau = \frac{\varepsilon}{2n}, \quad \hat{A} = \text{trunc}_\tau(A),$$

and assume the squared-magnitude denominator of $\hat{A}$ is positive. Assume also that the floating-point gamma factors γ_s and γ_{s+1} are valid. Under the canonical product law that samples entries from $\hat{A}_{ij}^2 / \|\hat{A}\|_F^2$,

$$\mathbb{P} \left[\left\| A - \text{fl}(\tilde{A}_\tau) \right\|_2 \leq \varepsilon + n \frac{\|\hat{A}\|_F^2}{\tau} \gamma_{s+1} \right] \geq 1 - \delta.$$

Since $\|\hat{A}\|_F \leq \|A\|_F$ and $\tau = \varepsilon/(2n)$, the fully source-level version is

$$\mathbb{P} \left[\left\| A - \text{fl}(\tilde{A}_\tau) \right\|_2 \leq \varepsilon + \frac{2n^2 \|A\|_F^2}{\varepsilon} \gamma_{s+1} \right] \geq 1 - \delta.$$

Here

$$\gamma_{s+1} = \frac{(s+1)u}{1 - (s+1)u},$$

with the usual validity condition $(s+1)u < 1$. Thus the final FP error term is displayed in the dimension n , the sample count s , the unit roundoff u , $\|A\|_F$, and ε , rather than hidden inside a budget matrix. The same statement is restated as the final Algorithm 1 equation (2) corollary at the end of the Algorithm 1 section.

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSpectral.lean  sqMagTraceErrorBudget_zero_init_truncated_
le_const

frobNormRect_sqMagTraceErrorBudget_zero_init_truncated_le_const_square

sqMagTraceProbability_eventProb_algorithm1FlTruncatedSpectralRadius_ge_one_sub_delta_source_
sample_budget_gamma_square

sqMagTraceProbability_eventProb_algorithm1FlTruncatedSpectralRadius_ge_one_sub_delta_source_
sample_budget_explicit_gamma_square
```

Corollary 62 (Rectangular source-budget and floating-point gamma radius). *Let $A \in \mathbb{R}^{m \times n}$, set*

$$R = \sqrt{mn}, \quad M = \max\{m, n\}, \quad C_{\text{rect}} = 2M + \frac{4\sqrt{2}}{3}R,$$

and choose the rectangular truncation threshold

$$\tau = \frac{\varepsilon}{2R}, \quad \hat{A} = \text{trunc}_\tau(A).$$

Assume $mn > 0$, $s > 0$, $0 < \delta \leq 1$, $\varepsilon > 0$, and $\|\hat{A}\|_F^2 > 0$. If

$$4 C_{\text{rect}} \|A\|_F^2 \log \left(\frac{2(m+n)}{\delta} \right) \leq s\varepsilon^2,$$

then under the exact product law that samples entries from $\hat{A}_{ij}^2 / \|\hat{A}\|_F^2$,

$$\mathbb{P} \left[\left\| A - \tilde{A}_\tau \right\|_2 \leq \varepsilon \right] \geq 1 - \delta.$$

If γ_s and γ_{s+1} are valid for the floating-point model, then the rounded sampled-update computation satisfies

$$\mathbb{P}\left[\left\|A - \mathfrak{fl}(\tilde{A}_\tau)\right\|_2 \leq \varepsilon + R \frac{\|\hat{A}\|_F^2}{\tau} \gamma_{s+1}\right] \geq 1 - \delta.$$

Using $\|\hat{A}\|_F \leq \|A\|_F$ and $\tau = \varepsilon/(2R)$, the fully source-level displayed radius is

$$\varepsilon + \frac{2mn\|A\|_F^2}{\varepsilon} \gamma_{s+1}.$$

Equivalently, since $\gamma_{s+1} = (s+1)u/(1 - (s+1)u)$, the FP additive term is

$$\Theta\left(\frac{mn\|A\|_F^2}{\varepsilon} su\right)$$

in the regime of fixed m, n, A, ε and small $(s+1)u$. The sample condition is

$$s = \Omega\left(\frac{(M+R)\|A\|_F^2}{\varepsilon^2} \log \frac{m+n}{\delta}\right) = \Theta\left(\frac{\max\{m, n\}\|A\|_F^2}{\varepsilon^2} \log \frac{m+n}{\delta}\right),$$

because $R = \sqrt{mn} \leq M$. The result is nonvacuous: for fixed dimensions, source parameters, and a fixed sample count satisfying the displayed budget, the floating-point radius tends to ε as $u \rightarrow 0$.

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSpectral.lean elementwiseTruncate_rect_error_frobNormRect_
le_half

sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_truncatedTraceResidual_ge_one_sub_
delta_source_sample_budget_sharp_rect

sqMagTraceProbability_eventProb_algorithm1ExactTruncatedSpectralEvent_ge_one_sub_delta_source_
sample_budget_sharp_rect

sqMagTraceProbability_eventProb_algorithm1FlTruncatedSpectralEvent_ge_one_sub_delta_source_
sample_budget_gamma_rect

frobNormRect_sqMagTraceErrorBudget_zero_init_truncated_le_const_rect

sqMagTraceProbability_eventProb_algorithm1FlTruncatedSpectralRadius_ge_one_sub_delta_source_
sample_budget_gamma_rect

sqMagTraceProbability_eventProb_algorithm1FlTruncatedSpectralRadius_ge_one_sub_delta_source_
sample_budget_explicit_gamma_rect
```

Lemma 63 (Strictly positive exponential–logarithm inverse). *Let A be a strictly positive complex $d \times d$ C^* -matrix. Then*

$$\exp(\log A) = A.$$

The same identity is formalized both for the normed-algebra exponential and for the real and complex continuous-functional-calculus exponentials:

$$\exp_{\text{cfc}, \mathbb{C}}(\log A) = A, \quad \exp_{\text{cfc}, \mathbb{R}}(\log A) = A.$$

Lean formalization.

```
Analysis/OperatorLog.lean cstarMatrix_normedRingExp_nonnegSpectrumClass

cstarMatrix_normedSpace_exp_log_of_isStrictlyPositive

cstarMatrix_cfc_complex_exp_log_of_isStrictlyPositive

cstarMatrix_cfc_real_exp_log_of_isStrictlyPositive
```

Lemma 64 (Finite-real to complex C^* -matrix algebraic and order bridge). *Let $M, N \in \mathbb{R}^{d \times d}$. Define*

$$M_{\mathbb{C}} = (M_{ij} : \mathbb{C})_{i,j}.$$

Then entrywise embedding into complex C^ -matrices preserves subtraction, the identity, and scalar identities:*

$$(M - N)_{\mathbb{C}} = M_{\mathbb{C}} - N_{\mathbb{C}}, \quad I_{\mathbb{C}} = I, \quad (aI)_{\mathbb{C}} = aI.$$

Moreover, if M is symmetric, then $M_{\mathbb{C}}$ is self-adjoint. If $M \succeq 0$ in the local finite quadratic-form sense, then $M_{\mathbb{C}} \succeq 0$ in complex C^ -matrix order. If M, N are symmetric and $M \preceq N$ locally, then*

$$M_{\mathbb{C}} \preceq N_{\mathbb{C}}$$

in the same complex C^ -matrix order.*

Lean formalization.

```
Analysis/CStarMatrixBridge.lean finiteComplexCStarMatrix
finiteComplexCStarMatrix_isSelfAdjoint_of_symmetric
finiteComplexCStarMatrix_sub
finiteComplexCStarMatrix_finiteIdMatrix
finiteComplexCStarMatrix_smul_finiteIdMatrix
finiteComplexCStarMatrix_nonneg_of_finitePSD
finiteComplexCStarMatrix_le_of_finiteLoewnerLe
```

Lemma 65 (Regularized finite-real operator-log bridge). *Let $M, N \in \mathbb{R}^{d \times d}$ be symmetric, assume $M \succeq 0$, and assume $M \preceq N$ in local finite Loewner order. For every $\eta > 0$,*

$$M_{\mathbb{C}} + \eta I \quad \text{is strictly positive}$$

as a complex C^ -matrix, and*

$$M_{\mathbb{C}} + \eta I \preceq N_{\mathbb{C}} + \eta I.$$

Consequently,

$$\log(M_{\mathbb{C}} + \eta I) \preceq \log(N_{\mathbb{C}} + \eta I).$$

Lean formalization.

```
Analysis/CStarMatrixBridge.lean cstarMatrix_pos_real_smul_one_isStrictlyPositive
finiteComplexCStarMatrix_add_pos_smul_one_isStrictlyPositive_of_finitePSD
finiteComplexCStarMatrix_add_smul_one_le_of_finiteLoewnerLe
Analysis/OperatorLog.lean finiteComplexCStarMatrix_regularized_log_le_log_of_finiteLoewnerLe
```

Lemma 66 (Complex C^* -matrix trace bridge). *For a complex $d \times d$ C^* -matrix X , define*

$$\text{tr}_{\mathbb{C}}(X) = \sum_{i=1}^d X_{ii}.$$

This trace is additive and homogeneous, and

$$\text{tr}_{\mathbb{C}}(I) = d, \quad \text{tr}_{\mathbb{C}}(aI) = ad.$$

It is also cyclic:

$$\mathrm{tr}_{\mathbb{C}}(XY) = \mathrm{tr}_{\mathbb{C}}(YX).$$

The trace is positive and monotone in real part for the C^* -spectral order:

$$0 \preceq X \implies 0 \leq \Re \mathrm{tr}_{\mathbb{C}}(X),$$

and

$$X \preceq Y \implies \Re \mathrm{tr}_{\mathbb{C}}(X) \leq \Re \mathrm{tr}_{\mathbb{C}}(Y).$$

If $M \in \mathbb{R}^{d \times d}$ is embedded as $M_{\mathbb{C}}$, then

$$\mathrm{tr}_{\mathbb{C}}(M_{\mathbb{C}}) = \mathrm{tr}(M).$$

Consequently, if $M \succeq 0$ in local finite PSD order, then

$$0 \leq \Re \mathrm{tr}_{\mathbb{C}}(M_{\mathbb{C}}),$$

and if $M \preceq N$ in local finite Loewner order, then

$$\Re \mathrm{tr}_{\mathbb{C}}(M_{\mathbb{C}}) \leq \Re \mathrm{tr}_{\mathbb{C}}(N_{\mathbb{C}}).$$

If $M_{\mathbb{C}} = M_{\mathbb{C}}^*$, then

$$\Im \mathrm{tr}_{\mathbb{C}}(M_{\mathbb{C}}) = 0.$$

Finally, for scalar identities, the complex continuous functional calculus normalizes the exponential trace:

$$\mathrm{tr}_{\mathbb{C}}(\exp_{\mathrm{cfc}}(aI)) = e^a d.$$

Lean formalization.

```
Analysis/CStarMatrixTrace.lean cstarMatrixTrace
cstarMatrixTrace_neg
cstarMatrixTrace_sub
cstarMatrixTrace_mul_comm
cstarMatrixTrace_star_mul_self_re_nonneg
cstarMatrixTrace_re_nonneg_of_nonneg
cstarMatrixTrace_re_mono
cstarMatrixTrace_im_eq_zero_of_isSelfAdjoint
cstarMatrixTrace_finiteComplexCStarMatrix
cstarMatrixTrace_finiteComplexCStarMatrix_re_nonneg_of_finitePSD
cstarMatrixTrace_finiteComplexCStarMatrix_re_mono_of_finiteLoewnerLe
cstarMatrixTrace_cfc_exp_algebraMap
cstarMatrixTrace_cfc_exp_real_smul_one
```

Lemma 67 (Complex C^* -matrix expectation, order, and Jensen bridge). *Let $(\Omega, \mathbb{P})$ be a finite probability space. For a complex-valued random variable $Z : \Omega \rightarrow \mathbb{C}$, define*

$$\mathbb{E}_{\mathbb{C}} Z = \sum_{\omega \in \Omega} \mathbb{P}(\omega) Z(\omega).$$

For a complex C^* -matrix-valued random variable $X : \Omega \rightarrow \mathbb{C}^{d \times d}$, define expectation entrywise:

$$(\mathbb{E}_{\mathbb{C}}^{\text{mat}} X)_{ij} = \mathbb{E}_{\mathbb{C}}(X_{ij}).$$

Then complex expectation is linear, agrees with the repository's real expectation after coercion $\mathbb{R} \hookrightarrow \mathbb{C}$, satisfies

$$\text{Re}(\mathbb{E}_{\mathbb{C}} Z) = \mathbb{E}_{\mathbb{R}} \text{Re}(Z),$$

and

$$\text{tr}_{\mathbb{C}}(\mathbb{E}_{\mathbb{C}}^{\text{mat}} X) = \mathbb{E}_{\mathbb{C}}(\text{tr}_{\mathbb{C}}(X)).$$

Moreover, if $M(\omega)$ is a repository-native finite real matrix, then

$$\mathbb{E}_{\mathbb{C}}^{\text{mat}}(M(\omega)_{\mathbb{C}}) = (\mathbb{E}_{\mathbb{R}} M(\omega))_{\mathbb{C}}.$$

The matrix expectation has both complex and real weighted-sum forms

$$\mathbb{E}_{\mathbb{C}}^{\text{mat}} X = \sum_{\omega \in \Omega} \mathbb{P}(\omega) X(\omega).$$

Consequently, if $X(\omega) \succeq 0$ for every ω , then

$$\mathbb{E}_{\mathbb{C}}^{\text{mat}} X \succeq 0.$$

If $X(\omega) \succ 0$ for every ω , then

$$\mathbb{E}_{\mathbb{C}}^{\text{mat}} X \succ 0.$$

If $X(\omega) \preceq Y(\omega)$ for every ω , then

$$\mathbb{E}_{\mathbb{C}}^{\text{mat}} X \preceq \mathbb{E}_{\mathbb{C}}^{\text{mat}} Y.$$

Finally, if $X(\omega) \succeq 0$ and $\varepsilon > 0$, then

$$\mathbb{E}_{\mathbb{C}}^{\text{mat}} X + \varepsilon I$$

is strictly positive.

In addition, if $f : S \rightarrow \mathbb{R}$ is concave on a convex set $S \subseteq \mathbb{C}^{d \times d}$ and $X(\omega) \in S$, then finite Jensen gives

$$\mathbb{E}_{\mathbb{R}}[f(X)] \leq f(\mathbb{E}_{\mathbb{C}}^{\text{mat}} X).$$

Lean formalization.

```

Analysis/CStarMatrixExpectation.lean
FiniteProbability.expectationComplex
FiniteProbability.expectationComplex_sum
FiniteProbability.expectationComplex_ofReal
FiniteProbability.expectationComplex_re
FiniteProbability.exists_prob_pos
FiniteProbability.expectationReal_le_of_concaveOn
FiniteProbability.expectationComplex_add
FiniteProbability.expectationComplex_smul
FiniteProbability.expectationCStarMatrix
FiniteProbability.cstarMatrixTrace_expectationCStarMatrix
FiniteProbability.expectationCStarMatrix_finiteComplexCStarMatrix

```

```

FiniteProbability.expectationCStarMatrix_add
FiniteProbability.expectationCStarMatrix_smul
FiniteProbability.expectationCStarMatrix_eq_sum_smul
FiniteProbability.expectationCStarMatrix_eq_sum_real_smul
FiniteProbability.expectationReal_le_of_concaveOn_expectationCStarMatrix
FiniteProbability.expectationReal_trace_cfc_exp_add_log_le_of_concaveOn
FiniteProbability.expectationCStarMatrix_nonneg
FiniteProbability.expectationCStarMatrix_isStrictlyPositive
FiniteProbability.expectationCStarMatrix_mono
FiniteProbability.expectationCStarMatrix_add_pos_smul_one_isStrictlyPositive

```

Lemma 68 (Strictly positive cone vocabulary for Lieb). *Let*

$$\mathcal{P}_{++} = \{A \in \mathbb{C}^{d \times d} : A \text{ is strictly positive}\}$$

in the complex C^ -matrix spectral order. This cone is convex over real scalars. Positive real scalar multiples preserve strict positivity, and nonnegative real scalar multiples preserve nonnegativity. The local Lieb trace functional is named as*

$$\Phi_H(A) = \operatorname{Re} \operatorname{tr} \exp_{\text{cfc}}(H + \log A).$$

For self-adjoint H , the matrix $\exp_{\text{cfc}}(H + \log A)$ is nonnegative in the complex C^ -spectral order, its trace is real-valued, and*

$$\Phi_H(A) \geq 0.$$

It also agrees with the standard normed-algebra exponential notation used in Tropp's trace-MGF theorem:

$$\Phi_H(A) = \operatorname{Re} \operatorname{tr} \exp(H + \log A),$$

and on the strictly positive cone it satisfies the normalization

$$\Phi_0(A) = \operatorname{Re} \operatorname{tr}(A).$$

Consequently the $H = 0$ special case of the local Lieb concavity target is proved: Φ_0 is affine, hence concave, on $\mathcal{P}_{++}$. The remaining Lieb theorem target is the assertion that Φ_H is concave on $\mathcal{P}_{++}$ for fixed self-adjoint H .

Corollary 69 (Scalar normalized-kernel representation of $x \log x$). *For $x > 0$, let*

$$g(x) = \begin{cases} 1, & x = 1, \\ x \log x / (x - 1), & x \neq 1. \end{cases}$$

Then

$$(x - 1)g(x) = x \log x$$

and, using the already formalized unit-interval representation of g ,

$$(x - 1) \int_0^1 \frac{x}{u + (1 - u)x} du = x \log x.$$

Theorem 70 (Correct unit-interval kernel for $x \log x$). *For $x > 0$,*

$$\int_0^1 \frac{x(x-1)}{u+(1-u)x} du = x \log x.$$

Moreover, for $0 \leq u < 1$,

$$\frac{x(x-1)}{u+(1-u)x} = \frac{x}{1-u} - \frac{1}{(1-u)^2} + \frac{u}{(1-u)^3} \left(x + \frac{u}{1-u} \right)^{-1}.$$

Remark. The auxiliary square-kernel identity below is still true and useful, but it is not the kernel whose integral reconstructs $x \log x$. The reconstruction kernel is the $x(x-1)$ kernel in the theorem above.

Lean formalization.

```
real_xlog_eq_sub_one_mul_realNormalizedLogKernel
real_xlog_eq_sub_one_mul_normalizedKernel_setIntegral
real_xlog_eq_unit_interval_xlog_kernel_integral
real_unit_interval_xlog_kernel_integrand_eq_affine_add_shifted_inv
cstarMatrix_cfc_unit_interval_xlog_kernel_integrand_eq_affine_add_shifted_inv
```

Lemma 71 (Auxiliary square-kernel decomposition). *For $x > 0$ and $0 \leq u < 1$,*

$$\frac{(x-1)^2}{u+(1-u)x} = \frac{x}{1-u} - \frac{2-u}{(1-u)^2} + \frac{1}{(1-u)^3} \left(x + \frac{u}{1-u} \right)^{-1}.$$

Lean formalization.

```
real_unit_interval_xlog_integrand_eq_affine_add_shifted_inv
cstarMatrix_cfc_unit_interval_xlog_integrand_eq_affine_add_shifted_inv
```

Lean formalization.

```
Analysis/LiebTrace.lean strictPositiveCStarMatrixCone
mem_strictPositiveCStarMatrixCone
cstarMatrix_isStrictlyPositive_pos_real_smul
cstarMatrix_nonneg_nonneg_real_smul
cstarMatrix_isStrictlyPositive_pos_nonneg_real_smul_add
strictPositiveCStarMatrixCone_convex
cstarMatrix_log_isSelfAdjoint
liebTraceArgument_isSelfAdjoint
liebTraceArgument_isStarNormal
liebTraceCfcExp_nonneg
liebTraceCfcExp_isStrictlyPositive
liebTraceFunctional_trace_im_eq_zero
liebTraceFunctional
liebTraceFunctional_nonneg
```

```

liebTraceFunctional_eq_normedSpace_exp
liebTraceFunctional_zero_eq_trace
liebTraceConcavityTarget
liebTraceConcavityTarget_zero

```

Corollary 72 (Conditional one-step Tropp–Jensen trace-MGF bound). *Assume the Lieb concavity statement*

$$\Phi_H(A) = \text{Re tr exp}(H + \log A) \quad \text{is concave on } \mathcal{P}_{++}$$

for the fixed self-adjoint matrix H . Let X be a finite-valued self-adjoint matrix random variable. Then the local formalization proves

$$\mathbb{E} \text{Re tr exp}(H + X) \leq \text{Re tr exp}(H + \log \mathbb{E} \exp X).$$

Lean formalization.

```

Analysis/LiebTrace.lean      FiniteProbability.expectationReal_trace_normed_exp_add_le_of_
liebTraceConcavityTarget

```

Definition 73 (Scalar and finite-vector relative entropy). For positive scalars a, b , define

$$d(a; b) = a(\log a - \log b) - (a - b).$$

For finite real vectors x, y indexed by a finite type I , define

$$D_{\text{vec}}(x; y) = \sum_{i \in I} d(x_i; y_i).$$

Lemma 74 (Scalar relative-entropy nonnegativity). *If $a > 0$ and $b > 0$, then*

$$d(a; b) \geq 0.$$

Corollary 75 (Finite-vector relative-entropy nonnegativity). *If $x_i > 0$ and $y_i > 0$ for every coordinate i , then*

$$D_{\text{vec}}(x; y) \geq 0.$$

Lean formalization.

```

Analysis/LiebTrace.lean realRelativeEntropy
realRelativeEntropy_nonneg
finiteRealRelativeEntropy
finiteRealRelativeEntropy_nonneg

```

Theorem 76 (Finite log-sum inequality). *Let $p_i > 0$ and $q_i > 0$ for all $i \in I$, with I finite and nonempty. Then*

$$\left(\sum_i p_i \right) \log \left(\frac{\sum_i p_i}{\sum_i q_i} \right) \leq \sum_i p_i \log \left(\frac{p_i}{q_i} \right).$$

Corollary 77 (Commutative joint convexity of relative entropy). *If $x, y, a, b > 0$, $\alpha, \beta \geq 0$, and $\alpha + \beta = 1$, then*

$$d(\alpha x + \beta y; \alpha a + \beta b) \leq \alpha d(x; a) + \beta d(y; b).$$

Consequently, if all vector coordinates are positive, then

$$D_{\text{vec}}(\alpha x + \beta y; \alpha a + \beta b) \leq \alpha D_{\text{vec}}(x; a) + \beta D_{\text{vec}}(y; b).$$

Lean formalization.

```
Analysis/LiebTrace.lean finite_log_sum_inequality
realRelativeEntropy_jointConvex_of_pos_weights
realRelativeEntropy_jointConvex
finiteRealRelativeEntropy_jointConvex
```

Definition 78 (C-star matrix relative entropy vocabulary). For finite complex C*-matrices A and B , the current Tropp-route vocabulary defines

$$D(A; B) = \operatorname{Re} \operatorname{tr} (A(\log A - \log B) - (A - B)).$$

This is the finite-dimensional relative-entropy expression used by the chosen route toward Lieb trace concavity.

Lemma 79 (Diagonal normalization of relative entropy). *For every finite complex C*-matrix A ,*

$$D(A; A) = 0.$$

Lean formalization.

```
Analysis/LiebTrace.lean cstarMatrixRelativeEntropy
cstarMatrixRelativeEntropy_self
```

Definition 80 (Real diagonal C-star matrices). For a finite index set I and a real vector $a : I \rightarrow \mathbb{R}$, write

$$\Delta(a) = \operatorname{diag}(a_i)_{i \in I}$$

for the corresponding complex C*-matrix.

Lemma 81 (Diagonal logarithm). *If $a_i \neq 0$ for every $i \in I$, then*

$$\log \Delta(a) = \Delta((\log a_i)_{i \in I}).$$

Corollary 82 (Diagonal matrix relative entropy). *If $a_i \neq 0$ and $b_i \neq 0$ for every $i \in I$, then*

$$D(\Delta(a); \Delta(b)) = \sum_{i \in I} (a_i(\log a_i - \log b_i) - (a_i - b_i)).$$

Consequently, if $a_i > 0$ and $b_i > 0$ for every i , then

$$D(\Delta(a); \Delta(b)) \geq 0.$$

Lean formalization.

```
Analysis/OperatorLog.lean complex_realContinuousFunctionalCalculus
piComplex_realContinuousFunctionalCalculus
Analysis/LiebTrace.lean cstarMatrixDiagonalStarAlgHom
cstarMatrixDiagonalStarAlgHom_continuous
cstarMatrixRealDiagonal
cstarMatrixTrace_realDiagonal
cstarMatrix_log_realDiagonal
cstarMatrixRelativeEntropy_realDiagonal
cstarMatrixRelativeEntropy_realDiagonal_nonneg
```

Corollary 83 (Real-diagonal matrix relative entropy is jointly convex). *Let $x_i, y_i, a_i, b_i > 0$ for all i , and let $\alpha, \beta \geq 0$ with $\alpha + \beta = 1$. Then*

$$\begin{aligned} D(\alpha\Delta(x) + \beta\Delta(y); \alpha\Delta(a) + \beta\Delta(b)) \\ \leq \alpha D(\Delta(x); \Delta(a)) + \beta D(\Delta(y); \Delta(b)). \end{aligned}$$

Lean formalization.

```
Analysis/LiebTrace.lean cstarMatrixRealDiagonal_smul_add
positive_weighted_sum_pos
cstarMatrixRelativeEntropy_realDiagonal_jointConvex
```

Lemma 84 (Scalar-identity matrix relative entropy). *Let I_d denote the identity matrix in a finite d -dimensional C^* -matrix algebra. For real scalars a, b ,*

$$D(aI_d; bI_d) = d d(a; b).$$

Consequently, if $a > 0$ and $b > 0$, then

$$D(aI_d; bI_d) \geq 0.$$

Lean formalization.

```
Analysis/LiebTrace.lean cstarMatrixRelativeEntropy_algebraMap_real
cstarMatrixRelativeEntropy_algebraMap_real_nonneg
```

Lemma 85 (Left/right multiplication and the ratio endomorphism). *For a finite complex C^* -matrix A , let*

$$L_A(Z) = AZ, \quad R_A(Z) = ZA$$

be endomorphisms of the complex vector space of finite C^ -matrices. Then $A \mapsto L_A$ and $A \mapsto R_A$ are affine for real weighted sums:*

$$L_{\alpha A + \beta B} = \alpha L_A + \beta L_B, \quad R_{\alpha A + \beta B} = \alpha R_A + \beta R_B.$$

Moreover left and right multiplication commute,

$$L_A R_B = R_B L_A,$$

and they obey the product rules

$$L_{AB} = L_A L_B, \quad R_{AB} = R_B R_A.$$

Consequently, for every natural number k ,

$$L_{A^k} = L_A^k, \quad R_{A^k} = R_A^k.$$

and if A is a unit, then L_A and R_A are units in the endomorphism ring. In particular this holds for strictly positive A .

If A is supplied as a unit, define the ratio endomorphism

$$T_{X,A} := L_X R_{A^{-1}}.$$

Then

$$T_{X,A}(Z) = X Z A^{-1}, \quad T_{X,A}(A) = X.$$

The second identity also holds when A is strictly positive, using the unit coming from strict positivity.

Lean formalization.

```

Analysis/LiebTrace.lean
cstarMatrixLeftMul
cstarMatrixRightMul
cstarMatrixLeftMul_mul
cstarMatrixRightMul_mul
cstarMatrixLeftMul_pow
cstarMatrixRightMul_pow
cstarMatrixLeftMul_real_smul_add
cstarMatrixRightMul_real_smul_add
cstarMatrixLeftRightMul_commute
cstarMatrixLeftMul_isUnit_of_isStrictlyPositive
cstarMatrixRightMul_isUnit_of_isStrictlyPositive
cstarMatrixLeftRightRatio
cstarMatrixLeftRightRatio_apply
cstarMatrixLeftRightRatio_apply_unit
cstarMatrixLeftRightRatio_apply_of_unit_eq
cstarMatrixLeftRightRatio_apply_of_isStrictlyPositive

```

Lemma 86 (Kronecker lifts for the perspective route). *Let $A, B, H \in \mathbb{C}^{n \times n}$, and define*

$$K_L(A) = A \otimes I_n, \quad K_R(H) = I_n \otimes H.$$

For real scalars α, β , the lifts are affine:

$$K_L(\alpha A + \beta B) = \alpha K_L(A) + \beta K_L(B), \quad K_R(\alpha A + \beta B) = \alpha K_R(A) + \beta K_R(B).$$

They also commute and multiply to the ordinary Kronecker product:

$$K_L(A)K_R(H) = A \otimes H = K_R(H)K_L(A).$$

If A is positive definite, then $K_L(A)$ and $K_R(A)$ are positive definite. Under product-index vectorization,

$$(A \otimes B^\top) \text{vec}(M) = \text{vec}(AMB).$$

For every integer $k \geq 0$,

$$(A \otimes B^\top)^k \text{vec}(M) = \text{vec}(A^k M B^k).$$

Finally,

$$\text{tr}(A \otimes H) = \text{tr}(A) \text{tr}(H), \quad \text{tr}(K_L(A)) = \text{tr}(K_R(A)) = n \text{tr}(A).$$

If $v_I = \text{vec}(I_n)$, then the vectorized-identity trace pairing is

$$v_I^*(A \otimes B^\top)v_I = \text{tr}(AB).$$

The powered trace-pairing identity is

$$v_I^*(A \otimes B^\top)^k v_I = \text{tr}(A^k B^k).$$

Consequently, for any finite $S \subset \mathbb{N}$ and coefficients $c_k \in \mathbb{C}$,

$$v_I^* \left(\sum_{k \in S} c_k (A \otimes B^\top)^k \right) v_I = \sum_{k \in S} c_k \operatorname{tr}(A^k B^k).$$

Equivalently, Lean's polynomial-evaluation form gives the same identity for $p(A \otimes B^\top)$, with the right-hand side expanded over the finite support of p . The domain side needed for real functional calculus is also closed: if $A = A^*$ and $B = B^*$, then $A \otimes B^\top$ is self-adjoint, and positive semidefiniteness or positive definiteness is preserved by the same Kronecker-transpose lift. In particular, if $X \succ 0$ and $A \succ 0$, then

$$X \otimes (A^{-1})^\top \succ 0.$$

Mathlib's Weierstrass approximation theorem supplies real polynomials approximating $\log x$, $x \log x$, and $x \log x - (x - 1)$ uniformly on every positive compact interval.

The source-faithful polynomial form of the Effros superoperator trace term is now formalized. For $R_C = I_n \otimes C^\top$,

$$v_I^* (A \otimes B^\top)^k R_C v_I = \operatorname{tr}(A^k C B^k),$$

and hence, for a polynomial p ,

$$v_I^* p(A \otimes B^\top) R_C v_I = \sum_k p_k \operatorname{tr}(A^k C B^k).$$

Specializing to the left-right ratio $X \otimes (A^{-1})^\top$ and $C = A$, the formalized real-polynomial CFC statement is

$$v_I^* p(L_X R_A^{-1}) R_A v_I = \sum_k p_k \operatorname{tr}(X^k A (A^{-1})^k).$$

The same finite-polynomial trace terms now have the explicit spectral-overlap form

$$\operatorname{Re} \operatorname{tr}(X^k A (A^{-1})^k) = \sum_{i,j} \lambda_i^k \mu_j \mu_j^{-k} |\langle u_i, v_j \rangle|^2,$$

and hence

$$\operatorname{Re} \sum_k p_k \operatorname{tr}(X^k A (A^{-1})^k) = \sum_{i,j} \mu_j p(\lambda_i / \mu_j) |\langle u_i, v_j \rangle|^2.$$

Lean formalization.

```
Analysis/LiebTrace.lean matrixVecId
matrixVec
matrixComplexQuadraticForm
continuous_matrixComplexQuadraticForm
finset_sum_product_diagonal
matrixVecId_inner_matrixVec
matrix_transpose_conjTranspose_eq_self_of_isSelfAdjoint
matrix_kronecker_transpose_isSelfAdjoint_of_isSelfAdjoint
matrix_kronecker_transpose_posSemidef
matrix_kronecker_transpose_posDef
matrix_kronecker_inv_transpose_posDef
```

matrixSelfAdjointCfc
 matrixSelfAdjointCfc_polynomial
 exists_realPolynomial_near_log_on_Icc
 exists_realPolynomial_near_xlog_on_Icc
 exists_realPolynomial_near_realEntropyKernel_on_Icc
 matrix_posDef_spectrum_real_pos
 matrix_posDef_spectrum_real_subset_Icc
 exists_realPolynomial_tendstoUniformly0n_realEntropyKernel_on_subset_Icc
 exists_realPolynomial_tendstoUniformly0n_realEntropyKernel_spectrum_of_posDef
 matrix_kronecker_transpose_mulVec_matrixVec
 matrix_kronecker_transpose_pow
 matrix_kronecker_transpose_pow_mulVec_matrixVec
 matrixVec_one
 matrixComplexQuadraticForm_vecId_kronecker_transpose_pow
 matrixComplexQuadraticForm_vecId_kronecker_transpose_pow_right
 matrixComplexQuadraticForm_sum
 matrixComplexQuadraticForm_smul
 matrixComplexQuadraticForm_vecId_kronecker_transpose_polynomial
 matrixComplexQuadraticForm_vecId_kronecker_transpose_polynomial_right
 matrixComplexQuadraticForm_vecId_kronecker_transpose_aeval
 matrixComplexQuadraticForm_vecId_kronecker_transpose_aeval_right
 matrixComplexQuadraticForm_vecId_matrixSelfAdjointCfc_kronecker_transpose_realPolynomial
 matrixComplexQuadraticForm_vecId_matrixSelfAdjointCfc_kronecker_inv_transpose_realPolynomial_right
 tendsto_matrixComplexQuadraticForm_matrixSelfAdjointCfc_mul
 tendsto_superoperator_entropyKernel_of_realPolynomial_uniform_approx
 exists_realPolynomial_tendsto_superoperator_entropyKernel_trace_of_posDef
 matrixComplexQuadraticForm_re_nonneg_of_posSemidef
 matrixComplexQuadraticForm_re_mono_of_posSemidef_sub
 matrixComplexQuadraticForm_re_mono_of_cstarMatrix_le
 matrixComplexQuadraticForm_add
 exists_realPolynomial_tendsto_superoperator_entropyKernel_trace_and_overlap_of_posDef
 matrixTrace_pow_mul_inv_pow_re_eq_sum
 matrixPolynomialTraceRatio_re_eq_sum
 tendsto_matrixPolynomialTraceRatio_overlap_sum_of_uniform_approx
 matrixSuperoperatorEntropyKernelTrace
 matrixSuperoperatorEntropyKernelOverlapExpansion
 matrixSuperoperatorEntropyKernelOverlapExpansion_all
 matrixRelativeEntropyTraceRepresentation_of_superoperator_overlap
 cstarMatrixRelativeEntropyTraceRepresentationBySuperoperator

```

cstarMatrixRelativeEntropyTraceRepresentationBySuperoperator_all
continuous_matrix_polynomial_aeval
matrixComplexQuadraticForm_vecId_kronecker_transpose
matrix_kronecker_left_identity_real_smul_add
matrix_kronecker_right_identity_real_smul_add
matrix_kronecker_right_identity_transpose_real_smul_add
cstarMatrixSuperoperatorLeftLift
cstarMatrixSuperoperatorRightLift
cstarMatrixSuperoperatorLeftLift_real_smul_add
cstarMatrixSuperoperatorRightLift_real_smul_add
cstarMatrixSuperoperatorLeftLift_isStrictlyPositive
cstarMatrixSuperoperatorRightLift_isStrictlyPositive
cstarMatrixSuperoperatorLeftLift_rightLift_commute
cstarMatrixSuperoperatorPositiveInvSqrtRightLift_commute_leftLift
cstarMatrixSuperoperatorPositiveSqrtRightLift_commute_leftLift
cstarMatrixSuperoperatorPerspective_normalizedArgument_reorder
cstarMatrixSuperoperatorPerspective_normalizedArgument_eq_leftLift_mul_rightLift_unit_inv
cstarMatrixSuperoperatorLeftLift_mul_rightLift_unit_inv_commute_positiveSqrtRightLift
cstarMatrixSuperoperatorPerspectiveTrace
cstarMatrixSuperoperatorPerspectiveTrace_jointConvex
cstarMatrixRelativeEntropyPerspectiveTraceRepresentation
cstarMatrixRelativeEntropyJointConvexOnStrictPositive_of_perspectiveTraceRepresentation
matrix_kronecker_left_identity_mul_right_identity
matrix_kronecker_right_identity_mul_left_identity
matrix_kronecker_left_right_commute
matrix_kronecker_posDef_left_identity
matrix_kronecker_posDef_right_identity
matrix_trace_kronecker
matrix_trace_kronecker_left_identity
matrix_trace_kronecker_right_identity

```

Definition 87 (Positive operator-convexity source target). For a real function f , the local ordinary positive-cone matrix-convexity target is

$$f(\alpha T_1 + \beta T_2) \preceq \alpha f(T_1) + \beta f(T_2)$$

whenever T_1, T_2 are strictly positive C^* -matrices, $\alpha, \beta \geq 0$, and $\alpha + \beta = 1$. The Effros route needs this target for

$$f(x) = x \log x.$$

Lemma 88 (Identity sanity case for ordinary convexity). *The positive-cone matrix-convexity target is formalized and proved for $f(x) = x$. This checks the scalar-weighted functional-calculus statement in the local C^* -matrix vocabulary. It is not a proof of nonlinear operator convexity.*

Definition 89 (Hansen–Pedersen transfer target). The Hansen–Pedersen source theorem is represented as the transfer

$$\begin{aligned} & \text{ordinary positive-cone operator convexity of } f \\ \implies & f(A^*T_1A + B^*T_2B) \preceq A^*f(T_1)A + B^*f(T_2)B. \end{aligned}$$

for strictly positive T_1, T_2 and $A^*A + B^*B = I$. The generic all-functions transfer theorem is still not formalized. For the concrete function $x \log x$, however, the repository now proves the two-point Jensen conclusion later in this section by a direct reflection-average argument, so no final theorem uses this transfer target as a hidden hypothesis.

Lemma 90 (Identity sanity case for two-point Jensen). *The assembled two-point Hansen–Pedersen target is also formalized and proved for $f(x) = x$. For the identity function, continuous functional calculus reduces both sides to*

$$A^*T_1A + B^*T_2B.$$

The nonlinear $x \log x$ Jensen statement is named as the downstream target and proved later in this section by the unit-interval kernel and reflection-average corner route.

Lemma 91 (Derivative monotonicity for the Bendat–Sherman route). *On strictly positive C^* -matrices, the formal derivative of $x \log x$ is operator-monotone:*

$$A \preceq B, \quad A \succ 0, \quad B \succ 0 \implies I + \log A \preceq I + \log B.$$

Definition 92 (Bendat–Sherman divided-difference route). For $c > 0$, let the first divided difference of $f(x) = x \log x$ at the base point c be

$$f_c(x) = \begin{cases} 1 + \log c, & x = c, \\ \frac{x \log x - c \log c}{x - c}, & x \neq c. \end{cases}$$

The source-faithful finite route separates into a monotonicity theorem for every f_c and the Bendat–Sherman bridge:

$$\begin{aligned} & \text{operator monotonicity of every } f_c, \\ & \text{plus the Bendat–Sherman divided-difference bridge} \\ \implies & \text{operator convexity of } x \log x. \end{aligned}$$

on the strictly positive finite C^* -matrix cone. The monotonicity part is now proved below; the Bendat–Sherman bridge itself remains a local open target, not a hidden hypothesis used by a final paper theorem.

Lemma 93 (Scalar divided-difference normalization). *For positive c, x with $x \neq c$, the repository proves the scalar identities*

$$f_c(x) = \log c + \frac{x \log(x/c)}{x - c} = \log c + \frac{(x/c) \log(x/c)}{x/c - 1}.$$

The diagonal value $f_c(c) = 1 + \log c$ is also formalized.

Lemma 94 (Scalar logarithmic integral identity). *Define the normalized scalar kernel*

$$g(t) = \begin{cases} 1, & t = 1, \\ \frac{t \log t}{t - 1}, & t \neq 1. \end{cases}$$

Then the first divided difference satisfies

$$f_c(x) = \log c + g(x/c) \quad (c > 0, x > 0).$$

Moreover, for $t > 0$ and $t \neq 1$,

$$\int_0^1 \frac{t}{u + (1-u)t} du = \frac{t \log t}{t-1}.$$

Lemma 95 (Fractional-kernel monotonicity). *Let $0 \preceq A \preceq B$ be finite complex C^* -matrices. Then the unital CFC kernels*

$$x \mapsto 1 - (1+x)^{-1}, \quad x \mapsto \frac{x}{1+x}$$

are operator-monotone on this nonnegative cone:

$$(I - (I + A)^{-1}) \preceq (I - (I + B)^{-1}), \quad \frac{A}{I + A} \preceq \frac{B}{I + B}$$

where the fractions are interpreted by continuous functional calculus. The repository also proves the spectrum wrapper needed to use nonnegativity of A and B inside these CFC congruence arguments. Moreover, for every $s > 0$, the scaled fractional kernel

$$x \mapsto \frac{x}{s+x}$$

is operator-monotone on the same nonnegative cone. Consequently, every finite nonnegative linear combination

$$x \mapsto \sum_{k \in F} w_k \frac{x}{\sigma_k + x}, \quad w_k \geq 0, \quad \sigma_k > 0,$$

is operator-monotone on the same cone. For $0 < u < 1$, the unit-interval integrand

$$x \mapsto \frac{x}{u + (1-u)x}$$

is also operator-monotone, by rewriting it as a positive multiple of the scaled kernel with $s = u/(1-u)$. On the strictly positive cone this is extended to every $u \in [0, 1]$: the endpoint $u = 0$ is the constant-one CFC kernel and the endpoint $u = 1$ is the identity kernel.

Lemma 96 (Concrete interval-integral side conditions). *Let $A \succ 0$, and let M be any real number with $\lambda \leq M$ for every $\lambda \in \sigma(A)$. For $u \in [0, 1]$ and $\lambda \in \sigma(A)$,*

$$\left| \frac{\lambda}{u + (1-u)\lambda} \right| \leq \max\{1, M\}.$$

The same estimate is formalized both for Lebesgue measure restricted to $[0, 1]$ and for the subtype $\{u : \mathbb{R} \mid u \in [0, 1]\}$. The subtype version is the one aligned with the current `cfc_integral` API, because the API asks for continuity on the whole integration type. The repository also proves the corresponding joint-continuity statement on the interval subtype and finite integrability of the constant bound for any finite measure on that subtype.

Theorem 97 (Normalized logarithmic-kernel monotonicity). *Define*

$$g(t) = \begin{cases} 1, & t = 1, \\ \frac{t \log t}{t-1}, & t \neq 1. \end{cases}$$

For every $t > 0$,

$$g(t) = \int_0^1 \frac{t}{u + (1-u)t} du.$$

Consequently, if $0 \prec A \preceq B$ are finite complex C^* -matrices, then

$$g(A) \preceq g(B),$$

where $g(A)$ and $g(B)$ are interpreted by real continuous functional calculus.

Theorem 98 (Divided-difference operator monotonicity for $x \log x$). *For every $c > 0$, the first divided difference*

$$f_c(x) = \begin{cases} 1 + \log c, & x = c, \\ \frac{x \log x - c \log c}{x - c}, & x \neq c \end{cases}$$

is operator-monotone on the strictly positive finite C^ -matrix cone:*

$$0 \prec A \preceq B \implies f_c(A) \preceq f_c(B).$$

Corollary 99 (Conditional Benda–Sherman closure). *If the finite Benda–Sherman divided-difference bridge is proved locally, then the concrete positive-cone operator-convexity target for $x \log x$ follows. This is still only a route adapter: the bridge theorem remains open, but the divided-difference monotonicity input is now proved.*

Theorem 100 (Finite inverse convexity by Schur complement). *Let $A, B \in \mathbb{C}^{n \times n}$ be positive definite and let $\alpha, \beta \geq 0$ with $\alpha + \beta = 1$. Then*

$$(\alpha A + \beta B)^{-1} \preceq \alpha A^{-1} + \beta B^{-1}.$$

Corollary 101 (Finite C^* -matrix inverse-kernel convexity). *Let A, B be strictly positive finite complex C^* -matrices, and let $\alpha, \beta \geq 0$ with $\alpha + \beta = 1$. Then the real continuous functional calculus satisfies*

$$f(\alpha A + \beta B) \preceq \alpha f(A) + \beta f(B), \quad f(x) = x^{-1}.$$

Equivalently, the unshifted inverse kernel is operator-convex in the finite C^ -matrix vocabulary used by the Lieb/Tropp route.*

Corollary 102 (Shifted inverse-kernel convexity). *Let A, B be nonnegative finite complex C^* -matrices, let $s > 0$, and let $\alpha, \beta \geq 0$ with $\alpha + \beta = 1$. Then*

$$g_s(\alpha A + \beta B) \preceq \alpha g_s(A) + \beta g_s(B), \quad g_s(x) = (s + x)^{-1}.$$

Theorem 103 (Direct operator convexity of $x \log x$). *Let A, B be strictly positive finite complex C^* -matrices and let $\alpha, \beta \geq 0$ with $\alpha + \beta = 1$. Then*

$$(\alpha A + \beta B) \log(\alpha A + \beta B) \preceq \alpha A \log A + \beta B \log B,$$

where the products are interpreted through the real continuous functional calculus $f(x) = x \log x$.

Corollary 104 (Closed ordinary $x \log x$ operator-convexity target). *The Lean target `cstarMatrixXLogXPositiveOperatorConvexTarget` is now closed locally by the direct unit-interval kernel route. The next open Hansen–Pedersen/Effros dependency is the perspective/relative-entropy joint-convexity layer, not ordinary positive-cone operator convexity of $x \log x$.*

Corollary 105 (All-finite source correction for Hansen–Pedersen). *The direct unit-interval proof also gives `cstarMatrixXLogXPositiveOperatorConvexAllFiniteTarget`: ordinary positive-cone operator convexity of $x \log x$ at every finite matrix size. This is the hypothesis shape needed by the standard Hansen–Pedersen block-matrix proof. Consequently, `cstarMatrixXLogXHansenPedersenJensenTarget_of_allFiniteTransfer` reduces the assembled two-point Jensen target to the source-faithful all-finite transfer theorem `cstarMatrixXLogXHansenPedersenTransferAllFiniteTarget`. The concrete two-point Jensen target is closed below by a direct reflection-average proof, so the all-finite transfer theorem remains only a generic source-route target.*

Lemma 106 (Block-compression algebra for Hansen–Pedersen). *Let*

$$V = \begin{bmatrix} A \\ B \end{bmatrix}, \quad D = \begin{bmatrix} T_1 & 0 \\ 0 & T_2 \end{bmatrix}.$$

Then, in the finite C^ -matrix model,*

$$V^*V = A^*A + B^*B, \quad DV = \begin{bmatrix} T_1A \\ T_2B \end{bmatrix},$$

and

$$V^*DV = A^*T_1A + B^*T_2B.$$

*In particular, if $A^*A + B^*B = I$, then $V^*V = I$. In addition, the block diagonal constructor preserves the star-algebra operations used by the proof: zero, identity, addition, subtraction, star, multiplication, and units. It also preserves positivity and strict positivity: if $T_1, T_2 \geq 0$, then*

$$\text{diag}(T_1, T_2) \geq 0,$$

and if T_1, T_2 are strictly positive, then

$$\text{diag}(T_1, T_2)$$

is strictly positive. Thus the block algebra and positive-cone bookkeeping used by the standard Hansen–Pedersen proof are now formalized; the nonlinear functional-calculus corner and concrete Jensen transfer are handled in the following lemmas.

The block-isometry range projection is also formalized. If $V = [A; B]$ and $P = VV^$, then under $V^*V = I$,*

$$P = P^*, \quad P^2 = P, \quad PV = V, \quad V^*P = V^*.$$

These are the algebraic projection identities used by the pinching/compression route to Hansen–Pedersen Jensen. The associated range reflection

$$R = 2P - I$$

is also formalized:

$$R = R^*, \quad R^2 = I, \quad R \text{ is a unitary unit}, \quad RV = V.$$

*The adjoint block row is also fixed by the same reflection, $V^*R = V^*$, and therefore the algebraic reflection average has the same compression as the original block matrix:*

$$V^* \frac{D + RDR}{2} V = V^*DV.$$

The same averaged block is invariant under the reflection and commutes with it:

$$R \frac{D + RDR}{2} R = \frac{D + RDR}{2}, \quad R \frac{D + RDR}{2} = \frac{D + RDR}{2} R.$$

It also commutes with the range projection $P = VV^*$. Consequently its action on the block column and adjoint block row factors through the same compressed corner as D :

$$\frac{D + RDR}{2} V = V(V^*DV), \quad V^* \frac{D + RDR}{2} = (V^*DV)V^*.$$

This closes the linear projection/reflection algebra and the algebraic pinching-average compression/invariance/projection-commutation/range-reduction identities. The nonlinear inequality side and the $x \log x$ corner of the pinching route are handled below.

Lean formalization.

```
Analysis/CStarMatrixBridge.lean cstarMatrixBlockDiagonal
cstarMatrixBlockDiagonal_zero_zero
cstarMatrixBlockDiagonal_one_one
cstarMatrixBlockDiagonal_add
cstarMatrixBlockDiagonal_neg
cstarMatrixBlockDiagonal_sub
cstarMatrixBlockDiagonal_star
cstarMatrixBlockDiagonal_isSelfAdjoint
cstarMatrixBlockDiagonal_mul
cstarMatrixBlockDiagonal_isUnit
cstarMatrixBlockDiagonal_left_nonneg
cstarMatrixBlockDiagonal_right_nonneg
cstarMatrixBlockDiagonal_nonneg
cstarMatrixBlockDiagonal_isStrictlyPositive
cstarMatrixBlockDiagonalStarAlgHom
cstarMatrixBlockDiagonalStarAlgHom_continuous
cstarMatrix_mul_assoc_rect
cstarMatrix_mul_add_rect
cstarMatrix_add_mul_rect
cstarMatrix_mul_smul_rect
cstarMatrix_smul_mul_rect
cstarMatrix_mul_one_rect
cstarMatrix_one_mul_rect
cstarMatrixColumnPair
cstarMatrixColumnPair_conjTranspose_mul_columnPair
cstarMatrixColumnPair_conjTranspose_mul_self
cstarMatrixBlockDiagonal_mul_columnPair
cstarMatrixColumnPair_conjTranspose_mul_blockDiagonal_mul_columnPair
```

cstarMatrixColumnPair_conjTranspose_mul_self_eq_one_of_sum
cstarMatrixColumnPairRangeProjection
cstarMatrixColumnPairRangeProjection_isSelfAdjoint
cstarMatrixColumnPairRangeProjection_mul_self_of_sum
cstarMatrixColumnPairRangeProjection_mul_columnPair_of_sum
cstarMatrixColumnPair_conjTranspose_mul_rangeProjection_of_sum
cstarMatrixProjectionReflection
cstarMatrixProjectionReflection_isSelfAdjoint_of_isSelfAdjoint
cstarMatrixProjectionReflection_mul_self_of_idempotent
cstarMatrixProjectionReflection_isUnit_of_idempotent
cstarMatrixProjectionReflection_mem_unitary_of_isSelfAdjoint_of_idempotent
cstarMatrixProjectionReflection_mul_of_mul_eq_self
cstarMatrix_mul_projectionReflection_of_mul_eq_self
cstarMatrix_reflectionAverage_compression_of_fixed
cstarMatrix_reflectionAverage_conj_of_involutive
cstarMatrix_reflectionAverage_commute_of_involutive
cstarMatrix_commute_projection_of_commute_reflection
cstarMatrixColumnPairRangeReflection
cstarMatrixColumnPairRangeReflection_isSelfAdjoint
cstarMatrixColumnPairRangeReflection_mul_self_of_sum
cstarMatrixColumnPairRangeReflection_isUnit_of_sum
cstarMatrixColumnPairRangeReflection_mem_unitary_of_sum
cstarMatrixColumnPairRangeReflection_mul_columnPair_of_sum
cstarMatrixColumnPair_conjTranspose_mul_rangeReflection_of_sum
cstarMatrixColumnPair_reflectionAverage_compression_of_sum
cstarMatrixColumnPair_reflectionAverage_conj_rangeReflection_of_sum
cstarMatrixColumnPair_reflectionAverage_commute_rangeReflection_of_sum
cstarMatrixColumnPair_reflectionAverage_commute_rangeProjection_of_sum
cstarMatrixColumnPair_mul_columnPair_eq_columnPair_compression_of_commute
cstarMatrixColumnPair_conjTranspose_mul_eq_compression_mul_conjTranspose_of_commute
cstarMatrixColumnPair_reflectionAverage_mul_columnPair_of_sum
cstarMatrixColumnPair_conjTranspose_mul_reflectionAverage_of_sum

Lemma 107 (Block-diagonal functional calculus). *Let T_1, T_2 be self-adjoint finite C^* -matrices, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous on $\sigma(T_1) \cup \sigma(T_2)$. Then continuous functional calculus commutes with the block diagonal:*

$$f\left(\begin{bmatrix} T_1 & 0 \\ 0 & T_2 \end{bmatrix}\right) = \begin{bmatrix} f(T_1) & 0 \\ 0 & f(T_2) \end{bmatrix}.$$

This closes the block-diagonal CFC part of the Hansen–Pedersen block proof. It does not by itself prove the compression inequality

$$f(V^*DV) \leq V^*f(D)V.$$

Lean formalization.

`Analysis/LiebTrace.lean cstarMatrixBlockDiagonal_cfc`

Lemma 108 (Unitary-conjugation functional calculus). *Let U be a unitary finite C^* -matrix, let $T = T^*$, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous on $\sigma(T)$. Then*

$$f(UTU^*) = Uf(T)U^*.$$

This closes the CFC-conjugation part of the range-reflection route. It does not prove the nonlinear CFC pinching inequality or the compression Jensen inequality.

Lean formalization.

`Analysis/LiebTrace.lean cstarMatrix_cfc_unitary_conj`

Lemma 109 (Reflection-average CFC pinching inequality). *Let $V = [A; B]$, assume $V^*V = I$, and put*

$$R = 2VV^* - I, \quad E = \frac{D + RDR}{2}.$$

Assume D is strictly positive and f is continuous on $\sigma(D)$. If f is ordinary operator-convex on the positive cone at every finite matrix size, then

$$f(E) \leq \frac{f(D) + Rf(D)R}{2}.$$

Moreover, after compression by V ,

$$V^*f(E)V \leq V^*f(D)V.$$

Lean formalization.

`Analysis/LiebTrace.lean cstarMatrix_compression_nonneg`

`cstarMatrix_compression_mono`

`cstarMatrixColumnPair_reflectionAverage_cfc_le_average_of_sum`

`cstarMatrixColumnPair_reflectionAverage_compressed_cfc_le_compressed_of_sum`

Remark. This is a genuine new closure in the Hansen–Pedersen route, but it is still not the final concrete $x \log x$ Jensen theorem. The next nonlinear corner step is

$$V^*f(E)V = f(V^*DV),$$

first for the shifted-inverse kernels and then for $x \log x$, or an alternative Effros/perspective proof that avoids this corner identity.

Lemma 110 (Shifted-inverse corner identity). *Let $V = [A; B]$, assume $V^*V = I$, and put*

$$R = 2VV^* - I, \quad E = \frac{D + RDR}{2}, \quad X = V^*DV.$$

If D is strictly positive and $s > 0$, then the shifted inverse kernel satisfies the nonlinear corner identity

$$V^*(sI + E)^{-1}V = (sI + X)^{-1}.$$

Equivalently, in real continuous-functional-calculus notation,

$$V^* [x \mapsto (s + x)^{-1}](E) V = [x \mapsto (s + x)^{-1}](V^*DV).$$

Lean formalization.

```
Analysis/CStarMatrixBridge.lean cstarMatrix_units_inv_mul_rect_eq_mul_units_inv_of_mul_eq
Analysis/LiebTrace.lean cstarMatrix_cfc_shifted_inv_mul_rect_eq_mul_cfc_shifted_inv_of_mul_eq
cstarMatrixColumnPair_reflectionAverage_shifted_inv_corner_of_sum
```

Remark. This closes the nonlinear corner identity for the shifted-inverse kernels used in the direct $x \log x$ integral representation. The next theorem assembles the affine and Bochner-integral pieces into the full $x \log x$ corner identity and then the concrete two-point Hansen–Pedersen Jensen statement.

Theorem 111 (Concrete Hansen–Pedersen Jensen for $x \log x$). *Let A, B, T_1, T_2 be finite complex C^* -matrices. Assume*

$$T_1 \succ 0, \quad T_2 \succ 0, \quad A^*A + B^*B = I.$$

Then

$$f(A^*T_1A + B^*T_2B) \preceq A^*f(T_1)A + B^*f(T_2)B, \quad f(x) = x \log x.$$

Equivalently, the concrete target `cstarMatrixXLogXHansenPedersenJensenTarget` is now proved locally.

Lean formalization.

```
Analysis/CStarMatrixBridge.lean cstarMatrix_complex_finiteDimensional
cstarMatrix_compression_add
cstarMatrix_compression_sub
cstarMatrix_compression_smul
cstarMatrix_compression_real_smul
cstarMatrixCompressionCLM
cstarMatrixCompressionCLM_apply
cstarMatrix_compression_one_of_conjTranspose_mul_self_eq_one
Analysis/LiebTrace.lean cstarMatrix_compression_setIntegral
cstarMatrix_compression_isStrictlyPositive_of_injective_mulVec
cstarMatrixColumnPair_compression_isStrictlyPositive_of_sum
cstarMatrixColumnPair_reflectionAverage_xlog_kernel_corner_of_sum
cstarMatrixColumnPair_reflectionAverage_xlog_corner_of_sum
cstarMatrixXLogXHansenPedersenJensenTarget_of_reflectionAverage_xlog_corner
cstarMatrixXLogXHansenPedersenJensenTarget_of_unit_interval_kernel
```

Remark. This closes the concrete two-point $x \log x$ Jensen dependency needed by the chosen Effros/Tropp source route. This lemma alone does not prove the perspective theorem or the trace representation of matrix relative entropy. The ordinary finite perspective theorem for the normalized entropy kernel is stated below; the source-faithful superoperator trace representation, matrix relative-entropy joint convexity, arbitrary- H Lieb trace concavity, trace-MGF domination, and matrix Bernstein remain open.

Corollary 112 (Normalized entropy-kernel Jensen). *Let*

$$\varphi(x) = x \log x - (x - 1).$$

The ordinary positive-cone operator-convexity target for φ holds at every finite matrix size. In addition, under the same hypotheses as the preceding theorem,

$$\varphi(A^*T_1A + B^*T_2B) \preceq A^*\varphi(T_1)A + B^*\varphi(T_2)B.$$

Lean formalization.

```
Analysis/LiebTrace.lean realEntropyKernel
cstarMatrixEntropyKernelPositiveOperatorConvexTarget
cstarMatrixEntropyKernelPositiveOperatorConvexAllFiniteTarget
cstarMatrixEntropyKernelPositiveOperatorConvexTarget_of_xlog
cstarMatrixEntropyKernelPositiveOperatorConvexTarget_of_unit_interval_kernel
cstarMatrixEntropyKernelPositiveOperatorConvexAllFiniteTarget_of_unit_interval_kernel
cstarMatrixEntropyKernelHansenPedersenJensenTarget
cstarMatrix_cfc_realEntropyKernel_eq_xlog_sub_id_add_one
cstarMatrixEntropyKernelHansenPedersenJensenTarget_of_xlog
cstarMatrixEntropyKernelHansenPedersenJensenTarget_of_unit_interval_kernel
```

Remark. This corollary still does not prove the finite operator-perspective theorem, the trace representation of matrix relative entropy, or the noncommutative joint-convexity theorem. It closes only the affine-corrected scalar kernel needed by that perspective route.

Lemma 113 (Perspective square-root substrate). *Let A be a strictly positive finite C^* -matrix. The real continuous functional calculus defines*

$$A^{1/2} := \sqrt{\cdot}(A), \quad A^{-1/2} := (\sqrt{\cdot})^{-1}(A).$$

Then

$$A^{1/2}A^{1/2} = A, \quad A^{-1/2}A^{1/2} = A^{1/2}A^{-1/2} = I, \quad A^{-1/2}AA^{-1/2} = I.$$

Moreover, if $X \succ 0$, then $A^{-1/2}XA^{-1/2} \succ 0$.

Lean formalization.

```
Analysis/LiebTrace.lean cstarMatrixPositiveSqrt
cstarMatrixPositiveInvSqrt
cstarMatrixPositiveSqrt_isSelfAdjoint
cstarMatrixPositiveInvSqrt_isSelfAdjoint
continuousOn_real_inv_sqrt_spectrum_of_isStrictlyPositive
cstarMatrixPositiveSqrt_mul_self
cstarMatrixPositiveInvSqrt_mul_sqrt
cstarMatrixPositiveSqrt_mul_invSqrt
cstarMatrixPositiveInvSqrt_isUnit
complex_ofReal_sqrt_mul_self_of_nonneg
```

```

cstarMatrixPositiveSqrt_isStrictlyPositive
cstarMatrixPositiveInvSqrt_mul_self_mul
cstarMatrixPositiveInvSqrt_mul_self_eq_unit_inv
cstarMatrixPositiveSqrt_commute_unit_inv
cstarMatrixPositiveInvSqrt_conj_isStrictlyPositive

```

Theorem 114 (Ordinary finite entropy-kernel perspective). *Let $f(x) = x \log x - (x - 1)$, and define the finite C^* -matrix perspective*

$$P_f(X, A) = A^{1/2} f\left(A^{-1/2} X A^{-1/2}\right) A^{1/2}.$$

If $X, Y, A, B \succ 0$, $a, b \geq 0$, and $a + b = 1$, then

$$P_f(aX + bY, aA + bB) \preceq aP_f(X, A) + bP_f(Y, B).$$

Lean formalization.

```

Analysis/LiebTrace.lean cstarMatrixPerspective
cstarMatrixPerspectiveWeight
cstarMatrixPerspectiveWeight_star_mul_self
cstarMatrixPerspectiveWeights_star_mul_self_add
cstarMatrixPerspectiveWeight_compress_normalized
cstarMatrixPerspectiveWeight_mul_positiveSqrt
cstarMatrixPositiveSqrt_mul_perspectiveWeight_star
cstarMatrixPerspectiveWeight_value_uncompress
cstarMatrixEntropyKernelPerspective_jointConvex

```

Remark. This is a dependency closure for the Effros perspective route. It proves the ordinary finite perspective theorem for the normalized entropy kernel, but it does not by itself prove the Umegaki relative-entropy trace representation. The source-faithful route for $D(X; A)$ still requires the superoperator perspective built from $L_X R_A^{-1}$ and the trace identity connecting that superoperator perspective to matrix relative entropy.

Lemma 115 (Unitary-conjugation domain preservation). *Let U be a unitary finite C^* -matrix and let T be strictly positive. Then UTU^* is strictly positive. In particular, if $V^*V = I$ and $R = 2VV^* - I$ is the range reflection, then RTR^* is strictly positive whenever T is strictly positive.*

This closes the strict-positive-domain side condition needed before applying continuous functional calculus along the reflection/pinching route. It does not prove the nonlinear corner CFC identity or compression Jensen inequality.

Lean formalization.

```

Analysis/CStarMatrixBridge.lean cstarMatrix_unitary_conj_isStrictlyPositive
cstarMatrixColumnPairRangeReflection_conj_isStrictlyPositive_of_sum

```


Lemma 116 (Strict positivity of the compressed block argument). *Let $V = [A; B]$ and assume $V^*V = A^*A + B^*B = I$. If T_1 and T_2 are strictly positive, then*

$$V^* \begin{bmatrix} T_1 & 0 \\ 0 & T_2 \end{bmatrix} V = A^*T_1A + B^*T_2B$$

is strictly positive.

*The proof bridges strict positivity to ordinary finite matrix positive definiteness, uses $V^*V = I$ to prove that the matrix action of V is injective, applies preservation of positive definiteness under injective compression, and bridges back to the C^* -matrix order. This is a domain lemma only; it does not prove the nonlinear Jensen inequality.*

Lean formalization.

```
Analysis/LiebTrace.lean cstarMatrix_isStrictlyPositive_of_matrix_posDef
cstarMatrixColumnPair_mulVec_injective_of_sum
cstarMatrixColumnPair_compress_blockDiagonal_isStrictlyPositive_of_sum
cstarMatrixHansenPedersenCompression_isStrictlyPositive_of_sum
```

Lean formalization.

```
Analysis/LiebTrace.lean cstarMatrixPositiveOperatorConvexTarget
cstarMatrixPositiveOperatorConvexAllFiniteTarget
cstarMatrixPositiveOperatorConvexTarget_id
cstarMatrixPositiveOperatorConvexAllFiniteTarget_id
cstarMatrixPositiveHansenPedersenTransferTarget
cstarMatrixPositiveHansenPedersenTransferAllFiniteTarget
cstarMatrixXLogXPositiveOperatorConvexTarget
cstarMatrixXLogXHansenPedersenTransferTarget
cstarMatrixXLogXPositiveOperatorConvexAllFiniteTarget
cstarMatrixXLogXHansenPedersenTransferAllFiniteTarget
cstarMatrixHansenPedersenJensenTwoPointTarget
cstarMatrixHansenPedersenJensenTwoPointTarget_id
cstarMatrixXLogXHansenPedersenJensenTarget
cstarMatrixXLogXHansenPedersenJensenTarget_of_positiveOperatorConvex_of_transfer
cstarMatrixXLogXHansenPedersenJensenTarget_of_transfer
cstarMatrixXLogXPositiveOperatorConvexAllFiniteTarget_of_unit_interval_kernel
cstarMatrixXLogXHansenPedersenJensenTarget_of_allFiniteTransfer
cstarMatrixXLogXPositiveOperatorConvexTarget_of_unit_interval_kernel
cstarMatrix_cfc_one_add_log_eq_one_add_log
cstarMatrixXLogXDerivativeMonotoneTarget
cstarMatrixXLogXDerivativeMonotoneTarget_of_log_monotone
cstarMatrixStrictPositiveOperatorMonotoneTarget
realXLogXDividedDifference
realXLogXDividedDifference_self
```

realXLogXDividedDifference_eq_log_add_ratio
 realXLogXDividedDifference_eq_log_add_normalized
 realNormalizedLogKernel
 realNormalizedLogKernel_eq_of_ne_one
 realNormalizedLogKernel_eq_mul_dslope_log
 continuousOn_realNormalizedLogKernel_Ioi
 realXLogXDividedDifference_eq_log_add_normalizedKernel
 real_normalizedLogKernel_offdiag_intervalIntegral
 realNormalizedLogKernel_setIntegral
 cstarMatrix_spectrum_nonneg_of_nonneg
 cstarMatrix_cfc_one_sub_one_add_inv_monotone
 cstarMatrix_cfc_pos_over_one_add_monotone
 cstarMatrix_cfc_pos_over_pos_add_monotone
 cstarMatrix_cfc_unit_interval_fractional_kernel_monotone
 cstarMatrix_cfc_unit_interval_fractional_kernel_monotone_of_mem_Icc
 continuousOn_uncurry_unit_interval_fractional_kernel_spectrum
 real_unit_interval_fractional_kernel_abs_le_max_of_le
 real_unit_interval_fractional_kernel_spectrum_norm_le_max
 ae_unit_interval_fractional_kernel_spectrum_norm_le_max
 hasFiniteIntegral_const_max_one_spectrum_bound
 continuousOn_uncurry_unit_interval_subtype_fractional_kernel_spectrum
 ae_unit_interval_subtype_fractional_kernel_spectrum_norm_le_max
 hasFiniteIntegral_unit_interval_subtype_const_max_one_spectrum_bound
 cstarMatrix_cfc_realNormalizedLogKernel_eq_unit_interval_integral
 cstarMatrix_setIntegral_mono_on
 cstarMatrix_cfc_realNormalizedLogKernel_monotone_of_spectrum_bound
 cstarMatrix_cfc_realNormalizedLogKernel_monotone
 cstarMatrix_cfc_realXLogXDividedDifference_eq_log_add_scaled_normalizedKernel
 cstarMatrix_cfc_finset_sum_nonneg_mul_pos_over_pos_add_monotone
 cfc_integral_mono_of_forall_of_bound
 cstarMatrixXLogXDividedDifferenceMonotoneTarget
 cstarMatrixXLogXDividedDifferenceMonotoneTarget_of_normalizedLogKernel
 cstarMatrixBendatShermanDividedDifferenceBridgeTarget
 cstarMatrixXLogXPositiveOperatorConvexTarget_of_bendatShermanDividedDifferenceBridge
 cstarMatrixBendatShermanDerivativeBridgeTarget
 cstarMatrixXLogXPositiveOperatorConvexTarget_of_bendatShermanDerivativeBridge
 matrix_posDef_inverse_schur_block
 matrix_weighted_inverse_schur_block
 matrix_posDef_weighted_sum
 matrix_inv_convex_posDef

```

cstarMatrix_nonneg_of_matrix_posSemidef
cstarMatrix_le_of_matrix_le
cstarMatrix_cfc_inv_convex_isStrictlyPositive
cstarMatrix_cfc_shifted_inv_eq_cfc_inv_add_smul_one
cstarMatrix_cfc_shifted_inv_convex_nonneg
real_xlog_eq_unit_interval_xlog_kernel_integral
real_unit_interval_xlog_kernel_integrand_eq_affine_add_shifted_inv
cstarMatrix_cfc_unit_interval_xlog_kernel_integrand_eq_affine_add_shifted_inv
continuousOn_uncurry_unit_interval_xlog_kernel_spectrum
real_unit_interval_xlog_kernel_spectrum_norm_le_max_sq
ae_unit_interval_xlog_kernel_spectrum_norm_le_max_sq
hasFiniteIntegral_const_max_one_spectrum_bound_sq
cstarMatrix_cfc_xlog_eq_unit_interval_xlog_kernel_integral
cstarMatrix_cfc_unit_interval_xlog_integrand_convex_of_pos_lt_one
cstarMatrix_cfc_unit_interval_xlog_kernel_integrand_convex_of_pos_lt_one

```

Definition 117 (Relative-entropy variational route). For fixed H , define the variational objective

$$\Psi_H(X, A) = \operatorname{Re} \operatorname{tr}(XH) - D(X; A) + \operatorname{Re} \operatorname{tr}(A).$$

The extra trace term matches the repository normalization

$$D(X; A) = \operatorname{Re} \operatorname{tr}(X(\log X - \log A) - (X - A)).$$

At this point in the development, the generalized Klein, variational-formula, relative-entropy joint-convexity, finite-dimensional Lieb, one-step Tropp trace-MGF, independent-sum trace-MGF iteration, and finite-real trace-exponential transport foundations are closed locally. The remaining concentration foundation is the scalar matrix-CGF/log-MGF bound that feeds the matrix Bernstein/Khintchine tail conversion.

Lemma 118 (Optimizer-candidate value). *With the normalized objective above, the candidate*

$$X_\star = \exp(H + \log A)$$

satisfies

$$\Psi_H(X_\star, A) = \Phi_H(A).$$

This is only the attainment algebra for the expected optimizer. The global upper bound $\Psi_H(X, A) \leq \Phi_H(A)$ for all $X \succ 0$ is supplied below through the local generalized Klein and relative-entropy nonnegativity theorem.

Lemma 119 (Variational formula from relative-entropy nonnegativity). *If matrix relative entropy is nonnegative on the strictly positive cone, then the normalized variational formula holds. Indeed, for $X_\star = \exp(H + \log A)$,*

$$\Psi_H(Y, A) = \Phi_H(A) - D(Y; X_\star).$$

Thus the missing maximality inequality $\Psi_H(Y, A) \leq \Phi_H(A)$ is exactly the nonnegativity of $D(Y; X_\star)$.

Lemma 120 (Relative-entropy nonnegativity from generalized Klein). *Let*

$$\Phi(X) = \operatorname{Re} \operatorname{tr}(X \log X - X).$$

Assume the generalized Klein first-order trace inequality on the strictly positive cone:

$$\Phi(A) + \operatorname{Re} \operatorname{tr}((\log A)(X - A)) \leq \Phi(X) \quad (X, A \succ 0).$$

Then the local matrix relative entropy is nonnegative:

$$D(X; A) = \operatorname{Re} \operatorname{tr}(X(\log X - \log A) - (X - A)) \geq 0.$$

Corollary 121 (Klein and local relative entropy are equivalent). *In the repository normalization*

$$D(X; A) = \operatorname{Re} \operatorname{tr}(X(\log X - \log A) - (X - A)),$$

the generalized Klein first-order trace inequality above is equivalent to nonnegativity of $D(X; A)$ on the strictly positive cone.

Lemma 122 (Hermitian spectral-overlap expansion). *Let A and B be Hermitian $d \times d$ complex matrices. For finite families of scalar functions $f_r, g_r : \mathbb{R} \rightarrow \mathbb{R}$,*

$$\operatorname{Re} \sum_r \operatorname{tr}(f_r(A)g_r(B)) = \sum_{j,k} \left(\sum_r f_r(\lambda_j(A))g_r(\lambda_k(B)) \right) |\langle u_j, v_k \rangle|^2,$$

where $\{u_j\}$ and $\{v_k\}$ are Hermitian eigenbases for A and B . Hence a nonnegative scalar kernel gives a nonnegative trace sum.

Corollary 123 (Entropy first-order kernel on positive spectra). *For $\phi(t) = t \log t - t$ and $\psi(t) = \log t$, if the eigenvalues of Hermitian A and B are positive, then*

$$0 \leq \operatorname{Re} \operatorname{tr}(\phi(A)I - I\phi(B) - A\psi(B) + I(\psi(B)B)).$$

Equivalently, this is the separated-kernel Hermitian form of $\operatorname{tr} \phi(A) - \operatorname{tr} \phi(B) - \operatorname{tr}(\psi(B)(A - B)) \geq 0$.

Theorem 124 (Compact Hermitian generalized Klein inequality). *Let X, A be Hermitian complex matrices whose eigenvalues are all strictly positive. Then*

$$\operatorname{Re} \operatorname{tr}(A \log A - A + \log(A)(X - A)) \leq \operatorname{Re} \operatorname{tr}(X \log X - X).$$

Lemma 125 (Strictly positive C-star matrices have positive spectra). *If M is nonnegative as a finite complex C^* -matrix, then its underlying Hermitian matrix is positive semidefinite. If M is strictly positive, then every Hermitian eigenvalue of the underlying matrix is strictly positive.*

Theorem 126 (Generalized Klein inequality on the strict cone). *For all strictly positive finite complex C^* -matrices X, A ,*

$$\Phi(A) + \operatorname{Re} \operatorname{tr}(\log(A)(X - A)) \leq \Phi(X), \quad \Phi(Y) = \operatorname{Re} \operatorname{tr}(Y \log Y - Y).$$

Corollary 127 (Matrix relative entropy is nonnegative). *For all strictly positive finite complex C^* -matrices X, A ,*

$$D(X; A) = \operatorname{Re} \operatorname{tr}(X(\log X - \log A) - (X - A)) \geq 0.$$

Corollary 128 (Normalized entropy variational formula). *For self-adjoint H , the normalized variational formula holds:*

$$\Phi_H(A) = \sup_{X \succ 0} \Psi_H(X, A), \quad \Psi_H(X, A) = \operatorname{Re} \operatorname{tr}(XH) - D(X; A) + \operatorname{Re} \operatorname{tr}(A).$$

Corollary 129 (Joint convexity alone now implies the local Lieb target). *Assume matrix relative entropy is jointly convex on the strictly positive cone. Then, for every self-adjoint H ,*

$$A \mapsto \operatorname{Re} \operatorname{tr} \exp(H + \log A)$$

is concave on the strictly positive cone.

Theorem 130 (Finite-dimensional relative entropy is jointly convex). *For finite complex matrices $X_1, X_2, A_1, A_2 \succ 0$ and $\lambda, \mu \geq 0$ with $\lambda + \mu = 1$,*

$$D(\lambda X_1 + \mu X_2; \lambda A_1 + \mu A_2) \leq \lambda D(X_1; A_1) + \mu D(X_2; A_2).$$

Corollary 131 (Finite-dimensional Lieb trace concavity). *If $H = H^*$, then*

$$A \mapsto \operatorname{Re} \operatorname{tr} \exp(H + \log A)$$

is concave on the strictly positive cone.

Corollary 132 (One-step Tropp trace-MGF inequality). *For a finite probability space, a self-adjoint H , and a self-adjoint matrix random variable X ,*

$$\mathbb{E} \operatorname{Re} \operatorname{tr} \exp(H + X) \leq \operatorname{Re} \operatorname{tr} \exp(H + \log \mathbb{E} \exp X).$$

Theorem 133 (Relative-entropy route implies Lieb concavity). *Assume H is self-adjoint. If matrix relative entropy is jointly convex on the strictly positive cone and the variational formula above holds for this H , then*

$$A \mapsto \operatorname{Re} \operatorname{tr} \exp(H + \log A)$$

is concave on the strictly positive cone.

Lemma 134 (Spectral-overlap form of the compact relative-entropy trace). *Let $X, A \in \mathbb{C}^{d \times d}$ be Hermitian. Write λ_i and μ_j for their eigenvalues, and write*

$$W = U_X^* U_A$$

for the unitary overlap between their eigenvector bases. Then the repository's compact trace expression satisfies

$$\operatorname{Re} \operatorname{tr}(X \log X - X \log A - (X - A)) = \sum_{i,j} (\lambda_i (\log \lambda_i - \log \mu_j) - (\lambda_i - \mu_j)) |W_{ij}|^2.$$

Lean formalization.

```
Analysis/LiebTrace.lean
cstarMatrixEntropyVariationalObjective
cstarMatrixEntropyVariationalObjective_liebOptimizer
cstarMatrixRelativeEntropyNonnegOnStrictPositive
cstarMatrixEntropyTraceFirstOrderConvexityOnStrictPositive
```

cstarMatrixRelativeEntropyNonnegOnStrictPositive_of_entropyTraceFirstOrder
cstarMatrixEntropyTraceFirstOrderConvexityOnStrictPositive_of_relativeEntropy_nonneg
cstarMatrixEntropyTraceFirstOrderConvexityOnStrictPositive_iff_relativeEntropy_nonneg
matrixTrace_diagonal_mul_mul_diagonal_mul_star
matrixTrace_sum_diagonal_mul_mul_diagonal_mul_star_re
matrixTrace_sum_hermitianCfc_mul_cfc_re
matrixTrace_sum_hermitianCfc_mul_cfc_nonneg_of_kernel_nonneg
matrixTrace_sum_hermitianCfc_mul_cfc_nonneg_of_eigen_kernel_nonneg
matrixTrace_hermitianCfc_firstOrderKernel_sum_nonneg
matrixTrace_hermitianCfc_firstOrderKernel_sum_nonneg_of_eigen
realEntropy_firstOrderKernel_nonneg
matrixTrace_hermitianCfc_entropy_firstOrder_sum_nonneg
matrix_isHermitian_cfc_const_one
matrix_isHermitian_cfc_const_neg_one
matrix_isHermitian_cfc_neg_id
matrix_isHermitian_cfc_id
matrix_isHermitian_cfc_entropy
matrix_isHermitian_cfc_xlog
matrix_isHermitian_cfc_log_mul_id
matrixTrace_hermitianCfc_relativeEntropy_re_eq_sum
matrixTrace_hermitianCfc_entropy_firstOrder_compact_nonneg
cstarMatrix_nonneg_to_matrix_posSemidef
cstarMatrix_isStrictlyPositive_to_matrix_posDef
cstarMatrixEntropyTraceFirstOrderConvexityOnStrictPositive_of_hermitianCfc
cstarMatrixRelativeEntropyNonnegOnStrictPositive_of_hermitianCfc
cstarMatrixEntropyVariationalFormula_of_relativeEntropy_nonneg
cstarMatrixEntropyVariationalFormula_of_hermitianCfc
cstarMatrixRelativeEntropyJointConvexOnStrictPositive
cstarMatrixSuperoperatorEntropyKernelCfc_ratio_commute_positiveSqrtRightLift
cstarMatrixSuperoperatorPerspective_outerSqrt_cfc_ratio_mul_outerSqrt
cstarMatrixSuperoperatorPerspective_eq_cfc_ratio_mul_rightLift
cstarMatrix_unit_inv_to_matrix
cstarMatrixSuperoperatorRightLift_unit_inv_to_matrix
cstarMatrixSuperoperatorPerspectiveTrace_eq_matrixSuperoperatorEntropyKernelTrace
cstarMatrixRelativeEntropyPerspectiveTraceRepresentation_all
cstarMatrixRelativeEntropyJointConvexOnStrictPositive_all
cstarMatrixEntropyVariationalFormula
liebTraceConcavityTarget_of_relativeEntropy_route
liebTraceConcavityTarget_of_relativeEntropy_jointConvex
liebTraceConcavityTarget_all
FiniteProbability.expectationReal_trace_normed_exp_add_le

Lemma 135 (Algorithm 1 one-step product-law adapter). *Let P_s be the canonical independent squared-magnitude trace law for Algorithm 1, and let P_1 be the corresponding one-sample law on sampled entries. For any fixed trace coordinate $t < s$, any complex-valued one-sample statistic f , and any C^* -matrix-valued one-sample statistic F ,*

$$\mathbb{E}_{P_s} f(\omega_t) = \mathbb{E}_{P_1} f, \quad \mathbb{E}_{P_s} F(\omega_t) = \mathbb{E}_{P_1} F.$$

Consequently, if H and every $X(x)$ are self-adjoint, then

$$\mathbb{E}_{P_s} \operatorname{Re} \operatorname{tr} \exp(H + X(\omega_t)) \leq \operatorname{Re} \operatorname{tr} \exp(H + \log \mathbb{E}_{P_1} \exp(X)).$$

This is the one-coordinate Algorithm 1 specialization of the closed finite-dimensional Tropp/Lieb one-step trace-MGF theorem.

Lean formalization.

```
sqMagSampleProbability
sqMagTraceProbability_expectationComplex_step_eq
sqMagTraceProbability_expectationCStarMatrix_step_eq
sqMagTraceProbability_expectationCStarMatrix_step_eq_sampleExpectation
sqMagTraceProbability_expectationReal_trace_normed_exp_add_step_le
```

Theorem 136 (Algorithm 1 iid trace-MGF domination). *Let P_s be the s -fold independent squared-magnitude trace law and let P_1 be its one-sample marginal. Let*

$$X : \{1, \dots, m\} \times \{1, \dots, n\} \longrightarrow \mathbb{C}^{d \times d}$$

be self-adjoint valued, and let H be self-adjoint. Define

$$K = \log \mathbb{E}_{P_1} \exp(X).$$

Then

$$\mathbb{E}_{P_s} \operatorname{Re} \operatorname{tr} \exp\left(H + \sum_{t=1}^s X(\omega_t)\right) \leq \operatorname{Re} \operatorname{tr} \exp\left(H + \sum_{t=1}^s K\right).$$

This is the independent-sum lift of the one-step Tropp/Lieb theorem for the canonical Algorithm 1 product sampler. The proof first factors the trace mass after appending a last sample, then conditions on the last coordinate, applies the one-step theorem, and iterates.

Lean formalization.

```
sqMagTraceProbMass_snoc
sqMagTraceProbability_expectationReal_succ_last_eq
cstarMatrix_finset_sum_isSelfAdjoint
sqMagTraceProbability_expectationReal_trace_normed_exp_add_sum_le
```

Corollary 137 (Algorithm 1 finite-real trace-MGF bridge). *Let $H \in \mathbb{R}^{d \times d}$ and every one-sample increment $X(x) \in \mathbb{R}^{d \times d}$ be symmetric. Under the same independent Algorithm 1 trace law P_s , define*

$$K = \log \mathbb{E}_{P_1} \exp(X_{\mathbb{C}}),$$

where $X_{\mathbb{C}}$ denotes the complex C^* -matrix embedding of the real matrix X . Then

$$\mathbb{E}_{P_s} \operatorname{tr}_{\mathbb{R}} \exp_{\text{mat}} \left(H + \sum_{t=1}^s X(\omega_t) \right) \leq \operatorname{Re} \operatorname{tr}_{\mathbb{C}} \exp \left(H_{\mathbb{C}} + \sum_{t=1}^s K \right).$$

The bridge uses the newly formalized identity

$$\operatorname{Re} \operatorname{tr}_{\mathbb{C}} \exp(M_{\mathbb{C}}) = \operatorname{tr}_{\mathbb{R}}(\exp_{\text{mat}}(M)),$$

proved by showing that the finite-real embedding is a continuous ring homomorphism and therefore commutes with the power-series matrix exponential. This is a genuine interface theorem between the C^* -matrix Lieb iteration and the repository's finite-real trace-exponential concentration layer. It does not yet bound K by a scalar variance proxy and therefore is not by itself the CACM equation (2) Bernstein/Khintchine tail bound.

Lean formalization.

```
finiteComplexCStarMatrixRingHom
finiteComplexCStarMatrixRingHom_continuous
finiteComplexCStarMatrix_finiteMatrixExp
cstarMatrixTrace_normed_exp_finiteComplexCStarMatrix
cstarMatrixTrace_normed_exp_finiteComplexCStarMatrix_re
finiteComplexCStarMatrix_add
finiteComplexCStarMatrix_finset_sum
sqMagTraceProbabilityFiniteRealTraceMGFLogBound
sqMagTraceProbability_expectationReal_finiteTrace_finiteMatrixExp_add_sum_le
```

Corollary 138 (Algorithm 1 residual-dilation trace-MGF target). *Let $Z(x)$ be the one-sample Algorithm 1 residual increment and let*

$$D(Z) = \begin{bmatrix} 0 & Z \\ Z^{\top} & 0 \end{bmatrix}$$

be its self-adjoint dilation. For every scalar θ , the finite-real trace-MGF bridge applies to the symmetric increments $\theta D(Z(x))$. Consequently,

$$\mathbb{E}_{P_s} \operatorname{tr}_{\mathbb{R}} \exp_{\text{mat}} \left(\theta D(A - \tilde{A}) \right) \leq \operatorname{Re} \operatorname{tr}_{\mathbb{C}} \exp \left(\sum_{t=1}^s \log \mathbb{E}_{P_1} \exp(\theta D(Z)_{\mathbb{C}}) \right).$$

The proof first instantiates the finite-real trace-MGF theorem with $\theta D(Z(x))$, then rewrites $\sum_t \theta D(Z(\omega_t)) = \theta D(A - \tilde{A})$ using the formal residual decomposition. This is the exact trace-exponential object needed by the matrix Bernstein route. The remaining unproved step is to bound the logarithmic mean increment by a scalar/variance proxy.

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSpectral.lean rectSelfAdjointDilation_
elementwiseSampleResidualIncrement_smul_symmetric
sqMagTraceProbability_expectationReal_finiteTrace_finiteMatrixExp_sum_rectSelfAdjointDilation_
sampleResidualIncrement_le
sqMagTraceProbability_expectationReal_finiteTrace_finiteMatrixExp_rectSelfAdjointDilation_
elementwiseTraceResidual_le
```


Corollary 139 (Algorithm 1 eigenvalue Markov step after trace-MGF). *With the notation of the preceding corollary, let $T \in \mathbb{R}$. The repository now also specializes the generic trace-exponential Markov interface to the scaled dilation residual:*

$$\mathbb{P}\left\{\exists a : T \leq \lambda_a\left(\theta D(A - \tilde{A})\right)\right\} \leq e^{-T} \operatorname{Re} \operatorname{tr}_{\mathbb{C}} \exp\left(\sum_{t=1}^s \log \mathbb{E}_{P_1} \exp(\theta D(Z)_{\mathbb{C}})\right).$$

It also proves the corresponding two-sided absolute-eigenvalue event by using the same trace-MGF theorem with θ and $-\theta$. This closes the trace-exponential-to-eigenvalue Markov dependency for the actual Algorithm 1 dilation residual. The later source-aligned sections add the scalar matrix-CGF/Bernstein constants and convert this dependency into the closed equation (2)-style theorem for the truncated squared-magnitude law.

Lean formalization.

```
Algorithms/RandNLA/ElementwiseSpectral.lean      sqMagTraceProbability_eventProb_exists_
finiteHermitianEigenvalue_scaled_rectSelfAdjointDilation_elementwiseTraceResidual_ge_le
sqMagTraceProbability_eventProb_forall_abs_finiteHermitianEigenvalue_scaled_
rectSelfAdjointDilation_elementwiseTraceResidual_lt_ge
```

Definition 140 (Truncated elementwise variant). For a threshold $\tau \geq 0$, define the hard-thresholded matrix

$$\hat{A}_{ij} = \begin{cases} 0, & |A_{ij}| < \tau, \\ A_{ij}, & \text{otherwise.} \end{cases}$$

The source-aligned truncated sketch samples from $\hat{A}$ but measures the final residual against the original matrix A .

Theorem 141 (Deterministic truncation cost). *Let $A \in \mathbb{R}^{n \times n}$, $n > 0$, and $\varepsilon \geq 0$. With $\tau = \varepsilon/(2n)$,*

$$\|A - \hat{A}\|_2 \leq \|A - \hat{A}\|_F \leq \frac{\varepsilon}{2}.$$

Thus, if the exact sampled residual of the truncated matrix satisfies

$$\|\hat{A} - \tilde{\hat{A}}\|_2 \leq \frac{\varepsilon}{2},$$

then the truncated sketch is within ε of the original matrix:

$$\|A - \tilde{\hat{A}}\|_2 \leq \varepsilon.$$

Corollary 142 (Exact truncated transfer). *Any high-probability half-budget theorem for sampling the truncated matrix $\hat{A}$ transfers immediately to the original-matrix event $\|A - \tilde{\hat{A}}\|_2 \leq \varepsilon$. This is the exact source-alignment layer. The rounded endpoint is not stated here with a generic perturbation matrix; the final Algorithm 1 corollary below expands the internal floating-point error into the explicit γ_{s+1} -radius.*

Theorem 143 (Support and bounded increments for the truncated route). *Assume $\tau > 0$, $s > 0$, and $\|\hat{A}\|_F^2 > 0$. Under the independent product law that samples entries with probabilities $\hat{A}_{ij}^2 / \|\hat{A}\|_F^2$,*

the event that every sampled entry has positive probability has probability one. On that event, each one-step residual increment

$$Z_X = \frac{\hat{A}}{s} - C_X$$

satisfies

$$\|Z_X\|_2 \leq \|Z_X\|_F \leq \frac{1}{s} \|\hat{A}\|_F + \frac{\|\hat{A}\|_F^2}{s\tau}.$$

Equivalently, its self-adjoint dilation $D(Z_X)$ satisfies the finite operator bound

$$\|D(Z_X)\|_2 \leq \sqrt{2} \left(\frac{1}{s} \|\hat{A}\|_F + \frac{\|\hat{A}\|_F^2}{s\tau} \right),$$

and therefore the squared-Loewner bound

$$D(Z_X)^2 \preceq 2 \left(\frac{1}{s} \|\hat{A}\|_F + \frac{\|\hat{A}\|_F^2}{s\tau} \right)^2 I.$$

The operator bound is also exposed in the one-sided and lower Loewner forms

$$D(Z_X) \preceq LI, \quad -D(Z_X) \preceq LI, \quad L = \sqrt{2} \left(\frac{1}{s} \|\hat{A}\|_F + \frac{\|\hat{A}\|_F^2}{s\tau} \right).$$

The corresponding “all increments are bounded”, “all increments are two-sided Loewner-bounded”, and “all squared increments are Loewner-bounded” trace events have probability one. Their intersection, the simultaneous Bernstein-boundedness event needed by a future matrix Bernstein theorem, also has probability one.

Corollary 144 (Support-aware truncated finite-test MGF). *Let $\mathcal{N}$ be a finite family of deterministic test vectors $z_a \in \mathbb{R}^{m+n}$, and set*

$$L_\tau = \sqrt{2} \left(\frac{1}{s} \|\hat{A}\|_F + \frac{\|\hat{A}\|_F^2}{s\tau} \right).$$

For every positive-probability sampled entry of the truncated law,

$$z_a^T D(Z_X) z_a \leq L_\tau \|z_a\|_2^2.$$

Therefore, for any $\lambda_a > 0$,

$$\mathbb{P}[z_a^T D(R_{\hat{A}}) z_a \leq T_a \text{ for all } a \in \mathcal{N}] \geq 1 - \sum_{a \in \mathcal{N}} \exp(s\lambda_a L_\tau \|z_a\|_2^2 - \lambda_a T_a).$$

This is a support-aware finite-test scalar tail. It removes the hidden “the sampled entry is retained” side condition from finite test-vector arguments, but it is not the matrix Bernstein half-budget event.

Lean formalization.

elementwiseTruncate
elementwiseTruncatedTraceResidual
fl_elementwiseTruncatedTraceResidual
elementwiseTruncate_error_entry_abs_le
elementwiseTruncate_abs_le
frobNormSqRect_elementwiseTruncate_le
frobNormRect_elementwiseTruncate_le
elementwiseTruncate_nonzero_abs_ge
frobNormRect_const_square
elementwiseTruncate_square_error_frobNormRect_le_half
elementwiseTruncate_square_error_rectOpNorm2Le_half
elementwiseTruncatedTraceResidual_rectOpNorm2Le_of_truncated
elementwiseTruncatedTraceResidual_square_rectOpNorm2Le_of_half
algorithm1ExactTruncatedSpectralEvent
algorithm1ExactSpectralEvent_truncated_half_subset_original
probability_algorithm1_exact_truncated_spectral_of_sampled_half
fl_elementwiseTruncatedTraceResidual_rectOpNorm2Le_of_truncated
algorithm1FlTruncatedSpectralEvent
algorithm1ExactSpectralEvent_truncated_half_subset_fl_original
probability_algorithm1_fl_truncated_spectral_of_sampled_half
entry_ne_zero_of_sqMagProb_pos
FiniteProbability.eventProb_le_one
elementwiseTracePositiveProb
sqMagTraceProbability_eventProb_elementwiseTracePositiveProb
frobNormRect_elementwiseSampleContribution_truncated_le
frobNormRect_elementwiseSampleResidualIncrement_truncated_le
rectOpNorm2Le_elementwiseSampleResidualIncrement_truncated
sqMagTraceProbability_eventProb_truncatedResidualIncrementsBoundedEvent_eq_one
finiteOpNorm2Le_rectSelfAdjointDilation_elementwiseSampleResidualIncrement_truncated
finiteQuadraticForm_rectSelfAdjointDilation_elementwiseSampleResidualIncrement_truncated_le
sqMagTraceProbability_eventProb_truncatedDilationIncrementsBoundedEvent_eq_one
finiteLoewnerLe_rectSelfAdjointDilation_elementwiseSampleResidualIncrement_truncated
finiteLoewnerLe_neg_rectSelfAdjointDilation_elementwiseSampleResidualIncrement_truncated
truncatedDilationIncrementLoewnerBoundedEvent
sqMagTraceProbability_eventProb_truncatedDilationIncrementLoewnerBoundedEvent_eq_one
truncatedDilationIncrementLoewnerBoundedEvent_subset_exactDilationUpperEvent_sum_bound
sqMagTraceProbability_eventProb_algorithm1ExactDilationUpperEvent_truncated_sum_bound_eq_one
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_truncated_sum_bound_eq_one

`finiteLoewnerLe_rectSelfAdjointDilation_square_elementwiseSampleResidualIncrement_truncated`
`truncatedDilationIncrementSquaresBoundedEvent`
`sqMagTraceProbability_eventProb_truncatedDilationIncrementSquaresBoundedEvent_eq_one`
`FiniteProbability.eventProb_inter_eq_one_of_eq_one`
`truncatedDilationBernsteinBoundedEvent`
`sqMagTraceProbability_eventProb_truncatedDilationBernsteinBoundedEvent_eq_one`
`sqMagTraceProbability_eventProb_forall_finiteQuadraticForm_rectSelfAdjointDilation_truncatedTraceResidual_le_ge_one_sub_sum_exp_of_support_bound`

Theorem 145 (Accumulated bounded-increment spectral bound). *Let $\hat{A} = T_\tau(A)$, and suppose each retained truncated self-adjoint dilation increment X_t satisfies the one-sided Loewner bound*

$$X_t \preceq LI.$$

Then the full exact truncated residual dilation satisfies

$$\sum_{t=1}^s X_t \preceq sL I.$$

Consequently, under the truncated squared-magnitude product law, the exact truncated sampled residual satisfies

$$\|\hat{A} - \tilde{\hat{A}}\|_2 \leq sL$$

with probability 1, for the explicit retained-increment bound

$$L = \sqrt{2} \left(\frac{\|\hat{A}\|_F}{s} + \frac{\|\hat{A}\|_F^2}{s\tau} \right).$$

Lean formalization.

`truncatedDilationIncrementLoewnerBoundedEvent_subset_exactDilationUpperEvent_sum_bound`
`sqMagTraceProbability_eventProb_algorithm1ExactDilationUpperEvent_truncated_sum_bound_eq_one`
`sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_truncated_sum_bound_eq_one`

Theorem 146 (One-step residual variance proxies). *Let X be one Algorithm 1 sampled entry drawn from the squared-magnitude law, and define the one-step residual increment*

$$Z_X = \frac{A}{s} - C_X,$$

where C_X is the one-sample contribution matrix. Then, for every entry (i, j) ,

$$\mathbb{E}(Z_X)_{ij}^2 \leq \frac{\|A\|_F^2}{s^2}.$$

Consequently,

$$\mathbb{E} \|Z_X\|_F^2 \leq mn \frac{\|A\|_F^2}{s^2}, \quad \mathbb{E} \|Z_X x\|_2^2 \leq mn \frac{\|A\|_F^2}{s^2} \|x\|_2^2.$$

Corollary 147 (Fixed-vector residual tail). *For the full s -sample residual $R = A - \tilde{A}$, every fixed vector x , and every $\eta > 0$,*

$$\mathbb{P}[\|Rx\|_2 \leq \eta] \geq 1 - \frac{(mn/s) \|A\|_F^2 \|x\|_2^2}{\eta^2}.$$

This is a fixed-vector Markov theorem. It is not uniform over all vectors and therefore is not the CACM equation (2) spectral-norm concentration theorem.

Corollary 148 (Finite test set residual tail). *Let $\mathcal{N}$ be any finite family of test vectors, with thresholds $\eta_x > 0$. Then the same proof plus the finite union bound gives*

$$\mathbb{P}[\|Rx\|_2 \leq \eta_x \text{ for every } x \in \mathcal{N}] \geq 1 - \sum_{x \in \mathcal{N}} \frac{(mn/s) \|A\|_F^2 \|x\|_2^2}{\eta_x^2}.$$

This is a future net-argument support theorem; it does not assert that $\mathcal{N}$ covers the unit sphere.

Corollary 149 (Finite-cover operator support). *Suppose now that the finite family $\mathcal{N}$ is supplied together with a radius ρ such that every vector in the unit ball is within Euclidean distance ρ of some vector in $\mathcal{N}$. Let $\eta > 0$ and $L > 0$. Combining the finite-test theorem with the Frobenius residual event gives*

$$\mathbb{P}[\|Rx\|_2 \leq (\eta + L\rho) \|x\|_2 \text{ for every } x] \geq 1 - \sum_{x \in \mathcal{N}} \frac{(mn/s) \|A\|_F^2 \|x\|_2^2}{\eta^2} - \frac{(mn/s) \|A\|_F^2}{L^2}.$$

This is a useful covering-route theorem. It assumes the finite cover and does not itself construct one, so it is not the route used for the closed source-aligned equation (2) theorem below.

Lemma 150 (Product-grid cover). *Let $G = \{g_a : a \in \alpha\} \subset \mathbb{R}$ be a finite one-dimensional grid such that for every $t \in [-1, 1]$ there is $a \in \alpha$ with*

$$|t - g_a| \leq \delta.$$

Assume $\delta \geq 0$. For dimension n , define the product grid

$$\mathcal{G}_n = \{(g_{a_1}, \dots, g_{a_n}) : (a_1, \dots, a_n) \in \alpha^n\}.$$

Then every $x \in \mathbb{R}^n$ with $\|x\|_2 \leq 1$ is within Euclidean distance $\sqrt{n}\delta$ of some point of $\mathcal{G}_n$. Moreover the formal index type has cardinality

$$|\alpha^n| = |\alpha|^n.$$

This closes the product-cover bookkeeping dependency for a covering-net route. It is not a concentration theorem and does not supply the sharp scalar tail bounds needed for equation (2).

Lean formalization.

```
abs_coord_le_vecNorm2
realUnitIntervalCover
rectUnitBallCover_product_grid
fintype_card_product_grid_index
```

Lean formalization.

```

vecNorm2_smul
rectMatMulVec_smul
rectMatMulVec_add
rectMatMulVec_sub
rectUnitBallCover
rectOpNorm2Le_of_unit_ball_bound
rectOpNorm2Le_of_unit_ball_cover
sqMagProb_sum_elementwiseSampleContribution_entry_sq_le
sqMagProb_sum_elementwiseSampleResidualIncrement_entry_sq_le
sqMagProb_sum_elementwiseSampleResidualIncrement_frob_sq_le
sqMagProb_sum_vecNorm2Sq_rectMatMulVec_elementwiseSampleResidualIncrement_le
sqMagTraceProbability_expectationReal_vecNorm2Sq_rectMatMulVec_elementwiseTraceResidual_le
sqMagTraceProbability_eventProb_vecNorm2_rectMatMulVec_elementwiseTraceResidual_le_ge_one_sub
fl_elementwiseTraceResidual_vecNorm2_le_of_exact_fixed_vector
sqMagTraceProbability_eventProb_fl_vecNorm2_rectMatMulVec_elementwiseTraceResidual_le_ge_one_sub
sqMagTraceProbability_eventProb_forall_vecNorm2_rectMatMulVec_elementwiseTraceResidual_le_ge_one_sub_sum
sqMagTraceProbability_eventProb_forall_fl_vecNorm2_rectMatMulVec_elementwiseTraceResidual_le_ge_one_sub_sum
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_ge_one_sub_cover
sqMagTraceProbability_eventProb_algorithm1FlSpectralEvent_ge_one_sub_cover

```

Theorem 151 (Exact Frobenius residual moment). *Under the canonical independent squared-magnitude product trace law for Algorithm 1, with $s > 0$,*

$$\mathbb{E} \left\| A - \tilde{A} \right\|_F^2 \leq \frac{mn}{s} \|A\|_F^2.$$

Equivalently, for every threshold $\eta > 0$,

$$\mathbb{P} \left[\left\| A - \tilde{A} \right\|_F \leq \eta \right] \geq 1 - \frac{(mn/s) \|A\|_F^2}{\eta^2}.$$

Lean formalization.

```

sqMagTraceProbMass_marginal_two_ne
sqMagTraceProbability_eventProb_sampleHits_pair_ne
hitCountPairwiseCenteredMoment_le_steps_mul
sqMagTraceProbability_expectationReal_elementwiseTraceResidual_entry_sq_le
sqMagTraceProbability_expectationReal_elementwiseTraceResidual_frob_sq_le
sqMagTraceProbability_eventProb_algorithm1ExactFrobEvent_ge_one_sub

```

Corollary 152 (Frobenius residual bridge). *If the exact Algorithm 1 residual satisfies*

$$\|R\|_F \leq \varepsilon,$$

then it also satisfies the rectangular vector-action operator event. The generic floating-point adapter for this weaker route remains a Lean support lemma, but the PDF does not advertise it as an Algorithm 1 endpoint because its radius is phrased through an internal perturbation matrix.

Lean formalization.

```
algorithm1ExactFrobEvent
algorithm1ExactFrobEvent_subset_exactSpectralEvent
probability_algorithm1_exact_spectral_of_frob
```

Corollary 153 (Nonconditional Frobenius-to-operator probability). *Combining the exact Frobenius residual theorem with the deterministic bridge, the canonical product trace law satisfies*

$$\mathbb{P}\left[\|(A - \tilde{A})x\|_2 \leq \eta \|x\|_2 \text{ for all } x\right] \geq 1 - \frac{(mn/s) \|A\|_F^2}{\eta^2}.$$

Lean formalization.

```
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_ge_one_sub_frob
```

Theorem 154 (Exact entrywise union-bound fallback, not equation (2)). *Let $\tau > 0$. Under the same canonical product trace law, each residual entry satisfies*

$$\mathbb{P}\left[|A_{ij} - \tilde{A}_{ij}| \leq \tau\right] \geq 1 - \frac{\|A\|_F^2}{s\tau^2}.$$

Consequently, by a finite union bound over all mn entries,

$$\mathbb{P}\left[|A_{ij} - \tilde{A}_{ij}| \leq \tau \text{ for all } i, j\right] \geq 1 - mn \frac{\|A\|_F^2}{s\tau^2}.$$

On this event,

$$\|(A - \tilde{A})x\|_2 \leq \|\tau \mathbf{1}\|_F \|x\|_2 \text{ for all } x,$$

where $\tau \mathbf{1}$ denotes the $m \times n$ matrix whose entries are all τ . This is an exact-arithmetic fallback theorem. It is deliberately not the Algorithm 1 equation (2) theorem, and the PDF does not advertise the internal floating-point adapter for this weaker route because that adapter is phrased through a generic perturbation budget.

Lean formalization.

```
FiniteProbability.eventProb_finset_forall_ge_one_sub_sum
FiniteProbability.eventProb_forall_ge_one_sub_sum
algorithm1ExactEntrywiseEvent
sqMagTraceProbability_eventProb_elementwiseTraceResidual_entry_abs_le_ge_one_sub
sqMagTraceProbability_eventProb_algorithm1ExactEntrywiseEvent_ge_one_sub
algorithm1ExactEntrywiseEvent_subset_exactSpectralEvent_const
probability_algorithm1_exact_spectral_of_entrywise_const
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_ge_one_sub_entrywise
```

Remark. This section is not the proof of equation (2) itself. The Frobenius and entrywise probability theorems are nonconditional Markov or union-bound routes with weaker dimension factors. The sharper source-aligned Bernstein/Khintchine-type theorem is proved later by the trace-MGF route.

Corollary 155 (Truncated element-wise sampler with explicit FP radius). *Let $A \in \mathbb{R}^{n \times n}$, $n > 0$, $s > 0$, $\varepsilon > 0$, and $0 < \delta \leq 1$. Set*

$$\tau = \frac{\varepsilon}{2n}, \quad \hat{A} = \text{trunc}_\tau(A),$$

where the truncation is the hard-thresholding map

$$\hat{A}_{ij} = \begin{cases} 0, & |A_{ij}| < \tau, \\ A_{ij}, & |A_{ij}| \geq \tau. \end{cases}$$

The value $\tau = \varepsilon/(2n)$ is chosen because A is $n \times n$: every discarded entry has magnitude at most τ , hence

$$\|A - \hat{A}\|_F \leq \sqrt{n^2 \tau^2} = n\tau = \frac{\varepsilon}{2}.$$

Thus deterministic truncation uses one half of the final ε -radius, leaving the other half for the sampled residual of $\hat{A}$. The same choice also gives $|\hat{A}_{ij}| \geq \tau$ for every retained nonzero sampled entry, which is what turns the internal division-by- $\hat{A}_{ij}$ stability budget into the explicit floating-point term below. Finally, assume the squared-magnitude denominator of $\hat{A}$ is positive. Assume the source sample budget

$$14n \|A\|_F^2 \log\left(\frac{2(2n)}{\delta}\right) \leq s\varepsilon^2.$$

Then exact source-aligned truncated element-wise sampling from $\hat{A}_{ij}^2 / \|\hat{A}\|_F^2$ satisfies

$$\mathbb{P}\left[\|A - \tilde{A}_\tau\|_2 \leq \varepsilon\right] \geq 1 - \delta.$$

If additionally γ_s and γ_{s+1} are valid for the floating-point model, then the rounded sketch satisfies

$$\mathbb{P}\left[\|A - \text{fl}(\tilde{A}_\tau)\|_2 \leq \varepsilon + \frac{2n^2 \|A\|_F^2}{\varepsilon} \gamma_{s+1}\right] \geq 1 - \delta,$$

where

$$\gamma_{s+1} = \frac{(s+1)u}{1 - (s+1)u}.$$

This is a theorem for the hard-thresholded element-wise sampler above. It is not the literal CACM Algorithm 1 sampler with distribution $p_{ij} = A_{ij}^2 / \|A\|_F^2$, because the sampling law has been changed to use $\hat{A}$. The term after ε is the explicit floating-point contribution; no budget matrix is hidden in the statement.

Lean formalization.

```
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_truncatedTraceResidual_ge_one_sub_delta_source_sample_budget_sharp_square

sqMagTraceProbability_eventProb_algorithm1ExactTruncatedSpectralEvent_ge_one_sub_delta_source_sample_budget_sharp_square

sqMagTraceProbability_eventProb_algorithm1FlTruncatedSpectralRadius_ge_one_sub_delta_source_sample_budget_explicit_gamma_square
```


Remark (Exact probability convention for Corollary 143). This corollary is an exact-law result: the sampler draws from the truncated squared-magnitude law $\widehat{A}_{ij}^2 / \|\widehat{A}\|_F^2$. Under the current convention no floating-point error is charged for computing that law. The FP radius in the formal theorem charges only the rounded sketch arithmetic after the sample has been drawn.

Corollary 156 (Literal Algorithm 1 without truncation). *Let $A \in \mathbb{R}^{n \times n}$, let $s > 0$, let $\varepsilon > 0$, and assume*

$$\|A\|_F^2 > 0.$$

Run Algorithm 1 with the literal squared-magnitude sampling distribution

$$p_{ij} = \frac{A_{ij}^2}{\|A\|_F^2}.$$

Assume the Frobenius/Markov sample budget

$$\frac{n^2 \|A\|_F^2}{s\varepsilon^2} \leq \delta.$$

For the floating-point part, assume there is a displayed nonzero-entry floor $\alpha > 0$ such that

$$A_{ij} \neq 0 \implies \alpha \leq |A_{ij}| \quad \text{for all } i, j.$$

This is not a truncation: it is only a stated condition on the actual input matrix A , used to make the division-by- A_{ij} floating-point term readable. If γ_{s+1} is valid for the floating-point model, then

$$\mathbb{P} \left[\left\| A - \text{fl}(\widetilde{A}) \right\|_2 \leq \varepsilon + n \frac{\|A\|_F^2}{\alpha} \gamma_{s+1} \right] \geq 1 - \delta,$$

where

$$\gamma_{s+1} = \frac{(s+1)u}{1 - (s+1)u}.$$

The probability is taken under the literal Algorithm 1 trace law above. The proof intersects the exact Frobenius/Markov spectral event with the probability-one support event for p_{ij} . Hence zero entries of A are never sampled on the event where the local FP division theorem is used, and no hard-thresholded matrix $\widehat{A}$ appears.

Lean formalization.

```
sqMagTraceErrorBudget_zero_init_le_const_of_entry_abs_ge
frobNormRect_sqMagTraceErrorBudget_zero_init_le_const_square_of_entry_abs_ge
sqMagTraceProbability_eventProb_algorithm1FlSpectralRadius_ge_one_sub_delta_frob_explicit_gamma_square
```

Corollary 157 (Nonconditional literal Algorithm 1 support-radius theorem). *Let $A \in \mathbb{R}^{m \times n}$, $s > 0$, $\|A\|_F^2 > 0$, and $0 < \delta \leq 1$. Run Algorithm 1 with the literal exact squared-magnitude law*

$$p_{ij} = \frac{A_{ij}^2}{\|A\|_F^2}.$$

The probability law is exact by convention. The computed non-probability quantities are the sampled-entry rescaling, the additions into the sketch, and the final rounded residual; their floating-point

contribution is charged below. The threshold quantities in the display are exact analysis parameters, not additional algorithm outputs. If an implementation also computes these thresholds for a stopping or certification rule, that separate computation is not part of this theorem.

Define the exact reciprocal-entry quantity

$$G(A) = \sum_{\{(i,j): A_{ij} \neq 0\}} \frac{\|A\|_F^2}{|A_{ij}|}, \quad H(A) = \|A\|_F + G(A).$$

For a chosen analysis parameter $\theta > 0$, define

$$L_{\text{lit}} = \frac{H(A)}{s}, \quad V_{\text{lit}} = \max\{m, n\} \frac{\|A\|_F^2}{s^2},$$

and

$$\beta_{\text{lit}}(\theta) = \frac{\exp(\theta L_{\text{lit}}) - \theta L_{\text{lit}} - 1}{L_{\text{lit}}^2}.$$

The exact spectral radius appearing in the formal theorem is

$$\varepsilon_{\text{lit}}(\theta, \delta) = \frac{\log\left(\frac{2(m+n)\exp(s\beta_{\text{lit}}(\theta)V_{\text{lit}})}{\delta}\right)}{\theta}.$$

Substituting all definitions gives the irreducible expanded form

$$\varepsilon_{\text{lit}}(\theta, \delta) = \frac{\log(2(m+n)/\delta)}{\theta} + \frac{\max\{m, n\} \|A\|_F^2 s}{\theta H(A)^2} \left(\exp\left(\frac{\theta H(A)}{s}\right) - \frac{\theta H(A)}{s} - 1 \right).$$

If γ_s and γ_{s+1} are valid for the floating-point model, then the rounded literal Algorithm 1 sketch satisfies

$$\mathbb{P}\left[\left\|A - \text{fl}(\tilde{A})\right\|_2 \leq T_{\text{lit}}^{\text{fp}}(\theta, \delta)\right] \geq 1 - \delta,$$

where the fully expanded final radius is

$$T_{\text{lit}}^{\text{fp}}(\theta, \delta) = \frac{\log(2(m+n)/\delta)}{\theta} + \frac{\max\{m, n\} \|A\|_F^2 s}{\theta H(A)^2} \left(\exp\left(\frac{\theta H(A)}{s}\right) - \frac{\theta H(A)}{s} - 1 \right) + \sqrt{mn} \gamma_{s+1} G(A),$$

with

$$\gamma_{s+1} = \frac{(s+1)u}{1 - (s+1)u}.$$

Thus, in the regime where $\theta > 0$ is fixed and $\theta H(A)/s = O(1)$, the same bound has order

$$T_{\text{lit}}^{\text{fp}}(\theta, \delta) = \Theta\left(\frac{\log((m+n)/\delta)}{\theta} + \frac{\max\{m, n\} \|A\|_F^2 s}{\theta H(A)^2} \left(\exp\left(\frac{\theta H(A)}{s}\right) - \frac{\theta H(A)}{s} - 1 \right) + \sqrt{mn} \gamma_{s+1} G(A)\right).$$

In the small-argument subregime $\theta H(A)/s \rightarrow 0$, this simplifies to

$$T_{\text{lit}}^{\text{fp}}(\theta, \delta) = \Theta\left(\frac{\log((m+n)/\delta)}{\theta} + \frac{\theta \max\{m, n\} \|A\|_F^2}{s} + \sqrt{mn} \gamma_{s+1} G(A)\right).$$

For fixed $A, m, n, s, \theta, \delta$, the floating-point part tends to zero as $\gamma_{s+1} \rightarrow 0$. The theorem is nonconditional and literal, but it is not the source-uniform CACM equation (2) rate: tiny nonzero entries enlarge $G(A)$, exactly as the formal support obstruction predicts.

The formalization now also removes the free parameter θ by applying the proved Bennett optimizer. Let

$$q_\delta = \log\left(\frac{2(m+n)}{\delta}\right), \quad W_{\text{lit}} = sV_{\text{lit}} = \frac{\max\{m, n\} \|A\|_F^2}{s}.$$

For any target exact radius $r > 0$, if

$$q_\delta \leq \frac{r^2}{2W_{\text{lit}} + \frac{2}{3}L_{\text{lit}}r},$$

equivalently, after substituting $L_{\text{lit}} = H(A)/s$,

$$s \geq \frac{q_\delta \left(2 \max\{m, n\} \|A\|_F^2 + \frac{2}{3}H(A)r\right)}{r^2},$$

then the exact literal sketch satisfies $\|A - \tilde{A}\|_2 \leq r$ with probability at least $1 - \delta$. The rounded implementation-facing version has the fully expanded scalar radius

$$T_{\text{lit},B}^{\text{fp}}(r) = r + \sqrt{mn} \gamma_{s+1} G(A).$$

Thus the direct Bennett sample requirement has order

$$s = \Theta\left(\frac{\left(\max\{m, n\} \|A\|_F^2 + H(A)r\right) \log((m+n)/\delta)}{r^2}\right),$$

and the final FP radius has order

$$T_{\text{lit},B}^{\text{fp}}(r) = \Theta(r + \sqrt{mn} \gamma_{s+1} G(A)).$$

This form is nonvacuous for fixed input and target radius as $\gamma_{s+1} \rightarrow 0$, and the sample condition is nonvacuous whenever the displayed lower bound on s can be met. The dependence on $G(A)$ and $H(A)$ is deliberate and irreducible for this support-radius route.

The formalization also gives a simpler floor-bound surface for readers who prefer not to inspect $G(A)$ directly. Suppose that for some $\alpha > 0$,

$$A_{ij} \neq 0 \implies \alpha \leq |A_{ij}| \quad \text{for all } i, j.$$

Then the exact deterministic simplifications

$$G(A) \leq \frac{mn \|A\|_F^2}{\alpha}, \quad H(A) \leq H_\alpha(A) := \|A\|_F + \frac{mn \|A\|_F^2}{\alpha}$$

are proved in Lean. Consequently, the stronger but more readable budget

$$q_\delta \leq \frac{r^2}{2W_{\text{lit}} + \frac{2}{3} \frac{H_\alpha(A)}{s} r}$$

implies the exact event

$$\mathbb{P}\left[\|A - \tilde{A}\|_2 \leq r\right] \geq 1 - \delta,$$

and the rounded implementation-facing event

$$\mathbb{P} \left[\|A - \mathfrak{fl}(\tilde{A})\|_2 \leq r + \sqrt{mn} \frac{mn \|A\|_F^2}{\alpha} \gamma_{s+1} \right] \geq 1 - \delta.$$

Equivalently, the alpha-floor sample budget has order

$$s = \Theta \left(\frac{\left(\max\{m, n\} \|A\|_F^2 + H_\alpha(A)r \right) \log((m+n)/\delta)}{r^2} \right),$$

and the displayed rounded radius has order

$$\Theta \left(r + \sqrt{mn} \frac{mn \|A\|_F^2}{\alpha} \gamma_{s+1} \right).$$

For fixed $A, m, n, s, r, \alpha, \delta$, the floating-point additive term tends to zero as $\gamma_{s+1} \rightarrow 0$. This entry-floor corollary is still not source-uniform, but it is often easier to interpret than the irreducible reciprocal-entry formula.

Lean formalization.

```

elementwiseLiteralContributionRadius
elementwiseLiteralResidualSupportRadius
elementwiseLiteralContributionRadius_le_of_entry_abs_ge
elementwiseLiteralResidualSupportRadius_le_of_entry_abs_ge
smul_elementwiseLiteralContributionRadius_le_of_entry_abs_ge
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_literalTraceResidual_ge_one_sub_
delta_scaled_radius_supportRadius
sqMagTraceProbability_eventProb_algorithm1FlSpectralEvent_literal_scaled_radius_gamma_
supportRadius
sqMagTraceErrorBudget_zero_init_le_literalContributionRadius
frobNormRect_sqMagTraceErrorBudget_zero_init_le_literalContributionRadius
sqMagTraceProbability_eventProb_algorithm1FlSpectralRadius_literal_scaled_radius_gamma_
supportRadius
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_literalTraceResidual_ge_one_sub_
delta_bennett_radius_supportRadius
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_literalTraceResidual_ge_one_sub_
delta_bernstein_denominator_two_thirds_supportRadius
sqMagTraceProbability_eventProb_algorithm1FlSpectralRadius_literal_ge_one_sub_delta_bernstein_
denominator_gamma_supportRadius
algorithm1LiteralBernsteinDenominatorBudget_of_entry_floor
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_literalTraceResidual_ge_one_sub_
delta_bernstein_denominator_entry_floor
sqMagTraceProbability_eventProb_algorithm1FlSpectralRadius_literal_ge_one_sub_delta_bernstein_
denominator_gamma_entry_floor

```

Corollary 158 (Literal Algorithm 1 at the source spectral rate). *Let $A \in \mathbb{R}^{n \times n}$, $n > 0$, $s > 0$, $\varepsilon > 0$, $0 < \delta \leq 1$, and $\|A\|_F^2 > 0$. Run Algorithm 1 with the literal distribution*

$$p_{ij} = \frac{A_{ij}^2}{\|A\|_F^2}.$$

Assume the source-rate sample budget

$$14n \|A\|_F^2 \log\left(\frac{2(2n)}{\delta}\right) \leq s\varepsilon^2.$$

Assume also that the source threshold would not remove any nonzero entry:

$$A_{ij} \neq 0 \implies \frac{\varepsilon}{2n} \leq |A_{ij}| \quad \text{for all } i, j.$$

Equivalently, $\text{trunc}_{\varepsilon/(2n)}(A) = A$. Therefore the theorem below is a literal Algorithm 1 statement; the sampling distribution is not changed.

Then the exact sketch satisfies

$$\mathbb{P}\left[\|A - \tilde{A}\|_2 \leq \varepsilon\right] \geq 1 - \delta.$$

If γ_s and γ_{s+1} are valid for the floating-point model, then the rounded sketch satisfies

$$\mathbb{P}\left[\|A - \text{fl}(\tilde{A})\|_2 \leq \varepsilon + \frac{2n^2 \|A\|_F^2}{\varepsilon} \gamma_{s+1}\right] \geq 1 - \delta,$$

where

$$\gamma_{s+1} = \frac{(s+1)u}{1 - (s+1)u}.$$

Thus, when $\varepsilon = \eta \|A\|_F$, the sampling condition has the paper-style size $s \gtrsim n \log(n/\delta)/\eta^2$, plus the displayed no-small-entry condition needed to make the source truncation identity and the floating-point denominator bound explicit.

The same specialization is formalized for rectangular $A \in \mathbb{R}^{m \times n}$. Define

$$R = \sqrt{mn}, \quad M = \max\{m, n\}, \quad C_{\text{rect}} = 2M + \frac{4\sqrt{2}}{3}R.$$

Assume $mn > 0$, $\|A\|_F^2 > 0$, and

$$4C_{\text{rect}} \|A\|_F^2 \log\left(\frac{2(m+n)}{\delta}\right) \leq s\varepsilon^2.$$

Assume also the rectangular source threshold is the identity on the literal input:

$$A_{ij} \neq 0 \implies \frac{\varepsilon}{2R} \leq |A_{ij}| \quad \text{for all } i, j.$$

Equivalently, $\text{trunc}_{\varepsilon/(2R)}(A) = A$, so the probability law remains the exact literal law $p_{ij} = A_{ij}^2 / \|A\|_F^2$. In exact arithmetic,

$$\mathbb{P}\left[\|A - \tilde{A}\|_2 \leq \varepsilon\right] \geq 1 - \delta.$$

With valid γ_s and γ_{s+1} , the computed non-probability operations in the rounded sampled updates and residual satisfy

$$\mathbb{P}\left[\left\|A - \text{fl}(\tilde{A})\right\|_2 \leq \varepsilon + \frac{2mn \|A\|_F^2}{\varepsilon} \gamma_{s+1}\right] \geq 1 - \delta.$$

In the small- $(s+1)u$ regime the rectangular FP additive term is

$$\Theta\left(\frac{mn \|A\|_F^2}{\varepsilon} su\right),$$

and, since $R \leq M$, the rectangular sample budget has order

$$\Theta\left(\frac{\max\{m, n\} \|A\|_F^2}{\varepsilon^2} \log \frac{m+n}{\delta}\right).$$

Lean formalization.

```
elementwiseTruncate_eq_self_of_forall_nonzero_entry_abs_ge
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_ge_one_sub_delta_source_sample_
budget_no_small_entries_square
sqMagTraceProbability_eventProb_algorithm1FlSpectralRadius_ge_one_sub_delta_source_sample_budget_
no_small_entries_square
sqMagTraceProbability_eventProb_algorithm1ExactSpectralEvent_ge_one_sub_delta_source_sample_
budget_no_small_entries_rect
sqMagTraceProbability_eventProb_algorithm1FlSpectralRadius_ge_one_sub_delta_source_sample_budget_
no_small_entries_rect
```

Remark (Exact probability convention for Corollaries 144 and 145). The preceding literal Algorithm 1 corollaries use the exact squared-magnitude sampling law

$$p_{ij} = \frac{A_{ij}^2}{\|A\|_F^2}.$$

Under the current convention there is no separate perturbation theorem for a rounded probability table. The implementation-facing FP radius is the radius already displayed in the literal corollaries: it charges the rounded sampled updates and residual arithmetic while keeping the probability law exact.

Remark (Literal Algorithm 1 status). The literal corollaries above are faithful to the Algorithm 1 distribution

$$p_{ij} = \frac{A_{ij}^2}{\|A\|_F^2}$$

and contain no changed sampling law. The Frobenius/Markov corollary is unconditional but weaker. The support-radius corollary is also unconditional and nonconditional, with no nonzero-entry floor, but it exposes the reciprocal-entry term $G(A)$. The source-rate corollary has the paper-style $n \log n$ sample budget, but it assumes that the source threshold $\varepsilon/(2n)$ would not remove any nonzero entry. The source-uniform literal matrix-Bernstein/Khintchine version of equation (2), with neither reciprocal-entry nor no-small-entry dependence, remains a separate theorem target if kept in scope.

4 Algorithm 2: Row Sampling with Equation (4)

For $A \in \mathbb{R}^{m \times n}$, define

$$r_i(A) = \|A_{i*}\|_2^2, \quad D(A) = \|A\|_F^2 = \sum_i r_i(A).$$

Equation (4) samples row i with probability

$$p_i = \frac{r_i(A)}{D(A)}.$$

If I_t is the row sampled at trial t , Algorithm 2 returns

$$\tilde{A}_{tj} = \frac{A_{I_t j}}{\sqrt{s p_{I_t}}}.$$

By current convention the row probabilities p_i are exact mathematical sampling laws. The implementation-facing formalization still allows computed scale denominators $\hat{d}_i \approx \sqrt{s p_i}$, with explicit denominator errors. The rounded-row sketch is then

$$\hat{\tilde{A}}_{tj} = \text{fl} \left(\frac{A_{I_t j}}{\hat{d}_{I_t}} \right),$$

so square-root denominator computation and row-rescaling division are visible in the theorem surface. No FP error is charged for row-probability construction, normalization, or sampling-law perturbation.

Lean formalization.

```
ComputedRowScaleDen
rowSampleSketchWithComputedDen
fl_rowSampleSketchWithComputedDen_error_bound
```

Theorem 159 (Unbiased sampled Gram matrix). *Assume $D(A) > 0$ and $s > 0$. Under the independent row sampler with probabilities $p_i = r_i(A)/D(A)$,*

$$\mathbb{E}[(\tilde{A}^T \tilde{A})_{jk}] = (A^T A)_{jk} \quad \text{for all } j, k.$$

Lean formalization.

```
rowSqNormTraceProbability_expectationReal_rowSampleGram_entry
```

Theorem 160 (Equation (5)). *Assume $D(A) > 0$, $s > 0$, and $\varepsilon > 0$. Then*

$$\mathbb{P} \left[\left\| \tilde{A}^T \tilde{A} - A^T A \right\|_F \leq \varepsilon D(A) \right] \geq 1 - \frac{1}{s\varepsilon^2}.$$

Lean formalization.

```
rowSqNormTraceProbability_expectationReal_rowSampleGram_frob_error_sq_le
rowSqNormTraceProbability_eventProb_rowSampleGram_frob_error_le_epsilon
```

Corollary 161 (Computed-denominator floating-point equation (5)). *Let*

$$d_i = \sqrt{s p_i}, \quad \hat{d}_i = \text{flsqrt}(\text{flmul}(s, p_i)), \quad \alpha(u) = \sqrt{1+u}u + u.$$

The exact probabilities $p_i = \|A_{i}\|_2^2 / D(A)$ are not rounded, but the non-probability denominator table is computed. On positive-probability rows, Lean proves*

$$|\hat{d}_i - d_i| \leq d_i \alpha(u), \quad \hat{d}_i \neq 0.$$

Define

$$\rho_A = \sum_{i: p_i > 0} \frac{d_i \alpha(u)}{|\hat{d}_i|}, \quad \eta_A = u + (1+u)\rho_A.$$

If $\hat{A}_{\text{comp}}$ is the sampled sketch with entries

$$(\hat{A}_{\text{comp}})_{tj} = \text{fldiv}(A_{Itj}, \hat{d}_{It}),$$

then, under the same sampling assumptions,

$$\mathbb{P}\left[\left\|\hat{A}_{\text{comp}}^T \hat{A}_{\text{comp}} - A^T A\right\|_F \leq \varepsilon D(A) + n(2\eta_A + \eta_A^2)D(A)\right] \geq 1 - \frac{1}{s\varepsilon^2}.$$

If each Gram entry is also computed by the repository length- s dot-product kernel, then

$$\mathbb{P}\left[\left\|\hat{G}_{\text{dot}} - A^T A\right\|_F \leq \varepsilon D(A) + n((2\eta_A + \eta_A^2) + \gamma_s(1 + \eta_A)^2) D(A)\right] \geq 1 - \frac{1}{s\varepsilon^2}.$$

The additive fully computed FP term has order

$$\Theta(nD(A)(\eta_A + \eta_A^2 + \gamma_s(1 + \eta_A)^2)),$$

and, in the small-roundoff regime $\eta_A, \gamma_s \rightarrow 0$,

$$\Theta(nD(A)(\eta_A + \gamma_s)).$$

For fixed A, m, n, s and nonzero computed positive-probability denominators, the additive FP term tends to zero as $u, \gamma_s \rightarrow 0$, so the result is non-vacuous. The exact-denominator theorems are recovered by setting $\rho_A = 0$ and $\eta_A = u$.

Lean formalization.

```

ComputedRowScaleDen.flMulThenSqrt
fl_rowSampleSketchWithComputedDen_total_error_bound_le_budget
rowSampleGramRelPerturbBudget_eq_nat_mul
rowSampleGramDotProductRelBudget_eq_nat_mul
fl_rowSampleGramWithComputedDen_perturb_bound
fl_rowSampleGramDotWithComputedDen_perturb_bound
rowSqNormTraceProbability_eventProb_fl_rowSampleGramWithComputedDen_frob_error_le_epsilon_add_
explicit_budget
rowSqNormTraceProbability_eventProb_fl_rowSampleGramDotWithComputedDen_frob_error_le_epsilon_add_
explicit_budget

```

Remark (Exact row probabilities for computed-denominator equation (5)). The displayed bound uses the exact row-norm sampling law $p_i = \|A_{i*}\|_2^2 / D(A)$. Under the current convention this law is exact, so no probability-construction or sampling-law transfer term appears. The computed non-probability denominator, final row-scaling division, and optional Gram dot-product kernel are all charged explicitly.

5 Algorithm 2: Leverage Scores and Equation (7)

Let $U \in \mathbb{R}^{m \times n}$ have orthonormal columns, so $U^T U = I$. Equation (6) is the specialization of equation (4) to U :

$$p_i = \frac{\|U_{i*}\|_2^2}{n}.$$

By current convention, the leverage probabilities in equation (6) are exact sampling laws. Floating-point construction of probabilities is not modeled in the Algorithm 2 leverage theorems. If a future implementation theorem needs a computed SVD/QR basis, that basis perturbation should be stated separately from probability construction and then transferred to the exact-law sampling theorem.

Theorem 162 (Equation (7), exact arithmetic). *Assume $U^T U = I$, $n > 0$, $s > 0$, and $\varepsilon > 0$. The leverage-score row sampler satisfies*

$$\mathbb{P}\left[\forall x, \left\|(\tilde{U}^T \tilde{U} - I)x\right\|_2 \leq \varepsilon \|x\|_2\right] \geq 1 - \frac{1}{s(\varepsilon/n)^2}.$$

Lean formalization.

```
leverageScoreProb_eq_rowNormSq_div_nat
leverageTraceProbability_eventProb_rowSampleGram_opNorm2_error_le_epsilon
```

Lemma 163 (One-step leverage covariance facts). *Let $u_i \in \mathbb{R}^n$ be row i of U , and let*

$$X_i = \begin{cases} u_i^T u_i / p_i, & p_i > 0, \\ 0, & p_i = 0. \end{cases}$$

For the leverage law $p_i = \|u_i\|_2^2 / n$, the one-sample estimator satisfies

$$X_i \succeq 0, \quad \sum_i p_i X_i = I_n, \quad X_i \preceq n I_n.$$

Lean formalization.

```
rowOuterGramSample_eq_zero_of_prob_zero
finiteQuadraticForm_rowOuterGramSample_eq_sq_div
finitePSD_rowOuterGramSample
leverage_rowOuterGramSample_finitePSD
leverage_rowOuterGramSample_mean_eq_id
leverage_rowOuterGramSample_finiteLoewnerLe_nat
```

Lemma 164 (Row-trace MGF product-law adapter). *Let X_i be any self-adjoint matrix observable attached to one sampled row, and let $\omega = (i_1, \dots, i_s)$ be drawn from the Algorithm 2 independent row-trace law. The no-hidden-Lieb one-step trace-MGF inequality iterates to*

$$\mathbb{E} \operatorname{tr} \exp\left(H + \sum_{t=1}^s X_{i_t}\right) \leq \operatorname{tr} \exp(H + s \log \mathbb{E} \exp(X_i)).$$

For finite real matrices this is also connected to the repository's 'finiteMatrixExp' trace interface. If the one-step logarithmic mean increment is bounded by cI , the right-hand side is at most

$$n \exp(sc).$$

Lean formalization.

```

rowSqNormSampleProbability
rowSqNormTraceProbability_expectationReal_trace_normed_exp_add_step_le
rowSqNormTraceProbability_expectationReal_succ_last_eq
rowSqNormTraceProbability_expectationReal_trace_normed_exp_add_sum_le
rowSqNormTraceProbability_expectationReal_finiteTrace_finiteMatrixExp_add_sum_le
rowSqNormTraceProbabilityFiniteRealTraceMGFLogBound_zero_le_card_mul_exp_of_log_le_real_smul_one

```

Lemma 165 (Centered leverage covariance log-CGF). *Let*

$$Y_i = X_i, \quad Z_i = Y_i - I_n,$$

where Y_i is the one-sample leverage outer-product estimator from the previous lemma. Under the leverage law $p_i = \|u_i\|_2^2/n$,

$$\mathbb{E}Z_i = 0, \quad Z_i = Z_i^T, \quad \lambda_{\max}(Z_i) \leq n.$$

The exact second-moment proxy is

$$\mathbb{E}[Z_i^2] = (n-1)I_n.$$

Consequently, for every $\theta \geq 0$,

$$\log \mathbb{E} \exp(\theta Z_i) \preceq \frac{\exp(\theta n) - \theta n - 1}{n^2} \mathbb{E}[Z_i^2].$$

The lower-tail observable $-Z_i = I_n - Y_i$ is handled separately:

$$\lambda_{\max}(-Z_i) \leq 1, \quad \log \mathbb{E} \exp(\theta(-Z_i)) \preceq (\exp \theta - \theta - 1) \mathbb{E}[Z_i^2].$$

Lean formalization.

```

Algorithms/RandNLA/RowSamplingLeverageMGF.lean
rowOuterGramSample_symmetric
rowOuterGramSample_centered_symmetric
leverage_rowOuterGramSample_centered_cstar_selfAdjoint
leverage_rowOuterGramSample_centered_expectationCStarMatrix_eq_zero
leverage_rowOuterGramSample_centered_finiteLoewnerLe_nat
leverage_rowOuterGramSample_centered_spectrum_le_nat
leverage_rowOuterGramSample_centered_square_expectationCStarMatrix_eq
leverage_rowOuterGramSample_centered_log_cgf_le
leverage_rowOuterGramSample_centered_log_cgf_le_scalar
leverage_rowOuterGramSample_neg_centered_log_cgf_le
leverage_rowOuterGramSample_neg_centered_log_cgf_le_scalar

```

Theorem 166 (Two-sided leverage row-sampling concentration). *Let $i_1, \dots, i_s$ be independent samples from the leverage law $p_i = \|u_i\|_2^2/n$, and set*

$$\widehat{G} = \frac{1}{s} \sum_{t=1}^s Y_{i_t}.$$

For every $\theta > 0$ and every $\varepsilon \in \mathbb{R}$, define

$$\beta_+(\theta) = \frac{\exp(\theta n) - \theta n - 1}{n^2}, \quad \beta_-(\theta) = \exp \theta - \theta - 1.$$

Then the repository proves

$$\begin{aligned} & \mathbb{P} \left[\widehat{G} - I_n \preceq \varepsilon I_n \text{ and } I_n - \widehat{G} \preceq \varepsilon I_n \right] \\ & \geq 1 - n \exp(-\theta s \varepsilon + s \beta_+(\theta)(n-1)) - n \exp(-\theta s \varepsilon + s \beta_-(\theta)(n-1)). \end{aligned}$$

Equivalently, if the two displayed scalar tails sum to at most δ , then the same event holds with probability at least $1 - \delta$.

Lean formalization.

```
leverage_rowSqNormTraceProbabilityFiniteRealTraceMGFLogBound_centered_le
leverage_rowSqNormTraceProbabilityFiniteRealTraceMGFLogBound_neg_centered_le
leverage_rowSqNormTraceProbability_eventProb_exists_finiteHermitianEigenvalue_centered_sum_ge_le_exp
leverageTraceProbability_eventProb_rowSampleGram_finiteLoewnerLe_upper_lt_ge_one_sub_exp
leverageTraceProbability_eventProb_rowSampleGram_finiteLoewnerLe_lower_lt_ge_one_sub_exp
leverageTraceProbability_eventProb_rowSampleGram_two_sided_finiteLoewnerLe_ge_one_sub_exp
leverageTraceProbability_eventProb_rowSampleGram_two_sided_finiteLoewnerLe_ge_one_sub_delta_of_tail_budget
```

Corollary 167 (Bennett sample-budget equation (7)). *Assume $n > 1$, $\varepsilon > 0$, $\delta > 0$, and*

$$\log \frac{2n}{\delta} \leq s \frac{\varepsilon^2}{2(n-1) + (2/3)n\varepsilon}, \quad \log \frac{2n}{\delta} \leq s \frac{\varepsilon^2}{2(n-1) + (2/3)\varepsilon}.$$

Then

$$\mathbb{P} \left[\widehat{G} - I_n \preceq \varepsilon I_n \text{ and } I_n - \widehat{G} \preceq \varepsilon I_n \right] \geq 1 - \delta.$$

Lean formalization.

```
real_bernstein_tail_le_half_delta_of_quadratic_budget
leverageTraceProbability_eventProb_rowSampleGram_finiteLoewnerLe_upper_ge_one_sub_delta_half_of_sample_budget
leverageTraceProbability_eventProb_rowSampleGram_finiteLoewnerLe_lower_ge_one_sub_delta_half_of_sample_budget
leverageTraceProbability_eventProb_rowSampleGram_two_sided_finiteLoewnerLe_ge_one_sub_delta_of_sample_budget
```

Corollary 168 (Computed-denominator floating-point Bennett equation (7)). *With the same sample-budget assumptions and γ_s validity, let $\tilde{G}_{\text{comp}}$ be the Gram matrix computed from the leverage sketch using the concrete denominator routine*

$$\hat{d}_i = \text{flsqrt} \left(\text{flmul} \left(s, \frac{\|U_{i*}\|_2^2}{n} \right) \right),$$

rounded row scaling, and rounded length- s Gram dot products. Define

$$\eta_U = u + (1 + u) \sum_{i: \|U_{i*}\|_2^2 > 0} \frac{\sqrt{s \|U_{i*}\|_2^2 / n} (\sqrt{1 + u} u + u)}{\left| \text{flsqrt} \left(\text{flmul} \left(s, \|U_{i*}\|_2^2 / n \right) \right) \right|}.$$

Since $U^T U = I_n$, $D(U) = n$. Hence the fully computed FP radius is

$$\tau_{\text{comp}}(U; s) = n^2 \left((2\eta_U + \eta_U^2) + \gamma_s(1 + \eta_U)^2 \right).$$

The theorem proved in Lean is

$$\mathbb{P} \left[\tilde{G}_{\text{comp}} - I_n \preceq (\varepsilon + \tau_{\text{comp}}(U; s)) I_n \text{ and } I_n - \tilde{G}_{\text{comp}} \preceq (\varepsilon + \tau_{\text{comp}}(U; s)) I_n \right] \geq 1 - \delta.$$

In the small-roundoff regime,

$$\tau_{\text{comp}}(U; s) = \Theta(n^2(\eta_U + \gamma_s)).$$

The exact-denominator comparison theorem is recovered by setting $\eta_U = u$.

Lean formalization.

```
unitRoundoff_lt_one_of_pos_gammaValid
leverageFlMulThenSqrtRowScaleDen
leverageFlMulThenSqrtRowScaleDen_den
leverage_fl_rowSampleGramDotWithComputedDen_perturb_bound
leverageTraceProbability_eventProb_fl_rowSampleGramDotWithComputedDen_two_sided_finiteLoewnerLe_
ge_one_sub_delta_of_sample_budget
leverageTraceProbability_eventProb_fl_rowSampleGramDotWithFlMulThenSqrtDen_two_sided_
finiteLoewnerLe_ge_one_sub_delta_of_sample_budget
```

Corollary 169 (Actual-input Bennett equation (7)). *Let $U \in \mathbb{R}^{m \times r}$ have orthonormal columns, let $C \in \mathbb{R}^{r \times q}$, and set $A = UC$. The exact leverage law is*

$$p_i^U = \|U_{i*}\|_2^2 / r.$$

The factors U and C are exact analysis witnesses in this theorem, not computed outputs. The implemented row sketch samples rows of the actual input A from this exact law. Under the Bennett sample-budget hypotheses with r replacing n , the exact sampled actual-input Gram satisfies

$$\mathbb{P} \left[\tilde{A}_U^T \tilde{A}_U - A^T A \preceq \varepsilon A^T A \text{ and } A^T A - \tilde{A}_U^T \tilde{A}_U \preceq \varepsilon A^T A \right] \geq 1 - \delta.$$

The concrete floating-point theorem computes

$$\hat{d}_i^U = \text{flsqrt}(\text{flmul}(s, p_i^U)),$$

rounds the sampled-row divisions on entries of A , and computes the Gram entries by length- s floating-point dot products. Define

$$\eta_U^A = u + (1 + u) \sum_{i: p_i^U > 0} \frac{\sqrt{sp_i^U} (\sqrt{1+u} u + u)}{|\text{flsqrt}(\text{flmul}(s, p_i^U))|}$$

and

$$T_A^{\text{fp}} = (\gamma_s(1 + \eta_U^A)^2 + 2\eta_U^A + (\eta_U^A)^2) \sum_{j=1}^q \left(\sum_{i: p_i^U > 0} \frac{|A_{ij}|}{\sqrt{p_i^U}} \right)^2.$$

Then the computed Gram $\widehat{G}_{\text{dot}}(A; U)$ satisfies

$$\mathbb{P} \left[\widehat{G}_{\text{dot}}(A; U) - A^T A \preceq \varepsilon A^T A + T_A^{\text{fp}} I_q \text{ and } A^T A - \widehat{G}_{\text{dot}}(A; U) \preceq \varepsilon A^T A + T_A^{\text{fp}} I_q \right] \geq 1 - \delta.$$

With $k_U = |\{i : p_i^U > 0\}|$, fixed U, C, m, r, q, s , and nonzero positive-probability computed denominators,

$$T_A^{\text{fp}} = \Theta \left((\gamma_s + u(1 + k_U)) \sum_{j=1}^q \left(\sum_{i: p_i^U > 0} \frac{|A_{ij}|}{\sqrt{p_i^U}} \right)^2 \right),$$

so the floating-point additive radius tends to zero as $u, \gamma_s \rightarrow 0$. A concrete QR, SVD, or rank-revealing routine that produces U or C must still be certified separately before it is composed with this endpoint.

Lean formalization.

```
leverageFactoredInputSampleGram
leverageTraceProbability_eventProb_factoredInputSampleGram_two_sided_finiteLoewnerLe_ge_one_sub_delta_of_sample_budget
leverageFactoredInputDenGramBudget
leverage_fl_rowSampleGramDotWithComputedDen_factoredInput_perturb_bound
leverageTraceProbability_eventProb_factoredInput_fl_rowSampleGramDotWithFlMulThenSqrtDen_two_sided_finiteLoewnerLe_ge_one_sub_delta_of_sample_budget
```

Corollary 170 (Stored-basis floating-point Bennett equation (7)). *The exact leverage law is still $p_i = \|U_{i*}\|_2^2/n$, and it remains an exact sampling law. The computed sketch may instead use a stored basis table*

$$\widehat{U}_{ij}^{(\text{mul1})} = \text{flmul}(U_{ij}, 1), \quad \widehat{U}_{ij}^{(\text{add0})} = \text{fladd}(U_{ij}, 0), \quad \widehat{U}_{ij}^{(\text{sub0})} = \text{flsub}(U_{ij}, 0),$$

each satisfying

$$|\widehat{U}_{ij}^{(\mu)} - U_{ij}| \leq u|U_{ij}|.$$

Let

$$\rho_U = \sum_{i: p_i > 0} \frac{\sqrt{sp_i} (\sqrt{1+u} u + u)}{|\text{flsqrt}(\text{flmul}(s, p_i))|}, \quad \lambda_U = 2u + u^2 + (1 + u)^2 \rho_U.$$

The fully expanded stored-basis radius can be written as

$$T_U^{\text{fp}} = (\gamma_s(1 + \lambda_U)^2 + 2\lambda_U + \lambda_U^2) \sum_{j=1}^n \left(\sum_{i:p_i>0} \frac{|U_{ij}|}{\sqrt{p_i}} \right)^2.$$

For each $\mu \in \{\text{mul1}, \text{add0}, \text{sub0}\}$, the Gram matrix computed from $\widehat{U}^{(\mu)}$, the concrete denominator $\text{flsqrt}(\text{flmul}(s, p_i))$, rounded row divisions, and rounded length- s dot products satisfies the same Bennett probability lower bound with event radius $\varepsilon + T_U^{\text{fp}}$. In the fixed problem small-roundoff regime,

$$T_U^{\text{fp}} = \Theta \left((\gamma_s + u(1 + |\{i : p_i > 0\}|)) \sum_{j=1}^n \left(\sum_{i:p_i>0} \frac{|U_{ij}|}{\sqrt{p_i}} \right)^2 \right),$$

and the radius tends to zero as $u, \gamma_s \rightarrow 0$. This closes the three modeled stored-table paths; computing U from a data matrix by QR, SVD, or another routine remains a separate computed-basis routine obligation.

Lean formalization.

```
RowSamplingLeverageComputedBasis.lean
leverageExactBasisSampleColumnAbsBudget
leverageComputedBasisSampleColumnErrorBudget
leverageComputedBasisDenGramBudget
leverageTraceProbability_eventProb_fl_rowSampleGramDotWithStoredBasisMulOneAndFlMulThenSqrtDen_
two_sided_finiteLoewnerLe_ge_one_sub_delta_of_sample_budget
leverageTraceProbability_eventProb_fl_rowSampleGramDotWithStoredBasisAddZeroRightAndFlMulThenSqrtDen_
two_sided_finiteLoewnerLe_ge_one_sub_delta_of_sample_budget
leverageTraceProbability_eventProb_fl_rowSampleGramDotWithStoredBasisSubZeroRightAndFlMulThenSqrtDen_
two_sided_finiteLoewnerLe_ge_one_sub_delta_of_sample_budget
```

Remark (Exact leverage probabilities for computed-denominator equation (7)). The displayed Bennett FP theorem samples from the exact leverage-score law defined by the ideal orthonormal basis U . Under the current convention, there is no rounded leverage-probability table and no sampling-law transfer loss. The computed-denominator theorem charges the non-probability denominator, rounded row scaling, and rounded Gram dot products. The stored-basis theorem additionally charges the three modeled rounded-copy basis tables. Any future QR/SVD-generated basis theorem must instantiate a concrete computed-basis routine rather than being hidden inside the leverage law.

6 Algorithm 3: Random-Projection Preconditioning

Algorithm 3 applies preprocessing matrices to an input matrix A :

$$\Pi_L A, \quad A \Pi_R, \quad \Pi_L A \Pi_R.$$

The random projection law is intentionally not fixed in the meta-algorithm. The Lean results are therefore deterministic after Π_L and Π_R are chosen.

For an implementation, the preprocessing matrices may themselves be generated or stored in floating point before the matrix products are evaluated. The implementation-facing surface records computed matrices $\widehat{\Pi}_L, \widehat{\Pi}_R$ with entrywise errors against the ideal preconditioners, and separates those transform-entry errors from the rounded matrix-multiplication error in $\text{fl}(\widehat{\Pi}_L A)$, $\text{fl}(A \widehat{\Pi}_R)$, and $\text{fl}(\widehat{\Pi}_L A \widehat{\Pi}_R)$. The uniform row sampler used after preconditioning has a separate certificate for the computed $\widehat{d} \approx \sqrt{s/m}$ scale denominator.

Lean formalization.

```

ComputedPreconditioner
fl_preconditionRowsWithComputedLeft_total_error_bound
fl_preconditionColumnsWithComputedRight_total_error_bound
fl_preconditionElementsWithComputed_error_bound
fl_preconditionElementsWithComputed_total_error_bound
ComputedUniformRowScaleDen
fl_uniformRowSampleSketch_computedDen_error_bound
uniformRowSampleIncrementWithComputedDen_ideal_error_bound
fl_uniformRowSampleSketch_computedDen_total_error_bound
fl_uniformRowSampleSketch_computedDen_total_error_bound_exact
ComputedUniformRowScaleDen.flDivThenSqrt
ComputedUniformRowScaleDen.flInvMulThenSqrt
ComputedUniformRowScaleDen.flSqrtDivSqrt
ComputedUniformRowScaleDen.flSqrtMulInvSqrt

```

Theorem 171 (Computed uniform-row denominator certificate). *Let*

$$d = \sqrt{s/m}, \quad \widehat{d} \neq 0, \quad |\widehat{d} - d| \leq \eta_d,$$

where $d \neq 0$. For an exact uniform row sample i and an implemented row scaling division using the computed denominator $\widehat{d}$,

$$B_{ij} = U_{ij}/d, \quad \widehat{B}_{ij}^{\widehat{d}} = \text{fl}(U_{ij}/\widehat{d}),$$

one has

$$\left| \widehat{B}_{ij}^{\widehat{d}} - B_{ij} \right| \leq \left| U_{ij}/\widehat{d} \right| u + \frac{|U_{ij}| \eta_d}{|\widehat{d}| |d|}.$$

Thus the row law remains the exact uniform law $1/m$, while the non-probability scalar computed by the algorithm is charged explicitly. In the exact-denominator specialization $\widehat{d} = d$ and $\eta_d = 0$, this reduces to the usual rounded-division bound

$$|\text{fl}(U_{ij}/d) - U_{ij}/d| \leq |U_{ij}/d| u.$$

Lean formalization.

```

ComputedUniformRowScaleDen.exact
uniformRowSampleIncrementWithComputedDen_ideal_error_bound
fl_uniformRowSampleSketch_computedDen_total_error_bound
fl_uniformRowSampleSketch_computedDen_total_error_bound_exact

```

Corollary 172 (Rounded-sqrt exact-input denominator certificate). *Assume $0 < u < 1$ in the floating-point model and that the scalar (s/m) is supplied exactly to the square-root primitive. If*

$$\widehat{d} = \text{fl}(\sqrt{s/m}),$$

then the square-root model gives

$$\widehat{d} \neq 0, \quad |\widehat{d} - d| \leq d u, \quad d = \sqrt{s/m}.$$

Consequently, the computed-denominator theorem applies with $\eta_d = d u$:

$$\left| \widehat{B}_{ij}^{\widehat{d}} - B_{ij} \right| \leq \left| U_{ij}/\widehat{d} \right| u + \frac{|U_{ij}| d u}{|\widehat{d}| |d|}.$$

This corollary charges the rounded square-root primitive. If the implementation also computes s/m by rounded division or multiplication, that earlier scalar routine must instantiate a separate certificate before this exact-input constructor is used.

Lean formalization.

```
ComputedUniformRowScaleDen.flSqrtExactInput
ComputedUniformRowScaleDen.flSqrtExactInput_den
ComputedUniformRowScaleDen.flSqrtExactInput_den_abs_error
```

Corollary 173 (Rounded division-then-square-root denominator certificate). *Assume $0 < u < 1$, $m > 0$, and $s > 0$. Let the implementation compute*

$$\widehat{x} = \text{fl}(s/m), \quad \widehat{d} = \text{fl}(\sqrt{\widehat{x}}), \quad d = \sqrt{s/m}.$$

Then $\widehat{d} \neq 0$ and the denominator error satisfies

$$|\widehat{d} - d| \leq d(\sqrt{1+u} u + u).$$

Consequently, for an exact uniform row sample i ,

$$\left| \text{fl}(U_{ij}/\widehat{d}) - U_{ij}/d \right| \leq \left| U_{ij}/\widehat{d} \right| u + \frac{|U_{ij}| d(\sqrt{1+u} u + u)}{|\widehat{d}| |d|}.$$

The row-sampling law remains the exact uniform law $1/m$. The two displayed losses charge only non-probability computations performed by the implementation: rounded scalar division in s/m , rounded square root, and the final rounded row-scaling division.

Lean formalization.

```
ComputedUniformRowScaleDen.abs_sqrt_one_add_sub_one_le_abs
ComputedUniformRowScaleDen.flDivThenSqrt
ComputedUniformRowScaleDen.flDivThenSqrt_den
ComputedUniformRowScaleDen.flDivThenSqrt_den_abs_error
```


Corollary 174 (Rounded reciprocal-multiply-square-root denominator certificate). Assume $0 < u < 1$, $m > 0$, and $s > 0$. Let the implementation compute

$$\hat{r} = \text{fl}(1/m), \quad \hat{x} = \text{fl}(s\hat{r}), \quad \hat{d} = \text{fl}(\sqrt{\hat{x}}), \quad d = \sqrt{s/m}.$$

Then $\hat{d} \neq 0$ and

$$|\hat{d} - d| \leq d \left(\sqrt{(1+u)(1+u)} u + 2u + u^2 \right).$$

Consequently, for an exact uniform row sample i ,

$$\left| \text{fl}(U_{ij}/\hat{d}) - U_{ij}/d \right| \leq \left| U_{ij}/\hat{d} \right| u + \frac{|U_{ij}| d \left(\sqrt{(1+u)(1+u)} u + 2u + u^2 \right)}{|\hat{d}| |d|}.$$

This corollary charges the reciprocal-as-division $1/m$, the rounded multiplication by s , the rounded square root, and the final rounded row-scaling division. It introduces no probability-construction, probability-normalization, or sampling-law transfer term: the row law remains the exact uniform law $1/m$.

Lean formalization.

```
ComputedUniformRowScaleDen.flInvMulThenSqrt
ComputedUniformRowScaleDen.flInvMulThenSqrt_den
ComputedUniformRowScaleDen.flInvMulThenSqrt_den_abs_error
```

Corollary 175 (Rounded split-square-root denominator certificate). Assume $0 < u < 1$, $m > 0$, and $s > 0$. Let the implementation compute

$$\hat{s}_0 = \text{fl}(\sqrt{s}), \quad \hat{m}_0 = \text{fl}(\sqrt{m}), \quad \hat{d} = \text{fl}(\hat{s}_0/\hat{m}_0), \quad d = \sqrt{s/m}.$$

Then $\hat{d} \neq 0$ and

$$|\hat{d} - d| \leq d \frac{3u + u^2}{1 - u}.$$

Consequently, for an exact uniform row sample i ,

$$\left| \text{fl}(U_{ij}/\hat{d}) - U_{ij}/d \right| \leq \left| U_{ij}/\hat{d} \right| u + \frac{|U_{ij}| d (3u + u^2)}{|\hat{d}| |d| (1 - u)}.$$

This corollary charges the two rounded square roots, the rounded scalar division used to form $\hat{d}$, and the final rounded row-scaling division. The row-sampling law remains the exact uniform law $1/m$.

Lean formalization.

```
ComputedUniformRowScaleDen.flSqrtDivSqrt
ComputedUniformRowScaleDen.flSqrtDivSqrt_den
ComputedUniformRowScaleDen.flSqrtDivSqrt_den_abs_error
```

Corollary 176 (Rounded square-root times reciprocal-square-root denominator certificate). Assume $0 < u < 1$, $m > 0$, and $s > 0$. Let the implementation compute

$$\hat{s}_0 = \text{fl}(\sqrt{s}), \quad \hat{m}_0 = \text{fl}(\sqrt{m}), \quad \hat{r}_m = \text{fl}(1/\hat{m}_0), \quad \hat{d} = \text{fl}(\hat{s}_0 \hat{r}_m), \quad d = \sqrt{s/m}.$$

Then $\widehat{d} \neq 0$ and

$$|\widehat{d} - d| \leq d \frac{4u + 3u^2 + u^3}{1 - u}.$$

Consequently, for an exact uniform row sample i ,

$$\left| \mathbb{E}(U_{ij}/\widehat{d}) - U_{ij}/d \right| \leq \left| U_{ij}/\widehat{d} \right| u + \frac{|U_{ij}| d(4u + 3u^2 + u^3)}{|\widehat{d}| |d| (1 - u)}.$$

This corollary charges the two rounded square roots, the rounded reciprocal of $\widehat{m}_0$, the final rounded multiplication used to form $\widehat{d}$, and the final rounded row-scaling division. It does not charge the construction of the uniform sampling law, which remains the exact law $1/m$.

Lean formalization.

```
ComputedUniformRowScaleDen.flSqrtMulInvSqrt
ComputedUniformRowScaleDen.flSqrtMulInvSqrt_den
ComputedUniformRowScaleDen.flSqrtMulInvSqrt_den_abs_error
```

Theorem 177 (Exact Frobenius stability for orthogonal preprocessing). *If $A, \Pi_L, \Pi_R \in \mathbb{R}^{n \times n}$ and the preprocessing matrices are orthogonal, then*

$$\|\Pi_L A\|_F = \|A\|_F, \quad \|A \Pi_R\|_F = \|A\|_F, \quad \|\Pi_L A \Pi_R\|_F = \|A\|_F.$$

Lean formalization.

```
preconditionRows_frobNorm_orthogonal
preconditionColumns_frobNorm_orthogonal
preconditionElements_frobNorm_orthogonal
```

Theorem 178 (Orthonormal-column invariance under orthogonal preprocessing). *Let $U \in \mathbb{R}^{m \times n}$ satisfy*

$$U^T U = I_n, \quad \text{i.e.} \quad \sum_{i=1}^m U_{ij} U_{ik} = \delta_{jk}.$$

If $\Pi_L \in \mathbb{R}^{m \times m}$ and $\Pi_R \in \mathbb{R}^{n \times n}$ are orthogonal, then

$$(\Pi_L U)^T (\Pi_L U) = I_n, \quad (U \Pi_R)^T (U \Pi_R) = I_n, \quad (\Pi_L U \Pi_R)^T (\Pi_L U \Pi_R) = I_n.$$

Corollary 179 (Leverage denominator after Algorithm 3 preprocessing). *Under the same hypotheses, the equation (6) leverage denominator remains the number of columns:*

$$\sum_{i=1}^m \|(\Pi_L U)_{i,*}\|_2^2 = \sum_{i=1}^m \|(U \Pi_R)_{i,*}\|_2^2 = \sum_{i=1}^m \|(\Pi_L U \Pi_R)_{i,*}\|_2^2 = n.$$

This is a deterministic basis-invariance statement. It does not say that a particular random projection law makes the individual leverage scores nearly uniform.

Lean formalization.

```
preconditionRows_hasOrthonormalColumns_of_orthogonal
preconditionColumns_hasOrthonormalColumns_of_orthogonal
preconditionElements_hasOrthonormalColumns_of_orthogonal
rowSqNormProbDen_preconditionRows_eq_nat_of_orthogonal
rowSqNormProbDen_preconditionColumns_eq_nat_of_orthogonal
rowSqNormProbDen_preconditionElements_eq_nat_of_orthogonal
```

Theorem 180 (Signed orthogonal preprocessing). *Let $H \in \mathbb{R}^{m \times m}$ be orthogonal and let $d \in \mathbb{R}^m$ satisfy $d_i^2 = 1$ for every i . Write $D = \text{diag}(d)$. Then D is orthogonal, HD is orthogonal, and for every $U \in \mathbb{R}^{m \times n}$ with $U^T U = I_n$,*

$$(HDU)^T(HDU) = I_n, \quad \sum_{i=1}^m \|(HDU)_{i,*}\|_2^2 = n.$$

Lean formalization.

```
IsOrthogonal.diagMatrix_of_sq_eq_one
signedOrthogonalPreconditioner_isOrthogonal
signedOrthogonalPreconditionRows_hasOrthonormalColumns
rowSqNormProbDen_signedOrthogonalPreconditionRows_eq_nat
```

This closes the deterministic signed-orthogonal part of the SRHT proof-source route. The finite Rademacher support and moment facts are stated next. Later sections close the source-sharp signed-Hadamard row-norm flattening theorem in explicit- t and logarithmic forms; the remaining Algorithm 3 obligations are computed-object instantiations for concrete transform/basis/projector routines, not probability-construction errors.

Corollary 181 (Probability-one signed preprocessing under Rademacher signs). *Let ω be uniformly distributed over sign traces in $\{\pm 1\}^m$, and let $D_\omega = \text{diag}(\omega)$. If $H \in \mathbb{R}^{m \times m}$ is orthogonal and $U^T U = I_n$, then*

$$\mathbb{P} \left[HD_\omega \text{ is orthogonal, } (HD_\omega U)^T(HD_\omega U) = I_n, \quad \sum_{i=1}^m \|(HD_\omega U)_{i,*}\|_2^2 = n \right] = 1.$$

Lean formalization.

```
FiniteProbability.eventProb_eq_one_of_forall
RademacherTrace
rademacherSignVector_sq
rademacherTraceProbability
rademacherTraceProbability_eventProb_signedOrthogonalPreconditionRows_eq_one
```

Theorem 182 (Finite Rademacher moments). *Let ω be uniformly distributed over $\{\pm 1\}^m$. Then for all indices a, b ,*

$$\mathbb{E} \omega_a = 0, \quad \mathbb{E} \omega_a \omega_b = \delta_{ab}.$$

Consequently, for every deterministic vector $c \in \mathbb{R}^m$,

$$\mathbb{E} \left(\sum_{k=1}^m c_k \omega_k \right)^2 = \sum_{k=1}^m c_k^2.$$

Lemma 183 (Exact Rademacher signed-linear-form MGF). *Let ω be uniformly distributed over $\{\pm 1\}^m$. For every deterministic vector $a \in \mathbb{R}^m$,*

$$\mathbb{E} \exp \left(\sum_{k=1}^m a_k \omega_k \right) = \prod_{k=1}^m \frac{\exp(a_k) + \exp(-a_k)}{2}.$$

Consequently, for every $\lambda > 0$ and threshold T ,

$$\mathbb{P} \left\{ \sum_{k=1}^m a_k \omega_k \leq T \right\} \geq 1 - \exp(-\lambda T) \prod_{k=1}^m \frac{\exp(\lambda a_k) + \exp(-\lambda a_k)}{2}.$$

Theorem 184 (Scalar Rademacher Hoeffding tail). *Let ω be uniformly distributed over $\{\pm 1\}^m$. For every $a \in \mathbb{R}^m$, every $\lambda \in \mathbb{R}$, and every $T \in \mathbb{R}$,*

$$\mathbb{E} \exp \left(\lambda \sum_{k=1}^m a_k \omega_k \right) \leq \exp \left(\frac{\lambda^2}{2} \sum_{k=1}^m a_k^2 \right).$$

Consequently, for $\lambda > 0$,

$$\mathbb{P} \left\{ \sum_{k=1}^m a_k \omega_k \geq T \right\} \leq \exp(-\lambda T) \exp \left(\frac{\lambda^2}{2} \sum_{k=1}^m a_k^2 \right),$$

and

$$\mathbb{P} \left\{ \left| \sum_{k=1}^m a_k \omega_k \right| \leq T \right\} \geq 1 - 2 \exp(-\lambda T) \exp \left(\frac{\lambda^2}{2} \sum_{k=1}^m a_k^2 \right).$$

Lean formalization.

```
real_rademacher_cosh_factor_le_exp_sq_div_two
rademacherTraceProbability_expectationReal_exp_sum_mul_sign_le_exp_sum_sq_div_two
rademacherTraceProbability_expectationReal_exp_lam_sum_mul_sign_le_exp_lam_sq_sum_sq_div_two
rademacherTraceProbability_eventProb_sum_mul_sign_ge_le_exp_sq_bound
rademacherTraceProbability_eventProb_sum_mul_sign_le_ge_one_sub_exp_sq_bound
rademacherTraceProbability_eventProb_abs_sum_mul_sign_le_ge_one_sub_two_mul_exp_sq_bound
```

Corollary 185 (Optimized scalar Rademacher Hoeffding tail). *Let ω be uniformly distributed over the exact finite Rademacher cube $\{\pm 1\}^m$. Let $a \in \mathbb{R}^m$, $T > 0$, and $\sigma > 0$. If*

$$\sum_{k=1}^m a_k^2 \leq \sigma^2,$$

then

$$\mathbb{P} \left\{ \left| \sum_{k=1}^m a_k \omega_k \right| \leq T \right\} \geq 1 - 2 \exp \left(-\frac{T^2}{2\sigma^2} \right).$$

The statement optimizes the preceding auxiliary-parameter bound with $\lambda = T/\sigma^2$. It is an exact-probability concentration primitive: no floating-point computation is involved, and it does not by itself close the Gaussian, FJLT, or input-sparsity Algorithm 3 transform laws.

Lean formalization.

```
rademacherTraceProbability_eventProb_abs_sum_mul_sign_le_ge_one_sub_two_mul_exp_neg_sq_div
```

Corollary 186 (Finite-family Rademacher Hoeffding tail). *Let ω be uniformly distributed over the exact finite Rademacher cube $\{\pm 1\}^m$. Let $\mathcal{I}$ be a finite index set. For each $\alpha \in \mathcal{I}$, let $a_\alpha \in \mathbb{R}^m$, $T_\alpha > 0$, and $\sigma_\alpha > 0$. If*

$$\sum_{k=1}^m a_{\alpha,k}^2 \leq \sigma_\alpha^2 \quad \text{for every } \alpha \in \mathcal{I},$$

then

$$\mathbb{P} \left\{ \left| \sum_{k=1}^m a_{\alpha,k} \omega_k \right| \leq T_\alpha \text{ for every } \alpha \in \mathcal{I} \right\} \geq 1 - \sum_{\alpha \in \mathcal{I}} 2 \exp \left(-\frac{T_\alpha^2}{2\sigma_\alpha^2} \right).$$

This is the finite union-bound packaging of the preceding exact-probability Rademacher tail. It is intended as a reusable primitive for later finite Rademacher/FJLT preprocessing branches, not as a completed transform law.

Lean formalization.

```
rademacherTraceProbability_eventProb_forall_abs_sum_mul_sign_le_ge_one_sub_sum_two_mul_exp_neg_sq_div
```

Corollary 187 (Rademacher signed-linear vector norm bound). *Let ω be uniformly distributed over the exact finite Rademacher cube $\{\pm 1\}^m$. For $j = 1, \dots, n$, let*

$$X_j(\omega) = \sum_{k=1}^m a_{j,k} \omega_k.$$

Let $T_j > 0$, $\sigma_j > 0$, and assume

$$\sum_{k=1}^m a_{j,k}^2 \leq \sigma_j^2 \quad \text{for every } j.$$

Then

$$\mathbb{P} \left\{ \sum_{j=1}^n X_j(\omega)^2 \leq \sum_{j=1}^n T_j^2 \right\} \geq 1 - \sum_{j=1}^n 2 \exp \left(-\frac{T_j^2}{2\sigma_j^2} \right).$$

The proof uses the finite-family coordinate event and the deterministic implication $|X_j| \leq T_j$ for all $j \Rightarrow \sum_j X_j^2 \leq \sum_j T_j^2$. This is a row-norm-shaped exact-probability primitive, still not a completed Algorithm 3 transform law.

Lean formalization.

```
rademacherTraceProbability_eventProb_vecNorm2Sq_sum_mul_sign_le_ge_one_sub_sum_two_mul_exp_neg_sq_div
```

Corollary 188 (Exact signed-mixing row-norm bound). *Let $G \in \mathbb{R}^{r \times m}$ be a deterministic exact mixing table and let $U \in \mathbb{R}^{m \times n}$ have orthonormal columns. Let $\omega \in \{\pm 1\}^m$ be an exact Rademacher sign vector and define the exact signed-mixing preconditioner*

$$\Pi_\omega(i, k) = G_{ik} \omega_k.$$

For row thresholds $T_{ij} > 0$ and row coefficient caps $\alpha_i > 0$, assume the deterministic exact coefficient bound

$$G_{ik}^2 \leq \alpha_i^2 \quad \text{for every } i, k.$$

Then

$$\mathbb{P} \left\{ \forall i, \quad \|(\Pi_\omega U)_{i,*}\|_2^2 \leq \sum_{j=1}^n T_{ij}^2 \right\} \geq 1 - \sum_{i=1}^r \sum_{j=1}^n 2 \exp \left(-\frac{T_{ij}^2}{2\alpha_i^2} \right).$$

In particular, if $T_{ij} = B_i > 0$ for all coordinates in row i , then

$$\mathbb{P} \left\{ \forall i, \quad \|(\Pi_\omega U)_{i,*}\|_2^2 \leq nB_i^2 \right\} \geq 1 - \sum_{i=1}^r \sum_{j=1}^n 2 \exp \left(-\frac{B_i^2}{2\alpha_i^2} \right).$$

When $\alpha_i = \alpha$ and $B_i = B$ are uniform, the displayed failure term is exactly $2rn \exp(-B^2/(2\alpha^2))$, hence has order

$$\Theta \left(rn \exp \left(-\frac{B^2}{2\alpha^2} \right) \right)$$

in the regime where $r, n, B/\alpha$ vary and the exact sampling law is fixed. This is an exact-probability and exact-arithmetic preprocessing result: the Rademacher law is exact by project convention, G and U are exact analysis objects in this corollary, and no floating-point storage or transform application is charged here. Floating-point use of a concrete G remains a separate computed-preconditioner obligation.

Lean formalization.

```
signedMixingRows
signedMixingRows_preconditionRows_entry
signedMixingRows_coeff_sq_sum_le_of_entry_sq_le
rademacherTraceProbability_eventProb_forall_rowNormSq_signedMixingRows_entry_sq_le_ge_one_sub_
sum_sum_two_mul_exp_neg_sq_div
rademacherTraceProbability_eventProb_forall_rowNormSq_signedMixingRows_entry_sq_le_uniform_ge_
one_sub_sum_sum_two_mul_exp_neg_sq_div
```

Corollary 189 (Exact signed-mixing one-row mean identity). *Let $G \in \mathbb{R}^{r \times m}$ be a deterministic exact mixing table and let $U \in \mathbb{R}^{m \times n}$ have orthonormal columns. Let $\omega \in \{\pm 1\}^m$ be an exact Rademacher sign vector and let i be an exact uniform row index in $\{1, \dots, r\}$, independent of ω . Define $\Pi_\omega(i, k) = G_{ik}\omega_k$. Assume the exact column-square normalization*

$$\sum_{i=1}^r G_{ik}^2 = 1 \quad \text{for every } k = 1, \dots, m.$$

This normalization is computable in exact arithmetic as an analysis assumption on G . The formalization also proves the direct adapter from the stronger rectangular-isometry hypothesis $G^T G = I_m$, since that hypothesis implies the displayed column-square identities. For the one-row estimator $r(\Pi_\omega U)_{i,}^T (\Pi_\omega U)_{i,*}$,*

$$\mathbb{E}_{\omega,i} [r(\Pi_\omega U)_{i,j} (\Pi_\omega U)_{i,\ell}] = \delta_{j\ell}.$$

Equivalently, the expected one-row outer product is I_n . This closes the exact first-moment/normalization layer for finite signed-mixing rows under column-square normalization, and directly under $G^T G = I_m$ by the adapter. It is not a concentration theorem, not a subspace-embedding theorem, and not a floating-point theorem: the law of (ω, i) is exact by project convention, and applying or storing G in floating point remains a separate computed-preconditioner obligation.

Lean formalization.

```

rademacherTraceProbability_expectationReal_sum_mul_sign_mul_sum_mul_sign_eq_sum_mul
signedMixingUniformRowSampleProbability
rademacherTraceProbability_expectationReal_signedMixingRows_entry_mul_eq
signedMixingUniformRowSampleProbability_expectationReal_uniformRowOuterGramSample_eq_id
signedMixingUniformRowSampleProbability_expectationReal_uniformRowOuterGramSample_eq_id_of_
hasOrthonormalColumns

```

Corollary 190 (Exact CountSketch first moment). *Let $r > 0$, and let*

$$\mathcal{H}_{r,m} = \{h : \{1, \dots, m\} \rightarrow \{1, \dots, r\}\}$$

be the finite set of exact CountSketch hash maps. Let h be drawn from the exact uniform law on $\mathcal{H}_{r,m}$, and let $\omega \in \{\pm 1\}^m$ be drawn from the exact Rademacher product law, independently of h . For exact arithmetic, define the sparse CountSketch preconditioner $S_{h,\omega} \in \mathbb{R}^{r \times m}$ by

$$(S_{h,\omega})_{ik} = \begin{cases} \omega_k, & h(k) = i, \\ 0, & h(k) \neq i. \end{cases}$$

Thus every input row k contributes to exactly one output bucket $h(k)$, with exact sign ω_k .

For any exact matrix $U \in \mathbb{R}^{m \times n}$,

$$\mathbb{E}_{h,\omega} [(S_{h,\omega}U)^T(S_{h,\omega}U)] = U^T U.$$

Equivalently, entrywise,

$$\mathbb{E}_{h,\omega} [\text{rowGram}(S_{h,\omega}U)_{j\ell}] = \text{rowGram}(U)_{j\ell}.$$

If $U^T U = I_n$, then the expected sketched Gram is I_n .

The proof first fixes the exact hash h . Rademacher signs make all collision cross terms have zero expectation, and summing over the r buckets counts each input row once. The final Lean theorem then packages the exact uniform hash law and exact Rademacher sign law.

The formalization also proves the deterministic collision-free foundation. If h is injective and the exact signs satisfy $\omega_k^2 = 1$, then

$$S_{h,\omega}^T S_{h,\omega} = I_m,$$

so for every exact matrix $A \in \mathbb{R}^{m \times n}$,

$$(S_{h,\omega}A)^T(S_{h,\omega}A) = A^T A.$$

Consequently, if $U^T U = I_n$, then $S_{h,\omega}U$ has orthonormal columns. This collision-free statement is exact arithmetic and deterministic; it does not itself supply the probability of the injective event.

The formalization now also proves the exact pair-collision probability for the hash law. For distinct input rows $a, b \in \{1, \dots, m\}$, define

$$E_{a,b} = \{h \in \mathcal{H}_{r,m} : h(a) = h(b)\}, \quad E_{a,b}^{\text{nc}} = \{h \in \mathcal{H}_{r,m} : h(a) \neq h(b)\}.$$

Under the exact iid uniform hash law,

$$\Pr_h(E_{a,b}) = \frac{1}{r}, \quad \Pr_h(E_{a,b}^{\text{nc}}) = 1 - \frac{1}{r}.$$

Thus the exact pair-collision probability has order

$$\Pr_h(E_{a,b}) = \Theta(r^{-1})$$

as $r \rightarrow \infty$ with the two input rows fixed. This is non-vacuous for $r > 1$, where $0 < 1 - 1/r < 1$, and the collision probability tends to zero as the number of buckets grows. This probability statement concerns only the exact sampling law; it does not compute, store, round, or apply the sparse matrix $S_{h,\omega}$.

The non-injective CountSketch second-moment route now has its exact sign/hash-moment foundation. The first new ingredient is the indicator identity

$$\mathbb{E}_h \mathbf{1}_{\{h(a)=h(b)\}} = \frac{1}{r} \quad (a \neq b),$$

and the same identity after marginalizing the independent exact Rademacher sign law:

$$\mathbb{E}_{h,\Omega} \mathbf{1}_{\{h(a)=h(b)\}} = \frac{1}{r}.$$

The second ingredient is the exact fourth-moment classifier for ordered distinct Rademacher sign pairs. If $a \neq b$ and $c \neq d$, then

$$\begin{aligned} \mathbb{E}_\Omega [\omega_a \omega_b \omega_c \omega_d] \\ = \begin{cases} 1, & a = c \text{ and } b = d, \\ 1, & a = d \text{ and } b = c, \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

Equivalently, for any exact coefficient table $c : \mathcal{P}_m \rightarrow \mathbb{R}$, where $\mathcal{P}_m = \{(a,b) : a \neq b\}$,

$$\begin{aligned} \mathbb{E}_\Omega \left(\sum_{(a,b) \in \mathcal{P}_m} c_{ab} \omega_a \omega_b \right)^2 \\ = \sum_{(a,b) \in \mathcal{P}_m} \sum_{(c,d) \in \mathcal{P}_m} c_{ab} c_{cd} [\mathbf{1}_{\{a=c, b=d\}} + \mathbf{1}_{\{a=d, b=c\}}]. \end{aligned}$$

This exact formula is the reusable local substrate for a standard non-collision-free CountSketch variance or Frobenius embedding proof: after the CountSketch Gram error is expanded as an ordered collision sum, the hash indicator mean contributes the $1/r$ factor and the Rademacher fourth-moment formula leaves only identical and reversed ordered pairs. No floating-point quantity is introduced in this layer, and no concentration event is claimed yet. The primitive probability factor has order $\Theta(r^{-1})$ for fixed input-row pair; the summed sign identity is exact and remains non-vacuous for arbitrary exact coefficients, including the degenerate zero-coefficient case.

The formalization also proves a global injectivity lower bound for the exact hash law. Let

$$I_{r,m} = \{h \in \mathcal{H}_{r,m} : h \text{ is injective}\}, \quad \mathcal{P}_m = \{(a,b) : a, b \in \{1, \dots, m\}, a \neq b\}.$$

The event $I_{r,m}$ is exact and analysis-level: it is not computed by the algorithm. Lean proves that $I_{r,m}$ is the same as the simultaneous pairwise no-collision event over all $(a,b) \in \mathcal{P}_m$, and therefore the pair-collision law and the finite union bound give

$$\Pr_h(I_{r,m}) \geq 1 - \sum_{(a,b) \in \mathcal{P}_m} \frac{1}{r}.$$

Using only the cardinality bound $|\mathcal{P}_m| \leq m^2$, Lean also proves the simplified closed form

$$\Pr_h(I_{r,m}) \geq 1 - \frac{m^2}{r}.$$

Equivalently, the proven failure radius is

$$1 - \Pr_h(I_{r,m}) \leq \sum_{(a,b) \in \mathcal{P}_m} \frac{1}{r} \leq \frac{m^2}{r}.$$

In order notation, for varying m, r this union-bound failure radius is

$$O\left(\frac{m^2}{r}\right),$$

and for fixed m it is $O(r^{-1})$ as $r \rightarrow \infty$. This lower bound is non-vacuous when $r > m^2$, and it tends to one in the asymptotic regime $m^2/r \rightarrow 0$. The formalized theorem is one-sided, so the order statement is big- O , not a matching Θ lower-and-upper estimate.

The formalization also proves the first concrete sparse floating-point apply bound for CountSketch. This is no longer an exact-arithmetic statement about the table alone; it is an implementation-facing statement for one computed output entry. The hash h , signs ω , and integer bucket selection are exact, while the selected signed products and the subsequent bucket accumulation are computed in floating point.

For an exact hash h , define the exact bucket

$$B_i(h) = \{k \in \{1, \dots, m\} : h(k) = i\}, \quad b_i = |B_i(h)|.$$

Let $k_1, \dots, k_{b_i}$ be the canonical finite enumeration of this bucket used by Lean. Exact arithmetic gives

$$(S_{h,\omega}A)_{ij} = \sum_{t=1}^{b_i} \omega_{k_t} A_{k_t j}.$$

The modeled sparse floating-point routine computes

$$\hat{z}_t = \text{fl}(\omega_{k_t} A_{k_t j})$$

for each selected row, initializes $\hat{y}^{(0)} = 0$, and then performs left-to-right floating-point addition

$$\hat{y}^{(t)} = \text{fl}(\hat{y}^{(t-1)} + \hat{z}_t), \quad t = 1, \dots, b_i, \quad \hat{y}_{ij} = \hat{y}^{(b_i)}.$$

When $b_i = 0$, the routine returns $\hat{y}_{ij} = 0$. If $\gamma_{b_i} = b_i u / (1 - b_i u)$ is valid, Lean proves

$$|\hat{y}_{ij} - (S_{h,\omega}A)_{ij}| \leq (u + \gamma_{b_i} + u\gamma_{b_i}) \sum_{t=1}^{b_i} |\omega_{k_t}| |A_{k_t j}|.$$

Under the exact sign-magnitude condition $|\omega_k| \leq 1$, this simplifies to the readable column-one-norm bound

$$|\hat{y}_{ij} - (S_{h,\omega}A)_{ij}| \leq (u + \gamma_{b_i} + u\gamma_{b_i}) \|A_{:j}\|_1.$$

The coefficient has the irreducible expansion

$$u + \gamma_{b_i} + u\gamma_{b_i} = u + \frac{b_i u}{1 - b_i u} + u \frac{b_i u}{1 - b_i u} = u + \frac{b_i u(1 + u)}{1 - b_i u}.$$

For $b_i \geq 1$, $b_i u < 1$, and small unit roundoff u , this coefficient has order

$$u + \frac{b_i u (1 + u)}{1 - b_i u} = \Theta(b_i u),$$

so the sparse-entry error bound is

$$\Theta \left(b_i u \sum_{t=1}^{b_i} |\omega_{k_t}| |A_{k_t j}| \right),$$

or, under $|\omega_k| \leq 1$,

$$O(b_i u \|A_{\cdot j}\|_1).$$

For the empty-bucket case $b_i = 0$, the exact and computed entries are both zero and the displayed sum-bound is exactly zero. The result is non-vacuous: for fixed bucket size and fixed exact input column, the right-hand side tends to zero as $u \rightarrow 0$ while $b_i u < 1$.

At matrix level, Lean defines the computed sparse apply

$$\hat{Y} = \text{flCountSketchApply}_{h,\omega}(A), \quad \hat{Y}_{ij} = \hat{y}_{ij},$$

entrywise by the same bucket routine. Therefore, if γ_{b_i} is valid for every output bucket i , Lean proves for all i, j

$$|\hat{Y}_{ij} - (S_{h,\omega} A)_{ij}| \leq (u + \gamma_{b_i} + u \gamma_{b_i}) \sum_{t=1}^{b_i} |\omega_{k_t}| |A_{k_t j}|,$$

and, under $|\omega_k| \leq 1$,

$$|\hat{Y}_{ij} - (S_{h,\omega} A)_{ij}| \leq (u + \gamma_{b_i} + u \gamma_{b_i}) \|A_{\cdot j}\|_1.$$

Thus the whole computed sparse matrix is now exposed entrywise; no additional matrix-apply perturbation event is assumed.

Lean now also closes the deterministic floating-point arithmetic for the sparse CountSketch Gram formed from this computed matrix. Define the exact sparse sketch

$$Y = S_{h,\omega} A, \quad Y_{ij} = (S_{h,\omega} A)_{ij},$$

and define the exact, nonnegative sparse-apply error radius

$$E_{ij} = \left(u + \frac{b_i u}{1 - b_i u} + u \frac{b_i u}{1 - b_i u} \right) \sum_{t=1}^{b_i} |\omega_{k_t}| |A_{k_t j}|.$$

Equivalently,

$$E_{ij} = \left(u + \frac{b_i u (1 + u)}{1 - b_i u} \right) \sum_{t=1}^{b_i} |\omega_{k_t}| |A_{k_t j}|.$$

The computation first forms $\hat{Y} = \text{flCountSketchApply}_{h,\omega}(A)$, as above, and then forms the displayed Gram matrix by rounded length- r dot products:

$$\hat{G}_{jk} = \text{flDot}_r \left((\hat{Y}_{1j}, \dots, \hat{Y}_{rj}), (\hat{Y}_{1k}, \dots, \hat{Y}_{rk}) \right).$$

The exact reference matrix is

$$G_{jk} = (Y^T Y)_{jk} = \sum_{i=1}^r Y_{ij} Y_{ik}.$$

If γ_r and all γ_{b_i} are valid, Lean proves

$$\|\widehat{G} - G\|_F \leq T_{\text{CSGram}}^{\text{fp}},$$

where the exact-only budget is

$$\begin{aligned} T_{\text{CSGram}}^{\text{fp}} = & \left\| \left[\gamma_r \sum_{i=1}^r (|Y_{ij}| + E_{ij})(|Y_{ik}| + E_{ik}) \right]_{j,k} \right\|_F \\ & + \left\| \left[\sum_{i=1}^r (E_{ij}|Y_{ik}| + |Y_{ij}|E_{ik} + E_{ij}E_{ik}) \right]_{j,k} \right\|_F. \end{aligned}$$

In compact matrix notation, with absolute values taken entrywise, this is

$$T_{\text{CSGram}}^{\text{fp}} = \|\gamma_r(|Y| + E)^T(|Y| + E)\|_F + \|E^T|Y| + |Y|^T E + E^T E\|_F.$$

This expression exposes the two floating-point sources separately: the first term is rounded Gram-dot-product error, and the second term propagates the already proved sparse-apply error into the Gram matrix.

For interpretation, define the exact nonnegative matrix

$$Q_{ij} = b_i \sum_{t=1}^{b_i} |\omega_{kt}| |A_{ktj}|.$$

When $ru < 1$, $b_i u < 1$ for all buckets, u is small, and the displayed leading matrix is not identically zero, the exact budget has leading order

$$T_{\text{CSGram}}^{\text{fp}} = \Theta(ru \| |Y|^T |Y| \|_F + u \| Q^T |Y| + |Y|^T Q \|_F),$$

with higher-order terms of order

$$O(ru^2 \| |Y|^T Q + Q^T |Y| \|_F + ru^3 \| Q^T Q \|_F + u^2 \| Q^T Q \|_F).$$

The exact formula, not the asymptotic display, handles degenerate cases such as $Y = 0$ or empty buckets. The result is non-vacuous: for fixed exact A, h, ω , fixed dimensions, and valid γ -values, the right-hand side tends to zero as $u \rightarrow 0$.

The non-injective CountSketch route now has an actual Frobenius/Markov sparse Gram endpoint, rather than only a moment substrate. Let $x = (h, \Omega)$ be drawn from the exact CountSketch hash-sign product law, let $\omega = \text{rademacherSignVector}(\Omega)$, and define the exact sketched Gram error

$$\Delta_{\text{CS}}(h, \omega; A) = (S_{h, \omega} A)^T (S_{h, \omega} A) - A^T A.$$

For each Gram entry, Lean first proves the exact identity

$$(\Delta_{\text{CS}})_{jl} = \sum_{(a,b) \in \mathcal{P}_m} \mathbf{1}_{\{h(a)=h(b)\}} \omega_a \omega_b A_{aj} A_{bl}, \quad \mathcal{P}_m = \{(a,b) : a \neq b\}.$$

Using the fourth-moment classifier and exact hash collision law gives

$$\mathbb{E}_{h, \Omega} [(\Delta_{\text{CS}})_{jl}^2] \leq \frac{2}{r} \sum_{(a,b) \in \mathcal{P}_m} (A_{aj} A_{bl})^2.$$

Summing entries yields the exact Frobenius second-moment bound

$$\mathbb{E}_{h,\Omega} \|\Delta_{\text{CS}}(h, \omega; A)\|_F^2 \leq \frac{2}{r} \sum_{j,l=1}^n \sum_{(a,b) \in \mathcal{P}_m} (A_{aj} A_{bl})^2.$$

The same exact sign/hash argument is now available at the fixed-vector quadratic-form level. For an exact test vector $z \in \mathbb{R}^n$, define

$$v := Az \in \mathbb{R}^m, \quad v_a = \sum_{j=1}^n A_{aj} z_j.$$

The formalization first proves the algebraic bridge

$$z^T \Delta_{\text{CS}}(h, \omega; A) z = \text{rowGram}(S_{h,\omega} v)_{11} - \text{rowGram}(v)_{11},$$

where v is regarded as a one-column matrix. Therefore the already formalized CountSketch entry second-moment theorem applies to the concrete one-column input $v = Az$, giving

$$\mathbb{E}_{h,\Omega} \left[(z^T \Delta_{\text{CS}}(h, \omega; A) z)^2 \right] \leq \frac{2}{r} \sum_{(a,b) \in \mathcal{P}_m} (v_a v_b)^2.$$

This is the exact coefficient form: every term is determined by the exact input matrix A , exact test vector z , and exact hash/sign laws. There is no computed perturbation object and no certificate-existence assumption in this exact-law layer.

Lean also proves the readable coefficient simplifications

$$\sum_{(a,b) \in \mathcal{P}_m} (v_a v_b)^2 \leq \|v\|_2^4 = \|Az\|_2^4 \leq \|A\|_F^4 \|z\|_2^4.$$

Consequently,

$$\mathbb{E}_{h,\Omega} \left[(z^T \Delta_{\text{CS}}(h, \omega; A) z)^2 \right] \leq \frac{2\|Az\|_2^4}{r} \leq \frac{2\|A\|_F^4 \|z\|_2^4}{r}.$$

For any exact threshold $\eta > 0$,

$$\begin{aligned} \Pr_{h,\Omega} \{ |z^T \Delta_{\text{CS}}(h, \omega; A) z| \leq \eta \} \\ \geq 1 - \frac{2}{r\eta^2} \sum_{(a,b) \in \mathcal{P}_m} (v_a v_b)^2. \end{aligned}$$

Equivalently, if

$$\frac{2}{r\eta^2} \sum_{(a,b) \in \mathcal{P}_m} (v_a v_b)^2 \leq \delta,$$

then this fixed-vector quadratic-form event has probability at least $1 - \delta$. The irreducible order is

$$\Theta \left(\frac{\sum_{a \neq b} ((Az)_a (Az)_b)^2}{r\eta^2} \right),$$

and the readable orders are

$$O \left(\frac{\|Az\|_2^4}{r\eta^2} \right), \quad O \left(\frac{\|A\|_F^4 \|z\|_2^4}{r\eta^2} \right).$$

When the displayed coefficient sum is nonzero and A, z, η are fixed, the first expression is $\Theta(r^{-1})$ as $r \rightarrow \infty$; in the zero case the exact proved loss is zero. This fixed-vector theorem is a sharper foundation for later finite-cover or subspace-embedding work, but it is not yet a uniform-over-all-vectors CountSketch embedding theorem.

The fixed-vector statement is also lifted to an arbitrary finite family of exact test vectors. Let $\mathcal{T}$ be a finite index set, let $z_\alpha \in \mathbb{R}^n$ and $\eta_\alpha > 0$ for $\alpha \in \mathcal{T}$, and set

$$v^\alpha = Az_\alpha, \quad B_\alpha = \frac{2}{r\eta_\alpha^2} \sum_{a \neq b} (v_a^\alpha v_b^\alpha)^2.$$

Then Lean proves

$$\begin{aligned} \Pr_{h, \Omega} \{ \forall \alpha \in \mathcal{T}, \quad |z_\alpha^T \Delta_{\text{CS}}(h, \omega; A) z_\alpha| \leq \eta_\alpha \} \\ \geq 1 - \sum_{\alpha \in \mathcal{T}} B_\alpha. \end{aligned}$$

Equivalently, if $\sum_\alpha B_\alpha \leq \delta$, then the same finite-test event has probability at least $1 - \delta$. This is a finite union-bound theorem, not a new probability assumption. Its irreducible loss is

$$\Theta \left(\sum_{\alpha \in \mathcal{T}} \frac{\sum_{a \neq b} ((Az_\alpha)_a (Az_\alpha)_b)^2}{r\eta_\alpha^2} \right)$$

when the displayed numerator is nonzero, with readable upper forms

$$O \left(\sum_{\alpha \in \mathcal{T}} \frac{\|Az_\alpha\|_2^4}{r\eta_\alpha^2} \right), \quad O \left(\sum_{\alpha \in \mathcal{T}} \frac{\|A\|_F^4 \|z_\alpha\|_2^4}{r\eta_\alpha^2} \right).$$

The event remains exact-law and exact-arithmetic. The finite set $\mathcal{T}$ is an analysis test set; applying this theorem to a net or cover requires a separate deterministic cover-to-Loewner step.

That deterministic cover step is now formalized without a supplied operator-radius certificate. Suppose the same finite family $\{z_\alpha\}_{\alpha \in \mathcal{T}}$ is a Euclidean unit-ball cover of $\mathbb{R}^n$ with radius $\rho \geq 0$. Choose common thresholds $\eta > 0$ and $L > 0$, and define

$$\tau_{\text{cover}} = \eta + L(2\rho + \rho^2).$$

The event

$$-\tau_{\text{cover}} I \preceq \Delta_{\text{CS}}(h, \omega; A) \preceq \tau_{\text{cover}} I$$

has probability at least

$$\begin{aligned} 1 - \sum_{\alpha \in \mathcal{T}} \frac{2}{r\eta^2} \sum_{a \neq b} ((Az_\alpha)_a (Az_\alpha)_b)^2 \\ - \frac{2}{rL^2} \sum_{j, \ell} \sum_{a \neq b} (A_{aj} A_{b\ell})^2. \end{aligned}$$

Equivalently, if the two displayed losses sum to at most δ , then the same two-sided finite-Loewner event has probability at least $1 - \delta$. Lean also proves the readable sufficient exact-cover bound

$$\Pr \{ -\tau_{\text{cover}} I \preceq \Delta_{\text{CS}} \preceq \tau_{\text{cover}} I \} \geq 1 - \left(\sum_{\alpha \in \mathcal{T}} \frac{2\|Az_\alpha\|_2^4}{r\eta^2} + \frac{2\|A\|_F^4}{rL^2} \right),$$

plus the corresponding direct $1 - \delta$ wrapper for that readable loss. The Lean names are

`countSketchProbability_eventProb_rowGram_twoSidedLoewner_cover_ge_one_sub_sum_vecNorm_add_frobNo`

and

`countSketchProbability_eventProb_rowGram_twoSidedLoewner_cover_ge_one_sub_delta_of_sum_vecNorm_a`

The proof intersects the finite-test event with the exact Frobenius event $\|\Delta_{\text{CS}}(h, \omega; A)\|_F \leq L$; the Frobenius event supplies the coarse radius required by the deterministic cover lemma. Thus no hidden “there exists a coarse operator certificate” assumption is introduced.

The irreducible failure order is

$$\Theta\left(\sum_{\alpha \in \mathcal{T}} \frac{\sum_{a \neq b} ((Az_\alpha)_a (Az_\alpha)_b)^2}{r\eta^2} + \frac{\sum_{j, \ell} \sum_{a \neq b} (A_{aj} A_{b\ell})^2}{rL^2}\right)$$

when the displayed numerator is nonzero, with readable upper order

$$O\left(\sum_{\alpha \in \mathcal{T}} \frac{\|Az_\alpha\|_2^4}{r\eta^2} + \frac{\|A\|_F^4}{rL^2}\right).$$

For fixed A , fixed cover, and fixed positive η, L , this loss tends to zero like $\Theta(r^{-1})$ whenever the numerator is nonzero, and is exactly zero when both coefficient sums vanish. This is still exact-law and exact-arithmetic infrastructure; it is not a floating-point endpoint.

The supplied-cover version has a constructive product-grid specialization. Let $\mathcal{I}$ be a finite exact index set and let $g : \mathcal{I} \rightarrow \mathbb{R}$ be an exact one-dimensional grid. Suppose g covers the interval $[-1, 1]$ with radius δ_{grid} , with $\delta_{\text{grid}} \geq 0$, and suppose

$$\sqrt{n} \delta_{\text{grid}} \leq \rho.$$

For each $a : \{1, \dots, n\} \rightarrow \mathcal{I}$, define the exact product-grid vector

$$z_a(j) = g(a(j)), \quad j = 1, \dots, n.$$

The Lean theorems

`finiteUnitBallCover_of_rectUnitBallCover`, `finiteUnitBallCover_product_grid`

prove that $\{z_a\}_{a: \{1, \dots, n\} \rightarrow \mathcal{I}}$ is an exact unit-ball cover of radius ρ . Its index cardinality is

$$\#\{a : \{1, \dots, n\} \rightarrow \mathcal{I}\} = |\mathcal{I}|^n,$$

as formalized by

`fintype_card_product_grid_index`.

Instantiating the CountSketch finite-cover theorem with this explicit grid gives

$$\Pr\{-\tau_{\text{cover}} I \preceq \Delta_{\text{CS}} \preceq \tau_{\text{cover}} I\} \geq 1 - \left(\sum_{a: \{1, \dots, n\} \rightarrow \mathcal{I}} \frac{2\|Az_a\|_2^4}{r\eta^2} + \frac{2\|A\|_F^4}{rL^2}\right),$$

where $\tau_{\text{cover}} = \eta + L(2\rho + \rho^2)$. The corresponding direct target-budget theorem replaces the parenthesized loss by any δ that upper-bounds it. The Lean theorem names are

`countSketchProbability_eventProb_rowGram_twoSidedLoewner_productGrid_ge_one_sub_sum_vecNorm_add_`
and

`countSketchProbability_eventProb_rowGram_twoSidedLoewner_productGrid_ge_one_sub_delta_of_sum_vec`

For fixed exact A , fixed grid g , and fixed positive η, L , the product-grid loss has readable order

$$\Theta\left(\frac{\sum_{a:\{1,\dots,n\}\rightarrow\mathcal{I}} \|Az_a\|_2^4/\eta^2 + \|A\|_F^4/L^2}{r}\right)$$

when the numerator is nonzero, and is exactly zero when the numerator is zero. The grid and its cardinality are exact analysis objects; no probability table or floating-point cover construction is computed by Algorithm 3.

Therefore, for any exact threshold $\eta > 0$,

$$\begin{aligned} & \Pr_{h,\Omega} \{ \|\Delta_{\text{CS}}(h, \omega; A)\|_F \leq \eta \} \\ & \geq 1 - \frac{2}{r\eta^2} \sum_{j,l=1}^n \sum_{(a,b) \in \mathcal{P}_m} (A_{aj}A_{bl})^2. \end{aligned}$$

This exact-object result is then transferred to the computed sparse Gram

$$\widehat{G}(h, \Omega) = \text{fl}(\text{flCountSketchApply}_{h,\omega}(A)^T \text{flCountSketchApply}_{h,\omega}(A)).$$

The computed event is

$$\left\{ (h, \Omega) : \|\widehat{G}(h, \Omega) - A^T A\|_F \leq \eta + T_{\text{CSGram}}^{\text{fp}}(h, \omega, A) \right\}.$$

It has the same probability lower bound as the exact event. The proof is a deterministic triangle inequality plus the already formalized concrete sparse Gram arithmetic theorem $\|\widehat{G} - (S_{h,\omega}A)^T(S_{h,\omega}A)\|_F \leq T_{\text{CSGram}}^{\text{fp}}(h, \omega, A)$; no perturbation-existence assumption or collision-free event is used.

The *CountSketch* sign-storage arithmetic is also now formalized as a deterministic implementation substrate. Let $\widehat{\omega}$ be any computed table certified against the exact realized signs ω , with entrywise storage radius e_k . If the sparse apply uses $\widehat{\omega}_k$ in the rounded products, then the entrywise sparse-apply radius is

$$E_{ij}^{\text{CS,store}} = (u + \gamma_{b_i} + u\gamma_{b_i}) \sum_{k:h(k)=i} |\widehat{\omega}_k| |A_{kj}| + \sum_{k:h(k)=i} e_k |A_{kj}|.$$

The second term is the exact storage deviation back to the ideal $S_{h,\omega}A$. Theorems

`preconditionRows_countSketchRows_storedSign_entry_error_bound`, `fl_countSketchSparseApplyEntry_w`

`fl_countSketchSparseApplyWithStoredSign_entry_error_bound`, `fl_countSketchSparseGramDotWithStore`

thread this radius through the same rounded length- r Gram-dot interface. For the concrete rounded-copy sign-storage modes

$$\widehat{\omega}_k \in \{\text{fl}(\omega_k \cdot 1), \text{fl}(\omega_k + 0), \text{fl}(\omega_k - 0)\},$$

theorems

`fl_countSketchSparseGramDotWithFlStoredSign_perturb_bound`, `fl_countSketchSparseGramDotWithFlSto`

instantiate $e_k = u$ using $|\omega_k| = 1$. In the regime of fixed realized h, ω, A and small roundoff, this stored-sign radius has order

$$\Theta(\gamma_r \| |Y|^T |Y| \|_F + \|(E^{\text{CS,store}})^T |Y| + |Y|^T E^{\text{CS,store}}\|_F), \quad E_{ij}^{\text{CS,store}} = \Theta\left((u + \gamma_{b_i}) \sum_{k:h(k)=i} |A_{kj}|\right)$$

for the three displayed storage modes, and it tends to zero as $u, \gamma_r, \gamma_{b_i} \rightarrow 0$.

The stored-sign radius has now also been threaded through the product-grid finite-cover probability theorem. The event

$$-(\eta + L(2\rho + \rho^2) + T_{\text{CSGram,store}}^{\text{fp}}(h, \omega, A))I \preceq \widehat{G}^{\text{store}}(h, \omega, A) - A^T A \preceq (\eta + L(2\rho + \rho^2) + T_{\text{CSGram,store}}^{\text{fp}}(h, \omega, A))I$$

has the same exact-law product-grid lower bound as the exact-sign computed sparse-Gram event:

$$1 - \left(\sum_{a:\text{Fin } n \rightarrow \alpha} \frac{2\|Az_a\|_2^4}{r\eta^2} + \frac{2\|A\|_F^4}{rL^2} \right).$$

Lean records the generic transfer as `countSketchRowGramTwoSidedLoewnerCoverEvent_subset_flSparseGramDot` and the product-grid wrappers as `countSketchProbability_eventProb_flSparseGramDotWithStoredSign_rowGram` and `countSketchProbability_eventProb_flSparseGramDotWithStoredSign_rowGram_twoSidedLoewner_produ`. The final concrete sign-copy endpoints are `countSketchProbability_eventProb_flSparseGramDotWithFlStoredS`, `countSketchProbability_eventProb_flSparseGramDotWithFlStoredSignAddZeroRight_rowGram_twoSidedLoe` and `countSketchProbability_eventProb_flSparseGramDotWithFlStoredSignSubZeroRight_rowGram_twoSided`. The probability-loss order remains

$$\Theta\left(\frac{\sum_a \|Az_a\|_2^4/\eta^2 + \|A\|_F^4/L^2}{r}\right)$$

in the fixed exact-input/grid/threshold regime when the numerator is nonzero; stored sign copies add only the deterministic roundoff radius, not a new probability-loss term.

The same product-grid stored-sign theorem is now closed for any fixed per-bucket memory order. To avoid overloading notation in this paragraph, write ρ_{grid} for the product-grid cover radius and ρ_{sig} for the selected sign-copy mode. For each realized exact hash h , write

$$B_i(h) = \{k : h(k) = i\}, \quad b_i(h) = |B_i(h)|,$$

and let

$$k_{i,1}, \dots, k_{i,b_i(h)}$$

be an exact ordering of this bucket. This order is a discrete index choice; it is not a floating-point real computation. For a generic rounded-copy mode ρ , the permuted sparse-apply entry radius is

$$E_{ij}^{\text{CS},\rho,\pi} = (u + \gamma_{b_i(h)} + u\gamma_{b_i(h)}) \sum_{t=1}^{b_i(h)} |\widehat{\omega}_{k_{i,t}}^\rho| |A_{k_{i,t}j}| + u \sum_{t=1}^{b_i(h)} |A_{k_{i,t}j}|.$$

The exact reference remains

$$Y_{ij} = \sum_{k:h(k)=i} \omega_k A_{kj}.$$

Lean records this endpoint through `fl_countSketchSparseApplyEntryTree_error_bound`, `fl_countSketchSparseGramDotTree_perturb_bound`, `fl_countSketchSparseApplyWithStoredSignTree_entry`, `fl_countSketchSparseGramDotWithStoredSignTree_perturb_bound`, `countSketchFlSparseGramDotWithStoredSignTreeRowGramTwoSidedLoewnerEvent`, `countSketchRowGramTwoSidedLoewnerCoverEvent_subset_flSparseGramDotWithStoredSignTreeRowGramTwoSidedLoewnerEvent`, `countSketchProbability_eventProb_flSparseGramDotWithStoredSignTree_rowGram_twoSidedLoewner_produ` and its target-budget and three concrete stored-sign copy wrappers. For fixed realized data,

$$E_{ij}^{\text{CS},\rho,\mathcal{T}} = \Theta \left((u + \gamma_{d_i(h)}) \sum_{t=1}^{b_i(h)} |A_{k_i,tj}| \right),$$

and balanced trees give the interpretable specialization

$$E_{ij}^{\text{CS},\rho,\mathcal{T}} = \Theta \left(u(1 + \log(b_i(h) + 1)) \sum_{t=1}^{b_i(h)} |A_{k_i,tj}| \right).$$

The probability loss is unchanged from the stored-sign product-grid theorem, $\Theta((\sum_a \|Az_a\|_2^4/\eta^2 + \|A\|_F^4/L^2)/r)$, and the deterministic radius tends to zero with $u, \gamma_r, \gamma_{d_i(h)}, e_k^\rho$. This closes tree-reduced sparse-Gram CountSketch layouts; the downstream uniform-row tree-reduced composition is closed below. Aliasing or parallel sparse-sketch memory layouts remain separate implementation routines.

The same stored-sign sparse-apply radius has now been lifted through the downstream iid uniform-row sampling stage. The final event uses the concrete computed denominator for $\sqrt{s/r}$, rounded row divisions, and rounded length- s sampled-Gram dot products, with radius

$$\eta + L(2\rho + \rho^2) + \eta_{\text{row}} + T_{\text{CS,row,store}}^{\text{fp}}(h, \omega, \sigma, A).$$

The exact product-grid plus row-sampling loss is unchanged from the exact-sign downstream theorem:

$$\sum_a \frac{2r^{-1} \sum_{p \neq q} ((Az_a)_p (Az_a)_q)^2}{\eta^2} + \frac{2r^{-1} \sum_{j,k} \sum_{p \neq q} (A_{pj} A_{qk})^2}{L^2} + \frac{rm^2 \|A\|_F^4}{s\eta_{\text{row}}^2}.$$

Lean records the generic stored-sign downstream wrappers as `countSketchUniformRowTraceProbability_eventProb_sparseStoredSignComputedPreconditioned_fl_uniform` and its target-budget variant, and the concrete final endpoints for $\mathfrak{fl}(\omega_i \cdot 1)$, $\mathfrak{fl}(\omega_i + 0)$, and $\mathfrak{fl}(\omega_i - 0)$ are the corresponding `sparseFlStoredSignComputedPreconditioned`, `sparseFlStoredSignAddZeroRightComputedPreconditioned`, and `sparseFlStoredSignSubZeroRightComputedPreconditioned` theorems. These are final implementation-facing results for the modeled stored-sign CountSketch-plus-uniform-row path; no probability-computation term, perturbation event, or certificate-existence assumption is added.

The stored-sign downstream theorem is also closed for fixed exact per-bucket permutation layouts. For each realized hash h , the exact bucket order π is a discrete index operation, while the computed sparse sketch visits the bucket in that order. The final event radius is

$$\eta + L(2\rho + \rho^2) + \eta_{\text{row}} + T_{\text{CS,row,store},\pi}^{\text{fp}}(h, \omega, \sigma, A),$$

and the exact product-grid plus row-sampling loss remains the same as in the canonical stored-sign downstream theorem. Lean records the new basis and budget

through `countSketchSparseComputedPreconditionedBasisWithStoredSignPermuted` and `countSketchSparseUniformRowComputedDenStoredSignPermutedPerturbBudget`, the probability-one perturbation theorem through `countSketchUniformRowTraceProbability_eventProb_sparseStoredSignPerm` and the final product-grid wrappers through the `sparseStoredSignPermutedComputedPreconditioned` and `concrete` `sparseFlStoredSignPermutedComputedPreconditioned`, `sparseFlStoredSignAddZeroRightPermutedComputedPreconditioned`, and `sparseFlStoredSignSubZeroRightPermutedComputedPreconditioned` theorems. For fixed realized $h, \omega, \sigma, \pi, A$ and $d > 0$,

$$T_{\text{CS,row,store},\pi}^{\text{fp}} = \Theta\left(\gamma_s \left\| |\widehat{B}|^T |\widehat{B}| \right\|_F + \left\| (E^{\text{den},\pi})^T |\widehat{B}| + |\widetilde{B}|^T E^{\text{den},\pi} \right\|_F + \left\| (F^\pi)^T |\widetilde{B}| + |B|^T F^\pi \right\|_F\right),$$

with

$$E_{qj}^{\text{den},\pi} = \Theta\left(u |\widehat{B}_{qj}| + \frac{|\widehat{Y}_{\sigma_{qj}}^\pi|}{|\widehat{d}|} \frac{4u + 3u^2 + u^3}{1 - u}\right), \quad F_{qj}^\pi = \Theta\left(\frac{u + \gamma_{b_{\sigma_q}(h)}}{d} \sum_t |A_{k_{\sigma_q,t}j}|\right).$$

Thus the exact bucket order adds no probability loss and no probability-law perturbation; it only selects the concrete order charged by the deterministic floating-point accumulation radius. Parallel reassociation and aliasing sparse-sketch layouts remain separate implementation routines.

The stored-sign downstream theorem is also closed for explicit binary tree-reduced bucket accumulation. For each realized hash h , the exact tree shapes $\mathcal{T}_i(h)$ and their trailing zero leaves are discrete implementation choices, while the computed sparse sketch stores signs, rounds signed products, reduces each bucket through the selected tree, forms the concrete denominator $\widehat{d}_\star = \text{fl}(\text{fl}(\sqrt{s}) \text{fl}(1/\text{fl}(\sqrt{r})))$, rounds sampled-row divisions, and rounds length- s sampled-Gram dot products. Lean records the basis and budget through `countSketchSparseComputedPreconditionedBasisWithStoredSignTree` and `countSketchSparseUniformRowComputedDenStoredSignTreePerturbBudget`, the probability-one perturbation theorem through `countSketchUniformRowTraceProbability_eventProb_sparseStoredSignTreeComp` and the final product-grid wrappers through the `sparseStoredSignTreeComputedPreconditioned` and `concrete` `sparseFlStoredSignTreeComputedPreconditioned`, `sparseFlStoredSignAddZeroRightTreeComputedPreconditioned`, and `sparseFlStoredSignSubZeroRightTreeComputedPreconditioned` theorems. The final radius is

$$\eta + L(2\rho + \rho^2) + \eta_{\text{row}} + T_{\text{CS,row,store},\mathcal{T}}^{\text{fp}}(h, \omega, \sigma, A),$$

and the exact product-grid plus row-sampling loss is unchanged. For fixed realized data and $d = \sqrt{s/r} > 0$,

$$T_{\text{CS,row,store},\mathcal{T}}^{\text{fp}} = \Theta\left(\gamma_s \left\| |\widehat{B}|^T |\widehat{B}| \right\|_F + \left\| (E^{\text{den},\mathcal{T}})^T |\widehat{B}| + |\widetilde{B}|^T E^{\text{den},\mathcal{T}} \right\|_F + \left\| (F^\mathcal{T})^T |\widetilde{B}| + |B|^T F^\mathcal{T} \right\|_F\right),$$

where

$$E_{qj}^{\text{den},\mathcal{T}} = \Theta\left(u |\widehat{B}_{qj}| + \frac{|\widehat{Y}_{\sigma_{qj}}^\mathcal{T}|}{|\widehat{d}_\star|} \frac{4u + 3u^2 + u^3}{1 - u}\right),$$

and

$$F_{qj}^\mathcal{T} = \Theta\left(\frac{u + \gamma_{d_{\sigma_q}(h)}}{d} \sum_t |A_{k_{\sigma_q,t}j}|\right).$$

Balanced bucket trees give

$$F_{qj}^\mathcal{T} = \Theta\left(\frac{u(1 + \log(b_{\sigma_q}(h) + 1))}{d} \sum_t |A_{k_{\sigma_q,t}j}|\right).$$

Exact hash/sign laws, exact row sampling, exact product-grid vectors, tree shapes, and $A^T A$ remain exact analysis/probability or discrete objects. All computed non-probability objects on this modeled path are charged, and no perturbation-event or certificate-existence assumption is used.

The same stored-sign downstream product-grid theorem is now instantiated with four additional concrete denominator routines for $d = \sqrt{s/r}$. The routes are

$$\begin{aligned}\widehat{d} &= \text{fl}_{\text{sqr}}(s/r) \quad \text{when the ratio } s/r \text{ is supplied exactly,} \\ \widehat{d} &= \text{fl}_{\text{sqr}}(\text{fl}_{\text{div}}(s, r)), \\ \widehat{d} &= \text{fl}_{\text{sqr}}(\text{fl}_{\text{mul}}(s, \text{fl}_{\text{div}}(1, r))), \\ \widehat{d} &= \text{fl}_{\text{div}}(\text{fl}_{\text{sqr}}(s), \text{fl}_{\text{sqr}}(r)).\end{aligned}$$

Their denominator relative radii are respectively

$$u, \quad \sqrt{1+u} u + u, \quad \sqrt{(1+u)(1+u)} u + 2u + u^2, \quad \frac{3u + u^2}{1-u}.$$

Thus the exact product-grid plus downstream row probability loss is unchanged, and the deterministic stored-sign row radius is obtained by substituting the chosen route-specific denominator factor. For fixed realized (h, ω, σ, A) and $d > 0$, the stored-sign downstream floating-point radius has order

$$\Theta\left(\gamma_s \left\| |\widehat{B}|^T |\widehat{B}| \right\|_F + \left\| (E^{\text{den}})^T |\widehat{B}| + |\widehat{B}|^T E^{\text{den}} \right\|_F + \left\| F^T |\widehat{B}| + |B|^T F \right\|_F \right),$$

with

$$E_{qj}^{\text{den}} = \Theta\left(u |\widehat{B}_{qj}| + \frac{|\widehat{Y}_{\sigma_{qj}}|}{|\widehat{d}|} \Delta_d\right),$$

where Δ_d is the selected relative denominator factor. The radius tends to zero as $u, \gamma_s, \gamma_{b_i(h)}, \Delta_d \rightarrow 0$; denominator computation changes only the deterministic radius, not the exact probability law.

The coefficient sum is also simplified in Lean:

$$\sum_{j,l=1}^n \sum_{(a,b) \in \mathcal{P}_m} (A_{aj} A_{bl})^2 \leq \|A\|_F^4.$$

Thus the readable computed sparse Gram endpoint is

$$\Pr_{h,\Omega} \left\{ \|\widehat{G}(h, \Omega) - A^T A\|_F \leq \eta + T_{\text{CSGram}}^{\text{fp}}(h, \omega, A) \right\} \geq 1 - \frac{2\|A\|_F^4}{r\eta^2}.$$

In the asymptotic regime with exact A and $\eta > 0$ fixed and $r \rightarrow \infty$, the failure term has order

$$\Theta\left(\frac{\|A\|_F^4}{r\eta^2}\right).$$

For an exact orthonormal analysis basis $U \in \mathbb{R}^{m \times n}$, $U^T U = I_n$ implies $\|U\|_F^2 = n$, and the bound becomes

$$\Pr_{h,\Omega} \left\{ \|\widehat{G}(h, \Omega) - I_n\|_F \leq \eta + T_{\text{CSGram}}^{\text{fp}}(h, \omega, U) \right\} \geq 1 - \frac{2n^2}{r\eta^2},$$

with failure order

$$\Theta\left(\frac{n^2}{r\eta^2}\right)$$

when n and η are fixed and $r \rightarrow \infty$. The result is non-vacuous when $r > 2\|A\|_F^4/\eta^2$, or $r > 2n^2/\eta^2$ in the orthonormal case, and the failure term tends to zero as $r \rightarrow \infty$. This is a fully computed sparse-Gram Frobenius/Markov endpoint for the non-injective route. It is still weaker than the standard CountSketch finite-Loewner subspace embedding and has not yet been composed with the downstream uniform-row sampling theorem.

The same non-injective theorem now has the finite-Loewner form needed by the geometric Algorithm 3 statements. For each realized $x = (h, \Omega)$, set

$$\tau_{\text{CS}}^{\text{fp}}(x; A, \eta) := \eta + T_{\text{CSGram}}^{\text{fp}}(h, \text{rademacherSignVector}(\Omega), A),$$

and define the computed two-sided Gram event

$$\mathcal{E}_{\text{CS,L}}^{\text{fp}}(\eta) = \left\{ x : \begin{array}{l} \widehat{G}(x) - A^T A \preceq \tau_{\text{CS}}^{\text{fp}}(x; A, \eta) I_n, \\ -(\widehat{G}(x) - A^T A) \preceq \tau_{\text{CS}}^{\text{fp}}(x; A, \eta) I_n \end{array} \right\}.$$

Here $\widehat{G}(x)$ is the computed sparse CountSketch Gram matrix above, while $A^T A$ is the exact analysis reference Gram. Lean proves the deterministic inclusion from the Frobenius event by the local bridge

$$\|M\|_F \leq \tau \implies -\tau I \preceq M \preceq \tau I.$$

Consequently,

$$\Pr_{h, \Omega} \{\mathcal{E}_{\text{CS,L}}^{\text{fp}}(\eta)\} \geq 1 - \frac{2}{r\eta^2} \sum_{j,l=1}^n \sum_{(a,b) \in \mathcal{P}_m} (A_{aj} A_{bl})^2 \geq 1 - \frac{2\|A\|_F^4}{r\eta^2}.$$

The failure term has the same readable order

$$\Theta\left(\frac{\|A\|_F^4}{r\eta^2}\right)$$

for fixed exact A and $\eta > 0$ as $r \rightarrow \infty$. If $U^T U = I_n$, then

$$\Pr_{h, \Omega} \{\mathcal{E}_{\text{CS,L}}^{\text{fp}}(\eta)\} \geq 1 - \frac{2n^2}{r\eta^2}, \quad \text{failure order } \Theta\left(\frac{n^2}{r\eta^2}\right).$$

This Loewner statement charges the same computed sparse apply and rounded Gram-dot arithmetic as the Frobenius statement. It does not add a collision-free event or an external certificate assumption; it is still a Markov/Frobenius-derived Loewner endpoint, not the optimal CountSketch subspace-embedding concentration theorem.

The exact finite-cover CountSketch Loewner theorem is now also transferred to the computed sparse Gram object. Let $\mathcal{A}$ be a finite index set and let $z_\alpha \in \mathbb{R}^n$, $\alpha \in \mathcal{A}$, be an exact unit-ball cover of radius ρ . For exact thresholds $\eta > 0$ and $L > 0$, define

$$\tau_{\text{cover}}(A; \rho, \eta, L) := \eta + L(2\rho + \rho^2)$$

and the exact probability loss

$$\begin{aligned} \mathcal{L}_{\text{cover}}(A, Z; \eta, L, r) &:= \sum_{\alpha \in \mathcal{A}} \frac{2}{r\eta^2} \sum_{(a,b) \in \mathcal{P}_m} ((Az_\alpha)_a (Az_\alpha)_b)^2 \\ &\quad + \frac{2}{rL^2} \sum_{j,l=1}^n \sum_{(a,b) \in \mathcal{P}_m} (A_{aj} A_{bl})^2. \end{aligned}$$

For each exact CountSketch realization $x = (h, \Omega)$, set

$$\tau_{\text{cover}}^{\text{fp}}(x; A, \rho, \eta, L) := \eta + L(2\rho + \rho^2) + T_{\text{CSGram}}^{\text{fp}}(h, \text{rademacherSignVector}(\Omega), A).$$

The theorem

`countSketchRowGramTwoSidedLoewnerCoverEvent_subset_flSparseGramDotRowGramTwoSidedLoewnerEvent` proves the deterministic transfer

$$-\tau_{\text{cover}}^{\text{fp}}(x)I_n \preceq \widehat{G}(x) - A^T A \preceq \tau_{\text{cover}}^{\text{fp}}(x)I_n$$

from the exact finite-cover event. Here $\widehat{G}(x)$ is computed by `flCountSketchSparseGramDot(fp, h,  $\omega$ , A)`; the exact reference $A^T A$, the finite cover, and the hash/sign probability laws remain analysis objects. Sparse signed products, bucket accumulation, and rounded Gram dot products are charged through the concrete $T_{\text{CSGram}}^{\text{fp}}$ budget, not by an assumed perturbation certificate. Consequently Lean proves

$$\Pr_{h, \Omega} \left\{ -\tau_{\text{cover}}^{\text{fp}}(x)I_n \preceq \widehat{G}(x) - A^T A \preceq \tau_{\text{cover}}^{\text{fp}}(x)I_n \right\} \geq 1 - \mathcal{L}_{\text{cover}}(A, Z; \eta, L, r),$$

with direct target-budget theorem

$$\mathcal{L}_{\text{cover}}(A, Z; \eta, L, r) \leq \delta \implies \Pr\{\text{the displayed computed-cover event}\} \geq 1 - \delta.$$

The Lean names are

`countSketchProbability_eventProb_flSparseGramDot_rowGram_twoSidedLoewner_cover_ge_one_sub_sum_co`

and

`countSketchProbability_eventProb_flSparseGramDot_rowGram_twoSidedLoewner_cover_ge_one_sub_delta`

Lean also proves the readable sufficient version

$$\Pr\{\text{the displayed computed-cover event}\} \geq 1 - \left(\sum_{\alpha \in \mathcal{A}} \frac{2\|Az_\alpha\|_2^4}{r\eta^2} + \frac{2\|A\|_F^4}{rL^2} \right),$$

and its target-budget wrapper

$$\sum_{\alpha \in \mathcal{A}} \frac{2\|Az_\alpha\|_2^4}{r\eta^2} + \frac{2\|A\|_F^4}{rL^2} \leq \delta \implies \Pr\{\text{the displayed computed-cover event}\} \geq 1 - \delta.$$

The Lean names are

`countSketchProbability_eventProb_flSparseGramDot_rowGram_twoSidedLoewner_cover_ge_one_sub_sum_ve`

and

`countSketchProbability_eventProb_flSparseGramDot_rowGram_twoSidedLoewner_cover_ge_one_sub_delta`

In the regime of fixed exact A , fixed cover Z , fixed positive η, L , and $r \rightarrow \infty$, the irreducible failure order is

$$\Theta \left(\sum_{\alpha \in \mathcal{A}} \frac{\sum_{(a,b) \in \mathcal{P}_m} ((Az_\alpha)_a (Az_\alpha)_b)^2}{r\eta^2} + \frac{\sum_{j,l=1}^n \sum_{(a,b) \in \mathcal{P}_m} (A_{aj} A_{bl})^2}{rL^2} \right)$$

when the numerator is nonzero. A readable upper order is

$$O\left(\sum_{\alpha \in \mathcal{A}} \frac{\|Az_\alpha\|_2^4}{r\eta^2} + \frac{\|A\|_F^4}{rL^2}\right).$$

The probability term is non-vacuous whenever $\mathcal{L}_{\text{cover}}(A, Z; \eta, L, r) < 1$, and tends to zero as $r \rightarrow \infty$ for fixed exact inputs and thresholds. The additional floating-point term is deterministic radius inflation and vanishes when the charged sparse-apply and Gram-dot roundoff budgets vanish.

The same product grid also has a computed sparse-Gram endpoint. Keep the finite exact grid $g : \mathcal{I} \rightarrow \mathbb{R}$, product vectors z_a , and $\sqrt{n} \delta_{\text{grid}} \leq \rho$ hypotheses from the preceding product-grid paragraph, and assume the floating-point model has valid γ_m and γ_r . Then Lean proves

$$\Pr_{h, \Omega} \left\{ -\tau_{\text{pg}}^{\text{fp}}(x) I_n \preceq \widehat{G}(x) - A^T A \preceq \tau_{\text{pg}}^{\text{fp}}(x) I_n \right\} \geq 1 - \left(\sum_{a: \{1, \dots, n\} \rightarrow \mathcal{I}} \frac{2\|Az_a\|_2^4}{r\eta^2} + \frac{2\|A\|_F^4}{rL^2} \right),$$

where

$$\tau_{\text{pg}}^{\text{fp}}(x) = \eta + L(2\rho + \rho^2) + T_{\text{CSGram}}^{\text{fp}}(h, \text{rademacherSignVector}(\Omega), A).$$

The direct $1 - \delta$ wrapper uses the same displayed product-grid loss. The Lean theorem names are

`countSketchProbability_eventProb_flSparseGramDot_rowGram_twoSidedLoewner_productGrid_ge_one_sub`

and

`countSketchProbability_eventProb_flSparseGramDot_rowGram_twoSidedLoewner_productGrid_ge_one_sub`

The probability order is the same product-grid $\Theta((\sum_a \|Az_a\|_2^4/\eta^2) + \|A\|_F^4/L^2)/r$ order above in the fixed exact-input/grid/threshold regime. The added $T_{\text{CSGram}}^{\text{fp}}$ term is deterministic radius inflation from the computed sparse signed products, bucket accumulation, and rounded Gram dot products; it is not a probability loss and vanishes as those charged roundoff budgets vanish.

The downstream iid uniform-row sampling foundation is now available without assuming that the matrix being sampled already has orthonormal columns. Let $M \in \mathbb{R}^{m \times n}$ be an exact matrix, let $s > 0$, and let $\sigma = (\sigma_1, \dots, \sigma_s)$ be exact iid uniform row indices in $\{1, \dots, m\}$. Define the exact sampled Gram average

$$\widehat{G}_{\text{unif}}(M, \sigma) := \frac{1}{s} \sum_{q=1}^s m M_{\sigma_q, *}^T M_{\sigma_q, *}.$$

For each row i , write

$$R_i(M) := \|M_{i, *}\|_2^2 = \sum_{j=1}^n M_{ij}^2, \quad \|M\|_F^2 = \sum_{i=1}^m R_i(M).$$

All objects in this paragraph are exact arithmetic objects; the only random object is the exact uniform row law. No floating-point sparse apply, denominator, row scaling, or Gram-dot routine is being charged here, because none is computed in this exact foundation theorem.

Lean first proves the one-row mean identity

$$\mathbb{E}_i [m M_{i, *}^T M_{i, *}] = M^T M,$$

and the iid sample-average variance identity for each Gram entry. Summing entries and bounding the centered second moment by the raw second moment gives

$$\mathbb{E}_\sigma \left\| \widehat{G}_{\text{unif}}(M, \sigma) - M^T M \right\|_F^2 \leq \frac{m}{s} \sum_{i=1}^m R_i(M)^2.$$

Since

$$\sum_{i=1}^m R_i(M)^2 \leq \left(\sum_{i=1}^m R_i(M) \right)^2 = \|M\|_F^4,$$

the readable bound is

$$\mathbb{E}_\sigma \left\| \widehat{G}_{\text{unif}}(M, \sigma) - M^T M \right\|_F^2 \leq \frac{m}{s} \|M\|_F^4.$$

Therefore, for any exact threshold $\eta > 0$,

$$\begin{aligned} \Pr_\sigma \left\{ \left\| \widehat{G}_{\text{unif}}(M, \sigma) - M^T M \right\|_F \leq \eta \right\} &\geq 1 - \frac{m \sum_{i=1}^m R_i(M)^2}{s \eta^2} \\ &\geq 1 - \frac{m \|M\|_F^4}{s \eta^2}. \end{aligned}$$

The exact row-fourth failure term has order

$$\Theta \left(\frac{m \sum_{i=1}^m R_i(M)^2}{s \eta^2} \right)$$

when $\sum_i R_i(M)^2 \neq 0$, and the simplified Frobenius upper radius has order

$$\Theta \left(\frac{m \|M\|_F^4}{s \eta^2} \right)$$

when $\|M\|_F \neq 0$, in the regime with exact M , m , and $\eta > 0$ fixed and $s \rightarrow \infty$. The bound is non-vacuous when $s > m \sum_i R_i(M)^2 / \eta^2$, or under the simplified form when $s > m \|M\|_F^4 / \eta^2$, and the failure term tends to zero as $s \rightarrow \infty$.

This exact theorem is not, by itself, a final implementation-facing CountSketch-plus-uniform-row theorem: it is the exact-law foundation that the computed sampled-Gram bridge below uses. The Lean theorem surface is:

```
uniform_rowOuterGramSample_mean_eq_rowGram
uniformRowTraceProbability_expectationReal_sampleAverage_sub_mean_sq
uniformRowTraceProbability_expectationReal_uniformRowSampleGram_entry_error_sq
uniformRowTraceProbability_expectationReal_uniformRowSampleGram_frob_error_sq_le
uniformRowTraceProbability_expectationReal_uniformRowSampleGram_frob_error_sq_le_frobNorm
uniformRowTraceProbability_eventProb_uniformRowSampleGram_frob_error_le_ge_one_sub
uniformRowTraceProbability_eventProb_uniformRowSampleGram_frob_error_le_ge_one_sub_frobNorm
```

The arbitrary-matrix uniform-row sampled Gram now also has a concrete floating-point bridge. Keep the same exact matrix M , exact iid uniform row samples σ , sample count $s > 0$, and exact reference Gram $M^T M$. The probability law for σ is exact by project convention. The non-probability operations charged in this theorem are: row scaling by the uniform sampling denominator, optional computation/storage of that denominator, and the length- s dot products used to form the sampled Gram.

Let

$$d := \sqrt{s/m}, \quad B_{qj} := \frac{M_{\sigma_q j}}{d}, \quad C_B := \left(\sum_{q=1}^s |B_{qj}| |B_{qk}| \right)_{j,k}.$$

Here B is the exact sampled-and-rescaled sketch used only as the reference object in the analysis. If the implementation divides each sampled entry by the exact denominator d in floating point and then computes every Gram entry by a rounded length- s dot product, Lean proves

$$\begin{aligned} \Pr_{\sigma} \left\{ \left\| \widehat{G}_{\text{unif}}^{\text{fp}}(M, \sigma) - M^T M \right\|_F \leq \eta + T_{\text{unif}}^{\text{fp}}(M, \sigma) \right\} &\geq 1 - \frac{m \sum_{i=1}^m R_i(M)^2}{s\eta^2} \\ &\geq 1 - \frac{m \|M\|_F^4}{s\eta^2}, \end{aligned}$$

where the exact expanded floating-point radius is

$$\begin{aligned} T_{\text{unif}}^{\text{fp}}(M, \sigma) &= \left\| \left((2u + u^2) \sum_{q=1}^s |B_{qj}| |B_{qk}| \right)_{j,k} \right\|_F \\ &\quad + \left\| \left(\gamma_s (1 + u)^2 \sum_{q=1}^s |B_{qj}| |B_{qk}| \right)_{j,k} \right\|_F \\ &= (2u + u^2 + \gamma_s (1 + u)^2) \|C_B\|_F. \end{aligned}$$

The first term is the rounded row-division contribution; the second is the rounded dot-product contribution. In the usual small-roundoff regime with su bounded away from 1, $\gamma_s = \Theta(su)$, and therefore

$$T_{\text{unif}}^{\text{fp}}(M, \sigma) = \Theta((u + \gamma_s) \|C_B\|_F) = \Theta(su \|C_B\|_F)$$

when $C_B \neq 0$. For fixed s , exact M , and fixed realized samples, the same expression is $\Theta(u \|C_B\|_F)$. It tends to zero as the unit roundoff tends to zero.

Lean also proves the implementation-facing variant where the denominator is itself computed. Let $\widehat{d}$ be the computed nonzero denominator, let $\Delta_d := |\widehat{d} - d|$, define

$$\widehat{B}_{qj} := \text{fl} \left(\frac{M_{\sigma_{qj}}}{\widehat{d}} \right), \quad E_{qj} := |\widehat{B}_{qj}| u + |M_{\sigma_{qj}}| \frac{\Delta_d}{|\widehat{d}| |d|}.$$

The quantity E_{qj} is fully computable from the modeled rounded sketch, the exact sampled input entry used in the analysis, and the proved denominator absolute-error certificate. With this computed denominator, Lean proves

$$\begin{aligned} \Pr_{\sigma} \left\{ \left\| \widehat{G}_{\text{unif,den}}^{\text{fp}}(M, \widehat{d}, \sigma) - M^T M \right\|_F \leq \eta + T_{\text{unif,den}}^{\text{fp}}(M, \widehat{d}, \sigma) \right\} &\geq 1 - \frac{m \sum_{i=1}^m R_i(M)^2}{s\eta^2} \\ &\geq 1 - \frac{m \|M\|_F^4}{s\eta^2}, \end{aligned}$$

with exact expanded radius

$$\begin{aligned} T_{\text{unif,den}}^{\text{fp}} &= \left\| \left(\sum_{q=1}^s (E_{qj} |\widehat{B}_{qk}| + |B_{qj}| E_{qk}) \right)_{j,k} \right\|_F \\ &\quad + \left\| \left(\gamma_s \sum_{q=1}^s |\widehat{B}_{qj}| |\widehat{B}_{qk}| \right)_{j,k} \right\|_F. \end{aligned}$$

Equivalently, using matrix notation with entrywise absolute values,

$$T_{\text{unif,den}}^{\text{fp}} = \|E^T|\widehat{B}| + |B|^T E\|_F + \gamma_s \| |\widehat{B}|^T |\widehat{B}| \|_F,$$

and its interpretable order is

$$\Theta \left(\|E^T|\widehat{B}| + |B|^T E\|_F + \gamma_s \| |\widehat{B}|^T |\widehat{B}| \|_F \right).$$

Substituting E_{qj} gives the irreducible order form

$$\Theta \left(\left\| (u|\widehat{B}| + \delta_d|M_\sigma|)^T |\widehat{B}| + |B|^T (u|\widehat{B}| + \delta_d|M_\sigma|) \right\|_F + \gamma_s \| |\widehat{B}|^T |\widehat{B}| \|_F \right),$$

where $\delta_d := \Delta_d / (|\widehat{d}| |d|)$ and $(M_\sigma)_{qj} := M_{\sigma_{qj}}$. This radius is non-vacuous in the floating-point sense: for fixed exact data and samples, it tends to zero as $u \rightarrow 0$, $\gamma_s \rightarrow 0$, and the denominator error $\Delta_d \rightarrow 0$.

The Lean theorem surface for this floating-point bridge is: `uniformRowSampleGramRowGramFrobErrorEvent`
`uniformRowFLSampleGramDotRowGramFrobErrorEvent`
`uniformRowFLSampleGramDotWithComputedDenRowGramFrobErrorEvent`
`uniformRowSampleGramRowGramFrobErrorEvent_subset_fLSampleGramDot`
`uniformRowSampleGramRowGramFrobErrorEvent_subset_fLSampleGramDotWithComputedDen`
`uniformRowTraceProbability_eventProb_fl_uniformRowSampleGramDot_rowGram_frob_error_le_ge_one_sub`
`uniformRowTraceProbability_eventProb_fl_uniformRowSampleGramDot_rowGram_frob_error_le_ge_one_sub_frobNorm`
`uniformRowTraceProbability_eventProb_fl_uniformRowSampleGramDotWithComputedDen_rowGram_frob_error_le_ge_one_sub`
`uniformRowTraceProbability_eventProb_fl_uniformRowSampleGramDotWithComputedDen_rowGram_frob_error_le_ge_one_sub_frobNorm`

The non-injective `CountSketch` branch is now composed with this downstream uniform-row machinery. Draw $x = (h, \Omega)$ from the exact `CountSketch` hash-sign law, set $\omega = \text{rademacherSignVector}(\Omega)$, and let σ be exact iid uniform samples from the r `CountSketch` output rows. Define

$$V(h, \Omega) := S_{h, \omega} A, \quad \widehat{V}(h, \Omega) := \text{flCountSketchSparseApply}(h, \omega, A).$$

The new deterministic `CountSketch` bucket identity proves

$$\sum_{i=1}^r \sum_{k: h(k)=i} f(k) = \sum_{k=1}^m f(k),$$

and, by Cauchy–Schwarz, the exact `CountSketch` output satisfies

$$\|V(h, \Omega)\|_F^2 \leq m \|A\|_F^2$$

for every realized hash and every exact sign vector with $|\omega_k| \leq 1$. This supplies the uniform, nonconditional row-sampling loss needed after non-injective `CountSketch`.

Let $\eta_{\text{CS}} > 0$ be the exact `CountSketch` Gram threshold and $\eta_{\text{row}} > 0$ the exact downstream row-sampling threshold. The exact product event is the intersection

$$\|V(h, \Omega)^T V(h, \Omega) - A^T A\|_F \leq \eta_{\text{CS}},$$

$$\|\widehat{G}_{\text{unif}}(V(h, \Omega), \sigma) - V(h, \Omega)^T V(h, \Omega)\|_F \leq \eta_{\text{row}}.$$

Lean proves

$$\Pr\{\mathcal{E}_{\text{exact}}^{\text{CS+row}}\} \geq 1 - \frac{2}{r\eta_{\text{CS}}^2} \sum_{j,k=1}^n \sum_{(a,b) \in \mathcal{P}_m} (A_{aj}A_{bk})^2 - \frac{r m^2 \|A\|_F^4}{s\eta_{\text{row}}^2}, \quad \mathcal{P}_m = \{(a,b) : a \neq b\}.$$

Using the already-proved coefficient simplification, the readable form is

$$\Pr\{\mathcal{E}_{\text{exact}}^{\text{CS+row}}\} \geq 1 - \frac{2\|A\|_F^4}{r\eta_{\text{CS}}^2} - \frac{r m^2 \|A\|_F^4}{s\eta_{\text{row}}^2}.$$

The corresponding computed event forms $\widehat{V}$, computes a denominator $\widehat{d}$ for $\sqrt{s/r}$, rounds sampled-row divisions, and rounds the length- s sampled-Gram dot products:

$$\left\| \widehat{G}_{\text{CS+row}}^{\text{fp}}(h, \Omega, \sigma) - A^T A \right\|_F \leq \eta_{\text{CS}} + \eta_{\text{row}} + T_{\text{CS,row}}^{\text{fp}}(h, \Omega, \sigma).$$

The same probability lower bound holds for this computed event. The realized floating-point radius is not an opaque certificate:

$$T_{\text{CS,row}}^{\text{fp}} = T_{\text{unif,den}}^{\text{fp}}(\widehat{V}, \widehat{d}, \sigma) + T_{\text{basis}}^{\text{fp}}(V, \widehat{V}, E^{\text{CS}}, \sigma),$$

where the first term is the expanded computed-denominator sampled-Gram radius displayed above with $M = \widehat{V}$, and

$$T_{\text{basis}}^{\text{fp}} = \left\| \left(\sum_{q=1}^s \left(\frac{E_{\sigma_{qj}}^{\text{CS}}}{d} |\widehat{B}_{qk}| + |B_{qj}| \frac{E_{\sigma_{qk}}^{\text{CS}}}{d} \right) \right)_{j,k} \right\|_F, \quad d = \sqrt{s/r}.$$

Here B is the exact sampled sketch from V , $\widehat{B}$ is the sampled sketch from $\widehat{V}$, and

$$E_{ij}^{\text{CS}} = (u + \gamma_{b_i} + u\gamma_{b_i}) \sum_{k:h(k)=i} |\omega_k| |A_{kj}|, \quad b_i = |\{k : h(k) = i\}|.$$

For fixed exact $A, m, \eta_{\text{CS}}, \eta_{\text{row}}$, the simplified failure loss has order

$$\Theta \left(\frac{\|A\|_F^4}{r\eta_{\text{CS}}^2} + \frac{r m^2 \|A\|_F^4}{s\eta_{\text{row}}^2} \right).$$

It is non-vacuous when the displayed loss is below one, and it tends to zero in the regime $r \rightarrow \infty$, $s/r \rightarrow \infty$. For fixed realized hashes, signs, and row samples, the FP radius tends to zero as $u \rightarrow 0$, $\gamma_s \rightarrow 0$, every $\gamma_{b_i} \rightarrow 0$, and the denominator error $|\widehat{d} - d| \rightarrow 0$. This closes a genuine non-injective CountSketch plus downstream row-sampling implementation endpoint; it is still Markov/Frobenius-derived and weaker than the optimal CountSketch subspace-embedding theorem.

The Lean theorem surface is: `countSketchBucket_sum_sum_eq`
`frobNormSqRect_preconditionRows_countSketchRows_le`
`countSketchUniformRowSampleGramRowGramFrobEvent`
`countSketchSparseComputedPreconditionedFlUniformRowSampleGramWithComputedDenRowGramFrobEvent`
`countSketchUniformRowTraceProbability_eventProb_uniformRowSampleGram_rowGram_frob_error_le_ge_one_sub`
`countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl_`
`uniformRowSampleGramDotWithComputedDen_rowGram_frob_error_le_ge_one_sub`

`countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl_`
`uniformRowSampleGramDotWithComputedDen_rowGram_frob_error_le_ge_one_sub_frobNorm`

The same fully computed non-injective CountSketch-plus-uniform-row endpoint now has a two-sided finite-Loewner form. For each realization, set

$$\tau_{\text{CS,row}}^{\text{fp}} := \eta_{\text{CS}} + \eta_{\text{row}} + T_{\text{CS,row}}^{\text{fp}}(h, \Omega, \sigma).$$

The computed two-sided event is

$$-\tau_{\text{CS,row}}^{\text{fp}} I_n \preceq \widehat{G}_{\text{CS+row}}^{\text{fp}}(h, \Omega, \sigma) - A^T A \preceq \tau_{\text{CS,row}}^{\text{fp}} I_n.$$

It is obtained deterministically from the Frobenius event by the locally proved bridge

$$\|M\|_F \leq \tau \implies -\tau I \preceq M \preceq \tau I.$$

Therefore the exact coefficient and readable probability losses are unchanged:

$$\begin{aligned} \Pr\{\mathcal{E}_{\text{CS+row,L}}^{\text{fp}}\} &\geq 1 - \frac{2}{r\eta_{\text{CS}}^2} \sum_{j,k=1}^n \sum_{(a,b) \in \mathcal{P}_m} (A_{aj} A_{bk})^2 \\ &\quad - \frac{r m^2 \|A\|_F^4}{s\eta_{\text{row}}^2} \\ &\geq 1 - \frac{2\|A\|_F^4}{r\eta_{\text{CS}}^2} - \frac{r m^2 \|A\|_F^4}{s\eta_{\text{row}}^2}. \end{aligned}$$

The simplified failure loss again has order

$$\Theta\left(\frac{\|A\|_F^4}{r\eta_{\text{CS}}^2} + \frac{r m^2 \|A\|_F^4}{s\eta_{\text{row}}^2}\right).$$

The finite-Loewner theorem is non-vacuous in the same regime: the displayed loss is below one, $r \rightarrow \infty$, $s/r \rightarrow \infty$, and the charged roundoff/denominator errors vanish. This is a geometry-shaped theorem for the fully computed downstream row-sampling path, but its probability rate is still inherited from Markov/Frobenius estimates rather than from the optimal CountSketch subspace-embedding concentration theorem.

The Lean theorem surface is:

`countSketchSparseComputedPreconditionedFlUniformRowSampleGramWithComputedDen`
`RowGramTwoSidedLoewnerEvent`
`countSketchSparseComputedPreconditionedFlUniformRowSampleGramWithComputedDenRowGramFrobEvent_subset_`
`twoSidedLoewnerEvent`
`countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl_`
`uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_ge_one_sub`
`countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl_`
`uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_ge_one_sub_frobNorm`

The corresponding sample-budget corollaries are also formalized. The sharper exact-coefficient wrapper states that if

$$\frac{2}{r\eta_{\text{CS}}^2} \sum_{j,k=1}^n \sum_{(a,b) \in \mathcal{P}_m} (A_{aj} A_{bk})^2 + \frac{r m^2 \|A\|_F^4}{s\eta_{\text{row}}^2} \leq \delta,$$

then the same fully computed two-sided event has probability at least $1 - \delta$. The first term is the exact CountSketch collision/sign coefficient loss; the second term is the proved downstream uniform-row Frobenius-growth loss for $S_{h,\Omega} A$.

The readable Frobenius-norm wrapper follows by bounding the coefficient sum by $\|A\|_F^4$. If

$$\frac{2\|A\|_F^4}{r\eta_{\text{CS}}^2} + \frac{r m^2 \|A\|_F^4}{s\eta_{\text{row}}^2} \leq \delta,$$

then the same fully computed two-sided event has probability at least $1 - \delta$. With the equal split $\eta_{\text{CS}} = \eta_{\text{row}} = \varepsilon/2$, the exact non-floating-point part of the Loewner radius is ε , so the realized radius is

$$\varepsilon + T_{\text{CS,row}}^{\text{fp}}(h, \Omega, \sigma).$$

The readable sufficient loss condition expands irreducibly to

$$\frac{8\|A\|_F^4}{r\varepsilon^2} + \frac{4r m^2 \|A\|_F^4}{s\varepsilon^2} \leq \delta.$$

The equal-radius exact-coefficient sufficient condition is

$$\frac{8}{r\varepsilon^2} \sum_{j,k=1}^n \sum_{(a,b) \in \mathcal{P}_m} (A_{aj} A_{bk})^2 + \frac{4r m^2 \|A\|_F^4}{s\varepsilon^2} \leq \delta.$$

Lean now exposes these two expanded equal-radius sufficient conditions as checked theorem surfaces, not only as prose simplifications of the $\eta = \varepsilon/2$ wrappers. Thus, for fixed exact $A, m, \varepsilon > 0$ in the small-roundoff regime and varying r, s , the equal-radius failure loss has order

$$\Theta\left(\frac{\|A\|_F^4}{r\varepsilon^2} + \frac{r m^2 \|A\|_F^4}{s\varepsilon^2}\right).$$

Keeping the exact coefficient sum gives the irreducible order

$$\Theta\left(\frac{1}{r\varepsilon^2} \sum_{j,k=1}^n \sum_{(a,b) \in \mathcal{P}_m} (A_{aj} A_{bk})^2 + \frac{r m^2 \|A\|_F^4}{s\varepsilon^2}\right),$$

with the displayed coefficient sum treated as fixed exact input data. The statement is non-vacuous when the displayed loss is below one; it tends to zero when $r \rightarrow \infty$, $s/r \rightarrow \infty$, and the charged roundoff and denominator errors vanish.

The finite-cover CountSketch downstream wrappers replace the Frobenius preprocessing radius η_{CS} by the exact finite-cover Loewner radius

$$\tau_{\text{CS}} = \eta + L(2\rho + \rho^2).$$

Let z_α be the exact finite unit-ball cover vectors. If

$$\sum_{\alpha} \frac{2r^{-1}}{\eta^2} \sum_{(a,b) \in \mathcal{P}_m} ((Az_\alpha)_a (Az_\alpha)_b)^2 + \frac{2r^{-1}}{L^2} \sum_{j,k=1}^n \sum_{(a,b) \in \mathcal{P}_m} (A_{aj} A_{bk})^2 + \frac{r m^2 \|A\|_F^4}{s\eta_{\text{row}}^2} \leq \delta,$$

then the fully computed downstream event has probability at least $1 - \delta$ and radius

$$\tau_{\text{CS}} + \eta_{\text{row}} + T_{\text{CS,row}}^{\text{fp}}(h, \Omega, \sigma).$$

Thus, in the fixed-cover, fixed- A , small-roundoff regime, the irreducible loss order is

$$\Theta\left(\sum_{\alpha} \frac{r^{-1}}{\eta^2} \sum_{(a,b) \in \mathcal{P}_m} ((Az_\alpha)_a (Az_\alpha)_b)^2 + \frac{r^{-1}}{L^2} \sum_{j,k=1}^n \sum_{(a,b) \in \mathcal{P}_m} (A_{aj} A_{bk})^2 + \frac{r m^2 \|A\|_F^4}{s\eta_{\text{row}}^2}\right).$$

This is still finite-cover Markov/Frobenius-derived, not the optimal CountSketch subspace-embedding concentration theorem, but it is a checked end-to-end implementation-facing theorem for the modeled CountSketch plus uniform-row path.

The same downstream finite-cover theorem is now instantiated by the exact product grid. Let $\mathcal{I}$ be a finite exact index set, let $g : \mathcal{I} \rightarrow \mathbb{R}$ cover $[-1, 1]$ with radius $\delta_{\text{grid}} \geq 0$, and assume

$$\sqrt{n} \delta_{\text{grid}} \leq \rho.$$

For $a : \{1, \dots, n\} \rightarrow \mathcal{I}$, set

$$z_a(j) = g(a(j)).$$

The product-grid downstream theorem has the same computed event radius

$$\eta + L(2\rho + \rho^2) + \eta_{\text{row}} + T_{\text{CS, row}}^{\text{fp}}(h, \Omega, \sigma),$$

but its exact target loss is displayed without a supplied cover:

$$\begin{aligned} & \sum_{a: \{1, \dots, n\} \rightarrow \mathcal{I}} \frac{2r^{-1}}{\eta^2} \sum_{(p, q) \in \mathcal{P}_m} ((Az_a)_p (Az_a)_q)^2 \\ & + \frac{2r^{-1}}{L^2} \sum_{j, k=1}^n \sum_{(p, q) \in \mathcal{P}_m} (A_{pj} A_{qk})^2 + \frac{r m^2 \|A\|_F^4}{s \eta_{\text{row}}^2}. \end{aligned}$$

If this loss is at most δ , then the fully computed downstream event has probability at least $1 - \delta$. In the fixed exact-input/grid/threshold regime, the irreducible order is

$$\Theta \left(\sum_{a: \{1, \dots, n\} \rightarrow \mathcal{I}} \frac{r^{-1}}{\eta^2} \sum_{(p, q) \in \mathcal{P}_m} ((Az_a)_p (Az_a)_q)^2 + \frac{r^{-1}}{L^2} \sum_{j, k=1}^n \sum_{(p, q) \in \mathcal{P}_m} (A_{pj} A_{qk})^2 + \frac{r m^2 \|A\|_F^4}{s \eta_{\text{row}}^2} \right).$$

The concrete product-grid denominator theorem selects the same

$$\text{fl_mul}(\text{fl_sqrt}(s), \text{fl_div}(1, \text{fl_sqrt}(r)))$$

routine for $d = \sqrt{s/r}$. The grid and product index set are exact analysis objects; sparse apply, denominator formation, row divisions, and sampled-Gram dot products are the computed non-probability quantities charged by $T_{\text{CS, row}}^{\text{fp}}$.

The corresponding Lean theorem surfaces are:

```
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_cover_ge_one_sub
_sum_coeff_add_frob_add_row
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_cover_ge_one_sub
_delta_of_sum_coeff_add_frob_add_row_budget
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_cover
_ge_one_sub_delta_of_sum_coeff_add_frob_add_row_budget
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_productGrid
_ge_one_sub_sum_coeff_add_frob_add_row
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
```

```

_uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_productGrid
_ge_one_sub_delta_of_sum_coeff_add_frob_add_row_budget
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_productGrid
_ge_one_sub_delta_of_sum_coeff_add_frob_add_row_budget

```

The sample-budget Lean theorem surface is:

```

countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_ge_one_sub
_delta_of_coeff_budget
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_ge_one_sub
_delta_of_frobNorm_budget
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_ge_one_sub
_delta_of_equal_radius_coeff_budget
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_ge_one_sub
_delta_of_equal_radius_frobNorm_budget
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_ge_one_sub
_delta_of_equal_radius_coeff_budget_expanded
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_ge_one_sub
_delta_of_equal_radius_frobNorm_budget_expanded

```

The final concrete-denominator wrappers for this non-injective finite-Loewner path instantiate

$$\widehat{d} = \text{fl_mul}(\text{fl_sqrt}(s), \text{fl_div}(1, \text{fl_sqrt}(r))),$$

so the exact denominator is $d = \sqrt{s/r}$ and the charged denominator radius is

$$\sqrt{s/r} \frac{4u + 3u^2 + u^3}{1 - u}.$$

The generic *WithComputedDen* statements remain reusable infrastructure for other denominator routines; the concrete theorem surface is:

```

countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_ge_one_sub
_delta_of_coeff_budget
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_ge_one_sub
_delta_of_equal_radius_coeff_budget
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_ge_one_sub
_delta_of_frobNorm_budget
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_ge_one_sub
_delta_of_equal_radius_frobNorm_budget
countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl
_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_ge_one_sub

```

`_delta_of_equal_radius_coeff_budget_expanded`
`countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl`
`_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_ge_one_sub`
`_delta_of_equal_radius_frobNorm_budget_expanded`

The collision-free exact hash route now also has a computed sparse-Gram probability endpoint. Assume the exact signs satisfy $\omega_k^2 = 1$ for all input rows and assume γ_m and γ_r are valid. The lemma $b_i \leq m$ lets the single assumption γ_m cover every bucket coefficient γ_{b_i} . Define the exact hash event

$$\mathcal{E}_{\text{fp}} = \left\{ h : \|\widehat{G}(h) - A^T A\|_F \leq T_{\text{CSGram}}^{\text{fp}}(h) \right\},$$

where $\widehat{G}(h)$ is computed by the sparse apply and rounded Gram-dot routine above, while $A^T A$ and $T_{\text{CSGram}}^{\text{fp}}(h)$ are exact mathematical quantities in the theorem statement. If h is injective, then the exact CountSketch table has orthonormal columns and therefore

$$(S_{h,\omega} A)^T (S_{h,\omega} A) = A^T A.$$

Combining this exact identity with the sparse-Gram FP bound gives

$$\{h : h \text{ is injective}\} \subseteq \mathcal{E}_{\text{fp}}.$$

The exact hash-law union bound then yields

$$\Pr_h(\mathcal{E}_{\text{fp}}) \geq \Pr_h(h \text{ is injective}) \geq 1 - \frac{m^2}{r}.$$

This probability statement has no floating-point probability computation in it: the probability is the exact hash law, while the event describes the deterministic rounded sparse apply and rounded Gram dot products. It is a collision-free sufficient route and is non-vacuous when $r > m^2$. It is deliberately weaker than the usual CountSketch concentration theorem.

The same collision-free route is now closed for the full exact CountSketch hash-sign product law. Draw $x = (h, \Omega)$ from `countSketchProbability`, and let $\omega = \text{rademacherSignVector}(\Omega)$. Define

$$\mathcal{E}_{\text{fp}}^{\text{prod}} = \left\{ (h, \Omega) : \|\widehat{G}(h, \Omega) - A^T A\|_F \leq T_{\text{CSGram}}^{\text{fp}}(h, \text{rademacherSignVector}(\Omega)) \right\},$$

where $\widehat{G}(h, \Omega)$ is computed by the same sparse apply and rounded Gram-dot routine using the realized exact Rademacher signs. The Rademacher sign theorem gives $\omega_k^2 = 1$ for every realized sign vector, so

$$\{(h, \Omega) : h \text{ is injective}\} \subseteq \mathcal{E}_{\text{fp}}^{\text{prod}}.$$

The product-law marginal identity for first-coordinate events and the exact hash union bound give the fully realized CountSketch statement

$$\Pr_{h, \Omega}(\mathcal{E}_{\text{fp}}^{\text{prod}}) \geq \Pr_h(h \text{ is injective}) \geq 1 - \frac{m^2}{r}.$$

Again, probabilities and sign/hash draws are exact mathematical laws. The only floating-point terms are the deterministic sparse signed products, bucket accumulations, and Gram dot products inside $\widehat{G}(h, \Omega)$. The bound is non-vacuous on the same collision-free sufficient regime $r > m^2$ and remains weaker than the usual CountSketch concentration/subspace-embedding theorem.

The collision-free CountSketch route now also composes with iid exact uniform row sampling and the computed uniform-row sample-Gram routine. Let $U \in \mathbb{R}^{m \times n}$ be an exact analysis basis

with $U^T U = I_n$, and draw (h, Ω, σ) from the exact product law consisting of the exact CountSketch hash/sign law and s exact iid uniform samples from the r CountSketch output rows. Put

$$V(h, \Omega) = S_{h, \text{rademacherSignVector}(\Omega)} U.$$

On the injective-hash event, $V(h, \Omega)$ has orthonormal columns; hence each exact preconditioned row has squared norm at most one. The row-sampling Loewner radius in the local uniform-row MGF theorem is therefore the explicit quantity

$$L_{\text{CS}} = r.$$

For $\theta > 0$, sample size $s > 0$, and sample failure budget $\delta_{\text{sample}} \geq 0$, assume the exact tail-budget inequality

$$\begin{aligned} n \exp\left(-\theta s \varepsilon + s \frac{\exp(\theta r) - \theta r - 1}{r^2} (r^2 + 1)\right) \\ + n \exp(-\theta s \varepsilon + s(\exp \theta - \theta - 1)(r^2 + 1)) \leq \delta_{\text{sample}}. \end{aligned}$$

Then the exact sample-Gram event

$$-\varepsilon I_n \preceq \text{Gram}_{\text{unif}}(V(h, \Omega), \sigma) - I_n \preceq \varepsilon I_n$$

has exact product-law probability at least

$$1 - \left(\frac{m^2}{r} + \delta_{\text{sample}} \right).$$

The computed sparse CountSketch/sample-Gram endpoint replaces V by

$$\widehat{V}(h, \Omega) = \text{flCountSketchSparseApply}(h, \text{rademacherSignVector}(\Omega), U),$$

and uses a computed uniform denominator $\widehat{d}$ for $\sqrt{s/r}$. Its visible perturbation radius is

$$\begin{aligned} T_{\text{CS, row}}^{\text{fp}}(h, \Omega, \sigma) &= \mathcal{B}_{\text{den}}^{\text{unif}}(fp, \widehat{V}, \widehat{d}, \sigma) \\ &\quad + \mathcal{B}_{\text{basis}}^{\text{unif}}(V, \widehat{V}, \mathcal{B}_{\text{CS}}^{\text{sparse}}(fp, h, \omega, U), \sigma), \end{aligned}$$

where $\omega = \text{rademacherSignVector}(\Omega)$, and the budget symbols are exactly the local Lean definitions

`uniformRowSampleGramComputedDenFullFpPerturbBudget`

`uniformRowSampleGramBasisPerturbBudget`

`countSketchSparseApplyFpAbsBudget`

with the arguments displayed in the formula above. The first term charges the computed denominator, rounded row divisions, and rounded length- s Gram dot products. The second term propagates the already proved sparse CountSketch application entry radii, which charge rounded signed products and rounded bucket accumulation. The final fully computed-object event is

$$-(\varepsilon + T_{\text{CS, row}}^{\text{fp}}) I_n \preceq \widehat{G}_{\text{row}}(h, \Omega, \sigma) - I_n \preceq (\varepsilon + T_{\text{CS, row}}^{\text{fp}}) I_n,$$

where $\widehat{G}_{\text{row}}$ is computed from $\widehat{V}(h, \Omega)$, $\widehat{d}$, the sampled rows, and rounded Gram dot products. The Lean theorem proves

$$\Pr_{h, \Omega, \sigma} \left[-(\varepsilon + T_{\text{CS, row}}^{\text{fp}}) I_n \preceq \widehat{G}_{\text{row}} - I_n \preceq (\varepsilon + T_{\text{CS, row}}^{\text{fp}}) I_n \right] \geq 1 - \left(\frac{m^2}{r} + \delta_{\text{sample}} \right).$$

The actual-input version is now closed for the same collision-free route. Let $A = UC$, where $U \in \mathbb{R}^{m \times q}$ is an exact analysis witness with $U^T U = I_q$ and $C \in \mathbb{R}^{q \times n}$ is an exact right factor. The factors U, C are not computed by this theorem; the implemented non-probability quantity is the sparse *CountSketch* of the actual input

$$\hat{Y}(h, \Omega) = \text{flCountSketchSparseApply}(h, \text{rademacherSignVector}(\Omega), A).$$

With the same computed denominator $\hat{d}$, define

$$\begin{aligned} T_{\text{CS},A}^{\text{fp}}(h, \Omega, \sigma) &= \mathcal{B}_{\text{den}}^{\text{unif}}(fp, \hat{Y}, \hat{d}, \sigma) \\ &\quad + \mathcal{B}_{\text{basis}}^{\text{unif}}(S_{h,\omega} A, \hat{Y}, \mathcal{B}_{\text{CS}}^{\text{sparse}}(fp, h, \omega, A), \sigma), \end{aligned}$$

where the three $\mathcal{B}$ -terms are the same concrete Lean budget definitions displayed above, now instantiated on the actual input matrix A . The final actual-input event is

$$-(\varepsilon A^T A + T_{\text{CS},A}^{\text{fp}} I_n) \preceq \hat{G}_A(h, \Omega, \sigma) - A^T A \preceq \varepsilon A^T A + T_{\text{CS},A}^{\text{fp}} I_n,$$

where $\hat{G}_A$ is computed from $\hat{Y}(h, \Omega)$, $\hat{d}$, the sampled rows, and rounded Gram dot products. Lean proves

$$\Pr_{h, \Omega, \sigma} [\text{the displayed actual-input event}] \geq 1 - \left(\frac{m^2}{r} + \delta_{\text{sample}} \right).$$

Equivalently, if

$$\frac{m^2}{r} + \delta_{\text{sample}} \leq \delta,$$

the same fully computed actual-input event has probability at least $1 - \delta$. This target-budget form is a separate Lean theorem, so later input-sparsity theorem surfaces do not need to manually recombine the collision-free and downstream row-sampling losses. The order of the collision-free preprocessing loss is $\Theta(m^2/r)$, and the combined failure budget is non-vacuous whenever $m^2/r + \delta_{\text{sample}} < 1$. The exact right-congruence proof transfers the orthonormal-basis event to $A^T A$; the floating-point perturbation event is still proved with probability one from the concrete sparse-apply, computed-denominator, row-division, and Gram-dot budgets. There is no additional perturbation-event probability or certificate-existence assumption: the concrete computed-denominator perturbation event is proved with probability one from the displayed local budgets. The probability laws remain exact mathematical laws.

The same actual-input collision-free endpoint is now closed when the realized *CountSketch* signs are stored before sparse application. For storage mode $\rho_{\text{sig}} \in \{\text{mul1}, \text{add0}, \text{sub0}\}$, the computed sign table satisfies

$$|\hat{\omega}_k^{\rho_{\text{sig}}} - \omega_k| \leq u,$$

and the sparse-apply radius is replaced by

$$\begin{aligned} E_{ij}^{\text{CS}, \rho_{\text{sig}}} &= (u + \gamma_{b_i(h)} + u \gamma_{b_i(h)}) \sum_{k: h(k)=i} |\hat{\omega}_k^{\rho_{\text{sig}}}||A_{kj}| \\ &\quad + \sum_{k: h(k)=i} |\hat{\omega}_k^{\rho_{\text{sig}}} - \omega_k| |A_{kj}|. \end{aligned}$$

This term then propagates through the same concrete denominator, sampled-row division, and sampled-Gram dot-product budget, giving a stored-sign radius $T_{\text{CS},A, \rho_{\text{sig}}}^{\text{fp}}$. Lean proves the same two-sided event centered at $A^T A$ with probability at least

$$1 - \left(\frac{m^2}{r} + \delta_{\text{sample}} \right),$$

and the three final wrappers instantiate the sign storage as $\mathbb{f}(\omega_k \cdot 1)$, $\mathbb{f}(\omega_k + 0)$, or $\mathbb{f}(\omega_k - 0)$. The probability loss and non-vacuity condition are unchanged; sign storage affects only the deterministic floating-point radius.

The actual-input collision-free route is also closed for fixed per-bucket stored-sign traversal orders. For each realized hash h , the order $\pi_i(h)$ for bucket i is exact discrete implementation data. Let

$$k_{i,t}^\pi(h) = \text{bucketIndex}(h, i, \pi_i(h)(t)).$$

The stored-sign sparse-apply radius becomes

$$\begin{aligned} E_{ij}^{\text{CS}, \rho_{\text{sig}}, \pi} &= (u + \gamma_{b_i(h)} + u\gamma_{b_i(h)}) \sum_{t=0}^{b_i(h)-1} |\hat{\omega}_{k_{i,t}^\pi(h)}^{\rho_{\text{sig}}}||A_{k_{i,t}^\pi(h),j}| \\ &\quad + \sum_{t=0}^{b_i(h)-1} |\hat{\omega}_{k_{i,t}^\pi(h)}^{\rho_{\text{sig}}} - \omega_{k_{i,t}^\pi(h)}||A_{k_{i,t}^\pi(h),j}|. \end{aligned}$$

This radius is propagated through the same concrete denominator, row-division, and sampled-Gram dot-product budget. The final event is centered at $A^T A$, has probability at least

$$1 - \left(\frac{m^2}{r} + \delta_{\text{sample}} \right),$$

and the three final wrappers instantiate the sign storage as $\mathbb{f}(\omega_k \cdot 1)$, $\mathbb{f}(\omega_k + 0)$, or $\mathbb{f}(\omega_k - 0)$. Exact objects are $U, C, A = UC$, the hash/sign/row-sampling laws, $A^T A$, and the bucket orders. Computed objects are the stored signs, sparse signed products, bucket accumulations in the selected order, denominator, row divisions, and sampled-Gram dot products. In fixed realized data regimes the sampled-basis part has order

$$\Theta \left(\frac{u + \gamma_{b_{\sigma_t}(h)}}{\sqrt{s/r}} \sum_{v=0}^{b_{\sigma_t}(h)-1} |A_{k_{\sigma_t,v}^\pi(h),j}| \right),$$

and the full deterministic radius tends to zero as $u, \gamma_s, \gamma_{b_i(h)} \rightarrow 0$. No additional δ_{comp} , perturbation-event assumption, or certificate-existence assumption is used.

The actual-input collision-free route is also closed for tree-reduced stored-sign bucket accumulation. The tree shape $\mathcal{T}_i(h)$ for each realized bucket is an exact discrete implementation choice, while arithmetic along that tree is computed. If $d_i^{\mathcal{T}}(h)$ is the tree depth, the stored-sign sparse-apply radius becomes

$$\begin{aligned} E_{ij}^{\text{CS}, \rho_{\text{sig}}, \mathcal{T}} &= (u + \gamma_{d_i^{\mathcal{T}}(h)} + u\gamma_{d_i^{\mathcal{T}}(h)}) \sum_{k:h(k)=i} |\hat{\omega}_k^{\rho_{\text{sig}}}||A_{kj}| \\ &\quad + \sum_{k:h(k)=i} |\hat{\omega}_k^{\rho_{\text{sig}}} - \omega_k||A_{kj}|. \end{aligned}$$

This radius is propagated through the same concrete denominator, row-division, and sampled-Gram dot-product budget. The final event is again centered at $A^T A$, has probability at least

$$1 - \left(\frac{m^2}{r} + \delta_{\text{sample}} \right),$$

and adds no probability loss for choosing the exact tree shapes. Balanced bucket trees give the readable sampled-basis order

$$\Theta \left(\frac{u(1 + \log(b_i(h) + 1))}{\sqrt{s/r}} \sum_{k:h(k)=i} |A_{kj}| \right)$$

inside the deterministic radius.

Separately, the non-injective sparse-Gram branch is now closed at the computed finite-Loewner Gram level. For an exact input matrix $A \in \mathbb{R}^{m \times n}$, an exact CountSketch hash h , and exact Rademacher trace Ω , write $\omega = \text{rademacherSignVector}(\Omega)$. The computed non-probability quantity is

$$\hat{G}_{\text{CS}}^{\text{fp}}(h, \Omega) = \text{flCountSketchSparseGramDot}(fp, h, \omega, A),$$

formed from rounded sparse signed products, rounded bucket accumulation, and rounded Gram dot products. The exact reference object is the analysis Gram $A^T A$. For $\eta > 0$, define the realized radius

$$\tau_{\text{CS}}^{\text{fp}}(h, \Omega, A, \eta) = \eta + T_{\text{CSGram}}^{\text{fp}}(h, \Omega, A), \quad T_{\text{CSGram}}^{\text{fp}} = \text{countSketchSparseGramFullFpPerturbBudget}(fp, h, \omega, A).$$

Lean proves the event

$$-\tau_{\text{CS}}^{\text{fp}} I_n \preceq \hat{G}_{\text{CS}}^{\text{fp}} - A^T A \preceq \tau_{\text{CS}}^{\text{fp}} I_n$$

under the exact CountSketch hash/sign product law with probability at least

$$1 - \frac{2}{r\eta^2} \sum_{j=1}^n \sum_{\ell=1}^n \sum_{\substack{a,b=1 \\ a \neq b}}^m (A_{aj} A_{b\ell})^2.$$

The readable Frobenius upper form is

$$\Pr[\text{displayed finite-Loewner event}] \geq 1 - \frac{2\|A\|_F^4}{r\eta^2},$$

and if $A = U$ with $U^T U = I_n$, the orthonormal specialization is

$$\Pr[\text{displayed finite-Loewner event}] \geq 1 - \frac{2n^2}{r\eta^2}.$$

The target-budget wrappers make these non-vacuity conditions direct theorem surfaces: for $0 < \delta < 1$, if the exact coefficient loss, the Frobenius loss $2\|A\|_F^4/(r\eta^2)$, or the orthonormal loss $2n^2/(r\eta^2)$ is at most δ , respectively, then the same computed finite-Loewner event has probability at least $1 - \delta$. The irreducible loss orders are

$$\Theta\left(\frac{\sum_{j,\ell} \sum_{a \neq b} (A_{aj} A_{b\ell})^2}{r\eta^2}\right), \quad \Theta\left(\frac{\|A\|_F^4}{r\eta^2}\right), \quad \Theta\left(\frac{n^2}{r\eta^2}\right)$$

in the coefficient, Frobenius, and orthonormal forms when the displayed numerators are nonzero and η is fixed. These bounds are non-vacuous whenever the chosen loss is < 1 , and the loss tends to zero as $r\eta^2$ grows with fixed exact input.

The product-grid finite-cover branch now also has an orthonormal-basis simplification. Let $U^T U = I_n$, let $\mathcal{I}$ be the exact one-dimensional grid index set, let $g : \mathcal{I} \rightarrow \mathbb{R}$ be the exact grid, and write $z_a(j) = g(a(j))$ for each product-grid vector $a : \{1, \dots, n\} \rightarrow \mathcal{I}$. Since $\|U z_a\|_2^2 = \|z_a\|_2^2$ and $\|U\|_F^2 = n$, the exact product-grid loss for the computed sparse-Gram event reduces to

$$\mathcal{L}_{\text{pg}}^{\text{orth}} = \sum_a \frac{2\|z_a\|_2^4}{r\eta^2} + \frac{2n^2}{rL^2}.$$

Lean proves both the exact-law sparse-Gram theorem and the implementation facing stored-sign theorem with this same probability loss. In the stored-sign path the computed non-probability objects are

the stored realized signs, sparse signed products, bucket accumulations, and rounded Gram dot products; their deterministic contribution remains the already proved $T_{\text{CSGram}, \rho_{\text{sig}}}^{\text{fp}}$ radius. The irreducible order is

$$\Theta\left(\frac{\sum_a \|z_a\|_2^4/\eta^2 + n^2/L^2}{r}\right)$$

for fixed exact grid and positive η, L when the numerator is nonzero, and the bound is non-vacuous whenever $\mathcal{L}_{\text{pg}}^{\text{orth}} < 1$.

The same orthonormal specialization is now composed through downstream uniform row sampling for the ordinary stored-sign CountSketch path. With exact iid downstream row indices, concrete denominator $\hat{d} = \text{fl}(\sqrt{s} \text{fl}(1/\sqrt{r}))$, and $\eta_{\text{row}} > 0$, the final event is centered at I_n and has computed radius

$$\eta + L(2\rho + \rho^2) + \eta_{\text{row}} + T_{\text{CS, row}, \rho_{\text{sig}}}^{\text{fp}}(h, \omega, \sigma; U).$$

Its simplified sufficient loss is

$$\mathcal{L}_{\text{pg, row}}^{\text{orth}} = \sum_a \frac{2\|z_a\|_2^4}{r\eta^2} + \frac{2n^2}{rL^2} + \frac{r(mn)^2}{s\eta_{\text{row}}^2},$$

with order

$$\Theta\left(\frac{\sum_a \|z_a\|_2^4/\eta^2 + n^2/L^2}{r} + \frac{rm^2n^2}{s\eta_{\text{row}}^2}\right)$$

for fixed exact grid and positive thresholds when the numerator is nonzero. The exact objects are U , the product grid, exact hash/sign laws, and exact downstream row laws. The computed non-probability objects are the stored signs, sparse signed products, bucket accumulations, the concrete $\sqrt{s/r}$ denominator, sampled-row divisions, and sampled-Gram dot products. No probability-computation term, perturbation event, auxiliary certificate, or generic probability-loss assumption is introduced; the result is non-vacuous whenever $\mathcal{L}_{\text{pg, row}}^{\text{orth}} < 1$ and the deterministic floating-point radius is small.

The same orthonormal downstream product-grid simplification is now closed for fixed per-bucket traversal orders and for exact supplied bucket trees. For the fixed-order route, the exact order is discrete implementation data and the computed radius charges stored signs, rounded sparse products, accumulation in that chosen order, the concrete $\sqrt{s/r}$ denominator, sampled-row divisions, and sampled-Gram dot products. For the tree-reduced route, the exact tree shapes are discrete implementation data and the computed radius replaces the flat bucket-accumulation factor by the tree-depth $\gamma_{d_i^T(h)}$ factors, under explicit validity hypotheses. Both routes use the same sufficient exact loss

$$\mathcal{L}_{\text{pg, row}}^{\text{orth}} = \sum_a \frac{2\|z_a\|_2^4}{r\eta^2} + \frac{2n^2}{rL^2} + \frac{r(mn)^2}{s\eta_{\text{row}}^2},$$

and the same irreducible order

$$\Theta\left(\frac{\sum_a \|z_a\|_2^4/\eta^2 + n^2/L^2}{r} + \frac{rm^2n^2}{s\eta_{\text{row}}^2}\right).$$

No probability-computation term, perturbation event, auxiliary certificate, or generic probability-loss assumption is introduced; the only change from the ordinary orthonormal downstream theorem is the layout-specific deterministic floating-point radius.

The collision-free row-sampling results above remain collision-free sufficient results. The non-injective sparse-Gram result is Markov/Frobenius-derived; it does not yet prove the standard optimal CountSketch concentration/subspace-embedding theorem or the complete input-sparsity Algorithm 3 theorem.

Lean formalization.

CountSketchHash
rowGram_preconditionRows_eq_of_left_hasOrthonormalColumns
preconditionRows_hasOrthonormalColumns_of_left_hasOrthonormalColumns
countSketchRows
countSketchHashProbability
countSketchHashPairCollision
countSketchHashPairNoCollision
countSketchProbability
uniformRowTraceProbability_eventProb_pair_collision_eq_inv
countSketchHashProbability_eventProb_pairCollision_eq_inv
FiniteProbability.expectationReal_indicator_eq_eventProb
countSketchHashProbability_expectationReal_pairCollisionIndicator_eq_inv
countSketchProbability_expectationReal_pairCollisionIndicator_eq_inv
countSketchHashProbability_eventProb_pairNoCollision_eq_one_sub_inv
CountSketchDistinctPair
rademacherTraceProbability_expectationReal_eq_zero_of_flip_neg
rademacherTraceProbability_expectationReal_sign_four_eq_zero_of_left_unpaired
rademacherTraceProbability_expectationReal_sign_four_eq_one_same_pair
rademacherTraceProbability_expectationReal_sign_four_eq_one_reversed_pair
rademacherTraceProbability_expectationReal_sign_pair_mul_sign_pair_eq
rademacherTraceProbability_expectationReal_sq_sum_distinctPair_mul_sign_pair_eq
countSketchDistinctPairSwap
countSketchDistinctPair_fourKernel_sum_le_two_sum_sq
rowGram_preconditionRows_countSketchRows_sub_rowGram_eq_distinctPair_sum
rademacherTraceProbability_expectationReal_countSketchRows_rowGram_entry_error_sq_eq
rademacherTraceProbability_expectationReal_countSketchRows_rowGram_entry_error_sq_le
countSketchHashProbability_expectationReal_collision_coeff_sq_eq_inv_mul
countSketchProbability_expectationReal_rowGram_entry_error_sq_le
countSketchProbability_expectationReal_rowGram_frob_error_sq_le
countSketchProbability_eventProb_rowGram_frob_error_le_ge_one_sub
countSketchRowGramFrobErrorEvent
countSketchFlSparseGramDotRowGramFrobErrorEvent
countSketchRowGramFrobErrorEvent_subset_flSparseGramDotRowGramFrobErrorEvent
countSketchProbability_eventProb_flSparseGramDot_rowGram_frob_error_le_ge_one_sub
countSketchDistinctPair_gramCoeffSq_sum_le_frobNormSqRect_sq
countSketchProbability_eventProb_flSparseGramDot_rowGram_frob_error_le_ge_one_sub_frobNorm
frobNormSqRect_eq_nat_of_hasOrthonormalColumns
countSketchProbability_eventProb_flSparseGramDot_rowGram_frob_error_le_ge_one_sub_orthonormal

countSketchFlSparseGramDotRowGramTwoSidedLoewnerEvent
countSketchFlSparseGramDotRowGramFrobErrorEvent_subset_twoSidedLoewnerEvent
countSketchProbability_eventProb_flSparseGramDot_rowGram_twoSidedLoewner_ge_one_sub
countSketchProbability_eventProb_flSparseGramDot_rowGram_twoSidedLoewner_ge_one_sub_frobNorm
countSketchProbability_eventProb_flSparseGramDot_rowGram_twoSidedLoewner_ge_one_sub_orthonormal
countSketchProbability_eventProb_flSparseGramDot_rowGram_twoSidedLoewner_ge_one_sub_delta_of_
coeff_budget
countSketchProbability_eventProb_flSparseGramDot_rowGram_twoSidedLoewner_ge_one_sub_delta_of_
frobNorm_budget
countSketchProbability_eventProb_flSparseGramDot_rowGram_twoSidedLoewner_ge_one_sub_delta_of_
orthonormal_budget
countSketchProbability_eventProb_rowGram_twoSidedLoewner_productGrid_ge_one_sub_sum_gridNorm_add_
nat_orthonormal
countSketchProbability_eventProb_rowGram_twoSidedLoewner_productGrid_ge_one_sub_delta_of_sum_
gridNorm_add_nat_orthonormal_budget
countSketchProbability_eventProb_flSparseGramDot_rowGram_twoSidedLoewner_productGrid_ge_one_sub_
sum_gridNorm_add_nat_orthonormal
countSketchProbability_eventProb_flSparseGramDot_rowGram_twoSidedLoewner_productGrid_ge_one_sub_
delta_of_sum_gridNorm_add_nat_orthonormal_budget
countSketchProbability_eventProb_flSparseGramDotWithStoredSign_rowGram_twoSidedLoewner_
productGrid_ge_one_sub_sum_gridNorm_add_nat_orthonormal
countSketchProbability_eventProb_flSparseGramDotWithStoredSign_rowGram_twoSidedLoewner_
productGrid_ge_one_sub_delta_of_sum_gridNorm_add_nat_orthonormal_budget
countSketchProbability_eventProb_flSparseGramDotWithFlStoredSign_rowGram_twoSidedLoewner_
productGrid_ge_one_sub_delta_of_sum_gridNorm_add_nat_orthonormal_budget
countSketchProbability_eventProb_flSparseGramDotWithFlStoredSignAddZeroRight_rowGram_
twoSidedLoewner_productGrid_ge_one_sub_delta_of_sum_gridNorm_add_nat_orthonormal_budget
countSketchProbability_eventProb_flSparseGramDotWithFlStoredSignSubZeroRight_rowGram_
twoSidedLoewner_productGrid_ge_one_sub_delta_of_sum_gridNorm_add_nat_orthonormal_budget
countSketchUniformRowTraceProbability_eventProb_sparseStoredSignComputedPreconditioned_fl_
uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_productGrid_ge_one_sub_delta_of_
sum_gridNorm_add_nat_add_row_orthonormal_budget
countSketchUniformRowTraceProbability_eventProb_sparseStoredSignComputedPreconditioned_fl_
uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_productGrid_ge_one_sub_
delta_of_sum_gridNorm_add_nat_add_row_orthonormal_budget
countSketchUniformRowTraceProbability_eventProb_sparseFlStoredSignComputedPreconditioned_fl_
uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_productGrid_ge_one_sub_
delta_of_sum_gridNorm_add_nat_add_row_orthonormal_budget
countSketchUniformRowTraceProbability_eventProb_sparseFlStoredSignSubZeroRightComputedPreconditioned_
fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_productGrid_ge_one_sub_
delta_of_sum_gridNorm_add_nat_add_row_orthonormal_budget
countSketchUniformRow_productGrid_orthonormal_coeff_add_frob_add_row_budget

countSketchUniformRowTraceProbability_eventProb_sparseStoredSignPermutedComputedPreconditioned_
 fl_uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_productGrid_ge_one_sub_delta_
 of_sum_gridNorm_add_nat_add_row_orthonormal_budget

 countSketchUniformRowTraceProbability_eventProb_sparseStoredSignPermutedComputedPreconditioned_
 fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_productGrid_ge_one_sub_
 delta_of_sum_gridNorm_add_nat_add_row_orthonormal_budget

 countSketchUniformRowTraceProbability_eventProb_sparseFlStoredSignPermutedComputedPreconditioned_
 fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_productGrid_ge_one_sub_
 delta_of_sum_gridNorm_add_nat_add_row_orthonormal_budget

 countSketchUniformRowTraceProbability_eventProb_sparseFlStoredSignAddZeroRightPermutedComputedPreconditioned_
 fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_productGrid_ge_one_sub_
 delta_of_sum_gridNorm_add_nat_add_row_orthonormal_budget

 countSketchUniformRowTraceProbability_eventProb_sparseFlStoredSignSubZeroRightPermutedComputedPreconditioned_
 fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_productGrid_ge_one_sub_
 delta_of_sum_gridNorm_add_nat_add_row_orthonormal_budget

 countSketchUniformRowTraceProbability_eventProb_sparseStoredSignTreeComputedPreconditioned_fl_
 uniformRowSampleGramDotWithComputedDen_rowGram_twoSidedLoewner_productGrid_ge_one_sub_delta_of_
 sum_gridNorm_add_nat_add_row_orthonormal_budget

 countSketchUniformRowTraceProbability_eventProb_sparseStoredSignTreeComputedPreconditioned_fl_
 uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_productGrid_ge_one_sub_
 delta_of_sum_gridNorm_add_nat_add_row_orthonormal_budget

 countSketchUniformRowTraceProbability_eventProb_sparseFlStoredSignTreeComputedPreconditioned_fl_
 uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_productGrid_ge_one_sub_
 delta_of_sum_gridNorm_add_nat_add_row_orthonormal_budget

 countSketchUniformRowTraceProbability_eventProb_sparseFlStoredSignAddZeroRightTreeComputedPreconditioned_
 fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_productGrid_ge_one_sub_
 delta_of_sum_gridNorm_add_nat_add_row_orthonormal_budget

 countSketchUniformRowTraceProbability_eventProb_sparseFlStoredSignSubZeroRightTreeComputedPreconditioned_
 fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_rowGram_twoSidedLoewner_productGrid_ge_one_sub_
 delta_of_sum_gridNorm_add_nat_add_row_orthonormal_budget

 countSketchHashInjectiveEvent

 countSketchHash_forall_pairNoCollision_iff_injective

 countSketchHashProbability_eventProb_injective_ge_one_sub_pair_sum

 countSketchDistinctPairBudget_le_square_inv

 countSketchHashProbability_eventProb_injective_ge_one_sub_square_inv

 countSketchRows_preconditionRows_entry

 CountSketchBucket

 countSketchBucketSize

 countSketchBucketIndex

 countSketchBucketExactTerm

 fl_countSketchBucketProduct

 fl_countSketchSparseApplyEntry

 countSketchRows_preconditionRows_bucket_sum_eq

 fl_countSketchSparseApplyEntry_error_bound

 countSketchBucketAbsSum_le_column_abs_sum

fl_countSketchSparseApplyEntry_error_bound_of_abs_sign_le_one
 fl_countSketchSparseApply
 fl_countSketchSparseApply_entry
 fl_countSketchSparseApply_entry_error_bound
 fl_countSketchSparseApply_entry_error_bound_of_abs_sign_le_one
 rowSketch_abs_perturbed_le_of_abs_error
 rowSketch_abs_perturbed_mul_sum_le_of_abs_error
 rowSketchGram_entry_abs_error_bound_exact_of_entrywise
 rowSketchGram_frob_abs_error_bound_exact_of_entrywise
 fl_rowSketchGramDot
 rowSketchGramDotRoundoffExactBudget
 rowSketchGramAbsPerturbExactBudget
 rowSketchGramFullAbsFpExactBudget
 fl_rowSketchGramDot_roundoff_bound_of_abs_error
 fl_rowSketchGramDot_abs_perturb_bound_exact
 countSketchSparseApplyEntryFpAbsBudget
 countSketchSparseApplyFpAbsBudget
 fl_countSketchSparseGramDot
 countSketchSparseGramDotRoundoffBudget
 countSketchSparseGramApplyPerturbBudget
 countSketchSparseGramFullFpPerturbBudget
 fl_countSketchSparseGramDot_perturb_bound
 countSketchBucketSize_le
 fl_countSketchSparseGramDot_rowGram_perturb_bound_of_hash_injective
 countSketchHashFlSparseGramDotRowGramPerturbEvent
 countSketchHashInjectiveEvent_subset_flSparseGramDot_rowGram_perturbEvent
 countSketchHashProbability_eventProb_flSparseGramDot_rowGram_perturb_ge_one_sub_square_inv
 countSketchFlSparseGramDotRowGramPerturbEvent
 countSketchProbability_injectiveFst_subset_flSparseGramDot_rowGram_perturbEvent
 countSketchProbability_eventProb_flSparseGramDot_rowGram_perturb_ge_one_sub_square_inv
 countSketchRows_hasOrthonormalColumns_of_hash_injective
 rowGram_preconditionRows_countSketchRows_eq_of_hash_injective
 countSketchRows_preconditionRows_hasOrthonormalColumns_of_hash_injective
 rowNormSq_le_one_of_hasOrthonormalColumns
 countSketchUniformRowTraceProbability
 countSketchUniformRowSampleGramTwoSidedEvent
 countSketchUniformRowTraceProbability_eventProb_uniformRowSampleGram_two_sided_finiteLoewnerLe_ge_one_sub_square_inv_add_delta
 countSketchComputedPreconditionedFlUniformRowSampleGramWithComputedDenTwoSidedEvent
 countSketchComputedPreconditionedFlUniformRowPerturbWithComputedDenEvent

countSketchSparseComputedPreconditionedBasis
 countSketchSparseUniformRowComputedDenPerturbBudget
 countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl_
 uniformRowComputedDenPerturbEvent_eq_one
 countSketchUniformRowTraceProbability_eventProb_computedPreconditioned_fl_
 uniformRowSampleGramDotWithComputedDen_two_sided_finiteLoewnerLe_ge_one_sub_of_exact_event
 countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl_
 uniformRowSampleGramDotWithComputedDen_two_sided_finiteLoewnerLe_ge_one_sub_square_inv_add_delta
 countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_fl_
 _uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_two_sided_finiteLoewnerLe_ge_one_sub
 _square_inv_add_delta
 uniformRowSampleGram_countSketchRectFactoredInput_error_eq_rightGramCongruence_error
 countSketchUniformRowFactoredInputSampleGramTwoSidedEvent
 countSketchUniformRowSampleGramTwoSidedEvent_subset_factoredInput
 countSketchUniformRowTraceProbability_eventProb_uniformRowFactoredInputSampleGram_two_sided_
 finiteLoewnerLe_ge_one_sub_square_inv_add_delta
 countSketchComputedPreconditionedFactoredInputFlUniformRowSampleGramWithComputedDenTwoSidedEvent
 countSketchUniformRowTraceProbability_eventProb_computedPreconditioned_factoredInput_fl_
 uniformRowSampleGramDotWithComputedDen_two_sided_finiteLoewnerLe_ge_one_sub_of_exact_event
 countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_factoredInput_fl_
 uniformRowSampleGramDotWithComputedDen_two_sided_finiteLoewnerLe_ge_one_sub_square_inv_add_delta
 countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_factoredInput_fl_
 uniformRowSampleGramDotWithComputedDen_two_sided_finiteLoewnerLe_ge_one_sub_delta_of_square_inv_
 budget
 countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_factoredInput_fl_
 _uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_two_sided_finiteLoewnerLe_ge_one_sub
 _square_inv_add_delta
 countSketchUniformRowTraceProbability_eventProb_sparseComputedPreconditioned_factoredInput_fl_
 _uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_two_sided_finiteLoewnerLe_ge_one_sub
 _delta_of_square_inv_budget
 rademacherTraceProbability_expectationReal_countSketchRows_entry_mul_eq
 rademacherTraceProbability_expectationReal_countSketchRows_rowGram_eq
 countSketchProbability_expectationReal_rowGram_eq
 countSketchProbability_expectationReal_rowGram_eq_id_of_hasOrthonormalColumns

Corollary 191 (Exact signed-mixing sample-Gram composition). *Let $G \in \mathbb{R}^{r \times m}$ and $U \in \mathbb{R}^{m \times n}$ be exact matrices with $r > 0$, $n > 0$, $G^T G = I_m$, and $U^T U = I_n$. Let $\omega \in \{\pm 1\}^m$ be exact Rademacher signs, define $\Pi_\omega(i, k) = G_{ik}\omega_k$, and after this preprocessing draw s exact iid uniform row samples from the r rows of $\Pi_\omega U$. Assume the exact entry-square cap*

$$G_{ik}^2 \leq \alpha^2, \quad \alpha > 0,$$

and fix $B > 0$, $\theta > 0$, $\varepsilon \in \mathbb{R}$, and $s > 0$. If

$$L = rnB^2,$$

and

$$\begin{aligned}
T_{\text{sm}}(B, \alpha, \theta, \varepsilon) &= 2rn \exp\left(-\frac{B^2}{2\alpha^2}\right) \\
&\quad + n \exp\left(-\theta s\varepsilon + s \frac{\exp(\theta rn B^2) - \theta rn B^2 - 1}{r^2 n^2 B^4} (r^2 n^2 B^4 + 1)\right) \\
&\quad + n \exp\left(-\theta s\varepsilon + s (\exp(\theta) - \theta - 1) (r^2 n^2 B^4 + 1)\right),
\end{aligned}$$

then the exact product law over signs and uniform row samples satisfies

$$\mathbb{P}\{-\varepsilon I_n \preceq \text{Gram}_s(\Pi_\omega U) - I_n \preceq \varepsilon I_n\} \geq 1 - T_{\text{sm}}(B, \alpha, \theta, \varepsilon).$$

Equivalently, the Lean theorem permits separate budgets δ_{pre} and δ_{samp} that dominate the first line and the two matrix-Chernoff lines, respectively, and proves the bound $1 - (\delta_{\text{pre}} + \delta_{\text{samp}})$.

In the asymptotic regime where r, n, s may vary, α, B, θ are positive parameters, and the displayed exponents are kept in a non-vacuous negative range, the failure expression has order

$$\Theta\left(rne^{-B^2/(2\alpha^2)} + ne^{-\theta s\varepsilon + s \frac{e^{\theta rn B^2} - \theta rn B^2 - 1}{r^2 n^2 B^4} (r^2 n^2 B^4 + 1)} + ne^{-\theta s\varepsilon + s(e^\theta - \theta - 1)(r^2 n^2 B^4 + 1)}\right).$$

The non-vacuity check is explicit: the probability statement is meaningful when this total failure term is below 1. This result is exact probability and exact arithmetic. The sampling laws are exact by project convention, while applying/storing G , forming $\Pi_\omega U$, and forming the sample Gram in floating point remain separate computed-quantity obligations.

Lean formalization.

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signedMixingRows_preconditionRows_hasOrthonormalColumns
signedMixingUniformRowTraceProbability
signedMixingUniformRowSampleGramTwoSidedEvent
signedMixingUniformRowTraceProbability_eventProb_uniformRowSampleGram_two_sided_finiteLoewnerLe_
ge_one_sub_delta_of_entry_sq_le_uniform

```

Corollary 192 (Computed finite signed-mixing preconditioner formation). *Let $G \in \mathbb{R}^{r \times m}$ be an exact deterministic mixing table and let $\omega \in \{\pm 1\}^m$ be the exact realized sign vector. The exact analysis object is*

$$\Pi_\omega(i, j) = G_{ij}\omega_j, \quad \text{equivalently} \quad \Pi_\omega = G \text{diag}(\omega).$$

The Lean theorem first proves this equivalence exactly:

$$\text{preconditionRows}(G, \text{diag}(\omega)) = \text{signedMixingRows}(G, \omega).$$

The computable floating-point quantity in this proposition is the stored matrix

$$\hat{\Pi}_\omega = \text{fl_matMul}(G, \text{diag}(\omega)),$$

formed by rounded matrix multiplication. The probabilities and the Rademacher law are exact by project convention; the displayed bound charges only this non-probability preconditioner-formation arithmetic. If γ_m is valid for the m -term product, then every entry satisfies

$$|\hat{\Pi}_{\omega, ij} - G_{ij}\omega_j| \leq \gamma_m \sum_{k=1}^m |G_{ik}| |\text{diag}(\omega)_{kj}|.$$

Since the diagonal has only one nonzero entry and $|\omega_j| = 1$, this displayed sum has the readable exact-arithmetic simplification

$$\gamma_m |G_{ij}|.$$

Thus, for fixed exact G and exact signs, the entrywise formation error has order

$$\Theta(\gamma_m |G_{ij}|),$$

and the non-vacuity check is immediate: it tends to zero as $\gamma_m \rightarrow 0$. The more general Lean constructor `ComputedPreconditioner.flSignedMixing` also accepts computed certificates for G and for the sign vector, so their storage/generation errors are included before the rounded product is formed.

Lean formalization.

```
preconditionRows_diagMatrix_eq_signedMixingRows
ComputedPreconditioner.flSignedMixing
ComputedPreconditioner.flSignedMixingExactFactors
ComputedPreconditioner.flSignedMixingExactFactors_entry_error_bound
```

Corollary 193 (Computed finite signed-mixing sample Gram). *In the exact signed-mixing setting above, assume additionally that the floating-point model has valid γ_m for length- m products and valid γ_s for length- s Gram dot products. The probability laws for ω and for the uniform row trace $S = (S_1, \dots, S_s)$ are exact. All non-probability quantities below are computed.*

For each realized sign vector define the exact analysis preconditioner $\Pi_\omega = G \text{diag}(\omega)$. The implementation first forms

$$\widehat{\Pi}_\omega = \text{fl}(G \text{diag}(\omega)),$$

with exact-factor componentwise certificate

$$e_{ia}^\Pi = \gamma_m \sum_{b=1}^m |G_{ib}| |\text{diag}(\omega)_{ba}| = \gamma_m |G_{ia}|.$$

It then computes

$$\widehat{V}_\omega = \text{fl}(\widehat{\Pi}_\omega U).$$

The fully expanded entrywise budget used by Lean for this second product is

$$E_{ij}^V = \gamma_m \sum_{a=1}^m |\widehat{\Pi}_{\omega,ia}| |U_{aj}| + \sum_{a=1}^m e_{ia}^\Pi |U_{aj}|.$$

Thus $|\widehat{V}_{\omega,ij} - (\Pi_\omega U)_{ij}| \leq E_{ij}^V$, and no existence assumption for this perturbation is used.

Let $d = \sqrt{s/r}$. For a sampled row trace S , define the exact rescaled sketches

$$B_{tj} = \frac{(\Pi_\omega U)_{S_t j}}{d}, \quad \widetilde{B}_{tj} = \frac{\widehat{V}_{\omega, S_t j}}{d}.$$

For the exact-denominator row-scaling path, the proved sample-dependent FP radius is

$$\begin{aligned} \tau_{\text{sm}}^{\text{fp}}(\omega, S) = & \left((2u + u^2) + \gamma_s (1 + u)^2 \right) \left\| \left[\sum_{t=1}^s |\widetilde{B}_{tj}| |\widetilde{B}_{tk}| \right]_{j,k=1}^n \right\|_F^n \\ & + \left\| \left[\sum_{t=1}^s \left(\frac{E_{S_t j}^V}{d} |\widetilde{B}_{tk}| + |B_{tj}| \frac{E_{S_t k}^V}{d} \right) \right]_{j,k=1}^n \right\|_F^n. \end{aligned}$$

The first line charges rounded row divisions and rounded length- s Gram dot products after $\widehat{V}_\omega$ has been formed. The second line charges the exact sampled-Gram drift from $\widehat{V}_\omega$ back to $\Pi_\omega U$.

If the denominator is itself computed, let $\widehat{d} \neq 0$ satisfy $|\widehat{d} - d| \leq \eta_d$. Define

$$\widehat{B}_{tj} = \text{fl}\left(\frac{\widehat{V}_{\omega, S_{tj}}}{\widehat{d}}\right), \quad E_{tj}^d = \left|\frac{\widehat{V}_{\omega, S_{tj}}}{\widehat{d}}\right|u + \frac{|\widehat{V}_{\omega, S_{tj}}|\eta_d}{|\widehat{d}|d}.$$

The computed-denominator radius is

$$\begin{aligned} \tau_{\text{sm}, \text{den}}^{\text{fp}}(\omega, S) = & \left\| \left[\sum_{t=1}^s \left(E_{tj}^d |\widehat{B}_{tk}| + |\widehat{B}_{tj}| E_{tk}^d \right) \right]_{j,k=1}^n \right\|_F \\ & + \left\| \left[\gamma_s \sum_{t=1}^s |\widehat{B}_{tj}| |\widehat{B}_{tk}| \right]_{j,k=1}^n \right\|_F \\ & + \left\| \left[\sum_{t=1}^s \left(\frac{E_{S_{tj}}^V}{d} |\widehat{B}_{tk}| + |B_{tj}| \frac{E_{S_{tk}}^V}{d} \right) \right]_{j,k=1}^n \right\|_F. \end{aligned}$$

Consequently, with the same exact preprocessing and sampling budgets δ_{pre} and δ_{samp} as the exact signed-mixing sample-Gram theorem,

$$\mathbb{P}\left\{ -(\varepsilon + \tau_{\text{sm}, \text{den}}^{\text{fp}}(\omega, S))I_n \preceq \widehat{G}_{\omega, S} - I_n \preceq (\varepsilon + \tau_{\text{sm}, \text{den}}^{\text{fp}}(\omega, S))I_n \right\} \geq 1 - (\delta_{\text{pre}} + \delta_{\text{samp}}),$$

where $\widehat{G}_{\omega, S}$ is the Gram matrix computed from $\widehat{V}_\omega$ using the computed denominator $\widehat{d}$, rounded row divisions, and rounded length- s dot products. There is no additional δ_{comp} : the computed perturbation event is proved with probability one from the displayed concrete budget.

For the final concrete denominator path, Lean instantiates

$$\widehat{d} = \text{fl}_{\text{mul}}(\text{fl}_{\text{sqrt}}(s), \text{fl}_{\text{div}}(1, \text{fl}_{\text{sqrt}}(r))), \quad d = \sqrt{s/r},$$

with

$$\eta_d = \sqrt{s/r} \frac{4u + 3u^2 + u^3}{1 - u}.$$

Thus the displayed $\tau_{\text{sm}, \text{den}}^{\text{fp}}$ has no hidden denominator variable in the concrete theorem surface; the older arbitrary- $\widehat{d}$ statement remains only reusable infrastructure for other denominator routines.

For fixed G, U, r, m, n, s and sampled rows, in the regime $u, \gamma_m, \gamma_s, \eta_d/d \rightarrow 0$ with $|\widehat{d}|$ bounded away from zero,

$$\tau_{\text{sm}, \text{den}}^{\text{fp}} = \Theta\left(\left(u + \gamma_s + \frac{\eta_d}{d} \right) \left\| \left[\sum_t |\widehat{B}_{tj}| |\widehat{B}_{tk}| \right]_{jk} \right\|_F + \gamma_m \left\| \left[\sum_t \left(\frac{C_{S_{tj}}}{d} |\widehat{B}_{tk}| + |B_{tj}| \frac{C_{S_{tk}}}{d} \right) \right]_{jk} \right\|_F \right),$$

where the primitive coefficient

$$C_{ij} = \sum_{a=1}^m (|\Pi_{\omega, ia}| + |G_{ia}|) |U_{aj}|$$

is the first-order expansion of E_{ij}^V/γ_m . The non-vacuity check is explicit: for fixed exact data, the FP radius tends to 0 as $u, \gamma_m, \gamma_s, \eta_d \rightarrow 0$, and the probability lower bound remains the exact signed-mixing lower bound.

Lean formalization.

```

signedMixingComputedLeftPreconditionedBasis
signedMixingExactFactorPreconditioner
signedMixingComputedLeftUniformRowPerturbBudget
signedMixingComputedLeftUniformRowComputedDenPerturbBudget
signedMixingUniformRowTraceProbability_eventProb_computedLeftPreconditioned_fl_
uniformRowPerturbEvent_eq_one
signedMixingUniformRowTraceProbability_eventProb_computedLeftPreconditioned_fl_
uniformRowComputedDenPerturbEvent_eq_one
signedMixingUniformRowTraceProbability_eventProb_exactFactorComputedLeftPreconditioned_fl_
uniformRowSampleGramDot_two_sided_finiteLoewnerLe_ge_one_sub_delta_of_entry_sq_le_uniform
signedMixingUniformRowTraceProbability_eventProb_exactFactorComputedLeftPreconditioned_fl_
uniformRowSampleGramDotWithComputedDen_two_sided_finiteLoewnerLe_ge_one_sub_delta_of_entry_sq_le_
uniform
signedMixingUniformRowTraceProbability_eventProb_exactFactorComputedLeftPreconditioned_fl_
uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_two_sided_finiteLoewnerLe_ge_one_sub_delta_of_
entry_sq_le_uniform

```

Corollary 194 (Computed actual-input finite signed-mixing sample Gram). *Let r, m, q, n, s be positive dimensions. Let $G \in \mathbb{R}^{r \times m}$ be an exact deterministic rectangular isometry, $G^T G = I_m$, satisfying the exact entry cap $G_{ia}^2 \leq \alpha^2$. Let the exact input matrix $A \in \mathbb{R}^{m \times n}$ admit the exact analysis factorization*

$$A = UC, \quad U \in \mathbb{R}^{m \times q}, \quad U^T U = I_q, \quad C \in \mathbb{R}^{q \times n}.$$

The matrices U and C are exact analysis witnesses in this theorem; the algorithmic computation below is performed with the actual input A . The Rademacher signs ω and the uniform row trace $S = (S_1, \dots, S_s)$ are sampled from exact probability laws. Every non-probability quantity displayed next is computed in floating point.

Define the exact signed-mixing preconditioner

$$\Pi_\omega = G \operatorname{diag}(\omega),$$

and compute

$$\hat{\Pi}_{\omega, ia} = \mathbb{E} \left(\sum_{b=1}^m G_{ib} \operatorname{diag}(\omega)_{ba} \right).$$

The exact-factor preconditioner-formation radius is

$$e_{ia}^\Pi = \gamma_m \sum_{b=1}^m |G_{ib}| |\operatorname{diag}(\omega)_{ba}| = \gamma_m |G_{ia}|.$$

The implementation then computes the preconditioned actual input

$$\hat{Y}_\omega = \mathbb{E}(\hat{\Pi}_\omega A).$$

Since $A_{aj} = \sum_{\ell=1}^q U_{a\ell} C_{\ell j}$, the fully expanded entrywise product budget is

$$\begin{aligned} E_{ij}^A &= \gamma_m \sum_{a=1}^m |\hat{\Pi}_{\omega,ia}| |A_{aj}| + \sum_{a=1}^m e_{ia}^\Pi |A_{aj}| \\ &= \gamma_m \sum_{a=1}^m \left(\left| \text{fl} \left(\sum_{b=1}^m G_{ib} \text{diag}(\omega)_{ba} \right) \right| + |G_{ia}| \right) \left| \sum_{\ell=1}^q U_{a\ell} C_{\ell j} \right|. \end{aligned}$$

Lean proves

$$|\hat{Y}_{\omega,ij} - (\Pi_\omega A)_{ij}| \leq E_{ij}^A;$$

this is an instantiated arithmetic bound, not a perturbation-existence assumption.

Set $d = \sqrt{s/r}$. For the sampled rows define

$$Y_{tj} = \frac{(\Pi_\omega A)_{S_{tj}}}{d}, \quad \tilde{Y}_{tj} = \frac{\hat{Y}_{\omega,S_{tj}}}{d}.$$

Let $\hat{d} \neq 0$ be the computed row-scale denominator with $|\hat{d} - d| \leq \eta_d$, and define

$$\hat{B}_{tj} = \text{fl} \left(\frac{\hat{Y}_{\omega,S_{tj}}}{\hat{d}} \right), \quad E_{tj}^d = \left| \frac{\hat{Y}_{\omega,S_{tj}}}{\hat{d}} \right| u + \frac{|\hat{Y}_{\omega,S_{tj}}| \eta_d}{|\hat{d}| d}.$$

In the final concrete denominator theorem this $\hat{d}$ is not an abstract input. It is

$$\hat{d} = \text{fl}_{\text{mul}}(\text{fl}_{\text{sqrt}}(s), \text{fl}_{\text{div}}(1, \text{fl}_{\text{sqrt}}(r))), \quad d = \sqrt{s/r}, \quad \eta_d = \sqrt{s/r} \frac{4u + 3u^2 + u^3}{1 - u}.$$

The formula below is therefore already irreducible after this substitution: the only remaining floating-point parameters are u, γ_m, γ_s and the explicitly displayed dimensions and exact data. The computed-denominator perturbation radius used by the theorem is

$$\begin{aligned} \tau_A^{\text{fp}}(\omega, S) &= \left\| \left[\sum_{t=1}^s \left(E_{tj}^d |\hat{B}_{tk}| + |\tilde{Y}_{tj}| E_{tk}^d \right) \right]_{j,k=1}^n \right\|_F \\ &\quad + \left\| \left[\gamma_s \sum_{t=1}^s |\hat{B}_{tj}| |\hat{B}_{tk}| \right]_{j,k=1}^n \right\|_F \\ &\quad + \left\| \left[\sum_{t=1}^s \left(\frac{E_{S_{tj}}^A}{d} |\tilde{Y}_{tk}| + |Y_{tj}| \frac{E_{S_{tk}}^A}{d} \right) \right]_{j,k=1}^n \right\|_F. \end{aligned}$$

The first line charges the computed denominator and rounded row divisions, the second line charges rounded length- s Gram dot products, and the third line charges the preconditioner/input matrix-product error.

Let $\hat{\mathcal{G}}_{\omega,S}$ be the matrix obtained by rounded Gram dot products from the sampled rows $\hat{B}_{t,*}$. With the same exact preprocessing and sampling failures δ_{pre} and δ_{samp} as the exact signed-mixing sample-Gram theorem,

$$\mathbb{P} \left\{ -\varepsilon A^T A - \tau_A^{\text{fp}}(\omega, S) I_n \preceq \hat{\mathcal{G}}_{\omega,S} - A^T A \preceq \varepsilon A^T A + \tau_A^{\text{fp}}(\omega, S) I_n \right\} \geq 1 - (\delta_{\text{pre}} + \delta_{\text{samp}}).$$

There is no additional δ_{comp} : the computed perturbation event is proved with probability one from the displayed radius. Equivalently, if $\delta_{\text{pre}} + \delta_{\text{samp}} \leq \delta$, the same fully computed event has probability at least $1 - \delta$; this total-budget form is now a separate Lean theorem, so later theorem surfaces do not need to recombine the two exact-probability losses manually.

For fixed exact G, U, C, r, m, q, n, s and fixed sampled rows, assume $|\hat{d}|$ is bounded below by a positive constant multiple of d . In the regime $u, \gamma_m, \gamma_s, \eta_d/d \rightarrow 0$,

$$\tau_A^{\text{fp}} = \Theta \left(\left(u + \gamma_s + \frac{\eta_d}{d} \right) \left\| \left[\sum_t |\hat{B}_{tj}| |\hat{B}_{tk}| \right]_{jk} \right\|_F + \gamma_m \left\| \left[\sum_t \left(\frac{K_{Stj}}{d} |\tilde{Y}_{tk}| + |Y_{tj}| \frac{K_{Stk}}{d} \right) \right]_{jk} \right\|_F \right),$$

where the irreducible coefficient is

$$K_{ij} = \sum_{a=1}^m \left(\left| \text{fl} \left(\sum_{b=1}^m G_{ib} \text{diag}(\omega)_{ba} \right) \right| + |G_{ia}| \right) \left| \sum_{\ell=1}^q U_{a\ell} C_{\ell j} \right|.$$

Equivalently, $E_{ij}^A = \gamma_m K_{ij}$. Hence the FP radius tends to 0 as $u, \gamma_m, \gamma_s, \eta_d \rightarrow 0$ for fixed exact data and a nonzero denominator, while the probability lower bound remains exactly the signed-mixing lower bound.

Lean formalization.

```
preconditionRows_preconditionColumns_assoc_rect
uniformRowSampleGram_rectFactoredInput_error_eq_rightGramCongruence_error
signedMixingUniformRowFactoredInputSampleGramTwoSidedEvent
signedMixingUniformRowSampleGramTwoSidedEvent_subset_factoredInput
signedMixingUniformRowTraceProbability_eventProb_uniformRowFactoredInputSampleGram_two_sided_
finiteLoewnerLe_ge_one_sub_delta_of_entry_sq_le_uniform
signedMixingComputedPreconditionedFactoredInputFlUniformRowSampleGramWithComputedDenTwoSidedEvent
signedMixingUniformRowTraceProbability_eventProb_computedPreconditioned_factoredInput_fl_
uniformRowSampleGramDotWithComputedDen_two_sided_finiteLoewnerLe_ge_one_sub_of_exact_event
signedMixingUniformRowTraceProbability_eventProb_exactFactorComputedLeftPreconditioned_
factoredInput_fl_uniformRowSampleGramDotWithComputedDen_two_sided_finiteLoewnerLe_ge_one_sub_
delta_of_entry_sq_le_uniform
signedMixingUniformRowTraceProbability_eventProb_exactFactorComputedLeftPreconditioned_
factoredInput_fl_uniformRowSampleGramDotWithComputedDen_two_sided_finiteLoewnerLe_ge_one_sub_
delta_of_total_budget
signedMixingUniformRowTraceProbability_eventProb_exactFactorComputedLeftPreconditioned_
factoredInput_fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_two_sided_finiteLoewnerLe_ge_
one_sub_delta_of_entry_sq_le_uniform
signedMixingUniformRowTraceProbability_eventProb_exactFactorComputedLeftPreconditioned_
factoredInput_fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_two_sided_finiteLoewnerLe_ge_
one_sub_delta_of_total_budget
```

Corollary 195 (Coordinate-Hoeffding row-norm bound). Assume $U \in \mathbb{R}^{m \times n}$ has orthonormal columns and

$$H_{ik}^2 = \frac{1}{m} \quad \text{for all } i, k.$$

For $D_\omega = \text{diag}(\omega)$, $B \geq 0$, and $\lambda > 0$, define

$$\eta(B, \lambda, m) = 2 \exp(-\lambda B) \exp\left(\frac{\lambda^2}{2m}\right).$$

Then

$$\mathbb{P}\left\{\forall i, \quad \|(HD_\omega U)_{i,*}\|_2^2 \leq nB^2\right\} \geq 1 - \sum_{i=1}^m \sum_{j=1}^n \eta(B, \lambda, m).$$

Lean formalization.

```
signedHadamard_entry_coeff_sum_sq_eq_inv
rademacherTraceProbability_eventProb_entry_abs_signedHadamard_le_ge_one_sub_exp_sq_bound
rademacherTraceProbability_eventProb_forall_entry_abs_signedHadamard_le_ge_one_sub_sum_exp_sq_bound
rademacherTraceProbability_eventProb_forall_rowNormSq_signedHadamard_le_ge_one_sub_sum_exp_sq_bound
```

Corollary 196 (Coordinate-Hoeffding leverage probabilities). *Assume, in addition, that H is orthogonal and $n > 0$. Let*

$$p_i(HD_\omega U) = \frac{\|(HD_\omega U)_{i,*}\|_2^2}{n}$$

be the equation (6) leverage probability of the preconditioned basis. With the same $\eta(B, \lambda, m)$ as above,

$$\mathbb{P}\left\{\forall i, \quad p_i(HD_\omega U) \leq B^2\right\} \geq 1 - \sum_{i=1}^m \sum_{j=1}^n \eta(B, \lambda, m).$$

In delta-budget form, if

$$\sum_{i=1}^m \sum_{j=1}^n \eta(B, \lambda, m) \leq \delta,$$

then the probability is at least $1 - \delta$.

Lean formalization.

```
rademacherTraceProbability_eventProb_forall_leverageScoreProb_signedHadamard_le_ge_one_sub_sum_exp_sq_bound
rademacherTraceProbability_eventProb_forall_leverageScoreProb_signedHadamard_le_ge_one_sub_delta
```

Lemma 197 (Uniform row one-step Loewner bound after preconditioning). *Let $V \in \mathbb{R}^{m \times n}$ have orthonormal columns. For uniform sampling of one row, define the rank-one estimator*

$$X_i(V) = m v_i v_i^T.$$

Then $X_i(V) \succeq 0$ and

$$\frac{1}{m} \sum_{i=1}^m X_i(V) = I_n.$$

Moreover, if the leverage probability

$$p_i(V) = \frac{\|v_i\|_2^2}{n}$$

satisfies $p_i(V) \leq B^2$, then

$$X_i(V) \preceq mnB^2 I_n.$$

Consequently, under the signed-Hadamard hypotheses of the previous corollary, if the coordinate-Hoeffding failure budget is at most δ , then

$$\mathbb{P} \{ \forall i, \quad X_i(HD_\omega U) \preceq mnB^2 I_n \} \geq 1 - \delta.$$

Lean formalization.

```
Algorithms/RandNLA/UniformRowSampling.lean uniformRowOuterGramSample
finiteQuadraticForm_uniformRowOuterGramSample_eq
finitePSD_uniformRowOuterGramSample
uniform_rowOuterGramSample_mean_eq_id
uniformRowOuterGramSample_finiteLoewnerLe_of_leverageScoreProb_le
Algorithms/RandNLA/Preconditioning.lean      rademacherTraceProbability_eventProb_forall_
uniformRowOuterGramSample_signedHadamard_finiteLoewnerLe_ge_one_sub_delta
```

Theorem 198 (Uniform row sample-average concentration). *Let $V \in \mathbb{R}^{m \times n}$ have orthonormal columns, and let*

$$X_i(V) = m v_i v_i^T.$$

Draw iid uniform rows $i_1, \dots, i_s$ and define the sampled Gram average

$$G_s(V) = \frac{1}{s} \sum_{t=1}^s X_{i_t}(V).$$

Assume that for every row i ,

$$X_i(V) \preceq L I_n, \quad m \|v_i\|_2^2 \leq L.$$

Let $\theta > 0$, $\varepsilon > 0$, and $s > 0$. Define

$$\beta_+(\theta, L) = \frac{\exp(\theta L) - \theta L - 1}{L^2}, \quad \beta_-(\theta) = \exp(\theta) - \theta - 1, \quad W = L^2 + 1.$$

If

$$e^{-\theta s \varepsilon} n e^{s \beta_+(\theta, L) W} + e^{-\theta s \varepsilon} n e^{s \beta_-(\theta) W} \leq \delta,$$

then

$$\mathbb{P} \{ G_s(V) - I_n \preceq \varepsilon I_n \text{ and } -(G_s(V) - I_n) \preceq \varepsilon I_n \} \geq 1 - \delta.$$

Lean formalization.

```
Algorithms/RandNLA/UniformRowSamplingMGF.lean uniformRowProb
uniformRowTraceProbability
uniformRowSampleGram
uniformRowSampleGram_sub_finiteIdMatrix_eq_centered_average
uniformRowOuterGramSample_centered_square_expectationCStarMatrix_le
uniformRowTraceProbabilityFiniteRealTraceMGFLogBound_centered_le
uniformRowTraceProbabilityFiniteRealTraceMGFLogBound_neg_centered_le
uniformRowTraceProbability_eventProb_uniformRowSampleGram_two_sided_finiteLoewnerLe_ge_one_sub_
delta_of_tail_budget
```

Corollary 199 (Signed-Hadamard preprocessing plus uniform row sampling). *Let $U \in \mathbb{R}^{m \times n}$ have orthonormal columns. Let $H \in \mathbb{R}^{m \times m}$ be orthogonal and flat, $H_{ik}^2 = 1/m$. Draw a Rademacher sign vector ω , set $V_\omega = HD_\omega U$, and then draw iid uniform rows $i_1, \dots, i_s$. Let*

$$G_s(V_\omega) = \frac{1}{s} \sum_{t=1}^s m v_{\omega, i_t} v_{\omega, i_t}^T.$$

Assume the coordinate-Hoeffding preprocessing budget

$$\sum_{i,j} 2 \exp(-\lambda B) \exp\left(\frac{\lambda^2}{2m}\right) \leq \delta_{\text{pre}}$$

and define

$$L = mnB^2, \quad \beta_+(\theta, L) = \frac{\exp(\theta L) - \theta L - 1}{L^2}, \quad \beta_-(\theta) = \exp(\theta) - \theta - 1.$$

If the uniform-sampling tail budget

$$e^{-\theta s \varepsilon} n e^{s\beta_+(\theta, L)(L^2+1)} + e^{-\theta s \varepsilon} n e^{s\beta_-(\theta)(L^2+1)} \leq \delta_{\text{sample}},$$

then

$$\mathbb{P}\{G_s(V_\omega) - I_n \preceq \varepsilon I_n \text{ and } -(G_s(V_\omega) - I_n) \preceq \varepsilon I_n\} \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

Lean formalization.

```
Analysis/FiniteProbability.lean FiniteProbability.prod
FiniteProbability.prod_eventProb_inter_dependent_ge_one_sub_add
Algorithms/RandNLA/UniformRowSamplingComposition.lean
signedHadamardUniformRowTraceProbability
signedHadamardUniformRowSampleGramTwoSidedEvent
signedHadamardUniformRowTraceProbability_eventProb_uniformRowSampleGram_two_sided_
finiteLoewnerLe_ge_one_sub_delta
```

Corollary 200 (Source-sharp SRHT preprocessing plus uniform row sampling). *Under the same signed-Hadamard/SRHT setup, let $t > 0$ and define*

$$S(t) = \sqrt{n/m} + t, \quad L(t) = mS(t)^2.$$

Assume the source-sharp preprocessing budget

$$m \exp\left(-\frac{mt^2}{8}\right) \leq \delta_{\text{pre}}$$

and define

$$\beta_+(\theta, L) = \frac{\exp(\theta L) - \theta L - 1}{L^2}, \quad \beta_-(\theta) = \exp(\theta) - \theta - 1.$$

If the uniform-sampling tail budget with $L = L(t)$,

$$e^{-\theta s \varepsilon} n e^{s\beta_+(\theta, L)(L^2+1)} + e^{-\theta s \varepsilon} n e^{s\beta_-(\theta)(L^2+1)} \leq \delta_{\text{sample}},$$

then

$$\mathbb{P}\{G_s(V_\omega) - I_n \preceq \varepsilon I_n \text{ and } -(G_s(V_\omega) - I_n) \preceq \varepsilon I_n\} \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

Lean formalization.

```
Algorithms/RandNLA/UniformRowSamplingComposition.lean
signedHadamardUniformRowTraceProbability_eventProb_uniformRowSampleGram_two_sided_
finiteLoewnerLe_ge_one_sub_delta_srht

signedHadamardUniformRowTraceProbability_eventProb_uniformRowSampleGram_two_sided_
finiteLoewnerLe_ge_one_sub_delta_srht_log_preprocess
```

Corollary 201 (Floating-point signed-Hadamard preprocessing plus uniform row sampling). *Under the hypotheses of the coordinate-Hoeffding signed-Hadamard preprocessing plus uniform row-sampling corollary above, assume also a floating-point model fp with γ_s valid. Let*

$$\widehat{G}_{\omega,S} = \text{fl}\left(\widehat{V}_{\omega,S}^T \widehat{V}_{\omega,S}\right)$$

be the Gram matrix obtained by forming the uniformly rescaled sampled rows with rounded division and computing each Gram entry with the local rounded dot product. Then the same event probability lower bound holds:

$$\mathbb{P}\left\{- (\varepsilon + \tau_{\omega,S})I_n \preceq \widehat{G}_{\omega,S} - I_n \preceq (\varepsilon + \tau_{\omega,S})I_n\right\} \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}},$$

where

$$\tau_{\omega,S} = \text{uniformRowSampleGramFullFpPerturbBudget}(\text{fp}, s, V_{\omega}, S)$$

is the explicit sample-dependent budget for rounded row scaling and rounded Gram dot products.

Lean formalization.

```
Analysis/MatrixAlgebra.lean finiteLoewnerLe_two_sided_add_of_frobNorm_le
Algorithms/RandNLA/UniformRowSamplingFP.lean uniformRowSampleScaleDen
uniformRowSampleSketch
fl_uniformRowSampleSketch
rowSketchGram_uniformRowSampleSketch_eq_uniformRowSampleGram
fl_uniformRowSampleGramDot
uniformRowSampleGramFullFpPerturbBudget
fl_uniformRowSampleGramDot_perturb_bound
signedHadamardFlUniformRowSampleGramTwoSidedEvent
signedHadamardUniformRowTraceProbability_eventProb_fl_uniformRowSampleGramDot_two_sided_
finiteLoewnerLe_ge_one_sub_delta
```

Corollary 202 (Computed-left $\widehat{V}$ replacement for Corollary 171). *The floating-point corollary above is an exact- $V_{\omega} = HD_{\omega}U$ sampling-law result: the row probabilities are uniform and no leverage table is computed, but the displayed $\tau_{\omega,S}$ only charges rounded row scaling and rounded Gram dot products after the exact preprocessed basis has been supplied. For an implementation that forms the basis, let*

$$P_{\omega} = HD_{\omega}, \quad V_{\omega} = P_{\omega}U,$$

and let $\widehat{P}_{\omega}$ be a computed transform with a ComputedPreconditioner certificate

$$|\widehat{P}_{\omega,ia} - P_{\omega,ia}| \leq E_{\omega,ia}^P, \quad E_{\omega,ia}^P \geq 0.$$

Define the actually formed basis and its entrywise error budget by

$$\begin{aligned}\widehat{V}_\omega &= \text{fl}(\widehat{P}_\omega U), \\ E_{\omega,ij}^V &= \gamma_m \sum_a |\widehat{P}_{\omega,ia}| |U_{aj}| + \sum_a E_{\omega,ia}^P |U_{aj}|.\end{aligned}$$

If γ_m is valid, then

$$|\widehat{V}_{\omega,ij} - V_{\omega,ij}| \leq E_{\omega,ij}^V \quad \text{for all } \omega, i, j.$$

For a sampled row trace S , set

$$\tau_{\omega,S}^{\text{left}} = \text{uniformRowSampleGramFullFpPerturbBudget}(\text{fp}, s, \widehat{V}_\omega, S) + \text{uniformRowSampleGramBasisPerturbBudget}(\text{fp}, s, \widehat{V}_\omega, S)$$

Then the computed-left perturbation event has probability one:

$$\mathbb{P} \left[\left\| \text{fl}(\widehat{V}_{\omega,S}^T \widehat{V}_{\omega,S}) - G_s(V_\omega) \right\|_F \leq \tau_{\omega,S}^{\text{left}} \right] = 1.$$

Consequently Corollary 171 has an implementation-facing form with

$$\mathbb{P} \left\{ -(\varepsilon + \tau_{\omega,S}^{\text{left}})I_n \preceq \text{fl}(\widehat{V}_{\omega,S}^T \widehat{V}_{\omega,S}) - I_n \preceq (\varepsilon + \tau_{\omega,S}^{\text{left}})I_n \right\} \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}},$$

once the exact signed-Hadamard/uniform-row event from Corollary 171 is available. The remaining implementation obligation is not hidden in this statement: a concrete SRHT/FHT/sign generator or stored transform must instantiate $\widehat{P}_\omega$ and its ComputedPreconditioner certificate.

Lean formalization.

```
flPreconditionRowsWithComputedLeftEntryErrorBudget
fl_preconditionRowsWithComputedLeft_entry_error_budget_bound
uniformRowSampleGram_frob_error_bound_of_basis_entrywise_abs
signedHadamardComputedLeftPreconditionedBasis
signedHadamardComputedLeftUniformRowPerturbBudget
signedHadamardUniformRowTraceProbability_eventProb_computedLeftPreconditioned_fl_
uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_computedPreconditioned_fl_
uniformRowSampleGramDot_two_sided_finiteLoewnerLe_ge_one_sub_delta
```

Corollary 203 (Computed-input $\widehat{U}$ replacement for Corollary 171). *In the preceding computed-left setting, suppose the basis or singular-vector table used by the implementation is itself computed. Let $\widehat{U}$ carry a ComputedMatrix certificate*

$$|\widehat{U}_{aj} - U_{aj}| \leq E_{aj}^U, \quad E_{aj}^U \geq 0.$$

The implemented basis is now

$$\widehat{V}_\omega = \text{fl}(\widehat{P}_\omega \widehat{U}).$$

If γ_m is valid, then for every ω, i, j ,

$$|\widehat{V}_{\omega,ij} - (H D_\omega U)_{ij}| \leq \gamma_m \sum_a |\widehat{P}_{\omega,ia}| |\widehat{U}_{aj}| + \sum_a E_{\omega,ia}^P |\widehat{U}_{aj}| + \sum_a |P_{\omega,ia}| E_{aj}^U.$$

Define $E_\omega^{V,\text{in}}$ by the right-hand side. Let

$$\begin{aligned} B_{\omega,S}^{\text{fp}} &:= \mathcal{B}_{\text{fp}}(\text{fp}, s, \widehat{V}_\omega, S), \\ B_{\omega,S}^{\text{basis}} &:= \mathcal{B}_{\text{basis}}(HD_\omega U, \widehat{V}_\omega, E_\omega^{V,\text{in}}, S), \\ \tau_{\omega,S}^{\text{in}} &:= B_{\omega,S}^{\text{fp}} + B_{\omega,S}^{\text{basis}}. \end{aligned}$$

Here $\mathcal{B}_{\text{fp}}$ and $\mathcal{B}_{\text{basis}}$ are the Lean budgets `uniformRowSampleGramFullFpPerturbBudget` and `uniformRowSampleGramBasisPerturbBudget`, respectively. Then the computed-input perturbation event has probability one under the same exact Rademacher/uniform-row law:

$$\mathbb{P}\left[\left\|\mathfrak{fl}(\widehat{V}_{\omega,S}^T \widehat{V}_{\omega,S}) - G_s(HD_\omega U)\right\|_F \leq \tau_{\omega,S}^{\text{in}}\right] = 1.$$

Thus a computed QR/SVD/singular-vector routine enters Algorithm 3 through the matrix certificate E^U , not through any probability-law perturbation.

Lean formalization.

```

ComputedMatrix
fl_preconditionRowsWithComputedLeftAndInput_total_error_bound
flPreconditionRowsWithComputedLeftInputEntryErrorBudget
fl_preconditionRowsWithComputedLeftInput_entry_error_budget_bound
signedHadamardComputedLeftInputPreconditionedBasis
signedHadamardComputedLeftInputUniformRowPerturbBudget
signedHadamardUniformRowTraceProbability_eventProb_computedLeftInputPreconditioned_fl_
uniformRowPerturbEvent_eq_one

```

Corollary 204 (Deterministic floating-point radius for the uniform row sketch). *In the setting of the preceding corollary, suppose a deterministic scalar τ dominates the sample-dependent floating-point budget for every joint preprocessing and sampling outcome:*

$$\text{uniformRowSampleGramFullFpPerturbBudget}(\text{fp}, s, V_\omega, S) \leq \tau.$$

Then the same probability lower bound holds with the fixed radius $\varepsilon + \tau$:

$$\mathbb{P}\left\{- (\varepsilon + \tau)I_n \preceq \widehat{G}_{\omega,S} - I_n \preceq (\varepsilon + \tau)I_n\right\} \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

Moreover, if $R \geq 0$ and every sampled row satisfies

$$\|(V_\omega)_{S_i,*}\|_2^2 \leq R,$$

then the sample-dependent budget is bounded deterministically by

$$\text{uniformRowSampleGramFullFpConstBudget}(\text{fp}, s, mR).$$

The two parts of this fixed budget have the closed forms

$$n(2u + u^2)mR \quad \text{and} \quad n\gamma_s(1 + u)^2mR,$$

for row scaling and dot products, respectively.

Lean formalization.

```

Analysis/MatrixAlgebra.lean finiteLoewnerLe_smul_finiteIdMatrix_mono
Algorithms/RandNLA/UniformRowSamplingFP.lean uniformRowSampleGramFpPerturbConstBudget
uniformRowSampleGramDotProductConstBudget
uniformRowSampleGramFullFpConstBudget
uniformRowSampleGramFpPerturbConstBudget_eq_nat_mul
uniformRowSampleGramDotProductConstBudget_eq_nat_mul
abs_mul_entry_le_rowNormSq
uniformRowSampleSketch_abs_mul_sum_le_of_rowNormSq_le
uniformRowSampleGramFullFpPerturbBudget_le_const_of_sample_rowNormSq_le
signedHadamardFlUniformRowSampleGramTwoSidedConstBudgetEvent
signedHadamardUniformRowTraceProbability_eventProb_fl_uniformRowSampleGramDot_two_sided_
finiteLoewnerLe_ge_one_sub_delta_constBudget

```

Corollary 205 (Computed- $\hat{V}$ deterministic-radius replacement for Corollary 172). *Let $\hat{V}_\omega$ be the implemented preprocessed basis. Suppose the exact signed-Hadamard/uniform-row event has probability at least*

$$1 - \delta_{\text{pre}} - \delta_{\text{sample}},$$

and suppose the constant-radius computed-basis perturbation event satisfies

$$\mathbb{P} \left[\left\| \text{fl}(\hat{V}_{\omega,S}^T \hat{V}_{\omega,S}) - G_s(V_\omega) \right\|_F \leq \tau_{\text{comp}} \right] \geq 1 - \delta_{\text{comp}}.$$

Then

$$\mathbb{P} \left\{ -(\varepsilon + \tau_{\text{comp}})I_n \preceq \text{fl}(\hat{V}_{\omega,S}^T \hat{V}_{\omega,S}) - I_n \preceq (\varepsilon + \tau_{\text{comp}})I_n \right\} \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}} - \delta_{\text{comp}}.$$

The radius τ_{comp} must dominate every computed quantity used to produce and use $\hat{V}_\omega$: transform generation or storage, Hadamard/sign arithmetic, any computed input basis, the computed $\sqrt{s/m}$ denominator when one is used, rounded row scaling, and the final Gram dot products.

Lean formalization.

```

signedHadamardComputedPreconditionedFlUniformRowPerturbConstBudgetEvent
signedHadamardUniformRowTraceProbability_eventProb_computedPreconditioned_fl_
uniformRowSampleGramDot_two_sided_finiteLoewnerLe_ge_one_sub_delta_constBudget

```

Corollary 206 (Source-sharp SRHT floating-point uniform row sampling). *In the source-sharp SRHT setting, let*

$$S(t) = \sqrt{n/m} + t, \quad \tau_{\text{SRHT}}(t) = \text{uniformRowSampleGramFullFpConstBudget}(\text{fp}, s, mS(t)^2).$$

Under the same preprocessing and sample trace-MGF budgets as the source-sharp exact theorem, the rounded sample Gram matrix satisfies

$$\mathbb{P} \left\{ -(\varepsilon + \tau_{\text{SRHT}}(t))I_n \preceq \hat{G}_{\omega,S} - I_n \preceq (\varepsilon + \tau_{\text{SRHT}}(t))I_n \right\} \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

The logarithmic version substitutes

$$t = \sqrt{8 \log(m/\delta_{\text{pre}})/m}.$$

Lean formalization.

```
Algorithms/RandNLA/UniformRowSamplingFP.lean  signedHadamardUniformRowTraceProbability_
eventProb_fl_uniformRowSampleGramDot_two_sided_finiteLoewnerLe_ge_one_sub_delta_constBudget_srht

signedHadamardUniformRowTraceProbability_eventProb_fl_uniformRowSampleGramDot_two_sided_
finiteLoewnerLe_ge_one_sub_delta_constBudget_srht_log_preprocess
```

Corollary 207 (Intermediate computed- $\hat{V}$ source-sharp SRHT transfer). *In the source-sharp SRHT setting, the exact theorem controls $V_\omega = HD_\omega U$. Let $\hat{V}_\omega$ be the basis produced by an implementation. This is reusable infrastructure for arbitrary $\hat{V}_\omega$; the final concrete actual-input SRHT theorem below instantiates the transform, the input multiplication, the denominator routine, the row scaling, and the Gram dot products. If the exact SRHT preprocessing and sample trace-MGF budgets give probability at least*

$$1 - \delta_{\text{pre}} - \delta_{\text{sample}},$$

and the implemented-basis perturbation event satisfies

$$\mathbb{P}\left[\left\|\mathfrak{fl}(\hat{V}_{\omega,S}^T \hat{V}_{\omega,S}) - G_s(V_\omega)\right\|_F \leq \tau_{\text{SRHT,comp}}(t)\right] \geq 1 - \delta_{\text{comp}},$$

then

$$\mathbb{P}\left\{-\left(\varepsilon + \tau_{\text{SRHT,comp}}(t)\right)I_n \preceq \mathfrak{fl}(\hat{V}_{\omega,S}^T \hat{V}_{\omega,S}) - I_n \preceq \left(\varepsilon + \tau_{\text{SRHT,comp}}(t)\right)I_n\right\} \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}} - \delta_{\text{comp}}.$$

The logarithmic source wrapper substitutes $t = \sqrt{8 \log(m/\delta_{\text{pre}})/m}$. The exact SRHT concentration result is unchanged; the new term $\tau_{\text{SRHT,comp}}(t)$ is the place where the concrete FFT/Hadamard routine, sign storage, scale denominators, computed basis or singular-vector construction, row scaling, and Gram dot products must be charged.

Lean formalization.

```
signedHadamardUniformRowTraceProbability_eventProb_computedPreconditioned_fl_
uniformRowSampleGramDot_two_sided_finiteLoewnerLe_ge_one_sub_delta_srht

signedHadamardUniformRowTraceProbability_eventProb_computedPreconditioned_fl_
uniformRowSampleGramDot_two_sided_finiteLoewnerLe_ge_one_sub_delta_constBudget_srht

signedHadamardUniformRowTraceProbability_eventProb_computedPreconditioned_fl_
uniformRowSampleGramDot_two_sided_finiteLoewnerLe_ge_one_sub_delta_constBudget_srht_log_
preprocess
```

Corollary 208 (Computed actual-input SRHT with computed uniform denominator). *Let $A = U_A C_A$ be the exact Algorithm 3 input factorization used only in the analysis, where $U_A^T U_A = I_r$. The algorithm does not compute U_A or C_A ; it computes with A . On the concrete scaled sign-pattern, stored-sign SRHT path, form*

$$\hat{P}_\omega = \mathfrak{fl}(\hat{H} \hat{D}_\omega), \quad \hat{Y}_\omega = \mathfrak{fl}(\hat{P}_\omega A), \quad \hat{G}_A(\omega, S) = \mathfrak{fl}(\hat{B}_{\omega,S}^T \hat{B}_{\omega,S}),$$

where $\hat{B}_{\omega,S}$ uses the computed nonzero denominator

$$\hat{d}_* = \mathfrak{fl}(\mathfrak{fl}(\sqrt{s}) \mathfrak{fl}(1/\mathfrak{fl}(\sqrt{m}))),$$

with certified relative denominator radius $(4u + 3u^2 + u^3)/(1 - u)$ for $\sqrt{s/m}$. The exact Rademacher and uniform-row laws remain exact. Every non-probability computation is charged by the explicit sample-dependent budget

$$\tau_A^{\text{cd}}(\omega, S) = \text{uniformRowSampleGramComputedDenFullFpPerturbBudget}(\text{fp}, \hat{Y}_\omega, \hat{d}, S) + \text{uniformRowSampleGramBasis}$$

where E_ω is the entrywise product budget for $\mathfrak{fl}(\hat{P}_\omega A)$. With

$$t = \sqrt{8 \log(m/\delta_{\text{pre}})/m}, \quad L = m \left(\sqrt{r/m} + t \right)^2,$$

and the usual exact uniform-sampling trace-MGF budget at dimension r , the fully computed input Gram satisfies

$$\mathbb{P} \left[(1 - \varepsilon) A^T A - \tau_A^{\text{cd}}(\omega, S) I_n \preceq \hat{G}_A(\omega, S) \preceq (1 + \varepsilon) A^T A + \tau_A^{\text{cd}}(\omega, S) I_n \right] \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

There is no perturbation-event hypothesis and no δ_{comp} : the computed-denominator perturbation event is proved with probability one for this concrete path.

Lean formalization.

```
rightGramCongruence
uniformRowSampleGram_factoredInput_error_eq_rightGramCongruence_error
signedHadamardUniformRowFactoredInputSampleGramTwoSidedEvent
signedHadamardUniformRowSampleGramTwoSidedEvent_subset_factoredInput
signedHadamardUniformRowTraceProbability_eventProb_uniformRowFactoredInputSampleGram_two_sided_
finiteLoewnerLe_ge_one_sub_delta_srht_log_preprocess
signedHadamardComputedPreconditionedFactoredInputFlUniformRowSampleGramWithComputedDenTwoSidedEvent
signedHadamardUniformRowTraceProbability_eventProb_computedPreconditioned_factoredInput_fl_
uniformRowSampleGramDotWithComputedDen_two_sided_finiteLoewnerLe_ge_one_sub_of_exact_event
signedHadamardUniformRowTraceProbability_eventProb_scaledPatternStoredSignComputedLeftPreconditioned_
factoredInput_fl_uniformRowSampleGramDotWithComputedDen_two_sided_finiteLoewnerLe_ge_one_sub_
delta_srht_log_preprocess
uniformRowFlSqrtMulInvSqrtScaleDen
uniformRowFlSqrtMulInvSqrtScaleDen_den_abs_error
signedHadamardUniformRowTraceProbability_eventProb_scaledPatternStoredSignComputedLeftPreconditioned_
fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_two_sided_finiteLoewnerLe_ge_one_sub_delta_
srht_log_preprocess
signedHadamardUniformRowTraceProbability_eventProb_scaledPatternStoredSignComputedLeftPreconditioned_
factoredInput_fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_two_sided_finiteLoewnerLe_ge_one_
sub_delta_srht_log_preprocess
```

Corollary 209 (Exact-stored computed-left SRHT certificate). *Let $P_\omega = H D_\omega$, and suppose the implementation supplies this realized preconditioner exactly as its stored transform:*

$$\hat{P}_\omega = P_\omega, \quad E_{\omega,ik}^P = 0.$$

Define the implemented preconditioned basis by

$$\hat{V}_\omega = \mathfrak{fl}(\hat{P}_\omega U).$$

For every entry,

$$|\hat{V}_{\omega,ij} - (P_\omega U)_{ij}| \leq \gamma_m \sum_{k=1}^m |(P_\omega)_{ik}| |U_{kj}|.$$

Consequently, under the same positive-dimension and γ -validity hypotheses as the computed-left transfer, the joint exact Rademacher/uniform-row probability law satisfies

$$\mathbb{P} \left[\left\| \mathfrak{H}(\widehat{V}_{\omega,S}^T \widehat{V}_{\omega,S}) - G_s(P_\omega U) \right\|_F \leq \tau_{\text{row}/\text{Gram}}(\widehat{V}_\omega, S) + \tau_{\text{basis}}(P_\omega U, \widehat{V}_\omega, E_\omega^{\text{exact}}, S) \right] = 1,$$

where $E_{\omega,ij}^{\text{exact}}$ is the displayed matrix-product rounding budget above. Thus this specialization charges rounded formation of $\widehat{V}_\omega$, rounded row scaling, and rounded Gram dot products, but it charges no preprocessing-matrix storage/generation error. Non-exact FHT routines, or sign-storage formats other than the modeled sign-copy certificates below, must still instantiate a nonzero $\widehat{P}_\omega - P_\omega$ certificate.

Lean formalization.

```
ComputedPreconditioner.exact
flPreconditionRowsWithComputedLeftEntryErrorBudget_exact
fl_preconditionRowsWithExactLeft_entry_error_bound
signedHadamardExactStoredPreconditioner
signedHadamardComputedLeftPreconditionedBasisEntryErrorBudget_exactStored
signedHadamardUniformRowTraceProbability_eventProb_exactStoredComputedLeftPreconditioned_fl_
uniformRowPerturbEvent_eq_one
```

Corollary 210 (Expected row norm after flat signed-Hadamard preprocessing). *Assume $U \in \mathbb{R}^{m \times n}$ has orthonormal columns and*

$$H_{ik}^2 = \frac{1}{m} \quad \text{for all } i, k.$$

For $D_\omega = \text{diag}(\omega)$, each row of $HD_\omega U$ has expected squared Euclidean norm

$$\mathbb{E} \|(HD_\omega U)_{i,*}\|_2^2 = \frac{n}{m}.$$

Consequently,

$$\mathbb{E} \|(HD_\omega U)_{i,*}\|_2 \leq \sqrt{\frac{n}{m}}.$$

Lean formalization.

```
rademacherTraceProbability_expectationReal_sign_eq_zero
rademacherTraceProbability_expectationReal_sign_mul_eq_ite
rademacherTraceProbability_expectationReal_sq_sum_mul_sign_eq_sum_sq
rademacherTraceProbability_expectationReal_exp_sum_mul_sign_eq_prod
rademacherTraceProbability_eventProb_sum_mul_sign_le_ge_one_sub_exp_mul_prod
HadamardFlat
signedHadamardPreconditionRows_entry
rademacherTraceProbability_expectationReal_rowNormSq_signedHadamard_eq
rademacherTraceProbability_expectationReal_sqrt_rowNormSq_signedHadamard_le
```

Lemma 211 (Signed-Hadamard row-norm convexity input). *For a fixed row i , define*

$$f_i(x) = \|(H \operatorname{diag}(x)U)_{i,*}\|_2, \quad x \in \mathbb{R}^m.$$

Then f_i is convex:

$$f_i(\theta x + (1 - \theta)y) \leq \theta f_i(x) + (1 - \theta)f_i(y) \quad (0 \leq \theta \leq 1).$$

Lean formalization.

```
Analysis/MatrixAlgebra.lean FiniteVecConvex
vecNorm2_linear_combination_convex
Algorithms/RandNLA/Preconditioning.lean signedHadamard_row_vecNorm2_convex
```

Lemma 212 (Unit-cube to Rademacher scaling). *Let $F : \mathbb{R}^m \rightarrow \mathbb{R}$ be convex and L -Lipschitz with $L > 0$. Define*

$$G(x) = \frac{1}{2L} F(2x - \mathbf{1}).$$

Then G is convex and 1-Lipschitz. Consequently, Ledoux's $[0, 1]^m$ upper-tail statement with exponent $\exp(-t^2/2)$ converts to Tropp's Rademacher-sign statement with exponent $\exp(-t^2/8)$.

Lean formalization.

```
Analysis/MatrixAlgebra.lean FiniteVecLipschitzWith
unitCubeToRademacherVec
finiteVecConvex_scaled_unitCubeToRademacher
finiteVecLipschitzWith_scaled_unitCubeToRademacher
```

Lemma 213 (Signed-Hadamard row-norm Lipschitz input). *Assume $U \in \mathbb{R}^{m \times n}$ has orthonormal columns and*

$$H_{ik}^2 = \frac{1}{m} \quad \text{for all } i, k.$$

For deterministic sign-like vectors $x, y \in \mathbb{R}^m$, define

$$f_i(x) = \|(H \operatorname{diag}(x)U)_{i,*}\|_2.$$

Then

$$|f_i(x) - f_i(y)| \leq \sqrt{\frac{1}{m}} \|x - y\|_2.$$

Lean formalization.

```
Algorithms/RandNLA/RowSamplingLeverage.lean hasOrthonormalColumns_vecNorm2Sq_mul_vec_eq
hasOrthonormalColumns_transpose_mul_vecNorm2Sq_le
Analysis/MatrixAlgebra.lean abs_vecNorm2_sub_le_vecNorm2_sub
Algorithms/RandNLA/Preconditioning.lean signedHadamard_row_vecNorm2_lipschitz
```

Lemma 214 (Signed-Hadamard row-norm self-bounding). *Assume $U \in \mathbb{R}^{m \times n}$ has orthonormal columns and*

$$H_{ik}^2 = \frac{1}{m} \quad \text{for all } i, k.$$

For a fixed row i and Rademacher sign trace ω , define

$$F_i(\omega) = \|(H \operatorname{diag}(\varepsilon(\omega))U)_{i,*}\|_2,$$

and let ω^k denote ω with only coordinate k flipped. Then the positive coordinate-flip drops satisfy

$$\sum_{k=1}^m \max\{F_i(\omega) - F_i(\omega^k), 0\}^2 \leq \frac{4}{m}.$$

Lean formalization.

```
Analysis/MatrixAlgebra.lean vecNorm2_inv_smul_self_of_pos
vecInnerProduct_inv_smul_self_eq_norm
vecNorm2_sub_le_inner_unit_diff
Algorithms/RandNLA/Preconditioning.lean rademacherSignVector_flip_self
rademacherSignVector_flip_of_ne
rademacherSignVector_sub_flip
signedHadamard_row_inner_sq_sum_eq_inv_mul
signedHadamard_row_vec_sub_flip
signedHadamard_row_vecNorm2_positive_flip_sq_sum_le
```

Lemma 215 (Positive-drop exponential-tilt bridge). *Let X be a real-valued function on the finite Rademacher cube and let ω^i denote coordinate i flipped. If, pointwise in ω ,*

$$\sum_i \max\{X(\omega) - X(\omega^i), 0\}^2 \leq B,$$

then for every $\lambda \geq 0$,

$$\begin{aligned} \sum_i \mathbb{E} \left[\left(e^{\lambda X(\omega)/2} - e^{\lambda X(\omega^i)/2} \right)^2 \right] \\ \leq \frac{B}{2} \lambda^2 \mathbb{E} e^{\lambda X(\omega)}. \end{aligned}$$

Lean formalization.

```
Analysis/FiniteProbability.lean real_exp_half_sub_sq_le_quarter_mul_sq_mul_exp_of_le
real_exp_half_sub_sq_le_lam_sq_quarter_pair_pos
Algorithms/RandNLA/Preconditioning.lean rademacherTraceProbability_flip_tilt_sq_sum_
bound_of_pointwise_posdiff_sq_sum_le
```

Theorem 216 (Source-sharp signed-Hadamard row-norm concentration). *Assume $m > 0$, $U^T U = I_n$, and $H_{ik}^2 = 1/m$. For a fixed row*

$$F_i(\omega) = \|(H \operatorname{diag}(\varepsilon(\omega))U)_{i,*}\|_2,$$

and every $t > 0$,

$$\mathbb{P}\{F_i(\omega) \leq \mathbb{E}F_i + t\} \geq 1 - \exp\left(-\frac{mt^2}{8}\right).$$

Consequently,

$$\mathbb{P}\left\{\forall i, \quad F_i(\omega) \leq \sqrt{\frac{n}{m}} + t\right\} \geq 1 - m \exp\left(-\frac{mt^2}{8}\right),$$

and

$$\mathbb{P}\left\{\forall i, \quad \ell_i(HD_\omega U) \leq \frac{(\sqrt{n/m} + t)^2}{n}\right\} \geq 1 - m \exp\left(-\frac{mt^2}{8}\right),$$

where $\ell_i(V) = \|V_{i,*}\|_2^2/n$ is the equation (6) leverage probability for an orthonormal-column matrix.

Lean formalization.

```
Algorithms/RandNLA/Preconditioning.lean      rademacherTraceProbability_flip_tilt_sq_sum_
bound_signedHadamard_row_vecNorm2

rademacherTraceProbability_eventProb_vecNorm2_signedHadamard_le_mean_add_ge_one_sub_exp_m_t_sq_
div_eight

rademacherTraceProbability_eventProb_forall_vecNorm2_signedHadamard_le_sqrt_add_ge_one_sub_m_exp_
m_t_sq_div_eight

rademacherTraceProbability_eventProb_forall_rowNormSq_signedHadamard_le_sqrt_add_sq_ge_one_sub_m_
exp_m_t_sq_div_eight

rademacherTraceProbability_eventProb_forall_leverageScoreProb_signedHadamard_le_sqrt_add_sq_div_
nat_ge_one_sub_m_exp_m_t_sq_div_eight

rademacherTraceProbability_eventProb_forall_leverageScoreProb_signedHadamard_le_sqrt_add_sq_div_
nat_ge_one_sub_delta
```

Corollary 217 (Logarithmic SRHT preprocessing radius). *Let $0 < \delta < m$ and choose*

$$t_\delta = \sqrt{\frac{8 \log(m/\delta)}{m}}.$$

Then the preceding theorem gives

$$\mathbb{P}\left\{\forall i, \quad \|(HD_\omega U)_{i,*}\|_2^2 \leq \left(\sqrt{\frac{n}{m}} + t_\delta\right)^2\right\} \geq 1 - \delta$$

and

$$\mathbb{P}\left\{\forall i, \quad \ell_i(HD_\omega U) \leq \frac{(\sqrt{n/m} + t_\delta)^2}{n}\right\} \geq 1 - \delta.$$

Lean formalization.

```
Analysis/MatrixConcentration.lean real_sqrt_eight_log_div_pos_of_pos_lt
real_mul_exp_neg_mul_sqrt_eight_log_div_sq_div_eight_eq

Algorithms/RandNLA/Preconditioning.lean      rademacherTraceProbability_eventProb_forall_
rowNormSq_signedHadamard_le_log_delta_ge_one_sub_delta

rademacherTraceProbability_eventProb_forall_leverageScoreProb_signedHadamard_le_log_delta_ge_one_
sub_delta
```

Corollary 218 (Weak all-row Markov bound for flat signed-Hadamard preprocessing). *Assume the hypotheses of the preceding corollary, and let $T > 0$. Then*

$$\mathbb{P} \left\{ \forall i, \quad \|(HD_\omega U)_{i,*}\|_2^2 \leq T \right\} \geq 1 - m \frac{n/m}{T}.$$

Equivalently, if $m((n/m)/T) \leq \delta$, then

$$\mathbb{P} \left\{ \forall i, \quad \|(HD_\omega U)_{i,*}\|_2^2 \leq T \right\} \geq 1 - \delta.$$

Lean formalization.

```
rademacherTraceProbability_eventProb_rowNormSq_signedHadamard_le_ge_one_sub
rademacherTraceProbability_eventProb_forall_rowNormSq_signedHadamard_le_ge_one_sub_sum
rademacherTraceProbability_eventProb_forall_rowNormSq_signedHadamard_le_ge_one_sub
rademacherTraceProbability_eventProb_forall_rowNormSq_signedHadamard_le_ge_one_sub_delta
```

The Markov/union theorem and the coordinate-Hoeffding theorem remain in the library as weaker alternatives. The source-sharp SRHT row-norm and leverage route is now also formalized in both explicit t -parameter and logarithmic target-failure forms, composed with the exact iid uniform-row sampling theorem, and transferred through the rounded sample-Gram computation with the fixed row-norm-derived floating-point budget.

Theorem 219 (One-sided floating-point preprocessing). *For $\Pi_L \in \mathbb{R}^{r \times m}$, $A \in \mathbb{R}^{m \times n}$, and valid γ_m ,*

$$|\mathfrak{fl}(\Pi_L A)_{ij} - (\Pi_L A)_{ij}| \leq \gamma_m \sum_k |(\Pi_L)_{ik}| |A_{kj}|.$$

For $A \in \mathbb{R}^{m \times n}$, $\Pi_R \in \mathbb{R}^{n \times q}$, and valid γ_n ,

$$|\mathfrak{fl}(A \Pi_R)_{ij} - (A \Pi_R)_{ij}| \leq \gamma_n \sum_k |A_{ik}| |(\Pi_R)_{kj}|.$$

Lean formalization.

```
fl_preconditionRows_error_bound
fl_preconditionColumns_error_bound
ComputedPreconditioner
fl_preconditionRowsWithComputedLeft_total_error_bound
fl_preconditionColumnsWithComputedRight_total_error_bound
```

Corollary 220 (Computed signed-Hadamard preconditioner). *Let $H \in \mathbb{R}^{m \times m}$ be the exact scaled Hadamard/FHT factor and let $\sigma \in \mathbb{R}^m$ be the realized sign vector drawn from the exact Rademacher law. Put $D = \text{diag}(\sigma)$. Suppose the implementation stores or computes $\hat{H}$ and $\hat{\sigma}$ with certificates*

$$|\hat{H}_{ik} - H_{ik}| \leq E_{ik}^H, \quad |\hat{\sigma}_k - \sigma_k| \leq E_k^\sigma, \quad E_{ik}^H, E_k^\sigma \geq 0.$$

Let $\hat{D} = \text{diag}(\hat{\sigma})$, and form the realized computed preconditioner by the rounded product

$$\hat{\Pi} = \mathfrak{fl}(\hat{H} \hat{D}).$$

If γ_m is valid, then for every i, j ,

$$|\hat{\Pi}_{ij} - (HD)_{ij}| \leq \gamma_m \sum_k |\hat{H}_{ik}| |\hat{D}_{kj}| + \sum_k E_{ik}^H |\hat{D}_{kj}| + \sum_k |H_{ik}| E_{kj}^D =: E_{ij}^\Pi,$$

where $E_{kj}^D = E_k^\sigma$ when $k = j$ and $E_{kj}^D = 0$ otherwise. Consequently, for any $A \in \mathbb{R}^{m \times n}$,

$$|\mathfrak{fl}(\hat{\Pi}A)_{ij} - (HDA)_{ij}| \leq \gamma_m \sum_k |\hat{\Pi}_{ik}| |A_{kj}| + \sum_k E_{ik}^\Pi |A_{kj}|.$$

The probability law of σ is not perturbed: this result charges only storage and arithmetic in the realized transform object.

Lean formalization.

```

ComputedVector
ComputedMatrix.diag
fl_computedMatrixProduct_total_error_bound
flComputedMatrixProductEntryErrorBudget
ComputedMatrix.flProduct
ComputedPreconditioner.flSignedHadamard
fl_preconditionRowsWithComputedLeft_total_error_bound

```

Corollary 221 (Exact-factor signed-Hadamard preconditioner). *In the setting of the preceding corollary, suppose instead that H and the realized sign vector σ are supplied exactly as mathematical factors. The probability law of σ is still the exact Rademacher law. The implemented preconditioner is nevertheless formed by rounded arithmetic:*

$$\hat{\Pi} = \mathfrak{fl}(HD), \quad D = \text{diag}(\sigma).$$

If γ_m is valid, then for every i, j ,

$$|\hat{\Pi}_{ij} - (HD)_{ij}| \leq \gamma_m \sum_k |H_{ik}| |D_{kj}| =: E_{ij}^{\Pi, \text{fac}}.$$

Consequently, for any $A \in \mathbb{R}^{m \times n}$,

$$|\mathfrak{fl}(\hat{\Pi}A)_{ij} - (HDA)_{ij}| \leq \gamma_m \sum_k |\hat{\Pi}_{ik}| |A_{kj}| + \sum_k E_{ik}^{\Pi, \text{fac}} |A_{kj}|.$$

Moreover, for the joint exact Rademacher/uniform-row law,

$$\mathbb{P} \left[\left\| \mathfrak{fl}(\hat{V}_{\omega, S}^T \hat{V}_{\omega, S}) - G_s(HD_\omega U) \right\|_F \leq \tau_{\omega, S}^{\text{fac}} \right] = 1,$$

where $\hat{V}_\omega = \mathfrak{fl}(\hat{\Pi}_\omega U)$ and $\tau_{\omega, S}^{\text{fac}}$ is the existing computed-left budget with $E^{\Pi, \text{fac}}$. This closes the exact supplied-factor path; a fast FHT recurrence/apply routine, or a sign-storage format other than the modeled sign-copy certificates below, must still instantiate the nonzero-error certificates above.

Lean formalization.

```

ComputedPreconditioner.flSignedHadamardExactFactors
ComputedPreconditioner.flSignedHadamardExactFactors_matrix
ComputedPreconditioner.flSignedHadamardExactFactors_abs_error
ComputedPreconditioner.flSignedHadamardExactFactors_entry_error_bound
signedHadamardExactFactorPreconditioner
signedHadamardUniformRowTraceProbability_eventProb_exactFactorComputedLeftPreconditioned_fl_
uniformRowPerturbEvent_eq_one

```

Corollary 222 (Rounded scaled-Hadamard sign-pattern table). *Let $S \in \mathbb{R}^{m \times m}$ be a supplied sign-pattern table satisfying*

$$S_{ij}^2 = 1 \quad \text{for all } i, j.$$

Define the ideal scaled Hadamard-style table and the computed table by

$$H_{ij} = \sqrt{m^{-1}} S_{ij}, \quad \hat{H}_{ij} = \text{flsqrt}(m^{-1}) S_{ij}.$$

Here the sign pattern S , the Rademacher law used later, and all sampling laws are exact mathematical inputs; the only charged computation in this corollary is the rounded square-root scale. For every i, j ,

$$|\hat{H}_{ij} - H_{ij}| \leq \sqrt{m^{-1}} u |S_{ij}| =: E_{ij}^H.$$

Moreover the ideal table H satisfies the deterministic SRHT flatness hypothesis

$$H_{ij}^2 = m^{-1} \quad \text{for all } i, j,$$

so it can be used as the exact analysis object in the SRHT row-norm concentration theorem. The certificate E^H can then be supplied to the computed signed-Hadamard preconditioner corollary above. This does not formalize a concrete fast FHT recurrence/apply routine. The next results handle the modeled rounded-copy sign-storage paths $\text{fl}(\sigma_i \cdot 1)$, $\text{fl}(\sigma_i + 0)$, and $\text{fl}(\sigma_i - 0)$; other sign-storage representations remain ledger obligations if an implementation does not use one of those modeled paths.

Lean formalization.

```

ComputedMatrix.flSqrtInvNatScaledPattern
ComputedMatrix.flSqrtInvNatScaledPattern_matrix
ComputedMatrix.flSqrtInvNatScaledPattern_abs_error
ComputedMatrix.flSqrtInvNatScaledPattern_entry_error_bound
HadamardSignPattern
hadamardFlat_sqrt_inv_nat_mul_signPattern

```

Corollary 223 (Scaled-pattern computed-left preconditioner event). *In the setting of the preceding corollary, let*

$$H_{ij} = \sqrt{m^{-1}} S_{ij}, \quad \hat{H}_{ij} = \text{flsqrt}(m^{-1}) S_{ij}.$$

For a realized Rademacher sign vector σ_ω , define

$$D_\omega = \text{diag}(\sigma_\omega), \quad \hat{\Pi}_\omega = \text{fl}(\hat{H} D_\omega), \quad \hat{V}_\omega = \text{fl}(\hat{\Pi}_\omega U).$$

The laws of ω and of the subsequent uniform row sample are exact mathematical laws. If the matrix-product γ_m and row-scaling γ_s hypotheses are valid, then the computed-left perturbation event holds with probability one:

$$\mathbb{P}\left[\left\|\mathfrak{fl}(\widehat{V}_{\omega,S}^T \widehat{V}_{\omega,S}) - G_s(HD_\omega U)\right\|_F \leq \tau_{\omega,S}^{\text{scale}}\right] = 1.$$

The radius $\tau_{\omega,S}^{\text{scale}}$ is the existing computed-left budget instantiated with the rounded scale-table certificate for $\widehat{H} - H$, the rounded product certificate for $\widehat{\Pi}_\omega - HD_\omega$, the rounded formation of $\widehat{V}$, the uniform row-scaling arithmetic, and the final Gram dot products. No probability-construction or sampling-law error is introduced.

Lean formalization.

```

ComputedPreconditioner.flSignedHadamardScaledPattern
ComputedPreconditioner.flSignedHadamardScaledPattern_matrix
ComputedPreconditioner.flSignedHadamardScaledPattern_abs_error
ComputedPreconditioner.flSignedHadamardScaledPattern_entry_error_bound
signedHadamardScaledPatternPreconditioner
signedHadamardUniformRowTraceProbability_eventProb_scaledPatternComputedLeftPreconditioned_fl_
uniformRowPerturbEvent_eq_one

```

Corollary 224 (Scaled-pattern event with rounded stored Rademacher signs). *In the setting of the preceding corollary, suppose the realized Rademacher sign vector is stored by the rounded copy routine*

$$\widehat{\sigma}_{\omega,i} = \mathfrak{fl}(\sigma_{\omega,i} \cdot 1).$$

Since the exact Rademacher law gives $|\sigma_{\omega,i}| = 1$, the stored sign certificate satisfies

$$|\widehat{\sigma}_{\omega,i} - \sigma_{\omega,i}| \leq u =: E_{\omega,i}^\sigma.$$

Let

$$\widehat{D}_\omega = \text{diag}(\widehat{\sigma}_\omega), \quad \widehat{\Pi}_\omega = \mathfrak{fl}(\widehat{H}\widehat{D}_\omega), \quad \widehat{V}_\omega = \mathfrak{fl}(\widehat{\Pi}_\omega U),$$

where $\widehat{H}_{ij} = \text{flsqrt}(m^{-1})S_{ij}$. If the matrix-product γ_m and row-scaling γ_s hypotheses are valid, then, under the exact Rademacher/uniform-row product law,

$$\mathbb{P}\left[\left\|\mathfrak{fl}(\widehat{V}_{\omega,S}^T \widehat{V}_{\omega,S}) - G_s(HD_\omega U)\right\|_F \leq \tau_{\omega,S}^{\text{store}}\right] = 1.$$

The radius $\tau_{\omega,S}^{\text{store}}$ is the existing computed-left budget instantiated with the rounded scale-table certificate $\widehat{H} - H$, the stored-sign certificate $\widehat{\sigma}_\omega - \sigma_\omega$, the diagonal lift $\widehat{D}_\omega - D_\omega$, the rounded product certificate for $\widehat{\Pi}_\omega - HD_\omega$, the rounded formation of $\widehat{V}$, the uniform row-scaling arithmetic, and the final Gram dot products. No probability-construction or sampling-law error is introduced.

Lean formalization.

```

ComputedVector.flMulOne
ComputedVector.flStoredSign
rademacherSignVector_abs
ComputedPreconditioner.flSignedHadamardScaledPatternStoredSign
ComputedPreconditioner.flSignedHadamardScaledPatternStoredSign_matrix

```

```

ComputedPreconditioner.flSignedHadamardScaledPatternStoredSign_abs_error
ComputedPreconditioner.flSignedHadamardScaledPatternStoredSign_entry_error_bound
signedHadamardScaledPatternStoredSignPreconditioner
signedHadamardUniformRowTraceProbability_eventProb_scaledPatternStoredSignComputedLeftPreconditioned_
fl_uniformRowPerturbEvent_eq_one

```

Corollary 225 (Scaled-pattern event with add-zero stored Rademacher signs). *In the setting of the preceding corollary, suppose the realized Rademacher sign vector is stored by the rounded add-zero copy routine*

$$\widehat{\sigma}_{\omega,i}^+ = \text{fl}(\sigma_{\omega,i} + 0).$$

Since the exact Rademacher law gives $|\sigma_{\omega,i}| = 1$, the stored sign certificate satisfies

$$|\widehat{\sigma}_{\omega,i}^+ - \sigma_{\omega,i}| \leq u =: E_{\omega,i}^{\sigma,+}.$$

Let

$$\widehat{D}_{\omega}^+ = \text{diag}(\widehat{\sigma}_{\omega}^+), \quad \widehat{\Pi}_{\omega}^+ = \text{fl}(\widehat{H}\widehat{D}_{\omega}^+), \quad \widehat{V}_{\omega}^+ = \text{fl}(\widehat{\Pi}_{\omega}^+U),$$

where $\widehat{H}_{ij} = \text{flsqrt}(m^{-1})S_{ij}$. If the matrix-product γ_m and row-scaling γ_s hypotheses are valid, then, under the exact Rademacher/uniform-row product law,

$$\mathbb{P}\left[\left\|\text{fl}((\widehat{V}_{\omega,S}^+)^T\widehat{V}_{\omega,S}^+) - G_s(HD_{\omega}U)\right\|_F \leq \tau_{\omega,S}^{\text{store},+}\right] = 1.$$

The radius $\tau_{\omega,S}^{\text{store},+}$ is the existing computed-left budget instantiated with the rounded scale-table certificate $\widehat{H} - H$, the add-zero stored-sign certificate $\widehat{\sigma}_{\omega}^+ - \sigma_{\omega}$, the diagonal lift $\widehat{D}_{\omega}^+ - D_{\omega}$, the rounded product certificate for $\widehat{\Pi}_{\omega}^+ - HD_{\omega}$, the rounded formation of $\widehat{V}^+$, the uniform row-scaling arithmetic, and the final Gram dot products. No probability-construction or sampling-law error is introduced.

Lean formalization.

```

ComputedVector.flAddZeroRight
ComputedVector.flStoredSignAddZeroRight
rademacherSignVector_abs
ComputedPreconditioner.flSignedHadamardScaledPatternStoredSignAddZeroRight
ComputedPreconditioner.flSignedHadamardScaledPatternStoredSignAddZeroRight_matrix
ComputedPreconditioner.flSignedHadamardScaledPatternStoredSignAddZeroRight_abs_error
ComputedPreconditioner.flSignedHadamardScaledPatternStoredSignAddZeroRight_entry_error_bound
signedHadamardScaledPatternStoredSignAddZeroRightPreconditioner
signedHadamardUniformRowTraceProbability_eventProb_scaledPatternStoredSignAddZeroRightComputedLeftPreconditioned_
fl_uniformRowPerturbEvent_eq_one

```

Corollary 226 (Scaled-pattern event with subtract-zero stored Rademacher signs). *In the setting of the scaled-pattern computed-left corollary, suppose the realized Rademacher sign vector is stored by the rounded subtract-zero copy routine*

$$\widehat{\sigma}_{\omega,i}^- = \text{fl}(\sigma_{\omega,i} - 0).$$

Since the exact Rademacher law gives $|\sigma_{\omega,i}| = 1$, the stored sign certificate satisfies

$$|\hat{\sigma}_{\omega,i}^- - \sigma_{\omega,i}| \leq u =: E_{\omega,i}^{\sigma,-}.$$

Let

$$\hat{D}_{\omega}^- = \text{diag}(\hat{\sigma}_{\omega}^-), \quad \hat{\Pi}_{\omega}^- = \text{fl}(\hat{H}\hat{D}_{\omega}^-), \quad \hat{V}_{\omega}^- = \text{fl}(\hat{\Pi}_{\omega}^-U),$$

where $\hat{H}_{ij} = \text{flsqrt}(m^{-1})S_{ij}$. If the matrix-product γ_m and row-scaling γ_s hypotheses are valid, then, under the exact Rademacher/uniform-row product law,

$$\mathbb{P}\left[\left\|\text{fl}((\hat{V}_{\omega,S}^-)^T \hat{V}_{\omega,S}^-) - G_s(HD_{\omega}U)\right\|_F \leq \tau_{\omega,S}^{\text{store},-}\right] = 1.$$

The radius $\tau_{\omega,S}^{\text{store},-}$ is the existing computed-left budget instantiated with the rounded scale-table certificate $\hat{H} - H$, the subtract-zero stored-sign certificate $\hat{\sigma}_{\omega}^- - \sigma_{\omega}$, the diagonal lift $\hat{D}_{\omega}^- - D_{\omega}$, the rounded product certificate for $\hat{\Pi}_{\omega}^- - HD_{\omega}$, the rounded formation of $\hat{V}^-$, the uniform row-scaling arithmetic, and the final Gram dot products. No probability-construction or sampling-law error is introduced.

Lean formalization.

```

ComputedVector.flSubZeroRight
ComputedVector.flStoredSignSubZeroRight
rademacherSignVector_abs
ComputedPreconditioner.flSignedHadamardScaledPatternStoredSignSubZeroRight
ComputedPreconditioner.flSignedHadamardScaledPatternStoredSignSubZeroRight_matrix
ComputedPreconditioner.flSignedHadamardScaledPatternStoredSignSubZeroRight_abs_error
ComputedPreconditioner.flSignedHadamardScaledPatternStoredSignSubZeroRight_entry_error_bound
signedHadamardScaledPatternStoredSignSubZeroRightPreconditioner
signedHadamardUniformRowTraceProbability_eventProb_scaledPatternStoredSignSubZeroRightComputedLeftPreconditioned_
fl_uniformRowPerturbEvent_eq_one

```

Corollary 227 (Generated Sylvester/Walsh scaled-pattern event). *Let $p \in \mathbb{N}$ and $m = 2^p$. For $i, j \in \{0, \dots, m-1\}$, define the exact bit-parity predicate*

$$b(i, j) = \text{odd}\left(\sum_{\ell=0}^{p-1} \mathbf{1}\{\text{bit}_{\ell}(i) = 1 \text{ and } \text{bit}_{\ell}(j) = 1\}\right)$$

and the generated Sylvester/Walsh sign pattern

$$S_{ij}^{\text{syl}} = \begin{cases} -1, & b(i, j), \\ 1, & \neg b(i, j). \end{cases}$$

The generation of S^{syl} is exact integer/Boolean arithmetic, not a sampling-probability computation. Hence

$$|S_{ij}^{\text{syl}}| = 1, \quad (S_{ij}^{\text{syl}})^2 = 1,$$

and the ideal scaled table

$$H_{ij}^{\text{syl}} = \sqrt{m^{-1}} S_{ij}^{\text{syl}}$$

satisfies the flatness hypothesis $(H_{ij}^{\text{syl}})^2 = m^{-1}$. The computed table is

$$\hat{H}_{ij}^{\text{syl}} = \text{flsqrt}(m^{-1}) S_{ij}^{\text{syl}},$$

and the rounded scale certificate is

$$|\hat{H}_{ij}^{\text{syl}} - H_{ij}^{\text{syl}}| \leq \sqrt{m^{-1}} u =: E_{ij}^{H,\text{syl}}.$$

For an exact Rademacher sign vector σ_ω , let

$$D_\omega = \text{diag}(\sigma_\omega), \quad \hat{\Pi}_\omega^{\text{syl}} = \text{fl}(\hat{H}^{\text{syl}} D_\omega), \quad \hat{V}_\omega^{\text{syl}} = \text{fl}(\hat{\Pi}_\omega^{\text{syl}} U).$$

If the matrix-product γ_m and row-scaling γ_s hypotheses are valid, then under the exact Rademacher/uniform-row product law,

$$\mathbb{P} \left[\left\| \text{fl}((\hat{V}_{\omega,S}^{\text{syl}})^T \hat{V}_{\omega,S}^{\text{syl}}) - G_s(H^{\text{syl}} D_\omega U) \right\|_F \leq \tau_{\omega,S}^{\text{syl}} \right] = 1.$$

The same probability-one statement holds for the three rounded sign-storage variants $\hat{\sigma}_{\omega,i} = \text{fl}(\sigma_{\omega,i} \cdot 1)$, $\hat{\sigma}_{\omega,i}^+ = \text{fl}(\sigma_{\omega,i} + 0)$, and $\hat{\sigma}_{\omega,i}^- = \text{fl}(\sigma_{\omega,i} - 0)$, with radii $\tau_{\omega,S}^{\text{syl,store}}$, $\tau_{\omega,S}^{\text{syl,store},+}$, and $\tau_{\omega,S}^{\text{syl,store},-}$, respectively. These radii charge the rounded scale table, optional rounded sign storage, the diagonal lift, the rounded HD_ω product, rounded $\hat{V}$ formation, uniform row-scaling arithmetic, and final Gram dot products. They do not introduce probability-construction, probability-normalization, or sampling-law transfer terms. A fast in-place FHT recurrence/apply implementation remains a separate not-proved ledger obligation unless it materializes this generated table.

Lean formalization.

```

sylvesterHadamardParity
sylvesterHadamardSignPattern
sylvesterHadamardSignPattern_abs
sylvesterHadamardSignPattern_isSignPattern
hadamardFlat_sqrt_inv_nat_mul_sylvesterSignPattern
ComputedMatrix.flSqrtInvNatScaledSylvesterPattern
ComputedMatrix.flSqrtInvNatScaledSylvesterPattern_entry_error_bound
ComputedPreconditioner.flSignedHadamardSylvesterPattern
ComputedPreconditioner.flSignedHadamardSylvesterPatternStoredSign
ComputedPreconditioner.flSignedHadamardSylvesterPatternStoredSignAddZeroRight
ComputedPreconditioner.flSignedHadamardSylvesterPatternStoredSignSubZeroRight
signedHadamardSylvesterPatternPreconditioner
signedHadamardSylvesterPatternStoredSignPreconditioner
signedHadamardSylvesterPatternStoredSignAddZeroRightPreconditioner
signedHadamardSylvesterPatternStoredSignSubZeroRightPreconditioner
signedHadamardUniformRowTraceProbability_eventProb_sylvesterPatternComputedLeftPreconditioned_fl_
uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_sylvesterPatternStoredSignComputedLeftPreconditioned_
fl_uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_sylvesterPatternStoredSignAddZeroRightComputedLeftPreconditioned_
fl_uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_sylvesterPatternStoredSignSubZeroRightComputedLeftPreconditioned_
fl_uniformRowPerturbEvent_eq_one

```

Corollary 228 (Sylvester/Walsh parity-count sign recurrence). *Let $p \in \mathbb{N}$, and let S^{syl} be the exact Sylvester/Walsh sign table defined by the bit-parity count above. For $i, i', j \in \{0, \dots, 2^p - 1\}$ and a Boolean q , suppose*

$$\sum_{\ell=0}^{p-1} \mathbf{1}\{\text{bit}_\ell(i') = 1 \text{ and } \text{bit}_\ell(j) = 1\} = \sum_{\ell=0}^{p-1} \mathbf{1}\{\text{bit}_\ell(i) = 1 \text{ and } \text{bit}_\ell(j) = 1\} + \mathbf{1}\{q\}.$$

Then

$$S_{i'j}^{\text{syl}} = \begin{cases} -S_{ij}^{\text{syl}}, & q, \\ S_{ij}^{\text{syl}}, & \neg q. \end{cases}$$

In particular, equal parity counts give equal signs, while increasing the parity count by one gives the negated sign. This is the exact algebraic bridge from the one-stage partner bit facts to the Sylvester/Walsh recurrence: once the bit-toggle proof supplies the displayed count identity, the sign recurrence follows immediately. It is deterministic integer/Boolean arithmetic, not a probability-law theorem and not a floating-point theorem. The Rademacher and uniform-row sampling laws remain exact mathematical inputs.

Lean formalization.

```
sylvesterHadamardSignPattern_eq_or_neg_of_parityWeight_eq_add_bool
sylvesterHadamardSignPattern_eq_of_parityWeight_eq
sylvesterHadamardSignPattern_neg_of_parityWeight_eq_add_one
```

Corollary 229 (Abstract stage-bit parity-count recurrence). *Let $p \in \mathbb{N}$, let $0 \leq r < p$, and let S^{syl} be the exact Sylvester/Walsh sign table. Let $i, i', j \in \{0, \dots, 2^p - 1\}$. Suppose that*

$$\text{bit}_r(i) = 0, \quad \text{bit}_r(i') = 1,$$

and that the two row indices agree in every non-stage bit:

$$\text{bit}_\ell(i') = \text{bit}_\ell(i) \quad \text{for every } 0 \leq \ell < p, \ell \neq r.$$

Then the parity-count identity needed by the FHT recurrence is

$$\sum_{\ell=0}^{p-1} \mathbf{1}\{\text{bit}_\ell(i') = 1 \text{ and } \text{bit}_\ell(j) = 1\} = \sum_{\ell=0}^{p-1} \mathbf{1}\{\text{bit}_\ell(i) = 1 \text{ and } \text{bit}_\ell(j) = 1\} + \mathbf{1}\{\text{bit}_r(j) = 1\}.$$

Consequently,

$$S_{i'j}^{\text{syl}} = \begin{cases} -S_{ij}^{\text{syl}}, & \text{bit}_r(j) = 1, \\ S_{ij}^{\text{syl}}, & \text{bit}_r(j) = 0. \end{cases}$$

This closes the abstract finite-sum stage-bit part of the generated-FHT realization proof. The concrete generated partners $i + 2^r$ and $i - 2^r$ are instantiated in the next corollary. The result is exact deterministic integer/Boolean arithmetic; it does not add floating-point error or probability-law transfer. The Rademacher and uniform row sampling laws remain exact mathematical inputs.

Lean formalization.

```
sylvesterHadamardParityWeight_partner_eq_add_stage_of_bits
sylvesterHadamardSignPattern_partner_eq_or_neg_of_stage_bit
```

Corollary 230 (Generated stage bit-toggle parity recurrences). *Let $p \in \mathbb{N}$, $0 \leq r < p$, and let $i, j \in \{0, \dots, 2^p - 1\}$. The generic bit lemmas prove: if $\text{bit}_r(a) = 0$, then*

$$\text{bit}_\ell(a + 2^r) = \text{bit}_\ell(a) \quad (\ell \neq r),$$

and if $\text{bit}_r(a) = 1$, then

$$\text{bit}_\ell(a - 2^r) = \text{bit}_\ell(a) \quad (\ell \neq r).$$

Consequently the generated lower-half block test

$$i \bmod (2 \cdot 2^r) < 2^r$$

implies the concrete parity-count recurrence

$$\sum_{\ell=0}^{p-1} \mathbf{1}\{\text{bit}_\ell(i + 2^r) = 1 \text{ and } \text{bit}_\ell(j) = 1\} = \sum_{\ell=0}^{p-1} \mathbf{1}\{\text{bit}_\ell(i) = 1 \text{ and } \text{bit}_\ell(j) = 1\} + \mathbf{1}\{\text{bit}_r(j) = 1\},$$

and therefore

$$S_{i+2^r, j}^{\text{syl}} = \begin{cases} -S_{ij}^{\text{syl}}, & \text{bit}_r(j) = 1, \\ S_{ij}^{\text{syl}}, & \text{bit}_r(j) = 0. \end{cases}$$

Similarly, the generated upper-half block test

$$2^r \leq i \bmod (2 \cdot 2^r)$$

implies

$$\sum_{\ell=0}^{p-1} \mathbf{1}\{\text{bit}_\ell(i) = 1 \text{ and } \text{bit}_\ell(j) = 1\} = \sum_{\ell=0}^{p-1} \mathbf{1}\{\text{bit}_\ell(i - 2^r) = 1 \text{ and } \text{bit}_\ell(j) = 1\} + \mathbf{1}\{\text{bit}_r(j) = 1\},$$

and therefore

$$S_{ij}^{\text{syl}} = \begin{cases} -S_{i-2^r, j}^{\text{syl}}, & \text{bit}_r(j) = 1, \\ S_{i-2^r, j}^{\text{syl}}, & \text{bit}_r(j) = 0. \end{cases}$$

This closes the concrete generated-partner bit arithmetic feeding the exact FHT realization proof. The remaining exact theorem is the one-stage parity-table recurrence that combines these sign recurrences with the lower/upper butterfly coordinate formulas, followed by the induction over stages. These are exact deterministic integer/FHT facts; no probability construction, probability normalization, sampling-law transfer, or floating-point arithmetic is introduced here.

Lean formalization.

```

nat_testBit_add_two_pow_eq_of_testBit_eq_false_of_ne
two_pow_le_of_testBit_eq_true
nat_testBit_sub_two_pow_eq_of_testBit_eq_true_of_ne
fhtSylvesterStage_upper_partner_testBit_eq_of_ne_of_mod_lt
fhtSylvesterStage_lower_partner_testBit_eq_of_ne_of_mod_ge
sylvesterHadamardParityWeight_upper_partner_eq_add_stage_of_mod_lt
sylvesterHadamardSignPattern_upper_partner_eq_or_neg_of_mod_lt
sylvesterHadamardParityWeight_upper_eq_lower_partner_add_stage_of_mod_ge
sylvesterHadamardSignPattern_upper_eq_or_neg_lower_partner_of_mod_ge

```

Corollary 231 (One-butterfly FHT arithmetic certificate). *Consider one scalar pair update in a fast Hadamard transform stage. The exact butterfly sends*

$$(a, b) \mapsto (y_+, y_-) = (a + b, a - b),$$

while the implemented floating-point butterfly is

$$(\widehat{y}_+, \widehat{y}_-) = (\text{fladd}(a, b), \text{flsub}(a, b)).$$

Under the floating-point model with unit roundoff u , the two computed outputs satisfy

$$|\widehat{y}_+ - y_+| \leq u |y_+|, \quad |\widehat{y}_- - y_-| \leq u |y_-|.$$

*This is an implementation-facing certificate for the non-probability arithmetic inside one FHT butterfly: it charges the rounded addition and rounded subtraction. It does not introduce probability-construction, probability-normalization, or sampling-law transfer terms; the Rademacher and uniform row laws remain exact mathematical laws. It is also not yet a full fast-Hadamard apply theorem. A complete FHT implementation must still compose this primitive across a concrete stage schedule and account for scaling, storage/overwrite effects, vector or matrix application, and the resulting *ComputedMatrix* or *ComputedPreconditioner* certificate.*

Lean formalization.

```
fhtButterflyExact
flFhtButterfly
flFhtButterfly_fst
flFhtButterfly_snd
flFhtButterfly_add_error_bound
flFhtButterfly_sub_error_bound
```

Corollary 232 (Vector-pair FHT update certificate). *Let $x \in \mathbb{R}^n$, and let $p, q \in \{0, \dots, n-1\}$ be one pair of coordinates selected by an FHT stage. Define the exact pair update x' by*

$$x'_i = \begin{cases} x_p + x_q, & i = p, \\ x_p - x_q, & i = q \text{ and } i \neq p, \\ x_i, & \text{otherwise,} \end{cases}$$

and define the computed pair update $\widehat{x}'$ by

$$\widehat{x}'_i = \begin{cases} \text{fladd}(x_p, x_q), & i = p, \\ \text{flsub}(x_p, x_q), & i = q \text{ and } i \neq p, \\ x_i, & \text{otherwise.} \end{cases}$$

Set the entrywise butterfly budget

$$B_i = \begin{cases} u |x_p + x_q|, & i = p, \\ u |x_p - x_q|, & i = q \text{ and } i \neq p, \\ 0, & \text{otherwise.} \end{cases}$$

Then $B_i \geq 0$ for every i , and

$$|\hat{x}'_i - x'_i| \leq B_i \quad \text{for every } i.$$

This lifts the one-butterfly scalar arithmetic certificate to the vector shape used by an in-place FHT stage. The unchanged coordinates are treated as exact reference entries; if an implementation copies or overwrites them through rounded storage, that storage path needs its own certificate. This result still does not compose stages or scaling, and it introduces no probability-law or probability-construction error term.

Lean formalization.

```
fhtPairUpdateExact
flFhtPairUpdate
fhtPairUpdateErrorBudget
fhtPairUpdateErrorBudget_nonneg
flFhtPairUpdate_error_bound
```

Corollary 233 (Propagated vector-pair FHT update certificate). *Let $x, \hat{x} \in \mathbb{R}^n$, let $p, q \in \{0, \dots, n-1\}$, and suppose the input to one FHT pair update already satisfies an entrywise certificate*

$$|\hat{x}_i - x_i| \leq E_i, \quad E_i \geq 0 \quad \text{for every } i.$$

Let x' be the exact pair update applied to x ,

$$x'_i = \begin{cases} x_p + x_q, & i = p, \\ x_p - x_q, & i = q \text{ and } i \neq p, \\ x_i, & \text{otherwise,} \end{cases}$$

and let $\hat{x}'$ be the rounded pair update applied to the already computed vector $\hat{x}$,

$$\hat{x}'_i = \begin{cases} \text{fladd}(\hat{x}_p, \hat{x}_q), & i = p, \\ \text{flsub}(\hat{x}_p, \hat{x}_q), & i = q \text{ and } i \neq p, \\ \hat{x}_i, & \text{otherwise.} \end{cases}$$

Define the propagated one-pair budget

$$B_i^{\text{prop}} = \begin{cases} u|\hat{x}_p + \hat{x}_q| + E_p + E_q, & i = p, \\ u|\hat{x}_p - \hat{x}_q| + E_p + E_q, & i = q \text{ and } i \neq p, \\ E_i, & \text{otherwise.} \end{cases}$$

Then $B_i^{\text{prop}} \geq 0$ and

$$|\hat{x}'_i - x'_i| \leq B_i^{\text{prop}} \quad \text{for every } i.$$

The terms $E_p + E_q$ propagate the already-present non-probability input error from earlier FHT stages, while the $u|\hat{x}_p \pm \hat{x}_q|$ terms charge the new rounded addition/subtraction in this pair update. The unchanged coordinates retain their previous entrywise certificate. This is still a one-pair certificate, not a full fast-Hadamard recurrence: a complete implementation-facing FHT theorem must compose these budgets over the concrete stage schedule and add scale, copy, overwrite, vector-apply, and matrix-apply certificates as needed. Sampling probabilities, Rademacher signs as random laws, and uniform row laws remain exact mathematical inputs.

Lean formalization.

```
abs_add_pair_sub_add_pair_le
abs_sub_pair_sub_sub_pair_le
flFhtButterfly_add_error_bound_of_input_error
flFhtButterfly_sub_error_bound_of_input_error
fhtPairUpdatePropagatedErrorBudget
fhtPairUpdatePropagatedErrorBudget_nonneg
flFhtPairUpdate_propagated_error_bound
```

Corollary 234 (Ordered FHT pair-schedule propagation certificate). *Let*

$$\mathcal{P} = ((p_1, q_1), \dots, (p_T, q_T))$$

be any finite ordered list of FHT pair updates on $\mathbb{R}^n$. Starting from exact and computed inputs $x^{(0)} = x$, $\hat{x}^{(0)} = \hat{x}$, and an entrywise initial budget $E^{(0)} = E$, define recursively

$$x^{(r+1)} = \text{PairExact}_{p_{r+1}, q_{r+1}}(x^{(r)}), \quad \hat{x}^{(r+1)} = \text{PairFl}_{p_{r+1}, q_{r+1}}(\hat{x}^{(r)}),$$

and set $E^{(r+1)} = B^{\text{prop}}(p_{r+1}, q_{r+1}, \hat{x}^{(r)}, E^{(r)})$, where B^{prop} is the one-pair budget from Corollary 203:

$$E_i^{(r+1)} = \begin{cases} u |\hat{x}_{p_{r+1}}^{(r)} + \hat{x}_{q_{r+1}}^{(r)}| + E_{p_{r+1}}^{(r)} + E_{q_{r+1}}^{(r)}, & i = p_{r+1}, \\ u |\hat{x}_{p_{r+1}}^{(r)} - \hat{x}_{q_{r+1}}^{(r)}| + E_{p_{r+1}}^{(r)} + E_{q_{r+1}}^{(r)}, & i = q_{r+1} \text{ and } i \neq p_{r+1}, \\ E_i^{(r)}, & \text{otherwise.} \end{cases}$$

If the initial budget satisfies $E_i \geq 0$ and

$$|\hat{x}_i - x_i| \leq E_i \quad \text{for every } i,$$

then every recursively propagated budget is nonnegative and the final exact and rounded schedule outputs satisfy

$$|\hat{x}_i^{(T)} - x_i^{(T)}| \leq E_i^{(T)} \quad \text{for every } i.$$

This composes the visible rounded-addition/subtraction and previously propagated non-probability error across any supplied ordered pair schedule. It still does not add the FHT normalization scale, prove a concrete Hadamard stage-list generator, model storage/copy paths beyond the functional update, or lift the vector schedule to a matrix/vector apply certificate. The sampling laws remain exact mathematical inputs.

Lean formalization.

```
fhtPairScheduleExact
flFhtPairSchedule
fhtPairSchedulePropagatedErrorBudget
fhtPairSchedulePropagatedErrorBudget_nonneg
flFhtPairSchedule_propagated_error_bound
```

Corollary 235 (Scaled ordered FHT pair-schedule certificate). *Let $\mathcal{P}$ be any finite ordered list of FHT pair updates on $\mathbb{R}^n$. Let*

$$y_i = \text{FHTSchedule}_{\mathcal{P}}(x)_i, \quad \hat{y}_i = \text{fFHTSchedule}_{\mathcal{P}}(\hat{x})_i,$$

and let $E_i^{\mathcal{P}}$ be the propagated schedule budget from Corollary 204. Assume

$$E_i \geq 0, \quad |\hat{x}_i - x_i| \leq E_i \quad \text{for every } i,$$

and suppose the FHT normalization scale is represented by an exact analysis value c , a computed value $\hat{c}$, and a certificate

$$|\hat{c} - c| \leq \eta, \quad \eta \geq 0.$$

Define the exact scaled schedule output and the rounded implementation output by

$$z_i = c y_i, \quad \hat{z}_i = \text{flmul}(\hat{c}, \hat{y}_i).$$

For each coordinate set

$$B_i^{\text{scale}} = u |\hat{c} \hat{y}_i| + |\hat{c}| E_i^{\mathcal{P}} + \eta (|\hat{y}_i| + E_i^{\mathcal{P}}).$$

Then $B_i^{\text{scale}} \geq 0$ and

$$|\hat{z}_i - z_i| \leq B_i^{\text{scale}} \quad \text{for every } i.$$

Thus the generic scaled FHT schedule charges the final rounded multiplication, the already-propagated schedule error, and the computed-scale radius. This is an implementation-facing non-probability arithmetic certificate: Rademacher and uniform row sampling laws remain exact mathematical inputs, and no probability-construction or law-transfer term is introduced. A full fast FHT implementation still needs a concrete stage-list generator, a concrete normalization-scale routine when $\hat{c}$ is computed by code, storage/copy/overwrite certificates for the chosen layout, vector/matrix apply wrappers, and a final `ComputedMatrix`/`ComputedPreconditioner` instantiation.

Lean formalization.

```
fhtScaledPairScheduleExact
flFhtScaledPairSchedule
fhtScaledPairScheduleErrorBudget
fhtScaledPairScheduleErrorBudget_nonneg
flFhtScaledPairSchedule_error_bound
```

Corollary 236 (Columnwise computed-matrix FHT schedule certificate). *Let $A \in \mathbb{R}^{m \times n}$ be the exact analysis input matrix and let $\hat{A}$ be a computed input matrix with entrywise certificate*

$$|\hat{A}_{ij} - A_{ij}| \leq E_{ij}^A, \quad E_{ij}^A \geq 0.$$

Let $\mathcal{P}$ be a supplied ordered list of pair updates on the row index $\{1, \dots, m\}$, and let the scale certificate satisfy

$$|\hat{c} - c| \leq \eta, \quad \eta \geq 0.$$

*For each column j , apply the exact scaled schedule to A_{*j} and the rounded scaled schedule to $\hat{A}_{*j}$:*

$$Y_{ij} = \text{FHTScheduleMatrix}_{\mathcal{P},c}(A)_{ij}, \quad \hat{Y}_{ij} = \text{fFHTScheduleMatrix}_{\mathcal{P},\hat{c}}(\hat{A})_{ij}.$$

Let B_{ij}^{mat} be the columnwise budget obtained from Corollary 205, namely

$$B_{ij}^{\text{mat}} = u |\widehat{c} \widehat{y}_{ij}| + |\widehat{c}| E_{ij}^{\mathcal{P}} + \eta (|\widehat{y}_{ij}| + E_{ij}^{\mathcal{P}}),$$

where $\widehat{y}_{ij}$ is the rounded unscaled schedule output in column j and $E_{ij}^{\mathcal{P}}$ is the propagated unscaled schedule budget for that column. Then $B_{ij}^{\text{mat}} \geq 0$ and

$$|\widehat{Y}_{ij} - Y_{ij}| \leq B_{ij}^{\text{mat}} \quad \text{for every } i, j.$$

Consequently the rounded columnwise scheduled FHT output is a **ComputedMatrix** certificate for the exact supplied-schedule transform. This closes the vector-to-matrix lift for a supplied pair list. It still does not prove that the pair list is the Sylvester/Hadamard stage generator, nor does the generic adapter itself choose a normalization-scale routine, layout-specific storage/copy/overwrite model, or final chosen-transform **ComputedPreconditioner**. The following corollary instantiates the rounded square-root normalization scale.

Lean formalization.

```
fhtScaledPairScheduleMatrixExact
flFhtScaledPairScheduleMatrix
fhtScaledPairScheduleMatrixErrorBudget
fhtScaledPairScheduleMatrixErrorBudget_nonneg
flFhtScaledPairScheduleMatrix_error_bound
ComputedMatrix.flScaledFhtPairScheduleColumns
ComputedMatrix.flScaledFhtPairScheduleColumns_entry_error_bound
```

Corollary 237 (Rounded square-root FHT scale certificate). *Let m be the row dimension and let the exact FHT normalization scale, computed scale, and scale-error radius be*

$$c_m = \sqrt{m^{-1}}, \quad \widehat{c}_m = \text{flsqrt}(m^{-1}), \quad \eta_m = c_m u.$$

Then

$$\eta_m \geq 0, \quad |\widehat{c}_m - c_m| \leq \eta_m.$$

Consequently the generic scaled-schedule certificates above may be instantiated with $c = c_m$, $\widehat{c} = \widehat{c}_m$, and $\eta = \eta_m$.

For a vector input, suppose

$$|\widehat{x}_i - x_i| \leq E_i, \quad E_i \geq 0,$$

and let $z_i = \text{FHTSchedule}_{\mathcal{P}, c_m}(x)_i$ and $\widehat{z}_i = \text{flFHTSchedule}_{\mathcal{P}, \widehat{c}_m}(\widehat{x})_i$. If $\widehat{y}_i$ is the rounded unscaled schedule output and $E_i^{\mathcal{P}}$ is the propagated unscaled schedule budget, then

$$|\widehat{z}_i - z_i| \leq u |\widehat{c}_m \widehat{y}_i| + |\widehat{c}_m| E_i^{\mathcal{P}} + \eta_m (|\widehat{y}_i| + E_i^{\mathcal{P}}).$$

For a matrix input, suppose

$$|\widehat{A}_{ij} - A_{ij}| \leq E_{ij}^A, \quad E_{ij}^A \geq 0,$$

and apply the same supplied pair schedule columnwise. Then

$$\left| \text{flFHTScheduleMatrix}_{\mathcal{P}, \widehat{c}_m}(\widehat{A})_{ij} - \text{FHTScheduleMatrix}_{\mathcal{P}, c_m}(A)_{ij} \right| \leq u |\widehat{c}_m \widehat{y}_{ij}| + |\widehat{c}_m| E_{ij}^{\mathcal{P}} + \eta_m (|\widehat{y}_{ij}| + E_{ij}^{\mathcal{P}}).$$

The columnwise constructor packages this as a *ComputedMatrix* certificate. This is a fully computed-object result for the square-root FHT normalization scale in a supplied-schedule implementation: the exact pair schedule and exact input are the analysis references, while the rounded pair arithmetic, rounded final scale multiplication, input perturbation, and rounded $\text{flsqrt}(m^{-1})$ scale are charged. Rademacher signs, uniform row laws, and all sampling probabilities remain exact mathematical inputs.

Lean formalization.

```
fhtSqrtInvNatScale
flFhtSqrtInvNatScale
fhtSqrtInvNatScaleErrorRadius
fhtSqrtInvNatScaleErrorRadius_nonneg
flFhtSqrtInvNatScale_error_bound
flFhtScaledPairSchedule_sqrtInvNatScale_error_bound
flFhtScaledPairScheduleMatrix_sqrtInvNatScale_error_bound
ComputedMatrix.flScaledFhtPairScheduleColumnsSqrtInvNat
ComputedMatrix.flScaledFhtPairScheduleColumnsSqrtInvNat_entry_error_bound
```

Corollary 238 (Concrete one-stage FHT pair-list generator). *Let n be a finite row dimension and let s be a stage stride. For indices $a, b \in \{0, \dots, n-1\}$, define the one-stage FHT pair predicate*

$$G_s(a, b) \iff 0 < s, \quad b = a + s, \quad a \bmod (2s) < s.$$

Let $\text{Pairs}(n, s)$ be the finite list of all pairs satisfying this predicate. Then, for every pair (a, b) ,

$$(a, b) \in \text{Pairs}(n, s) \iff G_s(a, b).$$

Consequently, any generated pair satisfies

$$0 < s, \quad b = a + s, \quad a \bmod (2s) < s, \quad a < b, \quad a \neq b.$$

The one-stage list also has the first partner-uniqueness and lower/upper-half separation facts needed for the later exact stage formula. If $(a, b), (a', b') \in \text{Pairs}(n, s)$, then

$$a = a' \implies b = b', \quad b = b' \implies a = a', \quad a \neq b', \quad b \neq a'.$$

Equivalently, equal first or equal second coordinates identify the generated pair itself:

$$a = a' \implies (a, b) = (a', b'), \quad b = b' \implies (a, b) = (a', b').$$

The modular arithmetic used for the separation is explicit: whenever

$$0 < s, \quad a \bmod (2s) < s,$$

one has

$$s \leq (a + s) \bmod (2s).$$

For the Sylvester/Walsh dimension $n = 2^p$ and stage r , the generated stage list uses stride 2^r . In this case

$$(a, b) \in \text{SylvesterPairs}(p, r) \iff b = a + 2^r, \quad a \bmod (2 \cdot 2^r) < 2^r.$$

This closes the exact finite-index list-generation layer for one FHT stage. It is not a probability law and it does not charge probability construction. The later partial-transform recurrence and range induction close the transform-correctness proof that the composed generated stages realize the desired Sylvester/Hadamard transform. What remains after that exact result is to combine the rounded arithmetic with storage/layout and final `ComputedPreconditioner` certificates for a concrete implementation.

Lean formalization.

```
fhtStagePairPredicate
fhtStagePairs
mem_fhtStagePairs_iff
fhtStagePairs_stride_pos
fhtStagePairs_second_val_eq
fhtStagePairs_first_val_mod_lt
fhtStagePairs_first_val_lt_second_val
fhtStagePairs_fst_ne_snd
fhtStagePairs_snd_eq_of_fst_eq
fhtStagePairs_fst_eq_of_snd_eq
nat_stride_add_mod_two_stride_ge_of_mod_lt
fhtStagePairs_fst_ne_snd_of_mem_mem
fhtStagePairs_snd_ne_fst_of_mem_mem
fhtStagePairs_eq_of_fst_eq
fhtStagePairs_eq_of_snd_eq
fhtSylvesterStagePairs
mem_fhtSylvesterStagePairs_iff
```

Corollary 239 (Generated one-stage FHT pair updates commute). *Let $T_{a,b}$ denote the exact vector butterfly update that replaces (x_a, x_b) by $(x_a + x_b, x_a - x_b)$ and leaves all other coordinates unchanged. If*

$$(a, b), (a', b') \in \text{Pairs}(n, s), \quad (a, b) \neq (a', b'),$$

then the generated pairs have disjoint coordinates:

$$a \neq a', \quad a \neq b', \quad b \neq a', \quad b \neq b'.$$

Consequently, for every vector x ,

$$T_{a,b}(T_{a',b'}x) = T_{a',b'}(T_{a,b}x).$$

The same disjoint-coordinate conclusion is now formalized for the rounded functional update $\hat{T}_{a,b}$ and for the three modified-coordinate writeback updates that store only the two butterfly outputs through `fl_add(y, 0)`, `fl_mul(y, 1)`, or `fl_sub(y, 0)`. Thus, for the modeled functional schedules, same-stage generated-pair order does not introduce an extra aliasing or memory-order hypothesis beyond the already charged rounded butterfly and writeback arithmetic. Concrete Sylvester/Walsh wrappers specialize the same no-alias/list-order facts to $\text{Pairs}_{\text{Syl}}(p, r)$, so downstream generated-FHT proofs can use the 2^p -dimensional stage interface directly without unfolding it to a generic stride

schedule. Lean also proves the companion no-touch property for the rounded base schedule and all three modified-coordinate writeback schedules: if a coordinate is not the first or second entry of any pair in the supplied schedule list, then that coordinate is returned exactly unchanged. The generated-stage wrappers apply the same statement to $\text{Pairs}(n, s)$. This preservation result is not asserted for all-coordinate copy/writeback routines, which deliberately rewrite every entry. The same no-touch statement is also formalized for the propagated error budgets: under the same hypothesis, the base rounded budget and the three modified-coordinate budgets return exactly the incoming radius E_i . Thus the displayed modified-coordinate recurrences do not hide any copy contribution on coordinates outside the active pair list. The concrete Sylvester/Walsh wrappers specialize these value and budget facts to the full generated list $\text{Pairs}_{\text{syl}}(p)$ and to a single generated stage $\text{Pairs}_{\text{syl}}(p, r)$. Therefore a downstream proof can use the concrete generated-FHT interfaces directly: if coordinate i is outside every pair in the relevant full schedule or stage, then the rounded base schedule and the three modified-coordinate schedules return $\hat{x}_i$, and the corresponding propagated budgets return exactly E_i . These wrappers add no probability-law change and no extra floating-point term; all-coordinate copy routines remain outside this result. More generally, let L be an exact schedule list such that every $(c, d) \in L$ belongs to $\text{Pairs}(n, s)$ and $(a, b) \neq (c, d)$. If S_L denotes the exact schedule obtained by applying the updates in L , then

$$S_L(T_{a,b}x) = T_{a,b}(S_Lx).$$

This is the exact list-order bridge needed before proving that one generated stage realizes the simultaneous lower/upper-half FHT butterfly map. It does not introduce any probability or sampling-law approximation.

Lean formalization.

```
fhtPairUpdateExact_commute_of_disjoint
flFhtPairUpdate_commute_of_disjoint
fhtStagePairs_disjoint_of_ne
fhtStagePairs_pairUpdateExact_commute_of_ne
fhtPairScheduleExact_commute_update_of_forall
fhtStagePairs_pairUpdateExact_commute_schedule_of_ne
flFhtPairSchedule_commute_update_of_forall
fhtStagePairs_flFhtPairUpdate_commute_of_ne
fhtStagePairs_flFhtPairUpdate_commute_schedule_of_ne
flFhtPairUpdateModifiedStoredAddZeroRight_commute_of_disjoint
flFhtPairUpdateModifiedStoredMulOne_commute_of_disjoint
flFhtPairUpdateModifiedStoredSubZeroRight_commute_of_disjoint
flFhtPairScheduleModifiedStoredAddZeroRight_commute_update_of_forall
flFhtPairScheduleModifiedStoredMulOne_commute_update_of_forall
flFhtPairScheduleModifiedStoredSubZeroRight_commute_update_of_forall
fhtStagePairs_flFhtPairUpdateModifiedStoredAddZeroRight_commute_of_ne
fhtStagePairs_flFhtPairUpdateModifiedStoredAddZeroRight_commute_schedule_of_ne
fhtStagePairs_flFhtPairUpdateModifiedStoredMulOne_commute_of_ne
fhtStagePairs_flFhtPairUpdateModifiedStoredMulOne_commute_schedule_of_ne
fhtStagePairs_flFhtPairUpdateModifiedStoredSubZeroRight_commute_of_ne
```

fhtStagePairs_flFhtPairUpdateModifiedStoredSubZeroRight_commute_schedule_of_ne
 fhtSylvesterStagePairs_disjoint_of_ne
 fhtSylvesterStagePairs_pairUpdateExact_commute_of_ne
 fhtSylvesterStagePairs_pairUpdateExact_commute_schedule_of_ne
 fhtSylvesterStagePairs_flFhtPairUpdate_commute_of_ne
 fhtSylvesterStagePairs_flFhtPairUpdate_commute_schedule_of_ne
 fhtSylvesterStagePairs_flFhtPairUpdateModifiedStoredAddZeroRight_commute_of_ne
 fhtSylvesterStagePairs_flFhtPairUpdateModifiedStoredAddZeroRight_commute_schedule_of_ne
 fhtSylvesterStagePairs_flFhtPairUpdateModifiedStoredMulOne_commute_of_ne
 fhtSylvesterStagePairs_flFhtPairUpdateModifiedStoredMulOne_commute_schedule_of_ne
 fhtSylvesterStagePairs_flFhtPairUpdateModifiedStoredSubZeroRight_commute_of_ne
 fhtSylvesterStagePairs_flFhtPairUpdateModifiedStoredSubZeroRight_commute_schedule_of_ne
 flFhtPairSchedule_apply_of_forall_not_mem
 flFhtPairScheduleModifiedStoredAddZeroRight_apply_of_forall_not_mem
 flFhtPairScheduleModifiedStoredMulOne_apply_of_forall_not_mem
 flFhtPairScheduleModifiedStoredSubZeroRight_apply_of_forall_not_mem
 fhtStagePairs_flFhtPairSchedule_apply_of_forall_not_mem
 fhtStagePairs_flFhtPairScheduleModifiedStoredAddZeroRight_apply_of_forall_not_mem
 fhtStagePairs_flFhtPairScheduleModifiedStoredMulOne_apply_of_forall_not_mem
 fhtStagePairs_flFhtPairScheduleModifiedStoredSubZeroRight_apply_of_forall_not_mem
 fhtPairSchedulePropagatedErrorBudget_apply_of_forall_not_mem
 fhtPairScheduleModifiedStoredAddZeroRightPropagatedErrorBudget_apply_of_forall_not_mem
 fhtPairScheduleModifiedStoredMulOnePropagatedErrorBudget_apply_of_forall_not_mem
 fhtPairScheduleModifiedStoredSubZeroRightPropagatedErrorBudget_apply_of_forall_not_mem
 fhtStagePairs_fhtPairSchedulePropagatedErrorBudget_apply_of_forall_not_mem
 fhtStagePairs_fhtPairScheduleModifiedStoredAddZeroRightPropagatedErrorBudget_apply_of_forall_not_mem
 mem
 fhtStagePairs_fhtPairScheduleModifiedStoredMulOnePropagatedErrorBudget_apply_of_forall_not_mem
 fhtStagePairs_fhtPairScheduleModifiedStoredSubZeroRightPropagatedErrorBudget_apply_of_forall_not_mem
 mem
 flFhtSylvesterSchedule_apply_of_forall_not_mem
 flFhtSylvesterScheduleModifiedStoredAddZeroRight_apply_of_forall_not_mem
 flFhtSylvesterScheduleModifiedStoredMulOne_apply_of_forall_not_mem
 flFhtSylvesterScheduleModifiedStoredSubZeroRight_apply_of_forall_not_mem
 fhtSylvesterSchedulePropagatedErrorBudget_apply_of_forall_not_mem
 fhtSylvesterScheduleModifiedStoredAddZeroRightPropagatedErrorBudget_apply_of_forall_not_mem
 fhtSylvesterScheduleModifiedStoredMulOnePropagatedErrorBudget_apply_of_forall_not_mem
 fhtSylvesterScheduleModifiedStoredSubZeroRightPropagatedErrorBudget_apply_of_forall_not_mem
 flFhtSylvesterStageScheduleModifiedStoredAddZeroRight
 flFhtSylvesterStageScheduleModifiedStoredMulOne

```

flFhtSylvesterStageScheduleModifiedStoredSubZeroRight
fhtSylvesterStageScheduleModifiedStoredAddZeroRightPropagatedErrorBudget
fhtSylvesterStageScheduleModifiedStoredMulOnePropagatedErrorBudget
fhtSylvesterStageScheduleModifiedStoredSubZeroRightPropagatedErrorBudget
flFhtSylvesterStageSchedule_apply_of_forall_not_mem
flFhtSylvesterStageScheduleModifiedStoredAddZeroRight_apply_of_forall_not_mem
flFhtSylvesterStageScheduleModifiedStoredMulOne_apply_of_forall_not_mem
flFhtSylvesterStageScheduleModifiedStoredSubZeroRight_apply_of_forall_not_mem
fhtSylvesterStageSchedulePropagatedErrorBudget_apply_of_forall_not_mem
fhtSylvesterStageScheduleModifiedStoredAddZeroRightPropagatedErrorBudget_apply_of_forall_not_mem
fhtSylvesterStageScheduleModifiedStoredMulOnePropagatedErrorBudget_apply_of_forall_not_mem
fhtSylvesterStageScheduleModifiedStoredSubZeroRightPropagatedErrorBudget_apply_of_forall_not_mem

```

Corollary 240 (Exact output of one generated FHT stage). *Let $\text{Pairs}(n, s)$ be the generated one-stage pair list above, and let $S_{n,s}$ be the exact ordered schedule obtained by applying all butterfly updates in that list. For every generated pair $(a, b) \in \text{Pairs}(n, s)$ and every vector x ,*

$$S_{n,s}(x)_a = x_a + x_b, \quad S_{n,s}(x)_b = x_a - x_b.$$

Equivalently, in the Sylvester/Walsh specialization, if $(a, b) \in \text{SylvesterPairs}(p, r)$, then the one-stage schedule $S_{p,r}$ satisfies

$$S_{p,r}(x)_a = x_a + x_b, \quad S_{p,r}(x)_b = x_a - x_b.$$

The proof uses only exact finite-list arithmetic: the generated stage list has no duplicate pairs, and every other pair in the same stage is disjoint from (a, b) , hence later updates leave the target coordinates unchanged. This is an exact-object theorem, not a probability-law theorem, and it introduces no probability-construction or sampling-law approximation.

Lean formalization.

```

fhtStagePairs_nodup
fhtPairScheduleExact_apply_of_forall_not_mem
fhtPairScheduleExact_apply_pair_of_mem_stage_list
fhtStagePairs_pairScheduleExact_apply_pair_of_mem
fhtStagePairs_pairScheduleExact_apply_fst_of_mem
fhtStagePairs_pairScheduleExact_apply_snd_of_mem
fhtSylvesterStageScheduleExact_apply_pair_of_mem
fhtSylvesterStageScheduleExact_apply_fst_of_mem
fhtSylvesterStageScheduleExact_apply_snd_of_mem

```

Corollary 241 (Coordinate formulas for one generated Sylvester FHT stage). *Let $p, r \in \mathbb{N}$, let $S_{p,r}$ be the exact generated stage with stride 2^r in dimension 2^p , and let $x \in \mathbb{R}^{2^p}$. If i is a lower-half coordinate in its stride block, meaning*

$$i + 2^r < 2^p, \quad i \bmod (2 \cdot 2^r) < 2^r,$$

then the generated stage satisfies the explicit coordinate formula

$$S_{p,r}(x)_i = x_i + x_{i+2^r}.$$

If i is treated as an upper-half coordinate with lower partner $\ell = i - 2^r$, meaning

$$2^r \leq i, \quad (i - 2^r) \bmod (2 \cdot 2^r) < 2^r,$$

then

$$S_{p,r}(x)_i = x_{i-2^r} - x_i.$$

Equivalently, the exact finite-index generator contains the constructed pairs

$$(i, i + 2^r) \in \text{SylvesterPairs}(p, r), \quad (i - 2^r, i) \in \text{SylvesterPairs}(p, r)$$

under the displayed hypotheses. The upper-bound and modulus hypotheses are kept explicit in this statement; the next corollary discharges them from the power-of-two block tests used by the generated stage. This is an exact arithmetic result and does not introduce probability-construction or sampling-law error.

Lean formalization.

```
fhtStagePairs_mem_lower_mk
fhtSylvesterStagePairs_mem_lower_mk
fhtSylvesterStagePairs_mem_upper_mk
fhtSylvesterStageScheduleExact_apply_lower_mk
fhtSylvesterStageScheduleExact_apply_upper_mk
```

Corollary 242 (Power-of-two block-test formulas for one generated Sylvester FHT stage). *Let $p, r \in \mathbb{N}$ with $r < p$, let $S_{p,r}$ be the exact generated stage with stride 2^r in dimension 2^p , and let $x \in \mathbb{R}^{2^p}$. If a coordinate i satisfies the lower-half block test*

$$i \bmod (2 \cdot 2^r) < 2^r,$$

then the upper partner is automatically in bounds,

$$i + 2^r < 2^p,$$

and the exact generated stage satisfies

$$S_{p,r}(x)_i = x_i + x_{i+2^r}.$$

If i satisfies the upper-half block test

$$2^r \leq i \bmod (2 \cdot 2^r),$$

then $2^r \leq i$, the lower partner $i - 2^r$ satisfies

$$(i - 2^r) \bmod (2 \cdot 2^r) < 2^r,$$

and

$$S_{p,r}(x)_i = x_{i-2^r} - x_i.$$

The proof uses only exact integer arithmetic: $2 \cdot 2^r \mid 2^p$, the lower-half in-bounds lemma, and the upper-half subtract-stride modulus lemma. It is therefore a deterministic exact-object FHT result. The Rademacher and uniform-row sampling laws remain exact mathematical inputs, and no probability-construction, probability-normalization, or law-transfer error is introduced.

Lean formalization.

```

nat_add_stride_lt_of_mod_lt_of_dvd
nat_sub_stride_mod_two_stride_lt_of_mod_ge
two_mul_two_pow_dvd_two_pow_of_lt
fhtSylvesterStage_lower_partner_lt_of_mod_lt
fhtSylvesterStage_upper_value_le_of_mod_ge
fhtSylvesterStage_upper_partner_mod_lt_of_mod_ge
fhtSylvesterStageScheduleExact_apply_lower_of_mod_lt
fhtSylvesterStageScheduleExact_apply_upper_of_mod_ge

```

Corollary 243 (Generated Sylvester stage block tests identify the stage bit). *Let $p, r \in \mathbb{N}$, let $i \in \{0, \dots, 2^p - 1\}$, and consider the same generated stage with stride 2^r . If i is in the lower half of its stride block,*

$$i \bmod (2 \cdot 2^r) < 2^r,$$

then the r -th bit of i is cleared:

$$\text{testBit}(i, r) = \text{false}.$$

If i is in the upper half of its stride block,

$$2^r \leq i \bmod (2 \cdot 2^r),$$

then the r -th bit is set:

$$\text{testBit}(i, r) = \text{true}.$$

Thus the lower/upper cases in the generated FHT stage match the exact bit predicate appearing in `sylvesterHadamardParity`. The proof is pure integer arithmetic using reduction modulo 2^{r+1} and quotienting by 2^r . It is deterministic and exact; no sampling-law or probability construction error is introduced.

Lean formalization.

```

nat_testBit_eq_false_of_mod_two_mul_two_pow_lt
nat_testBit_eq_true_of_two_pow_le_mod_two_mul_two_pow
fhtSylvesterStage_testBit_eq_false_of_mod_lt
fhtSylvesterStage_testBit_eq_true_of_mod_ge

```

Corollary 244 (Generated Sylvester stage partner bits). *Let $p, r \in \mathbb{N}$, let $i \in \{0, \dots, 2^p - 1\}$, and consider the generated Sylvester/Walsh stage with stride 2^r . If i is a lower-half coordinate,*

$$i \bmod (2 \cdot 2^r) < 2^r,$$

then the generated upper partner $i + 2^r$ has its r -th bit set:

$$\text{testBit}(i + 2^r, r) = \text{true}.$$

If i is an upper-half coordinate,

$$2^r \leq i \bmod (2 \cdot 2^r),$$

then the generated lower partner $i - 2^r$ has its r -th bit cleared:

$$\text{testBit}(i - 2^r, r) = \text{false}.$$

These two partner-bit facts are the exact integer bridge needed when the one-stage coordinate formulas $x_i + x_{i+2^r}$ and $x_{i-2^r} - x_i$ are rewritten as the Sylvester/Walsh parity recurrence. The theorem is deterministic exact arithmetic for the generated FHT object. The Rademacher and uniform-row sampling laws remain exact mathematical inputs; no probability-construction, normalization, repair, or law-transfer error is introduced.

Lean formalization.

```
fhtSylvesterStage_upper_partner_testBit_eq_true_of_mod_lt
fhtSylvesterStage_lower_partner_testBit_eq_false_of_mod_ge
```

Corollary 245 (Generated Sylvester FHT schedule with explicit FP radius). *Let $p \in \mathbb{N}$, and let the full generated Sylvester/Walsh FHT schedule be*

$$\mathcal{F}_p = \bigcup_{\text{ordered } r=0}^{p-1} \text{SylvesterPairs}(p, r),$$

implemented as the concatenated finite list of stages with strides 2^r . Then for every generated pair (a, b) ,

$$(a, b) \in \mathcal{F}_p \iff \exists r < p : \quad b = a + 2^r, \quad a \bmod (2 \cdot 2^r) < 2^r.$$

In particular every generated pair is ordered and non-self:

$$a < b, \quad a \neq b.$$

Let $x, \hat{x} : \{0, \dots, 2^p - 1\} \rightarrow \mathbb{R}$ and suppose

$$|\hat{x}_i - x_i| \leq E_i \quad \text{for all } i.$$

Let $Y_p(x)$ be the exact vector obtained by applying the generated schedule $\mathcal{F}_p$, and let $\hat{Y}_p(\hat{x})$ be the result of applying the rounded butterfly schedule, with each butterfly using (fadd, flsub). If

$$B_i = \text{FHTScheduleBudget}_{p,i}(\hat{x}, E),$$

then

$$|\hat{Y}_p(\hat{x})_i - Y_p(x)_i| \leq B_i, \quad E_i \geq 0 \ \forall i \implies B_i \geq 0.$$

For a matrix input $A, \hat{A} \in \mathbb{R}^{2^p \times n}$ with

$$|\hat{A}_{ij} - A_{ij}| \leq E_{ij}^A,$$

let the exact scaled output use $c = \sqrt{(2^p)^{-1}}$, and let the computed output use

$$\hat{c} = \text{flsqrt}((2^p)^{-1}).$$

Then the full generated columnwise implementation satisfies

$$\begin{aligned} & \left| \text{flFHTSylvesterMatrix}_{\hat{c}}(\hat{A})_{ij} - \text{FHTSylvesterMatrix}_c(A)_{ij} \right| \\ & \leq \text{FHTSylvesterMatrixBudget}_{ij}(\hat{c}, c u, \hat{A}, E^A). \end{aligned}$$

The corresponding *ComputedMatrix* constructor packages this entrywise radius for the full generated schedule.

Moreover, the generated flat schedule is exactly the stage-by-stage recurrence over the ordered stage list $(0, 1, \dots, p-1)$. If StageExact_p , StageFP_p , and StageBudget_p denote the exact, rounded, and propagated-budget recurrences that apply one generated stage at a time, then

$$Y_p(x) = \text{StageExact}_p(0, 1, \dots, p-1; x),$$

$$\hat{Y}_p(\hat{x}) = \text{StageFP}_p(0, 1, \dots, p-1; \hat{x}),$$

and

$$B = \text{StageBudget}_p(0, 1, \dots, p-1; \hat{x}, E).$$

Thus the transform-correctness theorem may be stated purely as a stage-recurrence identity against the Sylvester/Hadamard sign table; the later partial-transform recurrence and range induction below close that identity. No probability-law or sampling-distribution approximation is introduced in this handoff.

This is an implementation-facing non-probability arithmetic result: the stage indices are exact integer computations, and the Rademacher signs, uniform row laws, and all sampling probabilities remain exact mathematical inputs. The exact transform-correctness theorem is closed below; what remains open is layout/storage/overwrite effects and final *ComputedPreconditioner* wiring for a chosen implementation path.

Lean formalization.

```
fhtSylvesterSchedulePairs
mem_fhtSylvesterSchedulePairs_iff
mem_fhtSylvesterSchedulePairs_iff_stage_rule
fhtSylvesterSchedulePairs_stage_exists
fhtSylvesterSchedulePairs_first_val_lt_second_val
fhtSylvesterSchedulePairs_fst_ne_snd
fhtPairScheduleExact_append
flFhtPairSchedule_append
fhtPairSchedulePropagatedErrorBudget_append
fhtSylvesterScheduleExact
flFhtSylvesterSchedule
fhtSylvesterSchedulePropagatedErrorBudget
fhtSylvesterSchedulePropagatedErrorBudget_nonneg
flFhtSylvesterSchedule_propagated_error_bound
fhtSylvesterStageScheduleExact
flFhtSylvesterStageSchedule
fhtSylvesterStageSchedulePropagatedErrorBudget
fhtSylvesterStageScheduleListExact
flFhtSylvesterStageScheduleList
fhtSylvesterStageScheduleListPropagatedErrorBudget
fhtSylvesterStageScheduleListExact_append
```

```

flFhtSylvesterStageScheduleList_append
fhtSylvesterStageScheduleListPropagatedErrorBudget_append
fhtSylvesterStageScheduleListExact_range_succ
flFhtSylvesterStageScheduleList_range_succ
fhtSylvesterStageScheduleListPropagatedErrorBudget_range_succ
fhtSylvesterStageScheduleListPropagatedErrorBudget_nonneg
fhtPairScheduleExact_flatMap_sylvesterStagePairs
flFhtPairSchedule_flatMap_sylvesterStagePairs
fhtPairSchedulePropagatedErrorBudget_flatMap_sylvesterStagePairs
fhtSylvesterScheduleExact_eq_stageScheduleListExact
flFhtSylvesterSchedule_eq_stageScheduleList
fhtSylvesterSchedulePropagatedErrorBudget_eq_stageScheduleList
fhtScaledSylvesterScheduleMatrixExact
flFhtScaledSylvesterScheduleMatrix
fhtScaledSylvesterScheduleMatrixErrorBudget
fhtScaledSylvesterScheduleMatrixErrorBudget_nonneg
flFhtScaledSylvesterScheduleMatrix_error_bound
flFhtScaledSylvesterScheduleMatrix_sqrtInvNatScale_error_bound
ComputedMatrix.flScaledFhtSylvesterScheduleColumnsSqrtInvNat
ComputedMatrix.flScaledFhtSylvesterScheduleColumnsSqrtInvNat_matrix
ComputedMatrix.flScaledFhtSylvesterScheduleColumnsSqrtInvNat_abs_error
ComputedMatrix.flScaledFhtSylvesterScheduleColumnsSqrtInvNat_entry_error_bound

```

Corollary 246 (Stage-list append and range-succ recurrence for generated FHT). *Let $p \in \mathbb{N}$, let L_1, L_2 be finite lists of generated stage indices, and let $x, \hat{x}, E \in \mathbb{R}^{2^p}$. Write*

$$\text{StageExact}_p(L; x), \quad \text{StageFP}_p(L; \hat{x}), \quad \text{StageBudget}_p(L; \hat{x}, E)$$

for the exact stage recurrence, rounded stage recurrence, and propagated entrywise error budget. Then stage-list append composes exactly:

$$\text{StageExact}_p(L_1 ++ L_2; x) = \text{StageExact}_p(L_2; \text{StageExact}_p(L_1; x)),$$

$$\text{StageFP}_p(L_1 ++ L_2; \hat{x}) = \text{StageFP}_p(L_2; \text{StageFP}_p(L_1; \hat{x})),$$

and

$$\begin{aligned} & \text{StageBudget}_p(L_1 ++ L_2; \hat{x}, E) \\ &= \text{StageBudget}_p(L_2; \text{StageFP}_p(L_1; \hat{x}), \text{StageBudget}_p(L_1; \hat{x}, E)). \end{aligned}$$

In particular, with $R_r = (0, 1, \dots, r-1)$, the successor recurrence is

$$\text{StageExact}_p(R_{r+1}; x) = \text{StageExact}_p([r]; \text{StageExact}_p(R_r; x)),$$

$$\text{StageFP}_p(R_{r+1}; \hat{x}) = \text{StageFP}_p([r]; \text{StageFP}_p(R_r; \hat{x})),$$

and the propagated budget obeys the analogous one-stage update. Equivalently, the exact induction step is all previous generated stages followed by the single current stage r .

This closes the deterministic list-recursion spine needed for the generated FHT realization proof. The next recurrence corollary identifies the one-stage step with the partial Sylvester/Walsh table, using the lower/upper coordinate formulas and the concrete sign recurrences from the preceding corollaries. No probability construction, probability normalization, sampling-law transfer, or floating-point probability error is introduced.

Lean formalization.

```
fhtSylvesterStageScheduleListExact_append
f1FhtSylvesterStageScheduleList_append
fhtSylvesterStageScheduleListPropagatedErrorBudget_append
fhtSylvesterStageScheduleListExact_range_succ
f1FhtSylvesterStageScheduleList_range_succ
fhtSylvesterStageScheduleListPropagatedErrorBudget_range_succ
```

Corollary 247 (Partial Sylvester/Walsh table anchors for generated FHT). *For $p, t \in \mathbb{N}$, define the partial sign table*

$$S_{p,t}(i, j) = (-1)^{\sum_{\ell=0}^{t-1} \mathbf{1}\{\text{bit}_\ell(i)=1 \text{ and } \text{bit}_\ell(j)=1\}}.$$

The corresponding partial transform is

$$T_{p,t}(x)_i = \sum_{j: \lfloor j/2^t \rfloor = \lfloor i/2^t \rfloor} S_{p,t}(i, j) x_j.$$

Then the endpoints of the generated-stage induction are exact:

$$T_{p,0}(x) = x,$$

because the quotient support with $2^0 = 1$ forces $j = i$ and the empty parity sum has sign 1, and

$$T_{p,p}(x)_i = \sum_j S_p(i, j) x_j,$$

because every $i, j < 2^p$ satisfies $\lfloor i/2^p \rfloor = \lfloor j/2^p \rfloor = 0$, while the partial parity sum uses exactly the p bit coordinates of the concrete Sylvester/Walsh table.

Thus the generated-FHT realization proof has a precise induction target: one stage maps $T_{p,r}$ to $T_{p,r+1}$, and the range-succ stage-list recurrence plus the full-depth identity complete the proof. The next corollary closes this one-stage step. This is exact deterministic integer/Boolean arithmetic; no probability construction, sampling-law transfer, or floating-point probability error is introduced.

Lean formalization.

```
sylvesterHadamardPartialParity
sylvesterHadamardPartialSignPattern
sylvesterHadamardPartialUnscaledApply
sylvesterHadamardPartialSignPattern_zero
sylvesterHadamardPartialUnscaledApply_zero
sylvesterHadamardPartialSignPattern_full
sylvesterHadamardPartialUnscaledApply_full
```

Corollary 248 (One-stage partial Sylvester/Walsh recurrence for generated FHT). *Let $r < p$. For each $i \in \{0, \dots, 2^p - 1\}$, write*

$$\text{low}_r(i) \iff i \bmod (2 \cdot 2^r) < 2^r.$$

Then the exact generated FHT stage advances the partial table by one bit:

$$\text{Stage}_{p,r}(T_{p,r}(x)) = T_{p,r+1}(x).$$

More explicitly, in the lower half,

$$T_{p,r+1}(x)_i = T_{p,r}(x)_i + T_{p,r}(x)_{i+2^r}, \quad \text{low}_r(i),$$

and in the upper half,

$$T_{p,r+1}(x)_i = T_{p,r}(x)_{i-2^r} - T_{p,r}(x)_i, \quad \neg \text{low}_r(i).$$

The proof splits each quotient block $\lfloor j/2^{r+1} \rfloor = \lfloor i/2^{r+1} \rfloor$ into the two 2^r -blocks used by the butterfly, proves the partial sign recurrence

$$S_{p,r+1}(i, j) = \begin{cases} -S_{p,r}(i, j), & \text{bit}_r(i) = \text{bit}_r(j) = 1, \\ S_{p,r}(i, j), & \text{otherwise,} \end{cases}$$

and uses the generated-partner bit-agreement facts for $i \pm 2^r$. This is exact deterministic integer/FHT arithmetic and does not introduce any probability-construction, sampling-law, or FP-probability term.

Lean formalization.

```

nat_div_stride_eq_two_mul_block_div_add_mod_div
nat_div_stride_eq_two_mul_block_div_of_mod_lt
nat_div_stride_eq_two_mul_block_div_add_one_of_mod_ge
nat_add_stride_div_stride_eq_two_mul_block_div_add_one_of_mod_lt
nat_add_stride_div_two_stride_eq_of_mod_lt
nat_sub_stride_div_stride_eq_two_mul_block_div_of_mod_ge
nat_sub_stride_div_two_stride_eq_of_mod_ge
nat_div_two_mul_stride_eq_div_stride_div_two
nat_div_two_stride_eq_of_div_stride_eq
sylvesterHadamardPartialParityWeight_succ
sylvesterHadamardPartialSignPattern_succ_eq_or_neg
sylvesterHadamardPartialSignPattern_succ_eq_of_stage_bit_false
sylvesterHadamardPartialSignPattern_succ_eq_or_neg_of_stage_bit_true
sylvesterHadamardPartialParityWeight_eq_of_bits
sylvesterHadamardPartialSignPattern_eq_of_bits
sylvesterHadamardPartialSignPattern_upper_partner_eq_of_mod_lt
sylvesterHadamardPartialSignPattern_upper_eq_lower_partner_of_mod_ge
sylvesterHadamardPartialUnscaledApply_succ_lower
sylvesterHadamardPartialUnscaledApply_succ_upper
fhtSylvesterStageScheduleExact_partialUnscaledApply_eq_succ

```

Corollary 249 (Generated FHT realization and signed-preconditioner bridge). *For $p \in \mathbb{N}$, let S_p be the concrete Sylvester/Walsh sign table*

$$S_p(i, j) = \begin{cases} -1, & \text{if } \text{parity}(i \wedge j) = 1, \\ 1, & \text{otherwise.} \end{cases}$$

The exact generated FHT schedule realizes this table:

$$Y_p(x)_i = \sum_j S_p(i, j)x_j \quad \text{for every } x \text{ and } i,$$

where $Y_p(x)$ is the exact generated FHT schedule from the preceding corollary. The proof first shows, for every $t \leq p$,

$$\text{StageExact}_p(0, \dots, t-1; x) = T_{p,t}(x),$$

and then sets $t = p$, where $T_{p,p}$ is the full bit-parity transform. Consequently, for every matrix $A \in \mathbb{R}^{2^p \times n}$ and every exact scale c ,

$$\text{FHTSylvesterMatrix}_c(A)_{ij} = c \sum_k S_p(i, k)A_{kj}.$$

In particular, for a realized sign vector σ and an input basis U ,

$$\text{FHTSylvesterMatrix}_c((k, j) \mapsto \sigma_k U_{kj}) = \text{preconditionRows}((cS_p) \text{diag}(\sigma))U.$$

Thus the generated FHT implementation is tied to the exact HDU preconditioner used by the existing SRHT exact-law and floating-point transfer theorems without a remaining realization hypothesis. This result is exact and deterministic; it does not introduce any probability-construction, sampling-law normalization, or law-transfer error. Layout-specific rounded storage and overwrite effects remain separate implementation obligations.

Lean formalization.

```
sylvesterHadamardUnscaledApply
fhtSylvesterScheduleRealizesSignPattern
fhtSylvesterScheduleRealizesSignPattern_zero
fhtSylvesterStageScheduleListExact_range_eq_partialUnscaledApply
fhtSylvesterScheduleRealizesSignPattern_generated
sylvesterHadamardScaledMatrixApply
fhtScaledSylvesterScheduleMatrixExact_eq_sylvesterHadamardScaledMatrixApply_of_realizes
fhtScaledSylvesterScheduleMatrixExact_eq_sylvesterHadamardScaledMatrixApply
fhtScaledSylvesterScheduleMatrixExact_signed_eq_preconditionRows_of_realizes
fhtScaledSylvesterScheduleMatrixExact_signed_eq_preconditionRows
```

Corollary 250 (Fast generated-FHT computed preconditioner). *Let S_p be the concrete Sylvester/Walsh sign table in dimension 2^p , let $c = \sqrt{((2^p)^{-1})}$, and let $\sigma \in \{\pm 1\}^{2^p}$ be a realized Rademacher sign vector. The sampling law for σ is exact. Suppose the implementation stores or computes the sign vector as $\hat{\sigma}$ with entrywise certificate*

$$|\hat{\sigma}_i - \sigma_i| \leq E_i^\sigma, \quad E_i^\sigma \geq 0,$$

and forms the computed diagonal input

$$\widehat{D}_{ij} = \begin{cases} \widehat{\sigma}_i, & i = j, \\ 0, & i \neq j, \end{cases} \quad E_{ij}^D = \begin{cases} E_i^\sigma, & i = j, \\ 0, & i \neq j. \end{cases}$$

Let $\widehat{c} = \text{flsqrt}((2^p)^{-1})$ and $\eta = cu$. The fast generated-FHT preconditioner is the rounded column-wise schedule

$$\widehat{P} = \text{flFHTSylvesterMatrix}_{\widehat{c}}(\widehat{D}),$$

and the exact analysis preconditioner is

$$P = (cS_p) \text{diag}(\sigma).$$

Then for every i, j ,

$$|\widehat{P}_{ij} - P_{ij}| \leq B_{ij}^{\text{FHT}},$$

where B_{ij}^{FHT} is the generated-FHT propagated budget obtained from

$$\text{FHTBudget}_{p, \widehat{c}, \eta}(\widehat{D}, E^D)_{ij}.$$

Equivalently, the implementation supplies a **ComputedPreconditioner** certificate for P . The exact bridge used in the certificate is

$$\text{FHTSylvesterMatrix}_c(\text{diag}(\sigma)) = (cS_p) \text{diag}(\sigma).$$

Thus the implementation no longer silently uses an exact materialized HD matrix: sign storage, diagonal input error, butterfly arithmetic, and rounded normalization are all charged. Under the exact Rademacher/uniform-row product law, the resulting computed-left perturbation event for $\widehat{V} = \text{fl}(\widehat{P}U)$ has probability one; the downstream budget adds $\widehat{V}$ -formation, row-scaling, and Gram-dot-product errors.

If an implementation forms the signed input $D_\sigma U$ directly before the FHT stages, a separate rowwise multiplication certificate is available:

$$|\text{fl}(\widehat{\sigma}_i \widehat{U}_{ij}) - \sigma_i U_{ij}| \leq u|\widehat{\sigma}_i \widehat{U}_{ij}| + |\widehat{\sigma}_i| E_{ij}^U + E_i^\sigma |U_{ij}|.$$

For arbitrary array implementations, the remaining FHT obligation is layout-specific in-place storage, copy, aliasing, or overwrite behavior if the chosen routine differs from the functional pair-update schedule formalized here. The next result closes one concrete writeback/copy model.

Lean formalization.

```

ComputedMatrix.flRowSignMul
ComputedMatrix.flRowSignMul_entry_error_bound
fhtScaledSylvesterScheduleMatrixExact_diag_eq_matMul_diag
ComputedPreconditioner.flSignedHadamardSylvesterFhtSchedule
ComputedPreconditioner.flSignedHadamardSylvesterFhtSchedule_entry_error_bound
signedHadamardSylvesterFhtSchedulePreconditioner
signedHadamardSylvesterFhtScheduleStoredSignPreconditioner
signedHadamardUniformRowTraceProbability_eventProb_sylvesterFhtScheduleComputedLeftPreconditioned_
fl_uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_sylvesterFhtScheduleStoredSignComputedLeftPreconditioned_
fl_uniformRowPerturbEvent_eq_one

```

Theorem 251 (Generated-FHT SRHT endpoint without supplied orthogonality). *Fix $p, r, n, s \in \mathbb{N}$, let $m = 2^p$, and let*

$$S_p(i, k) = \text{sylvesterHadamardSignPattern}(p, i, k), \quad H_p(i, k) = \sqrt{m^{-1}} S_p(i, k).$$

Let $U \in \mathbb{R}^{m \times r}$ have orthonormal columns, let $C \in \mathbb{R}^{r \times n}$, and let the exact analysis input be

$$A = UC = \text{preconditionColumns}(U, C).$$

The Rademacher sign law and the downstream uniform row law are exact mathematical laws. For each realized sign vector, the computed non-probability objects are:

$$\hat{P}_\omega = \text{signedHadamardSylvesterFhtScheduleStoredSignPreconditioner}(fp, \omega), \quad \hat{d} = \text{uniformRowFlSqrtMul}$$

followed by the rounded product $\hat{Y}_\omega = \text{fl}(\hat{P}_\omega A)$, rounded sampled-row divisions by $\hat{d}$, and rounded sampled-Gram dot products. Assume the same positivity, γ_m , γ_s , θ , and SRHT sample-budget hypotheses as in the source-sharp SRHT logarithmic theorem, with

$$t = \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}}, \quad S_{\text{rad}} = \sqrt{r/m} + t, \quad L = m S_{\text{rad}}^2,$$

$$\beta_+ = \frac{\exp(\theta L) - \theta L - 1}{L^2}, \quad \beta_- = \exp(\theta) - \theta - 1,$$

and

$$e^{-\theta s \varepsilon} r \left(e^{s \beta_+ (L^2 + 1)} + e^{s \beta_- (L^2 + 1)} \right) \leq \delta_{\text{sample}}.$$

Then the computed two-sided sampled-Gram event centered at $A^T A$ satisfies

$$\mathbb{P} \left[-\varepsilon I \preceq \hat{G}_{\text{SRHT}}^{\text{fp}} - A^T A \preceq \varepsilon I \text{ with radius enlarged by } \text{signedHadamardComputedLeftUniformRowComputedDenPer} \right]$$

The exact objects in the theorem are U , C , $A = UC$, S_p , H_p , and the two sampling laws. The computed objects are $\hat{P}_\omega$, $\hat{Y}_\omega$, $\hat{d}$, the sampled scaled rows, and the Gram dot products. The normalized Sylvester/Walsh table is proved orthogonal inside Lean, so no external H -orthogonality or flatness hypothesis appears. The stored-input siblings close the implementation paths in which the actual input matrix is first realized in floating point before preconditioning:

$$\hat{A}_{ij}^{\text{in}} \in \{\text{flmul}(A_{ij}, 1), \text{fladd}(A_{ij}, 0), \text{flsub}(A_{ij}, 0)\}, \quad A = UC.$$

Here U, C, A, S_p, H_p and the sampling laws are exact analysis references, while $\hat{A}^{\text{in}}$, $\hat{P}_\omega$, $\hat{Y}_\omega^{\text{in}} = \text{fl}(\hat{P}_\omega \hat{A}^{\text{in}})$, the denominator, row divisions, and Gram dot products are computed non-probability objects. The new product budget adds the irreducible input storage term

$$\sum_a |(P_\omega)_{ia}| u |A_{aj}|$$

inside the entrywise bound for $\text{fl}(\hat{P}_\omega \hat{A}^{\text{in}}) - P_\omega A$. Thus these endpoints have no perturbation-event or certificate-existence assumption for the stored input matrix. The same final theorem is also instantiated for modeled writeback/copy routines. In the all-coordinate add-zero routine, after every generated FHT pair update, each output coordinate is stored as $\text{fladd}(y_i, 0)$. In the all-coordinate multiply-one and subtract-zero routines, each output coordinate is stored as $\text{flmul}(y_i, 1)$ or $\text{flsub}(y_i, 0)$. In the modified-coordinate add-zero, multiply-one, and subtract-zero routines, only the two coordinates changed by the butterfly receive the corresponding writeback. In all cases, the writeback radius is included in the same computed-preconditioner and downstream sampled-Gram budget.

Lean formalization.

```
fhtButterflyExact_sq_sum
fhtButterflyExact_inner_sum
fhtPairUpdateExact_inner_sum
vecNorm2Sq_fhtPairUpdateExact
sylvesterHadamardSignPattern_symm
exists_testBit_ne_of_fin_two_pow_ne
sylvesterStageBitFlip
sylvesterStageBitFlip_involutive
sylvesterStageBitFlip_bijective
sylvesterHadamardSignPattern_mul_stageBitFlip_eq_neg
sylvesterHadamardSignPattern_col_inner_eq_zero_of_ne
sylvesterHadamardSignPattern_col_inner
sylvesterHadamardSignPattern_row_inner
sylvesterHadamardScaled_col_inner
sylvesterHadamardScaled_row_inner
isOrthogonal_sqrt_inv_nat_mul_sylvesterSignPattern
ComputedMatrix.flMulOne
ComputedMatrix.flMulOne_entry_error_bound
ComputedMatrix.flAddZeroRight
ComputedMatrix.flAddZeroRight_entry_error_bound
ComputedMatrix.flSubZeroRight
ComputedMatrix.flSubZeroRight_entry_error_bound
signedHadamardComputedLeftInputUniformRowComputedDenPerturbBudget
signedHadamardUniformRowTraceProbability_eventProb_computedLeftInputPreconditioned_fl_
uniformRowComputedDenPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_
  sylvesterFhtScheduleStoredSignComputedLeftPreconditioned_factoredInput_
  fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_two_sided_finiteLoewnerLe_
  ge_one_sub_delta_srht_log_preprocess
signedHadamardUniformRowTraceProbability_eventProb_
  sylvesterFhtScheduleStoredSignStoredInputMulOneComputedLeftInputPreconditioned_
  factoredInput_fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_
  two_sided_finiteLoewnerLe_ge_one_sub_delta_srht_log_preprocess
signedHadamardUniformRowTraceProbability_eventProb_
  sylvesterFhtScheduleStoredSignStoredInputAddZeroRightComputedLeftInputPreconditioned_
  factoredInput_fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_
  two_sided_finiteLoewnerLe_ge_one_sub_delta_srht_log_preprocess
signedHadamardUniformRowTraceProbability_eventProb_
  sylvesterFhtScheduleStoredSignStoredInputSubZeroRightComputedLeftInputPreconditioned_
```

```

factoredInput_fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_
two_sided_finiteLoewnerLe_ge_one_sub_delta_srht_log_preprocess
signedHadamardUniformRowTraceProbability_eventProb_
sylvesterFhtScheduleStoredSignStoredAddZeroRightComputedLeftPreconditioned_
factoredInput_fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_
two_sided_finiteLoewnerLe_ge_one_sub_delta_srht_log_preprocess
signedHadamardUniformRowTraceProbability_eventProb_
sylvesterFhtScheduleStoredSignModifiedStoredAddZeroRightComputedLeftPreconditioned_
factoredInput_fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_
two_sided_finiteLoewnerLe_ge_one_sub_delta_srht_log_preprocess

```

Corollary 252 (Stored add-zero writeback for fast generated-FHT). *Use the notation and exact probability laws of the preceding corollary. Let $\mathcal{P}_p$ be the generated Sylvester/Walsh pair schedule in dimension 2^p . For one pair (r, s) , let*

$$y^{\text{pair}} = \text{FHTPairUpdate}_{r,s}(x), \quad \hat{y}^{\text{pair}} = \text{flFHTPairUpdate}_{r,s}(\hat{x}),$$

and define the stored output by

$$\hat{y}_i^{\text{store}} = \text{fl}_{\text{add}}(\hat{y}_i^{\text{pair}}, 0).$$

If the input certificate satisfies

$$|\hat{x}_i - x_i| \leq E_i, \quad E_i \geq 0,$$

then the stored pair update has the entrywise bound

$$|\hat{y}_i^{\text{store}} - y_i^{\text{pair}}| \leq B_{r,s,i}^{\text{pair}}(\hat{x}, E) + u |\hat{y}_i^{\text{pair}}|,$$

where $B_{r,s,i}^{\text{pair}}$ is the usual propagated rounded-butterfly budget. Equivalently,

$$E_{r,s,i}^{\text{store},+} = B_{r,s,i}^{\text{pair}}(\hat{x}, E) + u |\text{flFHTPairUpdate}_{r,s}(\hat{x})_i|$$

is a valid propagated radius for one rounded pair update followed by an add-zero output write-back/copy.

Composing this recurrence over $\mathcal{P}_p$, let

$$\hat{z}_i = \text{flFHTSchedule}_{\mathcal{P}_p}^{\text{store},+}(\hat{x})_i, \quad E_{p,i}^{\text{store},+} = \text{FHTScheduleBudget}_{p,i}^{\text{store},+}(\hat{x}, E).$$

For the scaled transform, let $c = \sqrt{((2^p)^{-1})}$, $\hat{c} = \text{flsqrt}((2^p)^{-1})$, and $\eta = cu$. Then

$$|\text{fl}_{\text{mul}}(\hat{c}, \hat{z}_i) - c \text{FHTSchedule}_{\mathcal{P}_p}(x)_i| \leq u |\hat{c} \hat{z}_i| + |\hat{c}| E_{p,i}^{\text{store},+} + \eta (|\hat{z}_i| + E_{p,i}^{\text{store},+}).$$

Columnwise, this gives a computed matrix certificate

$$\left| \text{flFHTSylvesterMatrix}_{\hat{c}}^{\text{store},+}(\hat{A})_{ij} - \text{FHTSylvesterMatrix}_c(A)_{ij} \right| \leq B_{ij}^{\text{store},+},$$

where $B^{\text{store},+}$ is

$$\text{FHTSylvesterMatrixBudget}_{p,\hat{c},\eta}^{\text{store},+}(\hat{A}, E).$$

In particular, with the computed diagonal sign input $\hat{D}$ and its entrywise sign-storage certificate E^D , the implemented preconditioner

$$\hat{P}^{\text{store},+} = \text{flFHTSylvesterMatrix}_{\hat{c}}^{\text{store},+}(\hat{D})$$

satisfies, for every i, j ,

$$|\hat{P}_{ij}^{\text{store},+} - ((cS_p) \text{diag}(\sigma))_{ij}| \leq B_{ij}^{\text{store},+}.$$

Thus this is a fully computed-object implementation-facing certificate for the modeled routine “rounded butterfly, then $\text{fl}_{\text{add}}(\cdot, 0)$ output writeback/copy.” The Rademacher and uniform-row sampling laws remain exact. Under their exact product law, the computed-left perturbation event for

$$\hat{V} = \text{fl}(\hat{P}^{\text{store},+} U)$$

has probability one, with the downstream radius additionally charging $\hat{V}$ -formation, row scaling, and Gram-dot-product errors. If the realized signs are also stored by the existing rounded sign-copy certificate, the same probability-one event holds with that sign-storage radius included in E^D . Memory order, aliasing, vectorized layout, or in-place overwrite semantics not represented by this pair-update plus add-zero writeback model remain separate implementation obligations.

Lean formalization.

```

flFhtPairUpdateStoredAddZeroRight
fhtPairUpdateStoredAddZeroRightPropagatedErrorBudget
flFhtPairUpdateStoredAddZeroRight_propagated_error_bound
flFhtPairScheduleStoredAddZeroRight
fhtPairScheduleStoredAddZeroRightPropagatedErrorBudget
flFhtPairScheduleStoredAddZeroRight_propagated_error_bound
flFhtSylvesterScheduleStoredAddZeroRight
flFhtSylvesterScheduleStoredAddZeroRight_propagated_error_bound
flFhtScaledPairScheduleStoredAddZeroRight
flFhtScaledPairScheduleStoredAddZeroRight_sqrtInvNatScale_error_bound
flFhtScaledPairScheduleMatrixStoredAddZeroRight
flFhtScaledPairScheduleMatrixStoredAddZeroRight_sqrtInvNatScale_error_bound
flFhtScaledSylvesterScheduleMatrixStoredAddZeroRight
flFhtScaledSylvesterScheduleMatrixStoredAddZeroRight_sqrtInvNatScale_error_bound
ComputedMatrix.flScaledFhtSylvesterScheduleColumnsSqrtInvNatStoredAddZeroRight
ComputedMatrix.flScaledFhtSylvesterScheduleColumnsSqrtInvNatStoredAddZeroRight_entry_error_bound
ComputedPreconditioner.flSignedHadamardSylvesterFhtScheduleStoredAddZeroRight
ComputedPreconditioner.flSignedHadamardSylvesterFhtScheduleStoredAddZeroRight_entry_error_bound
signedHadamardSylvesterFhtScheduleStoredAddZeroRightPreconditioner
signedHadamardSylvesterFhtScheduleStoredSignStoredAddZeroRightPreconditioner
signedHadamardUniformRowTraceProbability_eventProb_
  sylvesterFhtScheduleStoredAddZeroRightComputedLeftPreconditioned_
  fl_uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_
  sylvesterFhtScheduleStoredSignStoredAddZeroRightComputedLeftPreconditioned_
  fl_uniformRowPerturbEvent_eq_one

```

Corollary 253 (Generated-SRHT multiply-one/subtract-zero FHT writeback). *Fix one FHT butterfly pair (r, s) and let*

$$y^{\text{pair}} = \text{FHTPairUpdate}_{r,s}(x), \quad \hat{y}^{\text{pair}} = \text{flFHTPairUpdate}_{r,s}(\hat{x}).$$

Assume the input certificate

$$|\hat{x}_i - x_i| \leq E_i, \quad E_i \geq 0$$

for every coordinate i . Let

$$B_{r,s,i}^{\text{pair}}(\hat{x}, E)$$

be the existing propagated radius for one rounded pair update. Define two stored outputs

$$\hat{y}_i^{\times 1} = \text{fl}_{\text{mul}}(\hat{y}_i^{\text{pair}}, 1), \quad \hat{y}_i^{-0} = \text{fl}_{\text{sub}}(\hat{y}_i^{\text{pair}}, 0).$$

Then both all-coordinate writeback/copy variants satisfy the same explicit one-pair radius:

$$|\hat{y}_i^{\times 1} - y_i^{\text{pair}}| \leq B_{r,s,i}^{\text{pair}}(\hat{x}, E) + u |\hat{y}_i^{\text{pair}}|$$

and

$$|\hat{y}_i^{-0} - y_i^{\text{pair}}| \leq B_{r,s,i}^{\text{pair}}(\hat{x}, E) + u |\hat{y}_i^{\text{pair}}|.$$

Equivalently,

$$E_{r,s,i}^{\times 1} = B_{r,s,i}^{\text{pair}}(\hat{x}, E) + u |\text{flFHTPairUpdate}_{r,s}(\hat{x})_i|$$

and

$$E_{r,s,i}^{-0} = B_{r,s,i}^{\text{pair}}(\hat{x}, E) + u |\text{flFHTPairUpdate}_{r,s}(\hat{x})_i|$$

are valid nonnegative propagated radii for the two one-pair routines.

Now let $\mathcal{P}$ be any ordered list of FHT pairs. Define

$$\hat{z}^{\times 1} = \text{flFHTSchedule}_{\mathcal{P}}^{\times 1}(\hat{x}), \quad \hat{z}^{-0} = \text{flFHTSchedule}_{\mathcal{P}}^{-0}(\hat{x})$$

by applying the corresponding one-pair routine at each pair in the list. Let

$$B_{\mathcal{P},i}^{\times 1}(\hat{x}, E), \quad B_{\mathcal{P},i}^{-0}(\hat{x}, E)$$

be the recursively propagated schedules of the one-pair radii above. Then, under the same input certificate,

$$|\hat{z}_i^{\times 1} - \text{FHTSchedule}_{\mathcal{P}}(x)_i| \leq B_{\mathcal{P},i}^{\times 1}(\hat{x}, E)$$

and

$$|\hat{z}_i^{-0} - \text{FHTSchedule}_{\mathcal{P}}(x)_i| \leq B_{\mathcal{P},i}^{-0}(\hat{x}, E).$$

The two schedule budgets are nonnegative whenever the input budget E is coordinatewise nonnegative.

Specialize now to the generated Sylvester/Walsh schedule $\mathcal{P}_p$ in dimension 2^p . Let

$$c = \sqrt{((2^p)^{-1})}, \quad \hat{c} = \text{fl}_{\text{sqrt}}((2^p)^{-1}), \quad \eta = c u.$$

For a computed matrix $\hat{A}$ with certificate

$$|\hat{A}_{ij} - A_{ij}| \leq E_{ij}, \quad E_{ij} \geq 0,$$

define the two scaled generated-FHT outputs

$$\widehat{Z}^{\times 1} = \text{flScaledFHT}_p^{\times 1}(\widehat{c}, \widehat{A}), \quad \widehat{Z}^{-0} = \text{flScaledFHT}_p^{-0}(\widehat{c}, \widehat{A}).$$

Then there are explicit entrywise budgets

$$C_{p,ij}^{\times 1}(\widehat{A}, E), \quad C_{p,ij}^{-0}(\widehat{A}, E)$$

obtained by the recursive schedule budgets above plus the rounded final scale and scale-radius terms, and they satisfy

$$\begin{aligned} |\widehat{Z}_{ij}^{\times 1} - \text{ScaledFHT}_{p,c}(A)_{ij}| &\leq C_{p,ij}^{\times 1}(\widehat{A}, E), \\ |\widehat{Z}_{ij}^{-0} - \text{ScaledFHT}_{p,c}(A)_{ij}| &\leq C_{p,ij}^{-0}(\widehat{A}, E). \end{aligned}$$

The same inequalities package as computed-matrix certificates. Applying them to the diagonal sign matrix gives computed preconditioners

$$\widehat{\Pi}_\omega^{\times 1} \quad \text{and} \quad \widehat{\Pi}_\omega^{-0}$$

for the ideal signed-Hadamard matrix

$$\Pi_\omega = H_p D_\omega, \quad (H_p)_{ik} = c S_p(i, k).$$

The sign vector may be supplied exactly or through the existing rounded $\text{fl}_{\text{mul}}(\text{sign}_i, 1)$ sign-storage certificate.

Finally, with the exact Rademacher law for ω and the exact uniform-row law for the row trace, the computed-left perturbation event for the resulting preconditioned basis has probability one:

$$\Pr_{\omega, \rho}[\mathcal{E}_{\text{fp}}(\widehat{\Pi}_\omega^{\times 1}, U, \rho)] = 1, \quad \Pr_{\omega, \rho}[\mathcal{E}_{\text{fp}}(\widehat{\Pi}_\omega^{-0}, U, \rho)] = 1,$$

and the same probability-one statements hold for the stored-sign variants. These are implementation-facing non-probability arithmetic certificates: butterfly arithmetic, writeback/copy, rounded normalization, computed preconditioner formation, $\widehat{V}$ formation, row scaling, and Gram dot products are charged. The Rademacher sign law and the uniform row-sampling law remain exact mathematical inputs. Array memory order, aliasing, in-place overwrite, or vectorized routines with different semantics still require separate routine certificates.

Lean formalization.

```
flFhtPairUpdateStoredMulOne
fhtPairUpdateStoredMulOnePropagatedErrorBudget
fhtPairUpdateStoredMulOnePropagatedErrorBudget_nonneg
flFhtPairUpdateStoredMulOne_propagated_error_bound
flFhtPairUpdateStoredSubZeroRight
fhtPairUpdateStoredSubZeroRightPropagatedErrorBudget
fhtPairUpdateStoredSubZeroRightPropagatedErrorBudget_nonneg
flFhtPairUpdateStoredSubZeroRight_propagated_error_bound
flFhtPairScheduleStoredMulOne
fhtPairScheduleStoredMulOnePropagatedErrorBudget
fhtPairScheduleStoredMulOnePropagatedErrorBudget_nonneg
```

```

flFhtPairScheduleStoredMulOne_propagated_error_bound
flFhtPairScheduleStoredSubZeroRight
fhtPairScheduleStoredSubZeroRightPropagatedErrorBudget
fhtPairScheduleStoredSubZeroRightPropagatedErrorBudget_nonneg
flFhtPairScheduleStoredSubZeroRight_propagated_error_bound
flFhtScaledPairScheduleStoredMulOne_sqrtInvNatScale_error_bound
flFhtScaledPairScheduleStoredSubZeroRight_sqrtInvNatScale_error_bound
flFhtScaledPairScheduleMatrixStoredMulOne_sqrtInvNatScale_error_bound
flFhtScaledPairScheduleMatrixStoredSubZeroRight_sqrtInvNatScale_error_bound
flFhtScaledSylvesterScheduleMatrixStoredMulOne_sqrtInvNatScale_error_bound
flFhtScaledSylvesterScheduleMatrixStoredSubZeroRight_sqrtInvNatScale_error_bound
ComputedMatrix.flScaledFhtSylvesterScheduleColumnsSqrtInvNatStoredMulOne_entry_error_bound
ComputedMatrix.flScaledFhtSylvesterScheduleColumnsSqrtInvNatStoredSubZeroRight_entry_error_bound
ComputedPreconditioner.flSignedHadamardSylvesterFhtScheduleStoredMulOne_entry_error_bound
ComputedPreconditioner.flSignedHadamardSylvesterFhtScheduleStoredSubZeroRight_entry_error_bound
signedHadamardSylvesterFhtScheduleStoredMulOnePreconditioner
signedHadamardSylvesterFhtScheduleStoredSubZeroRightPreconditioner
signedHadamardSylvesterFhtScheduleStoredSignStoredMulOnePreconditioner
signedHadamardSylvesterFhtScheduleStoredSignStoredSubZeroRightPreconditioner
signedHadamardUniformRowTraceProbability_eventProb_
    sylvesterFhtScheduleStoredMulOneComputedLeftPreconditioned_
    fl_uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_
    sylvesterFhtScheduleStoredSubZeroRightComputedLeftPreconditioned_
    fl_uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_
    sylvesterFhtScheduleStoredSignStoredMulOneComputedLeftPreconditioned_
    fl_uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_
    sylvesterFhtScheduleStoredSignStoredSubZeroRightComputedLeftPreconditioned_
    fl_uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_
    sylvesterFhtScheduleStoredSignStoredMulOneComputedLeftPreconditioned_
    factoredInput_fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_
    two_sided_finiteLoewnerLe_ge_one_sub_delta_srht_log_preprocess
signedHadamardUniformRowTraceProbability_eventProb_
    sylvesterFhtScheduleStoredSignStoredSubZeroRightComputedLeftPreconditioned_
    factoredInput_fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_
    two_sided_finiteLoewnerLe_ge_one_sub_delta_srht_log_preprocess

```


Corollary 254 (Modified-coordinate add/multiply/subtract writeback for fast generated-FHT). *Use the one-pair notation above and the exact probability laws already stated for Algorithm 3. Let $\mathcal{P}_p$ be the generated Sylvester/Walsh pair schedule in dimension 2^p . For one pair (r, s) , let*

$$y^{\text{pair}} = \text{FHTPairUpdate}_{r,s}(x), \quad \hat{y}^{\text{pair}} = \text{flFHTPairUpdate}_{r,s}(\hat{x}).$$

For $\tau \in \{+, \times 1, -0\}$, define the rounded copy map

$$\psi_+(a) = \text{fl}_{\text{add}}(a, 0), \quad \psi_{\times 1}(a) = \text{fl}_{\text{mul}}(a, 1), \quad \psi_{-0}(a) = \text{fl}_{\text{sub}}(a, 0),$$

and define the modified-coordinate stored output by

$$\hat{y}_i^{\text{mod},\tau} = \begin{cases} \psi_\tau(\hat{y}_i^{\text{pair}}), & i = r \text{ or } i = s, \\ \hat{y}_i^{\text{pair}}, & i \neq r \text{ and } i \neq s. \end{cases}$$

If the input certificate satisfies

$$|\hat{x}_i - x_i| \leq E_i, \quad E_i \geq 0,$$

then, for each $\tau \in \{+, \times 1, -0\}$, the modified-coordinate stored pair update satisfies

$$|\hat{y}_i^{\text{mod},\tau} - y_i^{\text{pair}}| \leq B_{r,s,i}^{\text{pair}}(\hat{x}, E) + \mathbf{1}_{\{i=r \text{ or } i=s\}} u |\hat{y}_i^{\text{pair}}|.$$

Equivalently,

$$E_{r,s,i}^{\text{mod},\tau} = B_{r,s,i}^{\text{pair}}(\hat{x}, E) + \begin{cases} u |\text{flFHTPairUpdate}_{r,s}(\hat{x})_i|, & i = r \text{ or } i = s, \\ 0, & \text{otherwise} \end{cases}$$

is a valid propagated radius for one rounded pair update followed by a writeback only of the modified butterfly outputs. The copy map ψ_τ is the only part changed across the three variants.

Composing this recurrence over $\mathcal{P}_p$, let

$$\hat{z}_i^\tau = \text{flFHTSchedule}_{\mathcal{P}_p}^{\text{mod},\tau}(\hat{x})_i, \quad E_{p,i}^{\text{mod},\tau} = \text{FHTScheduleBudget}_{p,i}^{\text{mod},\tau}(\hat{x}, E).$$

For the scaled transform, let $c = \sqrt{((2^p)^{-1})}$, $\hat{c} = \text{flsqrt}((2^p)^{-1})$, and $\eta = cu$. Then for each τ ,

$$|\text{fl}_{\text{mul}}(\hat{c}, \hat{z}_i^\tau) - c \text{FHTSchedule}_{\mathcal{P}_p}(x)_i| \leq u |\hat{c} \hat{z}_i^\tau| + |\hat{c}| E_{p,i}^{\text{mod},\tau} + \eta (|\hat{z}_i^\tau| + E_{p,i}^{\text{mod},\tau}).$$

Columnwise, this gives computed matrix certificates

$$\left| \text{flFHTSylvesterMatrix}_{\hat{c}}^{\text{mod},\tau}(\hat{A})_{ij} - \text{FHTSylvesterMatrix}_c(A)_{ij} \right| \leq B_{ij}^{\text{mod},\tau},$$

where

$$B^{\text{mod},\tau} = \text{FHTSylvesterMatrixBudget}_{p,\hat{c},\eta}^{\text{mod},\tau}(\hat{A}, E).$$

With computed diagonal sign input $\hat{D}$ and sign-storage certificate E^D , the implemented preconditioners

$$\hat{P}^{\text{mod},\tau} = \text{flFHTSylvesterMatrix}_{\hat{c}}^{\text{mod},\tau}(\hat{D})$$

satisfy, for every i, j ,

$$|\hat{P}_{ij}^{\text{mod},\tau} - ((cS_p) \text{diag}(\sigma))_{ij}| \leq B_{ij}^{\text{mod},\tau}.$$

Thus these are fully computed-object implementation-facing certificates for the modeled routines “rounded butterfly, then $\text{fl}_{\text{add}}(\cdot, 0)$, $\text{fl}_{\text{mul}}(\cdot, 1)$, or $\text{fl}_{\text{sub}}(\cdot, 0)$ writeback only on the two modified pair

outputs.” The Rademacher and uniform-row sampling laws remain exact mathematical inputs. Under their exact product law, for each τ , the computed-left perturbation event for

$$\hat{V}^\tau = \mathfrak{fl}(\hat{P}^{\text{mod},\tau}U)$$

has probability one, with the downstream radius additionally charging $\hat{V}^\tau$ -formation, row scaling, and Gram-dot-product errors. If the realized signs are also stored by the rounded sign-copy certificate, the same probability-one event holds with that sign-storage radius included in E^D . Memory order, aliasing, vectorized layout, or in-place overwrite semantics not represented by these modified-coordinate writeback models remain separate implementation obligations.

Lean formalization.

```

flFhtPairUpdateModifiedStoredAddZeroRight
fhtPairUpdateModifiedStoredAddZeroRightPropagatedErrorBudget
flFhtPairUpdateModifiedStoredAddZeroRight_propagated_error_bound
flFhtPairUpdateModifiedStoredMulOne
fhtPairUpdateModifiedStoredMulOnePropagatedErrorBudget
flFhtPairUpdateModifiedStoredMulOne_propagated_error_bound
flFhtPairUpdateModifiedStoredSubZeroRight
fhtPairUpdateModifiedStoredSubZeroRightPropagatedErrorBudget
flFhtPairUpdateModifiedStoredSubZeroRight_propagated_error_bound
flFhtPairScheduleModifiedStoredAddZeroRight
fhtPairScheduleModifiedStoredAddZeroRightPropagatedErrorBudget
flFhtPairScheduleModifiedStoredAddZeroRight_propagated_error_bound
flFhtPairScheduleModifiedStoredMulOne
fhtPairScheduleModifiedStoredMulOnePropagatedErrorBudget
flFhtPairScheduleModifiedStoredMulOne_propagated_error_bound
flFhtPairScheduleModifiedStoredSubZeroRight
fhtPairScheduleModifiedStoredSubZeroRightPropagatedErrorBudget
flFhtPairScheduleModifiedStoredSubZeroRight_propagated_error_bound
flFhtSylvesterScheduleModifiedStoredAddZeroRight
flFhtSylvesterScheduleModifiedStoredAddZeroRight_propagated_error_bound
flFhtSylvesterScheduleModifiedStoredMulOne
flFhtSylvesterScheduleModifiedStoredMulOne_propagated_error_bound
flFhtSylvesterScheduleModifiedStoredSubZeroRight
flFhtSylvesterScheduleModifiedStoredSubZeroRight_propagated_error_bound
flFhtScaledPairScheduleModifiedStoredAddZeroRight
flFhtScaledPairScheduleModifiedStoredAddZeroRight_sqrtInvNatScale_error_bound
flFhtScaledPairScheduleModifiedStoredMulOne
flFhtScaledPairScheduleModifiedStoredMulOne_sqrtInvNatScale_error_bound
flFhtScaledPairScheduleModifiedStoredSubZeroRight
flFhtScaledPairScheduleModifiedStoredSubZeroRight_sqrtInvNatScale_error_bound

```

```

flFhtScaledPairScheduleMatrixModifiedStoredAddZeroRight
flFhtScaledPairScheduleMatrixModifiedStoredAddZeroRight_sqrtInvNatScale_error_bound
flFhtScaledPairScheduleMatrixModifiedStoredMulOne_sqrtInvNatScale_error_bound
flFhtScaledPairScheduleMatrixModifiedStoredSubZeroRight_sqrtInvNatScale_error_bound
flFhtScaledSylvesterScheduleMatrixModifiedStoredAddZeroRight
flFhtScaledSylvesterScheduleMatrixModifiedStoredAddZeroRight_sqrtInvNatScale_error_bound
flFhtScaledSylvesterScheduleMatrixModifiedStoredMulOne_sqrtInvNatScale_error_bound
flFhtScaledSylvesterScheduleMatrixModifiedStoredSubZeroRight_sqrtInvNatScale_error_bound
ComputedMatrix.flScaledFhtSylvesterScheduleColumnsSqrtInvNatModifiedStoredAddZeroRight
ComputedMatrix.flScaledFhtSylvesterScheduleColumnsSqrtInvNatModifiedStoredAddZeroRight_entry_
error_bound
ComputedMatrix.flScaledFhtSylvesterScheduleColumnsSqrtInvNatModifiedStoredMulOne_entry_error_
bound
ComputedMatrix.flScaledFhtSylvesterScheduleColumnsSqrtInvNatModifiedStoredSubZeroRight_entry_
error_bound
ComputedPreconditioner.flSignedHadamardSylvesterFhtScheduleModifiedStoredAddZeroRight
ComputedPreconditioner.flSignedHadamardSylvesterFhtScheduleModifiedStoredAddZeroRight_entry_
error_bound
ComputedPreconditioner.flSignedHadamardSylvesterFhtScheduleModifiedStoredMulOne_entry_error_bound
ComputedPreconditioner.flSignedHadamardSylvesterFhtScheduleModifiedStoredSubZeroRight_entry_
error_bound
signedHadamardSylvesterFhtScheduleModifiedStoredAddZeroRightPreconditioner
signedHadamardSylvesterFhtScheduleModifiedStoredMulOnePreconditioner
signedHadamardSylvesterFhtScheduleModifiedStoredSubZeroRightPreconditioner
signedHadamardSylvesterFhtScheduleStoredSignModifiedStoredAddZeroRightPreconditioner
signedHadamardSylvesterFhtScheduleStoredSignModifiedStoredMulOnePreconditioner
signedHadamardSylvesterFhtScheduleStoredSignModifiedStoredSubZeroRightPreconditioner
signedHadamardUniformRowTraceProbability_eventProb_
    sylvesterFhtScheduleModifiedStoredAddZeroRightComputedLeftPreconditioned_
    fl_uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_
    sylvesterFhtScheduleModifiedStoredMulOneComputedLeftPreconditioned_
    fl_uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_
    sylvesterFhtScheduleModifiedStoredSubZeroRightComputedLeftPreconditioned_
    fl_uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_
    sylvesterFhtScheduleStoredSignModifiedStoredAddZeroRightComputedLeftPreconditioned_
    fl_uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_
    sylvesterFhtScheduleStoredSignModifiedStoredMulOneComputedLeftPreconditioned_

```

```

fl_uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_
sylvesterFhtScheduleStoredSignModifiedStoredSubZeroRightComputedLeftPreconditioned_
fl_uniformRowPerturbEvent_eq_one
signedHadamardUniformRowTraceProbability_eventProb_
sylvesterFhtScheduleStoredSignModifiedStoredMulOneComputedLeftPreconditioned_
factoredInput_fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_
two_sided_finiteLoewnerLe_ge_one_sub_delta_srht_log_preprocess
signedHadamardUniformRowTraceProbability_eventProb_
sylvesterFhtScheduleStoredSignModifiedStoredSubZeroRightComputedLeftPreconditioned_
factoredInput_fl_uniformRowSampleGramDotWithFlSqrtMulInvSqrtDen_
two_sided_finiteLoewnerLe_ge_one_sub_delta_srht_log_preprocess

```

Corollary 255 (Computed-projector replacement for Theorem 183). *Let $Q \in \mathbb{R}^{m \times d}$ be the exact analysis basis or singular-vector table used to define the projection matrix*

$$\Pi = QQ^T, \quad \Pi_{ij} = \sum_a Q_{ia} Q_{ja}.$$

Suppose the implementation computes $\hat{Q}$ with certificate

$$|\hat{Q}_{ia} - Q_{ia}| \leq E_{ia}^Q, \quad E_{ia}^Q \geq 0,$$

and forms the projection matrix by the rounded product

$$\hat{\Pi} = \text{fl}(\hat{Q}\hat{Q}^T).$$

If γ_d is valid, then for every i, j ,

$$\begin{aligned} |\hat{\Pi}_{ij} - \Pi_{ij}| &\leq \gamma_d \sum_a |\hat{Q}_{ia}| |\hat{Q}_{ja}| \\ &\quad + \sum_a E_{ia}^Q |\hat{Q}_{ja}| + \sum_a |Q_{ia}| E_{ja}^Q =: E_{ij}^\Pi. \end{aligned}$$

Consequently the one-sided Algorithm 3 preprocessing bounds become

$$|\text{fl}(\hat{\Pi}A)_{ij} - (\Pi A)_{ij}| \leq \gamma_m \sum_k |\hat{\Pi}_{ik}| |A_{kj}| + \sum_k E_{ik}^\Pi |A_{kj}|,$$

and analogously on the right,

$$|\text{fl}(A\hat{\Pi})_{ij} - (A\Pi)_{ij}| \leq \gamma_m \sum_k |A_{ik}| |\hat{\Pi}_{kj}| + \sum_k |A_{ik}| E_{kj}^\Pi.$$

Thus a computed SVD/QR/singular-vector routine enters through the E^Q certificate and the rounded formation of QQ^T , not through a hidden exact projector.

Lean formalization.

```

basisColumnProjector
fl_basisColumnProjector
fl_basisColumnProjector_total_error_bound
flBasisColumnProjectorEntryErrorBudget
fl_basisColumnProjector_entry_error_budget_bound
ComputedPreconditioner.flBasisColumnProjector
fl_preconditionRowsWithComputedLeft_total_error_bound
fl_preconditionColumnsWithComputedRight_total_error_bound

```

Corollary 256 (Certified QR/SVD basis routine handoff for Theorem 183). *Let a concrete QR, SVD, singular-vector, or orthonormal-basis routine return $\widehat{Q} \in \mathbb{R}^{m \times d}$ together with an entrywise certificate against the exact analysis basis Q ,*

$$|\widehat{Q}_{ia} - Q_{ia}| \leq E_{ia}^Q, \quad E_{ia}^Q \geq 0.$$

Define

$$\widehat{\Pi} = \mathfrak{fl}(\widehat{Q}\widehat{Q}^T), \quad \Pi = QQ^T,$$

and

$$E_{ij}^\Pi = \gamma_d \sum_a |\widehat{Q}_{ia}| |\widehat{Q}_{ja}| + \sum_a E_{ia}^Q |\widehat{Q}_{ja}| + \sum_a |Q_{ia}| E_{ja}^Q.$$

If the length- d dot products used to form $\widehat{Q}\widehat{Q}^T$ are γ -valid, then

$$|\widehat{\Pi}_{ij} - \Pi_{ij}| \leq E_{ij}^\Pi \quad \text{for all } i, j.$$

Thus a routine-produced basis table feeds Algorithm 3 through the explicit certificate E^Q , not by replacing $\widehat{Q}$ with the exact analysis basis.

For the two-sided projector branch, suppose two routines return $\widehat{Q}_L, \widehat{Q}_R$ with entrywise radii E^{Q_L}, E^{Q_R} , and define the corresponding projector budgets E^{Π_L}, E^{Π_R} by the same formula. If the matrix-product dimensions are γ -valid, then for every i, j ,

$$\begin{aligned}
& |\mathfrak{fl}(\widehat{\Pi}_L A \widehat{\Pi}_R)_{ij} - (\Pi_L A \Pi_R)_{ij}| \\
& \leq \gamma_n \sum_t |\mathfrak{fl}(\widehat{\Pi}_L A)_{it}| |\widehat{\Pi}_{R,tj}| \\
& \quad + \sum_t \left(\gamma_m \sum_a |\widehat{\Pi}_{L,ia}| |A_{at}| \right) |\widehat{\Pi}_{R,tj}| \\
& \quad + \sum_t \left(\sum_a E_{ia}^{\Pi_L} |A_{at}| \right) |\widehat{\Pi}_{R,tj}| + \sum_t |(\Pi_L A)_{it}| E_{tj}^{\Pi_R}.
\end{aligned}$$

The displayed bound charges basis-routine error, projector formation, and the two rounded preprocessing products. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```

ComputedMatrix.ofEntrywiseBound
certifiedBasisProjectorEntryErrorBudget
certifiedBasisProjectorEntryErrorBudget_nonneg
fl_basisColumnProjector_of_certifiedBasis_entry_error_bound
ComputedPreconditioner.flBasisColumnProjectorOfCertifiedBasis
ComputedPreconditioner.flBasisColumnProjectorOfCertifiedBasis_entry_error_bound
fl_preconditionElementsWithCertifiedBasisProjectors_total_error_bound

```

Corollary 257 (Right-orthogonal QR/SVD reference handoff for Theorem 183). *Let $Q \in \mathbb{R}^{m \times d}$ be the exact analysis basis and let $O \in \mathbb{R}^{d \times d}$ be exact and orthogonal. Define the rotated reference table*

$$Q^O = QO.$$

Then

$$Q^O(Q^O)^T = QQ^T.$$

Consequently, if a concrete QR, SVD, singular-vector, or orthonormal-basis routine returns $\hat{Q}$ with a certificate against the rotated reference,

$$|\hat{Q}_{ia} - Q_{ia}^O| \leq E_{ia}^Q, \quad E_{ia}^Q \geq 0,$$

and

$$\hat{\Pi} = \mathfrak{H}(\hat{Q}\hat{Q}^T), \quad \Pi = QQ^T,$$

then the budget

$$\begin{aligned}
E_{ij}^{\Pi,O} &= \gamma_d \sum_a |\hat{Q}_{ia}| |\hat{Q}_{ja}| \\
&\quad + \sum_a E_{ia}^Q |\hat{Q}_{ja}| + \sum_a |Q_{ia}^O| E_{ja}^Q
\end{aligned}$$

satisfies

$$|\hat{\Pi}_{ij} - \Pi_{ij}| \leq E_{ij}^{\Pi,O} \quad \text{for all } i, j,$$

provided the length- d projector dot products are γ -valid.

The same correction applies to generated-then-stored basis tables. Suppose the routine returns Q^{raw} with

$$|Q_{ia}^{\text{raw}} - Q_{ia}^O| \leq E_{ia}^Q,$$

and the algorithm stores the table actually used in projector formation as

$$|Q_{ia}^{\text{store}} - Q_{ia}^{\text{raw}}| \leq C_{ia}^Q, \quad C_{ia}^Q \geq 0.$$

With

$$\hat{\Pi} = \mathfrak{H}(Q^{\text{store}}(Q^{\text{store}})^T), \quad \Pi = QQ^T,$$

the stored-table budget

$$\begin{aligned}
E_{ij}^{\Pi,O/\text{store}} &= \gamma_d \sum_a |Q_{ia}^{\text{store}}| |Q_{ja}^{\text{store}}| \\
&\quad + \sum_a (C_{ia}^Q + E_{ia}^Q) |Q_{ja}^{\text{store}}| \\
&\quad + \sum_a |Q_{ia}^O| (C_{ja}^Q + E_{ja}^Q)
\end{aligned}$$

satisfies

$$|\hat{\Pi}_{ij} - \Pi_{ij}| \leq E_{ij}^{\Pi, O/\text{store}} \quad \text{for all } i, j.$$

Thus QR/SVD sign choices or right-orthogonal basis rotations are handled as an exact projector equivalence, while all computed non-probability quantities $\hat{Q}, Q^{\text{raw}}, Q^{\text{store}}$ remain explicitly certified. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```
basisColumnProjector_matMulRectRight_orthogonal
fl_basisColumnProjector_entry_error_budget_bound_rightOrthogonalReference
fl_basisColumnProjector_of_rightOrthogonalCertifiedBasis_entry_error_bound
fl_basisColumnProjector_of_rightOrthogonalCertifiedStoredBasis_entry_error_bound
```

Corollary 258 (Generated-then-stored QR/SVD basis handoff for Theorem 183). *Let a concrete QR, SVD, singular-vector, or orthonormal-basis routine first return a raw table $Q^{\text{raw}} \in \mathbb{R}^{m \times d}$ together with an entrywise generation certificate against the exact analysis basis Q ,*

$$|Q_{ia}^{\text{raw}} - Q_{ia}| \leq E_{ia}^Q, \quad E_{ia}^Q \geq 0.$$

Suppose the implemented algorithm does not use Q^{raw} directly, but stores or copies it as the table Q^{store} used in the projector formation, with a separate storage certificate

$$|Q_{ia}^{\text{store}} - Q_{ia}^{\text{raw}}| \leq C_{ia}^Q, \quad C_{ia}^Q \geq 0.$$

Define

$$\hat{\Pi} = \text{fl}(Q^{\text{store}}(Q^{\text{store}})^T), \quad \Pi = QQ^T,$$

and

$$\begin{aligned} E_{ij}^{\Pi, \text{store}} &= \gamma_d \sum_a |Q_{ia}^{\text{store}}| |Q_{ja}^{\text{store}}| \\ &\quad + \sum_a (C_{ia}^Q + E_{ia}^Q) |Q_{ja}^{\text{store}}| \\ &\quad + \sum_a |Q_{ia}| (C_{ja}^Q + E_{ja}^Q). \end{aligned}$$

If the length- d dot products used to form $Q^{\text{store}}(Q^{\text{store}})^T$ are γ -valid, then

$$|\hat{\Pi}_{ij} - \Pi_{ij}| \leq E_{ij}^{\Pi, \text{store}} \quad \text{for all } i, j.$$

For the concrete rounded-copy routines

$$Q_{ia}^{\text{store}} = \text{fl_mul}(Q_{ia}^{\text{raw}}, 1), \quad Q_{ia}^{\text{store}} = \text{fl_add}(Q_{ia}^{\text{raw}}, 0), \quad Q_{ia}^{\text{store}} = \text{fl_sub}(Q_{ia}^{\text{raw}}, 0),$$

the formalized storage certificate instantiates

$$C_{ia}^Q = u |Q_{ia}^{\text{raw}}|.$$

For the two-sided projector branch, suppose left and right routines provide $(Q_L^{\text{raw}}, Q_L^{\text{store}}, E^{Q_L}, C^{Q_L})$ and $(Q_R^{\text{raw}}, Q_R^{\text{store}}, E^{Q_R}, C^{Q_R})$ satisfying the same generation and storage

certificates, and define $E^{\Pi_L, \text{store}}$ and $E^{\Pi_R, \text{store}}$ by the displayed formula. If the matrix-product dimensions are γ -valid, then for every i, j ,

$$\begin{aligned}
& |\text{fl}(\widehat{\Pi}_L A \widehat{\Pi}_R)_{ij} - (\Pi_L A \Pi_R)_{ij}| \\
& \leq \gamma_n \sum_t |\text{fl}(\widehat{\Pi}_L A)_{it}| |\widehat{\Pi}_{R,tj}| \\
& \quad + \sum_t \left(\gamma_m \sum_a |\widehat{\Pi}_{L,ia}| |A_{at}| \right) |\widehat{\Pi}_{R,tj}| \\
& \quad + \sum_t \left(\sum_a E_{ia}^{\Pi_L, \text{store}} |A_{at}| \right) |\widehat{\Pi}_{R,tj}| + \sum_t |(\Pi_L A)_{it}| E_{tj}^{\Pi_R, \text{store}}.
\end{aligned}$$

The result charges the raw basis-generation error, the actual table-storage error, floating-point projector formation, and the two rounded preprocessing products. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```

ComputedMatrix.ofEntrywiseBoundThenStorage
ComputedMatrix.ofEntrywiseBoundStoredMulOne
ComputedMatrix.ofEntrywiseBoundStoredAddZeroRight
ComputedMatrix.ofEntrywiseBoundStoredSubZeroRight
certifiedStoredBasisProjectorEntryErrorBudget
certifiedStoredBasisProjectorEntryErrorBudget_nonneg
fl_basisColumnProjector_of_certifiedStoredBasis_entry_error_bound
ComputedPreconditioner.flBasisColumnProjectorOfCertifiedStoredBasis
ComputedPreconditioner.flBasisColumnProjectorOfCertifiedStoredBasis_entry_error_bound
fl_preconditionElementsWithCertifiedStoredBasisProjectors_total_error_bound

```

Corollary 259 (Normwise generated-then-stored QR/SVD basis handoff for Theorem 183). *Let a concrete QR, SVD, singular-vector, or orthonormal-basis routine first return a raw table $Q^{\text{raw}} \in \mathbb{R}^{m \times d}$. Suppose the algorithm then stores or copies this table as Q^{store} , and the projector is formed from Q^{store} , with storage certificate*

$$|Q_{ia}^{\text{store}} - Q_{ia}^{\text{raw}}| \leq C_{ia}^Q, \quad C_{ia}^Q \geq 0.$$

Assume that the raw routine supplies one of the following generation certificates against the exact analysis basis Q :

$$\begin{aligned}
\text{Frobenius:} \quad & \|Q^{\text{raw}} - Q\|_F \leq \eta, \quad \eta \geq 0, \\
\text{columnwise:} \quad & \|Q_{:a}^{\text{raw}} - Q_{:a}\|_2 \leq \eta_a, \quad \eta_a \geq 0, \\
\text{operator:} \quad & \|(Q^{\text{raw}} - Q)x\|_2 \leq \eta \|x\|_2 \quad \text{for all } x, \quad \eta \geq 0.
\end{aligned}$$

Define the raw entry radius

$$\rho_{ia} = \begin{cases} \eta, & \text{in the Frobenius case,} \\ \eta_a, & \text{in the columnwise case,} \\ \eta, & \text{in the operator case.} \end{cases}$$

The formalized constructors prove

$$|Q_{ia}^{\text{store}} - Q_{ia}| \leq C_{ia}^Q + \rho_{ia}.$$

Consequently, with

$$\widehat{\Pi} = \text{fl}(Q^{\text{store}}(Q^{\text{store}})^T), \quad \Pi = QQ^T,$$

and

$$\begin{aligned} E_{ij}^{\Pi, \text{norm}/\text{store}} &= \gamma_d \sum_a |Q_{ia}^{\text{store}}| |Q_{ja}^{\text{store}}| \\ &\quad + \sum_a (C_{ia}^Q + \rho_{ia}) |Q_{ja}^{\text{store}}| \\ &\quad + \sum_a |Q_{ia}| (C_{ja}^Q + \rho_{ja}), \end{aligned}$$

if the length- d dot products used to form $Q^{\text{store}}(Q^{\text{store}})^T$ are γ -valid, then

$$|\widehat{\Pi}_{ij} - \Pi_{ij}| \leq E_{ij}^{\Pi, \text{norm}/\text{store}} \quad \text{for all } i, j.$$

The concrete rounded-copy paths from the previous corollary instantiate $C_{ia}^Q = u|Q_{ia}^{\text{raw}}|$ for $\text{fl_mul}(Q_{ia}^{\text{raw}}, 1)$, $\text{fl_add}(Q_{ia}^{\text{raw}}, 0)$, or $\text{fl_sub}(Q_{ia}^{\text{raw}}, 0)$.

For the two-sided projector branch, suppose left and right routines provide raw/stored tables with storage radii C^{Q_L}, C^{Q_R} and raw normwise radii ρ^L, ρ^R , where each ρ is obtained from a Frobenius, columnwise, or operator certificate as above. Let $E^{\Pi_L, \text{norm}/\text{store}}$ and $E^{\Pi_R, \text{norm}/\text{store}}$ be the corresponding displayed projector budgets. If the matrix-product dimensions are γ -valid, then for every i, j ,

$$\begin{aligned} &|\text{fl}(\widehat{\Pi}_L A \widehat{\Pi}_R)_{ij} - (\Pi_L A \Pi_R)_{ij}| \\ &\leq \gamma_n \sum_t |\text{fl}(\widehat{\Pi}_L A)_{it}| |\widehat{\Pi}_{R,tj}| \\ &\quad + \sum_t \left(\gamma_m \sum_a |\widehat{\Pi}_{L,ia}| |A_{at}| \right) |\widehat{\Pi}_{R,tj}| \\ &\quad + \sum_t \left(\sum_a E_{ia}^{\Pi_L, \text{norm}/\text{store}} |A_{at}| \right) |\widehat{\Pi}_{R,tj}| + \sum_t |(\Pi_L A)_{it}| E_{tj}^{\Pi_R, \text{norm}/\text{store}}. \end{aligned}$$

Thus the two-sided implementation-facing theorem charges the raw normwise basis-generation certificates, the actual table-storage errors, floating-point projector formation, and the two rounded preprocessing products. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```

ComputedMatrix.ofFrobeniusBoundThenStorage
ComputedMatrix.ofColumnVecNorm2BoundThenStorage
ComputedMatrix.ofRectOpNorm2BoundThenStorage
fl_basisColumnProjector_of_frobeniusCertifiedStoredBasis_entry_error_bound
fl_basisColumnProjector_of_columnwiseCertifiedStoredBasis_entry_error_bound
fl_basisColumnProjector_of_opNormCertifiedStoredBasis_entry_error_bound
ComputedPreconditioner.flBasisColumnProjectorOfFrobeniusCertifiedStoredBasis
ComputedPreconditioner.flBasisColumnProjectorOfColumnwiseCertifiedStoredBasis

```

```

ComputedPreconditioner.flBasisColumnProjectorOfOpNormCertifiedStoredBasis
fl_preconditionElementsWithFrobeniusCertifiedStoredBasisProjectors_total_error_bound
fl_preconditionElementsWithColumnwiseCertifiedStoredBasisProjectors_total_error_bound
fl_preconditionElementsWithOpNormCertifiedStoredBasisProjectors_total_error_bound

```

Corollary 260 (Normwise storage certificates for generated QR/SVD basis handoffs). *Let Q^{raw} be the table produced by a QR, SVD, singular-vector, or orthonormal-basis routine, and let Q^{store} be the table actually stored and used to form the projector. Suppose the routine and storage path provide one of the following certificate pairs against the exact analysis basis Q :*

$$\begin{aligned}
\text{entrywise/Frobenius storage: } & |Q_{ia}^{\text{raw}} - Q_{ia}| \leq E_{ia}, \quad \|Q^{\text{store}} - Q^{\text{raw}}\|_F \leq \sigma, \\
\text{Frobenius/Frobenius: } & \|Q^{\text{raw}} - Q\|_F \leq \eta, \quad \|Q^{\text{store}} - Q^{\text{raw}}\|_F \leq \sigma, \\
\text{columnwise/columnwise: } & \|Q_{:a}^{\text{raw}} - Q_{:a}\|_2 \leq \eta_a, \quad \|Q_{:a}^{\text{store}} - Q_{:a}^{\text{raw}}\|_2 \leq \sigma_a, \\
\text{operator/operator: } & \|(Q^{\text{raw}} - Q)x\|_2 \leq \eta\|x\|_2, \quad \|(Q^{\text{store}} - Q^{\text{raw}})x\|_2 \leq \sigma\|x\|_2.
\end{aligned}$$

Assume all displayed radii are nonnegative. Define entry radii

$$(\rho_{ia}, \tau_{ia}) = \begin{cases} (E_{ia}, \sigma), & \text{entrywise/Frobenius storage,} \\ (\eta, \sigma), & \text{Frobenius/Frobenius,} \\ (\eta_a, \sigma_a), & \text{columnwise/columnwise,} \\ (\eta, \sigma), & \text{operator/operator.} \end{cases}$$

The formalized constructors prove

$$|Q_{ia}^{\text{store}} - Q_{ia}| \leq \tau_{ia} + \rho_{ia}.$$

Consequently, with

$$\widehat{\Pi} = \text{fl}(Q^{\text{store}}(Q^{\text{store}})^T), \quad \Pi = QQ^T,$$

and

$$\begin{aligned}
E_{ij}^{\Pi, \text{norm}/\text{store2}} &= \gamma_d \sum_a |Q_{ia}^{\text{store}}| |Q_{ja}^{\text{store}}| \\
&\quad + \sum_a (\tau_{ia} + \rho_{ia}) |Q_{ja}^{\text{store}}| + \sum_a |Q_{ia}| (\tau_{ja} + \rho_{ja}),
\end{aligned}$$

if the length- d dot products used to form the projector are γ -valid, then

$$|\widehat{\Pi}_{ij} - \Pi_{ij}| \leq E_{ij}^{\Pi, \text{norm}/\text{store2}} \quad \text{for all } i, j.$$

For two-sided Algorithm 3 preprocessing, the new ‘ComputedMatrix’ constructors supply the left and right computed-basis certificates consumed by the existing computed-projector theorem $\text{fl}(\widehat{\Pi}_L A \widehat{\Pi}_R)$. Thus storage layouts with only normwise certificates are charged explicitly; no floating-point or law-transfer error is added for sampling probabilities or sampling laws.

Lean formalization.

```

ComputedMatrix.ofEntrywiseBoundThenFrobeniusStorage
ComputedMatrix.ofFrobeniusBoundThenFrobeniusStorage
ComputedMatrix.ofColumnVecNorm2BoundThenColumnVecNorm2Storage
ComputedMatrix.ofRectOpNorm2BoundThenRectOpNorm2Storage

```

```

fl_basisColumnProjector_of_frobeniusCertifiedFrobeniusStoredBasis_entry_error_bound
fl_basisColumnProjector_of_columnwiseCertifiedColumnwiseStoredBasis_entry_error_bound
fl_basisColumnProjector_of_opNormCertifiedOpNormStoredBasis_entry_error_bound
ComputedPreconditioner.flBasisColumnProjector
fl_preconditionElementsWithComputedBasisProjectors_total_error_bound

```

Corollary 261 (Frobenius-certified QR/SVD basis routine handoff for Theorem 183). *Let a concrete QR, SVD, singular-vector, or orthonormal-basis routine return $\hat{Q} \in \mathbb{R}^{m \times d}$ together with a normwise certificate against the exact analysis basis Q ,*

$$\|\hat{Q} - Q\|_F \leq \eta, \quad \eta \geq 0.$$

Define

$$\hat{\Pi} = \mathfrak{H}(\hat{Q}\hat{Q}^T), \quad \Pi = QQ^T,$$

and

$$\begin{aligned}
E_{ij}^{\Pi, F} &= \gamma_d \sum_a |\hat{Q}_{ia}| |\hat{Q}_{ja}| \\
&\quad + \sum_a \eta |\hat{Q}_{ja}| + \sum_a |Q_{ia}| \eta.
\end{aligned}$$

If the length- d dot products used to form $\hat{Q}\hat{Q}^T$ are γ -valid, then

$$|\hat{\Pi}_{ij} - \Pi_{ij}| \leq E_{ij}^{\Pi, F} \quad \text{for all } i, j.$$

The proof first uses the Frobenius entry bound $|\hat{Q}_{ia} - Q_{ia}| \leq \|\hat{Q} - Q\|_F \leq \eta$, then applies the entrywise certified-projector theorem. Thus a normwise QR/SVD certificate is not silently treated as exact singular-vector data; it is converted into the displayed implementation-facing projector radius.

For the two-sided projector branch, suppose two routines return $\hat{Q}_L, \hat{Q}_R$ with Frobenius radii η_L, η_R , and define $E^{\Pi_L, F}$ and $E^{\Pi_R, F}$ by the same formula. If the matrix-product dimensions are γ -valid, then for every i, j ,

$$\begin{aligned}
&|\mathfrak{H}(\hat{\Pi}_L A \hat{\Pi}_R)_{ij} - (\Pi_L A \Pi_R)_{ij}| \\
&\leq \gamma_n \sum_t |\mathfrak{H}(\hat{\Pi}_L A)_{it}| |\hat{\Pi}_{R, tj}| \\
&\quad + \sum_t \left(\gamma_m \sum_a |\hat{\Pi}_{L, ia}| |A_{at}| \right) |\hat{\Pi}_{R, tj}| \\
&\quad + \sum_t \left(\sum_a E_{ia}^{\Pi_L, F} |A_{at}| \right) |\hat{\Pi}_{R, tj}| + \sum_t |(\Pi_L A)_{it}| E_{tj}^{\Pi_R, F}.
\end{aligned}$$

The displayed bound charges the normwise basis-routine certificates, projector formation, and the two rounded preprocessing products. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```

ComputedMatrix.ofFrobeniusBound
frobeniusCertifiedBasisProjectorEntryErrorBudget

```

```

frobeniusCertifiedBasisProjectorEntryErrorBudget_nonneg
fl_basisColumnProjector_of_frobeniusCertifiedBasis_entry_error_bound
ComputedPreconditioner.flBasisColumnProjectorOfFrobeniusCertifiedBasis
ComputedPreconditioner.flBasisColumnProjectorOfFrobeniusCertifiedBasis_entry_error_bound
fl_preconditionElementsWithFrobeniusCertifiedBasisProjectors_total_error_bound

```

Corollary 262 (Columnwise-certified QR/SVD basis routine handoff for Theorem 183). *Let a concrete QR, SVD, singular-vector, or orthonormal-basis routine return $\hat{Q} \in \mathbb{R}^{m \times d}$ together with per-column Euclidean certificates against the exact analysis basis Q ,*

$$\|\hat{Q}_{:a} - Q_{:a}\|_2 \leq \eta_a, \quad \eta_a \geq 0, \quad a = 1, \dots, d.$$

Define

$$\hat{\Pi} = \mathfrak{fl}(\hat{Q}\hat{Q}^T), \quad \Pi = QQ^T,$$

and

$$\begin{aligned} E_{ij}^{\Pi, \text{col}} &= \gamma_d \sum_a |\hat{Q}_{ia}| |\hat{Q}_{ja}| \\ &\quad + \sum_a \eta_a |\hat{Q}_{ja}| + \sum_a |Q_{ia}| \eta_a. \end{aligned}$$

If the length- d dot products used to form $\hat{Q}\hat{Q}^T$ are γ -valid, then

$$|\hat{\Pi}_{ij} - \Pi_{ij}| \leq E_{ij}^{\Pi, \text{col}} \quad \text{for all } i, j.$$

Indeed, coordinate domination gives $|\hat{Q}_{ia} - Q_{ia}| \leq \|\hat{Q}_{:a} - Q_{:a}\|_2 \leq \eta_a$ for each column a , and the entrywise certified-projector theorem applies with this column-dependent radius.

For the two-sided projector branch, suppose two routines return $\hat{Q}_L, \hat{Q}_R$ with columnwise radii η_a^L, η_a^R , and define $E^{\Pi_L, \text{col}}$ and $E^{\Pi_R, \text{col}}$ by the same formula. If the matrix-product dimensions are γ -valid, then for every i, j ,

$$\begin{aligned} &|\mathfrak{fl}(\hat{\Pi}_L A \hat{\Pi}_R)_{ij} - (\Pi_L A \Pi_R)_{ij}| \\ &\leq \gamma_n \sum_t |\mathfrak{fl}(\hat{\Pi}_L A)_{it}| |\hat{\Pi}_{R,tj}| \\ &\quad + \sum_t \left(\gamma_m \sum_a |\hat{\Pi}_{L,ia}| |A_{at}| \right) |\hat{\Pi}_{R,tj}| \\ &\quad + \sum_t \left(\sum_a E_{ia}^{\Pi_L, \text{col}} |A_{at}| \right) |\hat{\Pi}_{R,tj}| + \sum_t |(\Pi_L A)_{it}| E_{tj}^{\Pi_R, \text{col}}. \end{aligned}$$

The displayed bound charges the per-column basis-routine certificates, projector formation, and the two rounded preprocessing products. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```

ComputedMatrix.ofColumnVecNorm2Bound
columnwiseCertifiedBasisProjectorEntryErrorBudget
columnwiseCertifiedBasisProjectorEntryErrorBudget_nonneg

```

```

fl_basisColumnProjector_of_columnwiseCertifiedBasis_entry_error_bound
ComputedPreconditioner.flBasisColumnProjectorOfColumnwiseCertifiedBasis
ComputedPreconditioner.flBasisColumnProjectorOfColumnwiseCertifiedBasis_entry_error_bound
fl_preconditionElementsWithColumnwiseCertifiedBasisProjectors_total_error_bound

```

Corollary 263 (Operator-certified QR/SVD basis routine handoff for Theorem 183). *Let a concrete QR, SVD, singular-vector, orthonormal-basis, or subspace perturbation routine return $\hat{Q} \in \mathbb{R}^{m \times d}$. Suppose it supplies the rectangular operator certificate*

$$\|(\hat{Q} - Q)x\|_2 \leq \eta \|x\|_2 \quad \text{for all } x \in \mathbb{R}^d, \quad \eta \geq 0,$$

against the exact analysis basis Q . Define

$$\hat{\Pi} = \mathfrak{fl}(\hat{Q}\hat{Q}^T), \quad \Pi = QQ^T,$$

and

$$\begin{aligned} E_{ij}^{\Pi,2} &= \gamma_d \sum_a |\hat{Q}_{ia}| |\hat{Q}_{ja}| \\ &\quad + \sum_a \eta |\hat{Q}_{ja}| + \sum_a |Q_{ia}| \eta. \end{aligned}$$

If the length- d dot products used to form $\hat{Q}\hat{Q}^T$ are γ -valid, then

$$|\hat{\Pi}_{ij} - \Pi_{ij}| \leq E_{ij}^{\Pi,2} \quad \text{for all } i, j.$$

Indeed, testing the operator certificate on the standard basis vector e_a gives $\|(\hat{Q} - Q)e_a\|_2 \leq \eta$; coordinate domination then gives $|\hat{Q}_{ia} - Q_{ia}| \leq \eta$. The entrywise certified-projector theorem applies with this uniform radius.

For the two-sided projector branch, suppose two routines return $\hat{Q}_L, \hat{Q}_R$ with operator radii η_L, η_R , and define $E^{\Pi_L,2}$ and $E^{\Pi_R,2}$ by the same formula. If the matrix-product dimensions are γ -valid, then for every i, j ,

$$\begin{aligned} &|\mathfrak{fl}(\hat{\Pi}_L A \hat{\Pi}_R)_{ij} - (\Pi_L A \Pi_R)_{ij}| \\ &\leq \gamma_n \sum_t |\mathfrak{fl}(\hat{\Pi}_L A)_{it}| |\hat{\Pi}_{R,tj}| \\ &\quad + \sum_t \left(\gamma_m \sum_a |\hat{\Pi}_{L,ia}| |A_{at}| \right) |\hat{\Pi}_{R,tj}| \\ &\quad + \sum_t \left(\sum_a E_{ia}^{\Pi_L,2} |A_{at}| \right) |\hat{\Pi}_{R,tj}| + \sum_t |(\Pi_L A)_{it}| E_{tj}^{\Pi_R,2}. \end{aligned}$$

The displayed bound charges the operator-norm routine certificates, projector formation, and the two rounded preprocessing products. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```

ComputedMatrix.ofRectOpNorm2Bound
opNormCertifiedBasisProjectorEntryErrorBudget
opNormCertifiedBasisProjectorEntryErrorBudget_nonneg

```

```

fl_basisColumnProjector_of_opNormCertifiedBasis_entry_error_bound
ComputedPreconditioner.flBasisColumnProjectorOfOpNormCertifiedBasis
ComputedPreconditioner.flBasisColumnProjectorOfOpNormCertifiedBasis_entry_error_bound
fl_preconditionElementsWithOpNormCertifiedBasisProjectors_total_error_bound

```

Corollary 264 (Add-zero stored basis before projector formation). *Let $Q \in \mathbb{R}^{m \times k}$ be the exact basis or singular-vector table used in the analysis, and suppose the implemented table is stored entrywise by the rounded arithmetic copy*

$$\tilde{Q}_{ia} = \text{fl}_{\text{add}}(Q_{ia}, 0), \quad |\tilde{Q}_{ia} - Q_{ia}| \leq u |Q_{ia}|.$$

Let the implemented projector be formed by rounded dot products,

$$\hat{\Pi}_{ij} = \text{fl} \left(\sum_{a=1}^k \tilde{Q}_{ia} \tilde{Q}_{ja} \right), \quad \Pi_{ij} = (QQ^T)_{ij}.$$

If the length- k dot products satisfy the usual γ_k validity condition, then for every i, j

$$\begin{aligned} |\hat{\Pi}_{ij} - \Pi_{ij}| &\leq \gamma_k \sum_{a=1}^k |\tilde{Q}_{ia}| |\tilde{Q}_{ja}| \\ &\quad + u \sum_{a=1}^k |Q_{ia}| |\tilde{Q}_{ja}| + u \sum_{a=1}^k |Q_{ia}| |Q_{ja}|. \end{aligned}$$

Thus the implemented projector path charges the arithmetic storage of the basis/singular-vector table and the rounded formation of $\tilde{Q}\tilde{Q}^T$. The sampling probabilities and sampling laws are unchanged exact mathematical inputs.

Lean formalization.

```

ComputedMatrix.flAddZeroRight
ComputedMatrix.flAddZeroRight_entry_error_bound
ComputedPreconditioner.flBasisColumnProjectorStoredBasisAddZeroRight
ComputedPreconditioner.flBasisColumnProjectorStoredBasisAddZeroRight_entry_error_bound

```

Corollary 265 (Multiply-one and subtract-zero stored bases before projector formation). *Let $Q \in \mathbb{R}^{m \times k}$ be the exact basis or singular-vector table used in the analysis. Define the two additional implemented storage paths*

$$\tilde{Q}_{ia}^{(\times)} = \text{fl}_{\text{mul}}(Q_{ia}, 1), \quad \tilde{Q}_{ia}^{(-)} = \text{fl}_{\text{sub}}(Q_{ia}, 0).$$

For each $\star \in \{(\times), (-)\}$, suppose

$$|\tilde{Q}_{ia}^{\star} - Q_{ia}| \leq u |Q_{ia}| \quad \text{for all } i, a,$$

and form

$$\hat{\Pi}_{ij}^{\star} = \text{fl} \left(\sum_{a=1}^k \tilde{Q}_{ia}^{\star} \tilde{Q}_{ja}^{\star} \right), \quad \Pi_{ij} = (QQ^T)_{ij}.$$

If the length- k projector dot products satisfy the usual γ_k validity condition, then for every i, j and every $\star \in \{(\times), (-)\}$,

$$\begin{aligned} |\widehat{\Pi}_{ij}^\star - \Pi_{ij}| &\leq \gamma_k \sum_{a=1}^k |\widetilde{Q}_{ia}^\star| |\widetilde{Q}_{ja}^\star| \\ &\quad + u \sum_{a=1}^k |Q_{ia}| |\widetilde{Q}_{ja}^\star| + u \sum_{a=1}^k |Q_{ia}| |Q_{ja}|. \end{aligned}$$

Thus the implementation-facing projector theorem now covers the three rounded copy paths $\text{fl}_{\text{mul}}(Q_{ia}, 1)$, $\text{fl}_{\text{add}}(Q_{ia}, 0)$, and $\text{fl}_{\text{sub}}(Q_{ia}, 0)$. These are non-probability storage operations; sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```
ComputedMatrix.flMulOne
ComputedMatrix.flMulOne_entry_error_bound
ComputedMatrix.flSubZeroRight
ComputedMatrix.flSubZeroRight_entry_error_bound
ComputedPreconditioner.flBasisColumnProjectorStoredBasisMulOne
ComputedPreconditioner.flBasisColumnProjectorStoredBasisMulOne_entry_error_bound
ComputedPreconditioner.flBasisColumnProjectorStoredBasisSubZeroRight
ComputedPreconditioner.flBasisColumnProjectorStoredBasisSubZeroRight_entry_error_bound
```

Corollary 266 (Two-sided floating-point preprocessing). *Let $B = \Pi_L A$ and $\widehat{B} = \text{fl}(\Pi_L A)$. If the computed two-sided branch is $\text{fl}(\widehat{B}\Pi_R)$, then*

$$\begin{aligned} &\left| \text{fl}(\widehat{B}\Pi_R)_{ij} - (\Pi_L A \Pi_R)_{ij} \right| \\ &\leq \gamma_n \sum_k |\widehat{B}_{ik}| |(\Pi_R)_{kj}| + \sum_k \left(\gamma_m \sum_a |(\Pi_L)_{ia}| |A_{ak}| \right) |(\Pi_R)_{kj}|. \end{aligned}$$

Lean formalization.

```
preconditionColumns_entry_error_bound_of_entrywise
fl_preconditionElements_error_bound
fl_preconditionElementsWithComputed_error_bound
fl_preconditionElementsWithComputed_total_error_bound
```

Corollary 267 (Computed-projector replacement for Corollary 184). *Let*

$$\Pi_L = Q_L Q_L^T, \quad \Pi_R = Q_R Q_R^T$$

be the exact analysis projectors, and let

$$\widehat{\Pi}_L = \text{fl}(\widehat{Q}_L \widehat{Q}_L^T), \quad \widehat{\Pi}_R = \text{fl}(\widehat{Q}_R \widehat{Q}_R^T)$$

be the projectors formed in the implementation from computed basis or singular-vector tables. Suppose the preceding computed-projector bound gives entrywise radii E^{Π_L} and E^{Π_R} . If the product dimensions are γ -valid, then for every i, j ,

$$\begin{aligned} & |\mathfrak{fl}(\widehat{\Pi}_L A \widehat{\Pi}_R)_{ij} - (\Pi_L A \Pi_R)_{ij}| \\ & \leq \gamma_n \sum_t |\mathfrak{fl}(\widehat{\Pi}_L A)_{it}| |\widehat{\Pi}_{R,tj}| \\ & \quad + \sum_t \left(\gamma_m \sum_a |\widehat{\Pi}_{L,ia}| |A_{at}| \right) |\widehat{\Pi}_{R,tj}| \\ & \quad + \sum_t \left(\sum_a E_{ia}^{\Pi_L} |A_{at}| \right) |\widehat{\Pi}_{R,tj}| + \sum_t |(\Pi_L A)_{it}| E_{tj}^{\Pi_R}. \end{aligned}$$

The first two lines are the rounded second and first products; the last two lines are the left and right computed-projector formation errors. Sampling probabilities and sampling laws remain exact under the current convention.

Lean formalization.

`fl_preconditionElementsWithComputedBasisProjectors_total_error_bound`

7 Equation (8): Least-Squares Sketching

Equation (8) is the least-squares problem

$$x_{\text{opt}} = \arg \min_{x \in \mathbb{R}^n} \|Ax - b\|_2.$$

For a sketched problem (SA, Sb) , define

$$f(x) = \|Ax - b\|_2^2, \quad \tilde{f}(x) = \|SAx - Sb\|_2^2.$$

Definition 268 (Residual-objective preserving sketch). The sketched problem preserves the least-squares objective with accuracy ε if, for every x ,

$$(1 - \varepsilon)f(x) \leq \tilde{f}(x) \leq (1 + \varepsilon)f(x).$$

Theorem 269 (Quadratic subspace-embedding bridge). Suppose every residual has coordinates $y(x) \in \mathbb{R}^d$ in an orthonormal residual coordinate system, so that

$$f(x) = \|y(x)\|_2^2.$$

Suppose further that, for a fixed sketch outcome,

$$\tilde{f}(x) = \|y(x)\|_2^2 + y(x)^T E y(x) \quad \text{for every } x,$$

and that the coordinate Gram-error matrix satisfies the vector-action operator bound

$$\|Ez\|_2 \leq \varepsilon \|z\|_2 \quad \text{for every } z \in \mathbb{R}^d.$$

Then the sketch preserves the least-squares residual objective:

$$(1 - \varepsilon)f(x) \leq \tilde{f}(x) \leq (1 + \varepsilon)f(x) \quad \text{for every } x.$$

Moreover, on any finite probability space, a proved lower bound for the operator event $\|Ez\|_2 \leq \varepsilon \|z\|_2$ transfers directly to the least-squares preservation event.

Lean formalization.

```
abs_vecInnerProduct_le_vecNorm2_mul
abs_vecInnerProduct_matMulVec_le_of_opNorm2Le
preservesLSObjective_of_coordinate_quadratic_error
eventProb_preservesLSObjective_of_coordinate_quadratic_error
preservesLSObjective_of_coordinate_finiteLoewner_error
eventProb_preservesLSObjective_of_coordinate_finiteLoewner_error
```

Theorem 270 (Finite-Loewner subspace-embedding bridge). *In the same coordinate model, suppose instead that the Gram-error matrix satisfies the two-sided Loewner event*

$$E \preceq \varepsilon I_d, \quad -E \preceq \varepsilon I_d.$$

Then the sketch preserves every residual objective with accuracy ε . On a finite probability space, any proved lower bound for this two-sided Loewner event transfers to the least-squares preservation event.

Theorem 271 (Sketched minimizer residual guarantee). *Assume $\varepsilon < 1$, the sketch preserves every residual objective as above, and $\hat{x}$ is an exact minimizer of the sketched problem. Then, for every comparison vector x_{ref} ,*

$$\|A\hat{x} - b\|_2^2 \leq \frac{1 + \varepsilon}{1 - \varepsilon} \|Ax_{\text{ref}} - b\|_2^2.$$

In particular, this applies to an exact minimizer x_{opt} of the original problem.

Lean formalization.

```
lsResidual
lsObjective
lsLeastSquaresMinimizer
PreservesLSObjective
lsObjective_le_of_sketch_preserves
```

Corollary 272 ($1 + \eta$ residual-objective form). *If additionally*

$$\frac{1 + \varepsilon}{1 - \varepsilon} \leq 1 + \eta,$$

then

$$\|A\hat{x} - b\|_2^2 \leq (1 + \eta) \|Ax_{\text{ref}} - b\|_2^2.$$

Lean formalization.

```
lsObjective_le_one_add_eta_of_sketch_preserves
```

Corollary 273 (Randomized sketched-minimizer transfer). *On any finite probability space, if the sketch-preservation event has probability at least α , and for every outcome ω the vector $\hat{x}(\omega)$ is an exact minimizer of the sketched problem $(S_\omega A, S_\omega b)$, then the same probability lower bound applies to the sketched-minimizer objective guarantee:*

$$\mathbb{P} \left[\|A\hat{x}(\omega) - b\|_2^2 \leq \frac{1 + \varepsilon}{1 - \varepsilon} \|Ax_{\text{ref}} - b\|_2^2 \right] \geq \alpha.$$

The $1 + \eta$ version follows by the same scalar comparison as above.

Lean formalization.

eventProb_lsObjective_le_of_preserves

eventProb_lsObjective_le_one_add_eta_of_preserves

eventProb_lsObjective_le_one_add_eta_of_coordinate_finiteLoewner_error

Corollary 274 (Leverage-score row sampling to least-squares objective). *Let $U \in \mathbb{R}^{m \times d}$ have orthonormal columns, and let the Algorithm 2 row trace be sampled from the leverage-score law*

$$p_i = \frac{\|U_{i,*}\|_2^2}{d}.$$

Define the canonical residual coordinates

$$y(x) = U^T(Ax - b).$$

Assume that every residual lies in the column span of U , equivalently

$$(Ax - b)_i = \sum_{\ell=1}^d U_{i\ell} y_\ell(x), \quad 1 \leq i \leq m.$$

The repository proves from this column-space condition and $U^T U = I$ that

$$\|Ax - b\|_2^2 = \|y(x)\|_2^2.$$

Let the concrete sampled least-squares instance use rows

$$(S_\omega A)_{t,*} = \frac{A_{i_t,*}}{\sqrt{s p_{i_t}}}, \quad (S_\omega b)_t = \frac{b_{i_t}}{\sqrt{s p_{i_t}}}.$$

Then the repository proves the sampled objective identity

$$\tilde{f}_\omega(x) = \|y(x)\|_2^2 + y(x)^T \left(\tilde{U}(\omega)^T \tilde{U}(\omega) - I \right) y(x).$$

Then the leverage-score event for equation (7) transfers to the least-squares sketched-minimizer guarantee. The older operator-event route gives

$$\mathbb{P} \left[\|\hat{A}x(\omega) - b\|_2^2 \leq (1 + \eta) \|Ax_{\text{opt}} - b\|_2^2 \right] \geq 1 - \frac{1}{s(\varepsilon/d)^2},$$

provided $\varepsilon < 1$ and $(1 + \varepsilon)/(1 - \varepsilon) \leq 1 + \eta$.

The sharper finite-Loewner theorem proved for equation (7) gives the source-aligned Bennett sample-budget version. For the augmented-span basis, let $d = \dim \text{span}\{A_{,j}, b\}$. If $d > 1$, $\varepsilon > 0$, $\delta > 0$, and*

$$\log \frac{2d}{\delta} \leq s \frac{\varepsilon^2}{2(d-1) + (2/3)d\varepsilon}, \quad \log \frac{2d}{\delta} \leq s \frac{\varepsilon^2}{2(d-1) + (2/3)\varepsilon},$$

then the same sketched-minimizer event holds with probability at least $1 - \delta$.

The fully floating-point version replaces $\tilde{U}^T \tilde{U} - I$ by the rounded Gram matrix and replaces the preservation radius ε by

$$\varepsilon + \tau_{\text{Gram}}^{\text{fp}}, \quad \tau_{\text{Gram}}^{\text{fp}} = \text{rowSampleGramFullFpPerturbBudget}(\text{fp}, s, U).$$

The budget is proved nonnegative locally.

Lean formalization.

```
rowSampleGramFullFpPerturbBudget_nonneg
rowSampleLSMatrixWithBasisScale
rowSampleLSVectorWithBasisScale
fl_rowSampleLSMatrixWithBasisScale
fl_rowSampleLSVectorWithBasisScale
fl_rowSampleLSMatrixWithBasisScale_error_bound
fl_rowSampleLSVectorWithBasisScale_error_bound
fl_rowSampleLSResidualWithBasisScale_error_bound
fl_rowSampleLSResidualWithBasisScale_error_bound_of_positiveProb
lsObjective_residual_difference_bound
lsObjective_residual_budget_bound
rowSampleLSResidualFpBudget
rowSampleLSResidualFpBudget_nonneg
rowSampleLSObjectiveFpBudget
rowSampleLSObjectiveFpBudget_nonneg
fl_rowSampleLSObjectiveWithBasisScale_error_bound_of_positiveProb
fl_rowSampleLSObjectiveWithBasisScale_error_bound_of_positiveProb_budget
lsObjective_le_of_sketch_preserves_with_objective_error
lsObjective_le_one_add_eta_of_sketch_preserves_with_objective_error
eventProb_lsObjective_le_one_add_eta_of_preserves_with_objective_error
eventProb_lsObjective_le_one_add_eta_of_preserves_with_pointwise_objective_error
eventProb_lsObjective_le_one_add_eta_of_preserves_with_objective_error_on_event
eventProb_lsObjective_le_one_add_eta_of_coordinate_finiteLoewner_error_with_objective_error_on_event
residualCoordinates
ResidualsInColumnSpace
ColumnsAndRhsInColumnSpace
residualCoordinatesFromColumns
residualsInColumnSpace_of_residual_representation
lsResidual_eq_basis_sum_of_columnsAndRhsInColumnSpace
residualsInColumnSpace_of_columnsAndRhsInColumnSpace
augmentedDataVector
augmentedDataSpan
augmentedSpanBasisMatrix
augmentedSpanColumnCoords
augmentedSpanRhsCoords
hasOrthonormalColumns_augmentedSpanBasisMatrix
columnsAndRhsInColumnSpace_augmentedSpanBasisMatrix
residualsInColumnSpace_augmentedSpanBasisMatrix
```

```

lsObjective_eq_vecNorm2Sq_residualCoordinates_of_residualsInColumnSpace
rowSampleLSResidualWithBasisScale_eq_coord
rowSampleLSObjectiveWithBasisScale_eq_coordinate_quadratic_error
eventProb_lsObjective_le_one_add_eta_of_coordinate_quadratic_error
eventProb_lsObjective_le_one_add_eta_of_coordinate_finiteLoewner_error
leverageTraceProbability_eventProb_lsObjective_le_one_add_eta_of_coordinate_quadratic_error
leverageTraceProbability_eventProb_rowSampleLSObjective_le_one_add_eta_of_residual_coordinates
leverageTraceProbability_eventProb_rowSampleLSObjective_le_one_add_eta_of_residualsInColumnSpace
leverageTraceProbability_eventProb_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan
leverageTraceProbability_eventProb_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_
budget
leverageTraceProbability_eventProb_fl_lsObjective_le_one_add_eta_of_coordinate_quadratic_error
leverageTraceProbability_eventProb_fl_lsObjective_le_one_add_eta_of_augmentedSpan
leverageTraceProbability_eventProb_fl_lsObjective_le_one_add_eta_of_augmentedSpan_sample_budget
leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error
leverageTraceProbability_eventProb_fl_lsObjective_le_one_add_eta_of_idBasis

```

Theorem 275 (Literal rounded sampled-row residual perturbation). *Let $S_\omega A$ and $S_\omega b$ denote the exact sampled and scaled least-squares matrix and right-hand side obtained from Algorithm 2 equation (6). Let $\mathfrak{fl}(S_\omega A)$ and $\mathfrak{fl}(S_\omega b)$ be the concrete rounded implementation that applies the repository's floating-point division primitive to each sampled/scaled entry. If the sampled scaling denominator at row t is nonzero, then*

$$\begin{aligned}
& \left| (\mathfrak{fl}(S_\omega A)x - \mathfrak{fl}(S_\omega b))_t - (S_\omega Ax - S_\omega b)_t \right| \\
& \leq \sum_j u_{\text{fp}} |(S_\omega A)_{tj}| |x_j| + u_{\text{fp}} |(S_\omega b)_t|.
\end{aligned}$$

Under the product sampling law, the positive-probability support event discharges this nonzero-denominator premise with probability one for sampled rows.

Lean formalization.

```

fl_rowSampleLSMatrixWithBasisScale
fl_rowSampleLSVectorWithBasisScale
fl_rowSampleLSMatrixWithBasisScale_error_bound
fl_rowSampleLSVectorWithBasisScale_error_bound
fl_rowSampleLSResidualWithBasisScale_error_bound
fl_rowSampleLSResidualWithBasisScale_error_bound_of_positiveProb

```

Corollary 276 (Literal rounded sampled-row objective perturbation). *For fixed sampled rows ω and a fixed vector x , define the rowwise budget*

$$B_t(x) = \sum_j u_{\text{fp}} |(S_\omega A)_{tj}| |x_j| + u_{\text{fp}} |(S_\omega b)_t|.$$

On the positive-probability support event of the row sampler,

$$\begin{aligned} & \left| \|\mathfrak{fl}(S_\omega A)x - \mathfrak{fl}(S_\omega b)\|_2^2 - \|S_\omega Ax - S_\omega b\|_2^2 \right| \\ & \leq 2 \|S_\omega Ax - S_\omega b\|_2 \|B(x)\|_2 + \|B(x)\|_2^2. \end{aligned}$$

Lean formalization.

```
vecNorm2Sq_le_of_abs_le
vecNorm2_le_of_abs_le
abs_vecNorm2Sq_add_sub_le
lsObjective_residual_difference_bound
lsObjective_residual_budget_bound
rowSampleLSResidualFpBudget
rowSampleLSResidualFpBudget_nonneg
rowSampleLSObjectiveFpBudget
rowSampleLSObjectiveFpBudget_nonneg
fl_rowSampleLSObjectiveWithBasisScale_error_bound_of_positiveProb
fl_rowSampleLSObjectiveWithBasisScale_error_bound_of_positiveProb_budget
```

Corollary 277 (High-probability literal rounded sampled-row minimizer). *Let U be the augmented-span leverage basis for $\text{span}\{A_{*,1}, \dots, A_{*,n}, b\}$, and let $d = \dim \text{span}\{A_{*,1}, \dots, A_{*,n}, b\} > 1$. Assume $\varepsilon > 0$, $\varepsilon < 1$, $\delta > 0$, and the two Bennett sample-budget inequalities*

$$\log \frac{2d}{\delta} \leq s \frac{\varepsilon^2}{2(d-1) + (2/3)d\varepsilon}, \quad \log \frac{2d}{\delta} \leq s \frac{\varepsilon^2}{2(d-1) + (2/3)\varepsilon}.$$

Let $\hat{x}(\omega)$ be an exact minimizer of the literal rounded sampled/scaled problem

$$\min_x \|\mathfrak{fl}(S_\omega A)x - \mathfrak{fl}(S_\omega b)\|_2^2.$$

For x , define

$$T_\omega(x) = 2 \|S_\omega Ax - S_\omega b\|_2 \|B_\omega(x)\|_2 + \|B_\omega(x)\|_2^2,$$

where $B_\omega(x)$ is the rowwise residual budget from the previous corollary. If, on the positive-probability support event,

$$T_\omega(\hat{x}(\omega)) + T_\omega(x_{\text{opt}}) \leq ((1 + \eta)(1 - \varepsilon) - (1 + \varepsilon)) \|Ax_{\text{opt}} - b\|_2^2,$$

then

$$\mathbb{P}\left[\|A\hat{x}(\omega) - b\|_2^2 \leq (1 + \eta) \|Ax_{\text{opt}} - b\|_2^2\right] \geq 1 - \delta.$$

Lean formalization.

```
rowSampleLSObjectiveFpBudget
rowSampleLSObjectiveFpBudget_nonneg
fl_rowSampleLSObjectiveWithBasisScale_error_bound_of_positiveProb_budget
lsObjective_le_of_sketch_preserves_with_objective_error
lsObjective_le_one_add_eta_of_sketch_preserves_with_objective_error
```

```

eventProb_lsObjective_le_one_add_eta_of_preserves_with_objective_error
eventProb_lsObjective_le_one_add_eta_of_preserves_with_pointwise_objective_error
eventProb_lsObjective_le_one_add_eta_of_preserves_with_objective_error_on_event
eventProb_lsObjective_le_one_add_eta_of_coordinate_finiteLoewner_error_with_objective_error_on_
event
leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error

```

Corollary 278 (High-probability literal rounded sampled-row approximate solver). *Use the notation and Bennett sample-budget assumptions of the previous corollary. Let $\hat{x}(\omega)$ be an additive-gap approximate minimizer of the literal rounded sampled/scaled problem:*

$$\|\mathfrak{f}(S_\omega A)\hat{x}(\omega) - \mathfrak{f}(S_\omega b)\|_2^2 \leq \|\mathfrak{f}(S_\omega A)y - \mathfrak{f}(S_\omega b)\|_2^2 + \sigma_\omega \quad \text{for all } y.$$

If, on the positive-probability support event,

$$T_\omega(\hat{x}(\omega)) + T_\omega(x_{\text{opt}}) + \sigma_\omega \leq ((1 + \eta)(1 - \varepsilon) - (1 + \varepsilon)) \|Ax_{\text{opt}} - b\|_2^2,$$

then

$$\mathbb{P}\left[\|A\hat{x}(\omega) - b\|_2^2 \leq (1 + \eta) \|Ax_{\text{opt}} - b\|_2^2\right] \geq 1 - \delta.$$

Lean formalization.

```

IsLeastSquaresApproxMinimizer
isLeastSquaresApproxMinimizer_of_minimizer
lsObjective_le_of_sketch_preserves_with_objective_error_and_solver_gap
lsObjective_le_one_add_eta_of_sketch_preserves_with_objective_error_and_solver_gap
eventProb_lsObjective_le_one_add_eta_of_preserves_with_objective_error_and_solver_gap_on_event
eventProb_lsObjective_le_one_add_eta_of_coordinate_finiteLoewner_error_with_objective_error_and_
solver_gap_on_event
leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_solver_gap

```

Corollary 279 (High-probability literal rounded sampled-row solver with a forward-error certificate). *Use the notation and Bennett sample-budget assumptions of the previous two corollaries. For each sample trace ω , let $x_*(\omega)$ be an exact minimizer of the literal rounded sampled/scaled problem, and let $\hat{x}(\omega)$ be the vector returned by a solver. Suppose the solver also provides a componentwise certificate $d_\omega \geq 0$ such that*

$$|\hat{x}_j(\omega) - x_{*,j}(\omega)| \leq d_{\omega,j}, \quad j = 1, \dots, n.$$

For the literal rounded sampled/scaled matrix and vector, define the induced residual budget

$$R_\omega(i) = \sum_{j=1}^n |\mathfrak{f}(S_\omega A)_{ij}| d_{\omega,j}, \quad i = 1, \dots, s,$$

and the induced objective gap

$$\Sigma_\omega = 2 \|\mathfrak{f}(S_\omega A)x_*(\omega) - \mathfrak{f}(S_\omega b)\|_2 \|R_\omega\|_2 + \|R_\omega\|_2^2.$$

If, on the positive-probability support event,

$$T_\omega(\hat{x}(\omega)) + T_\omega(x_{\text{opt}}) + \Sigma_\omega \leq ((1 + \eta)(1 - \varepsilon) - (1 + \varepsilon)) \|Ax_{\text{opt}} - b\|_2^2,$$

then

$$\mathbb{P}\left[\|A\hat{x}(\omega) - b\|_2^2 \leq (1 + \eta) \|Ax_{\text{opt}} - b\|_2^2\right] \geq 1 - \delta.$$

Lean formalization.

```
lsSolutionForwardResidualBudget
lsSolutionForwardObjectiveGap
lsResidual_difference_bound_of_solution_abs_le
lsObjective_solution_forward_error_bound
isLeastSquaresApproxMinimizer_of_solution_abs_le
leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_forward_error
```

Corollary 280 (High-probability literal rounded sampled-row solver with a perturbed Gram certificate). *Use the notation and Bennett sample-budget assumptions of the previous corollary. Let*

$$G_\omega = \text{fl}(S_\omega A)^T \text{fl}(S_\omega A), \quad g_\omega = \text{fl}(S_\omega A)^T \text{fl}(S_\omega b).$$

Suppose G_ω has inverse H_ω , the exact rounded minimizer $x_(\omega)$ satisfies the normal equations*

$$G_\omega x_*(\omega) = g_\omega,$$

and the computed solver output satisfies a perturbed Gram system

$$(G_\omega + \Delta G_\omega) \hat{x}(\omega) = g_\omega + \Delta g_\omega$$

with componentwise perturbation radii

$$|(\Delta G_\omega)_{jk}| \leq \varepsilon_\omega^G, \quad |(\Delta g_\omega)_j| \leq \varepsilon_\omega^g.$$

Define

$$d_{\omega,i} = \sum_{j=1}^n |(H_\omega)_{ij}| \left(\varepsilon_\omega^G \sum_{k=1}^n |\hat{x}_k(\omega)| + \varepsilon_\omega^g \right).$$

Then d_ω is a componentwise forward-error certificate for the previous corollary. Consequently, if the same support-event budget condition holds with d_ω in the induced Σ_ω , then

$$\mathbb{P}\left[\|A\hat{x}(\omega) - b\|_2^2 \leq (1 + \eta) \|Ax_{\text{opt}} - b\|_2^2\right] \geq 1 - \delta.$$

Lean formalization.

```
lsNormalMatrix
lsNormalRhs
gramForwardSolverDx
gramForwardSolverDx_nonneg
gram_forward_error_certificate_of_perturbed_gram_system
leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_perturbed_gram_solver
```

Corollary 281 (High-probability literal rounded sampled-row solver from the local QR backward-error spec). *Use the notation and Bennett sample-budget assumptions of the previous corollary. Suppose that, for each sample trace ω , the solver output $\hat{x}(\omega)$ satisfies the repository's local QR least-squares backward-error specification for the rounded normal equations. Concretely, there exist perturbations $\Delta G_\omega, \Delta g_\omega$ such that*

$$(G_\omega + \Delta G_\omega)\hat{x}(\omega) = g_\omega + \Delta g_\omega, \quad \|\Delta G_\omega\|_F \leq c_\omega^G, \quad |(\Delta g_\omega)_j| \leq c_\omega^g.$$

Define

$$d_{\omega,i} = \sum_{j=1}^n |(H_\omega)_{ij}| \left(c_\omega^G \sum_{k=1}^n |\hat{x}_k(\omega)| + \max\{c_\omega^g, 0\} \right).$$

Then d_ω is a componentwise forward-error certificate for the forward-error corollary. Hence, if the same support-event budget condition holds with this d_ω , then

$$\mathbb{P}\left[\|A\hat{x}(\omega) - b\|_2^2 \leq (1 + \eta) \|Ax_{\text{opt}} - b\|_2^2\right] \geq 1 - \delta.$$

Lean formalization.

```
abs_entry_le_frobNorm
lsQRSolveBackwardSolverDx
lsQRSolveBackwardSolverDx_nonneg
gram_forward_error_certificate_of_ls_qr_solve_backward_error
leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_ls_qr_backward_error_solver
```

Corollary 282 (High-probability rounded sampled-row solver from stored QR off-diagonal control). *Use the same equation (8) sampled least-squares setting and Bennett sample-budget assumptions. For each sample trace ω , suppose the rounded sampled/scaled matrix and right-hand side are fed to a stored Householder QR process whose final top block is*

$$R_\omega = \hat{A}_{\omega,n}(1:n, 1:n), \quad c_\omega = \hat{b}_{\omega,n}(1:n),$$

and whose reported coefficient vector is

$$\hat{x}(\omega) = \text{flBackSub}(R_\omega, c_\omega).$$

Assume the stored process satisfies the visible packaged route-1 invariant *StoredQROffDiagonalControlInvariant* for every sample trace, together with signed-alpha Householder steps and the final QR Gram/RHS radius equations. Equivalently, one may supply the source-shaped off-diagonal-control fields and derive the packaged invariant locally. Then the local stored-QR theorem supplies

$$LSQRSolveBackwardError(G_\omega, g_\omega, \hat{x}(\omega), c_\omega^G, c_\omega^g).$$

Consequently the preceding QR-spec corollary applies: if the same support-event objective-budget condition holds with the induced solver certificate, then

$$\mathbb{P}\left[\|A\hat{x}(\omega) - b\|_2^2 \leq (1 + \eta) \|Ax_{\text{opt}} - b\|_2^2\right] \geq 1 - \delta.$$

Lean formalization.

```

storedQRFinalR
storedQRFinalTopRhs
storedQRBackSubSolution

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_offDiagonalControl_solver

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_offDiagonalControl_solver_of_diagDominant_
finiteMaxSmallness

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_sourceOffDiagonalControl_solver

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_leadingBlock_det_ne_zero_normSqBudget_offdiag_
product_solver

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_rowBudgetControl_solver

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_rowBudgetControl_kappaInf_dualBudget_solver

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_rowBudgetControl_globalProduct_kappaInf_
dualBudget_solver

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_rowMaxDiagDefect_globalProduct_solver

```

Corollary 283 (Source-shaped stored QR off-diagonal-control certificate). *The packaged invariant in the preceding corollary may be discharged from the primitive source-shaped certificate **StoredQRSourceOffDiagonalControl**: local upper-triangular leading blocks, nonzero displayed leading diagonals, row-wise off-diagonal domination, and the stored-sequence compact-product inequality. Under these source-shaped fields, the same high-probability rounded sampled-row objective guarantee holds.*

Lean formalization.

```

StoredQROffDiagonalControlInvariant.of_sourceOffDiagonalControl

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_offDiagonalControl

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_sourceOffDiagonalControl

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_sourceOffDiagonalControl_solver

```

Lemma 284 (Rectangular perturbed normal equations feed the local QR specification). *Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and let $\Delta A, \Delta b$ be rectangular data perturbations. Define*

$$G = A^T A, \quad g = A^T b,$$

and the induced Gram and right-hand-side perturbations

$$\Delta G = (A + \Delta A)^T (A + \Delta A) - A^T A, \quad \Delta g = (A + \Delta A)^T (b + \Delta b) - A^T b.$$

If a computed vector $\hat{x}$ satisfies the perturbed rectangular normal equations

$$(A + \Delta A)^T (A + \Delta A) \hat{x} = (A + \Delta A)^T (b + \Delta b),$$

then it satisfies the local perturbed Gram system

$$(G + \Delta G) \hat{x} = g + \Delta g.$$

Consequently, if $\|\Delta G\|_F \leq c_G$ and $|(\Delta g)_j| \leq c_g$ for every j , then $\hat{x}$ satisfies the repository's *LSQRSolveBackwardError* specification with radii c_G, c_g .

Lean formalization.

```
rectLSGram
rectLSRhs
RectLSNormalEquations
rectLSGramPerturbation
rectLSRhsPerturbation
rectLSNormalEquations_perturbed_to_gram_system
LSQRSolveBackwardError.of_rectangular_perturbed_normal_equations
```

Lemma 285 (Explicit budgets for induced rectangular perturbations). *With the notation of the previous lemma, the induced perturbations satisfy the entrywise identities*

$$(\Delta G)_{jk} = \sum_{i=1}^m (A_{ij}(\Delta A)_{ik} + (\Delta A)_{ij}A_{ik} + (\Delta A)_{ij}(\Delta A)_{ik}),$$

$$(\Delta g)_j = \sum_{i=1}^m (A_{ij}(\Delta b)_i + (\Delta A)_{ij}b_i + (\Delta A)_{ij}(\Delta b)_i).$$

Consequently

$$|(\Delta G)_{jk}| \leq \sum_{i=1}^m (|A_{ij}| |(\Delta A)_{ik}| + |(\Delta A)_{ij}| |A_{ik}| + |(\Delta A)_{ij}| |(\Delta A)_{ik}|),$$

and the analogous bound holds for $|(\Delta g)_j|$. In particular, if $\|\Delta A\|_F \leq c_A$ and $\|\Delta b\|_2 \leq c_b$, then

$$|(\Delta G)_{jk}| \leq \sum_{i=1}^m (|A_{ij}|c_A + c_A|A_{ik}| + c_A^2), \quad |(\Delta g)_j| \leq \sum_{i=1}^m (|A_{ij}|c_b + c_A|b_i| + c_Ac_b).$$

The Frobenius norm of ΔG is bounded by the Frobenius norm of either displayed Gram-budget matrix.

Lean formalization.

```
rectLSGramPerturbationEntryBudget
rectLSRhsPerturbationEntryBudget
rectLSGramPerturbationNormBudget
rectLSRhsPerturbationNormBudget
```

```

rectLSGramPerturbation_eq_sum
rectLSRhsPerturbation_eq_sum
rectLSGramPerturbation_abs_le_entryBudget
rectLSRhsPerturbation_abs_le_entryBudget
rectLSGramPerturbation_frobNorm_le_entryBudget
rectLSGramPerturbation_abs_le_normBudget
rectLSRhsPerturbation_abs_le_normBudget
rectLSGramPerturbation_frobNorm_le_normBudget
LSQRSolveBackwardError.of_rectangular_perturbed_normal_equations_normBudget

```

Lemma 286 (Rectangular one-step orthogonal-transformation accumulation). *Let $A, \hat{A}, \Delta A \in \mathbb{R}^{m \times n}$, let $Q, P \in \mathbb{R}^{m \times m}$ be orthogonal, and suppose*

$$\hat{A} = Q^T(A + \Delta A).$$

If the next computed transformation is

$$A_{\text{next}} = (P + \Delta P)\hat{A} \quad \text{with} \quad \|\Delta P\|_F \leq c_{\text{step}},$$

then there are an orthogonal $Q' \in \mathbb{R}^{m \times m}$ and a perturbation $\Delta A' \in \mathbb{R}^{m \times n}$ such that

$$A_{\text{next}} = (Q')^T(A + \Delta A')$$

and

$$\|\Delta A'\|_F \leq \|\Delta A\|_F + c_{\text{step}}\|A + \Delta A\|_F.$$

Lean formalization.

Algorithms/QR/HouseholderQR.lean rect_orthogonal_sequence_one_step

Theorem 287 (Rectangular multi-step orthogonal-transformation accumulation). *Let $A \in \mathbb{R}^{m \times n}$, let $A_0 = A$, and suppose that for $k = 0, \dots, r-1$,*

$$A_{k+1} = (P_k + \Delta P_k)A_k, \quad P_k^T P_k = I, \quad \|\Delta P_k\|_F \leq c,$$

with $c \geq 0$. Then there are an orthogonal $Q \in \mathbb{R}^{m \times m}$ and a perturbation $\Delta A \in \mathbb{R}^{m \times n}$ such that

$$A_r = Q^T(A + \Delta A)$$

and

$$\|\Delta A\|_F \leq ((1 + c)^r - 1) \|A\|_F.$$

Lean formalization.

Algorithms/QR/HouseholderQR.lean rect_orthogonal_sequence_geometric

Theorem 288 (Right-hand-side orthogonal-transformation accumulation). *Let $b \in \mathbb{R}^m$, let $b_0 = b$, and suppose that for $k = 0, \dots, r-1$,*

$$b_{k+1} = (P_k + \Delta P_k)b_k, \quad P_k^T P_k = I, \quad \|\Delta P_k\|_F \leq c,$$

with $c \geq 0$. Then there are an orthogonal $Q \in \mathbb{R}^{m \times m}$ and a perturbation $\Delta b \in \mathbb{R}^m$ such that

$$b_r = Q^T(b + \Delta b)$$

and

$$\|\Delta b\|_2 \leq ((1 + c)^r - 1) \|b\|_2.$$

Lean formalization.

```
Algorithms/QR/HouseholderQR.lean orthogonal_vector_sequence_one_step
orthogonal_vector_sequence_geometric
```

Theorem 289 (Common- Q matrix/right-hand-side accumulation). *Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $A_0 = A$, and $b_0 = b$. Suppose that the same perturbed orthogonal transformations are applied to both data blocks:*

$$A_{k+1} = (P_k + \Delta P_k)A_k, \quad b_{k+1} = (P_k + \Delta P_k)b_k,$$

where $P_k^T P_k = I$, $\|\Delta P_k\|_F \leq c$, and $c \geq 0$. Then there are one common orthogonal matrix Q , a matrix perturbation ΔA , and a vector perturbation Δb such that

$$A_r = Q^T(A + \Delta A), \quad b_r = Q^T(b + \Delta b),$$

with

$$\|\Delta A\|_F \leq ((1+c)^r - 1)\|A\|_F, \quad \|\Delta b\|_2 \leq ((1+c)^r - 1)\|b\|_2.$$

Lean formalization.

```
Algorithms/QR/HouseholderQR.lean rect_orthogonal_matrix_vector_sequence_one_step
rect_orthogonal_matrix_vector_sequence_geometric
```

Lemma 290 (Top-block QR solve gives transformed normal equations). *Let $\hat{A} \in \mathbb{R}^{m \times n}$ and $\hat{b} \in \mathbb{R}^m$ be transformed QR data. Suppose there are $R \in \mathbb{R}^{n \times n}$ and $c \in \mathbb{R}^n$ such that*

$$\hat{A}_{i,*} = R_{i,*}, \quad \hat{b}_i = c_i \quad \text{for every } i < n,$$

and

$$\hat{A}_{i,*} = 0 \quad \text{for every } i \geq n.$$

If $\hat{x}$ solves the top system

$$R\hat{x} = c,$$

then $\hat{x}$ satisfies the rectangular normal equations for $(\hat{A}, \hat{b})$:

$$\hat{A}^T \hat{A} \hat{x} = \hat{A}^T \hat{b}.$$

No condition is imposed on the lower entries of $\hat{b}$; they are the residual block in the usual tall-QR least-squares solve and disappear because the corresponding rows of $\hat{A}$ are zero.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean RectLSNormalEquations.of_rowwise_normal
RectLSNormalEquations.of_top_solve_zero_bottom
```

Lemma 291 (Rounded top-block solve gives perturbed transformed normal equations). *Assume the top QR block $R \in \mathbb{R}^{n \times n}$ is nonsingular on the diagonal and upper triangular in the sense required by the repository's back-substitution theorem. Let*

$$\hat{x} = \text{flBackSub}(R, c).$$

Then there is a perturbation ΔR such that

$$|\Delta R_{ij}| \leq \gamma_n |R_{ij}|$$

and $\hat{x}$ satisfies the rectangular normal equations for the perturbed transformed QR data

$$\tilde{A} = \begin{bmatrix} R + \Delta R \\ 0 \end{bmatrix}, \quad \tilde{b}_i = c_i \quad (i < n),$$

with no condition on the lower entries of $\tilde{b}$.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean rectTopBlock
rectTopBlock_top
rectTopBlock_bottom
RectLSNormalEquations.exists_topBlock_of_fl_backSub
```

Lemma 292 (Orthogonal handoff for rectangular normal equations). *Let $U \in \mathbb{R}^{m \times m}$ be orthogonal, let $\hat{A} = UA$, and let $\hat{b} = Ub$. Then*

$$\hat{A}^T \hat{A} = A^T A, \quad \hat{A}^T \hat{b} = A^T b.$$

Consequently, if $\hat{x}$ satisfies the exact normal equations for $(\hat{A}, \hat{b})$, then $\hat{x}$ satisfies the exact normal equations for (A, b) .

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean rectLSGram_matMulRectLeft_orthogonal
rectLSRhs_matMulRectLeft_orthogonal
RectLSNormalEquations.of_orthogonal_left
```

Lemma 293 (Pulling the rounded top-block perturbation back to the original data). *Assume that transformed QR data are represented with one common orthogonal factor,*

$$\hat{A} = Q^T(A + \Delta A), \quad \hat{b} = Q^T(b + \Delta b),$$

and that $\hat{A}$ has top block R and zero lower block. If $\hat{x} = \text{flBackSub}(R, c)$, then there are a triangular-solve perturbation ΔR and an original-data matrix perturbation ΔA_{tot} such that

$$|\Delta R_{ij}| \leq \gamma_n |R_{ij}|, \quad \Delta A_{\text{tot}} = \Delta A + Q \begin{bmatrix} \Delta R \\ 0 \end{bmatrix},$$

$$\|\Delta A_{\text{tot}}\|_F \leq \|\Delta A\|_F + \left\| \begin{bmatrix} \Delta R \\ 0 \end{bmatrix} \right\|_F,$$

and $\hat{x}$ satisfies the rectangular normal equations for the original perturbed data:

$$(A + \Delta A_{\text{tot}})^T (A + \Delta A_{\text{tot}}) \hat{x} = (A + \Delta A_{\text{tot}})^T (b + \Delta b).$$

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean rectTopBlock_add
RectLSNormalEquations.exists_original_perturbation_of_commonQ_topBlock_fl_backSub
```

Corollary 294 (Gamma-budget version of the pulled-back top-block theorem). *Under the hypotheses of the previous lemma, the abstract top-block radius can be discharged using the triangular-solve entrywise budget:*

$$\left\| \begin{bmatrix} \Delta R \\ 0 \end{bmatrix} \right\|_F \leq \gamma_n \left\| \begin{bmatrix} R \\ 0 \end{bmatrix} \right\|_F.$$

Consequently the original-data perturbation can be chosen so that

$$\|\Delta A_{\text{tot}}\|_F \leq \|\Delta A\|_F + \gamma_n \left\| \begin{bmatrix} R \\ 0 \end{bmatrix} \right\|_F.$$

Lean formalization.

```
Analysis/MatrixAlgebra.lean frobNormSqRect_abs
frobNormRect_abs
Algorithms/LeastSquares/LSQRSolve.lean rectTopBlock_frobNorm_perturb_bound
rectTopBlock_frobNorm_perturb_bound_of_gamma
RectLSNormalEquations.exists_original_perturbation_of_commonQ_topBlock_fl_backSub_gamma_bound
```

Corollary 295 (Supplied-transform rectangular QR route into the local solver specification). *Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and suppose a floating-point orthogonal-transformation pipeline produces transformed data $\hat{A}, \hat{b}$ with one common exact orthogonal representation*

$$\hat{A} = Q^T(A + \Delta A), \quad \hat{b} = Q^T(b + \Delta b).$$

Assume that the final transformed matrix has QR top-block form

$$\hat{A} = \begin{bmatrix} R \\ 0 \end{bmatrix},$$

that the first n entries of $\hat{b}$ are c , and that R is upper triangular with nonzero diagonal. If back-substitution is performed as

$$\hat{x} = \text{flBackSub}(R, c),$$

then the local QR least-squares backward-error specification holds with matrix perturbation radius

$$c_A = \|\Delta A\|_F + \gamma_n \left\| \begin{bmatrix} R \\ 0 \end{bmatrix} \right\|_F$$

and right-hand-side radius $c_b = \|\Delta b\|_2$, provided the displayed radii are large enough to satisfy the induced Gram and right-hand-side norm-budget hypotheses of the local specification.

In particular, if $\hat{A}_r, \hat{b}_r$ are produced by a supplied sequence of r perturbed orthogonal transformations whose per-step Frobenius radius is c , then the same conclusion holds with

$$c_A = ((1 + c)^r - 1)\|A\|_F + \gamma_n \left\| \begin{bmatrix} R \\ 0 \end{bmatrix} \right\|_F, \quad c_b = ((1 + c)^r - 1)\|b\|_2.$$

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean    LSQRSolveBackwardError.of_commonQ_topBlock_fl_
backSub_gamma_bound_normBudget
LSQRSolveBackwardError.of_rect_orthogonal_sequence_topBlock_fl_backSub_gamma_bound_normBudget
```

Scope. This corollary is intentionally a supplied-transform theorem. It does not claim that a concrete implementation of Householder QR has been formalized. The remaining red bottleneck is the algorithm-level theorem that constructs the supplied reflectors/updates, proves the final $[R; 0]$ shape and transformed right-hand side, and discharges the per-step floating-point recurrences from actual rounded operations.

Lemma 296 (Exact embedded Householder reflector with a zero prefix). *Let*

$$P = I - \beta vv^T$$

be the full-size Householder reflector represented in the repository. If

$$v_i = 0 \quad \text{for every } i < k,$$

then P agrees with the identity on the first k rows and columns:

$$P_{ij} = \delta_{ij} \quad (i < k), \quad P_{ij} = \delta_{ij} \quad (j < k).$$

Consequently, for every vector x and every matrix A ,

$$(Px)_i = x_i \quad (i < k), \quad (PA)_{ij} = A_{ij} \quad (i < k).$$

The matrix statement is formalized both for square matrices and for rectangular matrices.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean householder_row_eq_id_of_zero_prefix
householder_col_eq_id_of_zero_prefix
matMulVec_householder_eq_self_of_zero_prefix
matMul_householder_eq_self_row_of_zero_prefix
matMulRectLeft_householder_eq_self_row_of_zero_prefix
```

Role in the bottleneck. This is a genuine exact foundation for the concrete rectangular Householder route: it proves that a trailing reflector does not disturb already-finished rows/columns. The next lemma closes the exact active-column zeroing algebra.

Lemma 297 (Exact active-column zeroing for the constructed Householder reflector). *Let $x \in \mathbb{R}^m$, choose a pivot index p , and define*

$$v = x - \alpha e_p, \quad \beta = \frac{2}{v^T v}.$$

Assume

$$\alpha^2 = \|x\|_2^2, \quad v^T v \neq 0.$$

Then the exact Householder reflector $P = I - \beta vv^T$ maps the active column vector to a single pivot entry:

$$Px = \alpha e_p.$$

Equivalently,

$$(Px)_p = \alpha, \quad (Px)_i = 0 \quad (i \neq p).$$

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean householderActiveVector
householderBeta
householderActiveVector_inner_x
householderActiveVector_inner_self
householderActiveVector_inner_self_eq_two_inner_x
householderBeta_mul_activeVector_inner_x
matMulVec_householder_activeVector_eq_alpha_basis
matMulVec_householder_activeVector_eq_zero_of_ne
```

Role in the bottleneck. This closes the exact active-column algebra for route A. It is still not a floating-point construction of the reflector and not a full QR implementation: the QR-specific construction also needs the reflector vector to have a zero prefix. The next theorem closes that exact trailing-shape point.

Definition 298 (Trailing Householder vector for rectangular QR). For a pivot index p , split a vector $x \in \mathbb{R}^m$ as

$$x = x_{<p} + x_{\geq p}, \quad (x_{<p})_i = \begin{cases} x_i, & i < p, \\ 0, & i \geq p, \end{cases} \quad (x_{\geq p})_i = \begin{cases} 0, & i < p, \\ x_i, & i \geq p. \end{cases}$$

The QR trailing active vector is

$$v = x_{\geq p} - \alpha e_p.$$

Thus $v_i = 0$ for all $i < p$. This is the source-faithful rectangular QR construction; it differs from the full-vector construction $x - \alpha e_p$ unless the prefix of x is already zero.

Lemma 299 (Exact trailing-column Householder zeroing). Let $v = x_{\geq p} - \alpha e_p$, let

$$\beta = \frac{2}{v^T v},$$

and assume

$$\alpha^2 = \|x_{\geq p}\|_2^2, \quad v^T v \neq 0.$$

For $P = I - \beta v v^T$,

$$(Px)_i = \begin{cases} x_i, & i < p, \\ \alpha, & i = p, \\ 0, & i > p. \end{cases}$$

Theorem 300 (Exact trailing Householder loop shape). Let $m \geq n$. Suppose an exact rectangular Householder recurrence applies, at step $k < n$, the trailing reflector built from the current pivot column $\hat{A}_k(:, k)$. Assume the corresponding normalization hypotheses

$$\alpha_k^2 = \|(\hat{A}_k(:, k))_{\geq k}\|_2^2, \quad v_k^T v_k \neq 0$$

for each step. Then after n steps,

$$j < i \implies \hat{A}_n(i, j) = 0.$$

Equivalently, the final transformed matrix is lower-zero/trapezoidal in the QR sense.

Corollary 301 (Top-block shape facts from the lower-zero invariant). *Define*

$$R_{ij} = \widehat{A}_n(i, j) \quad (1 \leq i, j \leq n), \quad (c_{\text{top}})_i = (\widehat{b}_n)_i \quad (1 \leq i \leq n).$$

If $\widehat{A}_n(i, j) = 0$ whenever $j < i$, then

$$\widehat{A}_n = \begin{bmatrix} R \\ 0 \end{bmatrix}, \quad (\widehat{b}_n)_{1:n} = c_{\text{top}}, \quad R_{ij} = 0 \quad (j < i).$$

The nonzero diagonal condition for triangular back substitution is not a consequence of this shape theorem; it remains a visible rank/nonbreakdown assumption unless separately proved from a rank hypothesis.

Definition 302 (Stored rounded Householder panel and RHS steps). For a pivot step k , let $\text{raw}_k(A)$ denote the compact rounded Householder panel update computed by the dot-scale-subtract primitive. The stored panel step is

$$A_{k+1}(i, j) = \begin{cases} A_k(i, j), & j < k, \\ 0, & j = k \text{ and } i > k, \\ \text{raw}_k(A_k)(i, j), & \text{otherwise.} \end{cases}$$

For the right-hand side, the stored step preserves rows above the active pivot and applies the compact update on the active tail:

$$b_{k+1}(i) = \begin{cases} b_k(i), & i < k, \\ \text{raw}_k(b_k)(i), & i \geq k. \end{cases}$$

Theorem 303 (Stored rounded Householder loop shape). *Suppose A_{k+1} is obtained from A_k by the stored panel step above for each $k < n$. Then the final stored matrix satisfies*

$$j < i \implies A_n(i, j) = 0.$$

Corollary 304 (Stored top-block shape facts). *Let*

$$R_{ij} = A_n(i, j) \quad (1 \leq i, j \leq n), \quad (c_{\text{top}})_i = (b_n)_i \quad (1 \leq i \leq n).$$

For a stored rounded loop,

$$A_n = \begin{bmatrix} R \\ 0 \end{bmatrix}, \quad (b_n)_{1:n} = c_{\text{top}}, \quad R_{ij} = 0 \quad (j < i).$$

This is a storage-shape theorem only. The perturbation contract for one stored step is stated next, under explicit preservation, zeroing, and budget hypotheses. Neither statement proves nonzero diagonal pivots.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean householderPrefixPart
householderTrailingPart
householderTrailingNorm2Sq
householderTrailingActiveVector
```

```

householderPrefixPart_add_trailingPart
householderPrefixPart_support
householderTrailingActiveVector_zero_prefix
matMulVec_householder_eq_self_of_zero_prefix_support
matMulVec_householder_trailingActiveVector_eq_prefix_alpha_zero
matMulVec_householder_trailingActiveVector_eq_zero_of_pivot_lt
Algorithms/QR/HouseholderQR.lean exact_trailing_householder_sequence_lower_zero
rectangular_topBlock_shape_facts_of_lower_zero
Algorithms/QR/HouseholderApply.lean fl_householderStoredPanelStep
fl_householderStoredRhsStep
Algorithms/QR/HouseholderQR.lean fl_householderStoredPanel_sequence_lower_zero
fl_householderStoredPanel_sequence_topBlock_shape_facts

```

Theorem 305 (Stored Householder step satisfies the columnwise perturbation contract). *Fix one pivot step k . Let $P = I - \beta vv^T$ be the exact Householder reflector and let $\hat{A}, \hat{b}$ be computed by the stored rounded panel and right-hand-side steps above. Assume:*

$$P^T P = I,$$

completed columns are exactly preserved by the reflector,

$$j < k \implies (PA_{\cdot j})_i = A_{ij},$$

the entries stored as zero in the pivot column are exact zeros of the reflected column,

$$i > k \implies (PA_{\cdot k})_i = 0,$$

the right-hand-side prefix is exactly preserved,

$$i < k \implies (Pb)_i = b_i,$$

and the explicit compact Householder budgets satisfy

$$\|B_j\|_2 \leq c\|A_{\cdot j}\|_2, \quad \|B_b\|_2 \leq c\|b\|_2.$$

Here B_j and B_b are the deterministic componentwise budgets produced by the rounded dot-scale-subtract primitive. Then

$$\text{HouseholderColumnwisePanelAppError}(P, A, \hat{A}, b, \hat{b}; c)$$

holds.

Lean formalization.

```

Algorithms/QR/HouseholderApply.lean householderCompactComponentBudget_nonneg
fl_householderStoredRhsStep_componentwise_error_bound
fl_householderStoredRhsStep_forward_error_bound
fl_householderStoredPanelStep_column_componentwise_error_bound
fl_householderStoredPanelStep_column_forward_error_bound
fl_householderStoredPanelStep_HouseholderColumnwisePanelAppError_of_budget

```

Corollary 306 (One stored trailing Householder QR step). *Let p be the pivot row and let v be the trailing Householder vector formed from the current pivot column. Suppose the current stored matrix has the lower-zero invariant for all completed columns:*

$$j < k, \quad j < i \quad \implies \quad A_{ij} = 0.$$

Assume the Householder denominator is nonzero and the compact component budgets for this step satisfy the same domination inequalities as in the previous theorem. Then the stored rounded trailing Householder step satisfies

$$\text{HouseholderColumnwisePanelAppError}(P, A, \hat{A}, b, \hat{b}; c).$$

Lean formalization.

`Algorithms/QR/HouseholderQR.lean` `fl_householderStoredTrailingPanelStep_`
`HouseholderColumnwisePanelAppError_of_budget`

Theorem 307 (Stored trailing Householder QR sequence). *Let $m \geq n$, $A_0 = A$, and $b_0 = b$. For each pivot $k < n$, form the trailing Householder vector from the current stored pivot column, set $\beta_k = 2/(v_k^T v_k)$, and define the stored rounded panel and right-hand side recurrences*

$$A_{k+1} = \text{StoredPanelStep}(A_k, k, v_k, \beta_k), \quad b_{k+1} = \text{StoredRhsStep}(b_k, k, v_k, \beta_k).$$

Assume that the Householder denominator $v_k^T v_k$ is nonzero, that the chosen sign parameter satisfies $\alpha_k^2 = \sum_{i \geq k} A_{k,ik}^2$, and that the compact component budgets for every stored panel column and right-hand side update are dominated by a common constant $c \geq 0$. Then there exist an orthogonal matrix Q , a data perturbation ΔA , and a right-hand-side perturbation Δb such that

$$A_n = Q^T(A + \Delta A), \quad b_n = Q^T(b + \Delta b),$$

and, for every column j ,

$$\|\Delta A_{:,j}\|_2 \leq ((1+c)^n - 1)\|A_{:,j}\|_2, \quad \|\Delta b\|_2 \leq ((1+c)^n - 1)\|b\|_2.$$

Lean formalization.

`Algorithms/QR/HouseholderQR.lean` `fl_householderStoredTrailingPanel_rect_orthogonal_`
`columnwise_vector_sequence_geometric`

Theorem 308 (Higham columnwise stored QR factorization). *Under the hypotheses of the stored trailing Householder QR sequence theorem, let*

$$R_{ij} = (A_n)_{ij}, \quad c_i = (b_n)_i, \quad 0 \leq i, j < n,$$

denote the top block of the final stored matrix and the top part of the final stored right-hand side. Then there exist an orthogonal matrix Q , a data perturbation ΔA , and a right-hand-side perturbation Δb such that

$$A_n = Q^T(A + \Delta A), \quad b_n = Q^T(b + \Delta b),$$

$$\|\Delta A_{:,j}\|_2 \leq ((1+c)^n - 1)\|A_{:,j}\|_2, \quad \|\Delta b\|_2 \leq ((1+c)^n - 1)\|b\|_2.$$

Moreover the final stored matrix has the block form

$$A_n = \begin{bmatrix} R \\ 0 \end{bmatrix},$$

the vector c is exactly the first n entries of b_n , and R is upper triangular.

Lean formalization.

Algorithms/QR/HouseholderQR.lean fl_householderStoredTrailingPanel_higham_columnwise_factorization

Corollary 309 (Stored trailing Householder QR solve certificate). *Let R be the $n \times n$ top block of the final stored matrix A_n , and let c_{top} be the first n entries of the final stored right-hand side b_n . Under the hypotheses of the stored trailing sequence theorem, assume in addition that R has nonzero diagonal, that the top-block arithmetic is gamma-valid, and that the induced Gram/RHS perturbation budgets are bounded by c_G, c_g . Then the computed triangular solve*

$$\hat{x} = \text{fl_backSub}(R, c_{\text{top}})$$

satisfies the repository's least-squares QR backward-error specification

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Lean formalization.

Algorithms/LeastSquares/LSQRSolve.lean LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_normBudget

Lemma 310 (Pivot-budget nonbreakdown for stored trailing QR). *At pivot k , let v_k be the trailing Householder vector, let $\beta_k = 2/(v_k^T v_k)$, and let B_k be the componentwise compact-update budget for the stored diagonal entry. If*

$$B_k < |\alpha_k|,$$

then the stored rounded diagonal entry written at that pivot is nonzero. If this inequality holds for every pivot, then the final stored top block R has nonzero diagonal.

Corollary 311 (Stored QR solve certificate with pivot-budget nonbreakdown). *In the stored trailing Householder QR solve certificate above, the hypothesis that R has nonzero diagonal may be replaced by the per-pivot inequalities*

$$B_k < |\alpha_k|, \quad 0 \leq k < n.$$

Under those inequalities, the same computed vector

$$\hat{x} = \text{fl_backSub}(R, c_{\text{top}})$$

satisfies

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Lean formalization.

Algorithms/QR/HouseholderQR.lean fl_householderStoredTrailingPanelStep_diag_nonzero_of_budget_lt_abs_alpha
fl_householderStoredPanel_sequence_diag_nonzero_of_step_diag_nonzero
fl_householderStoredTrailingPanel_sequence_diag_nonzero_of_budget_lt_abs_alpha
Algorithms/LeastSquares/LSQRSolve.lean LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_normBudget_of_pivot_error_lt_abs_alpha

Lemma 312 (Pivot-value nonbreakdown for the Householder denominator). *Let x be the current pivot column and*

$$v = x_{\text{tail}} - \alpha_k e_k$$

be the trailing Householder vector. If the active pivot entry differs from the chosen sign parameter,

$$x_k \neq \alpha_k,$$

then $v^T v \neq 0$. Thus the denominator needed to form $\beta_k = 2/(v^T v)$ follows from a scalar pivot nonbreakdown condition rather than from an unexplained sum-of-squares hypothesis.

Corollary 313 (Stored QR solve certificate with pivot-value nonbreakdown). *In the stored trailing Householder QR solve certificate with pivot-budget nonbreakdown, the Householder denominator assumptions may be replaced by*

$$A_k(k, k) \neq \alpha_k, \quad 0 \leq k < n,$$

where A_k is the stored panel before pivot k . Together with $B_k < |\alpha_k|$, the computed vector

$$\hat{x} = \text{fl_backSub}(R, c_{\text{top}})$$

satisfies the same

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean  householderTrailingActiveVector_inner_self_ne_zero_of_
pivot_ne_alpha
Algorithms/QR/HouseholderQR.lean    fl_householderStoredTrailingPanel_sequence_diag_nonzero_
of_pivot_ne_alpha
Algorithms/LeastSquares/LSQRSolve.lean  LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_normBudget_of_pivot_ne_alpha_and_pivot_
error_lt_abs_alpha
```

Lemma 314 (Sign-choice nonbreakdown for the Householder denominator). *Let x be the current pivot column and suppose the Householder parameter α_k satisfies*

$$\alpha_k^2 = \|x_{\text{tail}}\|_2^2, \quad \|x_{\text{tail}}\|_2^2 > 0, \quad \alpha_k x_k \leq 0.$$

Then $x_k \neq \alpha_k$, and hence the trailing Householder vector

$$v = x_{\text{tail}} - \alpha_k e_k$$

satisfies $v^T v \neq 0$.

Corollary 315 (Stored QR solve certificate with sign-choice nonbreakdown). *In the stored trailing Householder QR solve certificate with pivot-budget nonbreakdown, the Householder denominator assumptions may be replaced by the source-style local conditions*

$$\alpha_k^2 = \|A_k(k:m, k)\|_2^2, \quad \|A_k(k:m, k)\|_2^2 > 0, \quad \alpha_k A_k(k, k) \leq 0, \quad 0 \leq k < n.$$

Together with $B_k < |\alpha_k|$, the computed vector

$$\hat{x} = \text{fl_backSub}(R, c_{\text{top}})$$

satisfies the same

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean  householder_pivot_ne_alpha_of_trailingNorm2Sq_pos_mul_
nonpos

householderTrailingActiveVector_inner_self_ne_zero_of_trailingNorm2Sq_pos_mul_nonpos

Algorithms/QR/HouseholderQR.lean  fl_householderStoredTrailingPanel_sequence_diag_nonzero_
of_trailingNorm_pos_mul_nonpos

Algorithms/LeastSquares/LSQRSolve.lean  LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_normBudget_of_trailingNorm_pos_mul_nonpos_
and_pivot_error_lt_abs_alpha
```

Lemma 316 (Row sorting preserves rectangular least-squares data). *Let σ be a permutation of the m row indices. Define*

$$A_{ij}^\sigma = A_{\sigma(i),j}, \quad b_i^\sigma = b_{\sigma(i)}.$$

Then row sorting does not change the least-squares residual objective:

$$\|A^\sigma x - b^\sigma\|_2^2 = \|Ax - b\|_2^2.$$

It also preserves the normal-equation data:

$$(A^\sigma)^T A^\sigma = A^T A, \quad (A^\sigma)^T b^\sigma = A^T b.$$

The same permutation invariance is formalized for rectangular Frobenius norms and for rectangular matrix-vector multiplication.

Lean formalization.

```
Analysis/MatrixAlgebra.lean  vecPermute
rectPermuteRows
rectPermuteCols
vecNorm2Sq_permute
frobNormSqRect_permuteRows
frobNormSqRect_permuteCols
frobNormRect_permuteRows
frobNormRect_permuteCols
rectMatMulVec_permuteRows

Algorithms/LeastSquares/LSQRSolve.lean  rectLSGram_permuteRows
rectLSRhs_permuteRows

Algorithms/RandNLA/LeastSquaresSketch.lean  lsResidual_permuteRows
lsObjective_permuteRows
```

Lemma 317 (Column pivoting preserves rectangular least-squares data). *Let π be a permutation of the n column indices and define*

$$A_{ij}^\pi = A_{i,\pi(j)}.$$

Then column pivoting is a relabeling of the coefficient vector:

$$A^\pi x = A(x \circ \pi^{-1}), \quad \|A^\pi x - b\|_2^2 = \|A(x \circ \pi^{-1}) - b\|_2^2.$$

The normal-equation data are relabeled by the same column permutation,

$$(A^\pi)^T A_{jk}^\pi = (A^T A)_{\pi(j), \pi(k)}, \quad (A^\pi)^T b_j = (A^T b)_{\pi(j)}.$$

Consequently, a solution or exact minimizer of the column-pivoted least-squares problem maps back to the original coordinates by $x \mapsto x \circ \pi^{-1}$.

Lean formalization.

```
Analysis/MatrixAlgebra.lean vecPermute_symm_vecPermute
vecPermute_vecPermute_symm
rectMatMulVec_permuteCols
Algorithms/LeastSquares/LSQRSolve.lean rectLSGram_permuteCols
rectLSRhs_permuteCols
RectLSNormalEquations.of_permuteCols
Algorithms/RandNLA/LeastSquaresSketch.lean lsResidual_permuteCols
lsObjective_permuteCols
IsLeastSquaresMinimizer.of_permuteCols
```

Lemma 318 (Combined row sorting and column pivoting for least squares). *Let σ be a row permutation and π a column permutation. If*

$$\tilde{A}_{ij} = A_{\sigma(i), \pi(j)}, \quad \tilde{b}_i = b_{\sigma(i)},$$

then the row permutation disappears from the normal-equation data and the column permutation only relabels coefficient coordinates:

$$\tilde{A}^T \tilde{A}_{jk} = (A^T A)_{\pi(j), \pi(k)}, \quad \tilde{A}^T \tilde{b}_j = (A^T b)_{\pi(j)}.$$

Moreover,

$$\|\tilde{A}x - \tilde{b}\|_2^2 = \|A(x \circ \pi^{-1}) - b\|_2^2.$$

Consequently, any solution of the row-sorted, column-pivoted normal equations, and any exact minimizer of that row-sorted, column-pivoted least-squares problem, maps back to the original coordinates by $x \mapsto x \circ \pi^{-1}$.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean rectLSGram_permuteRowsCols
rectLSRhs_permuteRowsCols
RectLSNormalEquations.of_permuteRowsCols
Algorithms/RandNLA/LeastSquaresSketch.lean lsResidual_permuteRowsCols
lsObjective_permuteRowsCols
IsLeastSquaresMinimizer.of_permuteRowsCols
```

Lemma 319 (Cox–Higham signed Householder denominator bound). *Let x_{tail} be the active trailing pivot column segment and let*

$$v = x_{\text{tail}} - \alpha e_k.$$

If

$$\alpha^2 = \|x_{\text{tail}}\|_2^2 \quad \text{and} \quad \alpha x_k \leq 0,$$

then

$$2\|x_{\text{tail}}\|_2^2 \leq v^T v.$$

For the concrete signed Householder choice

$$\alpha = \text{signedHouseholderAlpha}(\sqrt{\|x_{\text{tail}}\|_2^2}, x_k),$$

the two hypotheses above are supplied by the local signed-alpha lemmas.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean    householderTrailingActiveVector_inner_self_ge_two_
trailingNorm2Sq_of_mul_nonpos
householderTrailingActiveVector_inner_self_ge_two_trailingNorm2Sq_signed
```

Lemma 320 (Cox–Higham active-column pivot comparison). *Fix a Householder step k and define*

$$\tau_j = \|A_{k:m,j}\|_2^2$$

for each remaining column $j \geq k$. There exists a remaining column j_ such that*

$$\tau_j \leq \tau_{j_*} \quad (j \geq k).$$

If $x = A_{:,j_}$, $y = A_{:,j}$, and*

$$v = x_{\text{tail}} - \alpha e_k,$$

then the pivot-max property and Cauchy–Schwarz give

$$|v^T y_{\text{tail}}| \leq \|v\|_2 \|y_{\text{tail}}\|_2 \leq \|v\|_2 \|x_{\text{tail}}\|_2.$$

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean    householderTrailingColumnNorm2Sq
exists_householderTrailingColumnNorm2Sq_active_max
abs_inner_householderTrailingActiveVector_trailingPart_le_vecNorm2_mul_sqrt
abs_inner_householderTrailingActiveVector_column_le_vecNorm2_mul_sqrt_of_pivot_max
```

Lemma 321 (Cox–Higham scalar update and one-step row growth). *With the same signed Householder vector v , exact coefficient*

$$\beta = \frac{2}{v^T v},$$

and a remaining column y , the previous two lemmas imply

$$|\beta v^T y_{\text{tail}}| \leq \sqrt{2}.$$

Consequently, if a row has

$$|a_{ij}| \leq B, \quad |a_{ik}| \leq B,$$

then the corresponding off-pivot Householder update with $\phi = \beta v^T y_{\text{tail}}$ satisfies

$$|a_{ij} - \phi a_{ik}| \leq (1 + \sqrt{2})B.$$

This is the single-step entry growth estimate used in Cox–Higham’s proof of equation (4.3). It is not yet the multi-stage row-sorting or row-pivoting accumulation theorem.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean abs_two_div_mul_le_sqrt_two_of_abs_le_sqrt_mul_sqrt
abs_householderBeta_mul_inner_trailingPart_le_sqrt_two_of_mul_nonpos
abs_householderBeta_mul_inner_trailingPart_le_sqrt_two_signed
abs_householderBeta_mul_inner_column_le_sqrt_two_of_signed_pivot_max
abs_sub_mul_le_one_add_sqrt_two_mul_bound
abs_householder_signed_pivot_update_entry_le_one_add_sqrt_two_mul_row_bound
```

Lemma 322 (Cox–Higham row-sorting stage accumulation). *Let $c = 1 + \sqrt{2}$. Suppose that the initial row-size bounds are sorted among the active rows, so that for every $r \geq k$,*

$$B_r^{(0)} \leq B_k^{(0)}.$$

If a fixed active entry grows at each previous Householder stage by at most the one-step factor c , then after k stages

$$|A_k(r, j)| \leq c^k B_k^{(0)}.$$

Equivalently, row sorting turns the one-step estimate $|A_{t+1}(r, j)| \leq (1 + \sqrt{2})|A_t(r, j)|$ into the accumulated Cox–Higham row bound used in the proof of Lemma 4.2.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean coxHighamGrowthFactor_nonneg
scalar_growth_iterate_bound
coxHigham_rowSorting_active_entry_bound_of_prior_growth
coxHigham_rowSorting_active_entry_bound_of_stage_growth
```

Lemma 323 (Cox–Higham pivot-row active-tail norm step). *Let $p = k$ be the current pivot row and suppose the next pivot-row entry in column j is bounded by the Euclidean norm of the active tail of the previous column,*

$$|A_{k+1}(k, j)| \leq \|A_k(k : m, j)\|_2.$$

If every active-tail entry of that column is bounded by B , then

$$|A_{k+1}(k, j)| \leq \sqrt{m} B.$$

Consequently, under the same row-sorting scale as above,

$$|A_{k+1}(k, j)| \leq \sqrt{m} (1 + \sqrt{2})^k B_k^{(0)}.$$

This is the repository’s ambient- $\mathbb{R}^m$ form of the norm step in Cox–Higham equations (4.4)–(4.5). The sharper source factor $\sqrt{m - k + 1}$ is not hidden; it remains a possible refinement once the library has a dedicated active-tail dimension lemma.

Lean formalization.

```
Analysis/MatrixAlgebra.lean vecNorm2_le_sqrt_card_mul_of_abs_le
Algorithms/QR/HouseholderSpec.lean coxHigham_pivot_row_entry_bound_of_active_tail_entry_bound
coxHigham_pivot_row_entry_bound_of_stage_entry_bound
```

Lemma 324 (Exact signed pivot-row same-reflector bridge). *Let p be the current pivot row and let the signed Householder reflector*

$$P = I - \beta v v^T, \quad v = A(:, k)_{\text{tail}} - \alpha e_p, \quad \alpha = -\text{sign}(A_{pk}) \|A(p : m, k)\|_2$$

be formed from a pivot column k with positive active trailing norm. Then for any updated column j ,

$$|(PA(:, j))_p| \leq \|A(p : m, j)\|_2.$$

Consequently, if the active entries of column j at stage p satisfy

$$|A_{ij}| \leq (1 + \sqrt{2})^p B_p^{(0)}, \quad i \geq p,$$

then

$$|(PA(:, j))_p| \leq \sqrt{m} (1 + \sqrt{2})^p B_p^{(0)}.$$

This supplies the exact same-reflector pivot-row field corresponding to Cox–Higham equations (4.4)–(4.5), in the repository’s ambient-dimension form.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean householderBeta_mul_inner_self_eq_two
abs_matMulVec_householder_pivot_le_trailing_vecNorm2_of_zero_prefix_orthogonal
coxHigham_signed_pivot_row_update_abs_le_trailing_vecNorm2
coxHigham_exact_signed_pivot_row_entry_bound_of_stage_entry_bound
```

Lemma 325 (Exact same-reflector row-growth bridge). *Let p be the current pivot row and k the current pivot column. Form the signed Householder vector*

$$v = A(:, k)_{\text{tail}} - \alpha e_p, \quad \alpha = -\text{sign}(A_{pk}) \|A(p : m, k)\|_2, \quad \beta = \frac{2}{v^T v}.$$

For any non-pivot active row $r > p$, the exact matrix-vector update of column j is the scalar Cox–Higham row update

$$((I - \beta v v^T) A(:, j))_r = A_{rj} - (\beta v^T A(p : m, j)) A_{rk}.$$

Consequently, if column k maximizes the active trailing column norm among remaining columns, $j \geq k$, and the active row entries obey

$$|A_{r\ell}| \leq B, \quad \ell \geq k,$$

then

$$|((I - \beta v v^T) A(:, j))_r| \leq (1 + \sqrt{2}) B.$$

This is the one-step exact same-reflector field needed by the direct stored rounded-panel recurrence for non-pivot active rows.

Lean formalization.

```
Algorithms/QR/HouseholderApply.lean matMulVec_householder_signed_pivot_update_entry_eq
coxHigham_exact_same_reflector_row_growth_of_signed_pivot_row_bound
```

Corollary 326 (One-step exact active-row signed-pivot bound). *In one signed pivoted Householder step, suppose the current row is active, $r \geq p$, the pivot column has positive active trailing norm, column k maximizes the active trailing column norm among the remaining columns, and the needed row and column active-entry bounds are bounded by $B \geq 0$. Then*

$$|(PA(:, j))_r| \leq \max\{1 + \sqrt{2}, \sqrt{m}\} B.$$

For $r > p$ this is the non-pivot Cox–Higham scalar row-growth bridge; for $r = p$ it is the signed pivot-row active-tail bridge. The statement is a one-step case split, not yet the concrete multi-stage sorted-loop theorem.

Lean formalization.

`Algorithms/QR/HouseholderApply.lean` `coxHigham_exact_signed_pivot_active_row_entry_bound_of_stage_bounds`

Corollary 327 (Active-block budget supplies the one-step signed-pivot fields). *In the preceding one-step signed-pivot setting, suppose a single active-block budget is available:*

$$|A_{i\ell}| \leq B \quad (i \geq p, \ell \geq k), \quad B \geq 0.$$

If the pivot column has positive active trailing norm and maximizes the active trailing column norm among the remaining columns, then for every active row $r \geq p$ and remaining column $j \geq k$,

$$|(PA(:, j))_r| \leq \max\{1 + \sqrt{2}, \sqrt{m}\} B.$$

Thus a block invariant of the form used in a sorted or pivoted loop supplies the separate row and column hypotheses in the one-step signed-pivot theorem. This is still a one-step field adapter: it does not prove that a particular sorting policy propagates the active-block budget over all stages.

Lean formalization.

`Algorithms/QR/HouseholderApply.lean` `coxHigham_exactSignedPivotPanelStep_active_block_bound_of_stage_bound`

Corollary 328 (Exact signed-pivot panel sequence with visible budgets). *Let A_t be the exact sequence generated by applying, at each stage t , the signed-pivot Householder panel step to the current matrix A_t . Let*

$$c_m = \max\{1 + \sqrt{2}, \sqrt{m}\}.$$

Suppose that, for every active stage $t < q$, the pivot column has positive active trailing norm, maximizes the active trailing norm among the remaining columns, the target column is remaining, and the row and column stage budgets are bounded by $B_t \geq 0$. If

$$c_m B_t \leq B_{t+1} \quad (0 \leq t < q), \quad |A_0(r, j)| \leq B_0,$$

then

$$|A_q(r, j)| \leq B_q.$$

In particular, if the source sorting/pivoting invariant supplies $B_t = c_m^t B_0$, then

$$|A_q(r, j)| \leq c_m^q B_0.$$

This is the concrete exact-loop budget propagation theorem needed by the Cox–Higham route. It still keeps the stage-budget/sorting fields visible; it does not yet prove that a particular sorting policy supplies those fields, and it is not the final QR/preconditioner theorem.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean coxHighamActiveRowGrowthFactor
coxHighamActiveRowGrowthFactor_nonneg
Algorithms/QR/HouseholderApply.lean exactSignedPivotHouseholderPanelStep
coxHigham_exactSignedPivotPanelStep_active_entry_bound_of_stage_bounds
coxHigham_exactSignedPivotPanel_sequence_active_entry_bound_of_stage_budgets
coxHigham_exactSignedPivotPanel_sequence_active_entry_bound_of_geometric_stage_budgets
```

Corollary 329 (Exact signed-pivot sequence with active-block budgets). *The preceding sequence theorem can be stated using one active-block budget at each stage. Suppose that, for every stage $t < q$,*

$$|A_t(i, \ell)| \leq B_t \quad (i \geq p_t, \ell \geq k_t),$$

and suppose the same positive active trailing norm, pivot maximality, stage recurrence $c_m B_t \leq B_{t+1}$, and initial bound assumptions hold. Then

$$|A_q(r, j)| \leq B_q.$$

In the geometric case, if the active block satisfies

$$|A_t(i, \ell)| \leq c_m^t B_0 \quad (i \geq p_t, \ell \geq k_t),$$

then

$$|A_q(r, j)| \leq c_m^q B_0.$$

This is the exact sequence theorem in the shape used by a sorting or pivoting invariant: one block bound supplies all row and column fields at every stage. It still does not prove that a particular sorting policy supplies the displayed active-block bounds.

Lean formalization.

```
Algorithms/QR/HouseholderApply.lean coxHigham_exactSignedPivotPanel_sequence_active_
block_bound_of_stage_budgets
coxHigham_exactSignedPivotPanel_sequence_active_block_bound_of_geometric_stage_budgets
```

Theorem 330 (Active-block budgets from monotone active windows). *Let A_t be the exact signed-pivot panel sequence above and let $c_m = \max\{1 + \sqrt{2}, \sqrt{m}\}$. Suppose the initial matrix has the entrywise bound*

$$|A_0(i, \ell)| \leq B_0 \quad (1 \leq i \leq m, 1 \leq \ell \leq n),$$

with $B_0 \geq 0$. Suppose also that the active row starts p_t and remaining-column starts k_t are monotone in t , and that every stage satisfies the visible positive active trailing-norm and column-pivot maximality conditions. Then, for every $t \leq q$,

$$|A_t(i, \ell)| \leq c_m^t B_0 \quad (i \geq p_t, \ell \geq k_t).$$

Lean formalization.

```
Algorithms/QR/HouseholderApply.lean coxHigham_exactSignedPivotPanel_sequence_active_
block_bound_of_initial_block_bound
```

Corollary 331 (Active-block budgets from nonzero active mass). *In the setting of the previous theorem, the positive active trailing-norm hypothesis can be replaced by the following source-shaped nonbreakdown condition: at each stage $t < q$, some entry in the remaining active block is nonzero,*

$$\exists \ell \geq k_t, \exists i \geq p_t : A_t(i, \ell) \neq 0.$$

Together with column-pivot maximality, this implies that the selected pivot column has positive active trailing norm. Hence the same conclusion holds:

$$|A_t(i, \ell)| \leq c_m^t B_0 \quad (t \leq q, i \geq p_t, \ell \geq k_t).$$

This is still not a proof that a concrete implementation supplies the nonzero-active-block and pivot-max fields; it removes only the direct positive-norm assumption from the exact sequence theorem.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean    householderTrailingColumnNorm2Sq_pos_of_pivot_max_
exists_pos
householderTrailingColumnNorm2Sq_pos_of_pivot_max_exists_active_entry_ne
Algorithms/QR/HouseholderApply.lean    coxHigham_exactSignedPivotPanel_sequence_active_
block_bound_of_initial_block_bound_of_active_block_nonzero
```

Corollary 332 (Active-block budgets from an active max-pivot policy). *In the setting of the preceding corollary, the raw column-pivot maximality field can be replaced by an explicit finite max-pivot policy. In the current exact-sequence formalism, the remaining-column start k_t denotes the current column after pivoting/sorting has placed the selected column in active position. The policy hypothesis says that this current column is exactly the output of `householderActiveMaxPivotColumn` for the displayed active block. Under this policy and the same nonzero-active-block condition,*

$$|A_t(i, \ell)| \leq c_m^t B_0 \quad (t \leq q, i \geq p_t, \ell \geq k_t).$$

This is still an exact-sequence dependency. It does not yet prove that the rounded stored implementation performs the corresponding pivoting/sorting, nor does it connect the result to the QR/preconditioner solve theorem.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean    householderActiveMaxPivotColumn
householderActiveMaxPivotColumn_ge
householderActiveMaxPivotColumn_pivot_max
Algorithms/QR/HouseholderApply.lean    coxHigham_exactSignedPivotPanel_sequence_active_
block_bound_of_initial_block_bound_of_active_max_pivot
```

Corollary 333 (Active-block budgets from positive active-block mass). *In the setting of the active max-pivot policy corollary, the remaining nonzero-active-block hypothesis can be replaced by a scalar active-block mass condition. Define*

$$\mathcal{M}_t = \sum_{i, \ell} \mathbf{1}_{\{i \geq p_t, \ell \geq k_t\}} A_t(i, \ell)^2.$$

If $\mathcal{M}_t > 0$ at every active stage and the current column is selected by `householderActiveMaxPivotColumn`, then

$$|A_t(i, \ell)| \leq c_m^t B_0 \quad (t \leq q, i \geq p_t, \ell \geq k_t).$$

This is still an exact-sequence dependency: a concrete pivoted/sorted loop must prove the positive active-block mass and policy fields.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean householderActiveBlockNorm2Sq
exists_active_entry_ne_of_householderActiveBlockNorm2Sq_pos
Algorithms/QR/HouseholderApply.lean      coxHigham_exactSignedPivotPanel_sequence_active_
block_bound_of_initial_block_bound_of_active_block_norm_pos
```

Corollary 334 (Active-block budgets after active max-column swapping). *Let q_t be the active column with maximal trailing column norm in a raw stage A_t^{raw} , and let the displayed stage be obtained by swapping columns k_t and q_t :*

$$A_t = \text{swapcols}(A_t^{\text{raw}}, k_t, q_t).$$

Then the displayed column k_t is pivot-maximal on the active suffix. Hence, under the same monotone active-window assumptions as above and the positive active-block mass condition for the displayed stages, the exact signed-pivot sequence satisfies

$$|A_t(i, \ell)| \leq c_m^t B_0 \quad (t \leq q, i \geq p_t, \ell \geq k_t).$$

This closes the concrete column-swap policy field for the exact Cox–Higham route. It still leaves the positive active-block mass/nonbreakdown invariant and the rounded QR/preconditioner solve connection open.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean householderSwapColumns
householderTrailingColumnNorm2Sq_swapColumns_left
householderTrailingColumnNorm2Sq_swapColumns_of_ne
householderSwapColumns_activeMaxPivotColumn_pivot_max
Algorithms/QR/HouseholderApply.lean      coxHigham_exactSignedPivotPanel_sequence_active_
block_bound_of_initial_block_bound_of_swapped_active_max_pivot
```

Corollary 335 (Raw active-block mass survives the active max-column swap). *In the previous corollary, the positive active-block mass condition may be placed before the column swap. If*

$$\mathcal{M}_t^{\text{raw}} = \sum_{i, \ell} \mathbf{1}_{\{i \geq p_t, \ell \geq k_t\}} A_t^{\text{raw}}(i, \ell)^2 > 0$$

and q_t lies in the active suffix, then

$$\mathcal{M}_t(A_t) > 0 \quad \text{for } A_t = \text{swapcols}(A_t^{\text{raw}}, k_t, q_t).$$

Consequently the exact swapped-stage Cox–Higham sequence again satisfies

$$|A_t(i, \ell)| \leq c_m^t B_0 \quad (t \leq q, i \geq p_t, \ell \geq k_t).$$

Thus the remaining nonbreakdown obligation is raw-stage active-block mass, not an artifact introduced after sorting.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean    householderActiveBlockNorm2Sq_pos_of_exists_active_
entry_ne

householderActiveBlockNorm2Sq_swapColumns_pos_of_pos

Algorithms/QR/HouseholderApply.lean    coxHigham_exactSignedPivotPanel_sequence_active_
block_bound_of_initial_block_bound_of_swapped_active_max_pivot_of_raw_active_block_norm_pos
```

Corollary 336 (Raw active-block mass from QR nonbreakdown). *Fix a QR stage k . Suppose the current column is not in the span of the previous columns, and suppose the previous columns span every vector supported above the active row. Then the raw active block has positive mass:*

$$\mathcal{M}_k(A) > 0.$$

Equivalently, the same conclusion follows from the concrete QR witnesses used elsewhere in the library: a previous-block coefficient matrix plus a leading column left inverse, local left inverses for the previous and current leading blocks, or nonzero determinants of the previous and current leading blocks.

In stagewise determinant form, if for every stored QR stage $k < n$ the previous leading block and current leading block are nonsingular, and the already completed columns have the stored lower-zero shape, then

$$\mathcal{M}_k(\hat{A}_k) > 0 \quad (k < n).$$

Thus the raw-stage nonbreakdown field in the Cox–Higham active-block theorem can be supplied by the existing QR rank/determinant infrastructure.

Lean formalization.

```
Algorithms/QR/HouseholderQR.lean    householderActiveBlockNorm2Sq_pos_of_column_
notInPreviousSpan

householderActiveBlockNorm2Sq_pos_of_leading_witnesses

householderActiveBlockNorm2Sq_pos_of_leading_block_leftInverses

householderActiveBlockNorm2Sq_pos_of_leading_block_det_ne_zero

householderActiveBlockNorm2Sq_pos_sequence_of_leading_block_det_ne_zero
```

Corollary 337 (Rounded active-block budgets for signed-pivot stored panels). *Let*

$$\hat{A}_{t+1} = \text{fl}((I - \beta_t v_t v_t^T) \hat{A}_t)$$

be the stored compact Householder panel update, where v_t and β_t are the signed-pivot Householder vector and coefficient built from $\hat{A}_t$, the active row p_t , and the pivot column k_t . Assume the active row and column windows are monotone, the pivot column has positive active trailing norm, the pivot column is maximal among the remaining active columns, completed columns are stored exactly, and the pivot-column storage equation has the required zero entries. Let $B_t \geq 0$ be an active-block budget for the stored matrix. If

$$c_m B_t + \eta_t(i, \ell) \leq B_{t+1}$$

for every next active entry (i, ℓ) , where

$$c_m = \max\{1 + \sqrt{2}, \sqrt{m}\}$$

and $\eta_t(i, \ell)$ is the compact Householder componentwise FP budget, then

$$|\hat{A}_t(i, \ell)| \leq B_t \quad (t \leq q, \ i \geq p_t, \ \ell \geq k_t).$$

Lean formalization.

```
Algorithms/QR/HouseholderApply.lean signedPivotHouseholderVector
signedPivotHouseholderBeta
coxHigham_storedPanel_sequence_active_block_bound_of_signed_pivot_stage_bounds
```

Lemma 338 (Cox–Higham row-wise accumulated perturbation). *Let $c = 1 + \sqrt{2}$. Suppose a computed row-entry sequence $\hat{A}_t(r, j)$ is compared against an exact sequence $A_t(r, j)$, and the entrywise discrepancy*

$$E_t = |\hat{A}_t(r, j) - A_t(r, j)|$$

satisfies the affine recurrence

$$E_{t+1} \leq cE_t + \eta_t, \quad 0 \leq t < k.$$

Define the accumulated additive budget recursively by

$$\mathcal{E}_0 = 0, \quad \mathcal{E}_{t+1} = c\mathcal{E}_t + \eta_t.$$

Then

$$E_k \leq c^k E_0 + \mathcal{E}_k.$$

Consequently, if the exact sequence satisfies the sorted Cox–Higham row bound $|A_k(r, j)| \leq c^k B_k^{(0)}$, then the computed sequence satisfies

$$|\hat{A}_k(r, j)| \leq c^k B_k^{(0)} + c^k E_0 + \mathcal{E}_k.$$

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean scalarAffineGrowthBudget
scalar_affine_growth_iterate_bound
coxHigham_rowSorting_active_entry_bound_of_stage_growth_with_additive
coxHigham_rowwise_error_accumulation_bound
coxHigham_abs_entry_le_exact_bound_add_error
coxHigham_rowSorting_active_entry_bound_with_accumulated_error
```

Lemma 339 (Concrete stored-panel FP budget for Cox–Higham row errors). *Consider one stored rounded compact Householder panel step*

$$\hat{A}_{t+1} = \text{fl}((I - \beta_t v_t v_t^T) \hat{A}_t)$$

with the repository’s stored-QR convention: completed columns are preserved and the current pivot column is stored with zeros below the pivot. If column j is active at stage t , then for each row r

$$\left| \hat{A}_{t+1}(r, j) - (I - \beta_t v_t v_t^T) \hat{A}_t(:, j)_r \right| \leq B_t^{\text{fp}}(r, j),$$

where $B_t^{\text{fp}}(r, j)$ is exactly the compact Householder component budget formalized in `householderCompactComponentBudget`. Consequently, if the exact Cox–Higham part supplies the visible Lipschitz field

$$\left| (I - \beta_t v_t v_t^T) \hat{A}_t(:, j)_r - A_{t+1}(r, j) \right| \leq (1 + \sqrt{2}) |\hat{A}_t(r, j) - A_t(r, j)|,$$

then

$$|\hat{A}_{t+1}(r, j) - A_{t+1}(r, j)| \leq (1 + \sqrt{2})|\hat{A}_t(r, j) - A_t(r, j)| + B_t^{\text{fp}}(r, j).$$

Iterating this recurrence gives the same affine accumulated budget

$$|\hat{A}_k(r, j) - A_k(r, j)| \leq (1 + \sqrt{2})^k |\hat{A}_0(r, j) - A_0(r, j)| + \mathcal{E}_k,$$

with $\mathcal{E}_{t+1} = (1 + \sqrt{2})\mathcal{E}_t + B_t^{\text{fp}}(r, j)$.

Lean formalization.

```
Algorithms/QR/HouseholderApply.lean      fl_householderStoredPanelStep_active_entry_
componentwise_error_bound
coxHigham_storedPanelStep_row_error_recurrence_of_exact_lipschitz
coxHigham_storedPanel_sequence_rowwise_error_accumulation_bound_of_exact_lipschitz
```

Lemma 340 (Direct rounded row-magnitude recurrence for stored panels). *Under the same stored-panel hypotheses, suppose the exact same-reflector update of the current stored panel satisfies the Cox–Higham row-growth estimate*

$$\left| (I - \beta_t v_t v_t^T) \hat{A}_t(:, j)_r \right| \leq (1 + \sqrt{2}) |\hat{A}_t(r, j)|.$$

Then the rounded stored entry satisfies

$$|\hat{A}_{t+1}(r, j)| \leq (1 + \sqrt{2}) |\hat{A}_t(r, j)| + B_t^{\text{fp}}(r, j).$$

Consequently, after k stored panel steps,

$$|\hat{A}_k(r, j)| \leq (1 + \sqrt{2})^k |\hat{A}_0(r, j)| + \mathcal{E}_k, \quad \mathcal{E}_{t+1} = (1 + \sqrt{2})\mathcal{E}_t + B_t^{\text{fp}}(r, j).$$

This is the source-shaped floating-point dependency for the Cox–Higham route: the remaining exact growth/sorting field is explicit, and the local FP contribution is exactly the repository’s compact Householder component budget.

Lean formalization.

```
Algorithms/QR/HouseholderApply.lean      coxHigham_storedPanelStep_active_entry_bound_of_
exact_growth
coxHigham_storedPanel_sequence_active_entry_bound_of_exact_growth
```

Corollary 341 (Stored-panel handoff for the active-row Cox–Higham factor). *Let*

$$c_m = \max\{1 + \sqrt{2}, \sqrt{m}\}.$$

Suppose the stored rounded panel loop satisfies the same concrete compact Householder update hypotheses as above, and suppose the exact same-reflector part now supplies the active-row growth field

$$\left| (I - \beta_t v_t v_t^T) \hat{A}_t(:, j)_r \right| \leq c_m |\hat{A}_t(r, j)|.$$

Then, with the same compact floating-point component budget $B_t^{\text{fp}}(r, j)$,

$$|\hat{A}_k(r, j)| \leq c_m^k |\hat{A}_0(r, j)| + \mathcal{E}_k, \quad \mathcal{E}_{t+1} = c_m \mathcal{E}_t + B_t^{\text{fp}}(r, j).$$

More generally, the Lean theorem is factor-parametric: any nonnegative exact same-reflector growth factor c can be used in place of c_m . This is the floating-point handoff needed by the exact signed-pivot panel sequence theorem; it still keeps the sorting-policy stage-budget field visible.

Lean formalization.

```
Algorithms/QR/HouseholderApply.lean      coxHigham_storedPanelStep_active_entry_bound_of_
exact_growth_factor
coxHigham_storedPanel_sequence_active_entry_bound_of_exact_growth_factor
coxHigham_storedPanel_sequence_active_entry_bound_of_exact_active_growth
```

Corollary 342 (Stage-budget handoff for signed-pivot stored panels). *Let $c_m = \max\{1 + \sqrt{2}, \sqrt{m}\}$. Suppose a stored rounded panel step uses the signed-pivot Householder vector for the current pivot column, and suppose the exact Cox-Higham stage fields hold at this step: positive active trailing norm, column-pivot maximality, active-row budget*

$$|A_t(r, \ell)| \leq B_t, \quad \ell \geq k,$$

and active-column budget

$$|A_t(i, j)| \leq B_t, \quad i \geq p.$$

Assume also the stored-panel convention for completed columns and the current pivot column. Then the rounded stored entry satisfies

$$|\hat{A}_{t+1}(r, j)| \leq c_m B_t + B_t^{\text{fp}}(r, j),$$

where $B_t^{\text{fp}}(r, j)$ is exactly the repository's compact Householder component budget for the same vector, scalar, and column. Consequently, if visible stage budgets satisfy

$$c_m B_t + B_t^{\text{fp}}(r, j) \leq B_{t+1}, \quad |\hat{A}_0(r, j)| \leq B_0,$$

then the stored rounded sequence satisfies

$$|\hat{A}_k(r, j)| \leq B_k.$$

Lean formalization.

```
Algorithms/QR/HouseholderApply.lean      coxHigham_storedPanelStep_active_entry_bound_of_
exact_stage_budget_factor
coxHigham_storedPanel_sequence_active_entry_bound_of_exact_stage_budgets_factor
coxHigham_storedPanel_sequence_active_entry_bound_of_exact_active_stage_budgets
coxHigham_storedPanelStep_active_entry_bound_of_signed_pivot_stage_bounds
```

Lemma 343 (Trailing-norm and pivot-budget scalar bridges). *Let x be the current pivot column and let x_{tail} denote its entries from row k onward. If some active trailing entry is nonzero, then*

$$\exists i \geq k, x_i \neq 0 \implies \|x_{\text{tail}}\|_2^2 > 0.$$

In particular, $x_k \neq 0$ is sufficient. Moreover, if

$$\alpha_k^2 = \|x_{\text{tail}}\|_2^2,$$

then

$$|\alpha_k| = \sqrt{\|x_{\text{tail}}\|_2^2}.$$

Consequently any bound

$$B_k < \sqrt{\|x_{\text{tail}}\|_2^2}$$

is exactly the stored-loop pivot-budget nonbreakdown condition

$$B_k < |\alpha_k|.$$

Finally, for any active trailing index $i \geq k$,

$$|x_i| \leq \sqrt{\|x_{\text{tail}}\|_2^2},$$

so a concrete entrywise lower bound

$$B_k < |x_i|$$

already implies the square-root trailing-norm budget.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean householderTrailingNorm2Sq_pos_of_exists_ne
householderTrailingNorm2Sq_pos_of_pivot_ne_zero
abs_alpha_eq_sqrt_trailingNorm2Sq
budget_lt_abs_alpha_of_lt_sqrt_trailingNorm2Sq
abs_entry_le_sqrt_householderTrailingNorm2Sq_of_pivot_le
budget_lt_sqrt_householderTrailingNorm2Sq_of_lt_abs_active_entry
```

Lemma 344 (Positive trailing norm does not force a nonzero pivot). *The previous lemma has no converse in the unpivoted Householder setting. There is a two-entry active column $x = (0, 1)$ with pivot index 0 such that*

$$\|x_{0:1}\|_2^2 > 0, \quad x_0 = 0.$$

Thus a proof route that tries to obtain the current leading pivot condition from positive active trailing norm alone is invalid; it needs pivoting, a structured leading-block invariant, or an explicit pivot/nonzero-domain assumption.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean householderTrailingPivotCounterexample2
householderTrailingPivotCounterexample2_pivot_zero
householderTrailingPivotCounterexample2_trailingNorm2Sq_pos
not_forall_trailingNorm2Sq_pos_implies_pivot_ne_zero
```

Definition 345 (Prefix-span QR nonbreakdown invariant). For the current stored QR panel A_k , write a_j for its j -th column. The local prefix-span nonbreakdown invariant at pivot k has two parts:

$$a_k \notin \text{span}\{a_0, \dots, a_{k-1}\},$$

and every vector supported on the first k rows is in that same span:

$$\{y : y_i = 0 \text{ for all } i \geq k\} \subseteq \text{span}\{a_0, \dots, a_{k-1}\}.$$

Lemma 346 (Leading-block inverse gives prefix coefficients). *Let $T_k \in \mathbb{R}^{k \times k}$ be the transpose of the leading block of the first k rows and previous k columns,*

$$T_k(j, s) = A_k(s, j), \quad 0 \leq j, s < k.$$

If $CT_k = I$, then C is a valid prefix-basis coefficient matrix:

$$\sum_{j < k} C_{rj} A_k(s, j) = \delta_{sr}, \quad 0 \leq r, s < k.$$

Lemma 347 (Leading-block inverse gives ambient column duals). *Let $S_k \in \mathbb{R}^{(k+1) \times (k+1)}$ be the leading block of the first $k+1$ rows and first $k+1$ columns,*

$$S_k(r, q) = A_k(r, q), \quad 0 \leq r, q \leq k.$$

If $DS_k = I$, define an ambient dual family by padding D with zeros:

$$L_{pi} = \begin{cases} D_{pi}, & i \leq k, \\ 0, & i > k. \end{cases}$$

Then L is a leading-column left inverse for the first $k+1$ columns:

$$\sum_{i < m} L_{pi} A_k(i, q) = \delta_{qp}, \quad 0 \leq p, q \leq k.$$

Lemma 348 (Leading-block coefficients imply prefix span). *Fix k , and suppose the first k previous columns have already acquired the QR lower-zero shape,*

$$A_k(i, j) = 0, \quad i \geq k, \quad j < k.$$

Assume there is a coefficient matrix $C \in \mathbb{R}^{k \times k}$ such that for every prefix coordinate basis vector e_r ,

$$\sum_{j < k} C_{rj} A_k(s, j) = \delta_{sr}, \quad 0 \leq r, s < k.$$

Then the previous columns span every prefix-supported vector. More precisely, if $y_i = 0$ for all $i \geq k$, then the coefficients

$$c_j = \sum_{r < k} y_r C_{rj}$$

satisfy

$$y_i = \sum_{j < k} c_j A_k(i, j), \quad 0 \leq i < m.$$

Lemma 349 (Leading-column left inverse gives column independence). *Suppose there is a dual coefficient family L for the first $k+1$ columns such that*

$$\sum_i L_{pi} A_k(i, q) = \delta_{qp}, \quad 0 \leq p, q \leq k.$$

Then the current pivot column is not in the span of previous columns:

$$a_k \notin \text{span}\{a_0, \dots, a_{k-1}\}.$$

Lemma 350 (Leading-block witnesses give active trailing nonbreakdown). *Fix a pivot k . Suppose the previous columns have the QR lower-zero shape,*

$$A_k(i, j) = 0, \quad i \geq k, \quad j < k,$$

suppose a coefficient matrix C reproduces every prefix coordinate basis vector using the previous columns,

$$\sum_{j < k} C_{rj} A_k(s, j) = \delta_{sr}, \quad 0 \leq r, s < k,$$

and suppose a dual family L is a left inverse for the first $k + 1$ columns,

$$\sum_i L_{pi} A_k(i, q) = \delta_{qp}, \quad 0 \leq p, q \leq k.$$

Then the active trailing part of the pivot column is nonzero:

$$\exists i \geq k, \quad A_k(i, k) \neq 0.$$

Consequently

$$\|A_k(k:m, k)\|_2^2 > 0.$$

Lemma 351 (Prefix-span nonbreakdown gives a nonzero active trailing entry). *If the prefix-span QR nonbreakdown invariant holds at pivot k , then the active trailing part of the pivot column is nonzero:*

$$\exists i \geq k, \quad A_k(i, k) \neq 0.$$

Consequently

$$\|A_k(k:m, k)\|_2^2 > 0.$$

Corollary 352 (Stored QR diagonal nonbreakdown from prefix-span and square-root budget). *In the stored trailing Householder QR sequence, assume at each pivot k the prefix-span QR nonbreakdown invariant, the sign-choice condition*

$$\alpha_k A_k(k, k) \leq 0,$$

the Householder normalization

$$\alpha_k^2 = \|A_k(k:m, k)\|_2^2,$$

and the explicit square-root pivot budget

$$B_k < \sqrt{\|A_k(k:m, k)\|_2^2}.$$

Then the stored triangular diagonal is nonzero:

$$\hat{R}_{ii} \neq 0, \quad 0 \leq i < n.$$

Corollary 353 (Stored QR diagonal nonbreakdown from prefix-span and active-entry budget). *In the stored trailing Householder QR sequence, assume at each pivot k the prefix-span QR nonbreakdown invariant, the sign-choice condition, and the Householder normalization*

$$\alpha_k^2 = \|A_k(k:m, k)\|_2^2.$$

Instead of assuming a square-root trailing-norm budget, suppose there is an active trailing row $i_k \geq k$ such that the compact diagonal update budget is dominated by that entry,

$$B_k < |A_k(i_k, k)|.$$

Then the stored triangular diagonal is nonzero:

$$\hat{R}_{ii} \neq 0, \quad 0 \leq i < n.$$

Lemma 354 (Dimensioned norm margin gives an active entry). *Let $x \in \mathbb{R}^m$, let $p < m$, and let $B \geq 0$. If*

$$mB^2 < \|x(p:m)\|_2^2,$$

then there exists an active index $i \geq p$ such that

$$B < |x_i|.$$

Lemma 355 (Leading dual norm lower bound for the trailing pivot column). *Fix a pivot k . Let L_k be a leading-column dual for the first $k+1$ columns, and let $L_{k,*}$ denote its row that selects the current pivot column. Assume:*

$$L_{k,*}^T A_k(:, k) = 1, \quad L_{k,*}^T A_k(:, j) = 0 \quad (j < k),$$

and assume the prefix-span invariant, so the prefix part of $A_k(:, k)$ lies in the span of the previous columns. If

$$\|L_{k,*}\|_2^2 \leq K_k, \quad K_k > 0,$$

then

$$\frac{1}{K_k} \leq \|A_k(k:m, k)\|_2^2.$$

Lemma 356 (Local inverse row norm lower bound). *Let C_k be a left inverse of the concrete leading block $S_k = A_k(0:k, 0:k)$. Pad the last row of C_k by zeros outside the first $k+1$ rows to obtain the ambient leading-column dual $L_{k,*}$. Then the padding preserves the squared norm:*

$$\|L_{k,*}\|_2^2 = \|C_k(k, :)\|_2^2.$$

Consequently, if

$$\|C_k(k, :)\|_2^2 \leq K_k, \quad K_k > 0,$$

and the prefix-span invariant holds, then

$$\frac{1}{K_k} \leq \|A_k(k:m, k)\|_2^2.$$

Corollary 357 (Stored QR diagonal nonbreakdown from a leading-dual budget). *In the stored trailing Householder QR sequence, assume prefix-span, sign choice, Householder normalization, and for every pivot $k < n$ a leading dual row satisfying*

$$\|L_{k,*}\|_2^2 \leq K_k, \quad K_k > 0.$$

If the compact diagonal update budget satisfies

$$mB_k^2 < \frac{1}{K_k},$$

then the stored triangular diagonal is nonzero:

$$\hat{R}_{ii} \neq 0, \quad 0 \leq i < n.$$

Corollary 358 (Stored QR diagonal nonbreakdown from a leading-block inverse row budget). *In the same stored trailing Householder QR sequence, replace the ambient dual row by a local left inverse C_k of the leading block S_k . Suppose*

$$\|C_k(k, :)\|_2^2 \leq K_k, \quad K_k > 0, \quad mB_k^2 < \frac{1}{K_k}.$$

Then the stored triangular diagonal is nonzero:

$$\hat{R}_{ii} \neq 0, \quad 0 \leq i < n.$$

Corollary 359 (Stored QR diagonal nonbreakdown from a leading-block inverse Frobenius budget). *In the same stored trailing Householder QR sequence, assume a local left inverse C_k of the leading block S_k at each pivot. If*

$$\|C_k\|_F^2 \leq K_k, \quad K_k > 0, \quad mB_k^2 < \frac{1}{K_k},$$

then the stored triangular diagonal is nonzero:

$$\hat{R}_{ii} \neq 0, \quad 0 \leq i < n.$$

Corollary 360 (Stored QR diagonal nonbreakdown from a leading-block inverse infinity-norm budget). *In the same stored trailing Householder QR sequence, assume a local left inverse C_k of the leading block S_k at each pivot. If*

$$(k+1)\|C_k\|_\infty^2 \leq K_k, \quad K_k > 0, \quad mB_k^2 < \frac{1}{K_k},$$

then the stored triangular diagonal is nonzero:

$$\hat{R}_{ii} \neq 0, \quad 0 \leq i < n.$$

Corollary 361 (Stored QR diagonal nonbreakdown from prefix-span and norm-square budget). *In the stored trailing Householder QR sequence, assume the prefix-span QR nonbreakdown invariant, sign choice, and Householder normalization at every pivot $k < n$. If the compact diagonal update budget satisfies the dimensioned margin*

$$mB_k^2 < \|A_k(k:m, k)\|_2^2,$$

then the stored triangular diagonal is nonzero:

$$\hat{R}_{ii} \neq 0, \quad 0 \leq i < n.$$

Corollary 362 (Stored QR diagonal nonbreakdown from leading-block witnesses). *In the stored trailing Householder QR sequence, assume that at every pivot k we have the leading-block coefficient witness C_k , the leading-column left-inverse witness L_k , the QR lower-zero shape for the previous columns, the sign-choice condition, the Householder normalization, and the explicit square-root pivot budget*

$$B_k < \sqrt{\|A_k(k:m, k)\|_2^2}.$$

Then the final stored triangular diagonal is nonzero:

$$\hat{R}_{ii} \neq 0, \quad 0 \leq i < n.$$

Corollary 363 (Stored QR diagonal nonbreakdown from local leading-block inverses). *In the stored trailing Householder QR sequence, assume that at every pivot k the transposed previous leading block has a local left inverse,*

$$C_k T_k = I,$$

and the actual leading $(k+1) \times (k+1)$ block has a local left inverse,

$$D_k S_k = I.$$

Assume also the QR lower-zero shape for the previous columns, the sign-choice condition, the Householder normalization, and the explicit square-root pivot budget

$$B_k < \sqrt{\|A_k(k:m, k)\|_2^2}.$$

Then the final stored triangular diagonal is nonzero:

$$\hat{R}_{ii} \neq 0, \quad 0 \leq i < n.$$

Lemma 364 (Nonzero determinant gives the local inverse predicate). *Let $T \in \mathbb{R}^{d \times d}$. If $\det T \neq 0$, then there exists a matrix $T^\dagger$ satisfying the repository predicate*

$$T^\dagger T = I.$$

The witness is Mathlib's nonsingular inverse, exported through the local function-shaped matrix representation.

Corollary 365 (Stored QR diagonal nonbreakdown from nonsingular leading blocks). *In the stored trailing Householder QR sequence, assume that for every pivot k the transposed previous leading block T_k and the current leading block $S_k \in \mathbb{R}^{(k+1) \times (k+1)}$ are nonsingular:*

$$\det T_k \neq 0, \quad \det S_k \neq 0.$$

Assume also the QR lower-zero shape for the previous columns, the sign-choice condition, the Householder normalization, and the explicit square-root pivot budget

$$B_k < \sqrt{\|A_k(k:m, k)\|_2^2}.$$

Then the final stored triangular diagonal is nonzero:

$$\hat{R}_{ii} \neq 0, \quad 0 \leq i < n.$$

Lemma 366 (Triangular local blocks have nonzero determinant). *Let $U \in \mathbb{R}^{d \times d}$ be upper triangular in the convention*

$$U_{ij} = 0 \quad \text{whenever } j < i.$$

If every diagonal entry is nonzero, then

$$\det U \neq 0.$$

The same statement holds for lower triangular matrices. Consequently, in a stored QR panel, the previous transposed leading block T_k and current leading block S_k have nonzero determinant whenever their visible triangular shape and local diagonal nonzero assumptions hold.

Lean formalization.

```
Analysis/MatrixAlgebra.lean nonsingInv
isLeftInverse_nonsingInv_of_det_isUnit
exists_isLeftInverse_of_det_ne_zero
det_ne_zero_of_upper_triangular_diag_ne_zero
det_ne_zero_of_lower_triangular_diag_ne_zero
Algorithms/QR/HouseholderQR.lean qrLeadingRow
```



```

qrColumnNotInPreviousSpan
qrPrefixSupportSpannedByPreviousColumns
qrPrefixBasisCoefficientMatrix
qrPreviousLeadingBlockTranspose
qrLeadingBlock
qrPrefixBasisCoefficientMatrix_of_leftInverse_previousLeadingBlockTranspose
qrPrefixBasisCoefficientMatrix_of_det_ne_zero_previousLeadingBlockTranspose
qrPrefixSupportSpannedByPreviousColumns_of_prefixBasisCoefficientMatrix
qrPrefixSupportSpannedByPreviousColumns_of_leftInverse_previousLeadingBlockTranspose
fl_householderStoredPanel_sequence_prefixSpan_of_leftInverse_previousLeadingBlockTranspose
qrPrefixSupportSpannedByPreviousColumns_of_det_ne_zero_previousLeadingBlockTranspose
qrLeadingColumnLeftInverse
qrLeadingColumnLeftInverse_of_leftInverse_leadingBlock
qrLeadingColumnLeftInverse_of_det_ne_zero_leadingBlock
qrPreviousLeadingBlockTranspose_det_ne_zero_of_upper_triangular_diag_ne_zero
qrLeadingBlock_det_ne_zero_of_upper_triangular_diag_ne_zero
qrColumnNotInPreviousSpan_of_leadingColumnLeftInverse
exists_active_trailing_entry_ne_of_column_notInPreviousSpan
householderTrailingNorm2Sq_pos_of_column_notInPreviousSpan
fl_householderStoredTrailingPanel_sequence_diag_nonzero_of_span_nonbreakdown_sqrt_budget
fl_householderStoredTrailingPanel_sequence_diag_nonzero_of_span_nonbreakdown_active_entry_budget
exists_active_entry_budget_lt_abs_of_dim_mul_budget_sq_lt_trailingNorm2Sq
budget_lt_sqrt_householderTrailingNorm2Sq_of_dim_mul_budget_sq_lt_trailingNorm2Sq
fl_householderStoredTrailingPanel_sequence_diag_nonzero_of_span_nonbreakdown_trailingNorm2Sq_
budget
householderTrailingNorm2Sq_ge_inv_leading_dual_norm_budget
dim_mul_budget_sq_lt_trailingNorm2Sq_of_leading_dual_norm_budget
fl_householderStoredTrailingPanel_sequence_diag_nonzero_of_span_nonbreakdown_leading_dual_norm_
budget
vecNorm2Sq_qrLeadingRow_padded_eq
qrLeadingColumnLeftInverse_padded_row_norm_sq_eq
Analysis/MatrixAlgebra.lean vecNorm2Sq_row_le_frobNormSq
vecNorm2Sq_row_le_frobNorm_sq
abs_coord_le_sum_abs
vecNorm2Sq_le_sum_abs_sq
frobNormSq_le_nat_mul_infNorm_sq
frobNorm_sq_le_nat_mul_infNorm_sq
infNorm_pos_of_det_ne_zero
isRightInverse_nonsingInv_of_det_isUnit
isInverse_nonsingInv_of_det_isUnit

```

```

isInverse_nonsingInv_of_det_ne_zero
exists_isInverse_of_det_ne_zero
Algorithms/InverseBounds.lean triInv_infNorm_upperBound
triInv_infNorm_sq_budget_of_diagDominantUpper
triInv_infNorm_sq_budget_of_diagDominantUpper_det_ne_zero
Algorithms/QR/HouseholderQR.lean householderTrailingNorm2Sq_ge_inv_leadingBlock_
leftInverse_row_norm_budget
dim_mul_budget_sq_lt_trailingNorm2Sq_of_leadingBlock_leftInverse_row_norm_budget
householderTrailingNorm2Sq_ge_inv_leadingBlock_leftInverse_frobNorm_budget
dim_mul_budget_sq_lt_trailingNorm2Sq_of_leadingBlock_leftInverse_frobNorm_budget
householderTrailingNorm2Sq_ge_inv_leadingBlock_leftInverse_infNorm_budget
dim_mul_budget_sq_lt_trailingNorm2Sq_of_leadingBlock_leftInverse_infNorm_budget
fl_householderStoredTrailingPanel_sequence_diag_nonzero_of_span_nonbreakdown_leadingBlock_
leftInverse_norm_budget
fl_householderStoredTrailingPanel_sequence_diag_nonzero_of_span_nonbreakdown_leadingBlock_
leftInverse_frobNorm_budget
fl_householderStoredTrailingPanel_sequence_diag_nonzero_of_span_nonbreakdown_leadingBlock_
leftInverse_infNorm_budget
fl_householderStoredPanel_sequence_prefixSpan_of_leftInverse_previousLeadingBlockTranspose
exists_active_trailing_entry_ne_of_leading_witnesses
householderTrailingNorm2Sq_pos_of_leading_witnesses
fl_householderStoredTrailingPanel_sequence_diag_nonzero_of_leading_witnesses_sqrt_budget
exists_active_trailing_entry_ne_of_leading_block_leftInverses
householderTrailingNorm2Sq_pos_of_leading_block_leftInverses
fl_householderStoredTrailingPanel_sequence_diag_nonzero_of_leading_block_leftInverses_sqrt_budget
exists_active_trailing_entry_ne_of_leading_block_det_ne_zero
householderTrailingNorm2Sq_pos_of_leading_block_det_ne_zero
fl_householderStoredTrailingPanel_sequence_diag_nonzero_of_leading_block_det_ne_zero_sqrt_budget
Analysis/PerturbationTheory.lean infNorm_eq_sup_row_sum
kappaInf_eq_infNorm_mul_infNorm
infNorm_inv_le_of_kappaInf_le_and_norm_lower
infNorm_sq_budget_of_kappaInf_le_and_norm_lower
infNorm_inv_le_of_kappaInf_le_and_det_ne_zero
infNorm_sq_budget_of_kappaInf_le_and_det_ne_zero
Algorithms/LeastSquares/LSQRSolve.lean LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_normBudget_of_span_nonbreakdown_active_
entry_budget
LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
normBudget_of_span_nonbreakdown_trailingNorm2Sq_budget
LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
normBudget_of_span_nonbreakdown_leading_dual_norm_budget

```



```

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_previousLeadingBlock_det_ne_zero_leadingBlock_det_ne_zero_
infNorm_budget

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_previousLeadingBlock_det_ne_zero_leadingBlock_det_ne_zero_
kappaInf_selfNorm_budget

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_upperTriangular_leadingDiag_ne_zero_kappaInf_selfNorm_
budget

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
normBudget_of_span_nonbreakdown_leadingBlock_det_ne_zero_kappaInf_budget

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
normBudget_of_span_nonbreakdown_leadingBlock_det_ne_zero_kappaInf_selfNorm_budget

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
normBudget_of_leading_blocks_det_ne_zero_kappaInf_selfNorm_budget

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
normBudget_of_span_nonbreakdown_diagDominant_leadingBlock_invNorm_budget

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
normBudget_of_span_nonbreakdown_diagDominant_leadingBlock_det_ne_zero_invNorm_budget

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
normBudget_of_leading_block_det_ne_zero_sqrt_budget

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
normBudget_of_triangular_leading_blocks_sqrt_budget

```

Lemma 367 (Full rank does not imply unpivoted leading-minor nonbreakdown). *There exists a real 2×2 matrix A with $\det A \neq 0$ but $A_{00} = 0$. Therefore a no-pivot Householder QR route cannot replace the current-pivot nonzero hypothesis by ordinary nonsingularity/full rank alone. Nor can it replace the per-pivot leading-block determinant hypotheses by whole-matrix nonsingularity: the first 1×1 unpivoted leading block of the same matrix has determinant zero. A final no-pivot theorem must therefore either assume no breakdown, use pivoting, or prove a stronger structured invariant that implies the displayed pivot and leading-minor conditions.*

Lean formalization.

```

Algorithms/QR/HouseholderQR.lean qrPivotCounterexample2_first_pivot_zero
qrPivotCounterexample2_det_ne_zero
not_forall_det_ne_zero_implies_first_pivot_ne_zero
qrPivotCounterexample2_first_leadingBlock_det_zero
not_forall_det_ne_zero_implies_all_leading_blocks_det_ne_zero

```

Corollary 368 (Structured current-pivot nonbreakdown from local leading determinants). *In the stored trailing Householder QR loop, let S_k denote the displayed $(k + 1) \times (k + 1)$ leading block of the stored matrix before pivot k . The stored-loop lower-zero theorem makes S_k upper triangular. If $\det S_k \neq 0$, then the current pivot $A_k(k, k)$ is nonzero. Consequently, the latest explicit compact QR least-squares certificate can replace its bare current-pivot nonzero hypothesis by the structured local condition*

$$\det S_k \neq 0 \quad (0 \leq k < n).$$

Under the same signed-alpha rule, dimensioned norm-square compact pivot margin, local κ_∞ hypotheses, dual/conditioning budget, and γ -validity assumptions as before, the computed back-substitution vector satisfies the same

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G^{\text{QR}}, c_g^{\text{QR}}).$$

Lean formalization.

```
Analysis/MatrixAlgebra.lean diag_ne_zero_of_upper_triangular_det_ne_zero
Algorithms/QR/HouseholderQR.lean      qrLeadingBlock_current_pivot_ne_zero_of_local_upper_
triangular_det_ne_zero
fl_householderStoredTrailingPanel_sequence_current_pivot_ne_zero_of_leadingBlock_det_ne_zero
Algorithms/LeastSquares/LSQRSolve.lean      LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
prefix_lower_zero_leadingBlock_det_ne_zero_kappaInf_selfNorm_normSqBudget
```

Corollary 369 (Structured QR certificate from local κ_∞ and dual budgets). *In the setting of the preceding corollary, the separate dimensioned norm-square compact pivot-margin hypothesis can be replaced by the following local conditioning budget. For every pivot $0 \leq k < n$, let S_k be the displayed local leading block. Assume*

$$\det S_k \neq 0, \quad \kappa_\infty(S_k, \text{nonsingInv}(S_k)) \leq \kappa_k,$$

$$(k+1) \left(\frac{\kappa_k}{\|S_k\|_\infty} \right)^2 \leq K_k, \quad mB_k^2 < \frac{1}{K_k},$$

where B_k is the compact componentwise diagonal-update budget used at pivot k . Under the same stored recurrences, signed-alpha rule, γ validity, and final explicit compact/final-radius definitions as before, the computed back-substitution vector satisfies

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G^{\text{QR}}, c_g^{\text{QR}}).$$

Lean formalization.

```
Algorithms/QR/HouseholderQR.lean      qrPrefixSupportSpannedByPreviousColumns_of_leadingBlock_
upper_det_ne_zero
Algorithms/LeastSquares/LSQRSolve.lean      LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
prefix_lower_zero_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget
```

Corollary 370 (Structured QR certificate from a direct inverse- ∞ budget). *In the same stored no-pivot QR setting, the local κ_∞ hypotheses in the preceding corollary can be replaced by a direct inverse-budget certificate. For each pivot $0 \leq k < n$, let S_k be the local leading block and assume*

$$\det S_k \neq 0, \quad (k+1) \|\text{nonsingInv}(S_k)\|_\infty^2 \leq K_k, \quad mB_k^2 < \frac{1}{K_k}.$$

Under the same stored recurrences, signed-alpha rule, γ -validity assumptions, and explicit compact/final-radius definitions, the computed back-substitution vector satisfies

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G^{\text{QR}}, c_g^{\text{QR}}).$$

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
prefix_lower_zero_leadingBlock_det_ne_zero_invNorm_dualBudget
```

Corollary 371 (Diagonal-dominant structured QR certificate). *In the setting of the preceding direct inverse-budget certificate, the displayed inverse- ∞ budget can itself be derived from Higham's triangular inverse bound. For each pivot $0 \leq k < n$, let S_k be the local leading block. Assume S_k is upper-triangular diagonally dominant, $\det S_k \neq 0$, and*

$$(k+1) \left(2^k \frac{1}{\min_{0 \leq r \leq k} |S_k(r, r)|} \right)^2 \leq K_k, \quad mB_k^2 < \frac{1}{K_k}.$$

Then, under the same stored recurrences, signed-alpha rule, γ -validity assumptions, and explicit compact/final-radius definitions, the computed back-substitution vector satisfies

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G^{\text{QR}}, c_g^{\text{QR}}).$$

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
prefix_lower_zero_diagDominant_leadingBlock_det_ne_zero_dualBudget
```

Corollary 372 (Concrete dual-budget diagonal-dominant QR certificate). *In the preceding diagonal-dominant certificate, define*

$$D_k := (k+1) \left(2^k \frac{1}{\min_{0 \leq r \leq k} |S_k(r, r)|} \right)^2.$$

Assume the same stored recurrences, signed-alpha rule, γ -validity conditions, local diagonal dominance, and $\det S_k \neq 0$. If for every pivot

$$mB_k^2 < \frac{1}{2D_k},$$

then the computed back-substitution vector satisfies the same

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G^{\text{QR}}, c_g^{\text{QR}}).$$

Lean formalization.

```
Algorithms/InverseBounds.lean diagDominantUpperInvBudgetExpr
diagDominantUpperInvBudgetExpr_pos
triInv_infNorm_sq_budget_of_diagDominantUpper_det_ne_zero_twice_budget
Algorithms/LeastSquares/LSQRSolve.lean      LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
prefix_lower_zero_diagDominant_leadingBlock_det_ne_zero_concreteDualBudget
```

Corollary 373 (Product-form concrete dual-budget QR certificate). *Under the hypotheses of the preceding corollary, it is enough to assume the dimensionless product inequality*

$$2D_k mB_k^2 < 1$$

for every pivot k . Then the same LSQRSolveBackwardError conclusion holds.

Lean formalization.

```
Algorithms/InverseBounds.lean mul_sq_lt_inv_two_mul_of_two_mul_mul_sq_lt_one
Algorithms/LeastSquares/LSQRSolve.lean LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
prefix_lower_zero_diagDominant_leadingBlock_det_ne_zero_concreteDualProductBudget
```

Corollary 374 (Stored-sequence product compact-smallness QR certificate). *Let c_{seq} be the deterministic stored QR compact sequence budget. In the product-form concrete dual-budget certificate, the raw local pivot compact budget B_k may be replaced by the computable upper bound*

$$c_{\text{seq}} \left\| (\hat{A}_k)_{:,k} \right\|_2.$$

Thus it is enough to assume, for every pivot k ,

$$2D_k m \left(c_{\text{seq}} \left\| (\hat{A}_k)_{:,k} \right\|_2 \right)^2 < 1.$$

Then the same LSQRSolveBackwardError conclusion holds.

Lean formalization.

```
Algorithms/QR/HouseholderQR.lean storedQRCompactPivotBudget_le_sequence_column_norm
Algorithms/InverseBounds.lean two_mul_mul_sq_lt_one_of_nonneg_le
Algorithms/LeastSquares/LSQRSolve.lean LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
prefix_lower_zero_diagDominant_leadingBlock_det_ne_zero_concreteDualProductSequenceBudget
```

Lemma 375 (Budget positivity alone does not imply compact smallness). *The product-form certificate cannot be closed from $D_k > 0$ alone. Even in the scalar case $D = 1$, $m = 1$, and $B = 1$, the product condition would read*

$$2 \cdot 1 \cdot (1 \cdot 1^2) < 1,$$

which is false. Thus a final QR theorem must either prove a real compact-update budget bound from the computed loop or keep that product smallness hypothesis visible.

Lean formalization.

```
Algorithms/InverseBounds.lean not_forall_pos_implies_two_mul_mul_sq_lt_one
```

Lemma 376 (Diagonal dominance does not imply compact smallness). *Even local diagonal dominance and the displayed Higham inverse budget D_k do not imply the product-form compact-smallness condition by themselves. In the scalar 1×1 case, take the local block $U = [1]$, so U is diagonally dominant and $D = 1$. With compact budget $B = 1$ and $m = 1$, the same product condition again reads*

$$2D m B^2 = 2 < 1,$$

which is false.

Lean formalization.

```
Algorithms/InverseBounds.lean not_forall_diagDominantUpper_implies_two_mul_budget_expr_mul_
sq_lt_one
```

Lemma 377 (Product compact smallness does not imply diagonal dominance). *Even adding the product compact-smallness inequality to triangular nonsingularity does not force diagonal dominance. For the upper-triangular matrix*

$$U = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix},$$

the displayed diagonal-dominant inverse budget D is finite. Choosing, for example, $B = 1/8$ and $m = 1$ makes

$$2DmB^2 < 1,$$

but U is still not diagonally dominant, since $2 \not\leq 1$. Thus the remaining QR assumptions cannot be collapsed into one another: diagonal dominance and compact-product smallness each need a real computed-loop or conditioning invariant, or else both must remain visible.

Lean formalization.

```
Algorithms/InverseBounds.lean    not_forall_upper_tri_det_ne_zero_product_budget_implies_
diagDominant
```

Lemma 378 (Finite conditioning alone does not imply compact smallness). *Upper-triangular nonsingularity and a finite local κ_∞ certificate also do not imply the product compact-smallness condition. For the same block*

$$U = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix},$$

choose the conditioning budget to be the exact formal value $\kappa_\infty(U)$. The conditioning hypothesis is then satisfied. But with compact budget $B = 1$ and $m = 1$, the displayed product inequality

$$2DmB^2 < 1$$

is false. Thus local conditioning information does not control the compact-update budget; that side condition still needs a computed-loop compactness invariant or must remain visible as a domain hypothesis.

Lean formalization.

```
Algorithms/InverseBounds.lean    not_forall_upper_tri_det_ne_zero_kappaInf_bound_implies_two_
mul_budget_expr_mul_sq_lt_one
```

Remark (Current rectangular QR route choice). After the preceding route eliminations, the remaining equation (8) QR/preconditioner gap is not another small adapter. The formalized counterexamples rule out deriving the visible assumptions from whole-matrix full rank, positive trailing norm, diagonal dominance alone, product compact-smallness alone, finite local κ_∞ , or any implication between diagonal dominance and product compact-smallness. The next theorem family must therefore be chosen explicitly: either prove a stronger computed-loop/off-diagonal-control invariant, switch to a Cox–Higham style pivoted or sorted row-wise weighted least-squares theorem, or keep the remaining nonbreakdown, conditioning, and compact-product hypotheses visible as domain assumptions.

Remark (Cox–Higham is not a drop-in closure). The Cox–Higham row-wise weighted least-squares route is deliberately not used to close the current unpivoted stored-QR theorem. Higham’s QR notes state the standard Householder result as a columnwise theorem, and then treat row-wise weighted least-squares stability as a separate question: in general the unpivoted answer is negative,

while row-wise bounds require column pivoting together with row pivoting or row sorting and the recommended sign convention. Cox–Higham prove precisely this pivoted/sorted row-wise theorem family. Consequently, switching to Cox–Higham would change the algorithmic hypotheses; it is not a small adapter for the current equation (8) implementation-backed theorem. For the present unpivoted route, the remaining honest alternatives are to prove a stronger computed-loop/off-diagonal-control invariant or to keep the local nonbreakdown, conditioning, diagonal-dominance, and compact-product assumptions visible.

Lemma 379 (Triangular nonsingularity and finite conditioning alone do not imply diagonal dominance). *The diagonal-dominant branch above cannot be justified from upper-triangular shape, nonzero diagonal entries, determinant nonzeroness, or the mere existence of a finite κ_∞ certificate. In Lean, the concrete matrix*

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

is upper triangular, has nonzero diagonal entries, and has determinant nonzero, but it is not diagonally dominant in the repository’s sense because $2 \not\leq 1$. Taking the conditioning budget to be the matrix’s own local κ_∞ value gives a formal finite conditioning certificate for the same counterexample. Therefore, any final theorem using the diagonal-dominant branch must either prove diagonal dominance from a stronger computed-loop or conditioning invariant or keep it as a visible domain hypothesis.

Lean formalization.

```
Algorithms/TriangularForwardBound.lean diagDominanceCounterexample2_det_ne_zero
not_forall_upper_tri_diag_nonzero_implies_diagDominant
not_forall_upper_tri_det_ne_zero_implies_diagDominant
exists_upper_tri_det_ne_zero_kappaInf_bound_not_diagDominant
not_forall_upper_tri_det_ne_zero_kappaInf_bound_implies_diagDominant
```

Lemma 380 (Exact QR shape alone does not imply diagonal dominance). *The obstruction above persists even if the matrix is already written in exact QR form. There exist matrices $A, Q, R \in \mathbb{R}^{2 \times 2}$ such that*

$$A = QR, \quad Q \text{ is orthogonal}, \quad R \text{ is upper triangular with nonzero diagonal},$$

but R is not diagonally dominant. In Lean the witness is

$$Q = I, \quad R = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Thus a final theorem cannot obtain the diagonal-dominance hypothesis from the final QR shape alone. The positive unpivoted route must either prove a stronger computed-loop/off-diagonal-control invariant or keep diagonal dominance as a visible domain assumption.

Lean formalization.

```
Algorithms/TriangularForwardBound.lean exists_orthogonal_upper_factorization_not_
diagDominant
not_forall_orthogonal_upper_factorization_implies_diagDominant
```

Lemma 381 (Exact no-pivot Householder QR need not give diagonal dominance). *The standard exact trailing Householder recurrence itself does not supply the diagonal-dominance hypothesis. There is a two-step unpivoted Householder QR sequence $A_0, A_1, A_2 \in \mathbb{R}^{2 \times 2}$ with signed Householder scalars α_0, α_1 such that, for $k = 0, 1$,*

$$A_{k+1} = H_k A_k, \quad \alpha_k^2 = \|A_k(k:2, k)\|_2^2, \quad v_k^T v_k \neq 0,$$

where H_k is the exact trailing Householder reflector generated from the k -th active column. The concrete Lean witness starts from

$$A_0 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

and reaches

$$A_2 = \begin{pmatrix} -1 & -2 \\ 0 & -1 \end{pmatrix}.$$

The final matrix is upper triangular and has nonzero diagonal, but it is not diagonally dominant because $|-2| > |-1|$. Thus diagonal dominance is not a generic consequence of the exact unpivoted Householder loop; a final theorem must add a genuine off-diagonal-control/pivoting invariant or keep diagonal dominance visible. The strengthened universal theorem says the same counterexample rules out deriving diagonal dominance together with any additional final-block property, such as the scalar finite-max smallness side condition, from the exact no-pivot recurrence alone.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean exactHouseholderQRDiagDominanceCounterexample_step
exactHouseholderQRDiagDominanceCounterexample_alpha_sq
exactHouseholderQRDiagDominanceCounterexample_den_ne_zero
exactHouseholderQRDiagDominanceCounterexample_FP
exactHouseholderQRDiagDominanceCounterexample_stored_step
exactHouseholderQRDiagDominanceCounterexample_signed_alpha_def
exactHouseholderQRDiagDominanceCounterexample_not_diagDominant
not_forall_exact_trailing_householder_sequence_implies_diagDominant
not_forall_exact_trailing_householder_sequence_implies_diagDominant_and_property
not_forall_signedAlphaDef_stored_trailing_sequence_actualU_sourceDen_implies_diagDominant
exactHouseholderQRDiagDominanceCounterexample_not_activeMaxPivotChoice
not_forall_signedAlphaDef_stored_trailing_sequence_actualU_sourceDen_implies_activeMaxPivotChoice
```

Lemma 382 (Active-block nonbreakdown does not imply off-diagonal control). *Even the stronger local data now available for the Cox–Higham raw-stage route do not imply the source-shaped off-diagonal domination field. There is a two-by-two displayed leading-block sequence such that, at each stage, the leading block is upper triangular, the leading-block determinant is nonzero, and the active-block mass is positive:*

$$\det S_k \neq 0, \quad \mathcal{M}_k(A_k) > 0, \quad k = 0, 1.$$

Nevertheless the row-wise domination condition

$$|(S_k)_{ij}| \leq |(S_k)_{ii}| \quad (i < j)$$

fails. The Lean witness again uses

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix},$$

whose first row has $2 > 1$. Therefore the rank/determinant active-block-mass bridge above cannot be used as a hidden substitute for the off-diagonal-control hypothesis required by *StoredQRSourceOffDiagonalControl*.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean not_forall_leadingBlock_upper_det_activeBlockPos_
implies_offdiag_le_diag
```

Lemma 383 (Row-growth budgets do not imply diagonal lower bounds). *The diagonal lower-bound half of the Cox–Higham row-budget split is also independent. There is a two-by-two displayed leading-block sequence and a row budget $R_k = 2$ such that, at each stage, the leading block is upper triangular, the leading-block determinant is nonzero, the active-block mass is positive, and every strict upper entry satisfies the row-growth budget*

$$|(S_k)_{ij}| \leq R_k \quad (i < j).$$

Nevertheless the matching diagonal lower-bound field

$$R_k \leq |(S_k)_{ii}| \quad (i < k)$$

fails. For the witness

$$S_1 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad R_1 = 2,$$

the strict upper entry is exactly bounded by R_1 , while the first displayed diagonal has magnitude 1. Hence $2 \leq 1$ would be required. The strengthened active-pivot variant shows that this obstruction persists even if the sequence also satisfies the active max-pivot policy. It uses

$$S_0 = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}, \quad S_1 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix},$$

so the first stage has the displayed active column maximal, while the second stage still violates the row-budget diagonal lower-bound field. The active-block-budget strengthening adds that the same row budget bounds every active trailing-block entry at each displayed stage:

$$|(A_k)_{r\ell}| \leq R_k \quad (k \leq r, k \leq \ell).$$

The same two-stage witness satisfies this extra active-block magnitude condition with $R_k = 2$, but still requires $2 \leq 1$ in the displayed diagonal lower-bound field. Finally, the meta strengthening adds an arbitrary auxiliary side property $P(\mathbf{A}_\bullet, \mathbf{R}_\bullet)$. Instantiating P with the always-true property shows that no universal theorem of this shape can repair the diagonal lower-bound failure by appending unrelated scalar or product conditions.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean not_forall_leadingBlock_upper_det_activeBlockPos_
offdiagBudget_implies_rowBudget_diag
activeMaxPivotRowBudgetDiagCounterexampleA0
```

```

activeMaxPivotRowBudgetDiagCounterexampleSeq

not_forall_leadingBlock_upper_det_activeBlockPos_activeMaxPivot_offdiagBudget_implies_rowBudget_
diag

not_forall_leadingBlock_upper_det_activeBlockPos_activeMaxPivot_activeBlockBudget_offdiagBudget_
implies_rowBudget_diag

not_forall_leadingBlock_upper_det_activeBlockPos_activeMaxPivot_activeBlockBudget_offdiagBudget_
property_implies_rowBudget_diag

```

Remark (Route choice after the exact-recurrence obstruction). The exact-recurrence counterexample turns the remaining equation (8) QR/preconditioner gap into a theorem-scope choice rather than another local adapter. For the current unpivoted stored-QR theorem, the false generic routes have been ruled out from full rank, positive trailing norm, finite local conditioning, diagonal dominance/product smallness alone, final exact QR shape, the standard exact no-pivot Householder recurrence, and positive active-block mass plus local leading-block nonsingularity. The row-budget split obstruction above also rules out deriving the required diagonal lower-bound field from a row-growth upper budget alone. A final positive theorem must therefore choose one of the following mathematical statements: add and prove a stronger computed-loop/off-diagonal-control and diagonal nonbreakdown invariant, change the algorithmic theorem family to a pivoted/sorted row-wise route, or keep the remaining nonbreakdown, conditioning, diagonal-dominance, off-diagonal-control, diagonal lower-bound, and compact-product assumptions visible as domain hypotheses.

Corollary 384 (QR solve certificate from an off-diagonal-control invariant). *Fix $n \leq m$, a floating-point model, and the stored trailing Householder recurrences for $\hat{A}_k, \hat{b}_k$ with the signed Householder choice α_k . Suppose that the loop satisfies the invariant $\mathcal{I}_{\text{off}}$ consisting, for every $k < n$, of:*

$$\det S_k \neq 0, \quad S_k \text{ is upper-triangular diagonally dominant,}$$

where S_k is the $(k+1) \times (k+1)$ displayed leading block of $\hat{A}_k$, and

$$2D_k m (B \|\hat{A}_k(:, k)\|_2)^2 < 1.$$

Here D_k is the formal Higham diagonal-dominant triangular inverse budget for S_k , and B is the repository's deterministic stored QR compact sequence budget for the whole loop. Then the computed triangular solve

$$\hat{x} = \text{fl_backSub}(\hat{R}, \hat{c})$$

satisfies

$$\text{LSQRSolveBackwardError}(n, A^T A, A^T b, \hat{x}, c_G(A, \hat{R}, B), c_g(A, b, \hat{R}, B)),$$

with $c_G = \text{qrSolveFinalGramBudget}$ and $c_g = \text{qrSolveFinalRhsBudget}$.

Lean formalization.

```

Algorithms/LeastSquares/LSQRSolve.lean StoredQROffDiagonalControlInvariant

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_offDiagonalControl

```

Corollary 385 (Finite-max diagonal-dominant route to the packaged invariant). *In the setting above, suppose every displayed leading block S_k is upper-triangular diagonally dominant. Let D_* be the canonical maximum of the diagonal-dominant inverse factors, let N_* be the canonical maximum of the pivot-column norms, and let B be the stored QR compact sequence relative budget. These*

three quantities are the repository definitions listed in the formalization block below. If the scalar finite-max smallness condition

$$2D_* m (BN_*)^2 < 1$$

holds, then the whole stored QR loop satisfies *StoredQROffDiagonalControlInvariant*. Consequently, the preceding QR solve certificate applies with the repository perturbation radii *qrSolveFinalGramBudget* and *qrSolveFinalRhsBudget*.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      StoredQROffDiagonalControlInvariant.of_
diagDominant_finiteMaxSmallness

storedQRCompactSequenceProductBudget_lt_one_of_diagDominant_finite_max_smallness

storedQRDiagDominantInvFactorBudget

storedQRPivotColumnNormBudget

storedQRCompactSequenceRelativeBudget

det_ne_zero_of_diagDominantUpper
```

Corollary 386 (Source-shaped off-diagonal control implies the QR certificate). *In the setting of the preceding corollary, it is enough to verify the following more primitive local control data. For every $k < n$, the displayed leading block S_k is upper triangular, its displayed diagonal is nonzero, and its off-diagonal entries in each row are dominated by the row diagonal:*

$$(S_k)_{ij} = 0 \quad (j < i), \quad (S_k)_{ii} \neq 0, \quad |(S_k)_{ij}| \leq |(S_k)_{ii}| \quad (i < j).$$

Assume also the same stored-sequence compact-product condition

$$2D_k m (B \|\hat{A}_k(:, k)\|_2)^2 < 1.$$

Then the stored QR loop satisfies $\mathcal{I}_{\text{off}}$, and therefore the same computed triangular solve satisfies the *LSQRSolveBackwardError* certificate with *qrSolveFinalGramBudget* and *qrSolveFinalRhsBudget*.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      StoredQRSourceOffDiagonalControl
StoredQROffDiagonalControlInvariant.of_sourceOffDiagonalControl

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_sourceOffDiagonalControl
```

Corollary 387 (Stored recurrence supplies the triangular source field). *In the same stored trailing Householder recurrence, the lower-zero shape theorem for completed columns implies the upper-triangular leading-block equalities*

$$(S_k)_{ij} = 0 \quad (j < i)$$

at every displayed leading block S_k . Therefore, to instantiate the source-shaped off-diagonal-control certificate above, it is enough to supply only the remaining three source-specific obligations:

$$(S_k)_{ii} \neq 0, \quad |(S_k)_{ij}| \leq |(S_k)_{ii}| \quad (i < j), \quad 2D_k m (B \|\hat{A}_k(:, k)\|_2)^2 < 1.$$

Under these three obligations, the same computed triangular solve satisfies the *LSQRSolveBackwardError* certificate with *qrSolveFinalGramBudget* and *qrSolveFinalRhsBudget*.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_diag_offdiag_product

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_diag_offdiag_product
```

Corollary 388 (Stored recurrence supplies previous diagonal nonzeros). *Assume the same stored trailing Householder recurrence and signed-alpha rule. Assume also the square-root nonbreakdown budget*

$$B_k^{\text{diag}} < \sqrt{\|A_k(k : m, k)\|_2^2}, \quad k < n,$$

where B_k^{diag} is the local compact component budget for the displayed pivot entry. If the current pivot entries $A_k(k, k)$ are nonzero, and if the row-wise off-diagonal domination and compact-product obligations of the source-shaped route hold, then

StoredQRSourceOffDiagonalControl

holds. Consequently the corresponding computed triangular solve satisfies the same *LSQRSolveBackwardError* certificate with *qrSolveFinalGramBudget* and *qrSolveFinalRhsBudget*.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_pivot_sqrtBudget_offdiag_product

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_pivot_sqrtBudget_offdiag_product
```

Corollary 389 (Stored recurrence supplies current pivots from leading determinants). *Assume the same stored trailing Householder recurrence, signed-alpha rule, and square-root nonbreakdown budget as in the preceding corollary. Replace the raw current-pivot condition $A_k(k, k) \neq 0$ by the structured local leading-block condition*

$$\det S_k \neq 0, \quad k < n,$$

where S_k is the displayed $(k+1) \times (k+1)$ leading block of A_k . If the row-wise off-diagonal domination and compact-product obligations of the source-shaped route also hold, then

StoredQRSourceOffDiagonalControl

holds, and the corresponding computed triangular solve satisfies the same *LSQRSolveBackwardError* certificate with *qrSolveFinalGramBudget* and *qrSolveFinalRhsBudget*.

Lean formalization.

```
Algorithms/QR/HouseholderQR.lean fl_householderStoredTrailingPanel_sequence_current_pivot_
ne_zero_of_leadingBlock_det_ne_zero

Algorithms/LeastSquares/LSQRSolve.lean      StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_leadingBlock_det_ne_zero_sqrtBudget_offdiag_product

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_sqrtBudget_offdiag_product
```

Corollary 390 (Norm-square nonbreakdown form of the source-shaped QR route). *In the preceding corollary, the square-root nonbreakdown budget may be replaced by the dimensioned trailing-norm-square margin*

$$m(B_k^{\text{diag}})^2 < \|A_k(k:m, k)\|_2^2, \quad k < n,$$

where B_k^{diag} is the same local compact component budget for the displayed pivot entry. Under the same local leading-block nonsingularity, row-wise off-diagonal domination, and compact-product hypotheses, the stored QR loop again satisfies

StoredQRSourceOffDiagonalControl,

and the corresponding computed triangular solve satisfies the same *LSQRSolveBackwardError* certificate with *qrSolveFinalGramBudget* and *qrSolveFinalRhsBudget*.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean  budget_lt_sqrt_householderTrailingNorm2Sq_of_dim_mul_
budget_sq_lt_trailingNorm2Sq
Algorithms/LeastSquares/LSQRSolve.lean  StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_leadingBlock_det_ne_zero_normSqBudget_offdiag_product
LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_offdiag_product
```

Corollary 391 (Row-budget form of the source-shaped QR route). *Keep the determinant, norm-square nonbreakdown, and compact-product hypotheses of the preceding corollary. Instead of assuming row-wise off-diagonal domination directly, suppose that for every displayed leading block S_k there are row budgets $R_{k,i}$ such that*

$$|(S_k)_{ij}| \leq R_{k,i} \leq |(S_k)_{ii}|, \quad 0 \leq i < j \leq k < n.$$

*Then S_k satisfies the row-wise off-diagonal domination field required by *StoredQRSourceOffDiagonalControl*, and the same computed triangular solve satisfies the local *LSQRSolveBackwardError* certificate.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean  StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_leadingBlock_det_ne_zero_normSqBudget_rowBudget_product
LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_rowBudget_product
StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_rowBudget_product_of_offdiag_rows
LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_rowBudget_product_of_
offdiag_rows
```

Corollary 392 (Packaged displayed row-budget control). *In the preceding row-budget setting, suppose the two row-budget obligations are provided as a single certificate:*

$$\mathcal{R}(A, R) = [|(S_k)_{ij}| \leq R_{k,i} \text{ for all } i < j, \quad R_{k,i} \leq |(S_k)_{ii}| \text{ for all } i < k].$$

Then the stored QR source-control theorem and the solver-level backward-error certificate follow from this packaged assumption, together with the same leading-block nonsingularity, norm-square nonbreakdown, and compact-product smallness hypotheses as above.

Lean formalization.

```

Algorithms/LeastSquares/LSQRSolve.lean      StoredQRDisplayedRowBudgetControl
StoredQRDisplayedRowBudgetControl.of_signed_stage_budgets_factor_of_normSqBudget

StoredQRDisplayedRowBudgetControl.of_signed_stage_uniformBudget_factor_of_normSqBudget

StoredQRDisplayedRowBudgetControl.of_signed_stage_uniformBudget_globalCompactBudget_of_
normSqBudget

StoredQRDisplayedRowBudgetControl.of_signed_stage_uniformBudget_globalCompactBudget_kappaInf_
dualBudget

StoredQRDisplayedRowBudgetControl.of_signed_stage_uniformBudget_globalCompactBudget_
activeMaxPivot_of_normSqBudget

StoredQRDisplayedRowBudgetControl.of_signed_stage_uniformBudget_globalCompactBudget_
activeMaxPivot_kappaInf_dualBudget

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_rowBudgetControl_product

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_rowBudgetControl_
product

```

Corollary 393 (Canonical row-max row-budget certificate). *Assume the stronger source-shaped off-diagonal-control field is available: for every displayed leading block S_k ,*

$$|(S_k)_{ij}| \leq |(S_k)_{ii}|, \quad i < j \leq k.$$

For each row $i < k$, define the canonical row budget

$$R_{k,i}^{\max} = \max_{i < j \leq k} |(S_k)_{ij}|.$$

Then the packaged displayed row-budget certificate holds with $R_{k,i} = R_{k,i}^{\max}$.

Lean formalization.

```

Algorithms/LeastSquares/LSQRSolve.lean      qrLeadingStrictUpperRowMaxBudget
qrLeadingStrictUpperRowMaxBudget_entry_le qrLeadingStrictUpperRowMaxBudget_le_diag_of_offdiag
StoredQRDisplayedRowBudgetControl.of_sourceOffDiagonalControl_rowMaxBudget

```

Corollary 394 (Route-1 row-max contraction handoff). *Let S_k be the displayed $(k+1) \times (k+1)$ leading block of the stored QR data at stage k , and let*

$$R_{k,i}^{\max} = \max_{i < j \leq k} |(S_k)_{ij}| \quad (i < k)$$

be the canonical strict-upper row maximum. Suppose a computed-loop invariant supplies one scalar $\rho \leq 1$ such that

$$R_{k,i}^{\max} \leq \rho |(S_k)_{ii}| \quad \text{for all } i < k < n.$$

Then the packaged displayed row-budget certificate holds with $R_{k,i} = R_{k,i}^{\max}$. Consequently, under the same stored recurrence, leading-block nonsingularity, norm-square nonbreakdown, and finite global compact-product hypotheses used in the row-budget route, the `StoredQRSourceOffDiagonalControl` certificate and the local `LSQRSolveBackwardError` theorem follow.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean StoredQRDisplayedRowBudgetControl.of_rowMaxBudget_
le_diag_factor

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_rowMaxBudgetFactor_globalProduct

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_rowMaxBudgetFactor_
globalProduct
```

Corollary 395 (Scalar row-max defect handoff). *With the notation of the previous corollary, define the finite scalar defect*

$$\Delta_{\max} = \max_{0 \leq k < n} \max_{0 \leq i < k} (R_{k,i}^{\max} - |(S_k)_{ii}|),$$

with value 0 if the index set is empty. If

$$\Delta_{\max} \leq 0,$$

then the packaged displayed row-budget certificate holds with $R_{k,i} = R_{k,i}^{\max}$. Under the same stored recurrence, leading-block nonsingularity, norm-square nonbreakdown, and finite global compact-product hypotheses, the source-control certificate and the local QR solve backward-error theorem follow.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean storedQRRowMaxDiagDefectBudget
storedQRRowMaxDiagDefect_le_budget StoredQRDisplayedRowBudgetControl.of_
rowMaxDiagDefectBudget_nonpos

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_rowMaxDiagDefect_globalProduct

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_rowMaxDiagDefect_
globalProduct
```

Corollary 396 (Actual-unit-roundoff validity for scalar row-max-defect route). *The scalar row-max-defect/global-product QR certificate can be stated without an abstract `gammaValid` hypothesis. The local solver theorem derives both `gammaValid fp m` and `gammaValid fp n` from the displayed actual-unit-roundoff guard*

$$m \text{ fp}.u < 1,$$

and the probability-level equation (8) theorem derives `gammaValid fp s` and `gammaValid fp n` from

$$s \text{ fp}.u < 1.$$

The determinant, norm-square nonbreakdown, scalar row-defect, and global compact-product hypotheses remain visible.

Remark. This is a validity-surface reduction only. It does not prove the scalar row-defect condition, local determinant or nonbreakdown facts, the global compact-product smallness condition, or the generic implementation-backed rectangular QR/preconditioner theorem.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
leadingBlock_det_ne_zero_normSqBudget_rowMaxDiagDefect_globalProduct_of_actualUnitRoundoff_no_
gammaValid

Algorithms/RandNLA/LeastSquaresSketch.lean    leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_rowMaxDiagDefect_globalProduct_solver_of_actualUnitRoundoff_no_gammaValid
```

Corollary 397 (Finite stage-diagonal lower-bound defect). *For the active/prefix Cox–Higham QR route, let B_k be the monotone stage budget and let S_k be the displayed $(k+1) \times (k+1)$ leading block. Define the finite scalar defect*

$$\Delta_{\text{stage}} = \max_{0 \leq k < n} \max_{0 \leq i < k} (B_k - |(S_k)_{ii}|),$$

with value 0 if the index set is empty. If

$$\Delta_{\text{stage}} \leq 0,$$

then every displayed off-diagonal row satisfies the diagonal lower-bound condition

$$B_k \leq |(S_k)_{ii}| \quad (i < k).$$

Conversely, if these displayed pointwise lower bounds hold for every off-diagonal row $i < k$, then the same finite maximum satisfies $\Delta_{\text{stage}} \leq 0$. Thus the scalar defect is now formalized as an exact finite package for the displayed stage-diagonal lower-bound family. Consequently the active/prefix global compact-step and finite global-product QR source-control theorem, and its local QR solve wrapper, can consume the single scalar condition $\Delta_{\text{stage}} \leq 0$ instead of the full pointwise diagonal-lower-bound family.

Remark. This is a bottleneck dependency reduction, not a proof of diagonal growth. The scalar condition $\Delta_{\text{stage}} \leq 0$ still has to be proved from a concrete pivoted/sorted/off-diagonal-controlled loop, or retained as a visible domain assumption.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      storedQRStageDiagLowerDefectBudget
storedQRStageDiagLowerDefect_le_budget      storedQRStageBudget_le_diag_of_
stageDiagLowerDefectBudget_nonpos

storedQRStageDiagLowerDefectBudget_nonpos_of_stageBudget_le_diag

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_signedStageUniformBudget_product_of_normSqBudget_activePrefix_activeBlockRecurrence_
globalCompactBudget_completedColumns_globalProduct_stageDiagDefect_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_
signedStageUniformBudget_product_of_normSqBudget_activePrefix_activeBlockRecurrence_
globalCompactBudget_completedColumns_globalProduct_stageDiagDefect_offdiag_rows

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_kappaInf_
selfNorm_dualBudget_signedStageUniformBudget_globalCompactBudget_completedColumns_globalProduct_
stageDiagDefect_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget_
signedStageUniformBudget_globalCompactBudget_completedColumns_globalProduct_stageDiagDefect_
offdiag_rows
```

Corollary 398 (Row-max bridge to stage-diagonal scalar nonpositivity). *Suppose the row-max scalar defect satisfies $\Delta_{\max} \leq 0$, and suppose the uniform stage budget is bounded by the displayed strict-upper row maximum:*

$$B_k \leq R_{k,i}^{\max} \quad (i < k).$$

Then the stage-diagonal scalar defect also satisfies

$$\Delta_{\text{stage}} \leq 0.$$

In particular, local displayed diagonal dominance implies the same conclusion under the same $B_k \leq R_{k,i}^{\max}$ comparison.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean storedQRStageDiagLowerDefectBudget_nonpos_of_
rowMaxDiagDefectBudget_nonpos_stageBudget_le_rowMax
storedQRStageDiagLowerDefectBudget_nonpos_of_diagDominant_stageBudget_le_rowMax
```

Corollary 399 (Finite stage-budget/row-max comparison defect). *For the same displayed off-diagonal rows $i < k$, define*

$$\Delta_{\text{cmp}} = \max_{0 \leq k < n} \max_{0 \leq i < k} (B_k - R_{k,i}^{\max}),$$

again with value 0 if the index set is empty. Nonpositivity of this scalar defect is equivalent, as a finite package, to the displayed comparison

$$B_k \leq R_{k,i}^{\max} \quad (i < k).$$

Consequently the row-max bridge can be stated using two scalar conditions:

$$\Delta_{\max} \leq 0, \quad \Delta_{\text{cmp}} \leq 0.$$

These imply the scalar stage-diagonal condition

$$\Delta_{\text{stage}} \leq 0.$$

Remark. This packages a listed dependency; it does not prove the comparison defect from the concrete pivoted/sorted QR loop. That scalar condition remains a visible invariant unless supplied by a stronger row-growth or ordering proof.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean storedQRStageRowMaxComparisonDefectBudget
storedQRStageRowMaxComparisonDefect_le_budget storedQRStageBudget_le_rowMax_of_
stageRowMaxComparisonDefectBudget_nonpos
storedQRStageRowMaxComparisonDefectBudget_nonpos_of_stageBudget_le_rowMax
storedQRStageDiagLowerDefectBudget_nonpos_of_rowMaxDiagDefectBudget_nonpos_
stageRowMaxComparisonDefectBudget_nonpos
storedQRStageDiagLowerDefectBudget_nonpos_of_diagDominant_stageRowMaxComparisonDefectBudget_
nonpos
```

Corollary 400 (Scalar-comparison active-pivot row-max QR surface). *The active-pivot row-max source-control, solver, and sampled equation (8) wrappers can now be stated using the two scalar row-max conditions*

$$\Delta_{\max} \leq 0, \quad \Delta_{\text{cmp}} \leq 0,$$

instead of the displayed comparison family $B_k \leq R_{k,i}^{\max}$. Internally, the proofs extract that family from $\Delta_{\text{cmp}} \leq 0$ and then reuse the already-formalized visible row-max active-pivot theorems. The explicit-validity horizon sampled sibling keeps the sampled γ -validity fields visible, but it calls the horizon row-max probability wrapper after the same finite comparison extraction, so the two-scalar surface no longer exposes samplewise global budget monotonicity. The actual-unit horizon sampled sibling derives those validity fields from $(s : \mathbb{R})u < 1$ and then calls the explicit-validity horizon wrapper, so it also removes sampled validity fields.

Remark. This is a theorem-interface reduction, not a proof of either scalar defect from a concrete pivoted or sorted loop. The remaining route still has to establish $\Delta_{\max} \leq 0$, $\Delta_{\text{cmp}} \leq 0$, determinant/conditioning fields, and compact-product smallness, or retain them as explicit assumptions.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean Algorithms/RandNLA/LeastSquaresSketch.lean
StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_kappaInf_
selfNorm_dualBudget_signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_
globalProduct_rowMaxDiagDefect_stageRowMaxComparisonDefect_offdiag_rows

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_kappaInf_
selfNorm_dualBudget_signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_
globalProduct_rowMaxDiagDefect_stageRowMaxComparisonDefect_offdiag_rows_of_actualUnitRoundoff_no_
gammaValid

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget_
signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_globalProduct_
rowMaxDiagDefect_stageRowMaxComparisonDefect_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget_
signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_globalProduct_
rowMaxDiagDefect_stageRowMaxComparisonDefect_offdiag_rows_of_actualUnitRoundoff_no_gammaValid

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_activeMaxPivot_
rowMaxDiagDefect_stageRowMaxComparisonDefect_kappaInf_dualBudget_solver

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_activeMaxPivot_
rowMaxDiagDefect_stageRowMaxComparisonDefect_kappaInf_dualBudget_solver_of_horizonBudget

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_activeMaxPivot_
rowMaxDiagDefect_stageRowMaxComparisonDefect_kappaInf_dualBudget_solver_of_actualUnitRoundoff_no_
gammaValid

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_activeMaxPivot_
rowMaxDiagDefect_stageRowMaxComparisonDefect_kappaInf_dualBudget_solver_of_actualUnitRoundoff_no_
gammaValid_of_horizonBudget
```

Corollary 401 (Diagonal-dominant scalar-comparison active-pivot surface). *For the local active-pivot source-control and solver routes, local displayed diagonal dominance can replace the separate leading-block determinant and row-max-defect fields. The proof first obtains leading-block nonsingularity from diagonal dominance, and then combines diagonal dominance with $\Delta_{\text{cmp}} \leq 0$ to obtain the scalar stage-diagonal condition.*

Remark. This is a local theorem-surface reduction. It does not prove local diagonal dominance, the scalar comparison defect, conditioning or dual compact-budget fields, active-pivot policy, compact-product smallness, or the final generic QR/preconditioner theorem from a concrete computed loop.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_diagDominant_kappaInf_selfNorm_dualBudget_signedStageUniformBudget_
globalCompactBudget_activeMaxPivot_completedColumns_globalProduct_stageRowMaxComparisonDefect_
offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_
bound_explicitCompactBudget_of_signed_alpha_diagDominant_kappaInf_selfNorm_dualBudget_
signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_globalProduct_
stageRowMaxComparisonDefect_offdiag_rows
```

Corollary 402 (Probability-level diagonal-dominant scalar-comparison surface). *The sampled equation (8) active-pivot theorem has the same diagonal-dominant scalar-comparison interface. For each sampled trace, local displayed diagonal dominance supplies the leading-block determinant and row-max scalar-defect fields, while the scalar comparison condition $\Delta_{\text{cmp}} \leq 0$ supplies the displayed stage-budget/row-max comparison through the finite-max extractor. The theorem then reuses the scalar-comparison active-pivot probability wrapper.*

Remark. This lifts the local dependency reduction to the high-probability rounded sampled-row objective theorem. The remaining visible obligations are unchanged: local diagonal dominance, the scalar comparison defect, conditioning and dual compact-budget data, active-pivot policy, compact-product smallness, and the final concrete QR/preconditioner theorem.

Lean formalization.

```
Algorithms/RandNLA/LeastSquaresSketch.lean    leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_activePrefix_globalProduct_activeMaxPivot_diagDominant_stageRowMaxComparisonDefect_kappaInf_
dualBudget_solver
```

Lemma 403 (Diagonal dominance does not force the scalar comparison defect). *Local displayed diagonal dominance alone does not imply*

$$\Delta_{\text{cmp}} \leq 0.$$

There is a checked two-by-two witness whose displayed leading blocks are diagonally dominant, but whose uniform stage budget still exceeds the displayed strict-upper row maximum.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      activeMaxPivotRowMaxComparisonCounterexample_
diagDominant

not_forall_diagDominant_implies_stageRowMaxComparisonDefectBudget_nonpos
```

Lemma 404 (Product smallness does not rescue diagonal dominance). *Adding the compact-product smallness inequality to local displayed diagonal dominance still does not imply*

$$\Delta_{\text{cmp}} \leq 0.$$

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean    activeMaxPivotRowMaxComparisonCounterexample_
productBudget
not_forall_diagDominant_product_budget_implies_stageRowMaxComparisonDefectBudget_nonpos
```

Lemma 405 (Active-pivot budget data does not rescue the comparison scalar). *Even adding the active-pivot budget surface to local displayed diagonal dominance and compact-product smallness still does not imply*

$$\Delta_{\text{cmp}} \leq 0.$$

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean    not_forall_diagDominant_product_activeBlockPos_
activeMaxPivot_activeBlockBudget_offdiagBudget_implies_stageRowMaxComparisonDefectBudget_nonpos
```

Corollary 406 (Finite-max product handoff for the active diagonal-dominant branch). *On the active-pivot diagonal-dominant scalar-comparison route, the raw product hypothesis*

$$\Pi_{\text{max}} < 1$$

may be replaced by the canonical finite-max smallness scalar involving the diagonal-dominant inverse-factor budget, the compact-sequence relative budget, and the pivot-column norm budget.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean    StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_diagDominant_kappaInf_selfNorm_dualBudget_signedStageUniformBudget_
globalCompactBudget_activeMaxPivot_completedColumns_finiteMaxSmallness_
stageRowMaxComparisonDefect_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_
bound_explicitCompactBudget_of_signed_alpha_diagDominant_kappaInf_selfNorm_dualBudget_
signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_finiteMaxSmallness_
stageRowMaxComparisonDefect_offdiag_rows

Algorithms/RandNLA/LeastSquaresSketch.lean    leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_activePrefix_finiteMaxSmallness_activeMaxPivot_diagDominant_stageRowMaxComparisonDefect_
kappaInf_dualBudget_solver
```

Corollary 407 (Concrete-dual finite-max handoff for the active diagonal-dominant branch). *On the active-pivot diagonal-dominant scalar-comparison route, the auxiliary κ_k , K_k , and dual compact-budget hypotheses can be removed from the finite-max theorem surface. It is enough to supply local diagonal dominance, the signed-stage recurrence budget, active-pivot policy, the scalar comparison defect, and the canonical finite-max product-smallness scalar.*

Lean formalization.

```

Algorithms/LeastSquares/LSQRSolve.lean  storedQRSignedStage_normSqBudget_of_leadingBlock_
det_ne_zero_diagDominant_concreteDualProductSequenceBudget

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_diagDominant_
signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_finiteMaxSmallness_
stageRowMaxComparisonDefect_concreteDual_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_diagDominant_signedStageUniformBudget_globalCompactBudget_
activeMaxPivot_completedColumns_finiteMaxSmallness_stageRowMaxComparisonDefect_concreteDual_
offdiag_rows

Algorithms/RandNLA/LeastSquaresSketch.lean      leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_activePrefix_finiteMaxSmallness_activeMaxPivot_diagDominant_stageRowMaxComparisonDefect_
concreteDual_solver

```

Corollary 408 (Actual-unit validity for the active concrete-dual finite-max handoff). *Let $n \leq m$. On the local active concrete-dual finite-max QR surface, assume the computed stored-QR data*

$$\hat{A}_{k+1} = \mathfrak{f}_{\text{panel}}(\hat{A}_k, \alpha_k), \quad \alpha_k = \alpha_{\text{sign}} \left(\sqrt{\|(\hat{A}_k)_{k:m,k}\|_2^2}, (\hat{A}_k)_{kk} \right)$$

for $k < n$, local diagonal dominance of every leading block

$$\text{IsDiagDominantUpper}((\hat{A}_k)_{0:k,0:k}),$$

the initial displayed-entry and full-entry bounds by B_0 , the signed global recurrence

$$\rho_m B_t + \text{storedQRSignedStageGlobalCompactBudget}(fp, \hat{A}, \alpha, t) \leq B_{t+1},$$

nonnegativity and monotonicity of B_t , the active-pivot policy

$$k = \text{householderActiveMaxPivotColumn}(\hat{A}_k; k),$$

the scalar comparison defect

$$\text{storedQRStageRowMaxComparisonDefectBudget}(\hat{A}, B) \leq 0,$$

and the finite-max concrete-dual smallness condition

$$2 D_{\text{dd}}(\hat{A}) \left(m (C_{\text{seq}}(fp, \hat{A}, \hat{b}, \alpha) N_{\text{piv}}(\hat{A}))^2 \right) < 1.$$

If the actual unit roundoff satisfies

$$m fp.u < 1,$$

then the source-control conclusion

$$\text{StoredQRSourceOffDiagonalControl}(n \leq m, fp, \hat{A}, \hat{b}, \alpha)$$

holds without an explicit gammaValid(fp, m) hypothesis.

If, in addition, the right-hand side recurrence

$$\hat{b}_{k+1} = \mathfrak{f}_{\text{rhs}}(\hat{b}_k, \hat{A}_k, \alpha_k), \quad \hat{A}_0 = A, \quad \hat{b}_0 = b,$$

is supplied, then the rounded QR/back-substitution solution satisfies the same local backward-error theorem as the explicit-validity concrete-dual finite-max handoff:

$$\text{LSQRSolveBackwardError}\left(n, A^\top A, A^\top b, \text{fl}_{\text{backSub}}(fp, \hat{R}, \hat{c}), c_G, c_g\right).$$

The triangular validity guard $\text{gammaValid}(fp, n)$ is derived from $n \leq m$ by monotonicity.

For the sampled equation (8) theorem, replace m by the sample size s in the actual-unit guard and require the same displayed QR hypotheses samplewise for the rounded sampled matrix and right-hand side. If

$$s fp.u < 1,$$

then the high-probability rounded objective conclusion remains

$$1 - \delta \leq \mathbb{P}\{\|Ax(\omega) - b\|_2^2 \leq (1 + \eta)\|Ax_\star - b\|_2^2\},$$

with the same objective-budget, exact sampling law, and exact probability measure as in the explicit-validity theorem. No probability-construction, normalization, alias-table, repair, or law-transfer error is introduced.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_diagDominant_signedStageUniformBudget_globalCompactBudget_activeMaxPivot_
completedColumns_finiteMaxSmallness_stageRowMaxComparisonDefect_concreteDual_offdiag_rows_of_
actualUnitRoundoff_no_gammaValid

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_diagDominant_signedStageUniformBudget_globalCompactBudget_
activeMaxPivot_completedColumns_finiteMaxSmallness_stageRowMaxComparisonDefect_concreteDual_
offdiag_rows_of_actualUnitRoundoff_no_gammaValid

Algorithms/RandNLA/LeastSquaresSketch.lean  leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_activePrefix_finiteMaxSmallness_activeMaxPivot_diagDominant_stageRowMaxComparisonDefect_
concreteDual_solver_of_actualUnitRoundoff_no_gammaValid
```

Corollary 409 (Active scalar-comparison source-denominator cap handoff). *On the active-pivot diagonal-dominant scalar-comparison route, the local source-control and solver surfaces need not expose the assembled finite-max compact-product scalar involving the compact-sequence relative budget. It is enough to supply local diagonal dominance, the signed-stage recurrence budget, active-pivot policy, the scalar comparison defect, source-denominator nonbreakdown, $fp.u \leq U_{\text{cap}}, (m : \mathbb{R})U_{\text{cap}} < 1$, and the canonical rational-gamma cap-smallness inequality.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_diagDominant_signedStageUniformBudget_globalCompactBudget_activeMaxPivot_
completedColumns_finiteMaxSourceDenURationalGammaCanonicalBounds_stageRowMaxComparisonDefect_
offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_diagDominant_signedStageUniformBudget_globalCompactBudget_
activeMaxPivot_completedColumns_finiteMaxSourceDenURationalGammaCanonicalBounds_
stageRowMaxComparisonDefect_offdiag_rows
```


Corollary 410 (Sampled active scalar-comparison source-denominator cap handoff). *The same source-denominator rational-gamma compact-product handoff lifts to the sampled equation (8) theorem. On each sampled stored-QR trace, it is enough to provide the active-pivot policy, local diagonal dominance, the scalar comparison defect, signed-stage recurrence budgets, source-denominator nonbreakdown, $fp.u \leq U_{\text{cap}}$, $(s : \mathbb{R})U_{\text{cap}} < 1$, and the canonical rational-gamma cap-smallness inequality.*

Lean formalization.

```
Algorithms/RandNLA/LeastSquaresSketch.lean      leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_activePrefix_finiteMaxSourceDenURationalGammaCanonicalBounds_activeMaxPivot_diagDominant_
stageRowMaxComparisonDefect_solver
```

Corollary 411 (Actual-unit source-denominator validity surface). *The active source-denominator handoff also has an actual-unit-roundoff form. On the local source-control and solver surfaces, the auxiliary cap may be specialized to $U_{\text{cap}} = fp.u$, and the validity guard is derived from $(m : \mathbb{R})fp.u < 1$. On the sampled equation (8) surface, the same validity guard is derived from $(s : \mathbb{R})fp.u < 1$.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean          StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_diagDominant_signedStageUniformBudget_globalCompactBudget_activeMaxPivot_
completedColumns_finiteMaxSourceDenURationalGammaCanonicalBounds_stageRowMaxComparisonDefect_
offdiag_rows_of_actualUnitRoundoff_no_gammaValid

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_diagDominant_signedStageUniformBudget_globalCompactBudget_
activeMaxPivot_completedColumns_finiteMaxSourceDenURationalGammaCanonicalBounds_
stageRowMaxComparisonDefect_offdiag_rows_of_actualUnitRoundoff_no_gammaValid

Algorithms/RandNLA/LeastSquaresSketch.lean      leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_activePrefix_finiteMaxSourceDenURationalGammaCanonicalBounds_activeMaxPivot_diagDominant_
stageRowMaxComparisonDefect_solver_of_actualUnitRoundoff_no_gammaValid
```

Corollary 412 (Stored-lower active source-denominator nonbreakdown). *On the actual-unit active source-denominator branch, the raw Householder source-denominator nonbreakdown hypothesis can be derived from the stored QR trace. It is enough to provide the stored panel recurrence, the signed-alpha definition, and local diagonal dominance.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean          storedQRSourceDenominator_ne_zero_of_diagDominant_
signedAlphaDef_stored_trailing_sequence

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_diagDominant_storedLower_
signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_
finiteMaxSourceDenURationalGammaCanonicalBounds_stageRowMaxComparisonDefect_offdiag_rows_of_
actualUnitRoundoff_no_gammaValid

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_diagDominant_storedLower_
signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_
finiteMaxSourceDenURationalGammaCanonicalBounds_stageRowMaxComparisonDefect_offdiag_rows_of_
actualUnitRoundoff_no_gammaValid
```

```
Algorithms/RandNLA/LeastSquaresSketch.lean      leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_activePrefix_storedLower_finiteMaxSourceDenURationalGammaCanonicalBounds_activeMaxPivot_
diagDominant_stageRowMaxComparisonDefect_solver_of_actualUnitRoundoff_no_gammaValid
```

Lemma 413 (Active-pivot budgets do not imply nonpositive scalar comparison defect). *The scalar comparison condition*

$$\Delta_{\text{cmp}} \leq 0$$

is not a consequence of the current active-pivot active/off-diagonal budget surface. Adding the product compact-smallness hypotheses used by the rectangular QR route still does not imply it.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean  not_forall_leadingBlock_upper_det_activeBlockPos_
activeMaxPivot_activeBlockBudget_offdiagBudget_implies_stageRowMaxComparisonDefectBudget_nonpos

not_forall_leadingBlock_upper_det_product_activeBlockPos_activeMaxPivot_activeBlockBudget_
offdiagBudget_implies_stageRowMaxComparisonDefectBudget_nonpos
```

Lemma 414 (Row-max defect does not force the scalar comparison defect). *Even after the scalar row-max condition*

$$\Delta_{\text{max}} \leq 0$$

is supplied, the active-pivot active/off-diagonal budget surface does not imply $\Delta_{\text{cmp}} \leq 0$. The same conclusion holds after adding the product compact-smallness hypotheses.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      activeMaxPivotRowMaxComparisonCounterexample_
rowMaxDiagDefectBudget_nonpos

activeMaxPivotRowMaxComparisonCounterexample_stageRowMaxComparisonDefectBudget_pos

not_forall_leadingBlock_upper_det_activeBlockPos_activeMaxPivot_activeBlockBudget_offdiagBudget_
rowMaxDiagDefect_implies_stageRowMaxComparisonDefectBudget_nonpos

not_forall_leadingBlock_upper_det_product_activeBlockPos_activeMaxPivot_activeBlockBudget_
offdiagBudget_rowMaxDiagDefect_implies_stageRowMaxComparisonDefectBudget_nonpos
```

Lemma 415 (Row-max defect alone does not force the stage-diagonal defect). *The scalar row-max condition $\Delta_{\text{max}} \leq 0$ does not by itself imply the downstream scalar stage-diagonal condition $\Delta_{\text{diag}} \leq 0$.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      activeMaxPivotRowMaxComparisonCounterexample_
stageDiagLowerDefectBudget_pos

not_forall_rowMaxDiagDefectBudget_implies_stageDiagLowerDefectBudget_nonpos
```

Lemma 416 (Product-active row-max data still do not force the stage-diagonal defect). *The active-pivot budget surface, even strengthened by compact-product smallness and a granted scalar row-max defect $\Delta_{\text{max}} \leq 0$, does not imply the downstream scalar stage-diagonal condition $\Delta_{\text{diag}} \leq 0$.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      not_forall_leadingBlock_upper_det_
activeBlockPos_activeMaxPivot_activeBlockBudget_offdiagBudget_rowMaxDiagDefect_implies_
stageDiagLowerDefectBudget_nonpos

not_forall_leadingBlock_upper_det_product_activeBlockPos_activeMaxPivot_activeBlockBudget_
offdiagBudget_rowMaxDiagDefect_implies_stageDiagLowerDefectBudget_nonpos
```

Lemma 417 (Active-pivot budgets do not imply nonpositive row-max defect). *The scalar row-max defect condition*

$$\Delta_{\max} \leq 0$$

is not a consequence of the current active-pivot budget surface. Even if displayed leading blocks are upper triangular and nonsingular, the active block has positive mass, the displayed pivot satisfies active max-pivoting, and one nonnegative budget controls both active-block entries and displayed strict-upper entries, the finite row-max defect can be positive.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      activeMaxPivotRowBudgetDiagCounterexample_
rowMaxDiagDefectBudget_pos

not_forall_leadingBlock_upper_det_activeBlockPos_activeMaxPivot_activeBlockBudget_offdiagBudget_
implies_rowMaxDiagDefectBudget_nonpos
```

Lemma 418 (Product smallness does not imply nonpositive row-max defect). *The product compact-smallness condition used by the diagonal-dominant rectangular QR route cannot replace the row-max scalar defect condition. Even with upper-triangular nonsingular displayed leading blocks and a satisfiable product compact-smallness inequality, it is possible that*

$$\Delta_{\max} > 0.$$

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      not_forall_upper_tri_det_ne_zero_product_budget_
implies_rowMaxDiagDefectBudget_nonpos
```

Lemma 419 (Product smallness does not imply active row-max defect control). *Even after adding product compact-smallness to the active-pivot budget surface, the nonpositive row-max scalar defect condition can fail. Thus the product condition and the current active-pivot budget fields cannot be combined as a hidden proof of*

$$\Delta_{\max} \leq 0.$$

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      not_forall_leadingBlock_upper_det_product_
activeBlockPos_activeMaxPivot_activeBlockBudget_offdiagBudget_implies_rowMaxDiagDefectBudget_
nonpos
```

Lemma 420 (Product smallness does not imply nonpositive stage-diagonal defect). *The same product compact-smallness condition also cannot replace the scalar stage-diagonal lower-bound condition. With the constant stage budget $B_k = 2$, there are upper-triangular nonsingular displayed leading blocks satisfying the compact-product inequality while*

$$\Delta_{\text{stage}} > 0.$$

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      activeMaxPivotRowBudgetDiagCounterexample_
stageDiagLowerDefectBudget_pos
not_forall_upper_tri_det_ne_zero_product_budget_implies_stageDiagLowerDefectBudget_nonpos
```

Lemma 421 (Product smallness does not imply active stage-diagonal defect control). *Even after adding product compact-smallness to the active-pivot budget surface, the scalar stage-diagonal lower-bound condition can fail. Thus the product condition and the current active-pivot budget fields cannot be combined as a hidden proof of*

$$\Delta_{\text{stage}} \leq 0.$$

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      not_forall_leadingBlock_upper_det_product_
activeBlockPos_activeMaxPivot_activeBlockBudget_offdiagBudget_implies_stageDiagLowerDefectBudget_
nonpos
```

Lemma 422 (Active-pivot budgets do not imply nonpositive stage-diagonal defect). *The active-pivot budget surface alone also cannot imply the scalar stage-diagonal lower-bound condition. Active max-pivoting and active/off-diagonal budget control therefore cannot be used as a hidden proof of*

$$\Delta_{\text{stage}} \leq 0.$$

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      not_forall_leadingBlock_upper_det_activeBlockPos_
activeMaxPivot_activeBlockBudget_offdiagBudget_implies_stageDiagLowerDefectBudget_nonpos
```

Lemma 423 (Product smallness does not imply the active stage-budget row-max comparison). *Even after adding product compact-smallness to the active-pivot budget surface, the comparison $B_k \leq R_{k,i}^{\max}$ required by the row-max bridge can fail. Thus the product condition cannot be used as a hidden proof of this comparison.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      not_forall_leadingBlock_upper_det_product_
activeBlockPos_activeMaxPivot_activeBlockBudget_offdiagBudget_implies_stageBudget_le_rowMax
```

Lemma 424 (Active-pivot budgets do not imply the stage-budget row-max comparison). *The comparison $B_k \leq R_{k,i}^{\max}$ required by the preceding row-max bridge is not a consequence of the current active-pivot budget surface. Even if displayed leading blocks are upper triangular and nonsingular, the active block has positive mass, the displayed pivot satisfies active max-pivoting, and one uniform stage budget controls both active-block entries and displayed strict-upper entries, the conclusion*

$$B_k \leq R_{k,i}^{\max}$$

can fail.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      not_forall_leadingBlock_upper_det_activeBlockPos_
activeMaxPivot_activeBlockBudget_offdiagBudget_implies_stageBudget_le_rowMax
```

Corollary 425 (Visible row-max assumptions for the active-pivot stage-diagonal QR route). *In the active max-pivot setting, the local source-control and solver theorems can expose row-max data instead of exposing the scalar stage-diagonal condition directly. Assume the finite row-max defect is nonpositive and that the uniform stage budget is bounded by the displayed strict-upper row maximum on every displayed off-diagonal row:*

$$B_k \leq R_{k,i}^{\max} \quad (i < k).$$

Then the scalar stage-diagonal lower-bound condition follows internally, and the active-pivot stage-diagonal QR route applies with the same stored recurrence, completed-column, global compact-step, determinant, κ_∞ /dual-budget, and global-product hypotheses.

Remark. This corollary does not prove the row-max scalar defect or the comparison $B_k \leq R_{k,i}^{\max}$ for a concrete pivoted loop. It makes those two fields visible on the theorem surface and prevents the downstream source-control/solver proof from re-assuming the stronger scalar $\Delta_{\text{stage}} \leq 0$ condition. Its actual-unit-roundoff source-control and solver siblings also remove the local abstract validity fields, deriving them from $(m : \mathbb{R})u < 1$, while leaving the same row-max, determinant, conditioning, dual-budget, and product-smallness fields visible.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget_signedStageUniformBudget_
globalCompactBudget_activeMaxPivot_completedColumns_globalProduct_rowMaxDiagDefect_
stageBudgetLeRowMax_offdiag_rows

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_kappaInf_
selfNorm_dualBudget_signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_
globalProduct_rowMaxDiagDefect_stageBudgetLeRowMax_offdiag_rows_of_actualUnitRoundoff_no_
gammaValid

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget_
signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_globalProduct_
rowMaxDiagDefect_stageBudgetLeRowMax_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget_
signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_globalProduct_
rowMaxDiagDefect_stageBudgetLeRowMax_offdiag_rows_of_actualUnitRoundoff_no_gammaValid
```

Corollary 426 (Active max-pivot stage-diagonal QR solve). *In the setting of the preceding corollary, suppose additionally that every displayed pivot column is chosen by the finite active max-pivot selector:*

$$k = \text{householderActiveMaxPivotColumn}(k, A_k).$$

Then the raw pivot-maximality inequality required by the active/prefix QR source-control theorem follows internally from that policy equation. Hence the local solver can consume the scalar stage-diagonal condition $\Delta_{\text{stage}} \leq 0$, the finite active-pivot policy, the stored recurrence, the global compact-step recurrence, the local determinant and κ_∞ /dual-budget nonbreakdown fields, and the scalar global-product condition, without a separate raw pivot-maximality hypothesis.

Remark. This is a pivot-policy reduction only. It does not prove $\Delta_{\text{stage}} \leq 0$, determinant or conditioning facts, nonbreakdown, compact-product smallness, or the final QR/preconditioner theorem from a concrete implementation.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean    storedQRActiveMaxPivotColumn_pivotMax
StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_signedStageUniformBudget_product_of_normSqBudget_activePrefix_activeBlockRecurrence_
globalCompactBudget_activeMaxPivot_completedColumns_globalProduct_stageDiagDefect_offdiag_rows

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_kappaInf_
selfNorm_dualBudget_signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_
globalProduct_stageDiagDefect_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget_
signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_globalProduct_
stageDiagDefect_offdiag_rows
```

Lemma 427 (Exact no-pivot QR does not imply nonpositive stage-diagonal defect). *There is a two-by-two exact unpivoted trailing Householder QR sequence $\hat{A}_k$, with valid signed Householder squared-norm identities and nonzero Householder denominators at each of its two stages, for which the constant stage budget $B_k = 2$ has*

$$\Delta_{\text{stage}}(\hat{A}, B) > 0.$$

Consequently, there is no universal theorem of the form

$$\begin{aligned} & \text{exact no-pivot trailing Householder recurrence} \\ & + \text{valid Householder squared norms} \\ & + \text{nonzero denominators} \\ \implies & \Delta_{\text{stage}}(\hat{A}, B) \leq 0 \quad (B_k \equiv 2). \end{aligned}$$

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean    exactHouseholderQRDiagDominanceCounterexample_
stageDiagLowerDefectBudget_pos

not_forall_exact_trailing_householder_sequence_implies_stageDiagLowerDefectBudget_nonpos
```

Lemma 428 (Exact no-pivot recurrence does not force the comparison scalar). *For the same exact no-pivot two-stage Householder witness, the constant stage budget $B_k \equiv 3$ gives*

$$\Delta_{\text{cmp}}(\hat{A}, B) > 0.$$

Consequently, there is no universal theorem of the form

$$\begin{aligned} & \text{exact no-pivot trailing Householder recurrence} \\ & + \text{valid Householder squared norms} \\ & + \text{nonzero denominators} \\ & + \text{nonnegative stage budgets} \\ \implies & \Delta_{\text{cmp}}(\hat{A}, B) \leq 0. \end{aligned}$$

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean    exactHouseholderQRDiagDominanceCounterexample_
stageRowMaxComparisonDefectBudget_pos

not_forall_exact_trailing_householder_sequence_implies_stageRowMaxComparisonDefectBudget_nonpos
```

Corollary 429 (Diagonal dominance implies nonpositive scalar defect). *If every displayed leading block S_k is locally diagonally dominant in the repository sense `IsDiagDominantUpper`, then*

$$\Delta_{\max} \leq 0.$$

Consequently the canonical row-max budget gives the packaged `StoredQRDisplayedRowBudgetControl` certificate.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      storedQRRowMaxDiagDefectBudget_nonpos_of_
diagDominant
StoredQRDisplayedRowBudgetControl.of_diagDominant
```

Lemma 430 (Exact no-pivot QR does not imply nonpositive scalar defect). *There is a two-by-two exact un pivoted trailing Householder QR sequence $\hat{A}_k$, with valid signed Householder squared-norm identities and nonzero Householder denominators at each of its two stages, for which*

$$\Delta_{\max}(\hat{A}) > 0.$$

Consequently, there is no universal theorem of the form

$$\begin{aligned} & \text{exact no-pivot trailing Householder recurrence} \\ & + \text{valid Householder squared norms} \\ & + \text{nonzero denominators} \\ \implies & \Delta_{\max}(\hat{A}) \leq 0. \end{aligned}$$

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      exactHouseholderQRDiagDominanceCounterexample_
rowMaxDiagDefectBudget_pos
not_forall_exact_trailing_householder_sequence_implies_rowMaxDiagDefectBudget_nonpos
```

Corollary 431 (Global compact-budget row-budget certificate). *Let B_t be a monotone signed-stage budget sequence for the stored QR panel. Suppose the initial sampled matrix entries are bounded by B_0 , and suppose the single finite compact-step recurrence*

$$c_m B_t + \Gamma_t \leq B_{t+1}$$

holds at every stage $t < n$, where c_m is the Cox–Higham active-row growth factor and Γ_t is the finite maximum of the rounded Householder compact-update component budget at stage t . If the signed Householder stages satisfy pivot maximality and the norm-square nonbreakdown margin, and if the same diagonal lower-bound field

$$B_k \leq |(S_k)_{ii}|, \quad i < k,$$

holds, then the packaged displayed row-budget certificate $\mathcal{R}(A, B)$ follows.

Moreover, the norm-square nonbreakdown margin can be replaced by the local κ_∞ /dual-budget hypotheses for the leading block S_k : local leading-block nonsingularity, a bound on $\kappa_\infty(S_k)$, a self-norm budget K_k , and the displayed dual compact-budget inequality.

There is also an active-pivot-policy version. Instead of assuming the raw pivot-maximality inequality at each stage, assume the displayed pivot column is the finite active max-pivot selector for the current active block. The same row-budget certificates follow, with or without the κ_∞ /dual budget replacement of the norm-square nonbreakdown margin.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean StoredQRDisplayedRowBudgetControl.of_signed_stage_
uniformBudget_globalCompactBudget_of_normSqBudget

StoredQRDisplayedRowBudgetControl.of_signed_stage_uniformBudget_globalCompactBudget_kappaInf_
dualBudget

StoredQRDisplayedRowBudgetControl.of_signed_stage_uniformBudget_globalCompactBudget_
activeMaxPivot_of_normSqBudget

StoredQRDisplayedRowBudgetControl.of_signed_stage_uniformBudget_globalCompactBudget_
activeMaxPivot_kappaInf_dualBudget
```

Lemma 432 (Cox–Higham stage budgets give displayed off-diagonal row budgets). *Fix a stored Householder panel sequence A_t , a nonnegative growth factor c , and displayed row budgets $R_{k,i}$. For each upper displayed entry (i, j) of the k th leading block, let*

$$\tau(k, j) = \begin{cases} j + 1, & j < k, \\ k, & j = k. \end{cases}$$

Assume a scalar Cox–Higham stage-budget recurrence for the same entry through stage $\tau(k, j)$: the initial entry is below B_0 , each exact same-reflector step is below cB_t , the concrete floating-point compact Householder component budget is absorbed by B_{t+1} , and the pivot-column zeroing equation holds when the entry’s column is the active pivot. If $B_{\tau(k,j)} \leq R_{k,i}$, then

$$|(S_k)_{ij}| \leq R_{k,i}, \quad 0 \leq i < j \leq k < n.$$

Lean formalization.

```
Algorithms/QR/HouseholderQR.lean qrLeadingOffdiagStop

storedQRSignedStageVector

storedQRSignedStageBeta

one_le_coxHighamActiveRowGrowthFactor

storedQRSignedStage_exact_same_reflector_bound_of_prefix_or_active_stage_bounds

fl_householderStoredPanel_sequence_completed_column_eq_pivot_succ

fl_householderStoredPanel_sequence_leadingBlock_offdiag_budget_of_exact_stage_budgets_factor

fl_householderStoredPanel_sequence_leadingBlock_offdiag_budget_of_signed_stage_budgets_factor

storedQRSignedStage_pivot_column_zero_below_of_trailingNorm_pos

storedQRSignedStage_pivot_zeroing_field_of_trailingNorm_pos

storedQRSignedStage_pivot_zeroing_field_of_normSqBudget

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_signedStageUniformBudget_product_of_normSqBudget_stage_entry_bounds

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_
signedStageUniformBudget_product_of_normSqBudget_stage_entry_bounds
```

Corollary 433 (Source-control certificate from stage budgets and diagonal lower bounds). *Keep the determinant, norm-square nonbreakdown, and compact-product hypotheses of the norm-square source-shaped QR route. Suppose also that the displayed upper entries of each leading block are*

equipped with Cox–Higham stage budgets $B_t^{(k,i,j)}$ satisfying the recurrence in the previous lemma, and that their terminal values are dominated by row budgets $R_{k,i}$:

$$B_{\tau(k,j)}^{(k,i,j)} \leq R_{k,i}, \quad 0 \leq i < j \leq k < n.$$

If the row budgets satisfy the diagonal lower-bound condition

$$R_{k,i} \leq |(S_k)_{ii}|, \quad 0 \leq i < k < n,$$

then the stored QR sequence satisfies

StoredQRSourceOffDiagonalControl.

Consequently, the corresponding computed triangular solve satisfies the local *LSQRSolveBackwardError* certificate with *qrSolveFinalGramBudget* and *qrSolveFinalRhsBudget*.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_leadingBlock_det_ne_zero_normSqBudget_stageBudget_rowBudget_product

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_
bound_explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_stageBudget_
rowBudget_product

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_stageBudget_rowBudget_product_of_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_
bound_explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_stageBudget_
rowBudget_product_of_offdiag_rows

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_signedStageBudget_rowBudget_product

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_signedStageBudget_
rowBudget_product

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_signedStageBudget_rowBudget_product_of_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_signedStageBudget_
rowBudget_product_of_offdiag_rows

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_signedStageBudget_rowBudget_product_of_normSqBudget

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_signedStageBudget_
rowBudget_product_of_normSqBudget

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_signedStageBudget_rowBudget_product_of_normSqBudget_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_signedStageBudget_
rowBudget_product_of_normSqBudget_offdiag_rows

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_rowBudgetControl_globalProduct
```

```

qrLeadingOffdiagStop_le

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_signedStageUniformBudget_product_of_normSqBudget

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_
signedStageUniformBudget_product_of_normSqBudget

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_signedStageUniformBudget_product_of_normSqBudget_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_
signedStageUniformBudget_product_of_normSqBudget_offdiag_rows

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_signedStageUniformBudget_product_of_normSqBudget_stage_entry_bounds_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_
signedStageUniformBudget_product_of_normSqBudget_stage_entry_bounds_offdiag_rows

```

Corollary 434 (Active/prefix split for the stage-entry QR certificate). *In the setting of the preceding stage-entry certificate, suppose the stage entry control is supplied in the following source-shaped form. For every stage $t < n$, all entries in the active suffix obey the single block budget*

$$|A_t(r, \ell)| \leq B_t \quad (t \leq r, t \leq \ell).$$

For a displayed strict upper entry $i < j$ in the k th leading block, the only separate row-growth hypothesis is the prefix case

$$|A_t(\rho_{k,i}, \ell)| \leq B_t \quad (t < \sigma_{k,j}, t \leq \ell, \rho_{k,i} < t),$$

where $\rho_{k,i}$ is the stored row index of displayed row i , and $\sigma_{k,j} = \text{qrLeadingOffdiagStop}(j)$. Together with the same signed-alpha recurrence, norm-square nonbreakdown, pivot-maximality, monotone stage budget, offdiag-row-only diagonal lower bounds $B_k \leq |(S_k)_{ii}|$ for $i < k$, and compact-product smallness as above, these hypotheses imply the same stored-QR source-control certificate and hence the same `LSQRSolveBackwardError` conclusion.

Lean formalization.

```

Algorithms/LeastSquares/LSQRSolve.lean      StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_leadingBlock_det_ne_zero_normSqBudget_signedStageUniformBudget_product_of_
normSqBudget_activePrefix_stage_entry_bounds_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_
signedStageUniformBudget_product_of_normSqBudget_activePrefix_stage_entry_bounds_offdiag_rows

```

Corollary 435 (Active-block recurrence for the active/prefix QR certificate). *In the setting of the active/prefix stage-entry certificate, the active-suffix block budget*

$$|A_t(r, \ell)| \leq B_t \quad (t \leq r, t \leq \ell)$$

need not be supplied as an independent hypothesis. It follows from the stored signed-stage recurrence, norm-square nonbreakdown, pivot-column maximality, completed old-column preservation, the initial full-block bound

$$|A_0(r, \ell)| \leq B_0,$$

and the Cox–Higham active-block recurrence inequality

$$\Gamma_m B_t + \Delta_t(r, \ell) \leq B_{t+1} \quad (t+1 < n, t+1 \leq r, t+1 \leq \ell),$$

where $\Gamma_m = \text{coxHighamActiveRowGrowthFactor}(m)$ and $\Delta_t(r, \ell)$ is the corresponding compact Householder component budget. Consequently the same stored-QR source-control certificate and *LSQRSolveBackwardError* conclusion follow with the active block propagated by recurrence. The separate prefix displayed-row budget, diagonal lower bounds, determinant/norm-square nonbreak-down, and compact-product smallness remain explicit.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_leadingBlock_det_ne_zero_normSqBudget_signedStageUniformBudget_product_of_
normSqBudget_activePrefix_activeBlockRecurrence_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_
signedStageUniformBudget_product_of_normSqBudget_activePrefix_activeBlockRecurrence_offdiag_rows
```

Corollary 436 (Prefix-row recurrence for the active/prefix QR certificate). *In the setting of the active-block recurrence certificate, the prefix displayed row budget*

$$|A_t(\rho, \ell)| \leq B_t \quad (\rho < t, t \leq \ell)$$

need not be supplied as a raw stage hypothesis. It follows by induction from the active-block bound, the signed-stage exact same-reflector prefix/active split, and the one-step prefix recurrence

$$\Gamma_m B_t + \Delta_t(\rho, \ell) \leq B_{t+1} \quad (\rho < t+1, t \leq \ell),$$

where $\Gamma_m = \text{coxHighamActiveRowGrowthFactor}(m)$. Consequently the stored-QR source-control certificate and the corresponding *LSQRSolveBackwardError* conclusion follow with both active and prefix stage-entry fields generated from one-step recurrences.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      storedQRSignedStage_active_block_bound_of_signed_
stage_budget

storedQRSignedStage_prefix_row_bound_of_active_block_and_prefix_budget

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_signedStageUniformBudget_product_of_normSqBudget_activePrefix_activeBlockRecurrence_
prefixRowRecurrence_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_
signedStageUniformBudget_product_of_normSqBudget_activePrefix_activeBlockRecurrence_
prefixRowRecurrence_offdiag_rows
```

Corollary 437 (Finite global compact-step budget for active/prefix QR). *For a signed stored-QR stage $t < n$, define*

$$\Delta_t^{\max} = \max_{r, \ell} \Delta_t(r, \ell),$$

where $\Delta_t(r, \ell)$ is the compact Householder floating-point component budget for entry (r, ℓ) at stage t . If the scalar recurrence

$$\Gamma_m B_t + \Delta_t^{\max} \leq B_{t+1}$$

holds at every stage, then all three one-step budget fields used by the active/prefix QR certificate follow: the displayed off-diagonal row recurrence, the active-suffix block recurrence, and the prefix displayed-row recurrence. Consequently the stored-QR source-control certificate and the corresponding *LSQRSolveBackwardError* conclusion follow without separately assuming these three one-step budget families.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean storedQRSignedStageGlobalCompactBudget
storedQRSignedStage_compact_component_le_globalBudget
storedQRSignedStage_active_prefix_budgets_of_globalCompactBudget

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_signedStageUniformBudget_product_of_normSqBudget_activePrefix_activeBlockRecurrence_
globalCompactBudget_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_
signedStageUniformBudget_product_of_normSqBudget_activePrefix_activeBlockRecurrence_
globalCompactBudget_offdiag_rows
```

Corollary 438 (Monotone signed-stage budgets from the global recurrence). *Assume the signed-stage finite global compact-step recurrence*

$$\Gamma_m B_t + \Delta_t^{\max} \leq B_{t+1}, \quad t < n,$$

and $B_t \geq 0$ for all t . Then the stage budget is monotone on the algorithmic QR horizon:

$$a \leq b \leq n \implies B_a \leq B_b.$$

This is only a stage-range consequence of the recurrence; it does not assert any behavior for budget indices beyond n .

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean storedQRSignedStageGlobalCompactBudget_nonneg
storedQRSignedStageBudget_mono_on_stages_of_globalCompactBudget
```

Corollary 439 (Horizon-clamped global-product QR budget surface). *In the completed-column, global-product active/prefix QR route, the separate hypothesis that the stage budget is monotone for all natural indices may be removed. Define the horizon-clamped budget*

$$\tilde{B}_t = \begin{cases} B_t, & t \leq n, \\ B_n, & t > n. \end{cases}$$

The recurrence above proves that $\tilde{B}_t$ is globally monotone, while $\tilde{B}_t = B_t$ at every algorithmic QR stage. Therefore the existing completed-column global-product source-control and solver certificates apply with no assumption about budget behavior beyond the QR horizon.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean qrStageHorizonBudget
qrStageHorizonBudget_eq_of_le
```

```

qrStageHorizonBudget_nonneg
qrStageHorizonBudget_mono_of_mono_on_stages

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_signedStageUniformBudget_product_of_normSqBudget_activePrefix_activeBlockRecurrence_
globalCompactBudget_completedColumns_globalProduct_offdiag_rows_of_horizonBudget

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_
signedStageUniformBudget_product_of_normSqBudget_activePrefix_activeBlockRecurrence_
globalCompactBudget_completedColumns_globalProduct_offdiag_rows_of_horizonBudget

```

Corollary 440 (Completed-column preservation for the active/prefix QR route). *In the same signed stored-QR loop, assume the rounded stored-panel recurrence for A_t . If $j < t$, then the exact signed Householder reflector used at stage t leaves the already completed column j unchanged:*

$$H_t A_t(:, j) = A_t(:, j).$$

Consequently the global compact-step source-control and solver certificates above no longer need a separate completed-column hypothesis; it is derived from the stored-loop recurrence itself.

Lean formalization.

```

Algorithms/QR/HouseholderQR.lean storedQRSignedStage_completed_column_preservation
Algorithms/LeastSquares/LSQRSolve.lean      StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_leadingBlock_det_ne_zero_normSqBudget_signedStageUniformBudget_product_of_
normSqBudget_activePrefix_activeBlockRecurrence_globalCompactBudget_completedColumns_offdiag_rows

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_
signedStageUniformBudget_product_of_normSqBudget_activePrefix_activeBlockRecurrence_
globalCompactBudget_completedColumns_offdiag_rows

```

Corollary 441 (Finite global compact-product budget for active/prefix QR). *In the same signed stored-QR loop, define for each pivot $k < n$*

$$\Pi_k = 2 D_k \left(m \left(c_{\text{seq}} \|A_k(:, k)\|_2 \right)^2 \right),$$

where

$$D_k = \text{diagDominantUpperInvBudgetExpr}(S_k, k), \quad c_{\text{seq}} = \text{storedQRCompactSequenceRelativeBudget}.$$

Let

$$\Pi_{\max} = \max_{0 \leq k < n} \Pi_k$$

with the vacuous value 0 when $n = 0$. If

$$\Pi_{\max} < 1,$$

then the per-pivot compact-product condition

$$\Pi_k < 1, \quad 0 \leq k < n,$$

used by the active/prefix source-control and solver certificates follows. Consequently the global compact-step and completed-column QR certificates no longer need a separate compact-product hypothesis at every pivot; they consume the single scalar smallness condition $\Pi_{\max} < 1$.

Conversely, the finite-maximum bookkeeping is now formalized in the other direction as well: if every $\Pi_k < 1$, then $\Pi_{\max} < 1$. Thus the global scalar assumption is exactly a finite packaging of the per-pivot compact-product smallness field; the remaining mathematical task is to prove those smallness inequalities from a concrete pivoted/sorted QR invariant or to keep them visible as domain assumptions.

There is one exact-arithmetic endpoint where the product-smallness side condition is automatic. If the stored compact relative budget vanishes,

$$c_{\text{seq}} = 0,$$

then every $\Pi_k = 0$, and hence $\Pi_{\max} < 1$. This closes the compact-product dependency for the exact/zero-compact regime. It is not a floating-point smallness theorem: when $c_{\text{seq}} > 0$, the positive-budget smallness inequality must still be proved from conditioning, pivoting, or a source/domain assumption.

A further scalar sufficient condition is also formalized. If

$$D_k \leq D_{\max}, \quad \|A_k(:, k)\|_2 \leq N_{\max}$$

for every pivot k , each local leading block is diagonally dominant, and

$$2D_{\max} (m(c_{\text{seq}} N_{\max})^2) < 1,$$

then $\Pi_{\max} < 1$. This records a single source-facing product budget; it does not prove the numerical smallness of $D_{\max}$, $N_{\max}$, or c_{seq} from the computed QR loop.

The global bounds themselves are now packaged canonically. Let

$$D_* = \max_{0 \leq k < n} D_k, \quad N_* = \max_{0 \leq k < n} \|A_k(:, k)\|_2,$$

again with the vacuous value 0 for an empty pivot set. Under local diagonal dominance, $D_* \geq 0$, $N_* \geq 0$, and every $D_k \leq D_*$ and $\|A_k(:, k)\|_2 \leq N_*$. Therefore the single canonical condition

$$2D_* (m(c_{\text{seq}} N_*)^2) < 1$$

implies $\Pi_{\max} < 1$. The remaining product-smallness task is no longer a family of global-bound hypotheses; it is the scalar inequality for these canonical finite maxima. This scalar condition is also composed into the main diagonal-dominant QR source-control and solver surfaces: those theorems now consume

$$2D_* (m(c_{\text{seq}} N_*)^2) < 1$$

directly rather than asking the caller to separately provide $\Pi_{\max} < 1$.

There is also a positive-budget threshold form. Let $c_{\max} \geq 0$ be an explicit cap such that

$$c_{\text{seq}} \leq c_{\max}.$$

If

$$2D_* (m(c_{\max} N_*)^2) < 1,$$

then again $\Pi_{\max} < 1$. This is the finite-max monotonicity step needed to replace the exact compact relative budget by a source-facing upper bound. It does not prove $c_{\text{seq}} \leq c_{\max}$, nor does it prove the scalar smallness inequality from machine precision or conditioning data.

One useful specialization is now also formalized at the stored-QR layer. Let c_{step} be a uniform one-step compact-panel relative budget:

$$\text{storedQRCompactStepRelativeBudget}_k \leq c_{\text{step}}, \quad 0 \leq k < n.$$

Since the sequence budget is the finite sum of these one-step budgets,

$$c_{\text{seq}} \leq n c_{\text{step}}.$$

Thus, if $c_{\text{step}} \geq 0$ and

$$2D_* \left(m \left((n c_{\text{step}}) N_* \right)^2 \right) < 1,$$

then $\Pi_{\text{max}} < 1$. This closes the finite-sum reduction from a uniform per-step cap to the positive-budget compact-product threshold. It still does not prove the one-step cap c_{step} , local diagonal dominance, or the final scalar inequality from a concrete computed QR recurrence.

The one-step cap itself has also been reduced to vector-level compact Householder caps. For a stored QR stage k , let

$$\begin{aligned} \rho_{k,j}^{\text{col}} &= \text{householderCompactRelativeBudget}(v_k, \beta_k, \hat{A}_k(:, j)), \\ \rho_k^{\text{rhs}} &= \text{householderCompactRelativeBudget}(v_k, \beta_k, \hat{b}_k). \end{aligned}$$

If

$$\rho_{k,j}^{\text{col}} \leq c_{\text{col}} \quad \text{for all } j, \quad \rho_k^{\text{rhs}} \leq c_{\text{rhs}},$$

then the stored-step panel budget satisfies

$$\text{storedQRCompactStepRelativeBudget}_k \leq n c_{\text{col}} + c_{\text{rhs}}.$$

Consequently

$$c_{\text{seq}} \leq n (n c_{\text{col}} + c_{\text{rhs}}).$$

Thus, if $c_{\text{col}}, c_{\text{rhs}} \geq 0$ and

$$2D_* \left(m \left((n (n c_{\text{col}} + c_{\text{rhs}})) N_* \right)^2 \right) < 1,$$

then $\Pi_{\text{max}} < 1$. This closes the finite-sum bridge from vector-level compact caps to the product threshold. It still does not prove the vector-level caps, local diagonal dominance, or the scalar product-smallness inequality from a concrete pivoted/sorted QR loop.

There is now a primitive norm-budget version of the same bridge. Define

$$\begin{aligned} B_{k,j}^{\text{col}} &= \text{householderCompactNormBudget}(v_k, \beta_k, \hat{A}_k(:, j)), \\ B_k^{\text{rhs}} &= \text{householderCompactNormBudget}(v_k, \beta_k, \hat{b}_k). \end{aligned}$$

If $c_{\text{col}}, c_{\text{rhs}} \geq 0$ and

$$B_{k,j}^{\text{col}} \leq c_{\text{col}} \|\hat{A}_k(:, j)\|_2 \quad \text{for all } j, \quad B_k^{\text{rhs}} \leq c_{\text{rhs}} \|\hat{b}_k\|_2,$$

then the definition of relative compact budget gives

$$\rho_{k,j}^{\text{col}} \leq c_{\text{col}}, \quad \rho_k^{\text{rhs}} \leq c_{\text{rhs}}.$$

Consequently the same sequence cap

$$c_{\text{seq}} \leq n (n c_{\text{col}} + c_{\text{rhs}})$$

and the same finite-max product-smallness conclusion hold. This is the primitive-budget bridge: it removes the need to assume already-normalized relative compact caps, but it still leaves the norm-budget inequalities and the scalar product-smallness inequality visible.

There is also a componentwise primitive version. Define the entrywise compact-budget quantities

$$\begin{aligned} E_{k,j,i}^{\text{col}} &= \text{householderCompactComponentBudget}(v_k, \beta_k, \hat{A}_k(:,j))_i, \\ E_{k,i}^{\text{rhs}} &= \text{householderCompactComponentBudget}(v_k, \beta_k, \hat{b}_k)_i. \end{aligned}$$

If $c_{\text{col}}, c_{\text{rhs}} \geq 0$ and

$$E_{k,j,i}^{\text{col}} \leq c_{\text{col}} |\hat{A}_k(i,j)| \quad \text{for all } i,j, \quad E_{k,i}^{\text{rhs}} \leq c_{\text{rhs}} |\hat{b}_k(i)| \quad \text{for all } i,$$

then local norm monotonicity and homogeneity give

$$B_{k,j}^{\text{col}} \leq c_{\text{col}} \|\hat{A}_k(:,j)\|_2, \quad B_k^{\text{rhs}} \leq c_{\text{rhs}} \|\hat{b}_k\|_2.$$

Thus the same sequence cap

$$c_{\text{seq}} \leq n(n c_{\text{col}} + c_{\text{rhs}})$$

and the same finite-max product-smallness conclusion hold. This closes the bookkeeping step from entrywise compact budgets to the product threshold, but it still does not prove the entrywise inequalities or the scalar smallness inequality from a concrete floating-point QR loop.

The valid primitive floating-point route is normwise rather than entrywise relative. For a reflector (v, β) , let

$$V = \|v\|_2, \quad \sigma_* = (1 + \gamma_m)V, \quad \tau_* = |\beta| \sigma_*(1 + u),$$

and let

$$e_* = u \tau_* + u |\beta| \sigma_* + |\beta| \gamma_m V, \quad z_* = \tau_*(1 + u).$$

The formal coefficient is

$$c_{\text{HH}}(v, \beta) = u + (e_* + uz_*) V.$$

For every vector x ,

$$\text{householderCompactNormBudget}(v, \beta, x) \leq c_{\text{HH}}(v, \beta) \|x\|_2.$$

The key component estimate behind this norm statement is

$$E_i(v, \beta, x) \leq (e_* + uz_*) \|x\|_2 |v_i| + u |x_i|.$$

Thus the nonlocal dot/scale/update error is charged to $\|x\|_2 |v_i|$, while only the final subtraction is charged to $|x_i|$. This is why the theorem does not require the generally false universal cap $E_i \leq c |x_i|$.

For a stored QR sequence define

$$c_k^{\text{HH}} = c_{\text{HH}}(v_k, \beta_k).$$

The Lean library first proves the conditional version: if $c_k^{\text{HH}} \leq c_{\text{HH}}^{\text{max}}$ for all k , then each compact stored step has relative budget at most

$$n c_{\text{HH}}^{\text{max}} + c_{\text{HH}}^{\text{max}},$$

and hence the global sequence budget is bounded by

$$c_{\text{seq}} \leq n(n c_{\text{HH}}^{\text{max}} + c_{\text{HH}}^{\text{max}}).$$

Therefore, under local diagonal dominance, if

$$2D_* \left(m \left((n(n c_{\text{HH}}^{\text{max}} + c_{\text{HH}}^{\text{max}})) N_* \right)^2 \right) < 1,$$

then $\Pi_{\text{max}} < 1$. This closes the concrete floating-point norm-coefficient reduction for the compact-product route.

The finite-max version removes the arbitrary uniform-coefficient field. Define

$$C_{\text{HH}}^* = \begin{cases} \max_{k \in \text{Fin}(n)} c_k^{\text{HH}}, & n > 0, \\ 0, & n = 0. \end{cases}$$

The formalization proves $0 \leq C_{\text{HH}}^*$ and $c_k^{\text{HH}} \leq C_{\text{HH}}^*$ for every stored stage. Hence, under local diagonal dominance, the scalar condition

$$2D_* \left(m \left((n(n C_{\text{HH}}^* + C_{\text{HH}}^*)) N_* \right)^2 \right) < 1$$

implies $\Pi_{\text{max}} < 1$. This is still not a machine-precision-only smallness theorem: diagonal dominance/off-diagonal control and the displayed scalar inequality for the canonical coefficient maximum remain visible numerical assumptions.

The coefficient maximum has a new per-pivot specialization. Define

$$F_m(u, \gamma_m) = u(1 + \gamma_m)(1 + u) + u(1 + \gamma_m) + \gamma_m + u(1 + \gamma_m)(1 + u)^2.$$

The compact coefficient has the exact form

$$c_{\text{HH}}(v, \beta) = u + (|\beta| \|v\|_2^2) F_m(u, \gamma_m).$$

Moreover $F_m(u, \gamma_m) \geq 0$ under the repository's γ -validity hypothesis. Hence a uniform signed-stage estimate

$$|\beta_k| \|v_k\|_2^2 \leq B_{\text{max}}$$

implies

$$C_{\text{HH}}^* \leq u + B_{\text{max}} F_m(u, \gamma_m).$$

Consequently, the finite-max product theorem may be checked with

$$C_{\text{max}} = u + B_{\text{max}} F_m(u, \gamma_m)$$

instead of an abstract coefficient maximum. The resulting scalar condition is

$$2D_{\text{max}} \left(m \left(n(n(u + B_{\text{max}} F_m) + (u + B_{\text{max}} F_m)) N_{\text{max}} \right)^2 \right) < 1.$$

This reduces the open coefficient-bound field to the familiar Householder normalization quantity $|\beta_k| \|v_k\|_2^2$. It still does not prove the B_{max} estimate, local diagonal dominance, or the displayed scalar smallness inequality from the concrete stored QR loop.

That B_{max} field is now closed for the exact Householder normalization used by the stored signed stages. If

$$\beta = \frac{2}{v^T v} \quad \text{and} \quad v^T v \neq 0,$$

then

$$|\beta| \|v\|_2^2 = 2.$$

Consequently, for every stored signed stage whose source-shaped Householder denominator is nonzero,

$$|\beta_k| \|v_k\|_2^2 = 2, \quad c_k^{\text{HH}} \leq u + 2F_m(u, \gamma_m),$$

and therefore

$$C_{\text{HH}}^* \leq u + 2F_m(u, \gamma_m).$$

The finite-max product and off-diagonal-control invariant may thus be checked with the concrete coefficient maximum

$$C_{\text{max}} = u + 2F_m(u, \gamma_m),$$

provided the signed-stage denominators are nonzero. The remaining fields are the scalar smallness inequality with this concrete C_{max} , local diagonal dominance or another off-diagonal-control invariant, inverse-factor and pivot-column estimates, and determinant/conditioning data needed by the chosen QR/preconditioner route.

The same active/prefix route now also has a structured nonbreakdown variant: instead of assuming the norm-square pivot margin directly, one may assume local leading-block nonsingularity together with a displayed κ_∞ self-norm budget and a dual compact-update budget. The previously formalized leading-block inverse-budget theorem then gives the dimensioned norm-square margin used by the signed-stage QR source-control proof.

The row-budget-control solver surface has the same finite-global-product packaging. A local **LSQRSolveBackwardError** theorem consumes the packaged displayed row-budget certificate and the scalar condition $\Pi_{\text{max}} < 1$; its κ_∞ /dual-budget sibling derives the norm-square nonbreakdown margin from the local leading-block inverse-budget route before applying the finite-global-product solver theorem.

There is now also a solver-facing active-pivot-policy version of this κ_∞ route. Instead of assuming the raw column-pivot maximality inequality, it assumes the displayed pivot column is the finite active max-pivot selector **householderActiveMaxPivotColumn**. The raw inequality is then recovered locally from **householderActiveMaxPivotColumn_pivot_max** before applying the existing global-product solver certificate. This closes the pivot-policy handoff at the local least-squares layer only; the diagonal lower-bound, local determinant/conditioning, dual compact-budget, and compact-product smallness assumptions remain explicit.

Corollary 442 (Source-facing scalar certificate). *With the hypotheses of the exact-normalization branch above, set*

$$C_{\text{max}} = u + 2F_m(u, \gamma_m).$$

If the displayed route constants satisfy

$$2D_{\text{max}} m (n(n+1)C_{\text{max}}N_{\text{max}})^2 < 1,$$

then the canonical compact-product smallness condition holds, and hence the same packaged off-diagonal-control invariant follows under the same local diagonal-dominance, source-nonbreakdown, D_{max} , and N_{max} hypotheses.

This corollary is an algebraic normalization of the scalar side condition. It does not prove the numerical inequality, diagonal dominance, inverse-factor or pivot-column bounds, or determinant/conditioning assumptions.

Corollary 443 (Capped source-facing scalar certificate). *Keep the hypotheses of the preceding corollary, but suppose that the unit-roundoff and coefficient factor are bounded by explicit caps*

$$u \leq U_{\text{max}}, \quad F_m(u, \gamma_m) \leq F_{\text{max}},$$

with $U_{\max}, F_{\max} \geq 0$. If

$$2D_{\max} m (n(n+1)(U_{\max} + 2F_{\max})N_{\max})^2 < 1,$$

then the same canonical compact-product smallness condition holds, and hence the same packaged off-diagonal-control invariant follows under the same local diagonal-dominance, source-nonbreakdown, $D_{\max}$, and $N_{\max}$ hypotheses.

This is the next scalar-smallness reduction in the LS.2g red-bottleneck route: it replaces the formula involving the concrete floating-point constants by a monotone cap certificate. It does not prove the caps, the $D_{\max}$ or $N_{\max}$ pointwise estimates, diagonal dominance, source nonbreakdown, or the final QR/preconditioner theorem.

Lemma 444 (Householder coefficient-factor cap). *Let*

$$F_m(u, \gamma_m) = u(1 + \gamma_m)(1 + u) + u(1 + \gamma_m) + \gamma_m + u(1 + \gamma_m)(1 + u)^2$$

be the compact Householder coefficient factor used above. Suppose γ_m is valid in the local floating-point model, and suppose

$$0 \leq U_{\max}, \quad 0 \leq G_{\max}, \quad u \leq U_{\max}, \quad \gamma_m \leq G_{\max}.$$

Then

$$F_m(u, \gamma_m) \leq U_{\max}(1 + G_{\max})(1 + U_{\max}) + U_{\max}(1 + G_{\max}) + G_{\max} + U_{\max}(1 + G_{\max})(1 + U_{\max})^2.$$

This lemma supplies one concrete source for the $F_{\max}$ cap in the preceding corollary. It does not prove the primitive caps on u or γ_m , nor the final scalar cap inequality.

Lemma 445 (Gamma cap from a unit-roundoff cap). *Suppose the unit roundoff satisfies $u \leq U_{\max}$, and suppose*

$$mU_{\max} < 1.$$

Then

$$mu < 1, \quad \gamma_m = \frac{mu}{1 - mu} \leq \frac{mU_{\max}}{1 - mU_{\max}}.$$

Consequently, if a displayed constant $G_{\max}$ satisfies

$$\frac{mU_{\max}}{1 - mU_{\max}} \leq G_{\max},$$

then $\gamma_m \leq G_{\max}$.

The first inequality is the Lean validity guard `gammaValid fp m`. Together with the preceding lemma, this supplies the gamma side of the coefficient-factor cap $F_m(u, \gamma_m) \leq F_{\max}$. It still leaves the actual unit-roundoff cap and the final scalar smallness inequality as explicit route obligations.

Corollary 446 (Coefficient-factor cap from a unit-roundoff cap). *Assume $0 \leq U_{\max}$, $u \leq U_{\max}$, and*

$$mU_{\max} < 1, \quad \frac{mU_{\max}}{1 - mU_{\max}} \leq G_{\max}.$$

Then the same displayed polynomial bound for $F_m(u, \gamma_m)$ in the Householder coefficient-factor cap holds with $U_{\max}$ and $G_{\max}$.

This corollary is the direct composition of the two preceding lemmas. It removes the independent hypothesis $\gamma_m \leq G_{\max}$ from the coefficient-factor cap route, but it still leaves the primitive unit-roundoff cap and the downstream scalar product-smallness inequality as explicit obligations.

Corollary 447 (Source-denominator cap route from unit-roundoff and gamma caps). *In the source-denominator scalar-smallness route above, replace the abstract factor cap $F_{\max}$ by the displayed polynomial bound from Corollary 322. Then the same scalar-smallness, compact-product, and packaged off-diagonal-control conclusions hold from*

$$u \leq U_{\max}, \quad mU_{\max} < 1, \quad \frac{mU_{\max}}{1 - mU_{\max}} \leq G_{\max},$$

and the displayed scalar inequality using

$$F_{\max} = U_{\max}(1 + G_{\max})(1 + U_{\max}) + U_{\max}(1 + G_{\max}) + G_{\max} + U_{\max}(1 + G_{\max})(1 + U_{\max})^2.$$

This is a theorem-surface reduction. It removes the separate $F_m(u, \gamma_m) \leq F_{\max}$ proof field from the least-squares source-denominator route; it does not prove the primitive unit-roundoff cap, the rational $G_{\max}$ domination, local diagonal dominance, pointwise inverse/pivot-column bounds, source nonbreakdown, or the final equation (8) QR/preconditioner theorem.

Corollary 448 (Rational gamma cap specialization). *In Corollary 323, choose*

$$G_{\max} = \frac{mU_{\max}}{1 - mU_{\max}}.$$

Then the same source-denominator scalar-smallness, compact-product, and packaged off-diagonal-control conclusions hold under the unit-roundoff cap $u \leq U_{\max}$, the validity condition $mU_{\max} < 1$, the pointwise $D_{\max}$ and $N_{\max}$ route bounds, source nonbreakdown, and the displayed scalar-smallness inequality with this value of $G_{\max}$ substituted into $F_{\max}$. Thus the rational $G_{\max}$ -domination hypothesis is not a separate route obligation when the rational cap itself is used.

This specialization does not prove the primitive unit-roundoff cap, the scalar-smallness inequality, local diagonal dominance, pointwise inverse or pivot-column estimates, source nonbreakdown, or the final equation (8) QR/preconditioner theorem.

Corollary 449 (Canonical finite-max rational gamma cap route). *In Corollary 324, choose*

$$D_{\max} = D_*(A_{\text{fl}}), \quad N_{\max} = N_*(A_{\text{fl}}),$$

where D_ is the repository finite maximum of the diagonal-dominant inverse-factor budgets and N_* is the repository finite maximum of the displayed pivot-column norms. Then the same source-denominator scalar-smallness, compact-product, and packaged off-diagonal-control conclusions hold with the scalar-smallness inequality written directly in terms of D_* , N_* , and the rational $G_{\max} = mU_{\max}/(1 - mU_{\max})$.*

Thus the rational-gamma route no longer needs separate pointwise certificates showing every local inverse factor is bounded by $D_{\max}$ and every pivot column norm is bounded by $N_{\max}$; those proofs are discharged by the finite-maximum lemmas. The result still assumes local diagonal dominance, source nonbreakdown, $u \leq U_{\max}$, $mU_{\max} < 1$, and the displayed scalar-smallness inequality, and it does not prove the final equation (8) QR/preconditioner theorem.

Corollary 450 (Solver and probability handoff for the canonical rational-gamma route). *Assume the hypotheses of Corollary 325 samplewise for the stored QR sequence: local diagonal dominance, source-shaped Householder denominator nonbreakdown, $u \leq U_{\max}$, $mU_{\max} < 1$, and the canonical finite-max scalar inequality with*

$$G_{\max} = \frac{mU_{\max}}{1 - mU_{\max}}, \quad D_{\max} = D_*(A_{\text{fl}}), \quad N_{\max} = N_*(A_{\text{fl}}).$$

Then the stored-QR solve certificate used by the equation (8) least-squares route holds. Composed with the already formalized Bennett sample-budget least-squares theorem, this gives the corresponding high-probability rounded sampled-row objective bound for Algorithm 2.

This corollary is a solver/probability assembly theorem. It does not prove local diagonal dominance, source denominator nonbreakdown, the primitive unit-roundoff cap, or the canonical scalar finite-max smallness inequality. Those remain the visible QR-domain obligations for this branch.

Lemma 451 (No universal unit-roundoff cap for the abstract model). *For every real number $U_{\max}$,*

$$\neg(\forall \text{fp} : \text{FPModel}, u(\text{fp}) \leq U_{\max}).$$

Indeed, exact arithmetic can be packaged as an FPModel with any chosen nonnegative declared unit roundoff. Therefore the hypothesis $u \leq U_{\max}$ in the preceding cap route is not a hidden consequence of the abstract floating-point model. It must be supplied as a domain assumption or derived from a more concrete machine model.

Lean formalization.

```
Model.lean FPModel.exactWithUnitRoundoff
FPModel.not_forall_u_le_cap
Analysis/Rounding.lean gamma_le_of_u_le_cap
gamma_le_Gcap_of_u_le_cap
Algorithms/QR/HouseholderSpec.lean abs_householderBeta_mul_vecNorm2_sq_eq_two
abs_householderBeta_mul_vecNorm2_sq_le_two
Algorithms/QR/HouseholderApply.lean householderAbsDotBudget_le_vecNorm2_mul
householderCompactUpdateCoeff
householderCompactNormBudgetCoeff
householderCompactNormBudgetCoeffFactor
householderCompactNormBudgetCoeff_eq_u_add_abs_beta_norm_sq_mul_factor
householderCompactNormBudgetCoeffFactor_nonneg
householderCompactNormBudgetCoeffFactor_le_of_u_gamma_le
householderCompactNormBudgetCoeffFactor_le_of_u_cap_gamma_cap
householderCompactNormBudgetCoeff_le_of_abs_beta_norm_sq_le
householderCompactUpdateCoeff_nonneg
householderCompactNormBudgetCoeff_nonneg
householderCompactComponentBudget_le_updateCoeff_mul_norm
householderCompactNormBudget_le_normBudgetCoeff_mul
householderCompactRelativeBudget_le_normBudgetCoeff
householderCompactPanelRelativeBudget_le_mul_add_normBudgetCoeff
householderCompactNormBudget_le_mul_of_componentBudget_le_mul_abs
householderCompactRelativeBudget_le_of_normBudget_le_mul
householderCompactRelativeBudget_le_of_componentBudget_le_mul_abs
householderCompactPanelRelativeBudget_le_mul_add_of_column_rhs_le
householderCompactPanelRelativeBudget_le_mul_add_of_normBudget_le_mul
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householderCompactPanelRelativeBudget_le_mul_add_of_componentBudget_le_mul_abs
 Algorithms/QR/HouseholderQR.lean storedQRCompactStepNormBudgetCoeff
 storedQRCompactStepNormBudgetCoeff_nonneg
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 storedQRSignedStage_abs_beta_norm_sq_eq_two_of_den_ne_zero
 storedQRSignedStage_abs_beta_norm_sq_le_two_of_den_ne_zero
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 storedQRCompactStepNormBudgetCoeff_le_of_forall_den_ne_zero
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 storedQRCompactStepRelativeBudget_le_mul_add_of_normBudget_le_mul
 storedQRCompactStepRelativeBudget_le_mul_add_of_componentBudget_le_mul_abs
 storedQRCompactStepRelativeBudget_le_mul_add_normBudgetCoeff
 storedQRCompactSequenceRelativeBudget_le_mul_of_step_le
 Algorithms/LeastSquares/LSQRSolve.lean storedQRCompactSequenceProductExpr
 storedQRCompactSequenceProductBudget
 storedQRCompactSequenceProductExpr_le_budget
 storedQRCompactSequenceProductBudget_lt_one_of_forall_expr_lt
 storedQRCompactSequenceProductBudget_lt_one_of_forall_pivot_product
 storedQRCompactSequenceProductBudget_lt_one_of_relativeBudget_eq_zero
 StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
 normSqBudget_rowBudgetControl_globalProduct
 StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
 normSqBudget_rowMaxBudgetFactor_globalProduct
 LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
 explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_rowBudgetControl_
 globalProduct
 LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
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 globalProduct
 LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
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 rowBudgetControl_globalProduct
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 storedQRDiagDominantInvFactorBudget
 storedQRDiagDominantInvFactor_le_budget
 storedQRDiagDominantInvFactorBudget_nonneg
 storedQRPivotColumnNormBudget
 storedQRPivotColumnNorm_le_budget
 storedQRPivotColumnNormBudget_nonneg
 storedQRCompactSequenceProductBudget_lt_one_of_diagDominant_finite_max_smallness

storedQRCompactSequenceProductBudget_lt_one_of_diagDominant_finite_max_relativeBudget_le
 storedQRCompactSequenceProductBudget_lt_one_of_diagDominant_finite_max_stepBudget_le
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 storedQRCompactStepNormBudgetCoeff_le_budget
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 storedQRPivotColumnNormBudget_le_of_forall_le
 storedQRCompactStepNormBudgetCoeffBudget_le_of_forall_le
 storedQRCompactStepNormBudgetCoeffBudget_le_of_forall_abs_beta_norm_sq_le
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 storedQRCompactSequenceProductBudget_lt_one_of_diagDominant_finite_max_normBudgetCoeffBudget
 storedQRCompactNormBudgetCoeffSmallness_of_bounds
 storedQRCompactSequenceProductBudget_lt_one_of_diagDominant_finite_max_normBudgetCoeffBudget_bounds
 storedQRCompactNormBudgetCoeffSmallness_of_pointwise_bounds
 storedQRCompactSequenceProductBudget_lt_one_of_diagDominant_finite_max_normBudgetCoeffBudget_pointwise_bounds
 storedQRCompactNormBudgetCoeffSmallness_of_pointwise_absBetaNormSq_bounds
 storedQRCompactNormBudgetCoeffSmallness_of_pointwise_source_den_ne_zero_bounds
 storedQRCompactNormBudgetCoeffSmallness_of_pointwise_source_den_ne_zero_simple_bounds
 storedQRCompactNormBudgetCoeffSmallness_of_pointwise_source_den_ne_zero_cap_bounds
 storedQRCompactNormBudgetCoeffSmallness_of_pointwise_source_den_ne_zero_uGammaCapBounds
 storedQRCompactNormBudgetCoeffSmallness_of_pointwise_source_den_ne_zero_uRationalGammaCapBounds
 storedQRCompactNormBudgetCoeffSmallness_of_source_den_ne_zero_uRationalGammaCanonicalBounds
 storedQRCompactSequenceProductBudget_lt_one_of_diagDominant_finite_max_absBetaNormSqPointwiseBounds
 storedQRCompactSequenceProductBudget_lt_one_of_diagDominant_finite_max_sourceDenPointwiseBounds
 storedQRCompactSequenceProductBudget_lt_one_of_diagDominant_finite_max_sourceDenSimpleBounds
 storedQRCompactSequenceProductBudget_lt_one_of_diagDominant_finite_max_sourceDenCapBounds
 storedQRCompactSequenceProductBudget_lt_one_of_diagDominant_finite_max_sourceDenUGammaCapBounds
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 StoredQROffDiagonalControlInvariant.of_diagDominant_finiteMaxSmallness
 StoredQROffDiagonalControlInvariant.of_diagDominant_finiteMaxNormBudgetCoeffSmallness

StoredQROffDiagonalControlInvariant.of_diagDominant_finiteMaxNormBudgetCoeffBoundedSmallness
 StoredQROffDiagonalControlInvariant.of_diagDominant_finiteMaxNormBudgetCoeffPointwiseBounds
 StoredQROffDiagonalControlInvariant.of_diagDominant_finiteMaxAbsBetaNormSqPointwiseBounds
 StoredQROffDiagonalControlInvariant.of_diagDominant_finiteMaxSourceDenPointwiseBounds
 StoredQROffDiagonalControlInvariant.of_diagDominant_finiteMaxSourceDenSimpleBounds
 StoredQROffDiagonalControlInvariant.of_diagDominant_finiteMaxSourceDenCapBounds
 StoredQROffDiagonalControlInvariant.of_diagDominant_finiteMaxSourceDenUGammaCapBounds
 StoredQROffDiagonalControlInvariant.of_diagDominant_finiteMaxSourceDenURationalGammaCapBounds
 StoredQROffDiagonalControlInvariant.of_diagDominant_finiteMaxSourceDenURationalGammaCanonicalBounds
 LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_
 fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_diagDominant_
 finiteMaxSourceDenURationalGammaCanonicalBounds
 Algorithms/RandNLA/LeastSquaresSketch.lean leverageTraceProbability_eventProb_fl_
 rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
 qr_diagDominant_finiteMaxSourceDenURationalGammaCanonicalBounds_solver
 Algorithms/LeastSquares/LSQRSolve.lean LSQRSolveBackwardError.of_stored_trailing_
 householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
 leadingBlock_det_ne_zero_diagDominant_finiteMaxNormBudgetCoeffPointwiseBounds_concreteDual
 LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
 explicitCompactBudget_of_signed_alpha_diagDominant_finiteMaxNormBudgetCoeffPointwiseBounds_
 concreteDual
 StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
 normSqBudget_diagDominant_finiteMaxSmallness
 StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_kappaInf_
 selfNorm_dualBudget_diagDominant_finiteMaxSmallness
 LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
 explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget_
 diagDominant_finiteMaxSmallness
 storedQRSignedStage_normSqBudget_of_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget
 StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
 normSqBudget_signedStageUniformBudget_product_of_normSqBudget_activePrefix_activeBlockRecurrence_
 globalCompactBudget_completedColumns_globalProduct_offdiag_rows
 LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
 gamma_bound_explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_normSqBudget_
 signedStageUniformBudget_product_of_normSqBudget_activePrefix_activeBlockRecurrence_
 globalCompactBudget_completedColumns_globalProduct_offdiag_rows
 StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_kappaInf_
 selfNorm_dualBudget_signedStageUniformBudget_globalCompactBudget_completedColumns_globalProduct_
 offdiag_rows
 LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
 explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget_
 signedStageUniformBudget_globalCompactBudget_completedColumns_globalProduct_offdiag_rows
 LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
 explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget_
 signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_globalProduct_
 offdiag_rows


```

Algorithms/RandNLA/LeastSquaresSketch.lean      leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_offDiagonalControl_solver_of_diagDominant_finiteMaxNormBudgetCoeffSmallness

Algorithms/LeastSquares/LSQRSolve.lean      not_forall_diagDominant_sourceDen_uCap_implies_
uRationalGammaCanonicalSmallness

not_forall_diagDominant_sourceDen_actualU_implies_uRationalGammaCanonicalSmallness

```

Corollary 452 (Canonical scalar smallness is not automatic). *The canonical rational-gamma route still has a genuine numerical smallness assumption. There is a one-dimensional witness with exact arithmetic packaged as `FPMODEL.exactWithUnitRoundoff(1/2)`, $A = [1]$, $\alpha = 0$, and $U_{\max} = 1/2$, for which the leading block is diagonally dominant, source nonbreakdown holds, $u \leq U_{\max}$, and $mU_{\max} < 1$, but the displayed canonical scalar condition is false. Consequently, diagonal dominance, source nonbreakdown, the unit-roundoff cap, and $mU_{\max} < 1$ do not by themselves discharge the scalar smallness obligation.*

The same obstruction remains if the displayed cap $U_{\max}$ is removed and the rational-gamma expression is written directly with the actual model roundoff u . Thus this is not just an artifact of introducing an explicit cap parameter: the canonical finite-max scalar inequality needs an independent scale, conditioning, or machine-model smallness argument.

Lean formalization.

```

Algorithms/LeastSquares/LSQRSolve.lean      not_forall_diagDominant_sourceDen_uCap_implies_
uRationalGammaCanonicalSmallness

not_forall_diagDominant_sourceDen_actualU_implies_uRationalGammaCanonicalSmallness

```

Corollary 453 (Source nonbreakdown from signed trailing norms). *In the canonical rational-gamma QR route, the raw Householder denominator condition $v_k^\top v_k \neq 0$ may be replaced by the signed-alpha source facts*

$$\alpha_k^2 = \|A_k(k : m, k)\|_2^2, \quad \|A_k(k : m, k)\|_2^2 > 0, \quad \alpha_k A_k(k, k) \leq 0.$$

Under these hypotheses, the canonical off-diagonal-control invariant, the stored-QR backward-error certificate, and the high-probability rounded sampled-row objective theorem are the same as in the previous canonical route, but no longer expose raw source denominator nonbreakdown.

Lean formalization.

```

Algorithms/LeastSquares/LSQRSolve.lean      storedQRSourceDenominator_ne_zero_of_
trailingNorm2Sq_pos_mul_nonpos

StoredQROffDiagonalControlInvariant.of_diagDominant_trailingNormPos_
finiteMaxSourceDenURationalGammaCanonicalBounds

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_diagDominant_trailingNormPos_
finiteMaxSourceDenURationalGammaCanonicalBounds

Algorithms/RandNLA/LeastSquaresSketch.lean      leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_diagDominant_trailingNormPos_finiteMaxSourceDenURationalGammaCanonicalBounds_solver

```

Corollary 454 (Source nonbreakdown from leading-block determinants). *In the previous canonical rational-gamma route, the positive trailing-norm hypothesis*

$$\|A_k(k : m, k)\|_2^2 > 0$$

may itself be derived from local determinant data. It is enough that, at each stage k , the previous transposed leading block and the current leading block have nonzero determinant, and the already-completed previous columns are zero below row k . Under these determinant and lower-zero hypotheses, the same canonical off-diagonal-control invariant, stored-QR backward-error certificate, and high-probability rounded sampled-row objective theorem hold.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      StoredQROffDiagonalControlInvariant.of_
diagDominant_leadingBlock_det_ne_zero_finiteMaxSourceDenURationalGammaCanonicalBounds

StoredQROffDiagonalControlInvariant.of_diagDominant_signedAlphaDef_leadingBlock_det_ne_zero_
finiteMaxSourceDenURationalGammaCanonicalBounds

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_diagDominant_leadingBlock_det_ne_zero_
finiteMaxSourceDenURationalGammaCanonicalBounds

Algorithms/RandNLA/LeastSquaresSketch.lean    leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_diagDominant_leadingBlock_det_ne_zero_finiteMaxSourceDenURationalGammaCanonicalBounds_solver
```

Corollary 455 (Current determinant from diagonal dominance). *In the determinant-facing canonical rational-gamma route above, the current leading-block determinant hypothesis may be omitted when the same current leading block is already assumed to satisfy IsDiagDominantUpper. The remaining determinant condition is only the previous transposed leading-block determinant; the current determinant follows from the local upper-triangular/nonzero-diagonal content of diagonal dominance.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      StoredQROffDiagonalControlInvariant.of_
diagDominant_signedAlphaDef_previousLeadingBlock_det_ne_zero_finiteMaxSourceDenURationalGammaCanonicalBounds

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_
bound_explicitCompactBudget_of_signed_alpha_diagDominant_previousLeadingBlock_det_ne_zero_
finiteMaxSourceDenURationalGammaCanonicalBounds

Algorithms/RandNLA/LeastSquaresSketch.lean    leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_diagDominant_previousLeadingBlock_det_ne_zero_finiteMaxSourceDenURationalGammaCanonicalBounds_
solver
```

Corollary 456 (Previous determinant from diagonal dominance). *In the same determinant-facing canonical rational-gamma route, the previous transposed leading-block determinant hypothesis may also be omitted. The previous $k \times k$ block is the transpose orientation of the top-left part of the current $(k + 1) \times (k + 1)$ leading block, so the upper-triangular and nonzero-diagonal components of IsDiagDominantUpper imply its nonsingularity. The remaining local shape assumption is the previous-column lower-zero field used by the trailing-norm nonbreakdown bridge.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      qrPreviousLeadingBlockTranspose_det_ne_zero_of_
diagDominant_leadingBlock

StoredQROffDiagonalControlInvariant.of_diagDominant_signedAlphaDef_lowerPrev_
finiteMaxSourceDenURationalGammaCanonicalBounds
```

```

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_
backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_diagDominant_lowerPrev_
finiteMaxSourceDenURationalGammaCanonicalBounds

Algorithms/RandNLA/LeastSquaresSketch.lean      leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_diagDominant_lowerPrev_finiteMaxSourceDenURationalGammaCanonicalBounds_solver

```

Corollary 457 (Previous lower-zero shape from the stored QR recurrence). *In the same canonical signed-alpha rational-gamma route, assume the stored Householder panel recurrence*

$$A_{k+1} = \text{fl_householderStoredPanelStep}(A_k, v_k, \beta_k), \quad k = 0, \dots, n-1.$$

Then the previous-column lower-zero field used above is not a separate mathematical hypothesis:

$$i \geq k, j < k \implies (A_k)_{i,j} = 0.$$

Consequently the determinant-facing solver and probability wrappers may be stated without the explicit lower-zero assumption. The remaining visible QR-domain assumptions on this route are local diagonal dominance/off-diagonal control, the displayed scalar smallness inequality, primitive unit-roundoff caps, conditioning fields, and the final concrete rectangular QR/preconditioner theorem.

Lean formalization.

```

Algorithms/LeastSquares/LSQRSolve.lean      storedQRPreviousColumnLowerZero_of_stored_
trailing_householder_sequence

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_
gamma_bound_explicitCompactBudget_of_signed_alpha_diagDominant_storedLower_
finiteMaxSourceDenURationalGammaCanonicalBounds

Algorithms/RandNLA/LeastSquaresSketch.lean  leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_diagDominant_storedLower_finiteMaxSourceDenURationalGammaCanonicalBounds_solver

```

Corollary 458 (Unit-roundoff cap nonnegativity is automatic). *In the stored-lower canonical rational-gamma route, the separate assumption*

$$0 \leq U_{\text{cap}}$$

is not an independent domain hypothesis. The FP model already contains $0 \leq u$, and therefore any primitive cap

$$u \leq U_{\text{cap}}$$

implies $0 \leq U_{\text{cap}}$. Consequently the solver and probability wrappers for this stored-lower route may be stated without an explicit nonnegativity field for U_{cap} .

Remark. This corollary removes only a proof-artifact hypothesis. It does not prove the primitive unit-roundoff cap $u \leq U_{\text{cap}}$, the scalar smallness inequality, local diagonal dominance/off-diagonal control, conditioning fields, or the final concrete rectangular QR/preconditioner theorem.

Lean formalization.

```

Algorithms/LeastSquares/LSQRSolve.lean      LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
diagDominant_storedLower_finiteMaxSourceDenURationalGammaCanonicalBounds_of_uCap

Algorithms/RandNLA/LeastSquaresSketch.lean  leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_diagDominant_storedLower_finiteMaxSourceDenURationalGammaCanonicalBounds_solver_of_uCap

```

Corollary 459 (Gamma-validity guards are automatic from the displayed cap). *In the same stored-lower canonical rational-gamma route, the cap hypotheses*

$$u \leq U_{\text{cap}}, \quad mU_{\text{cap}} < 1$$

imply `gammaValid fp m`. Since the rectangular solve also assumes $n \leq m$, monotonicity gives `gammaValid fp n`. At the sampled row-problem level the identical argument uses

$$u \leq U_{\text{cap}}, \quad sU_{\text{cap}} < 1, \quad n \leq s.$$

Thus the cap-based solver and probability statements need not list `gammaValid fp m`, `gammaValid fp s`, or `gammaValid fp n` as independent hypotheses.

Remark. This is still a hypothesis-surface cleanup. The primitive machine cap $u \leq U_{\text{cap}}$, the displayed scalar smallness inequality, local diagonal dominance/off-diagonal control, conditioning fields, and the final generic QR/preconditioner theorem remain visible or open.

Lean formalization.

```
Analysis/Rounding.lean gammaValid_of_u_le_cap
Algorithms/LeastSquares/LSQRSolve.lean      LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
diagDominant_storedLower_finiteMaxSourceDenURationalGammaCanonicalBounds_of_uCap_no_gammaValid
Algorithms/RandNLA/LeastSquaresSketch.lean    leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_
qr_diagDominant_storedLower_finiteMaxSourceDenURationalGammaCanonicalBounds_solver_of_uCap_no_
gammaValid
```

Corollary 460 (Stored-lower route at the actual unit roundoff). *In the stored-lower canonical rational-gamma route one may choose the displayed cap to be the model's actual unit roundoff:*

$$U_{\text{cap}} = u.$$

Then the primitive cap hypothesis $u \leq U_{\text{cap}}$ is discharged by reflexivity, and all U_{cap} -notation disappears from the theorem surface. The solver-level statement assumes `gammaValid fp m`; the sampled least-squares statement assumes `gammaValid fp s`. The remaining scalar smallness condition is the same canonical finite-max inequality, but with

$$G_m(u) = \frac{mu}{1 - mu}, \quad F_m(u) = u(1 + G_m(u))(1 + u) + u(1 + G_m(u)) + G_m(u) + u(1 + G_m(u))(1 + u)^2$$

in the solver theorem, and with m replaced by s in the sampled probability theorem.

Remark. This corollary is a theorem-surface reduction, not a proof of scalar smallness. The repository also contains a checked counterexample showing that the actual- u scalar smallness condition is not automatic from diagonal dominance, source-denominator nonbreakdown, and $mu < 1$. Local diagonal/off-diagonal control, conditioning fields, scalar smallness, and the final generic QR/preconditioner theorem therefore remain visible or open.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
diagDominant_storedLower_finiteMaxSourceDenURationalGammaCanonicalBounds_of_actualUnitRoundoff
Algorithms/RandNLA/LeastSquaresSketch.lean    leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_
stored_qr_diagDominant_storedLower_finiteMaxSourceDenURationalGammaCanonicalBounds_solver_of_
actualUnitRoundoff
```

Corollary 461 (Actual-unit-roundoff route with scalar validity guard). *In the stored-lower canonical rational-gamma QR route above, the ordinary `gammaValid` hypothesis can be replaced by the displayed scalar guard*

$$\mu < 1$$

for the local solver theorem, and by $su < 1$ for the sampled-row probability theorem. The proof chooses the actual unit roundoff $u = \text{fp}.u$ and reuses the cap-derived gamma-validity theorem with $U_{\max} = u$. The canonical finite-max scalar smallness inequality remains a separate visible hypothesis.

Remark. This is only a hypothesis-surface reduction: it does not prove diagonal dominance/off-diagonal control, scalar smallness, conditioning fields, or the generic implementation-backed rectangular QR/preconditioner theorem.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
diagDominant_storedLower_finiteMaxSourceDenURationalGammaCanonicalBounds_of_actualUnitRoundoff_
no_gammaValid

Algorithms/RandNLA/LeastSquaresSketch.lean    leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_
stored_qr_diagDominant_storedLower_finiteMaxSourceDenURationalGammaCanonicalBounds_solver_of_
actualUnitRoundoff_no_gammaValid
```

Corollary 462 (Signed-alpha scalar-smallness route elimination after validity reduction). *The scalar smallness hypothesis in the preceding actual-unit-roundoff route is not a consequence of the remaining signed Householder source fields. Even in a 1×1 exact-with-unit-roundoff model, local diagonal dominance, the concrete signed-alpha equation, source denominator nonbreakdown, and*

$$u < 1$$

do not force the canonical finite-max rational-gamma inequality. The `no_gammaValid` theorem states this after the validity guard has been replaced by the scalar condition $u = \text{fp}.u$, matching the post-LS.2g-gc surface.

Remark. This is a route-elimination theorem, not a positive scalar-smallness theorem: future route-1 progress must add stronger scale/conditioning hypotheses, prove a concrete machine-model smallness fact, or keep the finite-max scalar inequality visible.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean  not_forall_diagDominant_signedAlphaDef_sourceDen_
actualU_implies_uRationalGammaCanonicalSmallness

not_forall_diagDominant_signedAlphaDef_sourceDen_actualU_no_gammaValid_implies_
uRationalGammaCanonicalSmallness
```

Corollary 463 (Stored-recurrence scalar-smallness route elimination). *The scalar smallness hypothesis is still not forced after replacing the raw source-denominator field by the actual stored panel recurrence. There is a 1×1 exact-with-unit-roundoff witness satisfying the stored Householder panel recurrence, local diagonal dominance, the signed-alpha equation, and $u < 1$, while the canonical finite-max rational-gamma scalar is larger than 1. Thus the post-stored-lower route still needs an independent scale, conditioning, or concrete machine-model smallness theorem, or else the smallness condition must stay visible.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean    not_forall_diagDominant_signedAlphaDef_stored_
trailing_sequence_actualU_no_gammaValid_implies_uRationalGammaCanonicalSmallness
```

Corollary 464 (Stored-recurrence diagonal-dominance route elimination). *The local diagonal-dominance hypothesis is also not forced by the stored-panel actual-unit data. The existing 2×2 Householder diagonal-dominance counterexample follows the stored recurrence under exact floating-point arithmetic with $u = 0$, satisfies the signed-alpha equation and nonzero source denominators, and has $(2 : \mathbb{R})u < 1$. Its final displayed triangular block is still not diagonally dominant. Thus local diagonal dominance, or an equivalent off-diagonal-control invariant, must be proved from a stronger pivoted/sorted loop invariant or kept visible as a domain assumption.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean exactHouseholderQRDiagDominanceCounterexampleFP
exactHouseholderQRDiagDominanceCounterexample_stored_step
exactHouseholderQRDiagDominanceCounterexample_signed_alpha_def
not_forall_signedAlphaDef_stored_trailing_sequence_actualU_sourceDen_implies_diagDominant
```

Corollary 465 (Stored-recurrence active-pivot route elimination). *The displayed active-pivot policy is likewise not forced by the stored-panel actual-unit data. At the initial state of the same 2×2 witness, namely $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, the second column has active trailing norm square 5, while the displayed first pivot has active trailing norm square 1. Hence the finite selector theorem `householderActiveMaxPivotColumn\_pivot\_max` contradicts the claim that the displayed first pivot is `householderActiveMaxPivotColumn`. The universal stored-surface theorem therefore rules out deriving the active-pivot-policy equation from the stored recurrence, signed-alpha equation, nonzero source denominators, and $(2 : \mathbb{R})u < 1$. A positive proof must use a concrete pivoted/sorted loop invariant, or keep the active-pivot equation visible.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean exactHouseholderQRDiagDominanceCounterexample_not_
activeMaxPivotChoice
not_forall_signedAlphaDef_stored_trailing_sequence_actualU_sourceDen_implies_activeMaxPivotChoice
```

Corollary 466 (Stored-recurrence scalar row-budget route elimination). *The scalar row-budget fields in the active/prefix QR route are not forced by the stored-panel actual-unit data either. The same exact floating-point witness with $u = 0$ follows the stored recurrence, satisfies the signed-alpha equation and nonzero source denominators, and has $(2 : \mathbb{R})u < 1$. With the constant stage budget $B_k = 2$, the scalar stage-diagonal lower-bound defect is positive. With the nonnegative constant stage budget $B_k = 3$, the scalar stage-budget/row-max comparison defect is positive. Thus these scalar fields must be proved from a stronger pivoted/sorted/off-diagonal-control loop invariant or kept visible as source assumptions.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean    not_forall_signedAlphaDef_stored_trailing_
sequence_actualU_sourceDen_implies_stageDiagLowerDefectBudget_nonpos
not_forall_signedAlphaDef_stored_trailing_sequence_actualU_sourceDen_nonnegStageBudget_implies_
stageRowMaxComparisonDefectBudget_nonpos
```

Corollary 467 (Diagonal-dominant stored-recurrence scalar row-budget route elimination). *Even adding local diagonal dominance to the stored-panel actual-unit surface does not force the scalar row-budget fields. The exact stored sequence starting from $\begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}$ follows the stored recurrence under exact floating-point arithmetic with $u = 0$, satisfies the signed-alpha equation and nonzero source denominators, has $(2 : \mathbb{R})u < 1$, and is locally diagonally dominant at every displayed leading block. Still, the constant stage budget $B_k = 4$ makes the scalar stage-diagonal lower-bound defect positive, and the nonnegative constant stage budget $B_k = 3$ makes the scalar stage-budget/row-max comparison defect positive. Thus diagonal dominance does not remove the need to prove these scalar fields from a concrete loop invariant or keep them visible.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean storedDiagDominantComparisonCounterexample_stored_
step
storedDiagDominantComparisonCounterexample_diagDominant
not_forall_diagDominant_signedAlphaDef_stored_trailing_sequence_actualU_sourceDen_
nonnegStageBudget_implies_stageDiagLowerDefectBudget_nonpos
not_forall_diagDominant_signedAlphaDef_stored_trailing_sequence_actualU_sourceDen_
nonnegStageBudget_implies_stageRowMaxComparisonDefectBudget_nonpos
```

Corollary 468 (Active-pivot diagonal-dominant stored-recurrence comparison route elimination). *The same diagonal-dominant stored witness also satisfies the finite active max-pivot selector policy. Starting from $\begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}$, the exact stored sequence follows the stored recurrence with $u = 0$, satisfies the signed-alpha equation, source-denominator nonzero facts, actual-unit validity, local diagonal dominance, the active-pivot policy, and nonnegative stage budget $B_k = 3$. Nevertheless the scalar stage-budget/row-max comparison defect is positive. Thus active max-pivoting does not turn diagonal dominance into the scalar comparison invariant; that invariant must be proved from a stronger concrete loop argument or kept visible.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean storedDiagDominantComparisonCounterexample_
activeMaxPivotChoice
not_forall_diagDominant_activeMaxPivot_signedAlphaDef_stored_trailing_sequence_actualU_sourceDen_
nonnegStageBudget_implies_stageRowMaxComparisonDefectBudget_nonpos
```

Corollary 469 (Compact-product active-pivot diagonal-dominant stored-recurrence comparison route elimination). *Adding the stored right-hand-side recurrence and global compact-product smallness still does not force the scalar comparison invariant. Pair the same stored sequence from $\begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}$ with the zero right-hand-side trace. In exact arithmetic the RHS trace follows the stored RHS recurrence, the compact relative budget is zero, and hence the finite compact-product budget is strictly below one. The local diagonal dominance, active-pivot policy, signed-alpha equation, source-denominator facts, and actual-unit validity all remain available. Nevertheless $B_k = 3$ still makes the scalar stage-budget/row-max comparison defect positive.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean storedDiagDominantComparisonCounterexample_rhs_
step
```

```

storedDiagDominantComparisonCounterexample_compactSequenceRelativeBudget_eq_zero
storedDiagDominantComparisonCounterexample_compactSequenceProductBudget_lt_one
not_forall_diagDominant_activeMaxPivot_product_signedAlphaDef_stored_trailing_sequence_actualU_
sourceDen_nonnegStageBudget_implies_stageRowMaxComparisonDefectBudget_nonpos

```

Corollary 470 (Finite-max active-pivot diagonal-dominant stored-recurrence comparison route elimination). *The canonical finite-max compact-product scalar also does not force the scalar comparison invariant. For the same exact stored sequence and zero RHS trace, the compact relative budget is zero, so the displayed finite-max scalar used by the active branch reduces to $0 < 1$. The stored panel recurrence, stored RHS recurrence, signed-alpha equation, source-denominator facts, actual-unit validity, local diagonal dominance, active-pivot policy, and nonnegative stage budgets still hold. Nevertheless $B_k = 3$ leaves the scalar stage-budget/row-max comparison defect positive.*

Lean formalization.

```

Algorithms/LeastSquares/LSQRSolve.lean      storedDiagDominantComparisonCounterexample_
finiteMaxSmallness
not_forall_diagDominant_activeMaxPivot_finiteMaxSmallness_signedAlphaDef_stored_trailing_
sequence_actualU_sourceDen_nonnegStageBudget_implies_stageRowMaxComparisonDefectBudget_nonpos

```

Corollary 471 (Signed-stage global-budget finite-max active-pivot comparison route elimination). *Even adding the signed-stage finite global compact-budget recurrence does not force the scalar comparison invariant. Use the same exact stored sequence and zero RHS trace, but take the nonconstant nonnegative stage budget*

$$B_0 = 0, \quad B_1 = 3, \quad B_k = 10 \quad (k \geq 2).$$

The exact global compact-update budget is zero at both displayed stages, and the two-dimensional Cox–Higham factor is at most 3, so the finite global compact-budget recurrence holds. The stored panel recurrence, stored RHS recurrence, signed-alpha equation, source-denominator facts, actual-unit validity, local diagonal dominance, active-pivot policy, canonical finite-max compact-product scalar, and nonnegative stage budgets are all present. Nevertheless the stage-one scalar stage-budget/row-max comparison defect is positive.

Lean formalization.

```

Algorithms/LeastSquares/LSQRSolve.lean      storedDiagDominantComparisonCounterexample_
globalBudgetStageBudget_nonneg
storedDiagDominantComparisonCounterexample_globalCompactBudget_eq_zero
storedDiagDominantComparisonCounterexample_globalCompactBudget_recurrence
storedDiagDominantComparisonCounterexample_stageRowMaxComparisonDefectBudget_pos_
globalBudgetStageBudget
not_forall_diagDominant_activeMaxPivot_globalCompactBudget_finiteMaxSmallness_
signedAlphaDef_stored_trailing_sequence_actualU_sourceDen_nonnegStageBudget_implies_
stageRowMaxComparisonDefectBudget_nonpos

```

Corollary 472 (Row-max-granted signed-stage global-budget comparison route elimination). *The finite row-max/diagonal defect scalar is not enough to recover the missing stage-budget/row-max comparison scalar. For the same exact stored sequence, zero RHS trace, and nonconstant stage budget $B_0 = 0$, $B_1 = 3$, $B_k = 10$ for $k \geq 2$, local diagonal dominance packages*

$\text{storedQRRowMaxDiagDefectBudget} \leq 0$. The stored panel and RHS recurrences, signed-alpha equation, source-denominator facts, actual-unit validity, active-pivot policy, signed-stage global compact-budget recurrence, canonical finite-max compact-product scalar, row-max defect scalar, and non-negative stage budgets all hold together. Nevertheless the stage-one scalar stage-budget/row-max comparison defect is positive.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      storedDiagDominantComparisonCounterexample_
rowMaxDiagDefectBudget_nonpos

not_forall_diagDominant_activeMaxPivot_globalCompactBudget_finiteMaxSmallness_rowMaxDiagDefect_
signedAlphaDef_stored_trailing_sequence_actualU_sourceDen_nonnegStageBudget_implies_
stageRowMaxComparisonDefectBudget_nonpos
```

Corollary 473 (Determinant-granted signed-stage global-budget comparison route elimination). *Nonsingular displayed leading blocks are not enough to recover the missing stage-budget/row-max comparison scalar either. For the same exact stored sequence, zero RHS trace, and nonconstant stage budget $B_0 = 0$, $B_1 = 3$, $B_k = 10$ for $k \geq 2$, the displayed leading blocks have determinants 3 and -3 . The stored panel and RHS recurrences, signed-alpha equation, source-denominator facts, actual-unit validity, local diagonal dominance, active-pivot policy, row-max defect scalar, signed-stage global compact-budget recurrence, canonical finite-max compact-product scalar, determinant nonbreakdown, and nonnegative stage budgets all hold together. Nevertheless the stage-one scalar stage-budget/row-max comparison defect is positive.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      storedDiagDominantComparisonCounterexample_
leadingBlock_det_ne_zero

not_forall_diagDominant_activeMaxPivot_leadingBlock_det_ne_zero_globalCompactBudget_
finiteMaxSmallness_rowMaxDiagDefect_signedAlphaDef_stored_trailing_sequence_actualU_sourceDen_
nonnegStageBudget_implies_stageRowMaxComparisonDefectBudget_nonpos
```

Corollary 474 (Conditioning-granted signed-stage global-budget comparison route elimination). *Local conditioning and dual compact-budget data are not enough to recover the missing stage-budget/row-max comparison scalar either. For the same exact stored sequence, zero RHS trace, and nonconstant stage budget $B_0 = 0$, $B_1 = 3$, $B_k = 10$ for $k \geq 2$, the compact component budgets vanish in the exact arithmetic model. Taking the local κ_∞ budget to be the exact displayed condition expression and taking K to be the exact self-norm squared expression plus one supplies positive local conditioning and dual compact-budget fields. The stored panel and RHS recurrences, signed-alpha equation, source-denominator facts, actual-unit validity, local diagonal dominance, active-pivot policy, determinant nonbreakdown, row-max defect scalar, signed-stage global compact-budget recurrence, canonical finite-max compact-product scalar, and nonnegative stage budgets all hold together. Nevertheless the stage-one scalar stage-budget/row-max comparison defect is positive.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      storedDiagDominantComparisonCounterexample_
compactComponentBudget_eq_zero

storedDiagDominantComparisonCounterexampleKappaBudget
storedDiagDominantComparisonCounterexampleKappaNormSqBudget
storedDiagDominantComparisonCounterexample_kappaBudget_le
```

```

storedDiagDominantComparisonCounterexample_kappaNormSqBudget_pos
storedDiagDominantComparisonCounterexample_kappaNormSqBudget
storedDiagDominantComparisonCounterexample_dualBudget
not_forall_diagDominant_activeMaxPivot_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget_
globalCompactBudget_finiteMaxSmallness_rowMaxDiagDefect_signedAlphaDef_stored_trailing_sequence_
actualU_sourceDen_nonnegStageBudget_implies_stageRowMaxComparisonDefectBudget_nonpos

```

Corollary 475 (Final-surface signed-stage global-budget comparison route elimination). *Even the final active/prefix global-product surface does not recover the missing stage-budget/row-max comparison scalar automatically. Reuse the same exact stored sequence and zero RHS trace, but take*

$$B_0 = 3, \quad B_1 = 10, \quad B_k = 30 \quad (k \geq 2).$$

This final-surface budget satisfies the initial full-block and displayed row-budget fields, is non-negative and monotone, and still satisfies the exact signed-stage global compact-budget recurrence and global compact-product smallness. The determinant/nonbreakdown facts, local κ_∞ and self-norm budgets, dual compact-budget inequality, row-max defect scalar, active-pivot policy, and local diagonal dominance all remain available. Nevertheless the stage-one comparison scalar contains the positive displayed gap $10 - 2$. Thus the final surface must still prove $\text{storedQRStageRowMaxComparisonDefectBudget} \leq 0$ from a genuine loop invariant, or expose it as a source/domain assumption.

Lean formalization.

```

Algorithms/LeastSquares/LSQRSolve.lean storedDiagDominantComparisonCounterexampleFinalSurfaceStageBudget
storedDiagDominantComparisonCounterexample_finalSurfaceStageBudget_nonneg
storedDiagDominantComparisonCounterexample_finalSurfaceStageBudget_mono
storedDiagDominantComparisonCounterexample_finalSurface_init
storedDiagDominantComparisonCounterexample_finalSurface_initBlock
storedDiagDominantComparisonCounterexample_finalSurface_globalCompactBudget_recurrence
storedDiagDominantComparisonCounterexample_stageRowMaxComparisonDefectBudget_pos_
finalSurfaceStageBudget
not_forall_diagDominant_activeMaxPivot_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget_
init_globalProduct_globalCompactBudget_rowMaxDiagDefect_signedAlphaDef_stored_trailing_sequence_
actualU_sourceDen_nonneg_monoStageBudget_implies_stageRowMaxComparisonDefectBudget_nonpos

```

Corollary 476 (Final-surface source-denominator comparison route elimination). *The source-denominator rational-gamma finite-max route also does not recover the missing stage-budget/row-max comparison scalar automatically. On the same exact stored sequence and zero RHS trace, choose the source cap $U_{\max} = 0$. Then the rational-gamma term and the Householder-factor cap term vanish, so the canonical source-denominator scalar smallness inequality is true independently of the finite maximum budgets. All final-surface fields from the preceding corollary remain available, with the raw global-product smallness replaced by the displayed source-denominator cap-smallness fields. Nevertheless the stage-one comparison scalar still contains the positive gap $10 - 2$. Thus the canonical source-denominator scalar can be a valid product route assumption, but it cannot replace the separate $\text{storedQRStageRowMaxComparisonDefectBudget} \leq 0$ invariant.*

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      storedDiagDominantComparisonCounterexample_
finalSurface_sourceDenURationalGammaCanonicalSmallness

not_forall_diagDominant_activeMaxPivot_leadingBlock_det_ne_zero_kappaInf_selfNorm_
dualBudget_init_sourceDenURationalGammaCanonical_globalCompactBudget_rowMaxDiagDefect_
signedAlphaDef_stored_trailing_sequence_actualU_sourceDen_nonneg_monoStageBudget_implies_
stageRowMaxComparisonDefectBudget_nonpos
```

Corollary 477 (Horizon-clamped source-denominator QR handoff). *For the active/prefix diagonal-dominant source-denominator route, the stage-budget sequence need not be globally monotone on all natural indices. If the signed-stage compact recurrence and nonnegativity hold through the QR horizon, and if the scalar comparison defect*

$$\text{storedQRStageRowMaxComparisonDefectBudget} \leq 0$$

holds for the original stage budget, then the same comparison certificate holds after replacing the budget by qrStageHorizonBudget n . The clamped budget is globally monotone by construction and agrees with the original budget on every stage used by the QR loop. Consequently the source-denominator rational-gamma source-control theorem, and its sampled equation-(8), actual-unit, and stored-lower probability siblings, can be stated without a samplewise global stage-budget monotonicity hypothesis. The scalar comparison defect, local diagonal dominance, active-pivot policy, signed-stage recurrence, and source-denominator cap-smallness assumptions remain visible.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      storedQRStageRowMaxComparisonDefectBudget_nonpos_
of_horizonBudget

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_diagDominant_
signedStageUniformBudget_globalCompactBudget_activeMaxPivot_completedColumns_
finiteMaxSourceDenURationalGammaCanonicalBounds_stageRowMaxComparisonDefect_offdiag_rows_of_
horizonBudget

Algorithms/RandNLA/LeastSquaresSketch.lean      leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_activePrefix_finiteMaxSourceDenURationalGammaCanonicalBounds_activeMaxPivot_diagDominant_
stageRowMaxComparisonDefect_solver_of_horizonBudget

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_
augmentedSpan_sample_budget_of_objective_error_and_stored_qr_activePrefix_
finiteMaxSourceDenURationalGammaCanonicalBounds_activeMaxPivot_diagDominant_
stageRowMaxComparisonDefect_solver_of_actualUnitRoundoff_no_gammaValid_of_horizonBudget

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_
augmentedSpan_sample_budget_of_objective_error_and_stored_qr_activePrefix_storedLower_
finiteMaxSourceDenURationalGammaCanonicalBounds_activeMaxPivot_diagDominant_
stageRowMaxComparisonDefect_solver_of_actualUnitRoundoff_no_gammaValid_of_horizonBudget
```

Corollary 478 (Equation (8) assembly from active/prefix global-product QR). *Let A, b be the original least-squares data and let the row sampler use leverage probabilities from the augmented span basis of $[A \ b]$. Assume the Bennett sample-budget hypotheses for the two-sided finite-Loewner event, the positive-probability support event, and the rounded sampled-row objective slack condition. For each sample trace, suppose the rounded stored QR loop satisfies the signed-stage recurrences, either the norm-square nonbreakdown margin or the local κ_∞ /dual-budget hypotheses that imply*

it, local leading-block nonsingularity, the scalar global compact-step recurrence, the off-diagonal-row diagonal lower bounds, pivot maximality, and the global compact-product condition

$$\Pi_{\max} < 1.$$

If the reported solution is the back-substitution solution from the final stored QR block, and the reported perturbation radii are the repository `qrSolveFinalGramBudget` and `qrSolveFinalRhsBudget`, then with probability at least $1 - \delta$,

$$\|Ax_{\text{comp}} - b\|_2^2 \leq (1 + \eta) \|Ax_{\star} - b\|_2^2.$$

Corollary 479 (Equation (8) horizon-clamped active/prefix QR assembly). *In the active/prefix global-product equation (8) assembly above, the samplewise global stage-budget monotonicity hypothesis may be removed from the theorem surface. It is enough to assume the finite compact-step recurrence and nonnegative stage budgets for each sample trace, together with the same diagonal lower-bound, pivot-maximality, nonbreakdown, and global-product fields. The proof applies the horizon-clamped stored-QR source-control wrapper to each sample, so no budget behavior beyond the QR horizon is asserted or needed by the probability theorem. The same reduction also composes with the local κ_{∞} /dual-budget adapter that derives the norm-square nonbreakdown margin, and with the active-max-pivot wrapper that recovers raw pivot maximality from the finite selector.*

Lean formalization.

```
Algorithms/RandNLA/LeastSquaresSketch.lean      leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_activePrefix_globalProduct_solver_of_horizonBudget

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_kappaInf_dualBudget_
solver_of_horizonBudget

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_activeMaxPivot_
kappaInf_dualBudget_solver_of_horizonBudget

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_activeMaxPivot_
rowMaxDiagDefect_stageBudgetLeRowMax_kappaInf_dualBudget_solver_of_horizonBudget

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_activeMaxPivot_
rowMaxDiagDefect_stageBudgetLeRowMax_kappaInf_dualBudget_solver_of_actualUnitRoundoff_no_
gammaValid_of_horizonBudget
```

Lean formalization.

```
Algorithms/RandNLA/LeastSquaresSketch.lean      leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_activePrefix_globalProduct_solver

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_solver_of_horizonBudget

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_kappaInf_dualBudget_
solver
```

```

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_kappaInf_dualBudget_
solver_of_horizonBudget

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_activeMaxPivot_
kappaInf_dualBudget_solver

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_activeMaxPivot_
kappaInf_dualBudget_solver_of_horizonBudget

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_activeMaxPivot_
rowMaxDiagDefect_stageBudgetLeRowMax_kappaInf_dualBudget_solver

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_activeMaxPivot_
rowMaxDiagDefect_stageBudgetLeRowMax_kappaInf_dualBudget_solver_of_horizonBudget

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_activeMaxPivot_
rowMaxDiagDefect_stageBudgetLeRowMax_kappaInf_dualBudget_solver_of_actualUnitRoundoff_no_
gammaValid

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_activePrefix_globalProduct_activeMaxPivot_
rowMaxDiagDefect_stageBudgetLeRowMax_kappaInf_dualBudget_solver_of_actualUnitRoundoff_no_
gammaValid_of_horizonBudget

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_rowBudgetControl_globalProduct_kappaInf_
dualBudget_solver

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_rowBudgetControl_globalProduct_activeMaxPivot_
kappaInf_dualBudget_solver

```

Corollary 480 (Equation (8) assembly from diagonal-dominant QR). *Let A, b be the original least-squares data and use the same leverage sampling law on the augmented span basis of $[A \ b]$. Assume the Bennett sample-budget hypotheses, the positive-probability support event, and the rounded sampled-row objective slack condition from the preceding equation (8) theorem. For each sample trace, suppose the rounded stored QR loop satisfies the signed-stage recurrences. The compact-product assumption may be supplied either as the global condition $\Pi_{\max} < 1$, or in the canonical finite-max form*

$$2D_* (s (c_{\text{seq}} N_*)^2) < 1,$$

where s is the sampled row count, D_ is the maximum local diagonal-dominant inverse factor, N_* is the maximum displayed pivot-column norm, and c_{seq} is the stored QR sequence budget for that sample trace. Suppose further that every displayed local leading block S_k is upper-triangular and diagonally dominant in the repository sense*

$$IsDiagDominantUpper (k + 1) S_k,$$

and that the local determinant and compact-smallness hypotheses are available. There are two formalized theorem surfaces. The first keeps the local κ_∞ self-norm and dual compact-budget hypotheses explicit, so that the dimensioned norm-square nonbreakdown margin follows from the condition-number route. The second is the concrete-dual finite-max variant: it uses the repository's diagonal-dominant inverse-budget theorem to choose the auxiliary inverse budget internally, and therefore does

not expose separate κ_∞ , K_k , or dual compact-budget fields. In that concrete-dual form, the remaining QR-domain assumptions are local leading block nonsingularity, local diagonal dominance, and the displayed scalar finite-max smallness inequality. The final determinant-free concrete-dual form observes that `IsDiagDominantUpper` already contains upper triangular shape and nonzero diagonal entries, so local leading-block nonsingularity follows from diagonal dominance itself. In that form, the visible QR-domain assumptions are local diagonal dominance and the displayed scalar finite-max smallness inequality. Under any of these theorem surfaces, the stored-QR back-substitution solution $\hat{x}$ satisfies, with probability at least $1 - \delta$,

$$\|A\hat{x} - b\|_2^2 \leq (1 + \eta) \|Ax_\star - b\|_2^2.$$

Lean formalization.

```

Algorithms/LeastSquares/LSQRSolve.lean      StoredQRSourceOffDiagonalControl.of_stored_
trailing_sequence_leadingBlock_det_ne_zero_normSqBudget_diagDominant_globalProduct

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_kappaInf_
selfNorm_dualBudget_diagDominant_globalProduct

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget_
diagDominant_globalProduct

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_
normSqBudget_diagDominant_finiteMaxSmallness

StoredQRSourceOffDiagonalControl.of_stored_trailing_sequence_leadingBlock_det_ne_zero_kappaInf_
selfNorm_dualBudget_diagDominant_finiteMaxSmallness

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_kappaInf_selfNorm_dualBudget_
diagDominant_finiteMaxSmallness

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_leadingBlock_det_ne_zero_diagDominant_finiteMaxSmallness_
concreteDual

det_ne_zero_of_diagDominantUpper

LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitCompactBudget_of_signed_alpha_diagDominant_finiteMaxSmallness_concreteDual

Algorithms/RandNLA/LeastSquaresSketch.lean      leverageTraceProbability_eventProb_fl_
rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget_of_objective_error_and_stored_
qr_diagDominant_globalProduct_kappaInf_dualBudget_solver

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_offDiagonalControl_solver_of_diagDominant_
finiteMaxSmallness

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_diagDominant_finiteMaxSmallness_kappaInf_
dualBudget_solver

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_diagDominant_finiteMaxSmallness_concreteDual_
solver

leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_stored_qr_diagDominant_finiteMaxSmallness_concreteDual_
solver_of_diagDominant

```

Corollary 481 (Least-squares QR certificate from prefix-span and active-entry budget). *In the stored trailing Householder QR loop, assume the same rounded matrix and right-hand-side recurrences used in the stored-loop solver certificate. At every pivot $k < n$, suppose the prefix-span QR nonbreakdown invariant holds, the Householder normalization and sign choice hold, and there is an active row $i_k \geq k$ whose pivot-column magnitude dominates the compact diagonal update budget,*

$$B_k < |A_k(i_k, k)|.$$

Assume also the columnwise compact-update budget domination for the matrix panel and right-hand side, the γ -validity hypotheses, and the final Gram/RHS norm-budget bounds. Then the computed back-substitution vector

$$\hat{x} = \text{fl_backSub}(\hat{R}, \hat{c})$$

satisfies the repository's rectangular QR least-squares backward-error certificate,

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Corollary 482 (Least-squares QR certificate from prefix-span and norm-square budget). *In the same stored trailing Householder QR loop, assume prefix-span nonbreakdown, sign choice, Householder normalization, columnwise compact-update budget domination, γ -validity, and final Gram/RHS norm-budget bounds. Replace the active-entry witness by the dimensioned per-pivot margin*

$$mB_k^2 < \|A_k(k:m, k)\|_2^2.$$

Then the computed back-substitution vector

$$\hat{x} = \text{fl_backSub}(\hat{R}, \hat{c})$$

satisfies the same rectangular QR least-squares backward-error certificate,

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Corollary 483 (Least-squares QR certificate from prefix-span and leading-dual budget). *In the same stored trailing Householder QR loop, assume prefix-span, Householder normalization, sign choice, columnwise compact-update budget domination, γ -validity, and final Gram/RHS norm-budget bounds. For each pivot $k < n$, suppose there is a leading dual row satisfying*

$$\|L_{k,*}\|_2^2 \leq K_k, \quad K_k > 0,$$

and the compact diagonal update budget obeys

$$mB_k^2 < \frac{1}{K_k}.$$

Then the computed back-substitution vector

$$\hat{x} = \text{fl_backSub}(\hat{R}, \hat{c})$$

satisfies the same rectangular QR least-squares backward-error certificate,

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Corollary 484 (Prefix-span leading-dual QR certificate with repository budgets). *In the preceding leading-dual setting, the final perturbation radii may be chosen to be the repository radii*

$$c_G = \text{qrSolveFinalGramBudget}(A, \hat{R}, c_{\text{step}}), \quad c_g = \text{qrSolveFinalRhsBudget}(A, b, \hat{R}, c_{\text{step}}).$$

Moreover, if

$$c_{\text{step}} = \text{storedQRCompactSequenceRelativeBudget}(A_0, A_1, \dots, A_n; b_0, b_1, \dots, b_n; \alpha),$$

then the columnwise panel and RHS compact-update domination hypotheses are discharged by the local stored-loop budget lemmas. Thus the computed vector $\hat{x} = \text{fl_backSub}(\hat{R}, \hat{c})$ satisfies

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; \text{qrSolveFinalGramBudget}, \text{qrSolveFinalRhsBudget})$$

under the source-faithful prefix-span, leading-dual, sign-choice, and dual compact-smallness hypotheses.

Corollary 485 (Least-squares QR certificate from prefix-span and local inverse row budget). *In the same stored trailing Householder QR loop, replace the supplied ambient dual row by a local left inverse C_k of the leading block S_k . Assume prefix-span, Householder normalization, sign choice, columnwise compact-update budget domination, γ -validity, and final Gram/RHS norm-budget bounds. For every pivot $k < n$, suppose*

$$\|C_k(k, :)\|_2^2 \leq K_k, \quad K_k > 0, \quad mB_k^2 < \frac{1}{K_k}.$$

Then the computed back-substitution vector

$$\hat{x} = \text{fl_backSub}(\hat{R}, \hat{c})$$

satisfies the same rectangular QR least-squares backward-error certificate,

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Corollary 486 (Local inverse row-budget QR certificate with repository budgets). *In the preceding local-leading-block inverse row setting, the final radii may again be chosen as $\text{qrSolveFinalGramBudget}$ and $\text{qrSolveFinalRhsBudget}$. If the step constant is chosen as*

$$c_{\text{step}} = \text{storedQRCompactSequenceRelativeBudget}(A_0, A_1, \dots, A_n; b_0, b_1, \dots, b_n; \alpha),$$

then the columnwise panel and RHS compact-update domination hypotheses are also supplied by the stored compact-budget lemmas. Hence the local left inverse row-norm budget

$$\|C_k(k, :)\|_2^2 \leq K_k, \quad K_k > 0, \quad mB_k^2 < \frac{1}{K_k}$$

implies the same repository-budgeted $\text{LSQRSolveBackwardError}$ certificate.

Corollary 487 (Stored-prefix-span local inverse row-budget QR certificate). *In the preceding row-budget setting, the separate prefix-span hypothesis can be replaced by a previous-block local inverse hypothesis. For every pivot $k < n$, assume that the transposed previous leading block T_k has a local left inverse C_k^{prev} , and assume the current leading block S_k has the local left inverse C_k used in the row-budget condition. Then the stored panel recurrence supplies the lower-zero shape of the*

previous columns, and this shape plus $C_k^{\text{prev}}T_k = I$ implies the prefix-span invariant. Hence the repository-budgeted

LSQRSolveBackwardError

certificate follows from the row-norm budget

$$\|C_k(k, :)\|_2^2 \leq K_k, \quad K_k > 0, \quad mB_k^2 < \frac{1}{K_k},$$

sign choice, γ -validity, and the visible compact-smallness hypothesis, without assuming prefix-span separately.

Corollary 488 (Least-squares QR certificate from prefix-span and local inverse Frobenius budget). *In the same stored trailing Householder QR loop, suppose each leading block has a local left inverse C_k , and replace the row-norm condition above by*

$$\|C_k\|_F^2 \leq K_k, \quad K_k > 0, \quad mB_k^2 < \frac{1}{K_k}.$$

Under the same prefix-span, Householder, compact-update, γ -validity, and final Gram/RHS norm-budget hypotheses, the computed vector

$$\hat{x} = \text{fl_backSub}(\hat{R}, \hat{c})$$

satisfies

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Corollary 489 (Least-squares QR certificate from prefix-span and local inverse infinity-norm budget). *In the same stored trailing Householder QR loop, suppose each leading block has a local left inverse C_k , and replace the Frobenius condition above by*

$$(k+1)\|C_k\|_\infty^2 \leq K_k, \quad K_k > 0, \quad mB_k^2 < \frac{1}{K_k}.$$

Under the same prefix-span, Householder, compact-update, γ -validity, and final Gram/RHS norm-budget hypotheses, the computed vector

$$\hat{x} = \text{fl_backSub}(\hat{R}, \hat{c})$$

satisfies

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Corollary 490 (Repository-budget inverse-norm QR certificates). *In the Frobenius and infinity inverse-budget corollaries above, the final Gram/RHS radii may be chosen as $\text{qrSolveFinalGramBudget}$ and $\text{qrSolveFinalRhsBudget}$. If the step constant is chosen as*

$$c_{\text{step}} = \text{storedQRCompactSequenceRelativeBudget}(A_0, A_1, \dots, A_n; b_0, b_1, \dots, b_n; \alpha),$$

then the stored compact-budget lemmas also supply the columnwise panel and RHS compact-update domination hypotheses. Thus either

$$\|C_k\|_F^2 \leq K_k \quad \text{or} \quad (k+1)\|C_k\|_\infty^2 \leq K_k$$

together with $K_k > 0$, prefix-span, sign choice, and $mB_k^2 < 1/K_k$ gives the same repository-budgeted LSQRSolveBackwardError certificate.

Corollary 491 (Repository-budget inverse-norm certificates without separate prefix-span). *For the Frobenius and infinity branches in the previous corollary, the separate prefix-span hypothesis can also be removed when the stored panel recurrence is available and each previous transposed leading block has a local left inverse. Under those previous-block local inverse hypotheses, the same Frobenius or infinity inverse-budget assumptions imply the repository-budgeted LSQRSolveBackwardError certificate.*

Corollary 492 (Signed-alpha local-inverse row certificate). *In the stored-prefix-span local-inverse row certificate, the separate assumptions*

$$\alpha_k^2 = \|(\hat{A}_k)_{k:m,k}\|_2^2, \quad \alpha_k(\hat{A}_k)_{kk} \leq 0$$

may be replaced by the source Householder choice

$$\alpha_k = \text{signedHouseholderAlpha}(\|(\hat{A}_k)_{k:m,k}\|_2, (\hat{A}_k)_{kk}).$$

Under the same stored recurrence, previous/current local left-inverse, row-norm budget, and compact-smallness hypotheses, the repository-budgeted LSQRSolveBackwardError certificate follows.

Corollary 493 (Signed-alpha local-inverse inverse-norm certificates). *In the stored-prefix-span Frobenius and infinity local-inverse certificates, the same source Householder choice*

$$\alpha_k = \text{signedHouseholderAlpha}(\|(\hat{A}_k)_{k:m,k}\|_2, (\hat{A}_k)_{kk})$$

may replace the separate squared-alpha and sign-choice assumptions. Thus either inverse-norm budget

$$\|C_k\|_F^2 \leq K_k \quad \text{or} \quad (k+1)\|C_k\|_\infty^2 \leq K_k$$

together with the previous/current local left-inverse hypotheses, $K_k > 0$, and $mB_k^2 < 1/K_k$, gives the repository-budgeted LSQRSolveBackwardError certificate without assuming prefix-span, the squared-alpha identity, or the sign inequality separately.

Corollary 494 (Signed-alpha determinant local-inverse certificates). *Let P_k be the previous $k \times k$ transposed leading block and let S_k be the current $(k+1) \times (k+1)$ leading block of the stored Householder panel at step k . In the signed-alpha stored-prefix-span certificates above, the previous/current local left-inverse witnesses may be replaced by*

$$\det P_k \neq 0, \quad \det S_k \neq 0,$$

using the repository nonsingular inverses

$$C_k^{\text{prev}} = \text{nonsingInv}(P_k), \quad C_k = \text{nonsingInv}(S_k).$$

Thus any one of the visible inverse-budget assumptions

$$\|C_k(k, :)\|_2^2 \leq K_k, \quad \|C_k\|_F^2 \leq K_k, \quad (k+1)\|C_k\|_\infty^2 \leq K_k$$

together with $K_k > 0$, the signed Householder alpha rule, the stored panel recurrence, and $mB_k^2 < 1/K_k$, gives the repository-budgeted LSQRSolveBackwardError certificate. The theorem does not derive the determinant facts, inverse-budget inequality, or compact-smallness inequality from the computed loop.

Corollary 495 (Signed-alpha determinant certificate from κ_∞ budgets). *In the infinity-norm branch of the preceding corollary, the direct inverse budget may be replaced by a visible local condition-number budget. Suppose*

$$\kappa_\infty(S_k, \text{nonsingInv}(S_k)) \leq \kappa_k, \quad (k+1) \left(\frac{\kappa_k}{\|S_k\|_\infty} \right)^2 \leq K_k.$$

Together with $\det P_k \neq 0$, $\det S_k \neq 0$, $K_k > 0$, the signed Householder alpha rule, the stored panel recurrence, and $mB_k^2 < 1/K_k$, these hypotheses imply the same repository-budgeted LSQRsolveBackwardError certificate.

Corollary 496 (Signed-alpha triangular leading-block certificate). *In the previous corollary, the determinant hypotheses may be replaced by a visible triangular leading-block invariant. Suppose the stored panel at each step is upper triangular on its leading part and all displayed leading diagonal entries are nonzero:*

$$(\hat{A}_k)_{ij} = 0 \quad (j < i), \quad (\hat{A}_k)_{rr} \neq 0 \quad (0 \leq r \leq k).$$

Then the previous transposed leading block P_k and current leading block S_k are nonsingular, and the signed-alpha κ_∞ certificate above applies under the same visible condition-number, self-norm, and compact-smallness hypotheses.

Corollary 497 (Least-squares QR certificate from nonsingular local inverse-norm budgets). *In the previous corollary, the explicit local left-inverse witness C_k can be replaced by a determinant hypothesis and the repository nonsingular inverse. Let S_k be the local $(k+1) \times (k+1)$ leading block. Suppose*

$$\det S_k \neq 0, \quad (k+1) \|\text{nonsingInv}(S_k)\|_\infty^2 \leq K_k, \quad K_k > 0, \quad mB_k^2 < \frac{1}{K_k}.$$

Under the same prefix-span, Householder sign, compact-update, γ -validity, and final Gram/RHS norm-budget hypotheses as above, the computed vector

$$\hat{x} = \text{fl_backSub}(\hat{R}, \hat{c})$$

satisfies

$$\text{LSQRsolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Corollary 498 (Least-squares QR certificate from local κ_∞ budgets). *In the previous corollary, the inverse- ∞ budget may be derived from a condition-number route. Let S_k be the local $(k+1) \times (k+1)$ leading block. Suppose*

$$\det S_k \neq 0, \quad 0 < \rho_k \leq \|S_k\|_\infty, \quad \kappa_\infty(S_k, \text{nonsingInv}(S_k)) \leq \kappa_k,$$

and suppose the visible squared budget satisfies

$$(k+1) \left(\frac{\kappa_k}{\rho_k} \right)^2 \leq K_k, \quad K_k > 0, \quad mB_k^2 < \frac{1}{K_k}.$$

Under the same prefix-span, Householder sign, compact-update, γ -validity, and final Gram/RHS norm-budget hypotheses as above, the computed vector

$$\hat{x} = \text{fl_backSub}(\hat{R}, \hat{c})$$

satisfies

$$\text{LSQRsolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Corollary 499 (Least-squares QR certificate from local κ_∞ budgets without an auxiliary ρ_k). *In the preceding condition-number corollary, the auxiliary lower-bound parameter can be removed. Let S_k be the local $(k+1) \times (k+1)$ leading block. Suppose*

$$\det S_k \neq 0, \quad \kappa_\infty(S_k, \text{nonsingInv}(S_k)) \leq \kappa_k,$$

and suppose the visible squared budget satisfies

$$(k+1) \left(\frac{\kappa_k}{\|S_k\|_\infty} \right)^2 \leq K_k, \quad K_k > 0, \quad mB_k^2 < \frac{1}{K_k}.$$

Under the same prefix-span, Householder sign, compact-update, γ -validity, and final Gram/RHS norm-budget hypotheses as above, the computed vector

$$\hat{x} = \text{fl_backSub}(\hat{R}, \hat{c})$$

satisfies

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Corollary 500 (Self-norm κ_∞ QR certificate without abstract prefix span). *In the preceding self-norm condition-number corollary, the abstract prefix-span invariant may be replaced by determinant and shape hypotheses on the completed leading block. Let T_k be the transposed leading block of the first k rows and previous k columns, and let S_k be the current $(k+1) \times (k+1)$ leading block. Suppose*

$$\det T_k \neq 0, \quad A_k(i, j) = 0 \quad (i \geq k, j < k),$$

and also

$$\det S_k \neq 0, \quad \kappa_\infty(S_k, \text{nonsingInv}(S_k)) \leq \kappa_k, \quad (k+1) \left(\frac{\kappa_k}{\|S_k\|_\infty} \right)^2 \leq K_k.$$

Under the same Householder sign, compact-update, γ -validity, and final Gram/RHS norm-budget hypotheses as above, with $K_k > 0$ and

$$mB_k^2 < \frac{1}{K_k},$$

the computed vector

$$\hat{x} = \text{fl_backSub}(\hat{R}, \hat{c})$$

satisfies

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Lean formalization.

```
Algorithms/QR/HouseholderQR.lean  qrPrefixSupportSpannedByPreviousColumns_of_det_ne_zero_
previousLeadingBlockTranspose

Algorithms/LeastSquares/LSQRSolve.lean  LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_normBudget_of_leading_blocks_det_ne_zero_
kappaInf_selfNorm_budget
```

Corollary 501 (Triangular self-norm κ_∞ QR certificate). *In the preceding corollary, the hypotheses*

$$\det T_k \neq 0, \quad \det S_k \neq 0, \quad A_k(i, j) = 0 \quad (i \geq k, j < k)$$

may be replaced by the following visible triangular-domain assumptions. For each pivot k , suppose the current stored panel satisfies

$$A_k(i, j) = 0 \quad (j < i),$$

and suppose the diagonal entries of the previous and current leading blocks are nonzero. Then the same self-norm condition-number hypotheses

$$\kappa_\infty(S_k, \text{nonsingInv}(S_k)) \leq \kappa_k, \quad (k+1) \left(\frac{\kappa_k}{\|S_k\|_\infty} \right)^2 \leq K_k, \quad mB_k^2 < \frac{1}{K_k}$$

imply the same

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g)$$

conclusion under the same Householder sign, compact-update, γ -validity, and final Gram/RHS norm-budget hypotheses.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean      LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_normBudget_of_triangular_leading_blocks_
kappaInf_selfNorm_budget
```

Corollary 502 (Computed prefix-zero triangular self-norm QR certificate). *In the preceding triangular self-norm certificate, the broad intermediate-panel triangular-shape hypothesis can be replaced by the stored-loop prefix lower-zero invariant. For the stored Householder panel recurrence,*

$$A_{k+1} = \text{fl_householderStoredPanelStep}(A_k),$$

the repository proves that after k steps all entries below the diagonal in the first k completed columns are exactly zero:

$$j < k, \quad j < i \quad \implies \quad A_k(i, j) = 0.$$

Consequently, if the previous and current leading diagonal entries are nonzero, and if the same local κ_∞ , sign-choice, compact-update, and final Gram/RHS budget hypotheses hold, then

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g)$$

again follows.

Lean formalization.

```
Algorithms/QR/HouseholderQR.lean fl_householderStoredPanel_sequence_prefix_lower_zero
qrPreviousLeadingBlockTranspose_det_ne_zero_of_local_lower_triangular_diag_ne_zero
qrLeadingBlock_det_ne_zero_of_local_upper_triangular_diag_ne_zero
Algorithms/LeastSquares/LSQRSolve.lean      LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_normBudget_of_prefix_lower_zero_triangular_
leading_blocks_kappaInf_selfNorm_budget
```

Corollary 503 (Signed-alpha computed prefix-zero QR certificate). *Define the signed Householder scalar by*

$$\alpha(x_p, \rho) = \begin{cases} -\rho, & 0 \leq x_p, \\ \rho, & x_p < 0, \end{cases} \quad \rho = \sqrt{\sum_{i=k}^m A_k(i, k)^2}.$$

Then $\alpha^2 = \rho^2$ and $\alpha A_k(k, k) \leq 0$. Hence, in the computed prefix-zero triangular self-norm QR certificate above, the independent Householder squared-norm and sign-choice hypotheses may be replaced by the explicit source convention

$$\alpha_k = \alpha \left(A_k(k, k), \sqrt{\sum_{i=k}^m A_k(i, k)^2} \right).$$

The resulting least-squares QR backward-error certificate still keeps the remaining assumptions visible: nonzero local leading diagonals, local κ_∞ budgets, compact-update budgets, and final Gram/RHS solver budgets.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean signedHouseholderAlpha
signedHouseholderAlpha_sqrt_trailingNorm2Sq_sq
signedHouseholderAlpha_sqrt_trailingNorm2Sq_mul_pivot_nonpos
Algorithms/QR/HouseholderQR.lean fl_householderStoredTrailingPanel_sequence_diag_nonzero_
of_signed_alpha_trailingNorm_pos_sqrt_budget
Algorithms/LeastSquares/LSQRSolve.lean LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_normBudget_of_signed_alpha_prefix_lower_
zero_triangular_leading_blocks_kappaInf_selfNorm_budget
```

Corollary 504 (Prefix-local diagonal nonbreakdown from the stored loop). *In the signed-alpha stored trailing Householder QR loop above, suppose each completed pivot step has positive trailing norm and a square-root compact-update budget small enough to keep the written pivot nonzero. Then, after k stored steps, all previously written diagonal entries remain nonzero:*

$$A_k(r, r) \neq 0, \quad 0 \leq r < k.$$

Consequently, in the computed prefix-zero triangular self-norm QR certificate, the previous local diagonal hypotheses may be generated by the stored loop. Only the current leading pivot $A_k(k, k) \neq 0$ remains as the visible diagonal-domain condition for the current $(k+1) \times (k+1)$ principal block. The local κ_∞ , compact-update, and final Gram/RHS budgets remain explicit.

Lean formalization.

```
Algorithms/QR/HouseholderQR.lean fl_householderStoredPanel_sequence_prefix_diag_nonzero_
of_step_diag_nonzero
fl_householderStoredTrailingPanel_sequence_prefix_diag_nonzero_of_signed_alpha_trailingNorm_pos_
sqrt_budget
Algorithms/LeastSquares/LSQRSolve.lean LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_normBudget_of_signed_alpha_prefix_lower_
zero_pivot_nonzero_kappaInf_selfNorm_budget
```

Corollary 505 (Explicit final Gram/RHS radii for the prefix-local QR certificate). *Let $\hat{R} = A_n(1:n, 1:n)$, and define the final rectangular data and right-hand-side perturbation radii*

$$c_A^{\text{QR}} = ((1 + c_{\text{step}})^n - 1) \|A\|_F + \gamma_n \|\hat{R}; 0\|_F, \quad c_b^{\text{QR}} = ((1 + c_{\text{step}})^n - 1) \|b\|_2.$$

The induced final Gram and right-hand-side radii are

$$c_G^{\text{QR}} = \left\| \text{rectLSGramPerturbationNormBudget}(A, c_A^{\text{QR}}) \right\|_F.$$

$$c_g^{\text{QR}} = \sum_{j=1}^n \text{rectLSRhsPerturbationNormBudget}(A, b, c_A^{\text{QR}}, c_b^{\text{QR}})_j.$$

Under the hypotheses of the preceding prefix-local signed-alpha certificate, the computed vector satisfies

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G^{\text{QR}}, c_g^{\text{QR}}).$$

Thus the final Gram/RHS budget-domination assumptions are no longer separate side hypotheses in this route; they are instantiated by the displayed repository radii. The current-pivot nonzero condition, local κ_∞ budgets, compact-update budgets, square-root pivot budgets, and γ -validity remain explicit.

Lean formalization.

```
Algorithms/LeastSquares/LSQRSolve.lean qrSolveFinalDataPerturbationBudget
qrSolveFinalRhsPerturbationBudget
qrSolveFinalGramBudget
qrSolveFinalRhsBudget
rectLSRhsPerturbationNormBudget_le_sum
LSQRSolveBackwardError.of_stored_trailing_householder_sequence_topBlock_fl_backSub_gamma_bound_
explicitNormBudget_of_signed_alpha_prefix_lower_zero_pivot_nonzero_kappaInf_selfNorm_budget
```

Corollary 506 (Explicit compact-update and final radii for the prefix-local QR certificate). *For each stored QR step k , let d_k^{cmp} be the repository-defined compact-panel relative budget*

$$d_k^{\text{cmp}} = \text{householderCompactPanelRelativeBudget}(\hat{A}_k, \hat{b}_k, v_k, \beta_k).$$

Set

$$c_{\text{cmp}}^{\text{QR}} = \sum_{k=0}^{n-1} d_k^{\text{cmp}}.$$

Then every stored panel-column and right-hand-side compact-update budget is dominated by $c_{\text{cmp}}^{\text{QR}}$ times the corresponding input vector norm. Consequently, in the preceding prefix-local signed-alpha certificate one may take

$$c_{\text{step}} = c_{\text{cmp}}^{\text{QR}}$$

and the final radii $c_G^{\text{QR}}, c_g^{\text{QR}}$ from Corollary 226. Under the same current-pivot nonzero, local κ_∞ , square-root/diagonal budget, and γ -validity hypotheses, the computed vector satisfies

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G^{\text{QR}}, c_g^{\text{QR}}).$$

Thus neither the compact-update domination constant nor the final Gram/RHS radii are separate user-supplied side hypotheses in this route.

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean householderTrailingNorm2Sq_pos_of_nonneg_budget_lt_sqrt
Algorithms/QR/HouseholderApply.lean householderCompactRelativeBudget
householderCompactPanelRelativeBudget
Algorithms/QR/HouseholderQR.lean storedQRCompactStepRelativeBudget
storedQRCompactSequenceRelativeBudget
storedQRCompactSequenceRelativeBudget_column_bound
storedQRCompactSequenceRelativeBudget_rhs_bound
Algorithms/LeastSquares/LSQRSolve.lean LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
prefix_lower_zero_pivot_nonzero_kappaInf_selfNorm_budget
```

Corollary 507 (Explicit compact QR certificate from norm-square pivot margins). *In the setting of Corollary 227, replace the visible square-root pivot-budget condition by the dimensioned margin*

$$mB_k^2 < \|A_k(k:m, k)\|_2^2, \quad 0 \leq B_k,$$

where B_k is the repository compact diagonal-update budget for pivot k . Under the same current-pivot nonzero, local κ_∞ , dual/conditioning budget, and γ -validity hypotheses, the computed vector satisfies

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G^{\text{QR}}, c_g^{\text{QR}}).$$

Lean formalization.

```
Algorithms/QR/HouseholderSpec.lean budget_lt_sqrt_householderTrailingNorm2Sq_of_dim_mul_
budget_sq_lt_trailingNorm2Sq
Algorithms/LeastSquares/LSQRSolve.lean LSQRSolveBackwardError.of_stored_trailing_
householder_sequence_topBlock_fl_backSub_gamma_bound_explicitCompactBudget_of_signed_alpha_
prefix_lower_zero_pivot_nonzero_kappaInf_selfNorm_normSqBudget
```

Corollary 508 (Least-squares QR certificate from diagonal-dominant local inverse budgets). *In the same stored trailing Householder QR loop, let S_k be the $(k+1) \times (k+1)$ local leading block at pivot k , and let C_k be a full inverse of S_k . Suppose S_k is upper triangular and diagonally dominant in the Higham sense,*

$$S_k(i, j) = 0 \quad (j < i), \quad |S_k(i, j)| \leq |S_k(i, i)| \quad (i < j),$$

with nonzero diagonal. If the visible budget satisfies

$$(k+1) \left(2^k \frac{1}{\min_{0 \leq r \leq k} |S_k(r, r)|} \right)^2 \leq K_k, \quad K_k > 0, \quad mB_k^2 < \frac{1}{K_k},$$

then, under the same prefix-span, Householder, compact-update, γ -validity, and final Gram/RHS norm-budget hypotheses as above, the computed vector

$$\hat{x} = \text{fl_backSub}(\hat{R}, \hat{c})$$

satisfies

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Corollary 509 (Diagonal-dominant QR certificate from determinant). *In the previous corollary, the explicit full inverse witness C_k can be removed. It is enough to assume*

$$\det S_k \neq 0$$

for every local leading block, while keeping the same upper-triangular diagonal-dominance and diagonal-minimum budget. Lean instantiates $C_k = S_k^{-1}$ using the repository nonsingular inverse `nonsingInv`, proves it is a two-sided inverse from $\det S_k \neq 0$, and obtains the same

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g)$$

conclusion.

Corollary 510 (Least-squares QR certificate from nonsingular local leading blocks). *In the stored trailing Householder QR loop, assume the same rounded matrix and right-hand-side recurrences used in the stored-loop solver certificate. For each pivot $k < n$, suppose that the previous transposed leading block T_k and the current leading block S_k are nonsingular,*

$$\det T_k \neq 0, \quad \det S_k \neq 0,$$

that the previous columns have the QR lower-zero shape, that the Householder sign choice satisfies

$$\alpha_k A_k(k, k) \leq 0,$$

and that the compact rounded diagonal update budget satisfies

$$B_k < \sqrt{\|A_k(k:m, k)\|_2^2}.$$

Assume also the columnwise compact-update budget domination for the matrix panel and right-hand side, the γ -validity hypotheses, and the final Gram/RHS norm-budget bounds. Then the computed back-substitution vector

$$\hat{x} = \text{fl_backSub}(\hat{R}, \hat{c})$$

satisfies the repository's rectangular QR least-squares backward-error certificate,

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Corollary 511 (Least-squares QR certificate from triangular leading blocks). *In the stored trailing Householder QR loop, assume the same rounded matrix and right-hand-side recurrences, Householder normalizations, sign choices, compact panel/RHS budget domination, square-root pivot budgets, and final Gram/RHS norm-budget bounds as in the previous corollary. Instead of assuming $\det T_k \neq 0$ and $\det S_k \neq 0$ directly, assume that at every pivot $k < n$ the local panel has the visible upper-triangular shape used by the leading blocks and that the previous and current leading principal diagonals are nonzero:*

$$A_k(i, j) = 0 \quad (j < i), \quad A_k(r, r) \neq 0$$

for the relevant previous block T_k and current block S_k . Then the computed back-substitution vector again satisfies

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

Role in the bottleneck. This closes both the exact trailing-reflector shape theorem and the stored rounded loop shape theorem, and it closes the stored-step perturbation contract under explicit preservation, pivot-zeroing, RHS-prefix, and budget-domination hypotheses. It also closes the one-step trailing-reflector discharge of the exact preservation and zeroing hypotheses. The multi-step stored trailing sequence theorem and the stored-loop solver-spec handoff into LSQRSolveBackwardError are also closed. The diagonal condition is narrowed to the concrete per-pivot budget inequalities $B_k < |\alpha_k|$, and the Householder denominator condition is narrowed to the scalar pivot condition $A_k(k, k) \neq \alpha_k$, and then to the standard sign-choice conditions $\|A_k(k:m, k)\|_2^2 > 0$ and $\alpha_k A_k(k, k) \leq 0$. The positive-norm obligation is now reduced further: a left inverse for the transposed leading block gives the leading-block coefficient witness, while a left inverse for the actual leading $(k+1) \times (k+1)$ block pads to the ambient leading-column dual witness. The coefficient witness plus the QR lower-zero shape gives the prefix-span invariant; the leading-column dual witness gives column independence; prefix-span plus column independence gives a nonzero active trailing-column entry; and that entry gives the positive norm. One stored-loop wrapper consumes the concrete coefficient/left-inverse witnesses directly, and a second wrapper consumes the two local IsLeftInverse block witnesses directly. A third wrapper now consumes nonzero determinants for those two leading blocks and generates the local inverse witnesses internally. The new solver-facing wrapper feeds those determinant facts and the square-root pivot budget all the way into LSQRSolveBackwardError. The triangular-leading-block wrapper now pushes the visible triangular-principal-minor route through that same solver certificate. The prefix-span active-entry solver wrapper also pushes a different visible domain route through LSQRSolveBackwardError: it replaces the square-root budget by $B_k < |A_k(i_k, k)|$ for an active trailing entry. The diagonal-dominant triangular inverse wrapper now replaces the abstract inverse- ∞ -norm budget by Higham's explicit $2^k / \min_r |S_k(r, r)|$ estimate when each local leading block is diagonally dominant and a full inverse witness is supplied. The remaining source-faithful implementation work is either to keep the triangular/nonzero-diagonal and diagonal-dominance conditions visible as domain assumptions, or to prove the needed determinant/rank facts for the computed leading blocks and derive the usable budget lower bound from a formal full-rank, nonbreakdown, or conditioning assumption for the computed loop.

Definition 512 (Common rounded Householder panel/update contract). For a rectangular matrix panel $A \in \mathbb{R}^{m \times n}$, a right-hand side $b \in \mathbb{R}^m$, an exact reflector $P \in \mathbb{R}^{m \times m}$, and rounded outputs $\hat{A}, \hat{b}$, the contract $\text{HouseholderPanelAppError}(P, A, \hat{A}, b, \hat{b}; c)$ means:

$$P^T P = I$$

and there is one matrix ΔP such that

$$\|\Delta P\|_F \leq c, \quad \hat{A} = (P + \Delta P)A, \quad \hat{b} = (P + \Delta P)b.$$

Theorem 513 (Common panel contracts accumulate with one Q). Let $A_0 = A$, $b_0 = b$. Suppose that for every step $k < r$, the rounded Householder panel update from (A_k, b_k) to (A_{k+1}, b_{k+1}) satisfies

$$\text{HouseholderPanelAppError}(P_k, A_k, A_{k+1}, b_k, b_{k+1}; c), \quad c \geq 0.$$

Then there are a common orthogonal matrix Q , a matrix perturbation ΔA , and a vector perturbation Δb such that

$$A_r = Q^T(A + \Delta A), \quad b_r = Q^T(b + \Delta b),$$

and

$$\|\Delta A\|_F \leq ((1+c)^r - 1)\|A\|_F, \quad \|\Delta b\|_2 \leq ((1+c)^r - 1)\|b\|_2.$$

Lean formalization.

`Algorithms/QR/HouseholderSpec.lean` `HouseholderPanelAppError`

`Algorithms/QR/HouseholderQR.lean` `householderPanelAppError_rect_orthogonal_matrix_vector_`
`sequence_geometric`

Role in the bottleneck. This is a strong route-A adapter, but it is not the source-faithful standard Householder QR route. Higham's columnwise Householder QR analysis allows the per-step perturbation matrix to depend on the panel column. Therefore the active route has been corrected to the following columnwise theorem.

Definition 514 (Columnwise rounded Householder panel/update contract). For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exact reflector P , and rounded outputs $\hat{A}, \hat{b}$, `HouseholderColumnwisePanelAppError`($P, A, \hat{A}, b, \hat{b}; c$) means:

$$P^T P = I,$$

and for every panel column j there is a perturbation matrix ΔP_j such that

$$\|\Delta P_j\|_F \leq c, \quad \hat{A}(:, j) = (P + \Delta P_j)A(:, j),$$

while the right-hand side has its own perturbation matrix ΔP_b satisfying

$$\|\Delta P_b\|_F \leq c, \quad \hat{b} = (P + \Delta P_b)b.$$

Theorem 515 (Columnwise panel contracts accumulate with one Q). *Let $A_0 = A$, $b_0 = b$. Suppose that for every step $k < r$, the rounded Householder panel update from (A_k, b_k) to (A_{k+1}, b_{k+1}) satisfies*

$$\text{HouseholderColumnwisePanelAppError}(P_k, A_k, A_{k+1}, b_k, b_{k+1}; c), \quad c \geq 0.$$

Then there are an orthogonal matrix Q , a matrix perturbation ΔA , and a vector perturbation Δb such that

$$A_r = Q^T(A + \Delta A), \quad b_r = Q^T(b + \Delta b),$$

with columnwise and right-hand-side radii

$$\|\Delta A(:, j)\|_2 \leq ((1 + c)^r - 1)\|A(:, j)\|_2 \quad \text{for all } j,$$

$$\|\Delta b\|_2 \leq ((1 + c)^r - 1)\|b\|_2.$$

Lean formalization.

`Algorithms/QR/HouseholderSpec.lean` `HouseholderColumnwisePanelAppError`

`HouseholderColumnwisePanelAppError.of_vector_applications`

`Algorithms/QR/HouseholderQR.lean` `orthogonal_vector_sequence_one_step_fixedQ`

`rect_orthogonal_columnwise_vector_sequence_geometric`

`householderColumnwisePanelAppError_rect_orthogonal_columnwise_vector_sequence_geometric`

Lemma 516 (Rank-one norm bridge for vector residuals). *Let $e, b \in \mathbb{R}^m$ and suppose $\|b\|_2 \neq 0$. Then*

$$\left\| \frac{eb^T}{\|b\|_2^2} \right\|_F = \frac{\|e\|_2}{\|b\|_2}.$$

The proof first factors $\|xy^T\|_F = \|x\|_2\|y\|_2$, then applies the scalar homogeneity of the Frobenius norm.

Theorem 517 (Forward error gives a Householder backward-error contract). *Let $P \in \mathbb{R}^{m \times m}$ be orthogonal, let $b, \hat{y} \in \mathbb{R}^m$, and let $c \geq 0$. If*

$$\|\hat{y} - Pb\|_2 \leq c\|b\|_2,$$

then $\text{HouseholderAppError}(P, b, \hat{y}; c)$ holds. When $b \neq 0$, the perturbation witness is

$$\Delta P = \frac{(\hat{y} - Pb)b^T}{\|b\|_2^2}.$$

When $b = 0$, the forward-error hypothesis forces $\hat{y} - Pb = 0$, and the witness is $\Delta P = 0$. Thus the Lean theorem has no hidden nonzero input hypothesis.

Corollary 518 (Columnwise panel contract from forward-error bounds). *Let P be orthogonal, $c \geq 0$, and suppose each rounded panel column and the rounded right-hand side satisfy*

$$\|\hat{A}(:, j) - PA(:, j)\|_2 \leq c\|A(:, j)\|_2, \quad \|\hat{b} - Pb\|_2 \leq c\|b\|_2.$$

Then

$$\text{HouseholderColumnwisePanelAppError}(P, A, \hat{A}, b, \hat{b}; c)$$

holds. This is the adapter that turns future low-level rounded dot/scale/subtract forward-error estimates into the source-faithful columnwise QR accumulation theorem above.

Theorem 519 (Explicit-matrix rounded Householder application). *Let $P \in \mathbb{R}^{m \times m}$, $b \in \mathbb{R}^m$, and let $\hat{y} = \text{fl_matVec}(P, b)$. Under the usual γ_m -validity condition for the floating-point model,*

$$\|\hat{y} - Pb\|_2 \leq \gamma_m \|\text{abs}(P)\|_F \|b\|_2.$$

Equivalently, this is the normwise consequence of the repository's existing componentwise matrix-vector theorem

$$|\hat{y}_i - (Pb)_i| \leq \gamma_m \sum_j |P_{ij}| |b_j|.$$

The proof then uses

$$\|\text{abs}(P) \text{abs}(b)\|_2 \leq \|\text{abs}(P)\|_F \|\text{abs}(b)\|_2 = \|\text{abs}(P)\|_F \|b\|_2.$$

Corollary 520 (Explicit-matrix route to the columnwise Householder contract). *If P is orthogonal and every vector application in a panel is computed by the explicit-matrix routine above, then*

$$\text{HouseholderAppError}(P, b, \text{fl_matVec}(P, b); \gamma_m \|\text{abs}(P)\|_F)$$

holds, and the corresponding columnwise panel computation satisfies

$$\text{HouseholderColumnwisePanelAppError}(P, A, \hat{A}, b, \hat{b}; \gamma_m \|\text{abs}(P)\|_F).$$

This closes a concrete formed-reflector implementation route. It does not claim to be the compact Householder update that computes $b - \beta v(v^T b)$ by rounded dot, scale, and subtract operations.

Definition 521 (Compact Householder arithmetic budget). For the compact update, define

$$\sigma = v^T b, \quad S = \sum_j |v_j| |b_j|, \quad D = \gamma_m S.$$

Let

$$B_\sigma = S + D, \quad B_\tau = |\beta| B_\sigma (1 + u), \quad B_{z,i} = B_\tau |v_i| (1 + u).$$

The deterministic component budget is

$$E_i = u B_\tau |v_i| + u |\beta| B_\sigma |v_i| + |\beta| |v_i| D + u (|b_i| + B_{z,i}).$$

Theorem 522 (Compact rounded Householder dot-scale-subtract update). *Let*

$$\hat{\sigma} = \text{fl_dotProduct}(v, b), \quad \hat{\tau} = \text{fl_mul}(\beta, \hat{\sigma}),$$

and let the computed compact Householder application be

$$\hat{y}_i = \text{fl_sub}(b_i, \text{fl_mul}(\hat{\tau}, v_i)).$$

If $P = I - \beta v v^T$ and the γ_m -validity condition holds, then

$$Pb = b - \beta v (v^T b)$$

and, componentwise,

$$|\hat{y}_i - (Pb)_i| \leq E_i.$$

Consequently,

$$\|\hat{y} - Pb\|_2 \leq \|(E_i)_{i=1}^m\|_2.$$

The proof reuses the local dot-product theorem

$$|\hat{\sigma} - \sigma| \leq \gamma_m \sum_j |v_j| |b_j|$$

and the primitive floating-point model for the two multiplications and final subtraction.

Corollary 523 (Compact route to the columnwise Householder contract). *Suppose $P = I - \beta v v^T$ is orthogonal. If, for a vector b , the visible deterministic budget satisfies*

$$\|(E_i(b))_{i=1}^m\|_2 \leq c \|b\|_2,$$

then the compact rounded update satisfies

$$\text{HouseholderAppError}(P, b, \hat{y}; c).$$

If the same budget-domination condition holds for every panel column and for the right-hand side, the compact columnwise panel routine satisfies

$$\text{HouseholderColumnwisePanelAppError}(P, A, \hat{A}, b, \hat{b}; c).$$

This closes the low-level compact vector-application primitive. The budget-domination condition is an explicit deterministic side condition, not a probabilistic event or an assumed stability theorem.

Corollary 524 (Compact Householder sequence accumulation). *Let $P_k = I - \beta_k v_k v_k^T$ be a supplied sequence of orthogonal reflectors. For readability, write*

$$\begin{aligned} C_A(v, \beta, A) &:= \text{fl_householderApplyCompactPanel}(v, \beta, A), \\ C_b(v, \beta, b) &:= \text{fl_householderApplyCompact}(v, \beta, b). \end{aligned}$$

Assume each rounded panel update and right-hand-side update is computed by these compact routines:

$$\hat{A}_{k+1} = C_A(v_k, \beta_k, \hat{A}_k), \quad \hat{b}_{k+1} = C_b(v_k, \beta_k, \hat{b}_k),$$

and assume the explicit compact budget is dominated by the same c for every panel column and for the right-hand side. Then the existing columnwise geometric theorem gives $Q, \Delta A, \Delta b$ such that

$$\hat{A}_r = Q^T(A + \Delta A), \quad \hat{b}_r = Q^T(b + \Delta b),$$

with Q orthogonal and

$$\|\Delta A_{:,j}\|_2 \leq ((1+c)^r - 1)\|A_{:,j}\|_2, \quad \|\Delta b\|_2 \leq ((1+c)^r - 1)\|b\|_2.$$

This is still a supplied-sequence theorem: the concrete QR loop must prove the reflector construction, transformed $[R; 0]$ shape, and top-RHS linkage.

Corollary 525 (Compact Householder sequence to the LS QR solver specification). *Let $\eta_r = (1+c)^r - 1$. Under the hypotheses of Corollary 184, assume in addition that the final transformed data have the QR least-squares shape*

$$\hat{A}_r = \begin{bmatrix} R \\ 0 \end{bmatrix}, \quad (\hat{b}_r)_{1:n} = c_{\text{top}},$$

that R is upper triangular with nonzero diagonal, and that the induced Gram/RHS perturbation norm budgets are bounded by c_G, c_g after adding the rounded triangular-solve top-block radius

$$\eta_r \|A\|_F + \gamma_n \left\| \begin{bmatrix} R \\ 0 \end{bmatrix} \right\|_F.$$

Then the computed vector

$$\hat{x} = \text{fl_backSub}(R, c_{\text{top}})$$

satisfies the repository's QR least-squares backward-error specification

$$\text{LSQRSolveBackwardError}(A^T A, A^T b, \hat{x}; c_G, c_g).$$

The proof first converts the columnwise bounds $\|\Delta A_{:,j}\|_2 \leq \eta_r \|A_{:,j}\|_2$ into the Frobenius bound $\|\Delta A\|_F \leq \eta_r \|A\|_F$, then invokes the existing top-block triangular-solve pullback theorem.

Lean formalization.

```
Analysis/MatrixAlgebra.lean vecNorm2Sq_abs
vecNorm2_abs
frobNormSqRect_eq_sum_vecNorm2Sq_cols
frobNormRect_le_of_col_vecNorm2_le
frobNormSq_rankOne
```

```

frobNorm_rankOne
frobNorm_rankOne_smul
frobNorm_rankOne_div_vecNorm2Sq
Algorithms/QR/HouseholderSpec.lean HouseholderAppError.of_forward_error
HouseholderColumnwisePanelAppError.of_forward_errors
Algorithms/QR/HouseholderApply.lean fl_householderApplyExplicit_forward_error_bound
fl_householderApplyExplicit_HouseholderAppError
fl_householderApplyExplicitPanel_HouseholderColumnwisePanelAppError
matMulVec_householder_eq_compact
fl_householderApplyCompact_componentwise_error_bound
fl_householderApplyCompact_forward_error_bound
fl_householderApplyCompact_HouseholderAppError_of_budget
fl_householderApplyCompactPanel_HouseholderColumnwisePanelAppError_of_budget
Algorithms/QR/HouseholderQR.lean      fl_householderApplyCompactPanel_rect_orthogonal_
columnwise_vector_sequence_geometric
Algorithms/LeastSquares/LSQRSolve.lean  LSQRSolveBackwardError.of_compact_householder_
sequence_topBlock_fl_backSub_gamma_bound_normBudget

```

Source and remaining obligation. This theorem matches the columnwise structure in Higham’s QR notes, Section 4.1, Lemmas 4.3–4.4 and Theorem 4.5. The source is used only to choose the route; the local Lean theorems above prove the accumulation step. The front-loaded proof-source ledger also records the standard least-squares QR route in Higham’s *Accuracy and Stability of Numerical Algorithms*, second edition, Chapter 19, especially Lemma 19.3 and Theorems 19.4–19.6. That source is advisory here: it guides the remaining source-faithful route, but no paper-level theorem is closed by citation alone. The exact forward-error adapter, compact dot–scale–subtract primitive, compact supplied-sequence adapter, Frobenius/top-block bridge, and exact trailing-shape recurrence theorem are all closed. The remaining obligation is now narrower: assemble the rounded stored- R and transformed right-hand-side loop around the compact update, and prove or expose the nonzero-pivot/rank condition required by the triangular solve.

Corollary 526 (High-probability literal rounded sampled-row solver from normal equations and Cholesky). *Use the notation and Bennett sample-budget assumptions of Corollary 157, but compute the rounded sampled least-squares solution by the repository’s normal-equations/Cholesky pipeline. For each sample trace ω , let*

$$G_\omega = \text{fl}(S_\omega A)^T \text{fl}(S_\omega A), \quad g_\omega = \text{fl}(S_\omega A)^T \text{fl}(S_\omega b),$$

and let $\widehat{C}_\omega, \widehat{c}_\omega, \widehat{R}_\omega$ be the computed Gram matrix, computed right-hand side, and Cholesky factor. Suppose the local theorems

$$\text{GramProductError}, \quad \text{GramVecError}, \quad \text{CholeskyBackwardError}$$

hold for these objects, with the s -term Gram/vector product radius γ_s and the Cholesky radius γ_{n+1} . Define the computed solution by

$$\widehat{y}_\omega = \text{flSolve}(\widehat{R}_\omega^T, \widehat{c}_\omega), \quad \widehat{x}_\omega = \text{flSolve}(\widehat{R}_\omega, \widehat{y}_\omega).$$

The local normal-equations theorem supplies perturbations satisfying

$$(G_\omega + \Delta G_\omega)\hat{x}_\omega = g_\omega + \Delta g_\omega,$$

with the componentwise bounds

$$|(\Delta G_\omega)_{jk}| \leq \gamma_s B_{\omega,jk}^{\text{ATA}} + (\gamma_{n+1} + 2\gamma_n + \gamma_n^2) \sum_{\ell=1}^n |(\hat{R}_\omega)_{\ell j}| |(\hat{R}_\omega)_{\ell k}|,$$

and

$$|(\Delta g_\omega)_j| \leq \gamma_s B_{\omega,j}^{\text{ATb}}.$$

Consequently the componentwise certificate

$$d_{\omega,i} = \sum_{j=1}^n |(H_\omega)_{ij}| \left(\sum_{k=1}^n B_{\omega,jk}^G |\hat{x}_{\omega,k}| + B_{\omega,j}^g \right)$$

is valid, where $H_\omega = G_\omega^{-1}$ and B^G, B^g denote the two displayed perturbation radii. Hence, if the same support-event budget condition holds with this d_ω , then

$$\mathbb{P}[\|A\hat{x}_\omega - b\|_2^2 \leq (1 + \eta)\|Ax_{\text{opt}} - b\|_2^2] \geq 1 - \delta.$$

Lean formalization.

```
normalEqCholeskyXHat
normalEqCholeskyGramBound
normalEqCholeskyRhsBound
normalEqCholeskySolverDx
normalEqCholeskySolverDx_nonneg
normal_equations_cholesky_forward_error_certificate
leverageTraceProbability_eventProb_fl_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_
sample_budget_of_objective_error_and_normal_eq_cholesky_solver
```

Lemma 527 (Column representation implies residual representation). *Suppose the columns of A and the right-hand side b have coordinates in the columns of U :*

$$A_{ij} = \sum_{\ell=1}^d U_{i\ell} c_{j\ell}, \quad b_i = \sum_{\ell=1}^d U_{i\ell} d_\ell.$$

Then every residual has the induced coordinates

$$y_\ell(x) = \sum_{j=1}^n c_{j\ell} x_j - d_\ell, \quad (Ax - b)_i = \sum_{\ell=1}^d U_{i\ell} y_\ell(x).$$

If $U^T U = I$, the canonical coordinates $U^T(Ax - b)$ reconstruct the same residual.

Corollary 528 (Augmented-span leverage basis). *Let*

$$\mathcal{S} = \text{span}\{A_{*,1}, \dots, A_{*,n}, b\} \subseteq \mathbb{R}^m, \quad d = \dim \mathcal{S}.$$

The repository constructs an orthonormal basis $u_1, \dots, u_d$ of $\mathcal{S}$ using Mathlib’s finite-dimensional orthonormal-basis theorem, and forms $U = [u_1 \ \dots \ u_d]$. It also constructs coordinates

$$A_{*,j} = \sum_{\ell=1}^d u_{\ell} c_{j\ell}, \quad b = \sum_{\ell=1}^d u_{\ell} d_{\ell}.$$

Consequently every residual $Ax - b$ lies in the column span of U . If $d > 0$, equation (6) is well-defined for this basis and the concrete row-sampled least-squares theorem becomes

$$\mathbb{P}\left[\|A\hat{x}(\omega) - b\|_2^2 \leq (1 + \eta) \|Ax_{\text{opt}} - b\|_2^2\right] \geq 1 - \frac{1}{s(\varepsilon/d)^2}.$$

The positive-dimension hypothesis is the expected nondegeneracy condition for the leverage denominator. If $\mathcal{S} = \{0\}$, the least-squares problem is degenerate and the equation (6) probabilities are not defined by this basis. The source-aligned Bennett sample-budget theorem uses the sharper condition $d > 1$, matching the explicit $(d - 1)$ variance denominator in the formalized equation (7) concentration theorem.

The fully floating-point rounded-Gram transfer also specializes to this same augmented-span basis. In that theorem the preservation radius is

$$\varepsilon + \text{rowSampleGramFullFpPerturbBudget}(\text{fp}, s, U),$$

and the coordinate/original-objective hypotheses are discharged by the augmented-span construction. The theorem still explicitly requires the rounded sketched objective to have the rounded-Gram quadratic representation; it is an abstract objective transfer, not yet the theorem for the literal rounded sampled A, b implementation above.

Lean formalization.

```
augmentedDataVector
augmentedDataSpan
augmentedDataVector_mem_span
augmentedSpanBasisMatrix
augmentedSpanColumnCoords
augmentedSpanRhsCoords
hasOrthonormalColumns_augmentedSpanBasisMatrix
columnsAndRhsInColumnSpace_augmentedSpanBasisMatrix
residualsInColumnSpace_augmentedSpanBasisMatrix
leverageTraceProbability_eventProb_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan
leverageTraceProbability_eventProb_rowSampleLSObjective_le_one_add_eta_of_augmentedSpan_sample_budget
leverageTraceProbability_eventProb_fl_lsObjective_le_one_add_eta_of_augmentedSpan
leverageTraceProbability_eventProb_fl_lsObjective_le_one_add_eta_of_augmentedSpan_sample_budget
```

Corollary 529 (Identity-basis fallback). *Taking $U = I_m$ gives a completely concrete equation (6) instance:*

$$p_i = \frac{\|e_i^T\|_2^2}{m} = \frac{1}{m}.$$

Every residual lies in the span of I_m , and the canonical residual coordinates are just the residual itself. Therefore the previous corollary specializes to uniform row sampling with $d = m$:

$$\mathbb{P}\left[\|A\hat{x}(\omega) - b\|_2^2 \leq (1 + \eta) \|Ax_{\text{opt}} - b\|_2^2\right] \geq 1 - \frac{1}{s(\varepsilon/m)^2}.$$

This theorem is useful as a no-hidden-coordinate-assumption fallback. It is not the sharp survey theorem, because the paper’s leverage basis should have dimension equal to the residual subspace rank, not necessarily m . The analogous fully floating-point rounded-Gram objective transfer is also formalized for $U = I_m$, with the same $d = m$ denominator and the same explicit rounded-Gram representation hypothesis. The implementation-backed literal rounded A, b transfer is now proved for the augmented-span leverage law above under its explicit objective-budget slack condition; an I_m specialization can be derived separately if it becomes useful.

Lean formalization.

```
hasOrthonormalColumns_idMatrix
residualCoordinates_idMatrix
residualsInColumnSpace_idMatrix
leverageTraceProbability_eventProb_rowSampleLSObjective_le_one_add_eta_of_idBasis
leverageTraceProbability_eventProb_fl_lsObjective_le_one_add_eta_of_idBasis
```

Remark. This section now proves the deterministic least-squares implication, the coordinate quadratic-error and finite-Loewner bridges, the probability transfer to a random sketched minimizer, exact/FP leverage-score adapters, and the concrete sampled-row least-squares objective algebra using canonical residual coordinates. The augmented-span theorem constructs a low-dimensional orthonormal basis of $\text{span}\{A_{*,j}, b\}$ and closes the residual-coordinate hypothesis for that basis in the exact theorem, the Bennett sample-budget theorem, and the rounded-Gram FP transfer, while the identity-basis specialization remains a valid full-row-space exact/FP fallback. The note also proves the first implementation-backed rounded sampled-row foundation: rowwise residual perturbation and deterministic objective perturbation bounds for the literal rounded sampled/scaled A, b construction, and the corresponding support-aware high-probability rounded minimizer transfer under an explicit objective-budget slack condition. The same support-aware theorem is also proved for additive-gap approximate rounded minimizers, with the solver objective gap included explicitly in the slack budget. It does not yet prove a random-projection version or derive that gap from a concrete floating-point solver/preconditioner pipeline.

8 Hypotheses and Scope

- Algorithm 1 exact unbiasedness is proved for all entries under the canonical independent squared-magnitude trace law with $\text{steps} = s$ and $s \neq 0$.
- Algorithm 1 stability is entrywise and hit-counter based. The Chernoff final API proves the hit-count moment-generating-function bound from the canonical product trace law; it is not assumed.

- Algorithm 1 exact residuals are now decomposed as sums of one-sample mean-zero residual increments. Entrywise, Frobenius, and fixed-vector second-moment bounds are proved for these increments, and exact and floating-point fixed-vector, finite-test-set, and supplied finite-cover Markov tails are proved for the full residual. This is a matrix-concentration foundation, not by itself a uniform Bernstein or Khintchine spectral tail bound.
- Algorithm 1 also has a self-adjoint dilation layer: the rectangular residual is embedded as $\mathcal{D}(R) = \begin{bmatrix} 0 & R \\ R^T & 0 \end{bmatrix}$, dilation events transfer to rectangular exact and floating-point residual events, and the dilation residual has a proved zero-mean increment decomposition with squared-Frobenius, quadratic-form, Loewner-order, and PSD variance proxies, plus trace variance proxies with the explicit dilation dimension $m + n$. The finite square matrix layer also has product/trace vocabulary, PSD and Loewner-order vocabulary, basic trace algebra, trace monotonicity, Loewner sum/scaling algebra, symmetric-square PSD facts, the deterministic $M^2 \preceq L^2 I$ consequence of an operator bound, and the converse adapter from a squared Loewner event to an operator event. It also proves $\text{tr}(\mathcal{D}(M)^2) = 2 \|M\|_F^2$. These facts are now used by the source-aligned matrix-Bernstein proof route, while the generic rectangular Bernstein/Khintchine library remains a separate foundation target.
- Algorithm 1 has a deterministic floating-point transfer theorem in rectangular vector-action form. The repository also has a closed source-aligned square high-probability theorem for the hard-thresholded squared-magnitude distribution $\hat{A}_{ij}^2 / \|\hat{A}\|_F^2$, which is not the literal CACM Algorithm 1 distribution. The theorem derives the entrywise floating-point perturbation budget from the local gamma/hit-count stability library and the sampler's probability-one support event; it does not assume the concentration or perturbation event as a hypothesis. Its corollary expands that internal budget matrix into the explicit radius $\varepsilon + (2n^2 \|A\|_F^2 / \varepsilon) \gamma_{s+1}$, where $\gamma_{s+1} = ((s+1)u)/(1 - (s+1)u)$. The repository also has a faithful nontruncated Frobenius/Markov FP corollary under the literal $A_{ij}^2 / \|A\|_F^2$ law, with explicit radius $\varepsilon + n(\|A\|_F^2 / \alpha) \gamma_{s+1}$ under a visible nonzero-entry floor α . In addition, the literal support-radius theorem is closed with no nonzero-entry floor; it pays the reciprocal-entry term $G(A) = \sum_{A_{ij} \neq 0} \|A\|_F^2 / |A_{ij}|$ explicitly in the exact threshold and in the FP radius. The literal source-rate specialization is closed when every nonzero entry satisfies $\varepsilon/(2n) \leq |A_{ij}|$, so the source truncation is the identity. The source-uniform sharp literal equation (2) matrix-Bernstein law and a fully rectangular CACM-prose variant remain separate theorem targets.
- The Drineas-Zouzias truncation step for elementwise sparsification is now explicit: thresholding a square $n \times n$ matrix at $\varepsilon/(2n)$ costs at most $\varepsilon/2$ in Frobenius/operator norm, and exact or floating-point half-budget theorems for sampling $\hat{A}$ transfer to original-matrix residual events. This truncation layer is composed below with the source-aligned matrix-Bernstein theorem and the derived floating-point gamma budget.
- Algorithm 1 also has two older nonconditional weaker probability routes: Frobenius/Markov and scalar-entry/union-bound. They remain useful sanity checks, but the advertised source-aligned equation (2) theorem now uses the sharper trace-MGF/Bernstein route.
- The product-law scalar MGF and finite-family self-adjoint-dilation quadratic-form theorems are support results for supplied scalar statistics or supplied test vectors. The source-aligned equation (2) theorem uses the noncommutative trace-MGF/largest-eigenvalue route formalized later in the note; a reusable generic Bernstein/Khintchine API beyond that specialized route remains open.

- Algorithm 2 equation (5) is a Frobenius-norm high-probability theorem derived from a proved finite-probability second-moment bound and Markov’s inequality.
- Algorithm 2 equation (7) is stated as a vector-action operator-2 bound: for all x , $\|Mx\|_2 \leq c\|x\|_2$. The repository intentionally does not introduce a separate supremum-valued spectral norm object. The source-sharp Oliveira/Tropp route now has the one-step leverage covariance facts, row-trace MGF iteration, positive/negative centered log-CGF bounds, a two-sided finite-Loewner high-probability theorem with explicit exponential tails, Bennett sample-budget corollaries, and a matching floating-point finite-Loewner transfer. The theorem keeps the upper and lower scalar Bennett denominators explicit.
- Algorithm 3 random-projection concentration is distribution-specific. The formalization proves exact orthogonal Frobenius preservation, deterministic preservation of the orthonormal-column leverage basis and equation (6) denominator, the signed-diagonal orthogonality prerequisite for the SRHT route, the finite Rademacher sign-vector law with a probability-one signed-preprocessing event, finite Rademacher first/second moments, and the exact scalar signed-linear-form MGF factorization, plus the one-sided exponential-Markov tail skeleton, scalar Hoeffding MGF/tail bounds, and a weaker coordinate-Hoeffding row-norm theorem. The coordinate-Hoeffding theorem is also lifted to an equation (6) leverage-probability bound $p_i(HD_\omega U) \leq B^2$ under the same finite-entry failure budget. The deterministic-after-preconditioning iid uniform sample-average concentration theorem is also formalized in finite-Loewner tail-budget form. The signed-Hadamard preprocessing event and the uniform row-sampling event are then composed on the product probability law, giving the exact two-sided sample-Gram event with probability $1 - \delta_{\text{pre}} - \delta_{\text{sample}}$ for the scoped coordinate-Hoeffding route. It also proves the flat-Hadamard expected row-norm identity $\mathbb{E}\|(HD_\omega U)_{i,*}\|_2^2 = n/m$, together with the weak Markov/union consequence $\mathbb{P}\{\forall i, \|(HD_\omega U)_{i,*}\|_2^2 \leq T\} \geq 1 - m((n/m)/T)$. It also proves deterministic floating-point stability for the preprocessing products. The source-sharp SRHT row-norm branch is now closed by a positive-drop finite-cube route: one obtains the all-row cap $\|(HD_\omega U)_{i,*}\|_2 \leq \sqrt{n/m} + t$, the induced leverage cap $(\sqrt{n/m} + t)^2/n$, logarithmic preprocessing-budget wrappers, exact SRHT-plus-uniform-row composition, and a source-sharp FP constant-budget transfer for rounded uniform sketches. It does not prove a universal Gaussian/FJLT/input-sparsity random-projection uniformization theorem, nor concrete implementation certificates for generated/stored transforms, FHT or sign storage, computed bases/singular vectors/projectors, or scale denominators.
- Equation (8) now has a deterministic sketched least-squares objective theorem and a coordinate quadratic-error bridge from an operator-2 Gram-error event to residual-objective preservation. It also has the concrete sampled-row objective identity for rows of A and b scaled by leverage probabilities from U , using the canonical coordinates $U^T(Ax - b)$. The augmented-span theorem constructs an orthonormal basis of $\text{span}\{A_{*,j}, b\}$, proves coordinates for $[A \ b]$, and discharges the residual-column-space condition with dimension $d = \dim \text{span}\{A_{*,j}, b\}$ when $d > 0$. The same augmented-span and identity-basis coordinate discharges are now packaged for the fully floating-point rounded-Gram objective transfer. The identity-basis fallback also discharges the condition for uniform row sampling with $d = m$. The leverage-score Bennett sample-budget theorem is now closed in exact arithmetic and as a rounded-Gram FP transfer. The literal rounded sampled/scaled A, b construction now has residual and objective perturbation bounds and a support-aware high-probability rounded-minimizer transfer under an explicit objective-budget slack condition, including an additive solver-objective-gap variant. A concrete normal-equations/Cholesky solver transfer is formalized, and the rectangular

stored-QR route now reaches equation (8) through visible-domain certificates: local diagonal dominance or row-budget control, source-denominator/nonbreakdown, compact-product smallness, active-pivot/scalar-comparison fields, and rounded triangular solve budgets are explicit hypotheses rather than hidden conclusions. Random-projection equation (8) variants and the generic claim that an arbitrary computed QR/preconditioner loop supplies those source/domain fields remain open.

- The full CACM article also discusses matrix completion, Laplacian-based solvers, kernel methods, and other applications. Those are not explicit meta-algorithms like Algorithms 1–3, and their application-specific guarantees require additional foundations before they can be stated as Lean theorems.

9 Proof-Source Ledger

The full-paper automation now maintains a separate proof-source ledger:

`docs/RANDNLA_CACM_PROOF_SOURCE_LEDGER.md.`

Its purpose is to prevent repeated proof work around the same missing mathematical engine. For every source claim whose proof is incomplete in the CACM article, the ledger records the source location, the missing proof step, the external primary source chain, the intended Lean target, and whether the source has been formalized locally, rejected, kept advisory, or left open.

Definition 530 (Proof-source status). A source theorem is *formalized* only after the corresponding Lean theorem is proved locally or reused from an already formalized local dependency. A source theorem is *advisory* when it guides the statement, proof order, or constants but is not available as a Lean theorem. Advisory sources never close a paper-level theorem.

Theorem 531 (Current source chain for Algorithm 1 equation (2)). *The active proof-source chain for the Algorithm 1 spectral concentration claim is:*

CACM Algorithm 1 and equation (2)
 $\Downarrow$
Drineas–Zouzias Algorithm 1, Theorem 1, and Lemmas 1–4
 $\Downarrow$
matrix-valued Bernstein for the sampled truncated residual
 $\Downarrow$
Tropp Theorem 3.2, Corollary 3.3, Theorem 3.6, Corollary 3.7,
Theorem 1.4, Theorem 1.6, and the relative-entropy/Lieb route in
Tropp’s 2012 note and monograph, with Hansen–Pedersen Jensen,
Effros’s matrix perspective,
and Lindblad’s relative-entropy convexity theorem as source-chain
references for the joint-convexity step.

The local formalization has already proved the truncation transfer, self-adjoint dilation, variance/boundedness prerequisites, trace/eigenvalue Markov interfaces, C^ -matrix expectation/order bridges, and finite Jensen adapter. It has also named the strictly positive cone, proved that this cone is convex, named the local Lieb trace functional, and proved the local $\log(\exp X) = X$ and $\exp(\log A) = A$ functional-calculus inverse bridges needed by that route. The chosen relative-entropy route also now has scalar and finite-vector nonnegativity, the finite log-sum inequality, scalar and*

finite-vector joint convexity, local vocabulary $D(A; B)$, the diagonal normalization $D(A; A) = 0$, and the real scalar-identity case $D(aI; bI) \geq 0$ for $a, b > 0$, together with the real diagonal case $D(\Delta(a); \Delta(b)) = D_{\text{vec}}(a; b)$ and its positivity corollary for positive diagonals, plus the real-diagonal joint-convexity subcase, and the left/right multiplication endomorphism layer L_A, R_A with product/power algebra, and the explicit ratio endomorphism $L_X R_A^{-1}$ needed for the Effros perspective route, as well as the finite Kronecker lifts $A \otimes I$ and $I \otimes H$, their affine laws, commutation/product identities, and positive-definiteness, trace-normalization, product-index vectorization action, vectorized-identity trace-pairing facts, and the polynomial powered trace-pairing identity $v_I^*(A \otimes B^\top)^k v_I = \text{tr}(A^k B^k)$, and the finite-polynomial version of this trace identity, plus continuity of the finite quadratic form and finite polynomial evaluation in the matrix argument. It also now splits the Hansen–Pedersen/Effros source route into ordinary positive-cone operator convexity, Hansen–Pedersen transfer, and assembled two-point Jensen targets; it proves the identity-function sanity cases, the assembly adapters, the direct $x \log x$ operator-convexity input, and the concrete two-point $x \log x$ Jensen theorem by a reflection-average corner argument. It also proves the conditional relative-entropy route reduction: joint convexity of D plus the normalized entropy variational formula imply the finite-dimensional Lieb trace concavity theorem for

$$A \mapsto \text{Re tr exp}(H + \log A)$$

on the strictly positive cone. For that normalized variational formula, the optimizer-candidate equality $\Psi_H(\text{exp}(H + \log A), A) = \Phi_H(A)$ is also formalized, and the maximality part is reduced to matrix relative-entropy nonnegativity. On the nonnegativity side, the generalized Klein/nonnegativity equivalence, the Hermitian spectral-overlap expansion, and the positive-spectrum scalar entropy kernel are now formalized. The bridge from the separated Hermitian CFC theorem to the compact complex C^* -matrix logarithm form of generalized Klein is also formalized, so local matrix relative-entropy nonnegativity and the normalized variational formula are closed. Matrix relative-entropy joint convexity is also now formalized through the finite-dimensional Effros perspective route, and this gives finite-dimensional Lieb trace concavity and the one-step Tropp trace-MGF inequality without an explicit Lieb hypothesis. The Algorithm 1 product-law trace-MGF iteration, finite-real trace-exponential adapter, residual-dilation specialization, and trace-exponential Markov-to-eigenvalue step are also formalized. The remaining concentration foundation is the scalar matrix-CGF/log-MGF bound, with explicit Bernstein/Khinchine constants, needed for the Algorithm 1 equation (2) high-probability spectral-norm theorem. The proof-source ledger records Effros’s matrix-perspective proof route, the Hansen–Pedersen operator Jensen source behind that route, Lindblad’s original relative-entropy convexity lineage, and Bendat–Sherman’s operator monotonicity/convexity route as advisory primary sources. The concrete Hansen–Pedersen ‘ $x \log x$ ’ theorem is now represented by local Lean proofs, the CFC square-root algebra needed for perspective statements is local, and the ordinary finite perspective theorem for $f(x) = x \log x - (x - 1)$ is formalized. The finite-polynomial Kronecker trace-representation layer is also local, and an explicit uniform polynomial approximation hypothesis is now enough to transfer that layer to the normalized entropy-kernel CFC trace term. The positive-definite spectrum case now also constructs the required uniform polynomial approximation and specializes the trace convergence theorem to $X \otimes (A^{-1})^\top$. The common polynomial sequence also converges on every eigenvalue ratio $\lambda_i(X) \lambda_j(A)^{-1}$, so the limiting overlap theorem and the source-faithful Umegaki superoperator trace representation are now formalized. The earlier ordinary source-matrix perspective route is not used to close this step, because it is not the $L_X R_A^{-1}$ perspective required by Tropp–Effros. No theorem is closed by citation.

Corollary 532 (Current full-paper gate implication). *Matrix relative-entropy joint convexity, finite-dimensional Lieb trace concavity, and the one-step Tropp trace-MGF inequality are now formalized*

locally. The iid independent-sum trace-MGF domination theorem, its finite-real trace-exponential adapter, the actual Algorithm 1 self-adjoint dilation residual specialization, and the trace-exponential Markov-to-eigenvalue step are also formalized locally. The scalar Bernstein parabola with constants and the CFC step that lifts it under a real-spectrum upper bound to the operator quadratic bound $\exp(\theta X) \preceq I + \theta X + g(\theta, R)X^2$ are formalized as well. The generic centered one-sample log-CGF variance proxy

$$\log \mathbb{E} \exp(\theta X) \preceq g(\theta, R) \mathbb{E}[X^2]$$

is now also formalized locally, including a support-aware version. The truncated Algorithm 1 dilation-increment instantiation of that theorem is now formalized locally as well. The one-sample dilation variance proxy, the negative-support lower-tail companion, and the positive/negative scalar trace-MGF bounds now give parameterized upper-tail and two-sided trace-exponential eigenvalue bounds for the truncated dilation residual, plus an explicit $1 - \delta$ scaled-eigenvalue corollary obtained by choosing $T = \log(2B/\delta)$. The deterministic bridge from that scaled eigenvalue event to a rectangular spectral event at radius $\log(2B/\delta)/\theta$ is also formalized, and the scalar Bennett choice of θ now gives a radius- r high-probability theorem under the explicit Bennett budget. The proof-source ledger now marks the source denominator, source sample budget, deterministic truncation transfer, and support-aware floating-point transfer as closed for the hard-thresholded element-wise variant. The explicit entrywise perturbation budget used by the FP transfer theorem is now derived locally from the γ - and hit-count stability infrastructure on the sampler's probability-one support. The same support-aware mechanism is now composed with the literal, nontruncated Frobenius/Markov route and with the source-rate no-small-entry specialization recorded at the end of the Algorithm 1 section.

10 Low-Rank Approximation Foundations

Corollary 533 (Exact rank certificates for basis-product approximants). *Let $A \in \mathbb{R}^{m \times n}$, $V \in \mathbb{R}^{n \times q}$, and $U \in \mathbb{R}^{m \times q}$. Define the exact basis-product matrices*

$$P_V = VV^T, \quad P_U = UU^T,$$

and the projected approximants

$$B_R = AP_V, \quad B_L = P_U A.$$

Then B_R has the explicit rank factorization

$$B_R(i, j) = \sum_{a=1}^q \left(\sum_{k=1}^n A(i, k) V(k, a) \right) V(j, a),$$

so B_R has repository rank at most q . Similarly B_L has the explicit rank factorization

$$B_L(i, j) = \sum_{a=1}^q U(i, a) \left(\sum_{k=1}^m U(k, a) A(k, j) \right),$$

so B_L has repository rank at most q .

This is an exact-object foundation for the equation (9) low-rank route. It does not assume that U or V was computed, and it introduces no probability-construction or sampling-law error. If an implementation computes the bases, singular vectors, projectors, pseudoinverses, or downstream products, those non-probability computations remain subject to the existing computed-basis and computed-projector certificate rules.

Lean formalization.

```

RectRankFactorization
RectRankAtMost
lowRankResidualFrob
IsBestRankApproxFrob
rightBasisProjectorApprox
leftBasisProjectorApprox
rightBasisProjectorApproxFactorization
leftBasisProjectorApproxFactorization
rightBasisProjectorApprox_rankAtMost
leftBasisProjectorApprox_rankAtMost

```

Corollary 534 (Best-rank comparison for exact basis-product approximants). *Let $A, A_k \in \mathbb{R}^{m \times n}$, and suppose A_k carries a repository certificate $\text{BestRankFrob}(A, A_k, k)$, meaning that*

$$\text{rank}_{\text{cert}}(A_k) \leq k \quad \text{and} \quad \|A - A_k\|_F \leq \|A - B\|_F \quad \text{for every } B \text{ with } \text{rank}_{\text{cert}}(B) \leq k.$$

Then every exact repository-rank- k comparison matrix satisfies

$$\|A - A_k\|_F \leq \|A - B\|_F.$$

In particular, for $V \in \mathbb{R}^{n \times k}$ and $U \in \mathbb{R}^{m \times k}$, the rank certificates above give

$$\|A - A_k\|_F \leq \|A - A(VV^T)\|_F, \quad \|A - A_k\|_F \leq \|A - (UU^T)A\|_F.$$

This is still an exact-object theorem. It compares against exact analysis matrices $A(VV^T)$ and $(UU^T)A$; if an algorithm computes U , V , projectors, singular vectors, pseudoinverses, products, or downstream solver inputs, those non-probability computations require the explicit floating-point/inexact-arithmetic certificates recorded elsewhere. Sampling probabilities and laws remain exact mathematical inputs by the current convention.

Lean formalization.

```

IsBestRankApproxFrob.residual_le_of_rankAtMost
IsBestRankApproxFrob.residual_le_rightBasisProjectorApprox
IsBestRankApproxFrob.residual_le_leftBasisProjectorApprox

```

Corollary 535 (Column-sketch projector rank certificate for equation (9)). *Let $A \in \mathbb{R}^{m \times n}$, $Z \in \mathbb{R}^{n \times r}$, and define the exact column sketch*

$$B = AZ \in \mathbb{R}^{m \times r}.$$

Let $P_{AZ} \in \mathbb{R}^{m \times m}$ be an exact left multiplier that factors through the displayed columns of B : assume there is a coefficient matrix $C \in \mathbb{R}^{r \times m}$ such that

$$P_{AZ}(i, \ell) = \sum_{a=1}^r B(i, a)C(a, \ell).$$

Then the exact projected approximation $P_{AZ}A$ has the rank factorization

$$(P_{AZ}A)(i, j) = \sum_{a=1}^r B(i, a) \left(\sum_{\ell=1}^m C(a, \ell) A(\ell, j) \right),$$

and hence has repository rank at most r .

This is the exact rank surface needed for the source equation (9) projector P_{AZ} . It does not prove that P_{AZ} is the orthogonal projector onto $\text{range}(AZ)$, does not construct $(V_k^T Z)^+$, does not prove the full-rank condition on $V_k^T Z$, and does not prove the source residual bound involving $\Sigma_{k,\perp}(V_{k,\perp}^T Z)(V_k^T Z)^+$. If an algorithm computes Z , AZ , P_{AZ} , a pseudoinverse, or the product $P_{AZ}A$, those non-probability computations require explicit FP or inexact-arithmetic certificates. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
columnSketch
LeftFactorThrough
leftProductFactorizationOfLeftFactorThrough
preconditionRows_rankAtMost_of_leftFactorThrough
sketchColumnProjectorApprox_rankAtMost
```

Corollary 536 (Equation (9) residual-certificate surface). *Let $A \in \mathbb{R}^{m \times n}$, $Z \in \mathbb{R}^{n \times r}$, and $P_{AZ} \in \mathbb{R}^{m \times m}$. Assume first that P_{AZ} factors through the exact column sketch AZ , in the sense of the preceding corollary. Assume also that there are exact analysis scalars $T, C \geq 0$ such that the displayed equation (9) residual certificate holds:*

$$\|A - P_{AZ}A\|_F \leq T + C.$$

Then the exact candidate $P_{AZ}A$ has repository rank at most r , and

$$\|A - P_{AZ}A\|_F \leq T + C.$$

Moreover, if A_k carries a repository certificate $\text{BestRankFrob}(A, A_k, k)$, and if the source-side scalar comparison

$$T + C \leq \rho \|A - A_k\|_F$$

is supplied, then

$$\text{rankcert}(A_k) \leq k, \quad \text{rankcert}(P_{AZ}A) \leq r, \quad \|A - P_{AZ}A\|_F \leq \rho \|A - A_k\|_F.$$

This is not a proof of the CACM equation (9) SVD/pseudoinverse inequality. Rather, it makes the residual inequality an explicit exact-object certificate: the missing source instantiation must still prove that the tail and coupling terms T and C can be chosen from the singular-subspace expression $\|\Sigma_{k,\perp}\|$ and $\|\Sigma_{k,\perp}(V_{k,\perp}^T Z)(V_k^T Z)^+\|$. It therefore keeps rectangular SVD, the full-rank condition on $V_k^T Z$, pseudoinverse algebra, unitarily invariant norm infrastructure, and every computed non-probability object as separate obligations. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

Equation9ResidualCertificate

Equation9ResidualCertificate.tail_add_coupling_nonneg

equation9RankResidualSurface

equation9RelativeResidualSurface

Corollary 537 (Explicit coefficient handoff for the equation (9) projector). *Let $A \in \mathbb{R}^{m \times n}$, $Z \in \mathbb{R}^{n \times r}$, and let $C \in \mathbb{R}^{r \times m}$ be any exact coefficient table. Define*

$$P_C := (AZ)C, \quad P_C(i, j) = \sum_{a=1}^r (AZ)(i, a)C(a, j).$$

Then P_C factors through the exact column sketch AZ , and therefore

$$\text{rankcert}(P_C A) \leq r.$$

If, in addition, P_C carries an equation (9) residual certificate with exact terms $T, C_0 \geq 0$, then

$$\|A - P_C A\|_F \leq T + C_0.$$

Finally, if A_k is certified best rank k and $T + C_0 \leq \rho \|A - A_k\|_F$, then the relative residual conclusion

$$\|A - P_C A\|_F \leq \rho \|A - A_k\|_F$$

holds together with the displayed rank certificates.

This corollary is the algebraic handoff for the future choice $C = (AZ)^+$. It does not prove that the supplied C is a pseudoinverse, that P_C is the orthogonal projector onto $\text{range}(AZ)$, or that the equation (9) residual certificate follows from the source SVD hypotheses. Computed construction of C , P_C , a pseudoinverse, or products remains a non-probability FP/certificate obligation.

Lean formalization.

columnSketchLeftMultiplier

columnSketchLeftMultiplier_leftFactorThrough

columnSketchLeftMultiplier_rankAtMost

columnSketchLeftMultiplier_equation9RankResidualSurface

columnSketchLeftMultiplier_equation9RelativeResidualSurface

Corollary 538 (Generalized-inverse projector surface for the equation (9) sketch). *Let $A \in \mathbb{R}^{m \times n}$, $Z \in \mathbb{R}^{n \times r}$, and write*

$$B = AZ, \quad P_C = BC$$

for an exact coefficient table $C \in \mathbb{R}^{r \times m}$. Assume the displayed generalized-inverse condition

$$P_C B = B, \quad \text{equivalently} \quad \sum_{\ell=1}^m \left(\sum_{a=1}^r B(i, a)C(a, \ell) \right) B(\ell, b) = B(i, b)$$

for every row i and sketch column b . Then P_C reproduces the exact column sketch and is idempotent:

$$P_C B = B, \quad P_C^2 = P_C.$$

Moreover P_C factors through the exact sketch columns, and hence the projected approximation satisfies

$$\text{rankcert}(P_C A) \leq r.$$

This is the exact algebraic projector surface needed before a future pseudoinverse instantiation $C = (AZ)^+$. It does not prove the Moore–Penrose conditions, orthogonality or symmetry of P_C , existence of $C = (AZ)^+$, the full-rank condition on $V_k^T Z$, or the equation (9) SVD residual certificate. Computed construction of C , P_C , pseudoinverses, or products remains a non-probability FP/certificate obligation.

Lean formalization.

```
ColumnSketchGeneralizedInverse
columnSketchLeftMultiplier_reproducesSketch_of_generalizedInverse
columnSketchLeftMultiplier_idempotent_of_generalizedInverse
columnSketchLeftMultiplier_projectorSurface_of_generalizedInverse
```

Corollary 539 (Symmetric-idempotent projector surface for the equation (9) sketch). *In the setting of the preceding corollary, assume in addition that the exact multiplier $P_C = BC$ is symmetric:*

$$P_C(i, j) = P_C(j, i) \quad \text{for all } i, j.$$

Then P_C is a symmetric idempotent multiplier that reproduces the sketch:

$$P_C^T = P_C, \quad P_C^2 = P_C, \quad P_C B = B.$$

It also factors through the sketch columns, and so

$$\text{rankcert}(P_C A) \leq r.$$

This is the exact symmetric-idempotent surface expected of the source orthogonal projector onto $\text{range}(AZ)$. It does not construct the Moore–Penrose pseudoinverse, prove that $C = (AZ)^+$ satisfies the certificate, prove full rank of $V_k^T Z$, or supply the equation (9) residual certificate. Computed construction of C , P_C , pseudoinverses, or products remains a non-probability FP/certificate obligation.

Lean formalization.

```
ColumnSketchOrthogonalProjectorCertificate
columnSketchLeftMultiplier_symmetric_of_orthogonalProjectorCertificate
columnSketchLeftMultiplier_idempotent_of_orthogonalProjectorCertificate
columnSketchLeftMultiplier_orthogonalProjectorSurface
```

Corollary 540 (Moore–Penrose certificate handoff for the equation (9) sketch). *Let $A \in \mathbb{R}^{m \times n}$, $Z \in \mathbb{R}^{n \times r}$, and*

$$B = AZ, \quad P_C = BC, \quad Q_C = CB$$

for an exact coefficient table $C \in \mathbb{R}^{r \times m}$. Assume the four exact Moore–Penrose certificate equations

$$P_C B = B, \quad Q_C C = C, \quad P_C^T = P_C, \quad Q_C^T = Q_C.$$

Equivalently, the first two equations are $BCB = B$ and $CBC = C$. Then the first and third fields supply the generalized-inverse and symmetric-projector certificates for $P_C = (AZ)C$. Consequently

$$P_C B = B, \quad P_C^2 = P_C, \quad P_C^T = P_C, \quad \text{rankcert}(P_C A) \leq r.$$

The second and fourth fields are retained as the coefficient-side Moore–Penrose obligations $CBC = C$ and $(CB)^T = CB$, so later pseudoinverse algebra can use the full four-equation surface.

This corollary is a certificate handoff only. It does not construct $(AZ)^+$, prove that such a C exists, prove full rank of $V_k^T Z$, or supply the equation (9) residual certificate. Computed construction of C , P_C , Q_C , pseudoinverses, or products remains a non-probability FP/certificate obligation.

Lean formalization.

```
columnSketchRightMultiplier
ColumnSketchMoorePenroseCertificate
ColumnSketchMoorePenroseCertificate.to_generalizedInverse
ColumnSketchMoorePenroseCertificate.to_orthogonalProjectorCertificate
columnSketchRightMultiplier_reproducesCoeff_of_moorePenroseCertificate
columnSketchRightMultiplier_symmetric_of_moorePenroseCertificate
columnSketchLeftMultiplier_orthogonalProjectorSurface_of_moorePenroseCertificate
```

Corollary 541 (Gram-inverse Moore–Penrose instantiation for the equation (9) sketch). *Let $B = AZ$, let*

$$G = B^T B, \quad C = G_{\text{inv}} B^T, \quad C(a, i) = \sum_b G_{\text{inv}}(a, b) B(i, b).$$

Assume that G_{inv} is an exact two-sided inverse certificate for G , and assume that G_{inv} is symmetric. Then

$$CB = I, \quad CBC = C.$$

Moreover BC and CB are symmetric, and the generalized-inverse equation also holds:

$$BCB = B, \quad (BC)^T = BC, \quad (CB)^T = CB.$$

Therefore the coefficient table $C = G_{\text{inv}} B^T$ satisfies the four-equation Moore–Penrose certificate of the preceding corollary. In particular, for $P_C = BC$,

$$P_C B = B, \quad P_C^2 = P_C, \quad P_C^T = P_C, \quad \text{rankcert}(P_C A) \leq r.$$

This is still exact-object certificate algebra. It does not prove that $G = B^T B$ is invertible, construct G_{inv} , prove the full-rank condition for AZ or $V_k^T Z$, or supply the equation (9) residual certificate. If an implementation computes G , G_{inv} , C , P_C , or downstream products, those non-probability computations require explicit FP or inexact-arithmetic certificates. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
columnSketchGram
columnSketchGramInverseCoefficient
ColumnSketchGramInverseCertificate
columnSketchRightMultiplier_eq_id_of_gramInverseCertificate
```

`columnSketchGramInverseCoefficient_reproducesCoeff`
`columnSketchLeftMultiplier_symmetric_of_gramInverseCertificate`
`columnSketchRightMultiplier_symmetric_of_gramInverseCertificate`
`columnSketchGramInverseCoefficient_generalizedInverse`
`columnSketchGramInverseCoefficient_moorePenroseCertificate`
`columnSketchLeftMultiplier_orthogonalProjectorSurface_of_gramInverseCertificate`

Corollary 542 (Determinant-facing Gram-inverse route for the equation (9) sketch). *Let $B = AZ$ and $G = B^T B$. The exact Gram matrix is symmetric:*

$$G^T = G.$$

Assume the exact determinant condition

$$\det(G) \neq 0,$$

and define

$$G_{\text{ns}}^{-1} := \text{nonsingInv}(G), \quad C = G_{\text{ns}}^{-1} B^T.$$

Then G_{ns}^{-1} is a two-sided inverse certificate for G , and

$$(G_{\text{ns}}^{-1})^T = G_{\text{ns}}^{-1}.$$

Consequently the coefficient table C satisfies the same four Moore–Penrose equations as in the preceding corollary:

$$BCB = B, \quad CBC = C, \quad (BC)^T = BC, \quad (CB)^T = CB.$$

For $P_C = BC$, it follows that

$$P_C B = B, \quad P_C^2 = P_C, \quad P_C^T = P_C, \quad \text{rankcert}(P_C A) \leq r.$$

This is an exact-object determinant theorem. It does not prove the determinant condition from full column rank of AZ , does not connect that condition to the source hypothesis on $V_k^T Z$, and does not supply the equation (9) residual certificate. If an implementation computes G , G_{ns}^{-1} , C , P_C , or downstream products, those non-probability computations require explicit FP or inexact-arithmetic certificates. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

`nonsingInv_symmetric_of_symmetric`
`columnSketchGram_symmetric`
`columnSketchGramInverseCertificate_of_det_ne_zero`
`columnSketchGramInverseCoefficient_moorePenroseCertificate_of_det_ne_zero`
`columnSketchLeftMultiplier_orthogonalProjectorSurface_of_det_ne_zero`

Corollary 543 (Thin-factor Gram determinant bridge for the equation (9) sketch). *Let $B = AZ$. Suppose there are exact matrices*

$$U \in \mathbb{R}^{m \times r}, \quad R \in \mathbb{R}^{r \times r}$$

such that

$$B = UR, \quad U^T U = I, \quad \det(R) \neq 0.$$

Then the exact sketch Gram matrix satisfies

$$B^T B = R^T R,$$

and therefore

$$\det(B^T B) = \det(R)^2 \neq 0.$$

Consequently the determinant-facing corollary applies with

$$G = B^T B, \quad G_{\text{ns}}^{-1} := \text{nonsingInv}(G), \quad C = G_{\text{ns}}^{-1} B^T.$$

Thus C satisfies

$$BCB = B, \quad CBC = C, \quad (BC)^T = BC, \quad (CB)^T = CB,$$

and for $P_C = BC$,

$$P_C B = B, \quad P_C^2 = P_C, \quad P_C^T = P_C, \quad \text{rankcert}(P_C A) \leq r.$$

This is an exact-object thin-factor theorem. It does not construct such a thin factorization from rectangular QR or SVD, does not connect it to the source hypothesis on $V_k^T Z$, and does not supply the equation (9) residual certificate. If an implementation computes U , R , G , G_{ns}^{-1} , C , P_C , or downstream products, those non-probability computations require explicit FP or inexact-arithmetic certificates. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
ColumnSketchThinFactorCertificate
columnSketchGram_eq_factorGram_of_thinFactorCertificate
columnSketchGram_det_ne_zero_of_thinFactorCertificate
columnSketchGramInverseCertificate_of_thinFactorCertificate
columnSketchGramInverseCoefficient_moorePenroseCertificate_of_thinFactorCertificate
columnSketchLeftMultiplier_orthogonalProjectorSurface_of_thinFactorCertificate
```

Corollary 544 (Source-SVD head rank certificate for equation (9)). *Let*

$$U \in \mathbb{R}^{m \times r}, \quad \Sigma \in \mathbb{R}^{r \times r}, \quad V \in \mathbb{R}^{n \times r},$$

and define the exact source head

$$H = U \Sigma V^T, \quad H_{ij} = \sum_{a=1}^r U_{ia} \sum_{b=1}^r \Sigma_{ab} V_{jb}.$$

Then H factors through the displayed source dimension r :

$$H_{ij} = \sum_{a=1}^r U_{ia} R_{aj}, \quad R_{aj} = \sum_{b=1}^r \Sigma_{ab} V_{jb}.$$

Consequently

$$\text{rankcert}(H) \leq r.$$

This is an exact-object D3 source-SVD theorem. It supplies the rank certificate for the displayed source head only. It does not construct the rectangular SVD, prove source-tail orthogonality, prove singular-value comparison or Eckart–Young optimality, or certify computed SVD/singular-vector/projector/Gram data. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
sourceSVDFactorMatrixRankFactorization
sourceSVDFactorMatrix_rankAtMost
```

Corollary 545 (Source-split residual and best-rank handoff for equation (9)). *Let $A, H, T \in \mathbb{R}^{m \times n}$, let*

$$H = U\Sigma V^T,$$

and suppose the exact source split

$$A = H + T$$

holds entrywise. Then the source-head residual is exactly the tail norm:

$$\rho_F(A, H) := \|A - H\|_F = \|T\|_F.$$

More generally, for any supplied rectangular norm certificate ξ ,

$$\rho_\xi(A, H) := \xi(A - H) = \xi(T).$$

If, in addition, the Frobenius tail-optimality inequality

$$\|T\|_F \leq \rho_F(A, B) \quad \text{for every } B \text{ with } \text{rankcert}(B) \leq r$$

is supplied, then H is a Frobenius best rank- r approximation to A :

$$\text{BestRankFrob}_r(A, H).$$

Likewise, if

$$\xi(T) \leq \rho_\xi(A, B) \quad \text{for every } B \text{ with } \text{rankcert}(B) \leq r,$$

then H is a best rank- r approximation for the norm ξ :

$$\text{BestRank}_{\xi, r}(A, H).$$

This is an exact-object D3/D4 handoff. The residual equalities are proved from the displayed source split, and the rank side uses the preceding source head rank certificate. The Eckart–Young/tail-optimality inequalities remain explicit hypotheses until the rectangular SVD and singular-value comparison foundations are proved. Any implementation that computes U, Σ, V, T , or the norm data must supply non-probability FP or inexact-arithmetic certificates. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
lowRankResidualFrob_sourceSVDFactorMatrix_eq_tail
lowRankResidualNorm_sourceSVDFactorMatrix_eq_tail
sourceSVDFactorMatrix_isBestRankApproxFrob_of_tail_optimal
sourceSVDFactorMatrix_isBestRankApproxNorm_of_tail_optimal
```

Corollary 546 (Source-SVD Gram determinant bridge for the equation (9) sketch). *Let $A \in \mathbb{R}^{m \times n}$, $Z \in \mathbb{R}^{n \times r}$, and suppose there are exact matrices*

$$U \in \mathbb{R}^{m \times r}, \quad \Sigma \in \mathbb{R}^{r \times r}, \quad V \in \mathbb{R}^{n \times r}$$

such that

$$A = U\Sigma V^T, \quad U^T U = I.$$

Define the exact source cross factor and sketch right factor by

$$M := V^T Z, \quad R := \Sigma M = \Sigma(V^T Z).$$

If

$$\det(\Sigma) \neq 0, \quad \det(M) \neq 0,$$

then

$$AZ = UR, \quad \det(R) = \det(\Sigma) \det(M) \neq 0.$$

Consequently the thin-factor corollary applies, and hence

$$\det((AZ)^T(AZ)) \neq 0.$$

With

$$G = (AZ)^T(AZ), \quad G_{\text{ns}}^{-1} := \text{nonsingInv}(G), \quad C = G_{\text{ns}}^{-1}(AZ)^T,$$

the coefficient table C satisfies

$$(AZ)C(AZ) = AZ, \quad C(AZ)C = C, \quad ((AZ)C)^T = (AZ)C, \quad (C(AZ))^T = C(AZ).$$

For $P_C = (AZ)C$,

$$P_C(AZ) = AZ, \quad P_C^2 = P_C, \quad P_C^T = P_C, \quad \text{rankcert}(P_C A) \leq r.$$

This is an exact-object source-factor theorem. It assumes the exact factorization $A = U\Sigma V^T$, exact orthonormality of U , and exact determinant hypotheses on Σ and $V^T Z$. It does not prove rectangular SVD existence, Eckart–Young best-rank optimality, the source equation (9) residual inequality, or any computed SVD/singular-vector/projector/Gram/inverse/product routine. If an implementation computes U , Σ , V , M , R , G , G_{ns}^{-1} , C , P_C , or downstream products, those non-probability computations require explicit FP or inexact-arithmetic certificates. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
sourceSVDFactorMatrix
rightSketchCrossGram
sourceSVDSketchRightFactor
columnSketch_eq_sourceSVDfactorization
sourceSVDSketchRightFactor_det_ne_zero_of_det_ne_zero
columnSketchThinFactorCertificate_of_sourceSVD
columnSketchThinFactorCertificate_of_sourceSVD_det_factors
columnSketchGram_det_ne_zero_of_sourceSVD_det_factors
columnSketchGramInverseCertificate_of_sourceSVD_det_factors
columnSketchGramInverseCoefficient_moorePenroseCertificate_of_sourceSVD_det_factors
columnSketchLeftMultiplier_orthogonalProjectorSurface_of_sourceSVD_det_factors
```


Corollary 547 (Source-SVD head-tail residual bridge for equation (9)). *Let $A \in \mathbb{R}^{m \times n}$, $Z \in \mathbb{R}^{n \times r}$, and define the exact Gram-inverse projector*

$$B := AZ, \quad G := B^T B, \quad P_G := B \text{ nonsingInv}(G) B^T.$$

Assume the source-SVD determinant hypotheses of the previous corollary:

$$A = U \Sigma V^T, \quad U^T U = I, \quad \det(\Sigma) \neq 0, \quad \det(V^T Z) \neq 0.$$

Then P_G is symmetric, reproduces the sketch, is idempotent on itself, and $\text{rankcert}(P_G A) \leq r$.

Now suppose there are exact matrices

$$H, T \in \mathbb{R}^{m \times n}, \quad W \in \mathbb{R}^{r \times n},$$

and exact nonnegative radii $\tau, \kappa \geq 0$, such that

$$A = H + T, \quad H = BW, \quad \|T\|_F \leq \tau, \quad \|P_G T\|_F \leq \kappa.$$

Since $P_G B = B$, the head is reproduced:

$$P_G H = P_G B W = B W = H.$$

Therefore

$$A - P_G A = T - P_G T,$$

and the rectangular Frobenius subtraction inequality gives

$$\|A - P_G A\|_F = \|T - P_G T\|_F \leq \|T\|_F + \|P_G T\|_F \leq \tau + \kappa.$$

Equivalently, the data above instantiate the equation (9) residual certificate with tail radius τ and coupling radius κ . In particular,

$$\text{rankcert}(P_G A) \leq r, \quad \|A - P_G A\|_F \leq \tau + \kappa.$$

If A_k is supplied with an exact best-rank- k Frobenius certificate and

$$\tau + \kappa \leq \rho \|A - A_k\|_F,$$

then

$$\text{rankcert}(A_k) \leq k, \quad \text{rankcert}(P_G A) \leq r, \quad \|A - P_G A\|_F \leq \rho \|A - A_k\|_F.$$

This is an exact-object residual bridge. It does not construct H, T, W from rectangular SVD data, prove Eckart–Young optimality, prove the source $V_k^T Z$ pseudoinverse residual inequality, or supply computed SVD/projector/Gram/inverse/product error certificates. If an implementation computes $U, \Sigma, V, H, T, W, G, P_G$, or any downstream product, those non-probability computations require explicit FP or inexact-arithmetic certificates. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

`frobNormRect_neg`

`frobNormRect_sub_le`

`ColumnSketchHeadFactorization`

```

preconditionRows_reproduces_head_of_columnSketchHeadFactorization
Equation9HeadTailSketchCertificate
Equation9HeadTailSketchCertificate.to_residualCertificate
equation9HeadTailSketchRankResidualSurface
equation9HeadTailSketchRelativeResidualSurface
columnSketchGramInverseProjector
columnSketchGramInverseProjector_orthogonalProjectorSurface_of_sourceSVD_det_factors
columnSketchGramInverseProjector_sourceSVD_headTailRankResidualSurface
columnSketchGramInverseProjector_sourceSVD_headTailRelativeResidualSurface

```

Corollary 548 (Explicit-coefficient head-tail bridge for equation (9)). *Keep the exact source-SVD determinant hypotheses and the projector P_G from the preceding corollary. Let $W \in \mathbb{R}^{r \times n}$ be any exact coefficient table and define*

$$H_W := (AZ)W, \quad T_W := A - H_W.$$

Then H_W factors through the sketch by definition and

$$A = H_W + T_W.$$

Consequently, if exact radii $\tau, \kappa \geq 0$ satisfy

$$\|T_W\|_F \leq \tau, \quad \|P_G T_W\|_F \leq \kappa,$$

then the head-tail certificate of the preceding corollary is instantiated with $H = H_W$, $T = T_W$, and hence

$$\text{rankcert}(P_G A) \leq r, \quad \|A - P_G A\|_F \leq \tau + \kappa.$$

If A_k carries an exact best-rank- k Frobenius certificate and

$$\tau + \kappa \leq \rho \|A - A_k\|_F,$$

then

$$\text{rankcert}(A_k) \leq k, \quad \text{rankcert}(P_G A) \leq r, \quad \|A - P_G A\|_F \leq \rho \|A - A_k\|_F.$$

This is still an exact-object adapter. The remaining source proof must choose the pseudoinverse-derived W and prove the two displayed norm bounds from rectangular SVD, the $V_k^T Z$ condition, and unitarily invariant norm infrastructure. If an implementation computes W , T_W , P_G , or any source SVD/projector/Gram/inverse/product data, those non-probability computations require explicit FP or inexact-arithmetic certificates.

Lean formalization.

```

columnSketchHead
columnSketchTail
columnSketchHead_headFactorization
columnSketchHeadTail_split
equation9HeadTailSketchCertificate_of_columnSketchHead
columnSketchGramInverseProjector_sourceSVD_columnSketchHeadRankResidualSurface
columnSketchGramInverseProjector_sourceSVD_columnSketchHeadRelativeResidualSurface

```

Corollary 549 (Source-coefficient residual-tail bridge for equation (9)). *Let $U \in \mathbb{R}^{m \times r}$, $\Sigma \in \mathbb{R}^{r \times r}$, $V \in \mathbb{R}^{n \times r}$, $Z \in \mathbb{R}^{n \times r}$, and let*

$$M := V^T Z.$$

Assume

$$\det(M) \neq 0$$

and define the exact coefficient table

$$W := M^{-1} V^T.$$

Then

$$MW = V^T, \quad \Sigma MW = \Sigma V^T,$$

and the source head is reproduced from its own sketch:

$$((U \Sigma V^T) Z) W = U \Sigma V^T.$$

Now let $A, T \in \mathbb{R}^{m \times n}$ satisfy the exact source split

$$A = U \Sigma V^T + T.$$

Then the implemented coefficient-generated sketch head and sketch tail obey

$$(AZ)W = U \Sigma V^T + (TZ)W,$$

and

$$A - (AZ)W = T - (TZ)W.$$

Equivalently, with

$$T_{\text{src}} := T - (TZ)(V^T Z)^{-1} V^T,$$

the canonical sketch-tail object $T_W = A - (AZ)W$ is exactly

$$T_W = T_{\text{src}}.$$

Consequently, for any exact multiplier P_{AZ} that factors through the columns of AZ and reproduces AZ , if exact radii $\tau, \kappa \geq 0$ satisfy

$$\|T_{\text{src}}\|_F \leq \tau, \quad \|P_{AZ} T_{\text{src}}\|_F \leq \kappa,$$

then

$$\text{rankcert}(P_{AZ} A) \leq r, \quad \|A - P_{AZ} A\|_F \leq \tau + \kappa.$$

If A_k carries an exact best-rank- k Frobenius certificate and

$$\tau + \kappa \leq \rho \|A - A_k\|_F,$$

then

$$\text{rankcert}(A_k) \leq k, \quad \text{rankcert}(P_{AZ} A) \leq r, \quad \|A - P_{AZ} A\|_F \leq \rho \|A - A_k\|_F.$$

This is an exact-object source-coefficient theorem. It chooses the paper's pseudoinverse-derived coefficient table $W = (V^T Z)^{-1} V^T$ and rewrites the implemented sketch tail as the explicit source tail $T - (TZ)(V^T Z)^{-1} V^T$. It still does not prove the two Frobenius norm bounds, construct the exact orthogonal projector onto $\text{range}(AZ)$ for the head-plus-tail sketch, prove Eckart–Young optimality, or supply computed cross-product/inverse/coefficient/SVD/projector/product error certificates. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```

sourceSketchCoefficient
rightSketchCrossGram_sourceSketchCoefficient
sourceSVDSketchRightFactor_sourceSketchCoefficient
columnSketchHead_sourceSVDFactorMatrix_sourceSketchCoefficient
columnSketch_eq_add_of_eq_add
columnSketchHead_eq_add_of_columnSketch_eq_add
columnSketchHead_sourceHeadTail_sourceSketchCoefficient
sourceSketchResidualTail
columnSketchTail_sourceHeadTail_sourceSketchCoefficient
equation9HeadTailSketchCertificate_of_sourceHeadTail_sourceSketchCoefficient
equation9RankResidualSurface_of_sourceHeadTail_sourceSketchCoefficient
equation9RelativeResidualSurface_of_sourceHeadTail_sourceSketchCoefficient

```

Corollary 550 (Moore–Penrose source-tail projector bridge for equation (9)). *Keep the hypotheses and definitions of the preceding source-coefficient corollary:*

$$M = V^T Z, \quad \det(M) \neq 0, \quad W = M^{-1} V^T, \quad A = U \Sigma V^T + T,$$

and

$$T_{\text{src}} := T - (TZ)(V^T Z)^{-1} V^T.$$

Let $C \in \mathbb{R}^{r \times m}$ be an exact coefficient table for the full sketch $B := AZ$, and set

$$P_C := BC = (AZ)C.$$

Assume C satisfies the four exact Moore–Penrose certificate equations

$$BCB = B, \quad CBC = C, \quad (BC)^T = BC, \quad (CB)^T = CB.$$

Then P_C is symmetric, factors through AZ , reproduces AZ , and is idempotent:

$$P_C^T = P_C, \quad P_C(AZ) = AZ, \quad P_C^2 = P_C.$$

If exact radii $\tau, \kappa \geq 0$ satisfy

$$\|T_{\text{src}}\|_F \leq \tau, \quad \|P_C T_{\text{src}}\|_F \leq \kappa,$$

then

$$\text{rankcert}(P_C A) \leq r, \quad \|A - P_C A\|_F \leq \tau + \kappa.$$

If A_k carries an exact best-rank- k Frobenius certificate and

$$\tau + \kappa \leq \rho \|A - A_k\|_F,$$

then

$$\text{rankcert}(A_k) \leq k, \quad \text{rankcert}(P_C A) \leq r, \quad \|A - P_C A\|_F \leq \rho \|A - A_k\|_F.$$

This is an exact-object projector-certificate theorem. It removes the generic unnamed P_{AZ} hypothesis from the previous corollary and replaces it by the four displayed Moore–Penrose equations for the full sketch AZ . It still does not construct $C = (AZ)^+$, prove that a concrete Gram or pseudoinverse routine satisfies those equations for the head-plus-tail sketch, prove the two source-tail Frobenius bounds, or supply computed cross-product/inverse/ coefficient/SVD/projector/product error certificates. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

columnSketchLeftMultiplier_sourceHeadTail_sourceSketchCoefficientRankResidualSurface
columnSketchLeftMultiplier_sourceHeadTail_sourceSketchCoefficientRelativeResidualSurface

Corollary 551 (Concrete Gram-inverse source-tail projector bridge for equation (9)). *Keep the source-coefficient hypotheses*

$$M = V^T Z, \quad \det(M) \neq 0, \quad A = U \Sigma V^T + T,$$

and

$$T_{\text{src}} := T - (TZ)(V^T Z)^{-1} V^T.$$

Let $B := AZ$, $G := B^T B$, and assume the exact Gram determinant condition

$$\det(G) \neq 0.$$

Define the exact Gram-inverse coefficient table and projector

$$C_G := G^{-1} B^T, \quad P_G := B C_G = (AZ)((AZ)^T(AZ))^{-1}(AZ)^T.$$

Then the determinant-facing Gram route supplies the four exact Moore–Penrose equations

$$B C_G B = B, \quad C_G B C_G = C_G, \quad (B C_G)^T = B C_G, \quad (C_G B)^T = C_G B.$$

Consequently P_G is symmetric, factors through AZ , reproduces AZ , and is idempotent:

$$P_G^T = P_G, \quad P_G(AZ) = AZ, \quad P_G^2 = P_G.$$

If exact radii $\tau, \kappa \geq 0$ satisfy

$$\|T_{\text{src}}\|_F \leq \tau, \quad \|P_G T_{\text{src}}\|_F \leq \kappa,$$

then

$$\text{rankcert}(P_G A) \leq r, \quad \|A - P_G A\|_F \leq \tau + \kappa.$$

If A_k carries an exact best-rank- k Frobenius certificate and

$$\tau + \kappa \leq \rho \|A - A_k\|_F,$$

then

$$\text{rankcert}(A_k) \leq k, \quad \text{rankcert}(P_G A) \leq r, \quad \|A - P_G A\|_F \leq \rho \|A - A_k\|_F.$$

This is an exact-object Gram-inverse theorem. It instantiates the Moore–Penrose certificate C of the previous corollary with the concrete table $C_G = \text{nosingInv}((AZ)^T(AZ))(AZ)^T$, under the displayed exact determinant hypothesis. It still does not prove $\det((AZ)^T(AZ)) \neq 0$ from source SVD/full-rank hypotheses for the head-plus-tail sketch, prove the two source-tail Frobenius bounds, prove Eckart–Young optimality, build unitarily invariant norm infrastructure, or supply computed Gram/inverse/coefficient/SVD/projector/product error certificates. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

columnSketchGramInverseProjector_sourceHeadTail_sourceSketchCoefficientRankResidualSurface_of_det_ne_zero
columnSketchGramInverseProjector_sourceHeadTail_sourceSketchCoefficientRelativeResidualSurface_of_det_ne_zero

Corollary 552 (Source-tail orthogonality from exact tail factors for equation (9)). *Let*

$$T = U_{\perp} \Sigma_{\perp} V_{\perp}^T, \quad T_{ij} = \sum_{\beta} (U_{\perp})_{i\beta} \sum_{\gamma} (\Sigma_{\perp})_{\beta\gamma} (V_{\perp})_{j\gamma},$$

and let $U \in \mathbb{R}^{m \times r}$ be the exact source-head left basis. If the exact head and tail left bases are cross-orthogonal,

$$\sum_i U_{i\alpha} (U_{\perp})_{i\beta} = 0 \quad \text{for every } \alpha, \beta,$$

then the source-tail orthogonality predicate required by the head-tail sketch-Gram split holds:

$$U^T T = 0, \quad \sum_i U_{i\alpha} T_{ij} = 0 \quad \text{for every } \alpha, j.$$

This is an exact-object D3 theorem. It derives the displayed $U^T T = 0$ field from supplied tail factors and left-basis cross-orthogonality. It does not construct the rectangular SVD or the tail factors, prove their cross-orthogonality, prove singular-value comparison or Eckart–Young optimality, or certify computed SVD/singular-vector/projector/Gram data. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

`sourceTailLeftOrthogonal_of_tail_factor_left_cross_zero`

Corollary 553 (Orthogonal source head-tail sketch-Gram split for equation (9)). *Let*

$$H := U \Sigma V^T, \quad A = H + T,$$

where $U \in \mathbb{R}^{m \times r}$, $\Sigma \in \mathbb{R}^{r \times r}$, $V \in \mathbb{R}^{n \times r}$, $T \in \mathbb{R}^{m \times n}$, and $Z \in \mathbb{R}^{n \times r}$. Assume the exact source-tail left-orthogonality condition

$$U^T T = 0, \quad \text{i.e.} \quad \sum_i U_{i\alpha} T_{ij} = 0 \quad \text{for all } \alpha, j.$$

Define the sketched source-head and tail blocks

$$B_H := HZ, \quad B_T := TZ.$$

Then the sketched source head has the exact factorization

$$HZ = U(\Sigma(V^T Z)).$$

Moreover the two cross-Gram terms vanish:

$$B_H^T B_T = 0, \quad B_T^T B_H = 0.$$

Consequently the full sketch Gram matrix splits as

$$(AZ)^T (AZ) = (HZ)^T (HZ) + (TZ)^T (TZ).$$

Equivalently, for all sketch columns a, b ,

$$\sum_i (AZ)_{ia} (AZ)_{ib} = \sum_i (HZ)_{ia} (HZ)_{ib} + \sum_i (TZ)_{ia} (TZ)_{ib}.$$

This is an exact-object determinant-route theorem. It closes the first algebraic component of the active equation (9) determinant bottleneck: the head-plus-tail sketch Gram is now reduced to a source-head sketch Gram plus the tail sketch Gram under the displayed source-tail orthogonality hypothesis. It does not yet prove that the tail sketch Gram is positive semidefinite, that the source-head Gram is positive definite or has nonzero determinant from $\det(\Sigma) \neq 0$ and $\det(V^T Z) \neq 0$, or that adding the tail Gram preserves the nonzero determinant. It also does not supply source-tail norm bounds, SVD/Eckart–Young/unitarily invariant norm infrastructure, or computed SVD/Gram/product certificates. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
sourceTailLeftOrthogonal
columnSketch_sourceSVDFactorMatrix
columnSketch_sourceSVDFactorMatrix_tail_leftOrthogonal
columnSketch_tail_sourceSVDFactorMatrix_leftOrthogonal
columnSketchGram_sourceHeadTail_leftOrthogonal
```

Corollary 554 (Column-sketch Gram PSD and determinant preservation for equation (9)). *For any exact matrix $A \in \mathbb{R}^{m \times n}$ and sketch $Z \in \mathbb{R}^{n \times r}$, let*

$$B := AZ, \quad G_A := B^T B.$$

Then for every vector $x \in \mathbb{R}^r$,

$$x^T G_A x = \sum_i \left(\sum_a B_{ia} x_a \right)^2.$$

In particular $G_A \succeq 0$.

Now keep the orthogonal source split of the preceding corollary:

$$H = U \Sigma V^T, \quad A = H + T, \quad U^T T = 0,$$

and define

$$G_H := (HZ)^T (HZ), \quad G_T := (TZ)^T (TZ).$$

The previous corollary gives

$$G_A = G_H + G_T,$$

and the first part gives $G_T \succeq 0$. Therefore, if the exact source-head sketch Gram carries a positive-definite certificate

$$G_H \succ 0,$$

then

$$G_A = G_H + G_T \succ 0, \quad \det(G_A) \neq 0.$$

This is an exact-object determinant-preservation theorem. It closes the PSD tail-Gram and positive-definite-plus-PSD part of the active equation (9) determinant route. The remaining determinant obligation is now narrower: prove the displayed positive-definite certificate $G_H \succ 0$ from the source thin-factor/full-rank data, in particular from $\det(\Sigma) \neq 0$ and $\det(V^T Z) \neq 0$. The result still does not prove source-tail norm bounds, SVD/Eckart–Young/unitarily invariant norm infrastructure, or computed SVD/Gram/product certificates. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
finiteQuadraticForm_columnSketchGram_eq_sum_sq
columnSketchGram_finitePSD
matrix_det_ne_zero_of_posDef_add_posSemidef
columnSketchGram_sourceHeadTail_det_ne_zero_of_head_posDef
```

Corollary 555 (Source-factor positive-definite sketch Gram and determinant closure for equation (9)). *Let $R \in \mathbb{R}^{r \times r}$ be an exact square factor. If*

$$\det(R) \neq 0,$$

then

$$R^T R \succ 0.$$

Equivalently, for every nonzero $x \in \mathbb{R}^r$,

$$x^T R^T R x = \|Rx\|_2^2 > 0.$$

Consequently, if an exact sketch $B = AZ$ has a thin factorization

$$B = UR, \quad U^T U = I_r, \quad \det(R) \neq 0,$$

then

$$(AZ)^T (AZ) = R^T R \succ 0.$$

Now let

$$H = U \Sigma V^T, \quad A = H + T, \quad U^T T = 0, \quad U^T U = I_r,$$

and assume the exact source determinant hypotheses

$$\det(\Sigma) \neq 0, \quad \det(V^T Z) \neq 0.$$

Set

$$R := \Sigma(V^T Z).$$

Then $\det(R) \neq 0$, the source-head sketch satisfies

$$HZ = UR,$$

and therefore

$$G_H := (HZ)^T (HZ) \succ 0.$$

Combining this positive-definite source-head certificate with the preceding PSD-tail determinant bridge gives

$$\det((AZ)^T (AZ)) \neq 0.$$

This is an exact-object determinant theorem. It closes the exact determinant side of the active equation (9) bottleneck from source factor hypotheses: nonzero $\det(\Sigma)$, nonzero $\det(V^T Z)$, exact orthonormal source columns, and exact source-tail left orthogonality. The result still does not prove the source-tail norm bounds, rectangular SVD/Eckart–Young/unitarily invariant norm infrastructure, or implementation-facing certificates for computed SVD/singular-vector/projector/Gram/inverse/product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
matrix_transpose_mul_self_posDef_of_det_ne_zero
columnSketchGram_posDef_of_thinFactorCertificate
columnSketchGram_posDef_of_sourceSVD_det_factors
columnSketchGram_sourceHeadTail_det_ne_zero_of_sourceSVD_det_factors
```

Corollary 556 (Tail-factor determinant route for equation (9)). *Let*

$$A = U\Sigma V^T + T, \quad T = U_\perp \Sigma_\perp V_\perp^T,$$

where

$$T_{ij} = \sum_{\beta} (U_\perp)_{i\beta} \sum_{\gamma} (\Sigma_\perp)_{\beta\gamma} (V_\perp)_{j\gamma}.$$

Assume the exact left-basis cross-orthogonality and source-head orthonormality conditions

$$U^T U_\perp = 0, \quad U^T U = I_r.$$

Equivalently,

$$\sum_i U_{i\alpha} (U_\perp)_{i\beta} = 0, \quad \sum_i U_{i\alpha} U_{i\beta} = \delta_{\alpha\beta}.$$

If the exact source determinant hypotheses also hold,

$$\det(\Sigma) \neq 0, \quad \det(V^T Z) \neq 0,$$

then

$$\det((AZ)^T(AZ)) \neq 0.$$

The proof first derives $U^T T = 0$ from the displayed tail factorization and $U^T U_\perp = 0$, then invokes the source-factor determinant theorem above. Thus the determinant-facing route can consume exact tail factors directly instead of requiring a raw source-tail orthogonality hypothesis.

This is an exact-object D3-to-D1 theorem. It still does not construct the rectangular SVD or tail factors, prove left/right basis orthogonality from SVD existence, prove the source-tail norm bounds, prove Eckart–Young optimality, or certify computed SVD/singular-vector/projector/Gram/inverse/product routines. All of those are non-probability computed-object obligations in implementation-facing versions. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
columnSketchGram_sourceHeadTail_det_ne_zero_of_sourceSVD_det_factors_tail_factor_left_cross_zero
```

Corollary 557 (Diagonal singular-block determinant route for equation (9)). *Let*

$$A = U\Sigma V^T + T, \quad T = U_\perp \Sigma_\perp V_\perp^T,$$

with the same exact tail-factor interpretation as above. Assume

$$U^T U_\perp = 0, \quad U^T U = I_r, \quad \det(V^T Z) \neq 0.$$

Let $\sigma : \{1, \dots, r\} \rightarrow \mathbb{R}$ be the displayed singular-value list for the exact source block, and assume the exact diagonal identity

$$\Sigma_{\alpha\beta} = \begin{cases} \sigma_\alpha, & \alpha = \beta, \\ 0, & \alpha \neq \beta. \end{cases}$$

If

$$\sigma_\alpha \neq 0 \quad \text{for every } \alpha,$$

then

$$\det(\Sigma) = \prod_{\alpha=1}^r \sigma_\alpha \neq 0,$$

and therefore

$$\det((AZ)^T(AZ)) \neq 0.$$

In particular, the same conclusion holds under the stronger displayed singular-value hypothesis $\sigma_\alpha > 0$ for every α .

Thus the determinant theorem can consume the exact diagonal singular block and nonzero displayed singular values instead of a raw hypothesis $\det(\Sigma) \neq 0$. This is still an exact-object D3-to-D1 result: it does not construct the rectangular SVD, prove that an implementation computed the diagonal block or singular values with perturbation guarantees, establish singular-value ordering, prove source-tail norm bounds, prove Eckart–Young optimality, or certify computed SVD, singular-vector, projector, Gram, inverse, or product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
matrix_det_ne_zero_of_eq_diagonal_nonzero
```

```
matrix_det_ne_zero_of_eq_diagonal_pos
```

```
columnSketchGram_sourceHeadTail_det_ne_zero_of_sourceSVD_diagonal_tail_factor_left_cross_zero
```

Corollary 558 (Diagonal source-SVD sketch and projector route for equation (9)). *Let*

$$H = U\Sigma V^T, \quad U^T U = I_r, \quad \det(V^T Z) \neq 0,$$

and let the exact source singular block be diagonal,

$$\Sigma_{\alpha\beta} = \begin{cases} \sigma_\alpha, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases} \quad \sigma_\alpha \neq 0 \quad \text{for every } \alpha.$$

Then the exact source sketch right factor is nonsingular:

$$\det(\Sigma(V^T Z)) \neq 0.$$

Consequently HZ has the exact thin factorization

$$HZ = U \Sigma(V^T Z),$$

with orthonormal left factor and nonsingular square right factor. Hence

$$\det((HZ)^T(HZ)) \neq 0, \quad (HZ)^T(HZ) \succ 0.$$

If additionally $A = H + T$ and $U^T T = 0$, then the head-plus-tail sketch Gram satisfies

$$\det((AZ)^T(AZ)) \neq 0.$$

For the source-head matrix H , let

$$P_H := HZ((HZ)^T(HZ))^{-1}(HZ)^T$$

be the exact Gram-inverse column-sketch projector. The same diagonal singular-block hypotheses give the exact Gram-inverse certificate, the Moore–Penrose coefficient certificate, and the projector surface

$$P_H^T = P_H, \quad P_H^2 = P_H, \quad P_H(HZ) = HZ, \quad \text{rank}(P_H H) \leq r.$$

This exact-object result propagates the displayed diagonal singular-value hypotheses through the source-SVD sketch/projector algebra. It does not construct the rectangular SVD, prove the diagonal block or singular-value positivity/order from SVD existence, prove source-tail norm bounds, prove Eckart–Young optimality, or certify computed SVD, singular-vector, projector, Gram, inverse, or product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
sourceSVDSketchRightFactor_det_ne_zero_of_diagonal_nonzero
sourceSVDSketchRightFactor_det_ne_zero_of_diagonal_pos
columnSketchThinFactorCertificate_of_sourceSVD_diagonal_det_factors
columnSketchGram_det_ne_zero_of_sourceSVD_diagonal_det_factors
columnSketchGram_posDef_of_sourceSVD_diagonal_det_factors
columnSketchGram_sourceHeadTail_det_ne_zero_of_sourceSVD_diagonal_det_factors
columnSketchGramInverseCertificate_of_sourceSVD_diagonal_det_factors
columnSketchGramInverseCoefficient_moorePenroseCertificate_of_sourceSVD_diagonal_det_factors
columnSketchLeftMultiplier_orthogonalProjectorSurface_of_sourceSVD_diagonal_det_factors
columnSketchGramInverseProjector_orthogonalProjectorSurface_of_sourceSVD_diagonal_det_factors
```

Corollary 559 (Diagonal source-SVD residual surfaces for equation (9)). *Let*

$$H = U\Sigma V^T, \quad U^T U = I_r, \quad \det(V^T Z) \neq 0,$$

and let the exact source singular block be diagonal with nonzero displayed entries,

$$\Sigma_{\alpha\beta} = \begin{cases} \sigma_\alpha, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases} \quad \sigma_\alpha \neq 0 \quad \text{for every } \alpha.$$

Let

$$P_H := HZ((HZ)^T(HZ))^{-1}(HZ)^T$$

be the exact Gram-inverse column-sketch projector generated by the preceding corollary. If an exact equation-(9) head/tail certificate supplies matrices H_0, T_0 and radii τ, κ such that

$$\|H - (P_H H)\|_F \leq \tau + \kappa$$

through the repository head/tail certificate, then the concrete residual surface holds with the displayed diagonal singular-block hypotheses, not with a raw $\det(\Sigma) \neq 0$ assumption:

$$\rho_F(H, P_H H) \leq \tau + \kappa.$$

If A_k is a supplied best rank- k approximation certificate and

$$\tau + \kappa \leq \rho \rho_F(H, A_k),$$

then

$$\rho_F(H, P_H H) \leq \rho \rho_F(H, A_k).$$

The same diagonal replacement holds for the explicit-coefficient head/tail certificate obtained from a supplied exact coefficient table W .

For the source-head plus source-tail form, let

$$A = H + T, \quad U^T T = 0,$$

and let

$$P_A := AZ((AZ)^T(AZ))^{-1}(AZ)^T, \quad W = (V^T Z)^{-1}V^T.$$

The same exact diagonal singular-block hypotheses imply

$$\det((AZ)^T(AZ)) \neq 0,$$

so the concrete Gram-inverse projector is available without a raw sketch-Gram determinant hypothesis. If the exact source-tail residual and its projected image obey

$$\|T - (TZ)W\|_F \leq \tau, \quad \|P_A(T - (TZ)W)\|_F \leq \kappa,$$

then

$$\rho_F(A, P_A A) \leq \tau + \kappa.$$

If in addition A_k is a supplied best rank- k approximation certificate and

$$\tau + \kappa \leq \rho \rho_F(A, A_k),$$

then

$$\rho_F(A, P_A A) \leq \rho \rho_F(A, A_k).$$

This is an exact-object residual-surface theorem. It propagates the displayed diagonal singular-value hypotheses into the concrete equation-(9) residual and relative-residual wrappers. It still assumes the tail and coupling radii, and the relative form still assumes a supplied best-rank certificate. It does not construct the rectangular SVD, prove source-tail norm bounds, prove Eckart–Young optimality, derive randomness certificates, or certify computed SVD, singular-vector, projector, Gram, inverse, coefficient, or product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
columnSketchGramInverseProjector_sourceSVD_diagonal_headTailRankResidualSurface
columnSketchGramInverseProjector_sourceSVD_diagonal_headTailRelativeResidualSurface
columnSketchGramInverseProjector_sourceSVD_diagonal_columnSketchHeadRankResidualSurface
columnSketchGramInverseProjector_sourceSVD_diagonal_columnSketchHeadRelativeResidualSurface
columnSketchGramInverseProjector_sourceHeadTail_sourceSketchCoefficientRankResidualSurface_of_
sourceSVD_diagonal_det_factors
columnSketchGramInverseProjector_sourceHeadTail_sourceSketchCoefficientRelativeResidualSurface_
of_sourceSVD_diagonal_det_factors
```

Corollary 560 (Diagonal source-SVD scalar tail rate for equation (9)). *Let*

$$A = H + T, \quad H = U \Sigma_h V^T, \quad U^T U = I_r, \quad \det(V^T Z) \neq 0,$$

and let the exact source-head singular block be diagonal with nonzero displayed entries,

$$(\Sigma_h)_{\alpha\beta} = \begin{cases} \sigma_\alpha, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases} \quad \sigma_\alpha \neq 0 \quad \text{for every } \alpha.$$

Assume the exact source tail has the displayed SVD-style factorization

$$T = U_\perp \Sigma_\perp V_\perp^T, \quad U_\perp^T U_\perp = I_q, \quad U^T U_\perp = 0.$$

Assume also the exact right-basis fields

$$V_\perp^T V_\perp = I_q, \quad V^T V_\perp = 0, \quad V_\perp^T V = 0, \quad V^T V = I_r,$$

together with row completeness of the concatenated right-basis block $[V_\perp, V]$. If $\varepsilon \geq 0$ and the exact CACM cross-term certificate

$$\|\Sigma_\perp (V_\perp^T Z)(V^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_\perp\|_F$$

is supplied, then for the exact Gram-inverse column-sketch projector

$$P_A := AZ((AZ)^T(AZ))^{-1}(AZ)^T$$

the diagonal source-head determinant route and the source-tail/coupling certificates give

$$\text{rank}(P_A A) \leq r, \quad \rho_F(A, P_A A) \leq 2\sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F.$$

If A_k is a supplied best rank- k approximation certificate and

$$2\sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F \leq \rho \rho_F(A, A_k),$$

then

$$\rho_F(A, P_A A) \leq \rho \rho_F(A, A_k).$$

This is an exact-object scalar-rate assembly theorem. It removes the two opaque tail and coupling radii from the preceding source-head/tail residual surface by instantiating both of them from the exact source-tail factorization and exact cross-term certificate. It still does not construct the rectangular SVD/source-tail factors, prove singular-value ordering or Eckart–Young optimality, derive the cross-term certificate from randomness, or certify computed SVD, singular-vector, projector, Gram, inverse, coefficient, or product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRankResidualSurface_of_sourceSVD_
diagonal_crossTerm

columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
sourceSVD_diagonal_crossTerm
```

Corollary 561 (Source-tail-optimal scalar relative rate for equation (9)). *Under the hypotheses of the preceding corollary, write*

$$H = U\Sigma_h V^T, \quad T = U_\perp \Sigma_\perp V_\perp^T,$$

and assume in addition the supplied Frobenius tail-optimality inequality

$$\|T\|_F \leq \rho_F(A, B) \quad \text{for every } B \text{ with } \text{rank}(B) \leq r.$$

Assume also the scalar comparison

$$2\sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F \leq \rho \|T\|_F.$$

Then the exact source head is a best rank- r Frobenius approximation certificate,

$$H \in \underset{\text{rank}(B) \leq r}{\text{argmin}} \rho_F(A, B), \quad \rho_F(A, H) = \|T\|_F.$$

For the exact Gram-inverse column-sketch projector

$$P_A := AZ((AZ)^T(AZ))^{-1}(AZ)^T,$$

the same diagonal source-head determinant route, source-tail/coupling certificates, and tail-optimality handoff give

$$\text{rank}(P_A A) \leq r, \quad \rho_F(A, P_A A) \leq \rho \rho_F(A, H).$$

This is an exact-object D_4 handoff theorem. It composes the scalar tail-rate surface with a supplied tail-optimality inequality, so the old separate A_k /best-rank hypothesis is replaced by the exact source head H . It still does not prove the tail-optimality inequality from singular values or Eckart–Young, construct the rectangular SVD/source-tail factors, derive the cross-term certificate from randomness, or certify computed SVD, singular-vector, projector, Gram, inverse, coefficient, or product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
sourceSVD_diagonal_crossTerm_tailOptimal
```

Corollary 562 (Diagonal source-SVD certificate handoff for equation (9)). *Define a diagonal source-SVD tail certificate*

$$\text{DiagTail}(A, T, U, \Sigma_h, \sigma, V, U_\perp, \Sigma_\perp, V_\perp)$$

to consist of the following exact identities and orthogonality fields:

$$\begin{aligned} A &= U\Sigma_h V^T + T, & T &= U_\perp \Sigma_\perp V_\perp^T, \\ U_\perp^T U_\perp &= I, & U^T U_\perp &= 0, & U^T U &= I, \\ (\Sigma_h)_{ab} &= \begin{cases} \sigma_a, & a = b, \\ 0, & a \neq b, \end{cases} & \sigma_a &\neq 0 \text{ for all } a, \\ V_\perp^T V_\perp &= I, & V^T V_\perp &= 0, & V_\perp^T V &= 0, & V^T V &= I, \\ V_\perp V_\perp^T + V V^T &= I_n. \end{aligned}$$

This certificate implies the source-tail left-orthogonality field $U^T T = 0$. If the supplied Frobenius tail-optimality inequality

$$\|T\|_F \leq \rho_F(A, B) \quad \text{for every } B \text{ with } \text{rank}(B) \leq r$$

is also available, then it supplies the best-rank certificate for $H = U \Sigma_h V^T$.

Assume $\varepsilon \geq 0$,

$$\det(V^T Z) \neq 0, \quad \|\Sigma_\perp (V_\perp^T Z)(V^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_\perp\|_F.$$

For the exact Gram-inverse column-sketch projector

$$P_A := AZ((AZ)^T(AZ))^{-1}(AZ)^T,$$

the certificate gives

$$\text{rank}(P_A A) \leq r, \quad \rho_F(A, P_A A) \leq 2\sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F.$$

If, in addition,

$$2\sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F \leq \rho \|T\|_F,$$

then

$$H \in \underset{\text{rank}(B) \leq r}{\text{argmin}} \rho_F(A, B), \quad \rho_F(A, H) = \|T\|_F, \quad \rho_F(A, P_A A) \leq \rho \rho_F(A, H).$$

This is an exact-object D3/D4 certificate handoff. It packages the source-SVD data used by the previous two scalar-rate corollaries into one mathematical certificate, but it still does not construct that certificate from a rectangular SVD, prove singular-value positivity/order or Eckart–Young tail-optimality, derive the cross-term certificate from randomness, or certify computed SVD, singular-vector, projector, Gram, inverse, coefficient, or product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
DiagonalSourceSVDTailCertificate
DiagonalSourceSVDTailCertificate.sourceTailLeftOrthogonal
DiagonalSourceSVDTailCertificate.isBestRankApproxFrob_of_tail_optimal
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRankResidualSurface_of_
diagonalSourceSVDTailCertificate
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
diagonalSourceSVDTailCertificate
```

Corollary 563 (Block source-SVD certificate constructor for equation (9)). *Let*

$$L = [U, U_\perp], \quad R = [V_\perp, V]$$

be the concatenated left and right source blocks. Assume the exact block decomposition

$$A = U \Sigma_h V^T + U_\perp \Sigma_\perp V_\perp^T,$$

the exact block orthonormality and completeness conditions

$$L^T L = I_{r+q}, \quad R^T R = I_{q+r}, \quad R R^T = I_n,$$

and the exact diagonal nonsingular head block

$$(\Sigma_h)_{ab} = \begin{cases} \sigma_a, & a = b, \\ 0, & a \neq b, \end{cases} \quad \sigma_a \neq 0 \quad \text{for all } a.$$

Then, with

$$T := U_\perp \Sigma_\perp V_\perp^T,$$

these block hypotheses construct the diagonal source-SVD tail certificate

$$\text{DiagTail}(A, T, U, \Sigma_h, \sigma, V, U_\perp, \Sigma_\perp, V_\perp).$$

Equivalently, the left block certificate gives

$$U^T U = I, \quad U^T U_\perp = 0, \quad U_\perp^T U_\perp = I,$$

and the right block certificate gives

$$V_\perp^T V_\perp = I, \quad V^T V_\perp = 0, \quad V_\perp^T V = 0, \quad V^T V = I, \quad V_\perp V_\perp^T + V V^T = I_n.$$

Thus the source-tail left-orthogonality field $U^T T = 0$ follows from the block data.

Assume further that $\varepsilon \geq 0$,

$$\det(V^T Z) \neq 0, \quad \|\Sigma_\perp (V_\perp^T Z)(V^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_\perp\|_F.$$

For the exact Gram-inverse column-sketch projector

$$P_A := AZ((AZ)^T(AZ))^{-1}(AZ)^T,$$

the constructed certificate gives

$$\text{rank}(P_A A) \leq r, \quad \rho_F(A, P_A A) \leq 2\sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F.$$

If the supplied Frobenius tail-optimality inequality

$$\|U_\perp \Sigma_\perp V_\perp^T\|_F \leq \rho_F(A, B) \quad \text{for every } B \text{ with } \text{rank}(B) \leq r$$

and the scalar comparison

$$2\sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F \leq \rho \|U_\perp \Sigma_\perp V_\perp^T\|_F$$

are also available, then

$$U \Sigma_h V^T \in \underset{\text{rank}(B) \leq r}{\text{argmin}} \rho_F(A, B), \quad \rho_F(A, U \Sigma_h V^T) = \|U_\perp \Sigma_\perp V_\perp^T\|_F,$$

and

$$\rho_F(A, P_A A) \leq \rho \rho_F(A, U \Sigma_h V^T).$$

This is an exact-object D3 constructor for the preceding certificate theorem. It derives the certificate from primitive block decomposition and block orthonormality data, but it still does not prove rectangular SVD existence, singular-value ordering or positivity from an SVD construction, Eckart–Young tail optimality, randomness-derived cross-term certificates, or computed non-probability SVD, singular-vector, projector, Gram, inverse, coefficient, or product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```

leftBasisBlock
leftBasisBlock_component_orthonormal_fields_of_col_orthonormal
BlockDiagonalSourceSVDTailCertificate
BlockDiagonalSourceSVDTailCertificate.to_diagonalSourceSVDTailCertificate
BlockDiagonalSourceSVDTailCertificate.sourceTailLeftOrthogonal
BlockDiagonalSourceSVDTailCertificate.isBestRankApproxFrob_of_tail_optimal
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRankResidualSurface_of_
blockDiagonalSourceSVDTailCertificate
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
blockDiagonalSourceSVDTailCertificate

```

Corollary 564 (Square SVD split constructor for equation (9)). *Let $A \in \mathbb{R}^{(r+q) \times (r+q)}$. Suppose exact square SVD-style data are supplied:*

$$U_{\text{full}}^T U_{\text{full}} = I, \quad V_{\text{full}}^T V_{\text{full}} = I, \quad U_{\text{full}} U_{\text{full}}^T = I, \quad V_{\text{full}} V_{\text{full}}^T = I,$$

and

$$A_{ij} = \sum_{\ell=1}^{r+q} (U_{\text{full}})_{i\ell} \sigma_{\ell} (V_{\text{full}})_{j\ell}.$$

Let $h(a)$ denote the head index $a \in \{1, \dots, r\}$ embedded in $\{1, \dots, r+q\}$, and let $t(c)$ denote the tail index $r+c$. Define

$$U_a = (U_{\text{full}})_{\cdot, h(a)}, \quad U_{\perp, c} = (U_{\text{full}})_{\cdot, t(c)}, \quad V_a = (V_{\text{full}})_{\cdot, h(a)}, \quad V_{\perp, c} = (V_{\text{full}})_{\cdot, t(c)},$$

and diagonal blocks

$$(\Sigma_h)_{ab} = \begin{cases} \sigma_{h(a)}, & a = b, \\ 0, & a \neq b, \end{cases} \quad (\Sigma_{\perp})_{cd} = \begin{cases} \sigma_{t(c)}, & c = d, \\ 0, & c \neq d. \end{cases}$$

If the displayed head singular entries are nonzero,

$$\sigma_{h(a)} \neq 0 \quad \text{for all } a,$$

then the split exact SVD data construct

$$\text{BlockTail}(A, U, \Sigma_h, \sigma_h, V, U_{\perp}, \Sigma_{\perp}, V_{\perp}),$$

where

$$A = U \Sigma_h V^T + U_{\perp} \Sigma_{\perp} V_{\perp}^T.$$

In particular, the square SVD representation proves the head and tail expansions

$$U \Sigma_h V^T = \sum_{a=1}^r (U_{\text{full}})_{\cdot, h(a)} \sigma_{h(a)} (V_{\text{full}})_{\cdot, h(a)}^T,$$

and

$$U_{\perp} \Sigma_{\perp} V_{\perp}^T = \sum_{c=1}^q (U_{\text{full}})_{\cdot, t(c)} \sigma_{t(c)} (V_{\text{full}})_{\cdot, t(c)}^T.$$

Consequently, if $\varepsilon \geq 0$,

$$\det(V^T Z) \neq 0, \quad \|\Sigma_\perp(V_\perp^T Z)(V^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_\perp\|_F,$$

then the exact Gram-inverse column-sketch projector

$$P_A = AZ((AZ)^T(AZ))^{-1}(AZ)^T$$

satisfies

$$\text{rank}(P_A A) \leq r, \quad \rho_F(A, P_A A) \leq 2\sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F.$$

If, in addition, the supplied Frobenius tail-optimality inequality

$$\|U_\perp \Sigma_\perp V_\perp^T\|_F \leq \rho_F(A, B) \quad \text{for every } B \text{ with } \text{rank}(B) \leq r$$

and the scalar comparison

$$2\sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F \leq \rho \|U_\perp \Sigma_\perp V_\perp^T\|_F$$

hold, then

$$U \Sigma_h V^T \in \underset{\text{rank}(B) \leq r}{\text{argmin}} \rho_F(A, B), \quad \rho_F(A, U \Sigma_h V^T) = \|U_\perp \Sigma_\perp V_\perp^T\|_F,$$

and

$$\rho_F(A, P_A A) \leq \rho \rho_F(A, U \Sigma_h V^T).$$

This is an exact-object square-specialized D3 constructor. It does not prove existence of the SVD, singular-value order or positivity beyond the visible head-nonzero hypothesis, Eckart–Young tail optimality, randomness-derived cross-term certificates, or computed non-probability SVD, singular-vector, projector, Gram, inverse, coefficient, or product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```

squareSVDHeadLeft
squareSVDTailLeft
squareSVDHeadRight
squareSVDTailRight
squareSVDHeadDiagonal
squareSVDTailDiagonal
squareSVDHeadValues
sourceSVDFactorMatrix_squareSVDHeadDiagonal
sourceSVDFactorMatrix_squareSVDTailDiagonal
BlockDiagonalSourceSVDTailCertificate.of_squareSVD
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRankResidualSurface_of_squareSVD
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_squareSVD

```

Corollary 565 (Rectangular-thin SVD split constructor for equation (9)). *Let $A \in \mathbb{R}^{m \times (r+q)}$. Suppose exact thin rectangular SVD-style data are supplied:*

$$\sum_{i=1}^m (U_{\text{full}})_{ia} (U_{\text{full}})_{ib} = \begin{cases} 1, & a = b, \\ 0, & a \neq b, \end{cases}$$

for all $a, b \in \{1, \dots, r + q\}$, and suppose the right table is exactly orthogonal,

$$V_{\text{full}}^T V_{\text{full}} = I, \quad V_{\text{full}} V_{\text{full}}^T = I.$$

Assume the exact representation

$$A_{ij} = \sum_{\ell=1}^{r+q} (U_{\text{full}})_{i\ell} \sigma_{\ell} (V_{\text{full}})_{j\ell}.$$

Using the same head indices $h(a)$ and tail indices $t(c)$ as in the square-SVD split, define

$$U_a = (U_{\text{full}})_{\cdot, h(a)}, \quad U_{\perp, c} = (U_{\text{full}})_{\cdot, t(c)}, \quad V_a = (V_{\text{full}})_{\cdot, h(a)}, \quad V_{\perp, c} = (V_{\text{full}})_{\cdot, t(c)},$$

and the same diagonal blocks

$$(\Sigma_h)_{ab} = \begin{cases} \sigma_{h(a)}, & a = b, \\ 0, & a \neq b, \end{cases} \quad (\Sigma_{\perp})_{cd} = \begin{cases} \sigma_{t(c)}, & c = d, \\ 0, & c \neq d. \end{cases}$$

If the displayed head singular entries are nonzero,

$$\sigma_{h(a)} \neq 0 \quad \text{for all } a,$$

then these thin rectangular data construct

$$\text{BlockTail}(A, U, \Sigma_h, \sigma_h, V, U_{\perp}, \Sigma_{\perp}, V_{\perp}),$$

where

$$A = U \Sigma_h V^T + U_{\perp} \Sigma_{\perp} V_{\perp}^T.$$

The head and tail expansions are

$$U \Sigma_h V^T = \sum_{a=1}^r (U_{\text{full}})_{\cdot, h(a)} \sigma_{h(a)} (V_{\text{full}})_{\cdot, h(a)}^T,$$

and

$$U_{\perp} \Sigma_{\perp} V_{\perp}^T = \sum_{c=1}^q (U_{\text{full}})_{\cdot, t(c)} \sigma_{t(c)} (V_{\text{full}})_{\cdot, t(c)}^T.$$

Consequently, if $\varepsilon \geq 0$,

$$\det(V^T Z) \neq 0, \quad \|\Sigma_{\perp} (V_{\perp}^T Z) (V^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_{\perp}\|_F,$$

then the exact Gram-inverse column-sketch projector

$$P_A = AZ((AZ)^T(AZ))^{-1}(AZ)^T$$

satisfies

$$\text{rank}(P_A A) \leq r, \quad \rho_F(A, P_A A) \leq 2\sqrt{1 + \varepsilon^2} \|\Sigma_{\perp}\|_F.$$

If the supplied Frobenius tail-optimality inequality

$$\|U_{\perp} \Sigma_{\perp} V_{\perp}^T\|_F \leq \rho_F(A, B) \quad \text{for every } B \text{ with } \text{rank}(B) \leq r$$

and the scalar comparison

$$2\sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F \leq \rho \|U_\perp \Sigma_\perp V_\perp^T\|_F$$

hold, then

$$U\Sigma_h V^T \in \operatorname{argmin}_{\operatorname{rank}(B) \leq r} \rho_F(A, B), \quad \rho_F(A, U\Sigma_h V^T) = \|U_\perp \Sigma_\perp V_\perp^T\|_F,$$

and

$$\rho_F(A, P_A A) \leq \rho_F(A, U\Sigma_h V^T).$$

This is an exact-object thin-rectangular D3 constructor. It removes the square-left restriction from the preceding corollary, but it still does not prove rectangular SVD existence, singular-value order or positivity beyond the visible head-nonzero hypothesis, Eckart–Young tail optimality, randomness-derived cross-term certificates, or computed non-probability SVD, singular-vector, projector, Gram, inverse, coefficient, or product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```

rectangularThinSVDHeadLeft
rectangularThinSVDTailLeft
sourceSVDFactorMatrix_rectangularThinSVDHeadDiagonal
sourceSVDFactorMatrix_rectangularThinSVDTailDiagonal
BlockDiagonalSourceSVDTailCertificate.of_rectangularThinSVD
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRankResidualSurface_of_
rectangularThinSVD
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
rectangularThinSVD

```

Corollary 566 (Head-positive SVD split constructors for equation (9)). *In either of the two preceding supplied SVD-style settings, let*

$$h(a) \in \{1, \dots, r + q\}, \quad a = 1, \dots, r,$$

denote the displayed head singular-value indices. Assume the source-style strict positivity condition

$$\sigma_{h(a)} > 0 \quad \text{for every } a = 1, \dots, r.$$

Then the split head-value table is strictly positive,

$$(\sigma_h)_a > 0 \quad \text{for every } a,$$

and in particular the nonzero-head field required by the diagonal source-SVD constructors holds:

$$\sigma_{h(a)} \neq 0 \quad \text{for every } a.$$

Consequently, in the square SVD setting, the exact hypotheses

$$U_{\text{full}}^T U_{\text{full}} = I, \quad U_{\text{full}} U_{\text{full}}^T = I, \quad V_{\text{full}}^T V_{\text{full}} = I, \quad V_{\text{full}} V_{\text{full}}^T = I, \quad A = U_{\text{full}} \operatorname{diag}(\sigma) V_{\text{full}}^T$$

together with $\sigma_{h(a)} > 0$ construct

$$\text{BlockTail}(A, U, \Sigma_h, \sigma_h, V, U_\perp, \Sigma_\perp, V_\perp).$$

The same conclusion holds in the thin rectangular setting from

$$U_{\text{full}}^T U_{\text{full}} = I, \quad V_{\text{full}}^T V_{\text{full}} = I, \quad V_{\text{full}} V_{\text{full}}^T = I, \quad A = U_{\text{full}} \text{diag}(\sigma) V_{\text{full}}^T,$$

where $U_{\text{full}} \in \mathbb{R}^{m \times (r+q)}$ has exact orthonormal columns.

In both settings, if $\varepsilon \geq 0$,

$$\det(V^T Z) \neq 0, \quad \|\Sigma_{\perp}(V_{\perp}^T Z)(V^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_{\perp}\|_F,$$

then the exact Gram-inverse column-sketch projector satisfies

$$\text{rank}(P_A A) \leq r, \quad \rho_F(A, P_A A) \leq 2\sqrt{1 + \varepsilon^2} \|\Sigma_{\perp}\|_F.$$

If the supplied Frobenius tail-optimality inequality and scalar comparison from the preceding two corollaries are also available, then the same tail-optimal relative conclusion follows:

$$U \Sigma_h V^T \in \underset{\text{rank}(B) \leq r}{\text{argmin}} \rho_F(A, B), \quad \rho_F(A, P_A A) \leq \rho \rho_F(A, U \Sigma_h V^T).$$

This is an exact-object $D3/D4$ positivity handoff. It closes only the passage from positive displayed leading singular values to the nonzero singular block used by the determinant and split-constructor theorems. It still does not prove rectangular SVD existence, singular-value ordering, Eckart–Young tail optimality, randomness-derived cross-term certificates, or computed non-probability SVD, singular-vector, projector, Gram, inverse, coefficient, or product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
squareSVDHeadValues_pos
squareSVDHeadValues_nonzero_of_pos
BlockDiagonalSourceSVDTailCertificate.of_squareSVD_head_pos
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRankResidualSurface_of_squareSVD_head_pos
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_squareSVD_head_pos
BlockDiagonalSourceSVDTailCertificate.of_rectangularThinSVD_head_pos
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRankResidualSurface_of_rectangularThinSVD_head_pos
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_rectangularThinSVD_head_pos
```

Corollary 567 (Source-SVD tail Frobenius norm identity for equation (9)). *Let $U \in \mathbb{R}^{m \times q}$, $V \in \mathbb{R}^{n \times q}$, and $\Sigma \in \mathbb{R}^{q \times q}$ be exact analysis objects satisfying*

$$U^T U = I_q, \quad V^T V = I_q.$$

Then the exact source-SVD factor table satisfies

$$\|U \Sigma V^T\|_F^2 = \|\Sigma\|_F^2, \quad \|U \Sigma V^T\|_F = \|\Sigma\|_F.$$

Consequently, suppose the block source-SVD certificate of the previous equation-(9) route supplies

$$A = U\Sigma_h V^T + U_\perp \Sigma_\perp V_\perp^T$$

with exact block orthonormality for the displayed left and right factors. Then the source-tail matrix $T = U_\perp \Sigma_\perp V_\perp^T$ satisfies

$$\|T\|_F = \|\Sigma_\perp\|_F, \quad \rho_F(A, U\Sigma_h V^T) = \|\Sigma_\perp\|_F.$$

Moreover, if $\varepsilon \geq 0$, $\det(V^T Z) \neq 0$, and

$$\|\Sigma_\perp (V_\perp^T Z)(V^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_\perp\|_F,$$

and if the supplied tail-optimality and scalar-comparison hypotheses are already stated with the singular-tail norm,

$$\|\Sigma_\perp\|_F \leq \rho_F(A, B) \quad \text{for all } \text{rank}(B) \leq r,$$

$$2\sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F \leq \rho \|\Sigma_\perp\|_F,$$

then the exact Gram-inverse column-sketch projector satisfies

$$U\Sigma_h V^T \in \underset{\text{rank}(B) \leq r}{\text{argmin}} \rho_F(A, B), \quad \text{rank}(P_A A) \leq r,$$

$$\rho_F(A, P_A A) \leq \rho \rho_F(A, U\Sigma_h V^T).$$

This is still an exact-object theorem: it does not construct a rectangular SVD, prove singular-value ordering, prove Eckart–Young tail optimality from singular values, derive the cross-term radius from randomness, or certify computed non-probability SVD, singular-vector, projector, Gram, inverse, or product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
frobNormSqRect_sourceSVDFactorMatrix_orthonormal
frobNormRect_sourceSVDFactorMatrix_orthonormal
BlockDiagonalSourceSVDTailCertificate.tail_frobNorm_eq_sigma
BlockDiagonalSourceSVDTailCertificate.tail_lowRankResidual_eq_sigma
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
blockDiagonalSourceSVDTailCertificate_sigmaTail
```

Corollary 568 (Sigma-tail square and thin-rectangular SVD wrappers for equation (9)). *Let $r, q \in \mathbb{N}$. In the square case, suppose*

$$A = U_{\text{full}} \text{diag}(\sigma) V_{\text{full}}^T, \quad U_{\text{full}}^T U_{\text{full}} = V_{\text{full}}^T V_{\text{full}} = I_{r+q},$$

and every displayed head singular entry is nonzero. Let Σ_{tail} be the diagonal block $\text{diag}(\sigma_{r+1}, \dots, \sigma_{r+q})$. Then the split tail and source-head residual satisfy

$$\|U_{\text{tail}} \Sigma_{\text{tail}} V_{\text{tail}}^T\|_F = \|\Sigma_{\text{tail}}\|_F, \quad \rho_F(A, U_{\text{head}} \Sigma_{\text{head}} V_{\text{head}}^T) = \|\Sigma_{\text{tail}}\|_F.$$

Consequently, if $\varepsilon \geq 0$, $\det(V_{\text{head}}^T Z) \neq 0$,

$$\|\Sigma_{\text{tail}} (V_{\text{tail}}^T Z)(V_{\text{head}}^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_{\text{tail}}\|_F,$$

and

$$\|\Sigma_{\text{tail}}\|_F \leq \rho_F(A, B) \quad \text{for all } \text{rank}(B) \leq r, \quad 2\sqrt{1 + \varepsilon^2} \|\Sigma_{\text{tail}}\|_F \leq \rho \|\Sigma_{\text{tail}}\|_F,$$

then the exact Gram-inverse column-sketch projector obeys

$$U_{\text{head}} \Sigma_{\text{head}} V_{\text{head}}^T \in \underset{\text{rank}(B) \leq r}{\text{argmin}} \rho_F(A, B), \quad \rho_F(A, P_A A) \leq \rho \rho_F(A, U_{\text{head}} \Sigma_{\text{head}} V_{\text{head}}^T).$$

The same conclusion holds for the thin rectangular case

$$A = U_{\text{full}} \text{diag}(\sigma) V_{\text{full}}^T, \quad U_{\text{full}}^T U_{\text{full}} = I_{r+q}, \quad V_{\text{full}}^T V_{\text{full}} = I_{r+q},$$

with $A \in \mathbb{R}^{m \times (r+q)}$. In both cases, a strict-positive head condition $0 < \sigma_a$ may replace the nonzero-head hypothesis.

This corollary is exact-object algebra for supplied square or thin-rectangular SVD-style data. It does not construct the SVD, prove singular-value ordering, prove Eckart–Young tail optimality, derive the cross-term radius from randomness, or certify computed non-probability SVD, singular-vector, projector, Gram, inverse, or product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
frobNormRect_squareSVDTail_eq_sigmaTail
lowRankResidualFrob_squareSVDHead_eq_sigmaTail
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
squareSVD_sigmaTail
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
squareSVD_head_pos_sigmaTail
frobNormRect_rectangularThinSVDTail_eq_sigmaTail
lowRankResidualFrob_rectangularThinSVDHead_eq_sigmaTail
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
rectangularThinSVD_sigmaTail
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
rectangularThinSVD_head_pos_sigmaTail
```

Corollary 569 (Sigma-tail best-rank handoff for square and thin SVD data). *Under the square-SVD hypotheses of the previous corollary, assume the visible tail-optimality inequality is stated directly with the displayed tail singular block:*

$$\|\Sigma_{\text{tail}}\|_F \leq \rho_F(A, B) \quad \text{for every } B \text{ with } \text{rank}(B) \leq r.$$

Then the exact source head is a Frobenius best rank- r approximation:

$$U_{\text{head}} \Sigma_{\text{head}} V_{\text{head}}^T \in \underset{\text{rank}(B) \leq r}{\text{argmin}} \rho_F(A, B).$$

The same implication holds in the thin rectangular setting, and in both cases the nonzero-head hypothesis may be replaced by the strict-positive head hypothesis $0 < \sigma_a$.

This is only a D4 handoff from a visible Eckart–Young-style inequality to the repository certificate `IsBestRankApproxFrob`. It does not prove SVD existence, singular-value ordering, the Eckart–Young theorem that supplies the displayed inequality, randomness, or computed non-probability SVD, singular-vector, projector, Gram, inverse, or product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
isBestRankApproxFrob_of_squareSVD_sigmaTail_optimal
isBestRankApproxFrob_of_squareSVD_head_pos_sigmaTail_optimal
isBestRankApproxFrob_of_rectangularThinSVD_sigmaTail_optimal
isBestRankApproxFrob_of_rectangularThinSVD_head_pos_sigmaTail_optimal
```

Corollary 570 (Right-Gram singular-value order and nonnegativity for equation (9)). *Let $A \in \mathbb{R}^{m \times n}$ be an exact analysis matrix and define the right Gram matrix*

$$G = A^T A, \quad G_{jk} = \sum_{i=1}^m A_{ij} A_{ik}.$$

Then G is symmetric and positive semidefinite. Equivalently, for every $x \in \mathbb{R}^n$,

$$x^T G x = \sum_{i=1}^m \left(\sum_{j=1}^n A_{ij} x_j \right)^2 \geq 0.$$

Let $\lambda_j(A)$ be the ordered Hermitian eigenvalues of G , indexed in the repository's zero-based finite order, and define

$$\sigma_j(A) = \sqrt{\lambda_j(A)}.$$

Then

$$\begin{aligned} 0 \leq \lambda_j(A), \quad j \leq k \implies \lambda_k(A) \leq \lambda_j(A), \\ 0 \leq \sigma_j(A), \quad j \leq k \implies \sigma_k(A) \leq \sigma_j(A), \quad \sigma_j(A)^2 = \lambda_j(A). \end{aligned}$$

This is an exact spectral adapter for the equation-(9) SVD/Eckart–Young route. It supplies ordered nonnegative singular-value magnitudes from the right Gram $A^T A$, but it does not construct singular vectors, a rectangular SVD source split, or the Eckart–Young tail-optimality theorem. If an implementation computes $A^T A$, singular vectors, projectors, bases, or SVD data, those are non-probability computed quantities and still require explicit floating-point/inexact-arithmetic certificates. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRightGram
rectRightGram_symmetric
finiteQuadraticForm_rectRightGram_eq_sum_sq
rectRightGram_finitePSD
rectRightGram_matrix_posSemidef
rectSingularValueSq
rectSingularValue
rectSingularValueSq_nonneg
rectSingularValueSq_antitone
rectSingularValue_nonneg
rectSingularValue_antitone
rectSingularValue_sq_eq
```


Corollary 571 (Right-Gram eigenvector table and diagonalization for equation (9)). *Let $A \in \mathbb{R}^{m \times n}$ be an exact analysis matrix and let*

$$G = A^T A, \quad G_{jk} = \sum_{i=1}^m A_{ij} A_{ik}.$$

Let $\alpha_\ell(A)$ denote the basis-indexed Hermitian eigenvalues attached to a mathlib Hermitian eigenvector basis of G , and let

$$V_{j\ell}(A)$$

be the corresponding exact real eigenvector table. Define the basis-indexed singular values

$$\tau_\ell(A) = \sqrt{\alpha_\ell(A)}.$$

Then $V(A)$ is orthogonal:

$$\sum_{j=1}^n V_{ja}(A) V_{jb}(A) = \delta_{ab}, \quad \sum_{a=1}^n V_{ia}(A) V_{ja}(A) = \delta_{ij}.$$

Moreover

$$0 \leq \alpha_a(A), \quad 0 \leq \tau_a(A), \quad \tau_a(A)^2 = \alpha_a(A),$$

and each column of $V(A)$ is a right-Gram eigenvector:

$$\sum_{k=1}^n G_{jk} V_{ka}(A) = \alpha_a(A) V_{ja}(A).$$

Consequently,

$$\sum_{j=1}^n V_{ja}(A) \left(\sum_{k=1}^n G_{jk} V_{kb}(A) \right) = \begin{cases} \alpha_a(A), & a = b, \\ 0, & a \neq b, \end{cases}$$

equivalently the diagonal entries may be written as $\tau_a(A)^2$.

This is an exact-object eigenvector adapter for the equation-(9) SVD/Eckart–Young route. It is basis-indexed: it uses the index order of mathlib’s eigenvector basis, not the ordered zero-indexed singular-value sequence of the previous corollary. Thus it still does not construct the ordered head/tail rectangular SVD source split or prove the Eckart–Young tail-optimality theorem. If an implementation computes $A^T A$, the eigenvector table, singular vectors, projectors, bases, or SVD data, those are non-probability computed quantities and require explicit floating-point/inexact-arithmetic certificates. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRightGramEigenvalue
rectRightGramEigenbasis
rectRightGramBasisSingularValue
rectRightGramEigenbasis_isOrthogonal
rectRightGramEigenbasis_col_orthonormal
rectRightGramEigenbasis_row_orthonormal
```

```

rectRightGramEigenvalue_nonneg
rectRightGramBasisSingularValue_nonneg
rectRightGramBasisSingularValue_sq_eq
rectRightGramEigenbasis_eigenvector
rectRightGramEigenbasis_diagonalizes
rectRightGramEigenbasis_diagonalizes_singularValueSq

```

Corollary 572 (Full-positive right-Gram SVD reconstruction for equation (9)). *Let $A \in \mathbb{R}^{m \times n}$, let $G = A^T A$, and let (α_a, V_a, τ_a) be the basis-indexed eigenvalue, right-eigenvector, and singular-value data from the previous corollary:*

$$\tau_a = \sqrt{\alpha_a}, \quad GV_a = \alpha_a V_a, \quad V^T V = I_n, \quad VV^T = I_n.$$

Assume the full-positive hypothesis

$$0 < \tau_a \quad \text{for every } a \in \{1, \dots, n\}.$$

Define

$$y_{ia} = (AV_a)_i = \sum_{j=1}^n A_{ij} V_{ja}, \quad u_{ia} = \frac{y_{ia}}{\tau_a}.$$

Then the projected columns have diagonal Gram matrix

$$\sum_{i=1}^m y_{ia} y_{ib} = \begin{cases} \tau_a^2, & a = b, \\ 0, & a \neq b, \end{cases}$$

and the normalized left candidates are exactly column-orthonormal:

$$\sum_{i=1}^m u_{ia} u_{ib} = \delta_{ab}.$$

Moreover, the exact row-completeness of V gives

$$A_{ij} = \sum_{a=1}^n y_{ia} V_{ja} = \sum_{a=1}^n u_{ia} \tau_a V_{ja}.$$

This is an exact-object reconstruction theorem for the full-positive, basis-indexed case. It is not yet the ordered rank-deficient rectangular SVD source split used by the paper’s final Eckart–Young route: zero singular values, ordered head/tail selection, tail optimality, and the rank-deficient extension remain open. If an implementation computes $A^T A$, V , U , singular values, projectors, bases, or SVD data, those are non-probability computed quantities and require explicit floating-point/inexact-arithmetic certificates. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```

rectRightGramProjectedColumn
rectRightGramLeftSingularFromEigenbasis
rectRightGramBasisSingularDiagonal

```

```

rectRightGramProjectedColumn_dot
rectRightGramProjectedColumn_dot_diagonal
rectRightGramLeftSingularFromEigenbasis_col_orthonormal_of_pos
rectRightGramProjectedColumn_reconstruct
rectRightGramLeftSingularFromEigenbasis_factor_column_of_pos
rectRightGram_fullPositive_basisSVD_representation

```

Corollary 573 (Zero right-Gram projected columns for equation (9)). *Let $A \in \mathbb{R}^{m \times n}$, let $G = A^T A$, and let (α_a, V_a, τ_a) be the basis-indexed right-Gram data above. For*

$$y_{ia} = (AV_a)_i = \sum_{j=1}^n A_{ij} V_{ja},$$

one has the exact norm-square identities

$$\sum_{i=1}^m y_{ia}^2 = \tau_a^2 = \alpha_a.$$

Consequently,

$$\tau_a = 0 \implies y_{ia} = 0 \text{ for every } i,$$

and equivalently

$$\alpha_a = 0 \implies y_{ia} = 0 \text{ for every } i.$$

This is an exact-object rank-deficient adapter for the basis-indexed right-Gram route. It removes the obstruction that zero singular values would otherwise create in the full-positive normalization $u_a = y_a/\tau_a$, but it does not construct the ordered head/tail source split or prove Eckart–Young tail optimality. If an implementation computes $A^T A$, right singular vectors, singular values, left candidates, projectors, or SVD data, those are non-probability computed quantities and require explicit floating-point/inexact-arithmetic certificates. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```

rectRightGramProjectedColumn_normSq_eq_singularValue_sq
rectRightGramProjectedColumn_normSq_eq_eigenvalue
rectRightGramProjectedColumn_eq_zero_of_singularValue_eq_zero
rectRightGramProjectedColumn_eq_zero_of_eigenvalue_eq_zero

```

Corollary 574 (Zero-safe right-Gram reconstruction for equation (9)). *Let $A \in \mathbb{R}^{m \times n}$, let $G = A^T A$, and keep the basis-indexed right-Gram data (α_a, V_a, τ_a) . With*

$$y_{ia} = (AV_a)_i,$$

define the zero-safe left candidates by

$$\tilde{u}_{ia} = \begin{cases} 0, & \tau_a = 0, \\ y_{ia}/\tau_a, & \tau_a \neq 0. \end{cases}$$

Then every basis index satisfies

$$\tau_a \tilde{u}_{ia} = y_{ia}.$$

Consequently the exact right-Gram row-completeness identity gives the basis-indexed reconstruction

$$A_{ij} = \sum_{a=1}^n \tilde{u}_{ia} \tau_a V_{ja}.$$

This is an exact-object rank-deficient reconstruction theorem. It removes the full-positive hypothesis from the preceding reconstruction by using the zero projected-column result when $\tau_a = 0$. It is still basis-indexed rather than the ordered source head/tail split needed for the paper's final Eckart–Young route. If an implementation computes $A^T A$, singular vectors, singular values, left candidates, projectors, or SVD data, those are non-probability computed quantities and require explicit floating-point/inexact-arithmetic certificates. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRightGramLeftSingularZeroSafe
rectRightGramLeftSingularZeroSafe_eq_zero_of_singularValue_eq_zero
rectRightGramLeftSingularZeroSafe_eq_inv_mul_of_singularValue_ne_zero
rectRightGramLeftSingularZeroSafe_factor_column
rectRightGram_basisSVD_representation
```

Corollary 575 (Selected right-Gram head/tail split for equation (9)). *Let $A \in \mathbb{R}^{m \times n}$, let $G = A^T A$, and keep the basis-indexed right-Gram data (α_a, V_a, τ_a) and the zero-safe left candidates $\tilde{u}_{\cdot a}$ from the preceding corollary. Let $s \subseteq \{1, \dots, n\}$ be any finite selected set of basis indices. Define the exact selected head and complementary tail by*

$$H_{ij}^{(s)} = \sum_{a \in s} \tilde{u}_{ia} \tau_a V_{ja}, \quad T_{ij}^{(s)} = \sum_{a \notin s} \tilde{u}_{ia} \tau_a V_{ja}.$$

Then the zero-safe reconstruction partitions entrywise as

$$A_{ij} = H_{ij}^{(s)} + T_{ij}^{(s)} \quad \text{for every } i, j.$$

Moreover, if $\phi : \{1, \dots, |s|\} \rightarrow s$ is the increasing enumeration of the selected finite set, then the selected head has the explicit rank factorization

$$H_{ij}^{(s)} = \sum_{b=1}^{|s|} \tilde{u}_{i, \phi(b)} (\tau_{\phi(b)} V_{j, \phi(b)}),$$

and hence

$$\text{rank}(H^{(s)}) \leq |s|.$$

This is an exact-object selected-index source split. It proves the algebra needed once a later theorem identifies s with the ordered top singular directions, but it does not yet prove that ordered handoff, rectangular SVD existence, Eckart–Young tail optimality, randomness-derived certificates, or computed non-probability SVD/singular-vector/projector/Gram routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```

rectRightGramBasisSVDHead
rectRightGramBasisSVDTail
rectRightGramBasisSVD_head_add_tail
rectRightGramBasisSVD_head_tail_entry
rectRightGramBasisSVDHeadRankFactorization
rectRightGramBasisSVDHead_rankAtMost

```

Corollary 576 (Selected right-Gram sketch bridge for equation (9)). *Under the hypotheses of the preceding corollary, let $\phi : \{1, \dots, |s|\} \rightarrow s$ be the selected-index enumeration. Define the exact selected eigenvector sketch and coefficient table by*

$$(Z_s)_{jb} = V_{j,\phi(b)}, \quad (W_s)_{bj} = V_{j,\phi(b)}.$$

Then the exact column sketch AZ_s is the selected projected-column table:

$$(AZ_s)_{ib} = (AV_{\cdot,\phi(b)})_i.$$

Moreover the selected head from Corollary 549 is exactly the corresponding column-sketch head,

$$H_{ij}^{(s)} = ((AZ_s)W_s)_{ij} = \sum_{b=1}^{|s|} (AZ_s)_{ib} (W_s)_{bj}.$$

*Equivalently, the selected head has a formal **ColumnSketchHeadFactorization** through the displayed sketch columns AZ_s .*

This is an exact-object selected-head sketch-space theorem. It closes the algebraic statement that the arbitrary selected right-Gram head lies in the column space of the matching selected eigenvector sketch. It does not identify the selected set with the ordered top singular directions, prove rectangular SVD existence, prove Eckart–Young tail optimality, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/ Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```

rectRightGramBasisSketchMatrix
rectRightGramBasisSketchCoeff
columnSketch_rectRightGramBasisSketchMatrix
rectRightGramBasisSketch_head_eq
rectRightGramBasisSketchHeadFactorization
rectRightGramBasisSVDHead_columnSketchHeadFactorization

```

Corollary 577 (Selected right-Gram equation-(9) certificate adapter). *Let s be a finite selected set of basis-indexed right-Gram directions, and let*

$$Z_s, \quad H^{(s)}, \quad T^{(s)}$$

denote respectively the selected eigenvector sketch, selected head, and selected tail from Corollaries 549 and 550. Let $P_{AZ_s} \in \mathbb{R}^{m \times m}$ be an exact left multiplier. Suppose that the displayed radii τ, κ are nonnegative and satisfy

$$\|T^{(s)}\|_F \leq \tau, \quad \|P_{AZ_s} T^{(s)}\|_F \leq \kappa.$$

Then the selected data instantiate the exact equation-(9) head/tail sketch certificate:

$$A = H^{(s)} + T^{(s)}, \quad H^{(s)} \in \text{range}(AZ_s),$$

with tail radius τ and projected-tail coupling radius κ .

If, in addition, P_{AZ_s} factors through the selected sketch columns, meaning that for some coefficient table C ,

$$(P_{AZ_s})_{ij} = \sum_{b=1}^{|s|} (AZ_s)_{ib} C_{bj},$$

and P_{AZ_s} reproduces those columns,

$$P_{AZ_s}(AZ_s) = AZ_s,$$

then the exact projector candidate satisfies the equation-(9) rank and residual surface

$$\text{rank}(P_{AZ_s} A) \leq |s|, \quad \|A - P_{AZ_s} A\|_F \leq \tau + \kappa.$$

This is an exact-object adapter theorem. It closes the formal handoff from the arbitrary selected right-Gram split to the existing equation-(9) head/tail residual API. It does not prove the tail or projected-tail coupling bounds, does not prove that a concrete projector reproduces AZ_s , does not identify s with ordered top singular directions, does not prove rectangular SVD existence or Eckart–Young tail optimality, and does not certify computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
equation9HeadTailSketchCertificate_of_rectRightGramBasisSVDHead
equation9HeadTailSketchRankResidualSurface_of_rectRightGramBasisSVDHead
```

Corollary 578 (Selected cardinality rank handoff for equation (9)). *Let $B \in \mathbb{R}^{m \times n}$ have a displayed exact rank-factorization through r columns. If $r = k$, then the same factorization is a formal rank-at-most- k certificate:*

$$\text{rank}(B) \leq r, \quad r = k \implies \text{rank}(B) \leq k.$$

Consequently, for the selected right-Gram head from Corollary 549, if the selected set satisfies

$$|s| = k,$$

then

$$\text{rank}(H^{(s)}) \leq k.$$

Under the hypotheses of Corollary 551, and under the same cardinality certificate $|s| = k$, the selected equation-(9) projector candidate satisfies the paper-facing rank/residual surface

$$\text{rank}(P_{AZ_s} A) \leq k, \quad \|A - P_{AZ_s} A\|_F \leq \tau + \kappa.$$

This is an exact rank-index handoff. It closes the bookkeeping step that turns the arbitrary selected-set rank $|s|$ into a displayed paper rank k . It does not identify s with ordered top singular directions, does not prove that a concrete source-faithful top- k selection has cardinality k , does not prove rectangular SVD existence, tail/coupling bounds, projector reproduction, Eckart–Young, randomness certificates, or computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRankAtMost_of_eq_rank
rectRightGramBasisSVDHead_rankAtMost_of_card_eq
equation9HeadTailSketchRankResidualSurface_of_rectRightGramBasisSVDHead_card_eq
```

Corollary 579 (Injective selected-index handoff for equation (9)). *Let $e : \{1, \dots, k\} \hookrightarrow \{1, \dots, n\}$ be an injective map of displayed selected indices into the right-Gram basis indices. Define the selected set*

$$s_e = \text{image}(e).$$

Then

$$|s_e| = k.$$

Consequently, the selected right-Gram head induced by s_e satisfies

$$\text{rank}(H^{(s_e)}) \leq k.$$

Under the hypotheses of Corollary 551 with $s = s_e$, the exact projector candidate satisfies the embedding-selected equation-(9) rank/residual surface

$$\text{rank}(P_{AZ_{s_e}} A) \leq k, \quad \|A - P_{AZ_{s_e}} A\|_F \leq \tau + \kappa.$$

This is an exact injective-selection cardinality theorem. It removes the separate $|s| = k$ hypothesis when the selected directions are supplied by an injective displayed index map. It does not prove that e enumerates the ordered top singular directions, does not prove rectangular SVD existence, tail/coupling bounds, projector reproduction, Eckart–Young, randomness certificates, or computed non-probability SVD/singular-vector/projector/Gram/ sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRightGramSelectedIndexSet
rectRightGramSelectedIndexSet_card
rectRightGramBasisSVDHead_rankAtMost_of_embedding
equation9HeadTailSketchRankResidualSurface_of_rectRightGramBasisSVDHead_embedding
```

Corollary 580 (Semantic ordered-top embedding handoff for equation (9)). *Assume $k \leq n$, and let*

$$\iota_k^n : \{1, \dots, k\} \rightarrow \{1, \dots, n\}$$

be the displayed top-index inclusion. Let

$$e : \{1, \dots, k\} \hookrightarrow \{1, \dots, n\}$$

be an injective selected-index map into the right-Gram basis indices. Suppose the semantic ordered-top certificate holds:

$$\hat{\sigma}_{\text{basis}}(e(a)) = \sigma_{\text{ord}}(\iota_k^n(a)), \quad a = 1, \dots, k,$$

where $\hat{\sigma}_{\text{basis}}$ is the basis-indexed right-Gram singular value attached to *mathlib*'s exact right-Gram eigenbasis and σ_{ord} is the ordered zero-indexed right-Gram singular-value sequence. Then

$$\hat{\sigma}_{\text{basis}}(e(a))^2 = \lambda_{\text{ord}}(\iota_k^n(a)), \quad a = 1, \dots, k,$$

and the selected basis singular values are ordered:

$$a \leq b \implies \hat{\sigma}_{\text{basis}}(e(b)) \leq \hat{\sigma}_{\text{basis}}(e(a)).$$

Moreover, under the hypotheses of Corollary 553 for the same embedding e , the exact projector candidate satisfies

$$\text{rank}(P_{AZ_{s_e}} A) \leq k, \quad \|A - P_{AZ_{s_e}} A\|_F \leq \tau + \kappa.$$

This is an exact semantic-order handoff. It exposes the precise certificate needed to say that the embedding-selected basis directions are the ordered top right-Gram singular directions, and then composes that certificate with the embedding rank/residual surface. It does not construct the certificate from *mathlib*'s arbitrary right-Gram eigenbasis order, and it does not prove rectangular SVD existence, tail/coupling bounds, projector reproduction, Eckart–Young, randomness certificates, or computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```

rectTopIndex
rectTopIndex_le
RectRightGramOrderedTopEmbeddingCertificate
rectRightGramOrderedTopEmbeddingCertificate_selected_sq_eq
rectRightGramOrderedTopEmbeddingCertificate_selected_antitone
equation9HeadTailSketchRankResidualSurface_of_rectRightGramBasisSVDHead_orderedTopEmbedding

```

Corollary 581 (Constructed ordered-top embedding for equation (9)). *For the exact right-Gram matrix $A^T A$, let*

$$\phi_n : \text{Fin}(\# \text{Fin}(n)) \simeq \text{Fin}(n)$$

*be the same finite equivalence used by *mathlib* to reindex the ordered Hermitian eigenvalue sequence λ_{ord} into the basis-indexed right-Gram eigenvector table. Given $k \leq n$, define*

$$e_{\text{ord}}(a) = \phi_n(\iota_k^n(a)), \quad a = 1, \dots, k.$$

Then $e_{\text{ord}} : \{1, \dots, k\} \hookrightarrow \{1, \dots, n\}$ is injective and satisfies the ordered-top certificate

$$\hat{\sigma}_{\text{basis}}(e_{\text{ord}}(a)) = \sigma_{\text{ord}}(\iota_k^n(a)), \quad a = 1, \dots, k.$$

Consequently,

$$\hat{\sigma}_{\text{basis}}(e_{\text{ord}}(a))^2 = \lambda_{\text{ord}}(\iota_k^n(a)), \quad a \leq b \implies \hat{\sigma}_{\text{basis}}(e_{\text{ord}}(b)) \leq \hat{\sigma}_{\text{basis}}(e_{\text{ord}}(a)).$$

Under the hypotheses of Corollary 554 with $e = e_{\text{ord}}$, the exact projector candidate satisfies

$$\text{rank}(P_{AZ_{s_{e_{\text{ord}}}}} A) \leq k, \quad \|A - P_{AZ_{s_{e_{\text{ord}}}}} A\|_F \leq \tau + \kappa.$$

This closes the exact ordered-top embedding construction for the right-Gram eigenbasis chosen by `mathlib`. It does not prove rectangular SVD existence, tail/coupling bounds, projector reproduction, Eckart–Young, randomness certificates, or computed non-probability SVD/singular-vector/projector/Gram/ sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRightGramOrderedEigenbasisEquiv
rectRightGramOrderedTopEmbedding
rectRightGramOrderedTopEmbedding_certificate
equation9HeadTailSketchRankResidualSurface_of_rectRightGramBasisSVDHead_
constructedOrderedTopEmbedding
```

Corollary 582 (Constructed ordered top/complement singular-value dominance). *For the exact right-Gram matrix $A^T A$, let ϕ_n and e_{ord} be as in the preceding corollary. For a basis-indexed right-Gram direction $b \in \{1, \dots, n\}$, define its ordered coordinate*

$$j_{\text{ord}}(b) = \phi_n^{-1}(b),$$

viewed through the canonical cast back to the displayed ordered index set $\{1, \dots, n\}$. The basis-indexed singular value is exactly the ordered singular value at this coordinate:

$$\hat{\sigma}_{\text{basis}}(b) = \sigma_{\text{ord}}(j_{\text{ord}}(b)).$$

If $b \notin s_{e_{\text{ord}}}$, then

$$k \leq j_{\text{ord}}(b).$$

Consequently, for every displayed selected index $a = 1, \dots, k$,

$$\hat{\sigma}_{\text{basis}}(b) \leq \hat{\sigma}_{\text{basis}}(e_{\text{ord}}(a)).$$

This is an exact spectral-index dominance theorem for the constructed top- k right-Gram selection. It supplies the ordered head/complement comparison needed before a future Eckart–Young or tail-optimality proof, but it is not itself an Eckart–Young theorem, a rectangular SVD existence theorem, a tail/coupling bound, a projector-reproduction theorem, a randomness-derived certificate, or a computed non-probability SVD/singular-vector/projector/Gram/ sketch routine certificate. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRightGramBasisOrderedIndex
finCardIndex_rectRightGramBasisOrderedIndex
rectRightGramBasisSingularValue_eq_orderedIndex
rectRightGramOrderedTopEmbedding_not_mem_index_ge
rectTopIndex_le_rectRightGramBasisOrderedIndex_of_not_mem_orderedTopEmbedding
rectRightGramOrderedTopEmbedding_complement_singularValue_le_selected
```

Corollary 583 (Constructed top- k head positivity from the k th singular value). *Assume $0 < k \leq n$. Let*

$$a_\star = k - 1$$

be the last displayed index in the top- k block, and suppose the k th ordered right-Gram singular value is positive:

$$0 < \sigma_{\text{ord}}(\iota_k^n(a_\star)).$$

Then every displayed top- k ordered singular value is positive:

$$0 < \sigma_{\text{ord}}(\iota_k^n(a)), \quad a = 1, \dots, k.$$

Consequently, for the constructed ordered-top embedding e_{ord} , every selected basis-indexed head singular value is positive and hence nonzero:

$$0 < \hat{\sigma}_{\text{basis}}(e_{\text{ord}}(a)), \quad \hat{\sigma}_{\text{basis}}(e_{\text{ord}}(a)) \neq 0, \quad a = 1, \dots, k.$$

This is an exact selected-head positivity handoff. It turns the visible source-rank-style hypothesis that the k th ordered singular value is positive into the nonzero selected-head entries needed by source-SVD determinant and diagonal routes. It does not prove rectangular SVD existence, source-tail factor construction, tail/coupling bounds, projector reproduction, Eckart–Young, randomness certificates, or computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectTopLastIndex
le_rectTopLastIndex
rectTopIndex_le_last
rectSingularValue_top_pos_of_last_pos
rectRightGramOrderedTopEmbedding_selected_pos_of_last_pos
rectRightGramOrderedTopEmbedding_selected_nonzero_of_last_pos
```

Corollary 584 (Constructed top- k zero-safe left-basis orthonormality). *For a basis-indexed right-Gram direction a , let*

$$u_a^0 = \begin{cases} 0, & \hat{\sigma}_{\text{basis}}(a) = 0, \\ Av_a / \hat{\sigma}_{\text{basis}}(a), & \hat{\sigma}_{\text{basis}}(a) \neq 0, \end{cases}$$

be the zero-safe exact left candidate. If two displayed basis-indexed singular values are positive, then their zero-safe left candidates have the exact orthonormal dot product

$$\sum_{i=1}^m u_a^0(i) u_b^0(i) = \delta_{ab}.$$

Consequently, assume $0 < k \leq n$ and suppose, as in the preceding corollary, that the k th ordered right-Gram singular value is positive:

$$0 < \sigma_{\text{ord}}(\iota_k^n(a_\star)), \quad a_\star = k - 1.$$

For the constructed ordered-top embedding e_{ord} , define the selected zero-safe left candidate block

$$U_{\text{ord}}(i, a) = u_{e_{\text{ord}}(a)}^0(i), \quad a = 1, \dots, k.$$

Then

$$U_{\text{ord}}^T U_{\text{ord}} = I_k,$$

entrywise:

$$\sum_{i=1}^m u_{e_{\text{ord}}(a)}^0(i) u_{e_{\text{ord}}(b)}^0(i) = \delta_{ab}, \quad a, b = 1, \dots, k.$$

This is an exact source-SVD left-basis ingredient for the ordered top-k route. It does not construct the complementary tail factor, does not prove right-basis completeness for the source split, does not prove rectangular SVD existence or Eckart–Young tail optimality, and does not certify computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRightGramLeftSingularZeroSafe_col_orthonormal_of_pos
```

```
rectRightGramOrderedTopEmbedding_leftZeroSafe_col_orthonormal_of_last_pos
```

Corollary 585 (Constructed ordered source-head factorization). *Assume $k \leq n$, and let e_{ord} be the constructed ordered top-k right-Gram embedding. Define the exact ordered head factors*

$$U_{\text{ord}}(i, a) = u_{e_{\text{ord}}(a)}^0(i), \quad V_{\text{ord}}(j, a) = v_{e_{\text{ord}}(a)}(j),$$

and the diagonal head singular-value block

$$\Sigma_{\text{ord}}(a, b) = \begin{cases} \hat{\sigma}_{\text{basis}}(e_{\text{ord}}(a)), & a = b, \\ 0, & a \neq b. \end{cases}$$

Let s_{ord} be the selected finite set induced by e_{ord} , and let $H_{s_{\text{ord}}}$ be the selected right-Gram head. Then, entrywise,

$$H_{s_{\text{ord}}} = U_{\text{ord}} \Sigma_{\text{ord}} V_{\text{ord}}^T.$$

Equivalently,

$$H_{s_{\text{ord}}}(i, j) = \sum_{a=1}^k u_{e_{\text{ord}}(a)}^0(i) \hat{\sigma}_{\text{basis}}(e_{\text{ord}}(a)) v_{e_{\text{ord}}(a)}(j).$$

If $0 < k \leq n$ and the k th ordered singular value is positive, then the head left factor has exact orthonormal columns:

$$U_{\text{ord}}^T U_{\text{ord}} = I_k.$$

The head right factor has exact orthonormal columns without the positivity hypothesis:

$$V_{\text{ord}}^T V_{\text{ord}} = I_k.$$

This is an exact ordered source-head factorization theorem. It closes the head side of the constructed top-k source split, but it does not construct the complementary tail factor, prove row completeness for the full right-basis split, prove Eckart–Young tail optimality, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```

rectRightGramOrderedHeadLeft
rectRightGramOrderedHeadRight
rectRightGramOrderedHeadSingularDiagonal
sourceSVDFactorMatrix_rectRightGramOrderedHead_entry
rectRightGramBasisSVDHead_orderedTopEmbedding_eq_sourceSVDFactorMatrix
rectRightGramOrderedHeadLeft_col_orthonormal_of_last_pos
rectRightGramOrderedHeadRight_col_orthonormal

```

Corollary 586 (Constructed ordered source-tail factorization). *Assume $k \leq n$, let e_{ord} be the constructed ordered top- k right-Gram embedding, and let s_{ord} be its selected finite set. Put $q = |s_{\text{ord}}^c|$, and enumerate the complement by*

$$c \mapsto e_{\text{tail}}(c), \quad c = 1, \dots, q.$$

Define the exact ordered complement-tail factors

$$U_{\text{tail}}(i, c) = u_{e_{\text{tail}}(c)}^0(i), \quad V_{\text{tail}}(j, c) = v_{e_{\text{tail}}(c)}(j),$$

and

$$\Sigma_{\text{tail}}(c, d) = \begin{cases} \hat{\sigma}_{\text{basis}}(e_{\text{tail}}(c)), & c = d, \\ 0, & c \neq d. \end{cases}$$

Let $T_{s_{\text{ord}}}$ be the complementary right-Gram tail obtained by summing the zero-safe reconstruction terms over s_{ord}^c . Then, entrywise,

$$T_{s_{\text{ord}}} = U_{\text{tail}} \Sigma_{\text{tail}} V_{\text{tail}}^T,$$

or equivalently

$$T_{s_{\text{ord}}}(i, j) = \sum_{c=1}^q u_{e_{\text{tail}}(c)}^0(i) \hat{\sigma}_{\text{basis}}(e_{\text{tail}}(c)) v_{e_{\text{tail}}(c)}(j).$$

The exact right-tail table has orthonormal columns:

$$V_{\text{tail}}^T V_{\text{tail}} = I_q.$$

Together with the ordered source-head factorization, the exact ordered source split reconstructs A :

$$U_{\text{ord}} \Sigma_{\text{ord}} V_{\text{ord}}^T + U_{\text{tail}} \Sigma_{\text{tail}} V_{\text{tail}}^T = A.$$

This is an exact source-split algebra theorem. It does not assert that U_{tail} is an orthonormal rectangular-SVD tail basis when some tail singular values vanish; a nullspace completion is still required for that stronger certificate. It also does not prove full right-basis row completeness, Eckart–Young tail optimality, randomness-derived certificates, or computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```

rectRightGramBasisSVDTailLeft
rectRightGramBasisSVDTailRight

```

```

rectRightGramBasisSVDTailSingularDiagonal
rectRightGramOrderedTailLeft
rectRightGramOrderedTailRight
rectRightGramOrderedTailSingularDiagonal
rectRightGramBasisSVDTailRight_col_orthonormal
rectRightGramOrderedTailRight_col_orthonormal
sourceSVDFactorMatrix_rectRightGramBasisSVDTail_entry
rectRightGramBasisSVDTail_eq_sourceSVDFactorMatrix
rectRightGramBasisSVDTail_orderedTopEmbedding_eq_sourceSVDFactorMatrix
rectRightGramOrdered_source_head_add_tail

```

Corollary 587 (Constructed ordered right-basis block completeness). *Assume $k \leq n$, let s_{ord} be the selected finite set induced by the constructed ordered top- k right-Gram embedding, and put $q = |s_{\text{ord}}^c|$. Let V_{ord} and V_{tail} be the exact right factors from the ordered source-head and ordered source-tail constructions. Define the block right table*

$$Q_{\text{ord}}(j, \beta) = \begin{cases} V_{\text{tail}}(j, c), & \beta = \text{inl}(c), \\ V_{\text{ord}}(j, a), & \beta = \text{inr}(a). \end{cases}$$

Then Q_{ord} has exact orthonormal columns:

$$\sum_{j=1}^n Q_{\text{ord}}(j, \beta) Q_{\text{ord}}(j, \beta') = \delta_{\beta\beta'}, \quad \beta, \beta' \in \{1, \dots, q\} \sqcup \{1, \dots, k\}.$$

It is also row-complete:

$$\sum_{\beta} Q_{\text{ord}}(j, \beta) Q_{\text{ord}}(\ell, \beta) = \delta_{j\ell}, \quad j, \ell = 1, \dots, n.$$

Equivalently,

$$Q_{\text{ord}}^T Q_{\text{ord}} = I_{q+k}, \quad Q_{\text{ord}} Q_{\text{ord}}^T = I_n.$$

The proof uses only exact selected-set and complement enumeration, disjointness of a finite set and its complement, and exact column/row orthonormality of the right-Gram eigenbasis. This closes the constructed ordered right-basis block fields needed by the source-tail Frobenius route. It does not construct a nullspace-completed orthonormal tail-left basis, prove Eckart–Young tail optimality, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```

rectRightGramSelectedIndexSet_sum
rectRightGramComplement_sum_orderEmbOffFin
rectRightGramSelectedIndexSet_head_tail_cross_zero
rectRightGramSelectedIndexSet_tail_head_cross_zero
rectRightGramSelectedIndexSet_tail_head_row_complete
rectRightGramOrderedRightBasisBlock_col_orthonormal
rectRightGramOrderedRightBasisBlock_row_orthonormal
rectRightGramOrderedRightBasisBlock_col_row_orthonormal

```

Corollary 588 (Constructed ordered block source-SVD certificate). *Assume $k \leq n$. Let*

$$q = |s_{\text{ord}}^c|,$$

where s_{ord} is the selected finite set induced by the constructed ordered top- k right-Gram embedding. Let

$$H_{\text{ord}} = U_{\text{ord}} \Sigma_{\text{ord}} V_{\text{ord}}^T, \quad T_{\text{ord}} = U_{\text{tail}} \Sigma_{\text{tail}} V_{\text{tail}}^T$$

be the constructed ordered source-head and source-tail factors from the exact right-Gram eigenbasis. Suppose the remaining left block is certified:

$$L_{\text{ord}}(i, \alpha) = \begin{cases} U_{\text{ord}}(i, a), & \alpha = \text{inl}(a), \\ U_{\text{tail}}(i, c), & \alpha = \text{inr}(c), \end{cases} \quad L_{\text{ord}}^T L_{\text{ord}} = I_{k+q},$$

and suppose the displayed ordered head singular values are nonzero,

$$\sigma_a = \sigma_{\text{right}}(\text{ordTop}(a)) \neq 0, \quad a = 1, \dots, k.$$

Then the constructed ordered data instantiate the block diagonal source-SVD certificate:

$$A = U_{\text{ord}} \Sigma_{\text{ord}} V_{\text{ord}}^T + U_{\text{tail}} \Sigma_{\text{tail}} V_{\text{tail}}^T, \\ L_{\text{ord}}^T L_{\text{ord}} = I_{k+q}, \quad Q_{\text{ord}}^T Q_{\text{ord}} = I_{q+k}, \quad Q_{\text{ord}} Q_{\text{ord}}^T = I_n,$$

where $Q_{\text{ord}} = [V_{\text{tail}}, V_{\text{ord}}]$, and

$$(\Sigma_{\text{ord}})_{ab} = \delta_{ab} \sigma_a, \quad \sigma_a \neq 0.$$

Equivalently, the exact constructed ordered split supplies

$$\text{BlockDiagonalSourceSVDTailCertificate}(A, U_{\text{ord}}, \Sigma_{\text{ord}}, \sigma, V_{\text{ord}}, U_{\text{tail}}, \Sigma_{\text{tail}}, V_{\text{tail}}).$$

There is also a component-left form. It is enough to prove

$$U_{\text{ord}}^T U_{\text{ord}} = I_k, \quad U_{\text{ord}}^T U_{\text{tail}} = 0, \quad U_{\text{tail}}^T U_{\text{tail}} = I_q,$$

together with the same nonzero head singular values. If $k > 0$ and the k th ordered right-Gram singular value is positive, then the nonzero head field and $U_{\text{ord}}^T U_{\text{ord}} = I_k$ are supplied by the constructed ordered head-positivity theorem. In that case the only remaining left-side hypotheses are

$$U_{\text{ord}}^T U_{\text{tail}} = 0, \quad U_{\text{tail}}^T U_{\text{tail}} = I_q.$$

This is an exact-object certificate adapter. It closes the route from the constructed ordered right-Gram source split and right-basis block to the existing block source-SVD certificate surface. It does not construct the nullspace-completed orthonormal tail-left basis, prove Eckart–Young tail optimality, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
leftBasisBlock_col_orthonormal_of_component_orthonormal_fields
BlockDiagonalSourceSVDTailCertificate.of_rectRightGramOrderedSourceSplit
BlockDiagonalSourceSVDTailCertificate.of_rectRightGramOrderedSourceSplit_component_left
BlockDiagonalSourceSVDTailCertificate.of_rectRightGramOrderedSourceSplit_component_left_of_last_pos
```

Corollary 589 (Constructed ordered head-tail left cross field). *Assume $k > 0$, $k \leq n$, and the k th ordered right-Gram singular value is positive:*

$$0 < \sigma_k(A).$$

Let U_{ord} be the constructed ordered head-left zero-safe table and let U_{tail} be the constructed complement-tail zero-safe table. Then the exact head/tail left cross field vanishes:

$$U_{\text{ord}}^T U_{\text{tail}} = 0, \quad \sum_{i=1}^m U_{\text{ord}}(i, a) U_{\text{tail}}(i, c) = 0$$

for every displayed head index $a = 1, \dots, k$ and every complement-tail index $c = 1, \dots, q$.

More generally, if u_a and u_b are zero-safe left candidates built from right-Gram basis directions $a \neq b$, and the singular value for a is strictly positive, then

$$\sum_{i=1}^m u_a(i) u_b(i) = 0.$$

If the singular value for b is zero, this follows because $u_b = 0$; if it is nonzero, the existing positive-pair orthonormality theorem gives the off-diagonal zero.

Consequently, under the same k th-positivity hypothesis, the constructed ordered block source-SVD certificate from the preceding corollary only needs the remaining tail-left orthonormality field

$$U_{\text{tail}}^T U_{\text{tail}} = I_q.$$

The head-left orthonormality, head nonzero field, and head-tail left cross-orthogonality are all supplied by the constructed ordered right-Gram theorems.

This is still an exact-object D3 theorem. It does not construct a nullspace-completed orthonormal tail-left basis, prove Eckart–Young tail optimality, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRightGramLeftSingularZeroSafe_cross_zero_of_pos_ne
rectRightGramOrderedHeadTailLeft_cross_zero_of_last_pos
BlockDiagonalSourceSVDTailCertificate.of_rectRightGramOrderedSourceSplit_tail_left_of_last_pos
```

Corollary 590 (Positive-complement ordered tail-left certificate for equation (9)). *Let s be any selected right-Gram basis-index set and enumerate its complement by e_c , $c = 1, \dots, q$. Assume every complement singular value is strictly positive:*

$$0 < \tau_{e_c}(A), \quad c = 1, \dots, q.$$

Let U_{tail} be the complement-tail zero-safe left candidate table,

$$U_{\text{tail}}(i, c) = u_{e_c}(i).$$

Then the complement-tail left table has exact orthonormal columns:

$$U_{\text{tail}}^T U_{\text{tail}} = I_q, \quad \sum_{i=1}^m U_{\text{tail}}(i, c) U_{\text{tail}}(i, d) = \delta_{cd}.$$

For the constructed ordered top- k selected set, if additionally

$$k > 0, \quad k \leq n, \quad 0 < \sigma_k(A),$$

and every constructed complement singular value is strictly positive, then the constructed ordered block source-SVD certificate is instantiated entirely from the constructed exact right-Gram objects:

$$\text{BlockDiagonalSourceSVDTailCertificate}(A, U_{\text{ord}}, \Sigma_{\text{ord}}, \sigma_{\text{ord}}, V_{\text{ord}}, U_{\text{tail}}, \Sigma_{\text{tail}}, V_{\text{tail}}).$$

Here k th head positivity supplies the head nonzero and head-left fields, Corollary 563 supplies $U_{\text{ord}}^T U_{\text{tail}} = 0$, and complement positivity supplies $U_{\text{tail}}^T U_{\text{tail}} = I_q$.

This is an exact-object positive-complement branch. It does not construct a nullspace-completed tail-left basis when a complement singular value is zero, prove Eckart–Young tail optimality, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRightGramBasisSVDTailLeft_col_orthonormal_of_pos
rectRightGramOrderedTailLeft_col_orthonormal_of_complement_pos
BlockDiagonalSourceSVDTailCertificate.of_rectRightGramOrderedSourceSplit_all_tail_pos_of_last_pos
```

Corollary 591 (Zero-tail obstruction for the raw zero-safe tail-left table). *Let s be any selected right-Gram basis-index set and enumerate its complement by e_c , $c = 1, \dots, q$. If one complement singular value vanishes,*

$$\tau_{e_c}(A) = 0,$$

then the corresponding zero-safe tail-left column is the zero column:

$$U_{\text{tail}}(i, c) = 0, \quad i = 1, \dots, m.$$

Consequently its self-dot is zero,

$$\sum_{i=1}^m U_{\text{tail}}(i, c)^2 = 0,$$

and the raw zero-safe complement-tail table cannot be column orthonormal, because column orthonormality would require the same diagonal self-dot to be

$$(I_q)_{cc} = 1.$$

For the constructed ordered top- k selected set, the same obstruction rules out any block source-SVD certificate whose tail-left block is the raw constructed zero-safe table:

$$\neg \text{BlockDiagonalSourceSVDTailCertificate}(A, U_{\text{ord}}, \Sigma_{\text{ord}}, \sigma_{\text{ord}}, V_{\text{ord}}, U_{\text{tail}}, \Sigma_{\text{tail}}, V_{\text{tail}})$$

whenever a constructed complement singular value is zero.

This is a route-elimination result, not a replacement construction. It proves that the remaining zero-tail case genuinely needs a nullspace-completed orthonormal tail-left basis. It does not construct that basis, prove Eckart–Young tail optimality, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```

rectRightGramBasisSVDTailLeft_self_dot_eq_zero_of_singularValue_eq_zero
not_rectRightGramBasisSVDTailLeft_col_orthonormal_of_zero_singularValue
not_rectRightGramOrderedTailLeft_col_orthonormal_of_zero_complement_singularValue
not_BlockDiagonalSourceSVDTailCertificate_rectRightGramOrdered_zero_safe_tail_of_zero_complement_
singularValue

```

Corollary 592 (Replacement tail-left adapter for the zero-tail source split). *Let s be a selected right-Gram basis-index set with complement enumeration e_c , $c = 1, \dots, q$. Let $\tilde{U}_{\text{tail}} \in \mathbb{R}^{m \times q}$ be any replacement tail-left table satisfying the nonzero-direction agreement condition*

$$\tau_{e_c}(A) \neq 0 \implies \tilde{U}_{\text{tail}}(i, c) = U_{\text{tail}}(i, c) \quad \text{for all } i, c.$$

Then the exact complement-tail source factor is unchanged:

$$T_s = \tilde{U}_{\text{tail}} \Sigma_{\text{tail}} V_{\text{tail}}^T.$$

Indeed, if $\tau_{e_c}(A) = 0$, the diagonal entry of Σ_{tail} erases the c -th replacement column; if $\tau_{e_c}(A) \neq 0$, the replacement column agrees with the zero-safe column.

For the constructed ordered top- k selected set, assume

$$k > 0, \quad k \leq n, \quad 0 < \sigma_k(A),$$

and suppose $\tilde{U}_{\text{tail}}$ satisfies the same nonzero-direction agreement, the replacement orthonormality field

$$\tilde{U}_{\text{tail}}^T \tilde{U}_{\text{tail}} = I_q,$$

and the replacement head-tail cross field

$$U_{\text{ord}}^T \tilde{U}_{\text{tail}} = 0.$$

Then the constructed ordered block source-SVD certificate holds with $\tilde{U}_{\text{tail}}$ in place of the raw zero-safe tail-left table:

$$\text{BlockDiagonalSourceSVDTailCertificate}(A, U_{\text{ord}}, \Sigma_{\text{ord}}, \sigma_{\text{ord}}, V_{\text{ord}}, \tilde{U}_{\text{tail}}, \Sigma_{\text{tail}}, V_{\text{tail}}).$$

This is the exact adapter for a future nullspace-completed tail-left construction. It does not construct $\tilde{U}_{\text{tail}}$, prove Eckart–Young tail optimality, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```

sourceSVDFactorMatrix_rectRightGramBasisSVDTail_replacement_left_entry
rectRightGramBasisSVDTail_eq_sourceSVDFactorMatrix_replacement_left
rectRightGramOrdered_source_head_add_tail_replacement_left
BlockDiagonalSourceSVDTailCertificate.of_rectRightGramOrderedSourceSplit_replacement_tail_left_
of_last_pos

```

Corollary 593 (Left-block dimension guard for equation (9)). *Let $\mathcal{I}$ be a finite index set and let $X \in \mathbb{R}^{m \times \mathcal{I}}$ satisfy the exact column orthonormality equations*

$$\sum_{i=1}^m X(i, \alpha) X(i, \beta) = \begin{cases} 1, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases} \quad \alpha, \beta \in \mathcal{I}.$$

Then

$$|\mathcal{I}| \leq m.$$

Consequently, if the concatenated left source-SVD block

$$[U \ U_{\text{tail}}] \in \mathbb{R}^{m \times (r+q)}$$

is column orthonormal, then

$$r + q \leq m.$$

In particular, every exact block source-SVD certificate of the form

$$\text{BlockDiagonalSourceSVDTailCertificate}(A, U, \Sigma, \sigma, V, U_{\text{tail}}, \Sigma_{\text{tail}}, V_{\perp})$$

exposes the necessary dimension side condition $r + q \leq m$.

For the full right-Gram split used in the current ordered equation (9) route, whenever the selected and complement blocks partition n right directions so that $r+q = n$, this guard reads $n \leq m$. Thus the nullspace-completed replacement-tail construction must either carry this tall/thin condition explicitly or move to a rectangular SVD surface with only the appropriate number of left columns. This is an exact-object dimension guard. It does not construct the replacement basis, prove Eckart–Young tail optimality, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
colOrthonormal_fintype_card_le_rows
leftBasisBlock_col_orthonormal_card_le_rows
BlockDiagonalSourceSVDTailCertificate.left_column_count_le_row_dim
```

Corollary 594 (Partial left-block orthonormal completion for equation (9)). *Let $S \subseteq \{1, \dots, m\}$, and let $X \in \mathbb{R}^{m \times m}$ satisfy the exact partial column-orthonormality equations*

$$\sum_{i=1}^m X(i, a) X(i, b) = \begin{cases} 1, & a = b, \\ 0, & a \neq b, \end{cases} \quad a, b \in S.$$

Then there exists $Y \in \mathbb{R}^{m \times m}$ such that

$$Y(:, a) = X(:, a) \quad (a \in S)$$

and

$$\sum_{i=1}^m Y(i, a) Y(i, b) = \begin{cases} 1, & a = b, \\ 0, & a \neq b, \end{cases} \quad a, b \in \{1, \dots, m\}.$$

More generally, let

$$e : \{1, \dots, r\} \sqcup \{1, \dots, q\} \hookrightarrow \{1, \dots, m\}$$

embed the displayed head/tail block columns into the ambient left-column index set. Let $S \subseteq \{1, \dots, r\} \sqcup \{1, \dots, q\}$ contain every head index, and suppose the columns of the partially specified block

$$[U \ U_{\text{tail}}^{(0)}]$$

indexed by S are exactly orthonormal. Then there exists a replacement tail-left table $U_{\text{tail}} \in \mathbb{R}^{m \times q}$ such that

$$U_{\text{tail}}(:, a) = U_{\text{tail}}^{(0)}(:, a) \quad \text{for every tail index } a \text{ with } \text{tail}(a) \in S,$$

and

$$[U \ U_{\text{tail}}]^T [U \ U_{\text{tail}}] = I_{r+q}.$$

This is exact-object nullspace-completion infrastructure for the equation (9) route. It does not yet instantiate S with the ordered nonzero-tail directions, compose the completion with the replacement source-factor adapter, prove Eckart–Young tail optimality, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
partialColOrthonormal_exists_fullColOrthonormal
partialLeftBasisBlock_exists_replacement_tail
```

Corollary 595 (Ordered nonzero-tail completion for equation (9)). *Let $k \leq n$, $k > 0$, and suppose the k th ordered singular value of the exact right Gram route is positive:*

$$0 < \sigma_k.$$

Let S_{tail} be the complement of the constructed ordered top- k right-index set, enumerated by $c = 1, \dots, q$, and write

$$\tau_c = \text{rectRightGramBasisSingularValue}(A, S_{\text{tail}}(c)).$$

Define the partial left-block set

$$S_{\neq 0} = \{\text{head}(a) : 1 \leq a \leq k\} \cup \{\text{tail}(c) : 1 \leq c \leq q, \tau_c \neq 0\}.$$

For the exact zero-safe ordered left candidates

$$U_{\text{ord}} = \text{rectRightGramOrderedHeadLeft}(A), \quad U_{\text{tail}}^{(0)} = \text{rectRightGramOrderedTailLeft}(A),$$

the columns of $[U_{\text{ord}} \ U_{\text{tail}}^{(0)}]$ indexed by $S_{\neq 0}$ are exactly orthonormal:

$$\sum_{i=1}^m [U_{\text{ord}} \ U_{\text{tail}}^{(0)}](i, \alpha) [U_{\text{ord}} \ U_{\text{tail}}^{(0)}](i, \beta) = \begin{cases} 1, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases} \quad \alpha, \beta \in S_{\neq 0}.$$

Assume in addition that the full displayed head/tail left-block index set embeds into the ambient row-coordinate set,

$$e : \{1, \dots, k\} \sqcup \{1, \dots, q\} \hookrightarrow \{1, \dots, m\}.$$

Then there exists a replacement tail-left table $U_{\text{tail}} \in \mathbb{R}^{m \times q}$ such that

$$U_{\text{tail}}(:, c) = U_{\text{tail}}^{(0)}(:, c) \quad \text{whenever } \tau_c \neq 0,$$

$$[U_{\text{ord}} \ U_{\text{tail}}]^T [U_{\text{ord}} \ U_{\text{tail}}] = I_{k+q},$$

and the ordered block source-SVD certificate

$$\text{BlockDiagonalSourceSVDTailCertificate}(A, U_{\text{ord}}, \Sigma_{\text{ord}}, \sigma_{\text{ord}}, V_{\text{ord}}, U_{\text{tail}}, \Sigma_{\text{tail}}, V_{\text{tail}})$$

holds. Thus the replacement-tail source factor agrees with the ordered zero-safe tail source factor on all nonzero tail singular directions, while zero tail singular directions are supplied by orthonormal completion.

This is exact-object nullspace-completion infrastructure. It does not prove Eckart–Young tail optimality, randomness-derived cross-term certificates, or computed non-probability SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRightGramOrderedTailIndex
rectRightGramOrderedNonzeroTailPartialSet
rectRightGramOrderedNonzeroTailPartialSet_head
rectRightGramOrderedNonzeroTailPartialSet_tail_iff
rectRightGramOrderedNonzeroTailPartialSet_leftBasisBlock_col_orthonormal_of_last_pos
exists_rectRightGramOrdered_replacement_tail_left_block_certificate_of_last_pos
```

Corollary 596 (Ordered replacement-tail rank surface for equation (9)). *Keep the hypotheses of the ordered nonzero-tail completion corollary. Thus $k \leq n$, $k > 0$, the k th ordered right-Gram singular value is positive, and the full ordered head/tail left-block index set embeds in the ambient row coordinates. Let*

$$U_{\text{ord}}, \quad \Sigma_{\text{ord}}, \quad V_{\text{ord}}, \quad \Sigma_{\text{tail}}, \quad V_{\text{tail}}$$

be the exact constructed ordered right-Gram source factors, and let $Z \in \mathbb{R}^{n \times k}$ be the exact sketching matrix. Assume the exact source full-rank condition

$$\det(V_{\text{ord}}^T Z) \neq 0,$$

and assume the exact cross-term certificate

$$\|\Sigma_{\text{tail}}(V_{\text{tail}}^T Z)(V_{\text{ord}}^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_{\text{tail}}\|_F, \quad \varepsilon \geq 0.$$

Then there exists a replacement tail-left table U_{tail} such that

$$U_{\text{tail}}(:, c) = U_{\text{tail}}^{(0)}(:, c) \quad \text{for every nonzero complement-tail singular direction } c,$$

$$[U_{\text{ord}} \ U_{\text{tail}}]^T [U_{\text{ord}} \ U_{\text{tail}}] = I,$$

and the exact Gram-inverse column-sketch projector

$$P_{AZ} = AZ((AZ)^T(AZ))^{-1}(AZ)^T$$

satisfies the equation-(9) rank/residual surface

$$\text{rank}(P_{AZ}A) \leq k, \quad \rho_F(A, P_{AZ}A) \leq 2\sqrt{1 + \varepsilon^2} \|\Sigma_{\text{tail}}\|_F.$$

The theorem also returns the exact projector reproduction fields

$$P_{AZ}(AZ) = AZ, \quad P_{AZ}P_{AZ} = P_{AZ},$$

and the repository symmetric-projector certificate for P_{AZ} .

This is a new exact-object source-split-to-rank-surface result. It does not prove the relative Eckart–Young conclusion $\rho_F(A, P_{AZ}A) \leq \rho \rho_F(A, A_k)$, does not derive the displayed cross-term certificate from randomness, and does not certify computed non-probability SVD/singular-vector/projector/Gram/inverse/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
exists_columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRankResidualSurface_of_
rectRightGramOrdered_replacement_tail_left_of_last_pos
```

Corollary 597 (Ordered replacement-tail relative surface for equation (9)). *Keep the hypotheses and notation of the ordered replacement-tail rank-surface corollary. In addition, assume the exact tail-optimality inequality*

$$\|\Sigma_{\text{tail}}\|_F \leq \rho_F(A, B) \quad \text{for every exact } B \text{ with } \text{rank}(B) \leq k,$$

and assume the scalar comparison

$$2\sqrt{1 + \varepsilon^2} \|\Sigma_{\text{tail}}\|_F \leq \rho \|\Sigma_{\text{tail}}\|_F.$$

Then there exists a replacement tail-left table U_{tail} satisfying the same nonzero-direction agreement and block-orthonormality conditions as in the preceding corollary, and the constructed ordered source head

$$A_k^{\text{ord}} = U_{\text{ord}} \Sigma_{\text{ord}} V_{\text{ord}}^T$$

is an exact best rank- k approximation in the repository Frobenius sense:

$$A_k^{\text{ord}} \in \arg \min_{\text{rank}(B) \leq k} \rho_F(A, B), \quad \rho_F(A, A_k^{\text{ord}}) = \|\Sigma_{\text{tail}}\|_F.$$

The same exact Gram-inverse column-sketch projector

$$P_{AZ} = AZ((AZ)^T(AZ))^{-1}(AZ)^T$$

satisfies

$$\text{rank}(P_{AZ}A) \leq k, \quad \rho_F(A, P_{AZ}A) \leq \rho \rho_F(A, A_k^{\text{ord}}).$$

This is an exact-object relative theorem surface. It removes the constructed source-split handoff from the relative equation-(9) proof, but it still assumes the exact Eckart–Young tail-optimality inequality displayed above. It also does not derive the cross-term certificate from randomness and does not certify computed non-probability SVD/singular-vector/projector/Gram/inverse/sketch/ product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
exists_columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
rectRightGramOrdered_replacement_tail_left_of_last_pos
```

Corollary 598 (Constructed ordered tail-diagonal Frobenius expansion). *Let S_{ord} be the constructed ordered top- k right-Gram index set, let S_{ord}^c be its complement, and enumerate the complement by $\tau(c)$, $c = 1, \dots, q$. Define the exact ordered tail diagonal by*

$$(\Sigma_{\text{tail}})_{cd} = \begin{cases} \sigma_{\tau(c)}, & c = d, \\ 0, & c \neq d, \end{cases}$$

where $\sigma_{\tau(c)}$ is the exact basis-indexed right-Gram singular value in the complement direction. Then

$$\|\Sigma_{\text{tail}}\|_F^2 = \sum_{c=1}^q \sigma_{\tau(c)}^2, \quad \|\Sigma_{\text{tail}}\|_F = \left(\sum_{c=1}^q \sigma_{\tau(c)}^2 \right)^{1/2}.$$

More generally, for any exact diagonal table $D = \text{diag}(\alpha)$,

$$\|D\|_F^2 = \sum_c \alpha_c^2, \quad \|D\|_F = \left(\sum_c \alpha_c^2 \right)^{1/2}.$$

This is the exact-object diagonal-norm expansion needed before the remaining LR.1dt tail-optimality hypothesis can be discharged by the q -dimensional Eckart–Young lower-bound route. It does not prove the residual lower bound $\|\Sigma_{\text{tail}}\|_F \leq \rho_F(A, B)$, does not derive any randomized cross-term certificate, and does not certify computed non-probability SVD/singular-vector/projector/Gram/inverse/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
frobNormSq_diagonal_eq_sum
frobNorm_diagonal_eq_sqrt_sum
frobNormSq_rectRightGramOrderedTailSingularDiagonal_eq_sum
frobNorm_rectRightGramOrderedTailSingularDiagonal_eq_sqrt_sum
```

Corollary 599 (Constructed ordered head–tail cardinality bridge). *Let $e : \{1, \dots, k\} \hookrightarrow \{1, \dots, n\}$ be any injective selected-index map, and let*

$$S = \{e(a) : a = 1, \dots, k\}.$$

Then the exact selected set and its complement exhaust the ambient right coordinates:

$$|S| = k, \quad k + |S^c| = n.$$

In particular, for the constructed ordered top- k right-Gram embedding e_{ord} , if q denotes the cardinality of the ordered complement tail index type, then

$$k + q = n.$$

This is an exact-object reindexing dependency for the remaining Eckart–Young transport. It supplies only the head-plus-tail cardinality bridge needed before the q -dimensional lower-bound theorem can be moved to the original n -column ordered source split. It does not construct the column permutation/equivalence transport, does not prove the residual lower bound, does not derive randomness, and does not certify computed non-probability SVD/singular-vector/projector/Gram/inverse/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRightGramSelectedIndexSet_card_add_compl_card
rectRightGramOrderedTailIndex_card_add
```

Corollary 600 (Constructed ordered head–tail square gap). *Let $A \in \mathbb{R}^{m \times n}$, let $k \leq n$, and assume $0 < k$. Let*

$$e_{\text{ord}} : \{1, \dots, k\} \hookrightarrow \{1, \dots, n\}$$

be the constructed ordered top- k right-Gram embedding, and let

$$\tau : \{1, \dots, q\} \rightarrow \{1, \dots, n\}$$

enumerate the complement-tail right-Gram indices $\{e_{\text{ord}}(1), \dots, e_{\text{ord}}(k)\}^c$. If σ_ℓ denotes the exact right-Gram basis singular value associated with right coordinate ℓ , and a_ is the last selected ordered head index, define*

$$\eta = \sigma_{e_{\text{ord}}(a_*)}^2.$$

Then η separates the constructed ordered head squares from all complement-tail squares:

$$\eta \leq \sigma_{e_{\text{ord}}(a)}^2 \quad \text{for every } a = 1, \dots, k, \quad \sigma_{\tau(c)}^2 \leq \eta \quad \text{for every } c = 1, \dots, q.$$

The head inequality is proved by the exact selected-square equality for the constructed ordered embedding and antitonicity of ordered right-Gram singular-value squares. The tail inequality is proved by the complement-versus-selected singular-value comparison, nonnegativity of the basis singular values, and monotonicity of squaring on nonnegative reals. Thus the LR.1dw visible gap hypothesis is instantiated for the constructed ordered split without assuming that the complement-tail enumeration is internally sorted.

This is an exact-object gap theorem. It does not construct the original-column reindexing/equivalence transport, does not prove the LR.1dt residual lower-bound/tail-optimality discharge, does not derive randomness, and does not certify computed non-probability SVD/singular-vector/projector/Gram/inverse/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRightGramOrdered_head_tail_square_gap
```

Corollary 601 (Exact column-permutation transport for low-rank residuals). *Let $A, B \in \mathbb{R}^{m \times n}$, let r be a target rank, and let*

$$\pi : \{1, \dots, n\} \simeq \{1, \dots, n\}$$

be an exact column equivalence. Define the column-reindexed matrix

$$(A^\pi)_{ij} = A_{i, \pi(j)}.$$

If B has an exact rank factorization

$$B_{ij} = \sum_{a=1}^r L_{ia} R_{aj},$$

then B^π has the exact rank factorization

$$B_{ij}^\pi = \sum_{a=1}^r L_{ia} R_{a, \pi(j)}.$$

Consequently,

$$\text{rank}(B) \leq r \implies \text{rank}(B^\pi) \leq r, \quad \text{rank}(B^\pi) \leq r \implies \text{rank}(B) \leq r.$$

Moreover, if the same exact column equivalence is applied to the source and competitor, the Frobenius residual is unchanged:

$$\rho_F(A^\pi, B^\pi) = \rho_F(A, B).$$

This is exact-object column-reindexing infrastructure. It supplies the generic transport needed after the $k+q = n$ cardinality bridge, but this theorem itself does not construct the specific head-plus-complement-tail equivalence $\{1, \dots, k+q\} \simeq \{1, \dots, n\}$, does not discharge the LR.1dt tail-optimality inequality, does not derive randomness, and does not certify computed non-probability SVD/singular-vector/projector/Gram/inverse/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
RectRankFactorization.permuteCols
RectRankAtMost.permuteCols
RectRankAtMost.of_permuteCols
lowRankResidualFrob_permuteCols
```

Corollary 602 (Constructed ordered head–tail column equivalence). *Let $k \leq n$, let*

$$e_{\text{ord}} : \{1, \dots, k\} \hookrightarrow \{1, \dots, n\}$$

be the constructed ordered top- k embedding, and define

$$S = \{e_{\text{ord}}(1), \dots, e_{\text{ord}}(k)\}.$$

Let $q = |S^c|$, and let

$$\tau : \{1, \dots, q\} \rightarrow \{1, \dots, n\}$$

be the canonical increasing enumeration of the complement S^c . Define

$$\Phi : \{1, \dots, k\} \sqcup \{1, \dots, q\} \rightarrow \{1, \dots, n\}$$

by

$$\Phi(\text{inl } a) = e_{\text{ord}}(a), \quad \Phi(\text{inr } c) = \tau(c).$$

Then Φ is injective and surjective. Hence it induces an exact equivalence

$$\Phi_\Sigma : \{1, \dots, k\} \sqcup \{1, \dots, q\} \simeq \{1, \dots, n\}.$$

Composing with the standard finite sum equivalence gives the paper-facing column equivalence

$$\Phi_{\text{fin}} : \{1, \dots, k+q\} \simeq \{1, \dots, n\},$$

with the displayed action

$$\Phi_{\text{fin}}(\text{sumFin}(\text{inl } a)) = e_{\text{ord}}(a), \quad \Phi_{\text{fin}}(\text{sumFin}(\text{inr } c)) = \tau(c).$$

The injectivity proof uses disjointness of S and S^c , injectivity of the constructed ordered top embedding on the head block, and injectivity of the canonical complement enumeration on the tail

block. The surjectivity proof splits every original column coordinate into the selected set S or the complement S^c .

This is exact-object column-equivalence infrastructure. It constructs the specific head-plus-complement-tail equivalence needed after LR.1dv/LR.1dy, but it does not discharge the LR.1dt tail-optimality inequality, does not derive randomness, and does not certify computed non-probability SVD/singular-vector/projector/Gram/inverse/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRightGramOrderedHeadTailColumnMap
rectRightGramOrderedHeadTailColumnMap_injective
rectRightGramOrderedHeadTailColumnMap_surjective
rectRightGramOrderedHeadTailColumnSumEquiv
rectRightGramOrderedHeadTailColumnEquiv
rectRightGramOrderedHeadTailColumnEquiv_head
rectRightGramOrderedHeadTailColumnEquiv_tail
```

Corollary 603 (Cross-domain exact column-equivalence transport). *Let*

$$\pi : \{1, \dots, p\} \simeq \{1, \dots, n\}$$

be an exact column equivalence. For $A \in \mathbb{R}^{m \times n}$, define the pulled-back matrix

$$A_{ij}^\pi = A_{i, \pi(j)}, \quad i = 1, \dots, m, \quad j = 1, \dots, p.$$

If $B \in \mathbb{R}^{m \times n}$ has an exact rank factorization

$$B_{ij} = \sum_{a=1}^r L_{ia} R_{aj},$$

then $B^\pi \in \mathbb{R}^{m \times p}$ has the exact rank factorization

$$B_{ij}^\pi = \sum_{a=1}^r L_{ia} R_{a, \pi(j)}.$$

Consequently, exact rank-at-most certificates transport forward and backward:

$$\text{rank}(B) \leq r \implies \text{rank}(B^\pi) \leq r, \quad \text{rank}(B^\pi) \leq r \implies \text{rank}(B) \leq r.$$

Moreover,

$$\|A^\pi\|_F = \|A\|_F, \quad \rho_F(A^\pi, B^\pi) = \rho_F(A, B).$$

Specializing π to the constructed ordered head-tail equivalence Φ_{fin} from Corollary 590 gives the same rank and residual transport for the paper-facing reindexing

$$\{1, \dots, k + q\} \simeq \{1, \dots, n\}.$$

This is exact-object cross-domain reindexing infrastructure. It supplies the rank/residual transport needed before applying q -dimensional lower-bound theorems to the original n -column ordered source split, but it does not discharge the LR.1dt tail-optimality inequality, does not derive randomness, and does not certify computed non-probability SVD/singular-vector/projector/Gram/inverse/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```

rectReindexCols
RectRankFactorization.reindexCols
RectRankAtMost.reindexCols
RectRankAtMost.of_reindexCols
frobNormSqRect_reindexCols
frobNormRect_reindexCols
lowRankResidualFrob_reindexCols
RectRankAtMost.reindexCols_rectRightGramOrderedHeadTailColumnEquiv
RectRankAtMost.of_reindexCols_rectRightGramOrderedHeadTailColumnEquiv
lowRankResidualFrob_rectRightGramOrderedHeadTailColumnEquiv

```

Corollary 604 (Constructed ordered tail optimality for the LR.1dt surface). *Let $k \leq n$, $0 < k$, and suppose the last displayed ordered head singular value is positive:*

$$0 < \sigma_k(A).$$

Let S be the constructed ordered top- k right-Gram index set, let $q = |S^c|$, and let

$$e : \{1, \dots, k\} \sqcup \{1, \dots, q\} \hookrightarrow \{1, \dots, m\}$$

be the exact embedding used by the replacement-tail left-block completion. Let

$$\Sigma_{\text{tail}} = \text{rectRightGramOrderedTailSingularDiagonal}(A, k)$$

be the constructed complement-tail diagonal singular-value block. Then every exact rank-at-most- k competitor $B \in \mathbb{R}^{m \times n}$ satisfies

$$\|\Sigma_{\text{tail}}\|_F \leq \rho_F(A, B).$$

Equivalently, the visible LR.1dt tail-optimality hypothesis

$$\forall B, \quad \text{rank}(B) \leq k \implies \|\Sigma_{\text{tail}}\|_F \leq \rho_F(A, B)$$

is discharged for the constructed ordered replacement-tail source split.

The proof builds the head-first exact source factor on $\{1, \dots, k + q\}$. Its left block is orthonormal by the replacement-tail completion certificate, its right block is orthogonal after pulling rows back along Φ_{fin} , and the exact source factor equals

$$A_{ij}^{\Phi_{\text{fin}}} = A_{i, \Phi_{\text{fin}}(j)}.$$

The constructed head-tail square gap then feeds the q -dimensional source-factor lower-bound theorem, the tail diagonal norm is rewritten as the square root of the complement singular-square sum, and the Frobenius residual is transported back along Φ_{fin} .

Consequently, under the exact determinant hypothesis

$$\det(V_{\text{head}}^T Z) \neq 0,$$

the exact cross-term bound

$$\|\Sigma_{\text{tail}}(V_{\text{tail}}^T Z)(V_{\text{head}}^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_{\text{tail}}\|_F,$$

and the scalar comparison

$$2\sqrt{1 + \varepsilon^2} \|\Sigma_{\text{tail}}\|_F \leq \rho \|\Sigma_{\text{tail}}\|_F,$$

the exact Gram-inverse column-sketch projector satisfies the relative surface

$$\rho_F(A, P_{AZ}A) \leq \rho \rho_F(A, U_{\text{head}}\Sigma_{\text{head}}V_{\text{head}}^T),$$

without assuming a separate supplied tail-optimality hypothesis.

This is exact-object D_4 infrastructure for equation (9). It does not derive the cross-term radius from randomness, does not prove the scalar comparison, and does not certify computed non-probability SVD/singular-vector/projector/Gram/inverse/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```

rectRightGramOrderedHeadTailLeftFinBlock
rectRightGramOrderedHeadTailSigmaFin
rectRightGramOrderedHeadTailRightFinBlock
rectRightGramOrderedHeadTailRightOriginalFinBlock
rectRightGramOrderedHeadTailLeftFinBlock_col_orthonormal
rectRightGramOrderedHeadTailRightOriginalFinBlock_col_orthonormal
rectRightGramOrderedHeadTailRightOriginalFinBlock_row_orthonormal
rectRightGramOrderedHeadTailRightFinBlock_isOrthogonal
sourceSVDMatrix_rectRightGramOrderedHeadTailFinBlock_eq_reindexCols
frobNorm_rectRightGramOrderedTailSingularDiagonal_le_lowRankResidualFrob
exists_columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
rectRightGramOrdered_replacement_tail_left_of_last_pos_tailOptimal

```

Corollary 605 (Scalar-relative ordered tail-optimal surface). *Under the hypotheses of Corollary 592, the product-form scalar comparison*

$$2\sqrt{1 + \varepsilon^2} \|\Sigma_{\text{tail}}\|_F \leq \rho \|\Sigma_{\text{tail}}\|_F$$

may be replaced by the coefficient condition

$$2\sqrt{1 + \varepsilon^2} \leq \rho.$$

Indeed, for any nonnegative tail radius t ,

$$2(\sqrt{1 + \varepsilon^2} t) = (2\sqrt{1 + \varepsilon^2})t \leq \rho t.$$

Taking $t = \|\Sigma_{\text{tail}}\|_F$, which is nonnegative, gives the product-form comparison required by Corollary 592. Consequently, under

$$\det(V_{\text{head}}^T Z) \neq 0, \quad \|\Sigma_{\text{tail}}(V_{\text{tail}}^T Z)(V_{\text{head}}^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_{\text{tail}}\|_F, \quad 2\sqrt{1 + \varepsilon^2} \leq \rho,$$

the exact Gram-inverse column-sketch projector satisfies

$$\rho_F(A, P_{AZ}A) \leq \rho \rho_F(A, U_{\text{head}}\Sigma_{\text{head}}V_{\text{head}}^T),$$

with no supplied tail-optimality hypothesis and no raw product-form scalar comparison hypothesis.

This is exact-object D_4 theorem-surface cleanup for equation (9). It still does not derive the cross-term radius from randomness and does not certify computed non-probability SVD/singular-vector/projector/Gram/inverse/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
two_sqrt_one_add_sq_mul_tail_le_of_scalar
exists_columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
rectRightGramOrdered_replacement_tail_left_of_last_pos_tailOptimal_of_scalarRelative
```

Corollary 606 (Rank-nullity kernel vector for the equation (9) Eckart–Young route). *Let $B \in \mathbb{R}^{m \times (r+1)}$ have repository rank at most r , meaning that there are exact factors*

$$L \in \mathbb{R}^{m \times r}, \quad R \in \mathbb{R}^{r \times (r+1)}$$

with

$$B_{ij} = \sum_{a=1}^r L_{ia} R_{aj}.$$

Then there exists a nonzero vector $x \in \mathbb{R}^{r+1}$ such that

$$x \neq 0, \quad \sum_{j=1}^{r+1} B_{ij} x_j = 0 \quad \text{for every } i = 1, \dots, m.$$

Equivalently, every exact ‘RectRankAtMost $m(r+1)r$ ’ competitor has a nontrivial right-kernel vector on any displayed $r+1$ -coordinate right domain. The proof forms the exact linear map

$$x \longmapsto Rx : \mathbb{R}^{r+1} \rightarrow \mathbb{R}^r,$$

uses finite-dimensional rank-nullity to choose $x \neq 0$ with $Rx = 0$, and then pushes $Rx = 0$ through the factorization $B = LR$.

This is the first min–max/rank-nullity foundation for the Eckart–Young lower-bound route. It does not prove the singular-value lower bound, the Frobenius tail-optimality inequality, rectangular SVD existence, or any computed SVD/singular-vector/projector/Gram/sketch routine. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRankFactorization_exists_rightKernelVector_succ
rectRankAtMost_exists_rightKernelVector_succ
```

Corollary 607 (Kernel-to-residual lower bound for the equation (9) Eckart–Young route). *Let $A \in \mathbb{R}^{m \times (r+1)}$, let $\sigma \in \mathbb{R}$, and assume the exact vector-action lower bound*

$$\sigma \|x\|_2 \leq \|Ax\|_2 \quad \text{for every nonzero } x \in \mathbb{R}^{r+1}.$$

Let $B \in \mathbb{R}^{m \times (r+1)}$ have repository rank at most r . Then

$$\sigma \leq \|A - B\|_F.$$

Equivalently, the displayed source block has Frobenius distance at least σ from every exact rank- r competitor on the same $r+1$ right coordinates.

Indeed, the rank-nullity corollary gives a nonzero vector x with

$$Bx = 0.$$

Therefore

$$(A - B)x = Ax.$$

The exact Frobenius matrix-vector domination theorem gives

$$\|Ax\|_2 = \|(A - B)x\|_2 \leq \|A - B\|_F \|x\|_2.$$

Combining this with the lower-action hypothesis yields

$$\sigma \|x\|_2 \leq \|A - B\|_F \|x\|_2.$$

Since $x \neq 0$, $\|x\|_2 > 0$, and cancellation gives

$$\sigma \leq \|A - B\|_F.$$

This is an exact-object D4 min-max adapter. It closes the kernel-to-residual step after rank-nullity, but it does not prove the vector-action lower bound from an ordered singular-value block, does not prove the Frobenius Eckart–Young tail-optimality theorem, does not construct a rectangular SVD/source split, and does not certify computed SVD/singular-vector/projector/Gram/sketch routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectMatMulVec_sub_eq_left_of_rightKernel
rectRankAtMost_lowRankResidualFrob_ge_of_vector_lower_bound_succ
```

Corollary 608 (Diagonal source-block vector-action lower bound for equation (9)). *Let $U \in \mathbb{R}^{m \times q}$ have exact orthonormal columns,*

$$U^T U = I_q,$$

let $V \in \mathbb{R}^{q \times q}$ be exact orthogonal, and let $\Sigma \in \mathbb{R}^{q \times q}$ be diagonal:

$$\Sigma_{ab} = \begin{cases} \sigma_a, & a = b, \\ 0, & a \neq b. \end{cases}$$

Assume that $\sigma \geq 0$ and that every displayed diagonal entry satisfies

$$\sigma \leq \sigma_a.$$

Define the exact source block

$$A = U \Sigma V^T.$$

Then, for every $x \in \mathbb{R}^q$,

$$\sigma \|x\|_2 \leq \|Ax\|_2.$$

Indeed, set $y = V^T x$ and $z = \Sigma y$. Since V is orthogonal,

$$\|y\|_2 = \|x\|_2.$$

The diagonal hypothesis gives the squared lower bound

$$\|z\|_2^2 = \sum_a \sigma_a^2 y_a^2 \geq \sum_a \sigma^2 y_a^2 = \sigma^2 \|y\|_2^2.$$

Because $\sigma \geq 0$, this implies

$$\sigma \|y\|_2 \leq \|z\|_2.$$

Finally, exact left-column orthonormality gives

$$\|Uz\|_2 = \|z\|_2,$$

and $Ax = U\Sigma V^T x = Uz$. Combining the displayed identities yields

$$\sigma \|x\|_2 \leq \|Ax\|_2.$$

Thus an exact diagonal source block supplies the vector-action hypothesis used by the preceding min-max adapter. This corollary is exact-object spectral infrastructure: it does not instantiate the ordered singular-value monotonicity needed for the final Eckart–Young source block, does not prove the Frobenius tail-optimality theorem, does not construct a rectangular SVD/source split, and does not certify computed SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
vecNorm2Sq_leftOrthonormalFactor
vecNorm2Sq_diagonal_lower_bound
rectMatMulVec_sourceSVDFactorMatrix
sourceSVDFactorMatrix_diagonal_vector_action_lower_bound
```

Corollary 609 (Supplied square and thin SVD lower-action specializations for equation (9)). *Let $\sigma \geq 0$.*

Square case. Let $U, V \in \mathbb{R}^{q \times q}$ be exact orthogonal tables, and let σ_a be displayed diagonal entries with

$$\sigma \leq \sigma_a \quad \text{for every } a.$$

Define

$$\Sigma_{ab} = \begin{cases} \sigma_a, & a = b, \\ 0, & a \neq b, \end{cases} \quad A_{\text{sq}} = U\Sigma V^T.$$

Then for every $x \in \mathbb{R}^q$,

$$\sigma \|x\|_2 \leq \|A_{\text{sq}}x\|_2.$$

Thin-rectangular case. Let $U \in \mathbb{R}^{m \times q}$ have exact orthonormal columns, let $V \in \mathbb{R}^{q \times q}$ be exact orthogonal, and use the same diagonal Σ . Define

$$A_{\text{thin}} = U\Sigma V^T.$$

Then for every $x \in \mathbb{R}^q$,

$$\sigma \|x\|_2 \leq \|A_{\text{thin}}x\|_2.$$

Both conclusions are direct mathematical specializations of the previous diagonal source-block result. In the square case, the exact orthogonality of U supplies $U^T U = I_q$; in the thin case, this field is supplied as the explicit column-orthonormality hypothesis. The diagonal equality is the literal displayed matrix formula for Σ , and the lower-entry hypothesis is exactly $\sigma \leq \sigma_a$ for every displayed source coordinate.

This is an exact-object specialization for supplied SVD-style data. It does not prove existence of an SVD, does not prove singular-value ordering, does not prove Frobenius Eckart–Young tail optimality, and does not certify computed non-probability SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
squareSVD_diagonal_vector_action_lower_bound
rectangularThinSVD_diagonal_vector_action_lower_bound
```

Corollary 610 (Supplied diagonal SVD residual lower bound for equation (9)). *Let $\sigma \geq 0$.*

Square case. Let $U, V \in \mathbb{R}^{(r+1) \times (r+1)}$ be exact orthogonal tables, and let the displayed diagonal entries σ_a satisfy

$$\sigma \leq \sigma_a \quad \text{for every } a \in \{1, \dots, r+1\}.$$

With

$$\Sigma_{ab} = \begin{cases} \sigma_a, & a = b, \\ 0, & a \neq b, \end{cases} \quad A_{\text{sq}} = U \Sigma V^T,$$

every rank-at-most- r competitor $B \in \mathbb{R}^{(r+1) \times (r+1)}$ satisfies

$$\sigma \leq \|A_{\text{sq}} - B\|_F.$$

Thin-rectangular case. Let $U \in \mathbb{R}^{m \times (r+1)}$ have exact orthonormal columns, let $V \in \mathbb{R}^{(r+1) \times (r+1)}$ be exact orthogonal, and define $A_{\text{thin}} = U \Sigma V^T$. Then every rank-at-most- r competitor $B \in \mathbb{R}^{m \times (r+1)}$ satisfies

$$\sigma \leq \|A_{\text{thin}} - B\|_F.$$

The proof is the exact min–max composition. The preceding supplied-SVD lower-action result gives

$$\sigma \|x\|_2 \leq \|A_{\text{sq}} x\|_2 \quad \text{or} \quad \sigma \|x\|_2 \leq \|A_{\text{thin}} x\|_2$$

for every vector x . The rank-nullity residual adapter then chooses a nonzero right-kernel vector of any rank-at-most- r competitor B , rewrites $(A - B)x = Ax$, and uses Frobenius matrix-vector domination to conclude the displayed residual lower bound.

This is still an exact-object one-block lower-bound theorem. It does not prove the full Frobenius Eckart–Young tail-optimality theorem, singular-value ordering, SVD/source-split existence, randomness-derived certificates, or computed non-probability SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRankAtMost_lowRankResidualFrob_ge_of_squareSVD_diagonal_succ
rectRankAtMost_lowRankResidualFrob_ge_of_rectangularThinSVD_diagonal_succ
```

Corollary 611 (Ordered right-Gram head residual lower bound for equation (9)). *Let $A \in \mathbb{R}^{m \times n}$ and let $r+1 \leq n$. Let $\sigma_{r+1}^{\text{ord}}$ denote the last selected ordered right-Gram singular value in the constructed top- $r+1$ list, and assume*

$$\sigma_{r+1}^{\text{ord}} > 0.$$

Let $U_{\text{ord}} \in \mathbb{R}^{m \times (r+1)}$ be the constructed zero-safe ordered head-left table, and let

$$\Sigma_{\text{ord},ab} = \begin{cases} \tau_a, & a = b, \\ 0, & a \neq b, \end{cases}$$

where τ_a is the exact basis-indexed right-Gram singular value selected by the a -th ordered top embedding. Define the displayed coefficient block

$$H_{\text{coeff}} = U_{\text{ord}} \Sigma_{\text{ord}} I_{r+1}.$$

Then every repository rank-at-most- r competitor $B \in \mathbb{R}^{m \times (r+1)}$ satisfies

$$\sigma_{r+1}^{\text{ord}} \leq \|H_{\text{coeff}} - B\|_F.$$

The proof is an ordered instantiation of the previous one-block residual theorem. Positivity of $\sigma_{r+1}^{\text{ord}}$ gives exact column orthonormality of U_{ord} . The right block is the identity, hence orthogonal. Ordered singular-value monotonicity gives

$$\sigma_{r+1}^{\text{ord}} \leq \tau_a \quad \text{for every displayed head coordinate } a.$$

The thin-rectangular supplied-SVD residual lower-bound theorem therefore applies with $\sigma = \sigma_{r+1}^{\text{ord}}$.

This is an exact-object coefficient-block lower bound. It is not the full Frobenius Eckart–Young tail-optimality theorem, does not construct a full rectangular SVD/source split, does not derive randomness certificates, and does not certify computed non-probability SVD/singular-vector/projector/Gram/sketch or downstream product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

`rectRankAtMost_lowRankResidualFrob_ge_of_rectRightGramOrderedHeadDiagonal_succ`

Corollary 612 (Ordered one-step best-rank coefficient block for equation (9)). *Let $A \in \mathbb{R}^{m \times n}$, let $r+1 \leq n$, and keep the notation of the previous corollary. Assume again that the last selected ordered right-Gram singular value is positive:*

$$\sigma_{r+1}^{\text{ord}} > 0.$$

Let

$$\tau_a = \text{the exact basis-indexed right-Gram singular value selected by } a, \quad a \in \{1, \dots, r+1\},$$

and set

$$H_{\text{coeff}} = U_{\text{ord}} \text{diag}(\tau_1, \dots, \tau_{r+1}) I_{r+1}.$$

Let H_r be the supplied first- r truncation of this coefficient block,

$$H_r = U_{\text{ord}}(:, 1:r) \text{diag}(\tau_1, \dots, \tau_r) I_{r+1}(:, 1:r)^T.$$

Then H_r has rank at most r , and for every repository rank-at-most- r competitor $B \in \mathbb{R}^{m \times (r+1)}$,

$$\|H_{\text{coeff}} - H_r\|_F \leq \|H_{\text{coeff}} - B\|_F.$$

Equivalently, H_r is a Frobenius best rank- r approximant to the exact ordered top- $r+1$ coefficient block.

The one-entry tail identity used in the proof is

$$\|\text{diag}_{q=1}(\tau_{r+1})\|_F = \tau_{r+1} = \sigma_{r+1}^{\text{ord}}.$$

The first equality is the local diagonal Frobenius identity for a one-column tail, and the second equality is the ordered right-Gram embedding certificate. The preceding corollary gives the required tail-optimality inequality: every rank-at-most- r competitor has residual at least $\sigma_{r+1}^{\text{ord}}$. The local thin-rectangular supplied-SVD best-rank handoff then yields the displayed optimality statement.

This is an exact-object one-step Eckart–Young coefficient-block theorem. It does not prove the full multi-tail Frobenius theorem, rectangular SVD/source-split existence, randomness-derived certificates, or computed non-probability SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
frobNorm_squareSVDTailDiagonal_one
isBestRankApproxFrob_of_rectRightGramOrderedHeadDiagonal_succ
```

Corollary 613 (Multi-tail diagonal Frobenius identity for equation (9)). *Let $r, q \geq 0$ and let*

$$\tau : \{1, \dots, r + q\} \rightarrow \mathbb{R}$$

be the displayed exact singular-value table for a supplied square or thin-rectangular SVD-style split. Let

$$\Sigma_{\text{tail}} = \text{diag}(\tau_{r+1}, \dots, \tau_{r+q}) \in \mathbb{R}^{q \times q}.$$

Then

$$\|\Sigma_{\text{tail}}\|_F^2 = \sum_{c=1}^q \tau_{r+c}^2$$

and therefore

$$\|\Sigma_{\text{tail}}\|_F = \left(\sum_{c=1}^q \tau_{r+c}^2 \right)^{1/2}.$$

The proof is exact diagonal algebra: all off-diagonal entries of Σ_{tail} are zero, and the diagonal entries are precisely the tail singular values selected by the ‘Fin.natAdd’ tail embedding. This result supplies the displayed multi-tail norm used by the full Frobenius Eckart–Young route, but it does not prove the q -dimensional min-max/tail-optimality inequality itself. It also does not construct a rectangular SVD/source split, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
frobNormSq_squareSVDTailDiagonal_eq_sum
frobNorm_squareSVDTailDiagonal_eq_sqrt_sum
```

Corollary 614 (Q -dimensional right-kernel foundation for equation (9)). *Let $r, q \geq 0$, and let*

$$R : \{1, \dots, r\} \times \{1, \dots, r + q\} \rightarrow \mathbb{R}$$

be an exact right factor. Define the associated exact linear map

$$\mathcal{R}_R : \mathbb{R}^{r+q} \rightarrow \mathbb{R}^r, \quad (\mathcal{R}_R x)_a = \sum_{j=1}^{r+q} R_{aj} x_j.$$

Then the right kernel has dimension at least q :

$$q \leq \dim \ker(\mathcal{R}_R).$$

Indeed, rank-nullity gives

$$\dim \text{range}(\mathcal{R}_R) + \dim \ker(\mathcal{R}_R) = r + q,$$

while $\dim \text{range}(\mathcal{R}_R) \leq r$.

Now suppose an exact repository rank factorization of $B \in \mathbb{R}^{m \times (r+q)}$ is supplied:

$$B_{ij} = \sum_{a=1}^r L_{ia} R_{aj}.$$

If $x \in \ker(\mathcal{R}_R)$, then x is a right-kernel vector of the represented matrix:

$$\forall i, \quad \sum_{j=1}^{r+q} B_{ij} x_j = 0.$$

This follows by substituting the factorization and exchanging the two finite sums:

$$\sum_j B_{ij} x_j = \sum_a L_{ia} \left(\sum_j R_{aj} x_j \right) = 0.$$

This is exact-object q -dimensional rank-nullity infrastructure for the multi-tail min-max route. It does not select a q -dimensional witness family inside the kernel, prove the tail Frobenius lower bound, prove the full Eckart–Young theorem, construct rectangular SVD/source-split data, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRankRightFactorMap
rectRankRightFactorMap_ker_finrank_ge
rectRankFactorization_rightKernel_finrank_ge
rectRankFactorization_matrix_rightKernel_of_rightFactor_ker
```

Corollary 615 (Selected right-kernel family for equation (9)). *Let $B \in \mathbb{R}^{m \times (r+q)}$ be supplied with an exact repository rank factorization*

$$B_{ij} = \sum_{a=1}^r L_{ia} R_{aj}.$$

Let $\mathcal{R}_R : \mathbb{R}^{r+q} \rightarrow \mathbb{R}^r$ be the right-factor map from the preceding corollary. Then there exists a family

$$x_1, \dots, x_q \in \ker(\mathcal{R}_R)$$

that is linearly independent as a family in the kernel subtype, and every selected vector annihilates the represented competitor:

$$\forall c \in \{1, \dots, q\}, \forall i \in \{1, \dots, m\}, \quad \sum_{j=1}^{r+q} B_{ij}(x_c)_j = 0.$$

The proof uses the dimension lower bound

$$q \leq \dim \ker(\mathcal{R}_R)$$

from the preceding corollary and applies the finite-dimensional selection lemma for linearly independent families. Since each selected vector is a member of the right-factor kernel, the entrywise annihilation equation follows from the factorization bridge in the preceding corollary.

This is exact-object vector-selection infrastructure for the multi-tail min-max route. It does not prove the q -dimensional tail lower-bound argument, the full Frobenius Eckart–Young theorem, rectangular SVD/source-split construction, randomness-derived certificates, or computed non-probability SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

`rectRankFactorization_exists_rightKernelFamily`

Corollary 616 (Orthonormal right-kernel family for equation (9)). *Let $B \in \mathbb{R}^{m \times (r+q)}$ be supplied with an exact repository rank factorization*

$$B_{ij} = \sum_{a=1}^r L_{ia} R_{aj}.$$

Define the Euclidean-coordinate right-factor map

$$\mathcal{R}_R^E : \mathbb{R}_2^{r+q} \rightarrow \mathbb{R}^r, \quad (\mathcal{R}_R^E x)_a = \sum_{j=1}^{r+q} R_{aj} x_j.$$

Here $\mathbb{R}_2^{r+q}$ denotes the same coordinate vector space with the finite-dimensional inner-product structure used for orthonormal-basis selection. The coordinate formula is identical to the right-factor map in the preceding corollaries. Then

$$q \leq \dim \ker(\mathcal{R}_R^E),$$

and there exists an orthonormal family

$$x_1, \dots, x_q \in \ker(\mathcal{R}_R^E)$$

such that every selected vector annihilates the represented competitor:

$$\forall c \in \{1, \dots, q\}, \forall i \in \{1, \dots, m\}, \quad \sum_{j=1}^{r+q} B_{ij}(x_c)_j = 0.$$

The proof repeats the rank-nullity lower bound for $\mathcal{R}_R^E$, chooses the standard orthonormal basis of the kernel subtype, restricts it to the first q basis vectors using the dimension inequality, and uses injectivity of the restriction map to preserve orthonormality. Since each selected vector is in the Euclidean-coordinate right-factor kernel, substituting the exact factorization gives the same finite-sum annihilation identity as before:

$$\sum_j B_{ij}(x_c)_j = \sum_a L_{ia} \left(\sum_j R_{aj}(x_c)_j \right) = 0.$$

This is exact-object min-max-ready vector-selection infrastructure. It does not prove the Ky Fan/Courant–Fischer trace/Rayleigh lower bound, the full Frobenius Eckart–Young theorem, rectangular SVD/source-split construction, randomness-derived certificates, or computed non-probability SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRankRightFactorEuclideanMap
rectRankRightFactorEuclideanMap_ker_finrank_ge
rectRankFactorization_euclideanRightKernel_finrank_ge
rectRankFactorization_matrix_rightKernel_of_euclideanRightFactor_ker
rectRankFactorization_exists_orthonormalRightKernelFamily
```

Corollary 617 (Orthonormal right-kernel residual energy for equation (9)). *Let $M \in \mathbb{R}^{m \times n}$, and let*

$$x_1, \dots, x_q \in \mathbb{R}_2^n$$

be an exact orthonormal family. Then finite Bessel inequality, applied row by row, gives the Frobenius domination

$$\sum_{c=1}^q \|Mx_c\|_2^2 \leq \|M\|_F^2.$$

Equivalently, in the repository’s coordinate notation,

$$\sum_{c=1}^q \sum_{i=1}^m \left(\sum_{j=1}^n M_{ij}(x_c)_j \right)^2 \leq \sum_{i=1}^m \sum_{j=1}^n M_{ij}^2.$$

Consequently, let $A, B \in \mathbb{R}^{m \times n}$ and suppose the same orthonormal family lies in the exact right kernel of B :

$$\forall c, \forall i, \quad \sum_{j=1}^n B_{ij}(x_c)_j = 0.$$

Then $(A - B)x_c = Ax_c$ for every c , and hence

$$\sum_{c=1}^q \|Ax_c\|_2^2 \leq \|A - B\|_F^2.$$

In particular, if $B \in \mathbb{R}^{m \times (r+q)}$ is supplied with an exact repository rank factorization of inner dimension r , the orthonormal right-kernel family selected in the previous corollary satisfies

$$\sum_{c=1}^q \|Ax_c\|_2^2 \leq \|A - B\|_F^2.$$

This is the residual side of the q -dimensional trace/Rayleigh route toward the equation (9) tail lower bound. It does not prove the matching source-side tail-energy lower bound, the Ky Fan/Courant–Fischer comparison, the full Frobenius Eckart–Young theorem, rectangular SVD/source-split construction, randomness-derived certificates, or computed non-probability SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
sum_vecNorm2Sq_rectMatMulVec_le_frobNormSqRect_of_orthonormal
sum_vecNorm2Sq_rectMatMulVec_lowRankResidual_le_of_orthonormal_rightKernel
rectRankFactorization_exists_orthonormalRightKernelFamily_energy_le
```

Corollary 618 (Diagonal source-tail energy lower bound for equation (9)). *Let $r, q \geq 0$, let*

$$\tau : \{1, \dots, r+q\} \rightarrow \mathbb{R}, \quad D = \text{diag}(\tau_1, \dots, \tau_{r+q}),$$

and let $x_1, \dots, x_q \in \mathbb{R}_2^{r+q}$ be an exact orthonormal family. Suppose there is a visible gap parameter $\eta \in \mathbb{R}$ such that the displayed head and tail diagonal squares satisfy

$$\eta \leq \tau_a^2 \quad (1 \leq a \leq r), \quad \tau_{r+c}^2 \leq \eta \quad (1 \leq c \leq q).$$

Then the diagonal source action on the frame dominates the displayed tail energy:

$$\sum_{c=1}^q \tau_{r+c}^2 \leq \sum_{d=1}^q \|Dx_d\|_2^2.$$

Equivalently, writing

$$\alpha_j = \sum_{d=1}^q (x_d)_j^2,$$

the repository proves the expansion

$$\sum_{d=1}^q \|Dx_d\|_2^2 = \sum_{j=1}^{r+q} \tau_j^2 \alpha_j,$$

with coordinate-frame facts

$$0 \leq \alpha_j \leq 1, \quad \sum_{j=1}^{r+q} \alpha_j = q.$$

The scalar mass-transfer step is explicit: the nonnegative head coordinate mass equals the missing tail coordinate mass, and moving this mass from tail coordinates with square at most η to head coordinates with square at least η cannot decrease the weighted sum.

This is the source-side diagonal half of the q -dimensional trace/Rayleigh route. It does not yet instantiate the gap certificate from ordered singular values, transport the theorem through a non-trivial right singular-vector table, prove the full Ky Fan/Courant–Fischer theorem, close Frobenius Eckart–Young optimality, construct rectangular SVD/source-split data, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
headTail_weighted_tail_sum_le_of_gap
orthonormal_sum_coord_sq_le_one
orthonormal_sum_coord_sq_eq_card
sum_vecNorm2Sq_diagonal_rectMatMulVec_eq_weighted_coord_sq
sum_vecNorm2Sq_diagonal_rectMatMulVec_ge_tail_sq_of_orthonormal_gap
```

Corollary 619 (Ordered diagonal source-tail energy, positive-head case). *Let $r, q \geq 0$, assume $0 < r$, and let*

$$\tau : \{1, \dots, r + q\} \rightarrow \mathbb{R}, \quad D = \text{diag}(\tau_1, \dots, \tau_{r+q}).$$

Assume the displayed diagonal/singular-value squares are antitone in coordinate order:

$$i \leq j \implies \tau_j^2 \leq \tau_i^2 \quad (1 \leq i, j \leq r + q).$$

Let $x_1, \dots, x_q \in \mathbb{R}_2^{r+q}$ be an exact orthonormal family. Then the diagonal source action satisfies

$$\sum_{c=1}^q \tau_{r+c}^2 \leq \sum_{d=1}^q \|Dx_d\|_2^2.$$

The proof first constructs the visible gap parameter

$$\eta = \tau_r^2.$$

For every head coordinate $a \leq r$, antitonicity gives $\eta = \tau_r^2 \leq \tau_a^2$. For every tail coordinate $r + c$, antitonicity gives $\tau_{r+c}^2 \leq \tau_r^2 = \eta$. Thus the hypotheses of the preceding diagonal mass-transfer theorem hold, and the displayed tail-energy inequality follows.

This is an exact-object positive-head ordered-diagonal adapter. It removes the abstract gap parameter from the LR.1dk statement when the source supplies an ordered displayed diagonal table, but it does not cover the zero-head edge case, transport the result through a nontrivial right singular-vector table, prove the full Rayleigh/Ky Fan or Eckart–Young theorem, construct rectangular SVD/source-split data, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
diagonal_headTail_square_gap_of_antitone_head_pos
sum_vecNorm2Sq_diagonal_rectMatMulVec_ge_tail_sq_of_orthonormal_antitone_head_pos
```

Corollary 620 (Zero-head diagonal source-tail energy). *Let $q \geq 0$, let*

$$\tau : \{1, \dots, q\} \rightarrow \mathbb{R}, \quad D = \text{diag}(\tau_1, \dots, \tau_q),$$

and let $x_1, \dots, x_q \in \mathbb{R}_2^q$ be an exact orthonormal family. Then every coordinate has full squared mass across the frame:

$$\sum_{d=1}^q (x_d)_j^2 = 1 \quad (1 \leq j \leq q).$$

Consequently the diagonal source action is exactly the displayed zero-head tail energy:

$$\sum_{d=1}^q \|Dx_d\|_2^2 = \sum_{j=1}^q \tau_j^2,$$

and therefore

$$\sum_{j=1}^q \tau_j^2 \leq \sum_{d=1}^q \|Dx_d\|_2^2.$$

The proof combines the Bessel coordinate bound $\sum_d (x_d)_j^2 \leq 1$ with the total-mass identity $\sum_j \sum_d (x_d)_j^2 = q$. The nonnegative deficits $1 - \sum_d (x_d)_j^2$ sum to zero, so each deficit is zero. The diagonal energy expansion from the preceding source-tail theorem then gives the displayed equality.

This is the exact-object $r = 0$ companion to the positive-head ordered-gap adapter. It removes the zero-head exception from the diagonal source-side layer, but it does not transport the result through a nontrivial right singular-vector table, prove the full Rayleigh/Ky Fan or Eckart–Young theorem, construct rectangular SVD/source-split data, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
orthonormal_sum_coord_sq_eq_one_of_card_eq
orthonormal_sum_coord_sq_eq_one_of_full
sum_vecNorm2Sq_diagonal_rectMatMulVec_eq_sum_sq_of_orthonormal_full
sum_vecNorm2Sq_diagonal_rectMatMulVec_ge_tail_sq_of_orthonormal_zero_head
```

Corollary 621 (Combined ordered diagonal source-tail energy). *Let $r, q \geq 0$, let*

$$\tau : \{1, \dots, r + q\} \rightarrow \mathbb{R}, \quad D = \text{diag}(\tau_1, \dots, \tau_{r+q}),$$

and assume the displayed diagonal/singular-value squares are antitone:

$$i \leq j \implies \tau_j^2 \leq \tau_i^2 \quad (1 \leq i, j \leq r + q).$$

For every exact orthonormal family

$$x_1, \dots, x_q \in \mathbb{R}_2^{r+q},$$

the source-side diagonal lower bound holds:

$$\sum_{c=1}^q \tau_{r+c}^2 \leq \sum_{d=1}^q \|Dx_d\|_2^2.$$

The proof splits on the head count. If $0 < r$, the last head square τ_r^2 supplies the visible gap used in the positive-head ordered corollary. If $r = 0$, the frame is a full orthonormal frame in the whole q -dimensional coordinate space, so the zero-head equality above applies.

This is the closed exact-object ordered-diagonal source-side layer. It still does not transport the result through a nontrivial right singular-vector table, prove the full Rayleigh/Ky Fan or Eckart–Young theorem, construct rectangular SVD/source-split data, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

`sum_vecNorm2Sq_diagonal_rectMatMulVec_ge_tail_sq_of_orthonormal_antitone`

Corollary 622 (Source-factor ordered source-tail energy). *Let $r, q \geq 0$, let*

$$U \in \mathbb{R}^{m \times (r+q)}, \quad V \in \mathbb{R}^{(r+q) \times (r+q)}, \quad D = \text{diag}(\tau_1, \dots, \tau_{r+q}),$$

and assume

$$U^T U = I_{r+q}, \quad V^T V = V V^T = I_{r+q}.$$

Assume the displayed diagonal/singular-value squares are antitone:

$$i \leq j \implies \tau_j^2 \leq \tau_i^2 \quad (1 \leq i, j \leq r+q).$$

Then for every exact orthonormal family

$$x_1, \dots, x_q \in \mathbb{R}_2^{r+q},$$

the transported source-factor lower bound holds:

$$\sum_{c=1}^q \tau_{r+c}^2 \leq \sum_{d=1}^q \|U D V^T x_d\|_2^2.$$

Indeed, set $y_d = V^T x_d$. Exact right orthogonality of V gives

$$\langle y_c, y_d \rangle = \langle x_c, x_d \rangle,$$

so $y_1, \dots, y_q$ is again an exact orthonormal family. Exact left column-orthonormality of U gives, for each d ,

$$\|U D V^T x_d\|_2^2 = \|D y_d\|_2^2.$$

The result is therefore exactly the ordered diagonal source-tail theorem above applied to the transported frame y_d .

This is the closed exact-object right-basis transport layer. It still does not construct a rectangular SVD/source split, prove the full Rayleigh/Ky Fan or Eckart–Young theorem, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

`inner_matTranspose_mulVec_eq_of_isOrthogonal`

`orthonormal_matTranspose_mulVec_of_isOrthogonal`

`vecNorm2Sq_sourceSVDFactorMatrix_eq_diagonal_transpose_action`

`sum_vecNorm2Sq_sourceSVDFactorMatrix_ge_tail_sq_of_orthonormal_antitone`

Corollary 623 (Source-factor gap lower-bound bridge for equation (9)). *Let $r, q \geq 0$, let*

$$U \in \mathbb{R}^{m \times (r+q)}, \quad V \in \mathbb{R}^{(r+q) \times (r+q)}, \quad D = \text{diag}(\tau_1, \dots, \tau_{r+q}),$$

and assume

$$U^T U = I_{r+q}, \quad V^T V = V V^T = I_{r+q}.$$

Suppose there exists a visible separator $\eta \in \mathbb{R}$ such that

$$\eta \leq \tau_a^2 \quad (1 \leq a \leq r), \quad \tau_{r+c}^2 \leq \eta \quad (1 \leq c \leq q).$$

Then every exact orthonormal family

$$x_1, \dots, x_q \in \mathbb{R}_2^{r+q}$$

satisfies the source-factor gap lower bound

$$\sum_{c=1}^q \tau_{r+c}^2 \leq \sum_{d=1}^q \|U D V^T x_d\|_2^2.$$

Moreover, set $A = U D V^T$. For every exact rank-at-most- r competitor $B \in \mathbb{R}^{m \times (r+q)}$, the squared residual lower bound holds:

$$\sum_{c=1}^q \tau_{r+c}^2 \leq \|A - B\|_F^2,$$

and therefore the norm form holds:

$$\sqrt{\sum_{c=1}^q \tau_{r+c}^2} \leq \|A - B\|_F.$$

The proof of the first inequality sets $y_d = V^T x_d$. Exact right orthogonality of V makes $y_1, \dots, y_q$ an exact orthonormal family, and exact left column-orthonormality of U gives

$$\|U D V^T x_d\|_2^2 = \|D y_d\|_2^2.$$

The diagonal gap theorem then applies with the displayed separator η . For the residual statement, the exact rank factorization of B supplies a q -element orthonormal family in the right kernel of B . The residual energy theorem gives

$$\sum_{d=1}^q \|A x_d\|_2^2 \leq \|A - B\|_F^2,$$

and the source-factor gap lower bound gives the matching lower inequality for the same family. Chaining the inequalities proves the squared statement, and monotonicity of the square root gives the norm statement.

This is the exact-object form needed for constructed ordered source splits whose complement-tail enumeration is not itself sorted by singular value. It does not instantiate the separator η from the constructed right-Gram top set, build the original-column reindexing/equivalence transport, prove the remaining LR.1dt tail-optimality discharge, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch or product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
sum_vecNorm2Sq_sourceSVDFactorMatrix_ge_tail_sq_of_orthonormal_gap
rectRankAtMost_lowRankResidualFrob_sq_ge_tail_sum_of_sourceSVDFactorMatrix_gap
sqrt_tail_sum_le_lowRankResidualFrob_of_sourceSVDFactorMatrix_gap
```

Corollary 624 (Q-dimensional source-factor lower bound for Eckart–Young). *Use the hypotheses and notation of the source-factor ordered source-tail corollary above, including exact singular-square antitonicity, and set*

$$A = UDV^T.$$

Let $B \in \mathbb{R}^{m \times (r+q)}$ be any exact rank-at-most- r competitor. Then the squared residual lower bound holds:

$$\sum_{c=1}^q \tau_{r+c}^2 \leq \|A - B\|_F^2.$$

Equivalently,

$$\sqrt{\sum_{c=1}^q \tau_{r+c}^2} \leq \|A - B\|_F.$$

The proof selects, from the exact rank factorization of B , a q -element orthonormal family $x_1, \dots, x_q$ in the right kernel of B . The residual-energy theorem gives

$$\sum_{d=1}^q \|Ax_d\|_2^2 \leq \|A - B\|_F^2,$$

because $Bx_d = 0$. The source-factor ordered source-tail theorem gives the matching lower bound

$$\sum_{c=1}^q \tau_{r+c}^2 \leq \sum_{d=1}^q \|Ax_d\|_2^2.$$

Chaining the two inequalities proves the squared statement; monotonicity of the square root and nonnegativity of the Frobenius norm give the norm form.

This is an exact-object lower-bound bridge for supplied ordered source-factor data. It still does not prove that a source head realizes the matching tail norm, construct a rectangular SVD/source split, derive randomness certificates, or certify computed non-probability SVD/singular-vector/projector/Gram/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
rectRankAtMost_lowRankResidualFrob_sq_ge_tail_sum_of_sourceSVDFactorMatrix_antitone
sqrt_tail_sum_le_lowRankResidualFrob_of_sourceSVDFactorMatrix_antitone
```

Corollary 625 (Ordered supplied-SVD best-rank adapter for equation (9)). *Let $r, q \in \mathbb{N}$, let $\tau : \{1, \dots, r + q\} \rightarrow \mathbb{R}$, and assume the displayed singular-square table is antitone:*

$$i \leq j \implies \tau_j^2 \leq \tau_i^2.$$

For the square case, suppose $U, V \in \mathbb{R}^{(r+q) \times (r+q)}$ are exact orthogonal tables and

$$A_{ij} = \sum_{k=1}^{r+q} U_{ik} \tau_k V_{jk}.$$

Then every exact rank-at-most- r competitor B satisfies

$$\|D_{\text{tail}}\|_F = \sqrt{\sum_{c=1}^q \tau_{r+c}^2} \leq \|A - B\|_F,$$

where $D_{\text{tail}} = \text{diag}(\tau_{r+1}, \dots, \tau_{r+q})$. Consequently, if the displayed head singular entries are nonzero, the exact source head $U_{\text{head}} D_{\text{head}} V_{\text{head}}^T$ is a Frobenius best rank- r approximation to A . A strict-positive head version follows by deriving the nonzero-head field from positivity.

For the thin rectangular case, suppose $U \in \mathbb{R}^{m \times (r+q)}$ has exact orthonormal columns, V is exact orthogonal, and the same representation holds with $i = 1, \dots, m$. Then the same tail optimality inequality holds for every exact rank-at-most- r competitor $B \in \mathbb{R}^{m \times (r+q)}$, and the displayed thin source head is a Frobenius best rank- r approximation whenever the head entries are nonzero.

The proof identifies the paper-style diagonal SVD sum with the repository source-factor object $U \text{diag}(\tau) V^T$. The preceding q -dimensional lower-bound theorem supplies the residual lower bound, and the tail-diagonal identity rewrites $\|D_{\text{tail}}\|_F = \sqrt{\sum_c \tau_{r+c}^2}$. The existing supplied square/thin SVD best-rank constructors then consume this tail-optimality inequality.

This is an exact-object, exact-law result for supplied ordered SVD-style data. It does not prove existence of such an SVD, does not derive the ordering from a computed spectral routine, does not derive randomness certificates, and does not certify computed non-probability SVD/singular-vector/projector/Gram/sketch or product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
sourceSVDfactorMatrix_diagonal_eq_sum
squareSVD_sigmaTail_le_lowRankResidualFrob_of_antitone
isBestRankApproxFrob_of_squareSVD_antitone
isBestRankApproxFrob_of_squareSVD_head_pos_antitone
rectangularThinSVD_sigmaTail_le_lowRankResidualFrob_of_antitone
isBestRankApproxFrob_of_rectangularThinSVD_antitone
isBestRankApproxFrob_of_rectangularThinSVD_head_pos_antitone
```

Corollary 626 (Ordered supplied-SVD relative surfaces for equation (9)). *Use the square or thin rectangular hypotheses of the preceding supplied-SVD corollary, including exact singular-square antitonicity. Let Z be the exact sketching matrix and assume the exact full-rank source condition*

$$\det(V_{\text{head}}^T Z) \neq 0.$$

Assume also the exact cross-term certificate

$$\|\Sigma_{\text{tail}}(V_{\text{tail}}^T Z)(V_{\text{head}}^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_{\text{tail}}\|_F, \quad \varepsilon \geq 0,$$

and the visible scalar comparison

$$2\sqrt{1 + \varepsilon^2} \|\Sigma_{\text{tail}}\|_F \leq \rho \|\Sigma_{\text{tail}}\|_F.$$

Then the exact Gram-inverse column-sketch projector

$$P_{AZ} = AZ((AZ)^T(AZ))^{-1}(AZ)^T$$

satisfies the relative equation-(9) surface

$$\|A - P_{AZ}A\|_F \leq \rho \|A - A_r\|_F,$$

where A_r is the displayed square or thin supplied-SVD source head. The theorem also returns the exact best-rank certificate for A_r , the source-head residual identity

$$\|A - A_r\|_F = \|\Sigma_{\text{tail}}\|_F,$$

the exact projector reproduction fields, and the rank-at-most- r conclusion for $P_{AZ}A$.

The only change from the earlier sigma-tail relative surfaces is that the raw tail-optimality hypothesis is no longer assumed. It is supplied by the preceding ordered supplied-SVD best-rank adapter. The determinant, cross-term, and scalar comparison assumptions remain visible exact-object hypotheses.

This is an exact-law/exact-object theorem surface. It does not prove SVD existence, derive the ordering or cross-term certificate from randomness, or certify computed non-probability SVD/singular-vector/projector/Gram/inverse/sketch/product routines. Sampling probabilities and sampling laws remain exact mathematical inputs by the current convention.

Lean formalization.

```
columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
squareSVD_sigmaTail_antitone

columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
squareSVD_head_pos_sigmaTail_antitone

columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
rectangularThinSVD_sigmaTail_antitone

columnSketchGramInverseProjector_sourceHeadTail_sourceSVDTailRelativeResidualSurface_of_
rectangularThinSVD_head_pos_sigmaTail_antitone
```

Corollary 627 (Source-tail Frobenius norm reduction for equation (9)). *Let $U_{\perp} \in \mathbb{R}^{m \times q}$ have exact orthonormal columns,*

$$U_{\perp}^T U_{\perp} = I_q,$$

and let $C \in \mathbb{R}^{q \times n}$. Then

$$\|U_{\perp}C\|_F = \|C\|_F.$$

Equivalently, the squared Frobenius norms agree:

$$\|U_{\perp}C\|_F^2 = \|C\|_F^2.$$

Now suppose the exact source tail factors as

$$T = U_{\perp}T_c,$$

where $T_c \in \mathbb{R}^{q \times n}$. For the exact source coefficient table

$$W = (V^T Z)^{-1} V^T,$$

the source residual tail satisfies the exact factorization

$$T - (TZ)W = U_\perp (T_c - (T_c Z)W).$$

Consequently,

$$\|T - (TZ)(V^T Z)^{-1} V^T\|_F = \|T_c - (T_c Z)(V^T Z)^{-1} V^T\|_F.$$

This is an exact-object D2 reduction theorem. It moves the source-tail norm obligation from the ambient residual tail to the coordinate residual tail. It does not yet prove the coordinate residual bound from the $\Sigma_\perp$, $V_\perp^T Z$, and $V_k^T Z$ factors, does not prove unitarily invariant or spectral-norm comparison infrastructure, and does not certify computed SVD/cross-product/inverse/coefficient/projector/product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
frobNormSqRect_leftOrthonormalFactor
frobNormRect_leftOrthonormalFactor
sourceSketchResidualTail_leftFactor
frobNormSqRect_sourceSketchResidualTail_leftOrthonormalFactor
frobNormRect_sourceSketchResidualTail_leftOrthonormalFactor
```

Corollary 628 (Coordinate-tail residual factorization for equation (9)). Let $V_k \in \mathbb{R}^{n \times r}$, $V_\perp \in \mathbb{R}^{n \times q}$, and $Z \in \mathbb{R}^{n \times r}$. Set the exact source coefficient table

$$W = (V_k^T Z)^{-1} V_k^T$$

and the exact rectangular cross factor

$$M_\perp = V_\perp^T Z.$$

For the exact coordinate right-tail residual,

$$R_\perp = V_\perp^T - (V_\perp^T Z)(V_k^T Z)^{-1} V_k^T = V_\perp^T - M_\perp W,$$

the formal entrywise statement is

$$R_\perp(a, j) = (V_\perp)_{j,a} - \sum_{b=1}^r (M_\perp)_{a,b} W_{b,j}.$$

Now let $\Sigma_\perp \in \mathbb{R}^{q \times q}$ and define the coordinate tail

$$T_c = \Sigma_\perp V_\perp^T.$$

Then the exact residual induced by the paper's coefficient table factors as

$$T_c - (T_c Z)(V_k^T Z)^{-1} V_k^T = \Sigma_\perp (V_\perp^T - (V_\perp^T Z)(V_k^T Z)^{-1} V_k^T).$$

Consequently the repository proves the visible Frobenius bound

$$\|T_c - (T_c Z)(V_k^T Z)^{-1} V_k^T\|_F \leq \|\Sigma_\perp\|_F \|V_\perp^T - (V_\perp^T Z)(V_k^T Z)^{-1} V_k^T\|_F.$$

This is an exact-object D2 factorization theorem. It exposes the $\Sigma_\perp$, $V_\perp^T Z$, and $V_k^T Z$ factors in the CACM equation (9) residual. The displayed Frobenius bound is deliberately non-sharp: it uses rectangular Frobenius submultiplicativity through $\|\Sigma_\perp\|_F$. The sharp spectral/unitarily invariant source-tail bound, the right-orthogonal block invariance step, Eckart–Young, and computed SVD/cross-product/inverse/coefficient/projector/product certificates remain open. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
sourceRightBasisTranspose
rightSketchCrossGramRect
columnSketch_sourceRightBasisTranspose
columnSketchHead_sourceRightBasisTranspose
sourceSketchResidualTail_sourceRightBasisTranspose
sourceSketchResidualTail_leftSquareFactor
frobNormRect_sourceSketchResidualTail_leftSquareFactor_le
sourceSketchResidualTail_sigmaRightBasisTranspose_explicit
frobNormRect_sourceSketchResidualTail_sigmaRightBasisTranspose_le
```

Corollary 629 (Right-tail residual block products for equation (9)). *Let $V_k \in \mathbb{R}^{n \times r}$, $V_\perp \in \mathbb{R}^{n \times q}$, and $Z \in \mathbb{R}^{n \times r}$. Set the exact source coefficient table*

$$W = (V_k^T Z)^{-1} V_k^T$$

and the exact inverse cross factor

$$M_\perp^\# = (V_\perp^T Z)(V_k^T Z)^{-1}.$$

Assume the exact right-basis hypotheses

$$V_\perp^T V_\perp = I_q, \quad V_k^T V_\perp = 0, \quad V_\perp^T V_k = 0, \quad V_k^T V_k = I_r.$$

Then the coefficient table has the two exact block actions

$$W V_\perp = 0, \quad W V_k = (V_k^T Z)^{-1}.$$

For the exact right-tail residual

$$R_\perp = V_\perp^T - (V_\perp^T Z)W,$$

the repository proves the block products

$$R_\perp V_\perp = I_q, \quad R_\perp V_k = -M_\perp^\# = -(V_\perp^T Z)(V_k^T Z)^{-1}.$$

Equivalently, entrywise,

$$\sum_j R_\perp(a, j)(V_\perp)_{j,c} = \delta_{ac}, \quad \sum_j R_\perp(a, j)(V_k)_{j,c} = -(M_\perp^\#)_{ac}.$$

This is an exact-object D2 block-product theorem. It closes the displayed right-orthogonal algebra needed after the coordinate-tail factorization: $R_\perp$ is identity on the tail right-basis block and the negative inverse cross factor on the head right-basis block. It still does not prove the sharp Frobenius/unitarily invariant norm identity or bound that turns these block products into the CACM equation (9) source-tail rate. It also does not prove rectangular SVD/Eckart–Young foundations or implementation-facing certificates for computed SVD/singular vectors/cross-products/inverses/coefficient tables/products. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
rightSketchCrossGramRectInvFactor
sourceSketchCoefficient_mul_rightTailBasis_of_cross_zero
sourceSketchCoefficient_mul_headRightBasis_of_orthonormal
sourceRightResidual_mul_rightTailBasis_eq_id
sourceRightResidual_mul_headRightBasis_eq_neg_invFactor
```

Corollary 630 (Right-tail Frobenius block identity for equation (9)). *Let $V_k \in \mathbb{R}^{n \times r}$, $V_\perp \in \mathbb{R}^{n \times q}$, and $Z \in \mathbb{R}^{n \times r}$. Let*

$$R_\perp = V_\perp^T - (V_\perp^T Z)(V_k^T Z)^{-1}V_k^T, \quad M_\perp^\# = (V_\perp^T Z)(V_k^T Z)^{-1}.$$

Assume the exact right-basis hypotheses of the preceding corollary and the row-completeness identity for the concatenated right-basis block

$$[V_\perp, V_k][V_\perp, V_k]^T = I_n,$$

equivalently

$$\sum_c (V_\perp)_{j,c} (V_\perp)_{\ell,c} + \sum_b (V_k)_{j,b} (V_k)_{\ell,b} = \delta_{j\ell} \quad \text{for all } j, \ell.$$

Then for every exact square tail singular-value block $\Sigma_\perp \in \mathbb{R}^{q \times q}$,

$$\|\Sigma_\perp R_\perp\|_F^2 = \|\Sigma_\perp\|_F^2 + \|\Sigma_\perp M_\perp^\#\|_F^2.$$

Equivalently,

$$\|\Sigma_\perp R_\perp\|_F = \sqrt{\|\Sigma_\perp\|_F^2 + \|\Sigma_\perp (V_\perp^T Z)(V_k^T Z)^{-1}\|_F^2}.$$

The formal proof introduces the sum-indexed right-basis block $[V_\perp, V_k]$. Right multiplication by this exact row-complete block preserves squared Frobenius norm, and the preceding block-product theorem gives

$$R_\perp [V_\perp, V_k] = [I_q, -M_\perp^\#].$$

After left multiplication by $\Sigma_\perp$, the squared Frobenius norm of $[\Sigma_\perp, -\Sigma_\perp M_\perp^\#]$ splits into the two displayed summands.

This is an exact-object D2 Frobenius identity theorem. It closes the right-orthogonal Frobenius invariance step after the block-product corollary. It still does not prove the sharp spectral/unitarily invariant bound on $\Sigma_\perp (V_\perp^T Z)(V_k^T Z)^{-1}$, rectangular SVD/Eckart–Young foundations, or implementation-facing certificates for computed SVD/singular vectors/cross-products/inverses/coefficient tables/products. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```

finiteFrobNormSq_fin
finiteFrobNormSq_neg
finiteFrobNormSq_sumColumns
finiteFrobNormSq_rectRightOrthonormal
rightBasisBlock
rightBasisBlock_row_orthonormal_of_sum
sourceRightResidualBlock
sigmaRightResidualBlock
finiteFrobNormSq_sigmaRightResidualBlock
sourceRightResidual_rightBasisBlock_eq_block
sourceRightResidual_sigma_rightBasisBlock_eq_block
frobNormSqRect_sigma_sourceRightResidual_eq_block
frobNormRect_sigma_sourceRightResidual_eq_sqrt_block

```

Corollary 631 (Right-basis block orthonormality certificate for equation (9)). *Let the exact concatenated right-basis block be*

$$Q = [V_\perp, V_k], \quad Q_{j,\text{inl}(c)} = (V_\perp)_{j,c}, \quad Q_{j,\text{inr}(b)} = (V_k)_{j,b}.$$

Assume exact column orthonormality of this block:

$$\sum_j Q_{j,\mu} Q_{j,\nu} = \begin{cases} 1, & \mu = \nu, \\ 0, & \mu \neq \nu. \end{cases}$$

Then the component right-basis fields used by the source-tail residual proof follow:

$$V_\perp^T V_\perp = I_q, \quad V_k^T V_\perp = 0, \quad V_\perp^T V_k = 0, \quad V_k^T V_k = I_r.$$

Assume also exact row orthonormality of the same block,

$$\sum_\mu Q_{j,\mu} Q_{\ell,\mu} = \delta_{j\ell}.$$

Equivalently,

$$\sum_c (V_\perp)_{j,c} (V_\perp)_{\ell,c} + \sum_b (V_k)_{j,b} (V_k)_{\ell,b} = \delta_{j\ell}.$$

Consequently, if $\varepsilon \geq 0$ and the exact source cross-term radius satisfies

$$\|\Sigma_\perp (V_\perp^T Z) (V_k^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_\perp\|_F,$$

then

$$\|\Sigma_\perp (V_\perp^T - (V_\perp^T Z) (V_k^T Z)^{-1} V_k^T)\|_F \leq \sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F.$$

This is an exact-object D3-to-D2 theorem. It reduces the several right-basis component hypotheses in the source cross-term proof to one column/row orthonormality certificate for the exact block $[V_\perp, V_k]$. It still does not construct that block from a rectangular SVD, prove singular-value ordering or Eckart–Young optimality, derive the cross-term radius from randomness, or certify computed SVD/singular-vector/cross-product/inverse/product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```

rightBasisBlock_component_orthonormal_fields_of_col_orthonormal
rightBasisBlock_complete_sum_of_row_orthonormal
frobNormRect_sigma_sourceRightResidual_le_sqrt_one_add_eps_sq_of_rightBasisBlock_orthonormal

```

Corollary 632 (Right-basis block assembly from component fields for equation (9)). *Let $V_k \in \mathbb{R}^{n \times r}$ and $V_\perp \in \mathbb{R}^{n \times q}$, and define the exact concatenated block*

$$Q = [V_\perp, V_k], \quad Q_{j,\text{inl}(c)} = (V_\perp)_{j,c}, \quad Q_{j,\text{inr}(b)} = (V_k)_{j,b}.$$

Assume the exact component right-basis fields

$$V_\perp^T V_\perp = I_q, \quad V_k^T V_\perp = 0, \quad V_\perp^T V_k = 0, \quad V_k^T V_k = I_r.$$

Then Q is column orthonormal:

$$\sum_j Q_{j,\mu} Q_{j,\nu} = \begin{cases} 1, & \mu = \nu, \\ 0, & \mu \neq \nu. \end{cases}$$

If, in addition, the component row-completeness identity holds,

$$\sum_c (V_\perp)_{j,c} (V_\perp)_{\ell,c} + \sum_b (V_k)_{j,b} (V_k)_{\ell,b} = \delta_{j\ell} \quad \text{for all } j, \ell,$$

then Q is row orthonormal:

$$\sum_\mu Q_{j,\mu} Q_{\ell,\mu} = \delta_{j\ell}.$$

Consequently, if $Z \in \mathbb{R}^{n \times r}$, $\varepsilon \geq 0$, and the exact source cross-term radius satisfies

$$\|\Sigma_\perp (V_\perp^T Z) (V_k^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_\perp\|_F,$$

then the assembled block certificate gives

$$\|\Sigma_\perp (V_\perp^T - (V_\perp^T Z) (V_k^T Z)^{-1} V_k^T)\|_F \leq \sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F.$$

This is an exact-object D3 source-SVD assembly theorem. It proves the direction a rectangular SVD construction would naturally supply: separate component orthogonality and row-completeness fields assemble the block orthonormality certificate used by the preceding corollary. It still does not construct $V_\perp, V_k$ or $\Sigma_\perp$ from a rectangular SVD, prove singular-value ordering or Eckart–Young optimality, derive the cross-term radius from randomness, or certify computed SVD/singular-vector/cross-product/inverse/product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```

rightBasisBlock_col_orthonormal_of_component_orthonormal_fields
rightBasisBlock_col_row_orthonormal_of_component_fields
frobNormRect_sigma_sourceRightResidual_le_sqrt_one_add_eps_sq_of_component_block_assembly

```

Corollary 633 (Source cross-term certificate for equation (9)). *Keep the exact hypotheses and notation of the preceding corollary. Let $\varepsilon \geq 0$, and assume the Frobenius cross-term condition appearing in the CACM structural theorem:*

$$\|\Sigma_{\perp}(V_{\perp}^T Z)(V_k^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_{\perp}\|_F.$$

Then the exact right-tail residual satisfies the source-tail rate

$$\|\Sigma_{\perp}(V_{\perp}^T - (V_{\perp}^T Z)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + \varepsilon^2} \|\Sigma_{\perp}\|_F.$$

Equivalently, with

$$R_{\perp} = V_{\perp}^T - (V_{\perp}^T Z)(V_k^T Z)^{-1}V_k^T, \quad M_{\perp}^{\#} = (V_{\perp}^T Z)(V_k^T Z)^{-1},$$

the exact implication is

$$\|\Sigma_{\perp}M_{\perp}^{\#}\|_F \leq \varepsilon \|\Sigma_{\perp}\|_F \implies \|\Sigma_{\perp}R_{\perp}\|_F \leq \sqrt{1 + \varepsilon^2} \|\Sigma_{\perp}\|_F.$$

The proof combines the squared block identity above

$$\|\Sigma_{\perp}R_{\perp}\|_F^2 = \|\Sigma_{\perp}\|_F^2 + \|\Sigma_{\perp}M_{\perp}^{\#}\|_F^2$$

with the squared cross-term bound and monotonicity of the square root. This is an exact-object structural-certificate theorem: the cross-term inequality is a visible hypothesis from the source equation (9), not a consequence of the full-rank condition by itself. It does not prove the spectral-norm or general unitarily invariant norm variants, rectangular SVD/Eckart–Young foundations, or implementation-facing certificates for computed SVD/singular vectors/ cross-products/inverses/coefficient tables/products. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

`frobNormRect_sigma_sourceRightResidual_le_sqrt_one_add_eps_sq`

Corollary 634 (Ambient source-tail certificate for equation (9)). *Let $T \in \mathbb{R}^{m \times n}$ be an exact source-tail matrix, and assume that it has the exact source-tail factorization*

$$T = U_{\perp}\Sigma_{\perp}V_{\perp}^T, \quad U_{\perp}^T U_{\perp} = I_q.$$

Keep the exact right-basis hypotheses, row-completeness identity, and cross-term certificate of the preceding corollary:

$$\varepsilon \geq 0, \quad \|\Sigma_{\perp}(V_{\perp}^T Z)(V_k^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_{\perp}\|_F.$$

Let $W = (V_k^T Z)^{-1}V_k^T$. Then the exact ambient source-tail residual satisfies

$$\|T - (TZ)W\|_F = \|T - (TZ)(V_k^T Z)^{-1}V_k^T\|_F \leq \sqrt{1 + \varepsilon^2} \|\Sigma_{\perp}\|_F.$$

Equivalently, this result composes the three exact reductions

$$\|T - (TZ)W\|_F = \|\Sigma_{\perp}(V_{\perp}^T - (V_{\perp}^T Z)W)\|_F, \quad W = (V_k^T Z)^{-1}V_k^T,$$

with the displayed CACM cross-term certificate. It is an exact-object source-tail theorem: the left and right singular-vector blocks, the singular-value block, the inverse factor, and the products are supplied as exact mathematical objects. It does not construct a rectangular SVD/source split, prove Eckart–Young, prove the spectral-norm or general unitarily invariant variants, bound the projector-applied coupling tail, or certify any computed non-probability SVD/projector/cross-product/Gram/inverse/coefficient/ product routine. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

frobNormRect_sourceSketchResidualTail_sourceSVDTail_le_sqrt_one_add_eps_sq

Corollary 635 (Projected source-tail coupling certificate for equation (9)). *Let $P \in \mathbb{R}^{m \times m}$ be an exact symmetric idempotent row multiplier:*

$$P^T = P, \quad P^2 = P.$$

Then for every $R \in \mathbb{R}^{m \times n}$,

$$\|PR\|_F \leq \|R\|_F.$$

In particular, let $B = AZ$ and $P = (AZ)C$, and suppose that C carries an exact orthogonal column-sketch projector certificate for B , equivalently the multiplier P is symmetric, idempotent, and reproduces the sketch. The same contraction gives

$$\|PR\|_F \leq \|R\|_F \quad \text{for all } R \in \mathbb{R}^{m \times n}.$$

Now keep all exact source-tail hypotheses of the preceding corollary:

$$T = U_\perp \Sigma_\perp V_\perp^T, \quad U_\perp^T U_\perp = I_q,$$

the exact right-basis and row-completeness hypotheses, and the exact cross-term certificate

$$\varepsilon \geq 0, \quad \|\Sigma_\perp (V_\perp^T Z) (V_k^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_\perp\|_F.$$

With $W = (V_k^T Z)^{-1} V_k^T$, if C carries either the orthogonal-projector certificate for $P = (AZ)C$ or the four-equation Moore–Penrose certificate for the sketch $B = AZ$, then the projected coupling tail satisfies

$$\|P(T - (TZ)W)\|_F = \|P(T - (TZ)(V_k^T Z)^{-1} V_k^T)\|_F \leq \sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F.$$

This is an exact-object coupling theorem. It closes the Frobenius projector-applied tail hand-off for supplied exact source-tail and projector certificates. It does not construct a rectangular SVD/source split, prove Eckart–Young, derive the cross-term certificate from randomness, prove the spectral-norm or general unitarily invariant variants, or certify computed non-probability SVD/projector/Gram/cross-product/inverse/coefficient/product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

finiteVecNorm2_finiteMatVec_le_of_symmetric_idempotent

frobNormRect_preconditionRows_le_of_symmetric_idempotent

frobNormRect_preconditionRows_columnSketchLeftMultiplier_le_of_orthogonalProjectorCertificate

frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_orthogonalProjector_le_sqrt_one_add_eps_sq

frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_add_eps_sq

Corollary 636 (Transpose-action spectral certificate for equation (9)). *Let $M \in \mathbb{R}^{q \times r}$, let $\Sigma \in \mathbb{R}^{q \times q}$, and assume $\varepsilon \geq 0$. Suppose the exact transpose action of M has operator-2 radius ε :*

$$\|M^T x\|_2 \leq \varepsilon \|x\|_2 \quad \text{for every } x \in \mathbb{R}^q.$$

Then

$$\|\Sigma M\|_F \leq \varepsilon \|\Sigma\|_F.$$

Equivalently, the squared form is

$$\|\Sigma M\|_F^2 \leq \varepsilon^2 \|\Sigma\|_F^2.$$

In the CACM equation (9) notation, put

$$M = (V_\perp^T Z)(V_k^T Z)^{-1}.$$

If the exact transpose-action certificate

$$\|M^T x\|_2 \leq \varepsilon \|x\|_2 \quad (x \in \mathbb{R}^q)$$

holds, then the source cross-term hypothesis used above follows:

$$\|\Sigma_\perp (V_\perp^T Z)(V_k^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_\perp\|_F.$$

Consequently, under the exact source-tail hypotheses

$$T = U_\perp \Sigma_\perp V_\perp^T, \quad U_\perp^T U_\perp = I_q,$$

the exact right-basis and row-completeness hypotheses, and a Moore–Penrose certificate for C relative to $B = AZ$, the projected source-tail coupling bound can be stated directly from the transpose-action operator certificate:

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F.$$

This is an exact-object-plus-certificate theorem. The singular-vector blocks, cross products, inverse factor, projector coefficient table, and matrix products are exact mathematical objects with exact certificates. The result does not prove equivalence with an ordinary spectral-norm certificate for the non-transposed factor M , does not prove general unitarily invariant variants, does not derive the operator certificate from the randomness of Z , and does not certify computed non-probability SVD/projector/Gram/cross-product/inverse/coefficient/product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
frobNormSqRect_matMulRectLeft_le_sq_mul_of_transpose_rectOpNorm2Le
frobNormRect_matMulRectLeft_le_of_transpose_rectOpNorm2Le
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_transpose_rectOpNorm2Le
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_
add_eps_sq_of_transpose_rectOpNorm2Le
```

Corollary 637 (Ordinary operator certificate for equation (9)). *Let $M \in \mathbb{R}^{m \times n}$, let $\varepsilon \geq 0$, and suppose the ordinary rectangular operator certificate holds:*

$$\|Mx\|_2 \leq \varepsilon \|x\|_2 \quad \text{for every } x \in \mathbb{R}^n.$$

Then the transpose action has the same radius:

$$\|M^T y\|_2 \leq \varepsilon \|y\|_2 \quad \text{for every } y \in \mathbb{R}^m.$$

Consequently, for every $\Sigma \in \mathbb{R}^{m \times m}$,

$$\|\Sigma M\|_F \leq \varepsilon \|\Sigma\|_F.$$

For the CACM equation (9) cross factor

$$M = (V_\perp^T Z)(V_k^T Z)^{-1},$$

the non-transposed certificate

$$\|(V_\perp^T Z)(V_k^T Z)^{-1}x\|_2 \leq \varepsilon \|x\|_2 \quad (x \in \mathbb{R}^r)$$

therefore implies

$$\|\Sigma_\perp (V_\perp^T Z)(V_k^T Z)^{-1}\|_F \leq \varepsilon \|\Sigma_\perp\|_F.$$

Under the exact source-tail, right-basis, row-completeness, and Moore–Penrose sketch certificates of the preceding corollaries, it follows directly that

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + \varepsilon^2} \|\Sigma_\perp\|_F.$$

This is an exact-object operator-certificate theorem. It removes the transpose-action assumption from the previous corollary when the supplied certificate is the ordinary non-transposed rectangular operator predicate. It still does not prove general unitarily invariant variants, derive the operator certificate from the randomness of Z , construct the rectangular SVD/source split or Eckart–Young certificate, or certify computed non-probability SVD/projector/Gram/cross-product/inverse/coefficient/product routines. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
rectOpNorm2Le_finiteTranspose_of_rectOpNorm2Le
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_rectOpNorm2Le
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_
add_eps_sq_of_rectOpNorm2Le
```

Corollary 638 (Computed cross-factor perturbation for equation (9)). *Let $M \in \mathbb{R}^{q \times r}$ be the exact analysis cross factor and let $\widehat{M} \in \mathbb{R}^{q \times r}$ be the computed non-probability factor actually supplied by an implementation. Let $\varepsilon \geq 0$, and suppose that the computed factor has the ordinary operator certificate*

$$\|\widehat{M}x\|_2 \leq \varepsilon \|x\|_2 \quad \text{for every } x \in \mathbb{R}^r.$$

Suppose also that the computed factor approximates the exact analysis factor with Frobenius radius τ :

$$\|M - \widehat{M}\|_F \leq \tau.$$

Then for every $\Sigma \in \mathbb{R}^{q \times q}$,

$$\|\Sigma M\|_F \leq (\varepsilon + \tau) \|\Sigma\|_F.$$

In the CACM equation (9) notation, take

$$M = (V_\perp^T Z)(V_k^T Z)^{-1}.$$

If a concrete routine returns a computed factor $\widehat{M}$ satisfying

$$\|\widehat{M}x\|_2 \leq \varepsilon \|x\|_2 \quad (x \in \mathbb{R}^r), \quad \|(V_\perp^T Z)(V_k^T Z)^{-1} - \widehat{M}\|_F \leq \tau,$$

then the exact source cross-term certificate holds with enlarged radius:

$$\|\Sigma_\perp (V_\perp^T Z)(V_k^T Z)^{-1}\|_F \leq (\varepsilon + \tau) \|\Sigma_\perp\|_F.$$

Consequently, under the exact source-tail, right-basis, row-completeness, and Moore–Penrose sketch certificates of the preceding corollaries,

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + (\varepsilon + \tau)^2} \|\Sigma_\perp\|_F.$$

This is an implementation-facing certificate transfer for computed non-probability cross-factor data. The sampling probabilities and sampling law for Z remain exact mathematical inputs. The theorem does not claim that a particular floating-point routine has produced $\widehat{M}$, nor does it certify the computed singular vectors, projector, Gram matrix, inverse, or coefficient table; those concrete routine instantiations remain not-proved ledger obligations.

Lean formalization.

```
frobNormRect_sigma_exactFactor_le_of_computed_rectOpNorm2Le_of_frobNormRect_error
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_computed_rectOpNorm2Le_of_
frobNormRect_error
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_
add_eps_plus_tau_sq_of_computed_rectOpNorm2Le
```

Corollary 639 (Entrywise computed cross-factor radius for equation (9)). *Let $M, \widehat{M} \in \mathbb{R}^{q \times r}$, let $\eta \geq 0$, and suppose a concrete computed non-probability routine supplies the uniform entrywise certificate*

$$|M_{ab} - \widehat{M}_{ab}| \leq \eta \quad (1 \leq a \leq q, 1 \leq b \leq r).$$

Then

$$\|M - \widehat{M}\|_F \leq \sqrt{qr} \eta.$$

Consequently, if the computed factor also satisfies the ordinary operator certificate

$$\|\widehat{M}x\|_2 \leq \varepsilon \|x\|_2 \quad (x \in \mathbb{R}^r), \quad \varepsilon \geq 0,$$

then every $\Sigma \in \mathbb{R}^{q \times q}$ satisfies

$$\|\Sigma M\|_F \leq (\varepsilon + \sqrt{qr} \eta) \|\Sigma\|_F.$$

For the exact CACM cross factor

$$M = (V_\perp^T Z)(V_k^T Z)^{-1},$$

the same hypotheses on the computed factor $\widehat{M}$ give

$$\|\Sigma_\perp (V_\perp^T Z)(V_k^T Z)^{-1}\|_F \leq (\varepsilon + \sqrt{qr} \eta) \|\Sigma_\perp\|_F.$$

Under the exact source-tail, right-basis, row-completeness, and Moore–Penrose sketch certificates used above, this yields the projected equation (9) bound

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + (\varepsilon + \sqrt{qr} \eta)^2} \|\Sigma_\perp\|_F.$$

This is an implementation-facing entrywise-error transfer theorem for computed non-probability cross-factor data. Sampling probabilities and sampling laws remain exact mathematical inputs. The theorem does not instantiate a concrete floating-point routine for forming $\widehat{M}$; such a routine must still prove the entrywise budget, the computed operator certificate, and the computed SVD/singular-vector/projector/Gram/inverse/coefficient-table obligations recorded in the not-proved ledger.

Lean formalization.

```
frobNormRect_le_sqrt_mul_nat_of_entry_abs_le
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_computed_rectOpNorm2Le_of_entry_abs_error
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_add_eps_plus_entry_sq_of_computed_rectOpNorm2Le
```

Corollary 640 (Component-certified computed cross factor for equation (9)). *Let*

$$X = V_{\perp}^T Z, \quad Y = (V_k^T Z)^{-1}, \quad M = XY.$$

Let $\widehat{X}$, $\widehat{Y}$, and $\widehat{M}$ be the computed non-probability cross-gram, inverse factor, and rounded product supplied by an implementation. Suppose the component certificates

$$\sum_b |X_{ab} - \widehat{X}_{ab}| |Y_{bc}| \leq \alpha,$$

$$\sum_b |\widehat{X}_{ab}| |Y_{bc} - \widehat{Y}_{bc}| \leq \beta,$$

and

$$\left| \sum_b \widehat{X}_{ab} \widehat{Y}_{bc} - \widehat{M}_{ac} \right| \leq \rho$$

hold for every a, c , with $\alpha, \beta, \rho \geq 0$. Then

$$|M_{ac} - \widehat{M}_{ac}| \leq \alpha + \beta + \rho \quad (a, c).$$

Consequently, if $\widehat{M}$ also satisfies

$$\|\widehat{M}x\|_2 \leq \varepsilon \|x\|_2 \quad (x \in \mathbb{R}^r), \quad \varepsilon \geq 0,$$

then

$$\|\Sigma_{\perp} (V_{\perp}^T Z) (V_k^T Z)^{-1}\|_F \leq (\varepsilon + \sqrt{qr} (\alpha + \beta + \rho)) \|\Sigma_{\perp}\|_F.$$

Under the exact source-tail, right-basis, row-completeness, and Moore–Penrose sketch certificates used above,

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + (\varepsilon + \sqrt{qr} (\alpha + \beta + \rho))^2} \|\Sigma_{\perp}\|_F.$$

This is an implementation-facing component-certificate theorem for the computed cross factor. It charges the computed cross-gram, inverse factor, and rounded product, but it still does not instantiate a particular floating-point dot-product routine for $V_{\perp}^T Z$, a concrete inverse routine for $(V_k^T Z)^{-1}$, a final matrix-product routine, or the computed operator certificate for $\widehat{M}$. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```
rectMatMul_entry_abs_sub_computed_le_of_component_sums
rightSketchCrossGramRectInvFactor_entry_abs_error_le_of_component_sums
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_computed_rectOpNorm2Le_of_component_error
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_add_eps_plus_component_sq_of_computed_rectOpNorm2Le
```

Corollary 641 (Concrete floating-point cross Gram for equation (9)). *Let fp be a floating-point model, assume the Higham validity condition $\gamma(fp, n)$ for dot products of length n , and define the computed rectangular cross Gram*

$$\hat{X} = \text{fl_matMul}_{fp}(V_{\perp}^T, Z).$$

For each entry set

$$B_{ab} = \gamma(fp, n) \sum_j |(V_{\perp})_{ja}| |Z_{jb}|.$$

Then

$$|(V_{\perp}^T Z)_{ab} - \hat{X}_{ab}| \leq B_{ab}.$$

Consequently, with

$$Y = (V_k^T Z)^{-1}, \quad M = (V_{\perp}^T Z)Y,$$

suppose a computed inverse factor $\hat{Y}$, rounded product $\hat{M}$, and radii $\alpha, \beta, \rho \geq 0$ satisfy

$$\sum_b B_{ab} |Y_{bc}| \leq \alpha,$$

$$\sum_b |\hat{X}_{ab}| |Y_{bc} - \hat{Y}_{bc}| \leq \beta,$$

and

$$\left| \sum_b \hat{X}_{ab} \hat{Y}_{bc} - \hat{M}_{ac} \right| \leq \rho$$

for every a, c . Then

$$|M_{ac} - \hat{M}_{ac}| \leq \alpha + \beta + \rho.$$

If, in addition,

$$\|\hat{M}x\|_2 \leq \varepsilon \|x\|_2 \quad (x \in \mathbb{R}^r), \quad \varepsilon \geq 0,$$

then

$$\|\Sigma_{\perp}(V_{\perp}^T Z)(V_k^T Z)^{-1}\|_F \leq (\varepsilon + \sqrt{qr}(\alpha + \beta + \rho)) \|\Sigma_{\perp}\|_F.$$

Under the exact source-tail, right-basis, row-completeness, and Moore–Penrose sketch certificates used above,

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + (\varepsilon + \sqrt{qr}(\alpha + \beta + \rho))^2} \|\Sigma_{\perp}\|_F.$$

This is an implementation-facing theorem for the concrete floating-point computation of the rectangular cross Gram $V_{\perp}^T Z$. It still keeps the inverse routine for $(V_k^T Z)^{-1}$, the final rounded product $\hat{X}\hat{Y}$, the operator certificate for $\hat{M}$, the computed square cross Gram used as inverse input, and computed SVD/projector/Gram obligations as explicit remaining certificates. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```

flRightSketchCrossGramRect
rightSketchCrossGramRectDotBudget
rightSketchCrossGramRect_flMatMul_entry_abs_error_le
rightSketchCrossGramRect_flMatMul_component_left_error_le
rightSketchCrossGramRectInvFactor_entry_abs_error_le_of_flMatMul_crossGram_component_sums
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_computed_rectOpNorm2Le_of_flMatMul_
crossGram_component_error
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_
add_eps_plus_flMatMul_crossGram_component_sq_of_computed_rectOpNorm2Le

```

Corollary 642 (Concrete floating-point square cross Gram for equation (9)). *Let fp be a floating-point model satisfying the Higham validity condition $\gamma(fp, n)$ for length- n dot products, and define*

$$\widehat{G} = \text{fl_matMul}_{fp}(V_k^T, Z).$$

For each entry set

$$D_{ab} = \gamma(fp, n) \sum_j |(V_k)_{ja}| |Z_{jb}|.$$

Then the exact square cross Gram $G = V_k^T Z$ satisfies

$$|G_{ab} - \widehat{G}_{ab}| \leq D_{ab}.$$

If, for some $\omega \geq 0$,

$$D_{ab} \leq \omega \quad \text{for every } a, b,$$

then

$$\|G - \widehat{G}\|_F \leq \sqrt{r} r \omega.$$

This is an implementation-facing inverse-input theorem: it certifies the computed square cross Gram that an inverse routine would consume when forming an approximation to $(V_k^T Z)^{-1}$. It does not by itself prove inverse stability, a computed inverse-factor budget for $\widehat{Y}$, or the final rounded-product/operator certificates. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```

flRightSketchCrossGram
rightSketchCrossGramDotBudget
rightSketchCrossGram_flMatMul_entry_abs_error_le
frobNorm_rightSketchCrossGram_sub_flMatMul_le_of_dotBudget_le

```

Corollary 643 (Entrywise computed inverse factor for equation (9)). *Let*

$$\widehat{X} = \text{fl_matMul}_{fp}(V_\perp^T, Z), \quad Y = (V_k^T Z)^{-1},$$

and let $\widehat{Y}$ be the computed inverse-factor output supplied by an implementation. Suppose that the implementation supplies the entrywise inverse certificate

$$|Y_{bc} - \widehat{Y}_{bc}| \leq \eta \quad \text{for every } b, c, \quad \eta \geq 0,$$

and the computed left-factor row certificate

$$\sum_b |\hat{X}_{ab}| \leq \chi \quad \text{for every } a.$$

Then the right component sum used by the computed cross-factor theorem is

$$\sum_b |\hat{X}_{ab}| |Y_{bc} - \hat{Y}_{bc}| \leq \chi\eta \quad \text{for every } a, c.$$

Consequently, if the concrete cross-Gram left budget is bounded by α ,

$$\sum_b B_{ab} |Y_{bc}| \leq \alpha, \quad B_{ab} = \gamma(fp, n) \sum_j |(V_\perp)_{ja}| |Z_{jb}|,$$

and the final rounded product

$$\widehat{M}_{ac} \approx \sum_b \hat{X}_{ab} \hat{Y}_{bc}$$

has entrywise rounding certificate

$$\left| \sum_b \hat{X}_{ab} \hat{Y}_{bc} - \widehat{M}_{ac} \right| \leq \rho,$$

then

$$\left| ((V_\perp^T Z)(V_k^T Z)^{-1})_{ac} - \widehat{M}_{ac} \right| \leq \alpha + \chi\eta + \rho.$$

If, moreover, $\widehat{M}$ has operator certificate $\|\widehat{M}x\|_2 \leq \varepsilon\|x\|_2$, with $\varepsilon, \alpha, \chi, \eta, \rho \geq 0$, then

$$\|\Sigma_\perp (V_\perp^T Z)(V_k^T Z)^{-1}\|_F \leq (\varepsilon + \sqrt{qr}(\alpha + \chi\eta + \rho)) \|\Sigma_\perp\|_F.$$

Under the same exact source-tail, right-basis, row-completeness, and Moore–Penrose sketch certificates used above,

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + (\varepsilon + \sqrt{qr}(\alpha + \chi\eta + \rho))^2} \|\Sigma_\perp\|_F.$$

This is an implementation-facing inverse-factor adapter. It does not prove that a concrete inverse routine supplies the entrywise radius η ; that routine remains a not-proved obligation. It does prove that, once such a non-probability certificate is available, the previous abstract inverse component radius β may be chosen as $\chi\eta$. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```
rightSketchCrossGramRectInvFactor_inverse_component_sum_le_of_entry_abs_error
rightSketchCrossGramRectInvFactor_flMatMul_inverse_component_sum_le_of_entry_abs_error
rightSketchCrossGramRectInvFactor_entry_abs_error_le_of_flMatMul_crossGram_inverse_entry_abs_error
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_computed_rectOpNorm2Le_of_flMatMul_crossGram_inverse_entry_error
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_add_eps_plus_flMatMul_crossGram_inverse_entry_sq_of_computed_rectOpNorm2Le
```

Corollary 644 (Concrete rounded product for equation (9)). *Let*

$$\widehat{X} = \text{fl_matMul}_{fp}(V_{\perp}^T, Z), \quad Y = (V_k^T Z)^{-1},$$

let $\widehat{Y}$ be a computed inverse-factor output, and define the computed cross factor by the concrete rounded matrix product

$$\widehat{M} = \text{fl_matMul}_{fp}(\widehat{X}, \widehat{Y}).$$

For each a, c , set the visible product dot-budget

$$R_{ac} = \gamma(fp, r) \sum_b |\widehat{X}_{ab}| |\widehat{Y}_{bc}|.$$

Assume the non-probability inverse routine and computed left factor satisfy

$$|Y_{bc} - \widehat{Y}_{bc}| \leq \eta \quad \text{for every } b, c, \quad \sum_b |\widehat{X}_{ab}| \leq \chi \quad \text{for every } a,$$

and that the concrete cross-Gram left budget satisfies

$$\sum_b B_{ab} |Y_{bc}| \leq \alpha, \quad B_{ab} = \gamma(fp, n) \sum_j |(V_{\perp})_{ja}| |Z_{jb}|.$$

If

$$R_{ac} \leq \rho \quad \text{for every } a, c,$$

then the exact analysis factor

$$M = (V_{\perp}^T Z)(V_k^T Z)^{-1}$$

and the implemented rounded product $\widehat{M}$ satisfy

$$|M_{ac} - \widehat{M}_{ac}| \leq \alpha + \chi\eta + \rho \quad \text{for every } a, c.$$

If, moreover, the computed product has operator certificate

$$\|\widehat{M}x\|_2 \leq \varepsilon \|x\|_2 \quad \text{for every } x,$$

with $\varepsilon, \alpha, \chi, \eta, \rho \geq 0$, then

$$\|\Sigma_{\perp}(V_{\perp}^T Z)(V_k^T Z)^{-1}\|_F \leq (\varepsilon + \sqrt{qr}(\alpha + \chi\eta + \rho)) \|\Sigma_{\perp}\|_F.$$

Under the same exact source-tail, right-basis, row-completeness, and Moore–Penrose sketch certificates used above,

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + (\varepsilon + \sqrt{qr}(\alpha + \chi\eta + \rho))^2} \|\Sigma_{\perp}\|_F.$$

This is an implementation-facing rounded-product theorem. It instantiates the final product radius ρ by the concrete repository fl_matMul dot-budget R_{ac} . It does not prove that a concrete inverse routine supplies the entrywise radius η , and it does not prove the computed operator certificate for $\widehat{M}$. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```

flRightSketchCrossGramRectInvFactorProduct
rightSketchCrossGramRectInvFactorProductDotBudget
rightSketchCrossGramRectInvFactorProduct_flMatMul_entry_abs_error_le
rightSketchCrossGramRectInvFactor_entry_abs_error_le_of_flMatMul_crossGram_inverse_entry_abs_
error_flMatMul_product
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_computed_rectOpNorm2Le_of_flMatMul_
crossGram_inverse_entry_flMatMul_product
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_
add_eps_plus_flMatMul_crossGram_inverse_entry_flMatMul_product_sq_of_computed_rectOpNorm2Le

```

Corollary 645 (Computed-product Frobenius certificate for equation (9)). *With the notation and hypotheses of the preceding corollary, suppose the computed product itself has the visible Frobenius certificate*

$$\|\widehat{M}\|_F \leq \varepsilon, \quad \widehat{M} = \text{fl_matMul}_{fp}(\widehat{X}, \widehat{Y}).$$

Then $\varepsilon \geq 0$, and the ordinary rectangular operator certificate

$$\|\widehat{M}x\|_2 \leq \varepsilon \|x\|_2 \quad \text{for every } x$$

holds. Consequently, if the same concrete cross-Gram left budget, inverse entrywise certificate, computed-left row budget, and final product budget hold,

$$\sum_b B_{ab} |Y_{bc}| \leq \alpha, \quad |Y_{bc} - \widehat{Y}_{bc}| \leq \eta, \quad \sum_b |\widehat{X}_{ab}| \leq \chi, \quad R_{ac} \leq \rho,$$

with $\alpha, \chi, \eta, \rho \geq 0$, then

$$\|\Sigma_\perp (V_\perp^T Z) (V_k^T Z)^{-1}\|_F \leq (\varepsilon + \sqrt{qr} (\alpha + \chi\eta + \rho)) \|\Sigma_\perp\|_F.$$

Under the same exact source-tail, right-basis, row-completeness, and Moore–Penrose sketch certificates used above,

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + (\varepsilon + \sqrt{qr} (\alpha + \chi\eta + \rho))^2} \|\Sigma_\perp\|_F.$$

This is an implementation-facing computed-operator handoff. It replaces the abstract operator certificate for the computed product by the concrete, checkable Frobenius certificate $\|\widehat{M}\|_F \leq \varepsilon$. It does not prove that an inverse routine supplies η , nor that a randomized argument or a particular computation routine supplies the displayed Frobenius certificate. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```

rectOpNorm2Le_flRightSketchCrossGramRectInvFactorProduct_of_frobNormRect_le
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_frobNormRect_flMatMul_crossGram_
inverse_entry_flMatMul_product
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_
add_frobNormRect_flMatMul_crossGram_inverse_entry_flMatMul_product_sq

```

Corollary 646 (Product absolute-sum Frobenius certificate for equation (9)). *Let*

$$\widehat{M} = \text{fl_matMul}_{fp}(\widehat{X}, \widehat{Y}), \quad R_{ac} = \gamma(fp, r) \sum_b |\widehat{X}_{ab}| |\widehat{Y}_{bc}|.$$

Assume that the computed inputs to the final product satisfy the visible absolute-product budget

$$\sum_b |\widehat{X}_{ab}| |\widehat{Y}_{bc}| \leq \kappa \quad \text{for every } a, c,$$

and that the final rounded-product budget satisfies

$$R_{ac} \leq \rho \quad \text{for every } a, c.$$

Then

$$|\widehat{M}_{ac}| \leq \kappa + \rho \quad \text{for every } a, c.$$

If $\kappa, \rho \geq 0$, then

$$\|\widehat{M}\|_F \leq \sqrt{qr} (\kappa + \rho).$$

Consequently, any displayed radius ε satisfying

$$\sqrt{qr} (\kappa + \rho) \leq \varepsilon$$

supplies the ordinary rectangular operator certificate

$$\|\widehat{M}x\|_2 \leq \varepsilon \|x\|_2 \quad \text{for every } x.$$

With $\widehat{X} = \text{fl_matMul}_{fp}(V_\perp^T, Z)$, if the same concrete cross-Gram left budget, inverse entrywise certificate, computed-left row budget, and final product budget from the preceding corollaries hold,

$$\sum_b B_{ab} |Y_{bc}| \leq \alpha, \quad |Y_{bc} - \widehat{Y}_{bc}| \leq \eta, \quad \sum_b |\widehat{X}_{ab}| \leq \chi, \quad R_{ac} \leq \rho,$$

with $\alpha, \chi, \eta, \rho \geq 0$, then

$$\|\Sigma_\perp (V_\perp^T Z) (V_k^T Z)^{-1}\|_F \leq (\varepsilon + \sqrt{qr} (\alpha + \chi\eta + \rho)) \|\Sigma_\perp\|_F.$$

Under the same exact source-tail, right-basis, row-completeness, and Moore–Penrose sketch certificates used above,

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + (\varepsilon + \sqrt{qr} (\alpha + \chi\eta + \rho))^2} \|\Sigma_\perp\|_F.$$

This is an implementation-facing source for the product Frobenius/operator certificate. It charges only computed non-probability arithmetic: the final rounded product and the magnitudes of its computed inputs. It still does not prove that an inverse routine supplies η , nor does it derive randomness-generated certificates or computed SVD/projector/Gram certificates. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```
rightSketchCrossGramRectInvFactorProduct_entry_abs_le_of_product_sum_budget
frobNormRect_flRightSketchCrossGramRectInvFactorProduct_le_sqrt_mul_product_sum_budget
rectOpNorm2Le_flRightSketchCrossGramRectInvFactorProduct_of_product_sum_budget
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_product_sum_budget_flMatMul_crossGram_
inverse_entry_flMatMul_product
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_
add_product_sum_budget_flMatMul_crossGram_inverse_entry_flMatMul_product_sq
```

Corollary 647 (Perturbed-inverse certificate for equation (9)). *Let $A_0 \in \mathbb{R}^{r \times r}$ have nonzero determinant, and let $\hat{Y}$ and ΔA satisfy the perturbed-inverse equation*

$$(A_0 + \Delta A)\hat{Y} = I.$$

Assume the componentwise perturbation budget

$$|\Delta A_{ij}| \leq \varepsilon_{\text{inv}} |(A_0)_{ij}| \quad \text{for every } i, j,$$

and the visible inverse-sensitivity budget

$$\varepsilon_{\text{inv}} \sum_{k_1} |(A_0^{-1})_{ik_1}| \sum_{k_2} |(A_0)_{k_1 k_2}| |\hat{Y}_{k_2 j}| \leq \eta \quad \text{for every } i, j.$$

Then

$$|(A_0^{-1})_{ij} - \hat{Y}_{ij}| \leq \eta \quad \text{for every } i, j.$$

Specializing $A_0 = V_k^T Z$ gives the inverse-entry certificate

$$|((V_k^T Z)^{-1})_{bc} - \hat{Y}_{bc}| \leq \eta \quad \text{for every } b, c$$

consumed by the computed cross-factor results above.

Consequently, with $\hat{X} = \text{fl_matMul}_{fp}(V_\perp^T, Z)$, if the product absolute-sum certificate of the preceding corollary supplies $\|\hat{M}x\|_2 \leq \varepsilon_M \|x\|_2$, and if the same concrete cross-Gram left budget, computed-left row budget, and final product budget hold,

$$\sum_b B_{ab} |Y_{bc}| \leq \alpha, \quad \sum_b |\hat{X}_{ab}| \leq \chi, \quad R_{ac} \leq \rho,$$

with $\alpha, \chi, \eta, \rho \geq 0$, then

$$\|\Sigma_\perp (V_\perp^T Z) (V_k^T Z)^{-1}\|_F \leq (\varepsilon_M + \sqrt{qr} (\alpha + \chi\eta + \rho)) \|\Sigma_\perp\|_F.$$

Under the same exact source-tail, right-basis, row-completeness, and Moore–Penrose sketch certificates used above,

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + (\varepsilon_M + \sqrt{qr} (\alpha + \chi\eta + \rho))^2} \|\Sigma_\perp\|_F.$$

This is an implementation-facing inverse-radius source theorem. It uses the local Higham §13.1 forward-error theorem for computed inverses to replace the raw assumption $|Y - \hat{Y}| \leq \eta$ by a perturbed-inverse contract. It does not prove that a particular inversion loop supplies ΔA and $\hat{Y}$. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```
nonsingInv_entry_abs_sub_computed_inverse_le_of_perturbed_inverse_component_budget
rightSketchCrossGram_inverse_entry_abs_error_le_of_perturbed_inverse_component_budget
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_perturbed_inverse_product_sum_budget
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_
add_perturbed_inverse_product_sum_budget_sq
```

Corollary 648 (Method-A LU inverse certificate for equation (9)). *Let $A_0 = V_k^T Z \in \mathbb{R}^{r \times r}$ have nonzero determinant, and let $\hat{L}, \hat{U} \in \mathbb{R}^{r \times r}$ have nonzero diagonals. Assume that $\hat{L}, \hat{U}$ satisfy the componentwise LU backward-error certificate*

$$\text{LUBackwardError}_r(A_0, \hat{L}, \hat{U}, \gamma(fp, r)).$$

Define the Method-A computed inverse

$$\hat{Y} = \text{methodAComputedInverse}_{fp,r}(\hat{L}, \hat{U}),$$

meaning that the c -th column is computed by solving

$$\hat{L} \hat{y}_c = e_c, \quad \hat{U} \hat{x}_c = \hat{y}_c$$

with the repository floating-point forward- and back-substitution routines, and setting $\hat{Y}_{\cdot c} = \hat{x}_c$. Put

$$c_r = 3\gamma(fp, r) + \gamma(fp, r)^2.$$

Assume the visible Method-A inverse-sensitivity budget

$$c_r \sum_{k_1} |(A_0^{-1})_{bk_1}| \sum_{k_2} \left(\sum_{\ell} |\hat{L}_{k_1 \ell}| |\hat{U}_{\ell k_2}| \right) |\hat{Y}_{k_2 c}| \leq \eta \quad \text{for every } b, c.$$

Then

$$|(A_0^{-1})_{bc} - \hat{Y}_{bc}| \leq \eta \quad \text{for every } b, c.$$

Equivalently, specializing $A_0 = V_k^T Z$,

$$|((V_k^T Z)^{-1})_{bc} - \text{methodAComputedInverse}_{fp,r}(\hat{L}, \hat{U})_{bc}| \leq \eta.$$

Consequently, with $\hat{X} = \text{fl_matMul}_{fp}(V_{\perp}^T, Z)$ and $\hat{Y} = \text{methodAComputedInverse}_{fp,r}(\hat{L}, \hat{U})$, if the same product absolute-sum certificate supplies $\|\hat{M}x\|_2 \leq \varepsilon_M \|x\|_2$, and if

$$\sum_b B_{ab} |Y_{bc}| \leq \alpha, \quad \sum_b |\hat{X}_{ab}| \leq \chi, \quad R_{ac} \leq \rho,$$

with $\alpha, \chi, \eta, \rho \geq 0$, then

$$\|\Sigma_{\perp}(V_{\perp}^T Z)(V_k^T Z)^{-1}\|_F \leq (\varepsilon_M + \sqrt{qr}(\alpha + \chi\eta + \rho)) \|\Sigma_{\perp}\|_F.$$

Under the same exact source-tail, right-basis, row-completeness, and Moore–Penrose sketch certificates used above,

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + (\varepsilon_M + \sqrt{qr}(\alpha + \chi\eta + \rho))^2} \|\Sigma_{\perp}\|_F.$$

This is an implementation-facing inverse-solve source theorem for Method A. It charges the computed non-probability inverse columns through LU, forward-substitution, and back-substitution certificates. It is still conditional on the displayed LU backward-error certificate for the square cross Gram, and it does not prove computed SVD/projector/Gram or randomness-derived certificates. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```

methodAComputedInverse
methodA_column_backward_error_computed_inverse
methodA_computed_inverse_entry_abs_sub_nonsingInv_le_of_lu_budget
rightSketchCrossGram_inverse_entry_abs_error_le_of_methodA_lu_budget
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_methodA_inverse_product_sum_budget
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_
add_methodA_inverse_product_sum_budget_sq

```

Corollary 649 (Computed-input Method-A LU certificate for equation (9)). *Let $A_0 = V_k^T Z \in \mathbb{R}^{r \times r}$ have nonzero determinant, and let*

$$\hat{A}_0 = \text{flRightSketchCrossGram}_{fp}(V_k, Z) = \text{fl_matMul}_{fp}(V_k^T, Z)$$

be the rounded square cross Gram actually passed to the LU factorization. Assume that the computed factors $\hat{L}, \hat{U}$ satisfy

$$\text{LUBackwardError}_r(\hat{A}_0, \hat{L}, \hat{U}, \varepsilon_{\text{LU}}),$$

and that the rounded-input error is controlled in the same componentwise $|\hat{L}| |\hat{U}|$ weights:

$$\text{rightSketchCrossGramDotBudget}_{fp}(V_k, Z)_{bc} \leq \mu \sum_{\ell} |\hat{L}_{b\ell}| |\hat{U}_{\ell c}| \quad \text{for every } b, c.$$

Then the exact analysis matrix inherits the LU certificate

$$\text{LUBackwardError}_r(A_0, \hat{L}, \hat{U}, \varepsilon_{\text{LU}} + \mu).$$

Assume $\varepsilon_{\text{LU}} \geq 0$, $\mu \geq 0$, the triangular diagonals of $\hat{L}, \hat{U}$ are nonzero, and put

$$c_r^{\text{in}} = (\varepsilon_{\text{LU}} + \mu) + 2\gamma(fp, r) + \gamma(fp, r)^2.$$

With

$$\hat{Y} = \text{methodAComputedInverse}_{fp,r}(\hat{L}, \hat{U}),$$

assume the visible inverse-sensitivity budget

$$c_r^{\text{in}} \sum_{k_1} |(A_0^{-1})_{bk_1}| \sum_{k_2} \left(\sum_{\ell} |\hat{L}_{k_1\ell}| |\hat{U}_{\ell k_2}| \right) |\hat{Y}_{k_2 c}| \leq \eta \quad \text{for every } b, c.$$

Then

$$|(A_0^{-1})_{bc} - \hat{Y}_{bc}| \leq \eta \quad \text{for every } b, c.$$

Consequently, with $\hat{X} = \text{fl_matMul}_{fp}(V_{\perp}^T, Z)$, if the same product absolute-sum certificate supplies $\|\hat{M}x\|_2 \leq \varepsilon_M \|x\|_2$, and if

$$\sum_b B_{ab} |Y_{bc}| \leq \alpha, \quad \sum_b |\hat{X}_{ab}| \leq \chi, \quad R_{ac} \leq \rho,$$

with $\alpha, \chi, \eta, \rho \geq 0$, then

$$\|\Sigma_{\perp}(V_{\perp}^T Z)(V_k^T Z)^{-1}\|_F \leq (\varepsilon_M + \sqrt{qr}(\alpha + \chi\eta + \rho))\|\Sigma_{\perp}\|_F,$$

and under the same exact source-tail, right-basis, row-completeness, and Moore–Penrose sketch certificates used above,

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + (\varepsilon_M + \sqrt{qr}(\alpha + \chi\eta + \rho))^2} \|\Sigma_\perp\|_F.$$

This is an implementation-facing non-probability input-transfer theorem. It does not charge probability construction or the sampling law; those remain exact mathematical inputs. It does charge the rounded square cross Gram actually consumed by the LU factorization. It is still conditional on a concrete routine proving the displayed ‘LUBackwardError’ certificate for $\hat{A}_0 = \text{flRightSketchCrossGram}_{fp}(V_k, Z)$.

Lean formalization.

```
LUBackwardError.of_input_abs_error_le_absLUProduct
lu_solve_backward_error_factor_gamma
methodA_computed_inverse_entry_abs_sub_nonsingInv_le_of_lu_factor_budget
rightSketchCrossGram_LUBackwardError_of_flRightSketchCrossGram_input_budget
rightSketchCrossGram_inverse_entry_abs_error_le_of_methodA_fl_lu_input_budget
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_methodA_lu_factor_product_sum_budget
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_
add_methodA_lu_factor_product_sum_budget_sq
```

Corollary 650 (Doolittle-generated computed-input Method-A LU certificate for equation (9)).
Let $A_0 = V_k^T Z \in \mathbb{R}^{r \times r}$ have nonzero determinant, and let

$$\hat{A}_0 = \text{flRightSketchCrossGram}_{fp}(V_k, Z) = \text{fl_matMul}_{fp}(V_k^T, Z)$$

be the rounded square cross Gram consumed by the dense LU factorization. Assume that $\hat{L}, \hat{U}$ satisfy the Doolittle recurrence certificate

$$\text{DoolittleLU}_r(\hat{A}_0, \hat{L}, \hat{U}, fp).$$

Then the repository recurrence-to-backward-error theorem gives

$$\text{LUBackwardError}_r(\hat{A}_0, \hat{L}, \hat{U}, \gamma(fp, r)).$$

Assume moreover that the rounded square-cross-Gram input error is controlled in the same componentwise $|\hat{L}| |\hat{U}|$ weights:

$$\text{rightSketchCrossGramDotBudget}_{fp}(V_k, Z)_{bc} \leq \mu \sum_{\ell} |\hat{L}_{b\ell}| |\hat{U}_{\ell c}| \quad \text{for every } b, c.$$

Then the exact analysis matrix inherits

$$\text{LUBackwardError}_r(A_0, \hat{L}, \hat{U}, \gamma(fp, r) + \mu).$$

Assume $\mu \geq 0$, the triangular diagonals of $\hat{L}, \hat{U}$ are nonzero, and put

$$c_r^{\text{Doolittle}} = (\gamma(fp, r) + \mu) + 2\gamma(fp, r) + \gamma(fp, r)^2.$$

With

$$\hat{Y} = \text{methodAComputedInverse}_{fp,r}(\hat{L}, \hat{U}),$$

assume the visible inverse-sensitivity budget

$$c_r^{\text{Doolittle}} \sum_{k_1} |(A_0^{-1})_{bk_1}| \sum_{k_2} \left(\sum_{\ell} |\hat{L}_{k_1\ell}| |\hat{U}_{\ell k_2}| \right) |\hat{Y}_{k_2c}| \leq \eta \quad \text{for every } b, c.$$

Then

$$|(A_0^{-1})_{bc} - \hat{Y}_{bc}| \leq \eta \quad \text{for every } b, c.$$

Consequently, with $\hat{X} = \text{fl_matMul}_{fp}(V_{\perp}^T, Z)$, if the same product absolute-sum certificate supplies $\|\hat{M}x\|_2 \leq \varepsilon_M \|x\|_2$, and if

$$\sum_b B_{ab} |Y_{bc}| \leq \alpha, \quad \sum_b |\hat{X}_{ab}| \leq \chi, \quad R_{ac} \leq \rho,$$

with $\alpha, \chi, \eta, \rho \geq 0$, then

$$\|\Sigma_{\perp}(V_{\perp}^T Z)(V_k^T Z)^{-1}\|_F \leq (\varepsilon_M + \sqrt{qr}(\alpha + \chi\eta + \rho)) \|\Sigma_{\perp}\|_F,$$

and under the same exact source-tail, right-basis, row-completeness, and Moore–Penrose sketch certificates used above,

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + (\varepsilon_M + \sqrt{qr}(\alpha + \chi\eta + \rho))^2} \|\Sigma_{\perp}\|_F.$$

This is an implementation-facing, certificate-level dense-LU result. It charges the rounded square cross Gram, the Doolittle recurrence factorization, and the Method-A triangular solves. It does not charge probability construction or the sampling law; those remain exact mathematical inputs. It still requires a concrete executable dense-LU implementation to produce the displayed DoolittleLU witness.

Lean formalization.

```
DoolittleLU.to_LUBackwardError
doolittle_backward_error
rightSketchCrossGram_LUBackwardError_of_DoolittleLU_flRightSketchCrossGram
rightSketchCrossGram_inverse_entry_abs_error_le_of_methodA_doolittle_fl_input_budget
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_methodA_doolittle_fl_input_product_sum_budget
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_add_methodA_doolittle_fl_input_product_sum_budget_sq
```

Corollary 651 (Dense-loop Doolittle residual-compression certificate for equation (9)). *Let*

$$\hat{A}_0 = \text{flRightSketchCrossGram}_{fp}(V_k, Z) = \text{fl_matMul}_{fp}(V_k^T, Z)$$

be the rounded square cross Gram used by Method A. For $k \leq j$ define the literal rounded Doolittle upper-entry fold

$$\hat{u}_{kj} = \text{flDoolittleUEntry}_{fp}(\hat{A}_0, \hat{L}, \hat{U})_{kj},$$

and for $i > k$ define the rounded lower-entry numerator and division

$$\hat{n}_{ik} = \text{flDoolittleLNumerator}_{fp}(\hat{A}_0, \hat{L}, \hat{U})_{ik}, \quad \hat{\ell}_{ik} = \text{flDoolittleLEntry}_{fp}(\hat{A}_0, \hat{L}, \hat{U})_{ik}.$$

Assume the stored factors satisfy

$$\hat{U}_{kj} = \hat{u}_{kj} \quad (k \leq j), \quad \hat{L}_{ik} = \hat{\ell}_{ik} \quad (i > k),$$

with $\hat{L}$ unit lower triangular and $\hat{U}$ upper triangular. Assume also the two visible residual-compression budgets

$$\left| \hat{A}_{0,kj} - \sum_{s < k} \hat{L}_{ks} \hat{U}_{sj} - \hat{U}_{kj} \right| \leq \gamma(fp, r) |\hat{U}_{kj}| \quad (k \leq j),$$

and

$$\left| \hat{A}_{0,ik} - \sum_{s < k} \hat{L}_{is} \hat{U}_{sk} - \hat{L}_{ik} \hat{U}_{kk} \right| \leq \gamma(fp, r) |\hat{L}_{ik} \hat{U}_{kk}| \quad (i > k).$$

Then these data form a

$$\text{DoolittleDenseLoopCertificate}_r(\hat{A}_0, \hat{L}, \hat{U}, fp),$$

and the residual-compression theorem constructs the compact recurrence certificate

$$\text{DoolittleLU}_r(\hat{A}_0, \hat{L}, \hat{U}, fp).$$

If $\text{gammaValid}(fp, r)$, then

$$\text{LUBackwardError}_r(\hat{A}_0, \hat{L}, \hat{U}, \gamma(fp, r)).$$

Consequently, under the same determinant, triangular-diagonal, rounded-input budget

$$\text{rightSketchCrossGramDotBudget}_{fp}(V_k, Z)_{bc} \leq \mu \sum_{\ell} |\hat{L}_{b\ell}| |\hat{U}_{\ell c}|,$$

Method-A inverse-sensitivity budget with

$$c_r^{\text{loop}} = (\gamma(fp, r) + \mu) + 2\gamma(fp, r) + \gamma(fp, r)^2,$$

and product absolute-sum hypotheses from the previous Doolittle corollary, the computed inverse

$$\hat{Y} = \text{methodAComputedInverse}_{fp,r}(\hat{L}, \hat{U})$$

satisfies

$$|(V_k^T Z)_{bc}^{-1} - \hat{Y}_{bc}| \leq \eta \quad \text{for every } b, c,$$

and the equation-(9) cross-term and projected-tail bounds are

$$\|\Sigma_{\perp}(V_{\perp}^T Z)(V_k^T Z)^{-1}\|_F \leq (\varepsilon_M + \sqrt{qr}(\alpha + \chi\eta + \rho)) \|\Sigma_{\perp}\|_F,$$

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + (\varepsilon_M + \sqrt{qr}(\alpha + \chi\eta + \rho))^2} \|\Sigma_{\perp}\|_F.$$

This is an implementation-facing non-probability theorem surface. Sampling probabilities and sampling laws remain exact mathematical inputs. The theorem does not yet prove the residual-compression budgets from a concrete pivoting or no-cancellation regime; it states exactly the certificate that such an implementation proof must instantiate.

Lean formalization.

```

flDoolittleUEntry
flDoolittleLNUmberator
flDoolittleLEntry
DoolittleDenseLoopCertificate
DoolittleDenseLoopCertificate.to_DoolittleLU
DoolittleDenseLoopCertificate.to_LUBackwardError
rightSketchCrossGram_LUBackwardError_of_DoolittleDenseLoopCertificate_flRightSketchCrossGram
rightSketchCrossGram_inverse_entry_abs_error_le_of_methodA_doolittleDenseLoop_fl_input_budget
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_methodA_doolittleDenseLoop_fl_input_
product_sum_budget
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_
add_methodA_doolittleDenseLoop_fl_input_product_sum_budget_sq

```

Corollary 652 (Absolute-budget dense-loop Doolittle handoff for equation (9)). *Let $\hat{A}_0 = \text{flRightSketchCrossGram}_{fp}(V_k, Z)$ be the rounded square cross Gram used by Method A, and let $\hat{L}, \hat{U} \in \mathbb{R}^{r \times r}$ be stored dense-Doolittle factors with $\hat{L}$ unit lower triangular and $\hat{U}$ upper triangular. Suppose the stored entries agree with the literal rounded Doolittle folds:*

$$\hat{U}_{kj} = \text{flDoolittleUEntry}_{fp}(\hat{A}_0, \hat{L}, \hat{U})_{kj} \quad (k \leq j), \quad \hat{L}_{ik} = \text{flDoolittleLEntry}_{fp}(\hat{A}_0, \hat{L}, \hat{U})_{ik} \quad (i > k).$$

Let $B^U, B^L \in \mathbb{R}^{r \times r}$ be absolute residual budgets satisfying

$$\left| \hat{A}_{0,kj} - \sum_{s < k} \hat{L}_{ks} \hat{U}_{sj} - \hat{U}_{kj} \right| \leq B_{kj}^U \quad (k \leq j),$$

$$\left| \hat{A}_{0,ik} - \sum_{s < k} \hat{L}_{is} \hat{U}_{sk} - \hat{L}_{ik} \hat{U}_{kk} \right| \leq B_{ik}^L \quad (i > k).$$

Assume further that these absolute budgets are dominated by the relative compression radii

$$B_{kj}^U \leq \gamma(fp, r) |\hat{U}_{kj}| \quad (k \leq j), \quad B_{ik}^L \leq \gamma(fp, r) |\hat{L}_{ik} \hat{U}_{kk}| \quad (i > k).$$

Then the data form an implementation-facing certificate

$$\text{DoolittleDenseLoopAbsBudgetCertificate}_r(\hat{A}_0, \hat{L}, \hat{U}, fp, B^U, B^L),$$

which yields

$$\text{DoolittleDenseLoopCertificate}_r(\hat{A}_0, \hat{L}, \hat{U}, fp), \quad \text{DoolittleLU}_r(\hat{A}_0, \hat{L}, \hat{U}, fp).$$

If $\text{gammaValid}(fp, r)$, then

$$\text{LUBackwardError}_r(\hat{A}_0, \hat{L}, \hat{U}, \gamma(fp, r)).$$

Consequently, under the determinant and nonzero triangular-diagonal hypotheses from the previous corollary, the rounded-input budget

$$\text{rightSketchCrossGramDotBudget}_{fp}(V_k, Z)_{bc} \leq \mu \sum_{\ell} |\hat{L}_{b\ell}| |\hat{U}_{\ell c}|,$$

the Method-A inverse-sensitivity budget with

$$c_r^{\text{abs}} = (\gamma(fp, r) + \mu) + 2\gamma(fp, r) + \gamma(fp, r)^2,$$

and the same product absolute-sum, final-product, and left-cross-Gram budgets, the computed inverse

$$\hat{Y} = \text{methodAComputedInverse}_{fp,r}(\hat{L}, \hat{U})$$

satisfies

$$|(V_k^T Z)_{bc}^{-1} - \hat{Y}_{bc}| \leq \eta \quad \text{for all } b, c,$$

and the equation-(9) cross-term and projected-tail bounds are

$$\|\Sigma_{\perp}(V_{\perp}^T Z)(V_k^T Z)^{-1}\|_F \leq (\varepsilon_M + \sqrt{qr}(\alpha + \chi\eta + \rho))\|\Sigma_{\perp}\|_F,$$

$$\|(AZ)C(T - (TZ)(V_k^T Z)^{-1}V_k^T)\|_F \leq \sqrt{1 + (\varepsilon_M + \sqrt{qr}(\alpha + \chi\eta + \rho))^2}\|\Sigma_{\perp}\|_F.$$

This is a fully computed-object theorem surface for the non-probability dense-Doolittle arithmetic once the absolute residual budgets and dominance inequalities are supplied. Sampling probabilities and sampling laws remain exact mathematical inputs. The theorem does not yet prove B^U , B^L or the dominance inequalities from a concrete pivoting or no-cancellation regime; that routine instantiation remains in the not-proved ledger.

Lean formalization.

```
DoolittleDenseLoopAbsBudgetCertificate
DoolittleDenseLoopAbsBudgetCertificate.to_denseLoopCertificate
DoolittleDenseLoopAbsBudgetCertificate.to_DoolittleLU
DoolittleDenseLoopAbsBudgetCertificate.to_LUBackwardError
rightSketchCrossGram_LUBackwardError_of_DoolittleDenseLoopAbsBudgetCertificate_
flRightSketchCrossGram
rightSketchCrossGram_inverse_entry_abs_error_le_of_methodA_doolittleDenseLoopAbsBudget_fl_input_
budget
frobNormRect_sigma_rightSketchCrossGramRectInvFactor_le_of_methodA_doolittleDenseLoopAbsBudget_
fl_input_product_sum_budget
frobNormRect_preconditionRows_sourceSketchResidualTail_sourceSVDTail_moorePenrose_le_sqrt_one_
add_methodA_doolittleDenseLoopAbsBudget_fl_input_product_sum_budget_sq
```

Corollary 653 (Rounded-product Doolittle fold budgets for equation (9)). *Let $m \in \mathbb{N}$, let $s \in \mathbb{R}$, and let $t_0, \dots, t_{m-1} \in \mathbb{R}$ be the rounded terms supplied to a sequential floating-point subtraction fold. If $\text{gammaValid}(fp, m)$, then*

$$\left| \left(s - \sum_{a=0}^{m-1} t_a \right) - \text{foldl}_{a < m}(x \mapsto \text{fl_sub}_{fp}(x, t_a))(s) \right| \leq \gamma(fp, m) \left(|s| + \sum_{a=0}^{m-1} |t_a| \right).$$

Specializing this to the dense-Doolittle upper-entry fold, with

$$\hat{p}_{ksj}^U = \text{fl_mul}_{fp}(\hat{L}_{ks}, \hat{U}_{sj}),$$

gives

$$\left| \hat{A}_{0,kj} - \sum_{s < k} \hat{p}_{ksj}^U - \text{flDoolittleUEntry}_{fp}(\hat{A}_0, \hat{L}, \hat{U})_{kj} \right| \leq \gamma(fp, k) \left(|\hat{A}_{0,kj}| + \sum_{s < k} |\hat{p}_{ksj}^U| \right).$$

Specializing it to the lower-entry numerator fold, with

$$\hat{p}_{isk}^L = \text{fl_mul}_{fp}(\hat{L}_{is}, \hat{U}_{sk}),$$

gives

$$\left| \hat{A}_{0,ik} - \sum_{s < k} \hat{p}_{isk}^L - \text{flDoolittleLNumerator}_{fp}(\hat{A}_0, \hat{L}, \hat{U})_{ik} \right| \leq \gamma(fp, k) \left(|\hat{A}_{0,ik}| + \sum_{s < k} |\hat{p}_{isk}^L| \right).$$

This is an implementation-facing non-probability theorem for the subtraction accumulation that the dense Doolittle loop actually performs. It does not yet turn rounded products $\hat{p}^U, \hat{p}^L$ into exact products $\hat{L}\hat{U}$, does not charge the lower rounded division and subsequent multiplication by $\hat{U}_{kk}$, and does not prove the dominance inequalities required by the absolute-budget certificate above. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```
fl_sub_sum_error_init_abs_residual_le
flDoolittleUEntry_rounded_residual_abs_le
flDoolittleLNumerator_rounded_residual_abs_le
```

Corollary 654 (Exact-product Doolittle fold budgets for equation (9)). *For the dense-Doolittle upper-entry fold, define the exact and rounded non-probability products*

$$p_{ksj}^U = \hat{L}_{ks} \hat{U}_{sj}, \quad \hat{p}_{ksj}^U = \text{fl_mul}_{fp}(\hat{L}_{ks}, \hat{U}_{sj}).$$

The primitive multiplication model gives

$$|\text{fl_mul}_{fp}(x, y) - xy| \leq u_{fp}|xy|.$$

Consequently, if $\text{gammaValid}(fp, k)$, then

$$\begin{aligned} & \left| \hat{A}_{0,kj} - \sum_{s < k} p_{ksj}^U - \text{flDoolittleUEntry}_{fp}(\hat{A}_0, \hat{L}, \hat{U})_{kj} \right| \\ & \leq \gamma(fp, k) \left(|\hat{A}_{0,kj}| + \sum_{s < k} |\hat{p}_{ksj}^U| \right) \\ & \quad + u_{fp} \sum_{s < k} |p_{ksj}^U|. \end{aligned}$$

For the lower-entry numerator fold, with

$$p_{isk}^L = \hat{L}_{is} \hat{U}_{sk}, \quad \hat{p}_{isk}^L = \text{fl_mul}_{fp}(\hat{L}_{is}, \hat{U}_{sk}),$$

the same hypotheses give

$$\begin{aligned}
& \left| \widehat{A}_{0,ik} - \sum_{s < k} p_{isk}^L - \text{flDoolittleLNumerator}_{fp}(\widehat{A}_0, \widehat{L}, \widehat{U})_{ik} \right| \\
& \leq \gamma(fp, k) \left(|\widehat{A}_{0,ik}| + \sum_{s < k} |\widehat{p}_{isk}^L| \right) \\
& \quad + u_{fp} \sum_{s < k} |p_{isk}^L|.
\end{aligned}$$

This closes the product-transfer part of the dense-Doolittle absolute-budget source: the residuals are now measured against the exact products $\widehat{L}\widehat{U}$, not only against the rounded products supplied to the subtraction fold. It still does not package the $s < k$ sums into the masked n -term sums used by `DoolittleDenseLoopAbsBudgetCertificate`, does not charge the lower rounded division and multiplication by $\widehat{U}_{kk}$, and does not prove the dominance inequalities required by that certificate. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```

fl_mul_abs_sub_mul_le
flDoolittleUEntry_exact_product_residual_abs_le
flDoolittleLNumerator_exact_product_residual_abs_le

```

Corollary 655 (Masked-prefix Doolittle fold budgets for equation (9)). *For every $k < n$ and every table f_s ,*

$$\sum_{s=0}^{n-1} \mathbf{1}_{\{s < k\}} f_s = \sum_{s < k} f_s.$$

Thus the exact-product Doolittle fold budgets above can be stated in the masked n -term recurrence shape used by the dense-loop certificate. Let

$$p_{ksj}^U = \widehat{L}_{ks} \widehat{U}_{sj}, \quad \widehat{p}_{ksj}^U = \text{fl_mul}_{fp}(\widehat{L}_{ks}, \widehat{U}_{sj}).$$

If $\text{gammaValid}(fp, k)$, then

$$\begin{aligned}
& \left| \widehat{A}_{0,kj} - \sum_{s=0}^{n-1} \mathbf{1}_{\{s < k\}} p_{ksj}^U - \text{flDoolittleUEntry}_{fp}(\widehat{A}_0, \widehat{L}, \widehat{U})_{kj} \right| \\
& \leq \gamma(fp, k) \left(|\widehat{A}_{0,kj}| + \sum_{s < k} |\widehat{p}_{ksj}^U| \right) \\
& \quad + u_{fp} \sum_{s < k} |p_{ksj}^U|.
\end{aligned}$$

Likewise, with

$$p_{isk}^L = \widehat{L}_{is} \widehat{U}_{sk}, \quad \widehat{p}_{isk}^L = \text{fl_mul}_{fp}(\widehat{L}_{is}, \widehat{U}_{sk}),$$

the lower-entry numerator satisfies

$$\begin{aligned}
& \left| \hat{A}_{0,ik} - \sum_{s=0}^{n-1} \mathbf{1}_{\{s < k\}} p_{isk}^L - \text{flDoolittleLNumerator}_{fp}(\hat{A}_0, \hat{L}, \hat{U})_{ik} \right| \\
& \leq \gamma(fp, k) \left(|\hat{A}_{0,ik}| + \sum_{s < k} |\hat{p}_{isk}^L| \right) \\
& \quad + u_{fp} \sum_{s < k} |p_{isk}^L|.
\end{aligned}$$

This is an implementation-facing non-probability theorem for the recurrence shape: the computed multiplication and subtraction-fold errors are charged, and the prefix sums are reindexed into the masked n -term form. It still does not charge the lower rounded division and multiplication by the computed pivot $\hat{U}_{kk}$, and it does not prove the dominance inequalities needed to instantiate the full dense-Doolittle absolute-budget certificate. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```

finMaskedPrefixSum_eq_finSum
flDoolittleUEntry_masked_exact_product_residual_abs_le
flDoolittleLNumerator_masked_exact_product_residual_abs_le

```

Corollary 656 (Lower rounded-division Doolittle budget for equation (9)). *Let*

$$\hat{n}_{ik} = \text{flDoolittleLNumerator}_{fp}(\hat{A}_0, \hat{L}, \hat{U})_{ik}, \quad \hat{\ell}_{ik} = \text{flDoolittleLEntry}_{fp}(\hat{A}_0, \hat{L}, \hat{U})_{ik}.$$

If the computed pivot is nonzero, $\hat{U}_{kk} \neq 0$, then the primitive rounded-division model gives

$$|\hat{n}_{ik} - \hat{\ell}_{ik} \hat{U}_{kk}| \leq u_{fp} |\hat{n}_{ik}|.$$

Consequently, if also $\text{gammaValid}(fp, k)$ and the stored lower entry is the literal rounded Doolittle update,

$$\hat{L}_{ik} = \text{flDoolittleLEntry}_{fp}(\hat{A}_0, \hat{L}, \hat{U})_{ik},$$

then, with

$$p_{isk}^L = \hat{L}_{is} \hat{U}_{sk}, \quad \hat{p}_{isk}^L = \text{fl_mul}_{fp}(\hat{L}_{is}, \hat{U}_{sk}),$$

the lower-entry residual after multiplication by the computed pivot satisfies

$$\begin{aligned}
& \left| \hat{A}_{0,ik} - \sum_{s=0}^{n-1} \mathbf{1}_{\{s < k\}} p_{isk}^L - \hat{L}_{ik} \hat{U}_{kk} \right| \\
& \leq \left[\gamma(fp, k) \left(|\hat{A}_{0,ik}| + \sum_{s < k} |\hat{p}_{isk}^L| \right) + u_{fp} \sum_{s < k} |p_{isk}^L| \right] \\
& \quad + u_{fp} |\hat{n}_{ik}|.
\end{aligned}$$

This is an implementation-facing non-probability theorem for the lower-entry division and computed-pivot multiplication in dense Doolittle. It closes the lower rounded-division/pivot source for the absolute-budget certificate. It still does not prove the dominance inequality

$$B_{ik}^L \leq \gamma(fp, n) |\hat{L}_{ik} \hat{U}_{kk}|$$

or any concrete pivot/no-cancellation regime. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```
flDoolittleLEntry_mul_pivot_sub_numerator_abs_le
flDoolittleLEntry_masked_exact_product_residual_abs_le
```

Corollary 657 (Literal Doolittle source-budget certificate for equation (9)). *Define the concrete upper and lower dense-Doolittle absolute budgets*

$$B_{kj}^U = \gamma(fp, k) \left(|\hat{A}_{0,kj}| + \sum_{s < k} |\hat{p}_{ksj}^U| \right) + u_{fp} \sum_{s < k} |p_{ksj}^U|,$$

$$B_{ik}^L = \gamma(fp, k) \left(|\hat{A}_{0,ik}| + \sum_{s < k} |\hat{p}_{isk}^L| \right) + u_{fp} \sum_{s < k} |p_{isk}^L| + u_{fp} |\hat{n}_{ik}|.$$

Here

$$p_{ksj}^U = \hat{L}_{ks} \hat{U}_{sj}, \quad p_{isk}^L = \hat{L}_{is} \hat{U}_{sk}, \quad \hat{p} = \text{fl_mul}_{fp}(p \text{'s factors}),$$

and $\hat{n}_{ik} = \text{flDoolittleLNumerator}_{fp}(\hat{A}_0, \hat{L}, \hat{U})_{ik}$. Assume the stored factors have the required triangular shape, the upper and lower stored entries agree with the literal rounded Doolittle updates, all computed pivots are nonzero, and $\text{gammaValid}(fp, n)$. If the visible dominance inequalities

$$B_{kj}^U \leq \gamma(fp, n) |\hat{U}_{kj}| \quad (k \leq j), \quad B_{ik}^L \leq \gamma(fp, n) |\hat{L}_{ik} \hat{U}_{kk}| \quad (k < i)$$

hold, then the concrete budgets B^U, B^L instantiate

$$\text{DoolittleDenseLoopAbsBudgetCertificate}(\hat{A}_0, \hat{L}, \hat{U}, fp, B^U, B^L).$$

Consequently all downstream dense-loop Method-A handoffs from the absolute budget certificate apply with these explicit non-probability arithmetic budgets.

This closes the source-budget assembly for literal dense Doolittle. The only remaining dense-Doolittle arithmetic bottleneck is proving the displayed dominance inequalities from a concrete pivot/no-cancellation regime or recording that routine-specific regime as an implementation obligation. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```
doolittleUAbsBudget
doolittleLAbsBudget
DoolittleDenseLoopAbsBudgetCertificate.of_literal_doolittle_source_budgets
```

Corollary 658 (Componentwise Doolittle dominance for equation (9)). *For the literal dense-Doolittle budgets above define the upper component quantities*

$$W_{kj}^U = |\hat{A}_{0,kj}| + \sum_{s < k} \left| \text{fl_mul}_{fp}(\hat{L}_{ks}, \hat{U}_{sj}) \right|, \quad E_{kj}^U = \sum_{s < k} |\hat{L}_{ks} \hat{U}_{sj}|,$$

and the lower component quantities

$$W_{ik}^L = |\hat{A}_{0,ik}| + \sum_{s < k} \left| \text{fl_mul}_{fp}(\hat{L}_{is}, \hat{U}_{sk}) \right|, \quad E_{ik}^L = \sum_{s < k} |\hat{L}_{is} \hat{U}_{sk}|, \quad N_{ik}^L = |\hat{n}_{ik}|.$$

Assume $\text{gammaValid}(fp, n)$. If $k \leq j$ and

$$W_{kj}^U \leq |\widehat{U}_{kj}|, \quad E_{kj}^U \leq |\widehat{U}_{kj}|,$$

then

$$B_{kj}^U \leq (\gamma(fp, k) + u_{fp})|\widehat{U}_{kj}| \leq \gamma(fp, n)|\widehat{U}_{kj}|.$$

If $k < i$ and

$$W_{ik}^L \leq |\widehat{L}_{ik}\widehat{U}_{kk}|, \quad E_{ik}^L \leq |\widehat{L}_{ik}\widehat{U}_{kk}|, \quad N_{ik}^L \leq |\widehat{L}_{ik}\widehat{U}_{kk}|,$$

then

$$B_{ik}^L \leq (\gamma(fp, k) + u_{fp} + u_{fp})|\widehat{L}_{ik}\widehat{U}_{kk}| \leq \gamma(fp, n)|\widehat{L}_{ik}\widehat{U}_{kk}|.$$

Consequently, under the same triangular-shape, literal-update, nonzero-pivot, and $\text{gammaValid}(fp, n)$ hypotheses as the previous corollary, these componentwise no-cancellation inequalities instantiate

$$\text{DoolittleDenseLoopAbsBudgetCertificate}(\widehat{A}_0, \widehat{L}, \widehat{U}, fp, B^U, B^L).$$

This closes the dense-Doolittle dominance route from visible componentwise no-cancellation hypotheses. It still does not derive those component inequalities from a concrete pivoting or no-cancellation implementation regime; that routine-specific source remains an implementation obligation. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```
doolittleUWorkAbs
doolittleUProductAbs
doolittleLWorkAbs
doolittleLProductAbs
doolittleLNumeratorAbs
doolittleUAbsBudget_le_compression_of_component_dominance
doolittleLAbsBudget_le_compression_of_component_dominance
DoolittleDenseLoopAbsBudgetCertificate.of_literal_doolittle_component_dominance
```

Corollary 659 (Exact-product Doolittle margins for equation (9)). *For every real pair x, y , the primitive multiplication model gives the rounded-product growth estimate*

$$|\text{fl_mul}_{fp}(x, y)| \leq (1 + u_{fp})|xy|.$$

Hence the componentwise dominance hypotheses in the previous corollary may be checked from exact-product no-cancellation margins. For an upper entry define

$$E_{kj}^U = \sum_{s < k} |\widehat{L}_{ks}\widehat{U}_{sj}|.$$

If $k \leq j$ and

$$|\widehat{A}_{0,kj}| + (1 + u_{fp})E_{kj}^U \leq |\widehat{U}_{kj}|,$$

then

$$W_{kj}^U \leq |\widehat{U}_{kj}|, \quad E_{kj}^U \leq |\widehat{U}_{kj}|.$$

For a lower entry define

$$E_{ik}^L = \sum_{s < k} |\widehat{L}_{is} \widehat{U}_{sk}|.$$

If $k < i$ and

$$|\widehat{A}_{0,ik}| + (1 + u_{fp}) E_{ik}^L \leq |\widehat{L}_{ik} \widehat{U}_{kk}|,$$

then

$$W_{ik}^L \leq |\widehat{L}_{ik} \widehat{U}_{kk}|, \quad E_{ik}^L \leq |\widehat{L}_{ik} \widehat{U}_{kk}|.$$

Consequently, under the same triangular-shape, literal-update, nonzero-pivot, and $\text{gammaValid}(fp, n)$ hypotheses as above, if the exact-product margins hold for all upper and lower entries and the rounded lower numerator also satisfies

$$N_{ik}^L \leq |\widehat{L}_{ik} \widehat{U}_{kk}| \quad (k < i),$$

then the literal dense-Doolittle absolute-budget certificate is instantiated:

$$\text{DoolittleDenseLoopAbsBudgetCertificate}(\widehat{A}_0, \widehat{L}, \widehat{U}, fp, B^U, B^L).$$

This closes the product-growth handoff from exact-product no-cancellation margins to the componentwise dense-Doolittle dominance route. It is an implementation-facing non-probability arithmetic theorem: products computed inside the fold pay the explicit $(1 + u_{fp})$ growth factor, while sampling probabilities and sampling laws remain exact mathematical inputs. Proving the displayed exact-product margins and the lower rounded-numerator dominance from a concrete pivoting/no-cancellation implementation regime remains a separate routine obligation.

Lean formalization.

```
fl_mul_abs_le_one_add_u_mul_abs_mul
doolittleUWorkAbs_le_of_exact_product_margin
doolittleUProductAbs_le_of_exact_product_margin
doolittleLWorkAbs_le_of_exact_product_margin
doolittleLProductAbs_le_of_exact_product_margin
DoolittleDenseLoopAbsBudgetCertificate.of_literal_doolittle_exact_product_margins
```

Corollary 660 (Exact-product lower-numerator margins for equation (9)). *Let $k < i$, assume $\text{gammaValid}(fp, k)$, and define the exact-product lower prefix magnitude*

$$E_{ik}^L = \sum_{s < k} |\widehat{L}_{is} \widehat{U}_{sk}|,$$

together with the rounded lower numerator magnitude

$$N_{ik}^L = \left\lceil \text{flDoolittleLNumerator}_{fp}(\widehat{A}_0, \widehat{L}, \widehat{U}, i, k) \right\rceil.$$

If the stronger exact-product numerator margin

$$(|\widehat{A}_{0,ik}| + E_{ik}^L) + \left\lceil \gamma(fp, k) (|\widehat{A}_{0,ik}| + (1 + u_{fp}) E_{ik}^L) + u_{fp} E_{ik}^L \right\rceil \leq |\widehat{L}_{ik} \widehat{U}_{kk}|,$$

then

$$N_{ik}^L \leq |\widehat{L}_{ik} \widehat{U}_{kk}|.$$

Consequently, under triangular-shape hypotheses, literal flDoolittleUEntry and flDoolittleLEntry update equalities, nonzero computed pivots $\widehat{U}_{kk}$, and $\text{gammaValid}(fp, n)$, if the upper exact-product margins

$$|\widehat{A}_{0,kj}| + (1 + u_{fp})E_{kj}^U \leq |\widehat{U}_{kj}| \quad (k \leq j),$$

the lower exact-product margins

$$|\widehat{A}_{0,ik}| + (1 + u_{fp})E_{ik}^L \leq |\widehat{L}_{ik}\widehat{U}_{kk}| \quad (k < i),$$

and the lower exact-product numerator margins above hold for all required entries, then

$$\text{DoolittleDenseLoopAbsBudgetCertificate}(\widehat{A}_0, \widehat{L}, \widehat{U}, fp, B^U, B^L).$$

Thus the lower rounded-numerator dominance hypothesis in the previous corollary can be discharged by an explicit source-shaped margin. The theorem charges all non-probability arithmetic computed by the literal dense-Doolittle fold: rounded products, the lower subtraction fold, and the lower division surface inherited from the source-budget certificate. Sampling probabilities and sampling laws remain exact mathematical inputs. Proving these margins from a concrete pivoting/no-cancellation implementation regime remains an open routine obligation.

Lean formalization.

`doolittleLNumeratorAbs_le_of_exact_product_numerator_margin`

`DoolittleDenseLoopAbsBudgetCertificate.of_literal_doolittle_exact_product_numerator_margins`

Corollary 661 (Exact-target Doolittle gaps for equation (9)). *For the literal dense-Doolittle routine define the exact pre-rounded upper and lower targets*

$$T_{kj}^U = \widehat{A}_{0,kj} - \sum_{s < k} \widehat{L}_{ks} \widehat{U}_{sj}, \quad T_{ik}^L = \widehat{A}_{0,ik} - \sum_{s < k} \widehat{L}_{is} \widehat{U}_{sk}.$$

Let

$$E_{kj}^U = \sum_{s < k} |\widehat{L}_{ks} \widehat{U}_{sj}|, \quad E_{ik}^L = \sum_{s < k} |\widehat{L}_{is} \widehat{U}_{sk}|.$$

The rounded-product growth estimates for the products actually used by the subtraction folds are

$$\sum_{s < k} |\text{fl_mul}_{fp}(\widehat{L}_{ks}, \widehat{U}_{sj})| \leq (1 + u_{fp})E_{kj}^U, \quad \sum_{s < k} |\text{fl_mul}_{fp}(\widehat{L}_{is}, \widehat{U}_{sk})| \leq (1 + u_{fp})E_{ik}^L.$$

Consequently the literal upper fold satisfies the exact-target residual bound

$$|T_{kj}^U - \widehat{U}_{kj}| \leq R_{kj}^U := \gamma(fp, k)(|\widehat{A}_{0,kj}| + (1 + u_{fp})E_{kj}^U) + u_{fp}E_{kj}^U,$$

assuming $\widehat{U}_{kj} = \text{flDoolittleUEntry}_{fp}(\widehat{A}_0, \widehat{L}, \widehat{U}, k, j)$. For the lower numerator,

$$|T_{ik}^L - \text{flDoolittleLNumerator}_{fp}(\widehat{A}_0, \widehat{L}, \widehat{U}, i, k)| \leq R_{ik}^N := \gamma(fp, k)(|\widehat{A}_{0,ik}| + (1 + u_{fp})E_{ik}^L) + u_{fp}E_{ik}^L.$$

If the computed pivot is nonzero and

$$\widehat{L}_{ik} = \text{flDoolittleLEntry}_{fp}(\widehat{A}_0, \widehat{L}, \widehat{U}, i, k),$$

then the lower entry multiplied back by the computed pivot satisfies

$$|T_{ik}^L - \widehat{L}_{ik}\widehat{U}_{kk}| \leq R_{ik}^L := R_{ik}^N + u_{fp} \left| \text{flDoolittleLNumerator}_{fp}(\widehat{A}_0, \widehat{L}, \widehat{U}, i, k) \right|.$$

Thus exact-target no-cancellation gaps imply the source-shaped margin hypotheses used above. If $k \leq j$ and

$$|\hat{A}_{0,kj}| + (1 + u_{fp})E_{kj}^U + R_{kj}^U \leq |T_{kj}^U|,$$

then

$$|\hat{A}_{0,kj}| + (1 + u_{fp})E_{kj}^U \leq |\hat{U}_{kj}|.$$

If $k < i$, $\hat{U}_{kk} \neq 0$, and

$$|\hat{A}_{0,ik}| + (1 + u_{fp})E_{ik}^L + R_{ik}^L \leq |T_{ik}^L|,$$

then

$$|\hat{A}_{0,ik}| + (1 + u_{fp})E_{ik}^L \leq |\hat{L}_{ik}\hat{U}_{kk}|.$$

Finally, if the stronger exact-target lower gap

$$(|\hat{A}_{0,ik}| + E_{ik}^L + R_{ik}^N) + R_{ik}^L \leq |T_{ik}^L|$$

holds, then

$$|\hat{A}_{0,ik}| + E_{ik}^L + R_{ik}^N \leq |\hat{L}_{ik}\hat{U}_{kk}|.$$

Therefore, under triangular-shape hypotheses, literal `flDoolittleUEntry` and `flDoolittleLEntry` update equalities, nonzero computed pivots, and `gammaValid(fp, n)`, if the displayed exact-target gaps hold for all required entries, then

$$\text{DoolittleDenseLoopAbsBudgetCertificate}(\hat{A}_0, \hat{L}, \hat{U}, fp, B^U, B^L).$$

This is an implementation-facing non-probability arithmetic result: it charges rounded products, subtraction folds, lower rounded division, and multiplication by the computed pivot. It strictly reduces the remaining dense-Doolittle obligation to proving the exact-target gaps from a concrete pivoting/off-diagonal-control/no-cancellation invariant. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```
doolittleUExactTarget
doolittleLEExactTarget
doolittleUExactTargetResidualBudget
doolittleLEExactTargetNumeratorResidualBudget
doolittleLEExactTargetEntryResidualBudget
doolittleURoundedProductAbsSum_le_one_add_u_productAbs
doolittleLRoundedProductAbsSum_le_one_add_u_productAbs
doolittleUExactTarget_residual_abs_le_of_literal
doolittleLEExactTarget_numerator_residual_abs_le_of_literal
doolittleLEExactTarget_entry_residual_abs_le_of_literal
doolittleUExactProductMargin_of_exactTarget_gap
doolittleLEExactProductMargin_of_exactTarget_gap
doolittleLEExactProductNumeratorMargin_of_exactTarget_gap
DoolittleDenseLoopAbsBudgetCertificate.of_literal_doolittle_exact_target_gaps
```

Corollary 662 (Triangle obstruction for exact-target Doolittle gaps). *The exact-target transfer above is a valid conditional theorem, but its source-facing exact-target gaps are not an ordinary no-cancellation condition in nondegenerate floating point. The exact targets always obey the triangle upper bounds*

$$|T_{kj}^U| \leq |\hat{A}_{0,kj}| + E_{kj}^U, \quad |T_{ik}^L| \leq |\hat{A}_{0,ik}| + E_{ik}^L.$$

Consequently, if the upper exact-target gap from the previous corollary holds, then

$$u_{fp}E_{kj}^U + R_{kj}^U \leq 0.$$

If the lower entry exact-target gap holds, then

$$u_{fp}E_{ik}^L + R_{ik}^L \leq 0.$$

If the stronger lower numerator exact-target gap holds, then

$$R_{ik}^N + R_{ik}^L \leq 0.$$

Equivalently, the corresponding exact-target gap is contradictory whenever the displayed excess term is positive.

Thus LR.1bp remains useful as a conditional transfer from already-supplied exact-target gaps to stored-entry margins, but LR.1bq rules out using ordinary pivoting/no-cancellation or off-diagonal-control estimates to supply those gaps when the FP residual excess is positive. The live dense-Doolittle D5 route must instead prove stored-entry/component dominance directly, revise the compression surface, or move to sharper product/operator certificates. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```
doolittleUExactTarget_abs_le_source_plus_productAbs
doolittleLExactTarget_abs_le_source_plus_productAbs
doolittleUExactTarget_gap_excess_nonpos
doolittleUExactTarget_gap_false_of_positive_excess
doolittleLExactTarget_gap_excess_nonpos
doolittleLExactTarget_gap_false_of_positive_excess
doolittleLExactTarget_numerator_gap_excess_nonpos
doolittleLExactTarget_numerator_gap_false_of_positive_excess
```

Corollary 663 (Norm-generic head-tail surface for equation (9)). *Let $\xi_{m,n}$ be a rectangular norm-like functional on $\mathbb{R}^{m \times n}$ satisfying*

$$\xi_{m,n}(X) \geq 0, \quad \xi_{m,n}(X - Y) \leq \xi_{m,n}(X) + \xi_{m,n}(Y) \quad \text{for all } X, Y \in \mathbb{R}^{m \times n}.$$

Let $Z \in \mathbb{R}^{n \times r}$, let $P \in \mathbb{R}^{m \times m}$, and suppose that P factors through the exact column sketch:

$$P_{ij} = \sum_{a=1}^r (AZ)_{ia} C_{aj}.$$

Assume also that P exactly reproduces the sketch columns,

$$P(AZ) = AZ,$$

and that A has an exact head-tail split

$$A = H + T, \quad H = (AZ)W$$

for some coefficient table W . If the two visible norm bounds

$$\xi_{m,n}(T) \leq \tau, \quad \xi_{m,n}(PT) \leq \kappa$$

hold with $\tau, \kappa \geq 0$, then

$$\text{rank}(PA) \leq r, \quad \xi_{m,n}(A - PA) \leq \tau + \kappa.$$

Consequently, if A_k is a certified best rank- k approximation with respect to $\xi_{m,n}$ and

$$\tau + \kappa \leq \rho \xi_{m,n}(A - A_k),$$

then

$$\text{rank}(A_k) \leq k, \quad \text{rank}(PA) \leq r, \quad \xi_{m,n}(A - PA) \leq \rho \xi_{m,n}(A - A_k).$$

This is an exact-object, norm-abstract theorem surface. It proves the equation (9) rank/residual algebra for any supplied norm-like functional and head-tail certificate. It does not instantiate a concrete unitarily invariant norm, prove right/left orthogonal invariance, prove the singular-value comparison for the cross term, or prove Eckart–Young for that norm. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```

RectNormLike
lowRankResidualNorm
IsBestRankApproxNorm
Equation9ResidualNormCertificate
Equation9HeadTailSketchNormCertificate
Equation9HeadTailSketchNormCertificate.to_residualNormCertificate
equation9RankResidualNormSurface
equation9RelativeResidualNormSurface
equation9HeadTailSketchNormRankResidualSurface
equation9HeadTailSketchNormRelativeResidualSurface

```

Corollary 664 (Frobenius instance of the norm-generic equation (9) surface). *Define*

$$\xi_{m,n}^F(X) = \|X\|_F, \quad X \in \mathbb{R}^{m \times n}.$$

Then $\xi_{m,n}^F$ is a rectangular norm-like functional:

$$\xi_{m,n}^F(X) \geq 0, \quad \xi_{m,n}^F(X - Y) \leq \xi_{m,n}^F(X) + \xi_{m,n}^F(Y).$$

Moreover, for every orthogonal $U \in \mathbb{R}^{m \times m}$ and orthogonal $V \in \mathbb{R}^{n \times n}$,

$$\xi_{m,n}^F(UX) = \xi_{m,n}^F(X), \quad \xi_{m,n}^F(XV) = \xi_{m,n}^F(X).$$

Let $Z \in \mathbb{R}^{n \times r}$, let $P \in \mathbb{R}^{m \times m}$, and suppose that P factors through the exact column sketch and reproduces it:

$$P_{ij} = \sum_{a=1}^r (AZ)_{ia} C_{aj}, \quad P(AZ) = AZ.$$

Assume that

$$A = H + T, \quad H = (AZ)W,$$

and that the visible Frobenius bounds

$$\|T\|_F \leq \tau, \quad \|PT\|_F \leq \kappa, \quad \tau, \kappa \geq 0$$

hold. Then

$$\text{rank}(PA) \leq r, \quad \|A - PA\|_F \leq \tau + \kappa.$$

Consequently, if A_k is a certified best rank- k Frobenius approximation and

$$\tau + \kappa \leq \rho \|A - A_k\|_F,$$

then

$$\text{rank}(A_k) \leq k, \quad \text{rank}(PA) \leq r, \quad \|A - PA\|_F \leq \rho \|A - A_k\|_F.$$

This is a concrete Frobenius instantiation of the norm-generic exact-object surface above. It closes the ‘RectNormLike’ bridge for the repository Frobenius norm and exposes the already-formalized left/right orthogonal invariance facts. It still does not prove the all-unitarily-invariant norm theorem, singular-value comparison for the equation (9) cross term, or Eckart–Young. Sampling probabilities and laws remain exact mathematical inputs.

Lean formalization.

```
frobRectNormLike
frobRectNormLike_norm
lowRankResidualNorm_frobRectNormLike
IsBestRankApproxFrob.to_norm_frobRectNormLike
Equation9ResidualCertificate.to_norm_frobRectNormLike
Equation9HeadTailSketchCertificate.to_norm_frobRectNormLike
frobRectNormLike_orthogonal_left
frobRectNormLike_orthogonal_right
equation9HeadTailSketchFrobNormRankResidualSurface
equation9HeadTailSketchFrobNormRelativeResidualSurface
```

Corollary 665 (Unitary-invariant norm API for equation (9)). *Let $\xi_{m,n}$ be a rectangular norm-like functional equipped with exact left and right orthogonal invariance:*

$$\xi_{m,n}(UX) = \xi_{m,n}(X), \quad \xi_{m,n}(XV) = \xi_{m,n}(X)$$

for every orthogonal $U \in \mathbb{R}^{m \times m}$, every orthogonal $V \in \mathbb{R}^{n \times n}$, and every $X \in \mathbb{R}^{m \times n}$. The repository packages this data as $\text{UnitaryInvariantRectNormLike}_{m,n}$, extending $\text{RectNormLike}_{m,n}$. The Frobenius norm instance satisfies

$$\text{frobUnitaryInvariantRectNormLike}_{m,n} \longmapsto \text{frobRectNormLike}_{m,n}, \quad \xi_{m,n}(X) = \|X\|_F.$$

For any supplied UnitaryInvariantRectNormLike certificate, the norm-generic equation (9) rank/residual and relative-residual surfaces apply verbatim to its underlying RectNormLike part. In particular, if the exact head/tail sketch certificate and the visible norm bounds of the previous norm-generic corollary are supplied, then

$$\text{rank}(PA) \leq r, \quad \xi_{m,n}(A - PA) \leq \tau + \kappa,$$

and if A_k is a certified best rank- k approximation for $\xi_{m,n}$ and

$$\tau + \kappa \leq \rho \xi_{m,n}(A - A_k),$$

then

$$\text{rank}(A_k) \leq k, \quad \text{rank}(PA) \leq r, \quad \xi_{m,n}(A - PA) \leq \rho \xi_{m,n}(A - A_k).$$

This closes the supplied-norm API layer for unitarily invariant equation (9) surfaces. It still does not prove singular-value comparison, Eckart–Young, or the construction of best-rank certificates from rectangular SVD data. Sampling probabilities and sampling laws remain exact mathematical inputs.

Lean formalization.

```
UnitaryInvariantRectNormLike
UnitaryInvariantRectNormLike.norm_matMulRectLeft
UnitaryInvariantRectNormLike.norm_matMulRectRight
frobUnitaryInvariantRectNormLike
frobUnitaryInvariantRectNormLike_toRectNormLike
frobUnitaryInvariantRectNormLike_norm
equation9RankResidualUnitaryNormSurface
equation9HeadTailSketchUnitaryNormRankResidualSurface
equation9RelativeResidualUnitaryNormSurface
equation9HeadTailSketchUnitaryNormRelativeResidualSurface
```

Remark. The paper-level structural relative-error theorem behind equation (9) is still open. The source-SVD and head-tail bridges above narrow the determinant and residual-certificate sides to exact source factor data plus explicit head/tail radii. The explicit-coefficient and source-coefficient bridges now make the pseudoinverse-derived choice of W concrete, and the Moore–Penrose bridge replaces the generic projector hypothesis by four exact equations for a coefficient table C . The concrete Gram-inverse bridge then instantiates that table as $C_G = \text{nonsingInv}((AZ)^T(AZ))(AZ)^T$ under the exact determinant condition on the sketch Gram matrix, and the orthogonal source head-tail Gram split reduces the determinant side to a source-head sketch Gram plus the tail sketch Gram. The PSD/determinant bridge proves that this split gives a nonzero full sketch-Gram determinant as soon as the source-head sketch Gram is certified positive definite, and the source-factor positive-definiteness bridge now derives that certificate from $\det(\Sigma) \neq 0$ and $\det(V^T Z) \neq 0$. Thus the exact determinant side is closed for supplied source certificates. The source-tail norm reduction now also moves the ambient residual-tail Frobenius obligation to a coordinate residual-tail obligation under a supplied orthonormal left tail basis, and the coordinate-tail factorization exposes the exact $\Sigma_{\perp}(V_{\perp}^T - (V_{\perp}^T Z)(V_k^T Z)^{-1}V_k^T)$ residual with a non-sharp Frobenius submultiplicative bound. The

right-tail block-product theorem now proves $R_\perp V_\perp = I$ and $R_\perp V_k = -(V_\perp^T Z)(V_k^T Z)^{-1}$. The Frobenius block identity then proves

$$\|\Sigma_\perp R_\perp\|_F^2 = \|\Sigma_\perp\|_F^2 + \|\Sigma_\perp(V_\perp^T Z)(V_k^T Z)^{-1}\|_F^2$$

under exact row-completeness of $[V_\perp, V_k]$. The cross-term certificate corollary now converts the supplied Frobenius condition $\|\Sigma_\perp(V_\perp^T Z)(V_k^T Z)^{-1}\|_F \leq \varepsilon\|\Sigma_\perp\|_F$ into the displayed $\sqrt{1 + \varepsilon^2}$ coordinate source-tail rate, and the ambient source-tail certificate now transports that rate through a supplied left-orthonormal $U_\perp$ factor for $T = U_\perp \Sigma_\perp V_\perp^T$. Finally, the projected source-tail coupling certificate uses exact symmetric-idempotent Frobenius contractivity to pass the same bound through a supplied exact orthogonal or Moore–Penrose column-sketch projector $P = (AZ)C$. The transpose-action spectral certificate then proves the Frobenius cross-term condition from the exact right-acting operator hypothesis $\|((V_\perp^T Z)(V_k^T Z)^{-1})^T x\|_2 \leq \varepsilon\|x\|_2$, and composes that certificate through the projected Moore–Penrose source-tail coupling theorem. The ordinary operator certificate corollary now derives that transpose-action hypothesis from the non-transposed certificate $\|(V_\perp^T Z)(V_k^T Z)^{-1}x\|_2 \leq \varepsilon\|x\|_2$. The computed cross-factor perturbation corollary now allows a computed non-probability factor $\widehat{M}$ with operator radius ε and Frobenius distance τ from the exact $(V_\perp^T Z)(V_k^T Z)^{-1}$ factor to feed the same projected Moore–Penrose theorem with radius $\sqrt{1 + (\varepsilon + \tau)^2}$. Its entrywise specialization converts a uniform computed-entry budget $|M_{ab} - \widehat{M}_{ab}| \leq \eta$ into the Frobenius radius $\sqrt{qr}\eta$, giving projected radius $\sqrt{1 + (\varepsilon + \sqrt{qr}\eta)^2}$. The component-certified computed cross-factor corollary now derives that entrywise budget from separate computed cross-gram, inverse-factor, and rounded-product radii α, β, ρ , replacing η by $\alpha + \beta + \rho$. The concrete floating-point cross-Gram corollary now instantiates the $V_\perp^T Z$ side of that component certificate with the repository ‘fl_matMul’ dot-product budget $\gamma(fp, n) \sum_j |(V_\perp)_{ja}| |Z_{jb}|$. The square cross-Gram corollary similarly certifies the computed inverse input $V_k^T Z$ and its uniform Frobenius radius $\sqrt{rr}\omega$. The inverse-entry adapter now turns a supplied entrywise computed-inverse certificate $|Y - \widehat{Y}| \leq \eta$ and computed-left row budget χ into the component radius $\chi\eta$, hence replaces $\alpha + \beta + \rho$ by $\alpha + \chi\eta + \rho$ in the same projected computed-factor bound. The concrete rounded-product corollary now instantiates the remaining ρ component by the repository fl_matMul budget $\gamma(fp, r) \sum_b |\widehat{X}_{ab}| |\widehat{Y}_{bc}|$, so the final product arithmetic is closed for supplied inverse and operator certificates. The computed-product Frobenius certificate corollary then replaces that abstract operator certificate by the visible condition $\|\widehat{M}\|_F \leq \varepsilon$, using the deterministic Frobenius-to-operator handoff. The product absolute-sum Frobenius certificate corollary further supplies that condition from $\sum_b |\widehat{X}_{ab}| |\widehat{Y}_{bc}| \leq \kappa$ and the same final product radius ρ , giving $\|\widehat{M}\|_F \leq \sqrt{qr}(\kappa + \rho)$. The perturbed-inverse certificate corollary similarly replaces the raw inverse-entry assumption $|Y - \widehat{Y}| \leq \eta$ by the Higham §13.1 contract $(A_0 + \Delta A)\widehat{Y} = I$, a componentwise perturbation budget, and a visible inverse-sensitivity bound. The Method-A LU inverse certificate corollary goes one step further for a specific computed inverse surface: the columns of $\widehat{Y} = \text{methodAComputedInverse}_{fp,r}(\widehat{L}, \widehat{U})$ are generated by the repository forward/back substitution routines, and an ‘LUBackwardError’ certificate for $V_k^T Z$ plus the visible $(3\gamma + \gamma^2)$ -budget supplies the same η radius. The computed-input Method-A corollary now removes the silent exact-input assumption from that LU surface: if the LU routine is certified for the rounded square cross Gram $\widehat{A}_0 = \text{flRightSketchCrossGram}_{fp}(V_k, Z)$, the rounded-input dot-product budget transfers the certificate to $V_k^T Z$ at coefficient $\varepsilon_{LU} + \mu$, and the Method-A sensitivity budget uses $(\varepsilon_{LU} + \mu) + 2\gamma + \gamma^2$. The Doolittle-generated computed-input corollary now discharges that displayed LU certificate from a dense Doolittle recurrence certificate for $\widehat{A}_0$, giving coefficient $(\gamma + \mu) + 2\gamma + \gamma^2$ for Method A. The dense-loop Doolittle residual-compression corollary then exposes the literal rounded upper-entry fold, lower-entry numerator fold, and rounded division that a dense implementation computes, and proves that the two displayed compression inequalities are

sufficient to produce the same ‘DoolittleLU’ witness. The absolute-budget dense-loop corollary lowers that certificate one step further: a concrete implementation may prove ordinary absolute residual budgets B^U, B^L , then prove their dominance by the displayed relative γ -radii, and the same Method-A equation-(9) bounds follow. The rounded-product Doolittle fold-budget corollary now supplies the subtraction-accumulation part of those absolute budgets for the literal upper fold and lower numerator fold, measured against the rounded products actually subtracted by the implementation. The subsequent exact-product, masked-prefix, lower-division, product-margin, and lower-numerator-margin corollaries pay the rounded-product transfer, prefix reindexing, computed-pivot division surface, and source-shaped stored-entry margins. The exact-target gap corollary gives a valid conditional transfer from exact pre-rounded target gaps to those stored-entry margins, with explicit residual budgets for all non-probability arithmetic in the literal routine. The following triangle-obstruction proposition then shows that this exact-target source route is too strong whenever the FP excess is positive, so it should not be treated as the remaining ordinary dense-Doolittle no-cancellation proof path. The norm-generic head-tail corollary now closes the exact algebraic rank/residual theorem surface for any supplied ξ satisfying the displayed nonnegativity and triangle assumptions. The Frobenius-instance corollary now instantiates that surface for $\xi(X) = \|X\|_F$, transports the Frobenius best-rank and head-tail certificates into the norm-generic API, and exposes the existing left/right orthogonal-invariance handoffs. The concrete all-unitarily-invariant norm theorem still requires the missing norm API, singular-value comparison, and Eckart–Young instantiations. The next missing foundations are the rectangular SVD existence/Eckart–Young layer, general unitarily invariant variants of the cross-term condition, randomness-derived certificates for the displayed operator condition, a non-vacuous stored-entry/component-dominance or revised-compression source for the dense-Doolittle Method-A route, and computed SVD/projector/Gram, coefficient, and downstream product routines.

11 What Is Not Proved

The repository does not yet prove all RandNLA accuracy theorems mentioned in the CACM survey. The open paper-level items are tracked in `docs/RANDNLA_CACM_NOT_PROVED_LEDGER.md`. The active bottleneck ledger is `docs/RANDNLA_CACM_BOTTLENECK_LEDGER.md`; it names the exact blocking theorem family, closed dependencies, open dependencies, and rejected proof routes for stalled paper-level rows. The live theorem ledger is `docs/RANDNLA_CACM_THEOREM_LEDGER.md`; it records extracted claims, events, hypotheses, Lean names, and next proof steps. The proof-source ledger is `docs/RANDNLA_CACM_PROOF_SOURCE_LEDGER.md`; it records external primary sources and chosen proof routes for incomplete paper proofs. In particular, the repository still does not prove:

- the untruncated/general-rectangular version of Algorithm 1 equation (2). The source-aligned square theorem with truncation $\tau = \varepsilon/(2n)$, source sample budget, deterministic truncation transfer, and derived floating-point γ -budget transfer is now formalized for the nonzero truncated squared-magnitude denominator case. Earlier weaker fixed-vector, finite-test-set, supplied finite-cover/Markov, Frobenius/Markov, and scalar-entry/union-bound routes remain available as fallback statements. The residual has been expressed as a sum of zero-mean increments with second-moment bounds, and a self-adjoint dilation of that residual is now formalized with trace-of-square identities plus quadratic-form, Loewner, PSD, and trace variance proxies. The one-step zero-mean and variance proxies have also been lifted through the independent product trace law to give a mean-zero full dilation residual and trace-law expected squared-increment variance matrix bounds. A generic scalar MGF factorization, exponential-Markov tail for sums of arbitrary one-step trace statistics, no-hidden-Lieb iid C^* trace-MGF

domination, finite-real trace-exponential adapter, residual-dilation trace-MGF instantiation, trace-exponential Markov-to-eigenvalue step, scalar Bernstein parabola constants, and explicit scalar-to-operator CFC Bernstein lift are also formalized. The generic one-sample matrix-CGF/log-MGF variance-proxy bound $K \preceq g(\theta, R)V$, its support-aware variant, and its truncated Algorithm 1 dilation-increment instantiation are now formalized too. The one-sample C^* dilation variance proxy, the negative-support lower-tail companion, the positive and negative scalar trace-MGF bounds, and the parameterized upper-tail and two-sided eigenvalue tails are also formalized, including the explicit $1 - \delta$ scaled-eigenvalue corollary with $T = \log(2B/\delta)$. The scaled event is converted to a rectangular spectral event at radius $\log(2B/\delta)/\theta$, and the exact Bennett optimizer $\theta = \log(1 + Lr/W)/L$ gives a radius- r theorem under the corresponding Bennett budget. The square source route now also has the sharp $V = n\|\hat{A}\|_F^2/s^2$ variance proxy and $\text{no-}\sqrt{2}$ truncated support radius threaded through the two-sided $1 - \delta$ eigenvalue skeleton, plus matching source-sharp scaled-radius and Bennett-radius rectangular spectral-event corollaries. Both the conservative denominator bridge $q \leq r^2/(2W + Lr)$ and the source denominator bridge $q \leq r^2/(2W + \frac{2}{3}Lr)$ are closed. The final source sample-budget theorem, deterministic truncation transfer, and support-aware floating-point gamma-budget transfer are also closed. The product-grid cover reduction and cardinality $|\alpha^n| = |\alpha|^n$ are formalized, but the covering route still lacks an explicit fine one-dimensional interval grid and sharp finite-net scalar tails; moreover, the primary Drineas–Zouzias theorem uses a truncated matrix $\hat{A}$; the deterministic truncation transfer, probability-one support, bounded-increment, two-sided Loewner boundedness, bounded-square, and simultaneous Bernstein-boundedness prerequisites for sampling $\hat{A}$, plus a weak accumulated sL spectral consequence and a support-aware finite-test MGF theorem for truncated dilation quadratic forms, a one-sided dilation Loewner event-transfer theorem, local-to-mathlib symmetry/PSD/Loewner bridges, a mathlib-backed finite Hermitian eigenvalue/trace/PSD bridge, and an eigenvalue-event plus per-eigenvalue union-bound transfer layer, as well as the scalar matrix-exponential normalization $\text{tr}(\exp(LI)) = de^L$, diagonal trace-exponential normalization, matrix-exponential symmetry preservation, and the Hermitian CFC trace identity $\text{tr}(E_{\text{cfc}}(M)) = \sum_i e^{\lambda_i(M)}$, plus the repository-native matrix-exponential trace identity $\text{tr}(\text{Exp}(M)) = \sum_i e^{\lambda_i(M)}$, its lower-tail companion $\text{tr}(\text{Exp}(-M)) = \sum_i e^{-\lambda_i(M)}$, and the upper- and lower-tail MGF-to-eigenvalue tail/complement interfaces $\mathbb{P}\{\exists i : \lambda_i(M) \geq T\} \leq e^{-T}\mathbb{E}[\text{tr}(\text{Exp}(M))]$, $\mathbb{P}\{\exists i : \lambda_i(M) \leq T\} \leq e^T\mathbb{E}[\text{tr}(\text{Exp}(-M))]$, together with the two-sided $\mathbb{P}\{\forall i : |\lambda_i(M)| < T\}$ combiner and the scalar Loewner-to-trace-exponential bound $M \preceq cI \Rightarrow \text{tr}(\text{Exp}(M)) \leq de^c$, plus the complex C^* -matrix embedding of finite real matrices and its subtraction, identity, scalar-identity, self-adjointness, local PSD, local Loewner-order preservation, strict-positive scalar-identity regularization, and regularized operator-log preservation lemmas, together with the complex C^* -matrix trace bridge back to the repository-native finite real trace, cyclic trace algebra $\text{tr}_{\mathbb{C}}(XY) = \text{tr}_{\mathbb{C}}(YX)$, spectral-order trace positivity/monotonicity, and scalar CFC-exponential trace normalization $\text{tr}_{\mathbb{C}}(\exp_{\text{cfc}}(aI)) = e^ad$, plus finite-probability expectation for complex scalars and complex C^* -matrix-valued random variables, trace-expectation interchange, compatibility with embedded finite real matrices, complex and real expectation weighted-sum forms, C^* -order nonnegativity, strict positivity, and monotonicity preservation, positive-support existence for finite probability laws, strict-positive regularization after adding εI , and the finite Jensen adapter for concave real-valued functions of C^* -matrix random variables, plus the self-adjoint $\log(\exp X) = X$ identities in normed-space, complex CFC, and real CFC exponential forms, self-adjoint exponential strict positivity in normed-space, complex CFC, and real CFC forms, and the strictly positive $\exp(\log A) = A$ identities in normed-space, complex CFC, and real CFC exponential forms, together with the normalization of the local

Lieb functional against the standard normed-algebra exponential and the $H = 0$ trace case, plus the affine $H = 0$ special case of the local Lieb concavity target, and the conditional and unconditional one-step Tropp–Jensen trace-MGF inequalities, together with scalar and finite-vector relative-entropy nonnegativity, the finite log-sum inequality, scalar and finite-vector relative-entropy joint convexity, relative-entropy vocabulary $D(A; B)$, the diagonal normalization $D(A; A) = 0$, and scalar-identity matrix relative-entropy nonnegativity $D(aI; bI) \geq 0$ for $a, b > 0$, as well as the real diagonal matrix reduction $D(\Delta(a); \Delta(b)) = D_{\text{vec}}(a; b)$ and its nonnegativity and joint convexity for positive diagonals, plus the left/right multiplication endomorphism substrate L_A, R_A , their product/power algebra, and their unit/strict-positive invertibility, plus the explicit ratio endomorphism $L_X R_A^{-1}$ and its base-point normalization, plus the finite Kronecker lift substrate $A \otimes I, I \otimes H$, its affine laws, commutation/product identities, positive-definiteness preservation, trace normalizations, product-index vectorization action, and vectorized-identity trace pairing and polynomial powered trace-pairing identity, plus its finite-polynomial packaging, plus continuity of the finite quadratic form and polynomial evaluation in the matrix argument, plus the Hansen–Pedersen source-route split into ordinary positive-cone operator convexity, transfer, and assembled Jensen targets, including identity-function sanity checks, the assembly adapter from the concrete inputs to the assembled target, and the direct unit-interval-kernel proof of the ordinary $x \log x$ operator-convexity input, plus Bendat–Sherman-route derivative monotonicity of $1 + \log x$ on the strictly positive cone, plus the source-faithful first-divided-difference route names and scalar normalizations, plus the unital fractional-kernel monotonicity theorem for $x \mapsto 1 - (1 + x)^{-1} = x/(1 + x)$ and the scaled theorem for $x \mapsto x/(s + x)$, $s > 0$, plus finite nonnegative-combination closure for such kernels on the nonnegative C^* -matrix cone, plus the conditional adapter from divided-difference monotonicity and the Bendat–Sherman divided-difference bridge to operator convexity, plus the base-point CFC normalization and the full divided-difference monotonicity theorem, plus finite matrix inverse convexity on the positive-definite cone by Schur complements and its finite C^* -matrix CFC inverse-kernel bridge, plus shifted inverse-kernel convexity for $x \mapsto (s + x)^{-1}$, $s > 0$, plus the corrected scalar $x(x - 1)/(u + (1 - u)x)$ kernel identity, its CFC affine-plus-shifted-inverse decomposition, the CFC integral representation of $x \log x$, and the integrated direct proof of ordinary positive-cone operator convexity of $x \log x$, plus the conditional relative-entropy route theorem showing that joint convexity and the normalized variational formula imply the local Lieb trace-concavity target, and the optimizer-candidate equality part of that normalized formula, plus the reduction of variational maximality to relative-entropy nonnegativity, plus the reduction of relative-entropy nonnegativity to the generalized Klein first-order trace inequality, the equivalence between that first-order inequality and local relative-entropy nonnegativity, the Hermitian spectral-overlap expansion, and the positive-spectrum scalar entropy first-order specialization, plus the bridge from the separated Hermitian entropy statement to the compact complex C^* -matrix logarithm form of generalized Klein, local relative-entropy nonnegativity, the normalized entropy variational formula, and the reduction from joint convexity alone to the local Lieb target, the generic CFC Bochner-integral order lift under the `cfc_integral` side conditions, the scalar-integral-to-CFC equality for the normalized logarithmic kernel, the scalar normalization back to $x \log x$, the auxiliary square-kernel affine-plus-shifted-inverse decomposition, and normalized-kernel CFC monotonicity, the all-finite-size packaging of ordinary $x \log x$ operator convexity, and the Hansen–Pedersen block-compression algebra plus range-projection algebra plus block-diagonal star/order substrate plus block-diagonal CFC decomposition plus unitary CFC conjugation, strict-positive domain preservation, block-compression domain preservation, rectangular compression order preservation, the reflection-average CFC pinching inequality, and the shifted-

inverse kernel nonlinear corner identity, the compression-integral assembly, the full $x \log x$ corner identity, and the concrete two-point Hansen–Pedersen Jensen theorem, plus the affine-corrected normalized entropy-kernel ordinary operator-convexity and Jensen theorems, plus the CFC square-root and inverse-square-root perspective substrate, plus the ordinary finite perspective theorem for the normalized entropy kernel, plus the finite-polynomial Kronecker trace-representation layer and the source-faithful right-multiplication polynomial identity for $v_I^* p(L_X R_A^{-1}) R_A v_I$, including finite-matrix polynomial CFC, Weierstrass approximation wrappers, and the explicit uniform-approximation transfer plus the positive-spectrum construction of the approximating polynomial sequence for the entropy-kernel CFC trace term, the finite-polynomial overlap formulas for $\text{tr}(X^k A (A^{-1})^k)$ and real-polynomial sums, the common limiting overlap theorem, and the source-faithful superoperator trace-representation theorem, plus the product-index L_X/R_A lift domain lemmas and the real quadratic-form monotonicity needed to extract scalar inequalities from product-index Loewner inequalities, plus joint convexity of the ordinary product-index perspective trace $v_I^* P_f(L_X, R_A) v_I$, and the first CFC-commutation step showing that L_X commutes with $R_A^{-1/2}$ so the normalized perspective argument can be reordered, plus the inverse-square-root square identity putting that normalized argument in the $L_X R_A^{-1}$ shape and proving this ratio commutes with $R_A^{1/2}$, the final CFC transport and product-index equality bridge $v_I^* P_f(L_X, R_A) v_I = v_I^* f(L_X R_A^{-1}) R_A v_I$, finite-dimensional relative-entropy joint convexity, arbitrary- H Lieb trace concavity, and the one-step Tropp trace-MGF inequality, together with the one-coordinate Algorithm 1 product-law adapter from this one-step inequality to the squared-magnitude trace sampler and the iid sampled-sum trace-MGF iteration over the product trace law, and the source-aligned scalar matrix-CGF/Bernstein constants proving the half-budget event for sampling $\hat{A}$, are now formalized. The advisory Bendat–Sherman bridge from divided-difference monotonicity to operator convexity remains advisory, and the exact untruncated squared-magnitude law remains a separate theorem target. A locality check also found that mathlib currently supplies operator-log monotonicity but explicitly lists operator-log concavity, operator convexity of $x \log x$, and operator concavity of t^p as TODOs in its continuous-functional-calculus files. The repository now supplies the $x \log x$ operator-convexity theorem and finite-dimensional Lieb theorem locally, the iid trace-MGF iteration locally, and the source-aligned Algorithm 1 Bernstein specialization locally; no generic matrix Bernstein theorem is used as a hidden hypothesis;

- a reusable general-purpose matrix Bernstein or Khintchine API beyond the specialized Algorithm 1 and Algorithm 2 source-aligned routes;
- a cosmetic one-hypothesis restatement of the sharper Algorithm 2 leverage-score equation (7) sample budget. The current Lean theorem already proves the source-aligned finite-Loewner high-probability event and the matching floating-point finite-Loewner transfer from two explicit Bennett sample-budget inequalities, one for the upper tail and one for the lower tail. What remains here is only a presentation simplification if one wants to replace those two sufficient inequalities by a single more conservative displayed condition;
- the rectangular QR/preconditioner FP theorem for equation (8). The sampled-row objective algebra itself is now formalized with canonical coordinates, the augmented-span basis theorem is formalized for $d = \dim \text{span}\{A_{*,j}, b\}$, the exact augmented-span Bennett sample-budget least-squares theorem is formalized for $d > 1$, the abstract rounded-Gram FP transfer is formalized with the explicit Gram perturbation radius, the full identity-basis fallback is formalized for $d = m$, and the literal rounded sampled/scaled A, b construction has rowwise residual, deterministic objective perturbation, high-probability rounded-minimizer,

additive solver-gap, forward-error-certificate, perturbed-Gram-system transfer, local QR backward-error-spec transfer, and a concrete normal-equations/Cholesky solver transfer. The deterministic rectangular perturbed-normal-equation adapter, the explicit norm-budget handoff into the local QR specification, and the induced Gram/RHS perturbation-budget lemmas are also formalized. As route-A substrate, left and right multiplication of a rectangular matrix by a compatible square orthogonal matrix are now formalized to preserve the rectangular Frobenius norm, and compatible square left/right factors now satisfy rectangular Frobenius submultiplicativity; rectangular square-left action also has the needed identity, associativity, and additivity lemmas. The rectangular one-step and multi-step orthogonal-transformation accumulation theorems are formalized as well, with the multi-step theorem using a geometric perturbation radius; the matching right-hand-side/vector accumulation theorem is also formalized with the same geometric radius; and the orthogonal handoff into rectangular normal equations is closed. The exact zero-prefix and active-column Householder substrates are also closed. The strong common panel/update contract `HouseholderPanelAppError` is formalized, but the active source-faithful route is now the columnwise contract `HouseholderColumnwisePanelAppError`, matching Higham’s columnwise Householder QR proof in which the perturbation matrix may depend on the panel column. The columnwise accumulation theorem now supplies one final Q , per-column perturbation bounds, and the right-hand-side perturbation bound. The formed-reflector route via `fl_householderApplyExplicit` is proved using the local matrix-vector product theorem. The compact dot/scale/subtract route via `fl_householderApplyCompact` is also proved with an explicit deterministic component budget and a visible budget-domination adapter into `HouseholderAppError` and the columnwise panel contract. The compact supplied-sequence adapter is also closed, so compact panel/RHS updates now feed the columnwise geometric accumulation theorem. The stored trailing loop also has formalized $[R; 0]$ /top-RHS shape facts, top-block triangular-solve handoffs into `LSQRSolveBackwardError`, and several visible-domain nonbreakdown routes: active-entry, dimensioned trailing-norm, leading-dual, local inverse row/Frobenius/ ∞ -norm, nonsingular leading blocks, triangular principal minors, and diagonal-dominant triangular inverse estimates. The final Higham-style columnwise stored QR factorization assembly is closed: the Lean theorem now packages the common orthogonal factor, perturbation bounds, final $[R; 0]$ block form, top RHS, and upper triangularity of R . Still open is the solver/preconditioner theorem that discharges the remaining visible nonzero-diagonal, determinant/rank, conditioning or inverse-budget, and compact-smallness hypotheses from the actual computed loop, rather than keeping them as domain assumptions. The proof-source route is now a genuine mathematical choice beyond this columnwise factorization theorem: Higham’s standard Householder QR theorem is columnwise/normwise, while the stronger row-wise weighted least-squares theorem of Cox–Higham requires pivoting or row sorting plus the correct sign convention. The user-selected off-diagonal-control route now has a named Lean invariant `StoredQROffDiagonalControlInvariant` and a solver-facing theorem showing that this invariant yields the local `LSQRSolveBackwardError` certificate with repository final Gram/RHS budgets. That invariant has also been reduced to the source-shaped data `StoredQRSourceOffDiagonalControl`: local triangular leading blocks, nonzero local diagonals, row-wise off-diagonal domination, and the stored-sequence product bound. The triangular leading-block field is now derived from the stored recurrence using the repository’s prefix-lower-zero theorem, so the remaining open source-shaped obligations are nonzero displayed diagonals, row-wise off-diagonal domination, and the compact-product bound. The nonzero-diagonal obligation has been reduced further: previous displayed diagonals follow from the stored signed-alpha prefix-diagonal theorem and a square-root

nonbreakdown budget, leaving the current pivot nonzero condition visible. The current-pivot condition has now also been reduced to nonsingularity of the displayed local leading block S_k by the stored lower-zero determinant bridge, and the square-root budget has been replaced by the dimensioned norm-square margin using the repository's Householder norm-square bridge. The row-wise off-diagonal domination field is now decomposed into a row-growth budget plus a diagonal lower-bound obligation, matching the Cox–Higham pivoted/sorted route. The row-growth half now has a QR-layer propagation theorem from Cox–Higham stage budgets to displayed leading-block off-diagonal row budgets, and a solver-facing handoff now composes those stage budgets with the source-control and `LSQRSolveBackwardError` certificates under explicit diagonal lower bounds. The stage-entry assumptions have also been split into one-step active-block and prefix displayed-row recurrences, matching the zero-prefix/active-row split in the exact signed-stage reflector proof. A finite global compact-step budget now supplies the displayed off-diagonal, active-block, and prefix-row one-step fields from a single scalar recurrence $\Gamma_m B_t + \Delta_t^{\max} \leq B_{t+1}$. The same recurrence, together with nonnegative budgets, now also supplies monotonicity on the QR stage horizon $a \leq b \leq n$. The completed-column preservation field is now derived from the stored-loop lower-zero invariant and the zero-prefix Householder support lemma. The per-pivot compact-product hypotheses are now also packaged by a finite global product budget $\Pi_{\max} < 1$, and the equation (8) objective theorem is assembled from this active/prefix global-product QR route under the remaining visible fields. Still open is proving local leading-block nonsingularity, the norm-square nonbreakdown margin, diagonal lower bounds, row-max scalar defect and row-max comparison fields, and global compact-product smallness from genuine pivoting, ordering, or off-diagonal-growth assumptions, or explicitly keeping them visible as domain hypotheses. The local route-elimination theorems already rule out deriving these visible fields from full rank, whole-matrix determinant nonzeroness, positive trailing norm, exact final QR shape, finite local conditioning, diagonal dominance alone, product smallness alone, active-pivot plus active/off-diagonal budgets alone, or the ordinary exact no-pivot recurrence. For the current unpivoted theorem family, the repository keeps these fields as explicit source/domain assumptions. This scopes the implemented conditional QR stability theorem, but it does not close the generic paper-level claim that the ordinary unpivoted loop supplies them. Adjacent no-pivot QR adapters remain frozen by the bottleneck ledger. The positive QR route has therefore switched to the Cox–Higham pivoted/sorted weighted least-squares theorem family: the signed Householder denominator lower bound $2\|x_{\text{tail}}\|_2^2 \leq v^T v$ and the column-pivoting active-column norm comparison are now formalized. The scalar endpoint $|\beta v^T y_{\text{tail}}| \leq \sqrt{2}$ and the one-step row-growth estimate $|a_{ij} - \phi a_{ik}| \leq (1 + \sqrt{2})B$ are also formalized. The row-sorting stage-accumulation step is now formalized as a scalar induction and sorted-active-row transfer, and the pivot-row active-tail norm step from equations (4.4)–(4.5) is formalized with the explicit ambient factor $\sqrt{m}$. The scalar row-wise accumulated perturbation bound is also formalized as an affine recurrence plus an exact/computed triangle-inequality transfer, and the concrete stored rounded panel step now supplies the per-step compact Householder FP budgets for that recurrence. A direct stored-panel row-magnitude adapter is also formalized, so the source-shaped route can combine exact same-reflector row growth with the concrete compact FP budget without routing through a separate exact-sequence error field. The non-pivot active-row exact same-reflector bridge is also formalized: the signed Householder matrix-vector update is rewritten into Cox–Higham's scalar row update before applying the row-growth bound. The exact signed pivot-row same-reflector bridge is now formalized as well, in the same ambient-dimension form as the local pivot-row norm step. The unified active-row case split and the exact signed-pivot panel sequence propagation theorem are

now formalized too, with the stage budgets and their geometric $c_m^t B_0$ form kept explicit. The active-block one-step adapter is also formalized: a single bound over the active rows and remaining columns supplies the separate row and column fields for one signed-pivot step. The exact sequence now also has active-block wrappers, so the separate row and column stage fields can be replaced by one active-block budget family at every stage. The active-block family itself is now propagated from an initial entrywise bound under monotone active row/column windows, and the positive-norm field is reduced to a nonzero-active-block witness plus pivot-maximality; the raw pivot-maximality inequality is in turn supplied by the finite active max-pivot selector. The nonzero-active-block witness can now be supplied by positivity of the displayed active-block sum of squares, and the concrete column-swap bridge proves that swapping the selected active max column into the displayed active position supplies the post-swap pivot-maximality field. The stored-panel handoff is now also available in stage-budget form: a signed-pivot exact step bounded by $c_m B_t$ transfers to the rounded stored step with the concrete compact Householder FP budget. The rounded active-block recurrence is now formalized under visible signed-pivot nonbreakdown, pivot-maximality, storage, active-window, and compact-budget fields, and the prefix displayed rows are now propagated from a one-step prefix compact-budget recurrence using the active-block theorem and the exact same-reflector prefix/active split. A finite global compact-step budget now packages the off-diagonal, active-block, and prefix-row compact-update recurrences into one scalar stage-budget condition. The raw-stage positive active-block mass field is now supplied from local QR rank/determinant nonbreakdown data, including a stagewise determinant wrapper. A checked route-elimination counterexample also shows that this positive active-block mass, even together with upper-triangular nonsingular leading blocks, does not imply the row-wise off-diagonal-control field. A second checked counterexample shows that adding a valid strict-upper-entry row-growth budget still does not imply the diagonal lower-bound/nonbreakdown field: the local witness has row budget 2 and first diagonal magnitude 1. The Lean structure `StoredQRDisplayedRowBudgetControl` now names this residual row-budget plus diagonal-lower-bound package explicitly, and the solver wrapper consumes it as a visible source/domain assumption; it is not advertised as a proof of the package. A newer route-1 handoff shows that a scalar row-max contraction $R_{k,i}^{\max} \leq \rho |(S_k)_{ii}|$ with $\rho \leq 1$ is sufficient to build the package and feed the local solver theorem, but the contraction itself is still a source/domain invariant to prove for the chosen concrete loop. The active/prefix stage-diagonal lower-bound family is now similarly packaged by the finite scalar defect `storedQRStageDiagLowerDefectBudget`: nonpositivity of that scalar supplies $B_k \leq |(S_k)_{ii}|$ for every displayed off-diagonal row $i < k$, and the converse theorem packages that same pointwise family back into scalar nonpositivity. The local source-control and solver wrappers consume the scalar condition directly. The active max-pivot sibling now also derives the raw pivot-maximality field from the finite selector policy on this scalar route. A row-max bridge now proves this scalar condition from nonpositive row-max defect, or from local diagonal dominance, provided the remaining comparison $B_k \leq R_{k,i}^{\max}$ is supplied for each displayed off-diagonal row. That comparison is now also packaged by the finite scalar defect Δ_{cmp} : nonpositivity extracts $B_k \leq R_{k,i}^{\max}$, and the converse theorem packages a pointwise comparison proof back into scalar nonpositivity. The scalar row-max bridge therefore has a two-scalar form. The local source-control, local solver, and probability-level active-pivot wrappers now consume this two-scalar form directly, so the displayed comparison family no longer appears at those theorem surfaces; nevertheless both scalar row-max conditions still need concrete proof or visible hypotheses. The diagonal-dominant scalar-comparison local source-control and solver siblings sharpen that interface when local diagonal dominance

is supplied: diagonal dominance now gives leading-block determinant nonzeroness, and diagonal dominance together with the scalar comparison defect gives the scalar stage-diagonal condition. Thus the separate determinant and row-max scalar-defect fields no longer appear on that local branch, but local diagonal dominance itself, the scalar comparison defect, the conditioning/dual fields on the κ -budget branch, active-pivot policy, compact-product smallness, and the final generic QR/preconditioner theorem still remain visible or open. The probability-level equation (8) sibling now exposes this same narrowed surface at the sampled objective theorem: samplewise diagonal dominance supplies the determinant and row-max scalar-defect fields, and $\Delta_{\text{cmp}} \leq 0$ supplies the displayed comparison internally. This is a lift of the local reduction only; it still leaves local diagonal dominance, the scalar comparison defect, conditioning/dual compact-budget data, active-pivot policy, raw compact-product smallness, and the final generic QR/preconditioner theorem visible or open. The raw product field on this active diagonal-dominant scalar-comparison branch is now reducible to the canonical finite-max smallness scalar. On the concrete-dual finite-max sibling, the auxiliary κ_k , K_k , and dual compact-budget fields are also removed by deriving norm-square nonbreakdown from diagonal dominance and the per-pivot product-sequence budget. The finite-max scalar smallness inequality itself remains a visible numerical/domain obligation. A further local source-denominator rational-gamma sibling reduces that active local source-control/solver product surface to source-denominator nonbreakdown, a unit-roundoff cap, $(m : \mathbb{R})U_{\text{cap}} < 1$, and the canonical rational-gamma cap-smallness scalar, but those source-denominator and cap fields are still visible obligations. The sampled equation (8) sibling applies the same source-control certificate per trace, so the probability theorem has the same source-denominator/cap surface instead of the assembled finite-max compact-product scalar. The source-control, local solver, and sampled actual-unit siblings specialize the cap to $fp.u$ and derive the former `gammaValid` guards from $(m : \mathbb{R})fp.u < 1$ or $(s : \mathbb{R})fp.u < 1$, but the actual-unit scalar smallness, source-denominator, diagonal-dominance, scalar-comparison, active-pivot, and signed-stage obligations remain visible. On the stored-lower actual-unit branch, the raw source-denominator field is now derived from the stored recurrence, signed-alpha definition, and local diagonal dominance, leaving actual-unit scalar smallness, local diagonal-dominance/off-diagonal control, scalar comparison, active-pivot, and signed-stage loop invariants as the remaining visible QR/preconditioner obligations. The stored-recurrence scalar-smallness route is also now ruled out: those stored-loop, diagonal-dominance, signed-alpha, and actual-unit validity fields do not by themselves force the canonical rational-gamma smallness inequality. The stored-recurrence diagonal-dominance route is likewise ruled out: stored recurrence, signed-alpha choice, nonzero source denominators, and actual-unit validity do not by themselves force local diagonal dominance. A new route-elimination theorem shows that local diagonal dominance alone cannot discharge the scalar comparison defect: the diagonally dominant witness with displayed strict-upper row maximum 2 and stage budget 3 has $\Delta_{\text{cmp}} > 0$. The same witness satisfies the displayed compact-product inequality with budget $B = 1/16$, so product smallness does not rescue this diagonal-dominance shortcut. The scalar comparison defect is not forced by the current active-pivot active/off-diagonal budget surface, and product compact-smallness does not rescue that route: the finite-max extractor would otherwise imply the already-refuted displayed comparison $B_k \leq R_{k,i}^{\text{max}}$. Nor is it forced after adding diagonal dominance and product compact-smallness to that same active surface: the diagonally dominant witness with $B = 1/16$ still satisfies active max-pivot and active/off-diagonal budgets while $\Delta_{\text{cmp}} > 0$. Even granting the scalar row-max defect does not make the scalar comparison defect automatic; the checked diagonally safe witness has $R_{1,0}^{\text{max}} = 2$, displayed diagonal magnitude 3, and stage budget 3, so $\Delta_{\text{max}} \leq 0$ but

$\Delta_{\text{cmp}} > 0$. Product compact-smallness still does not rescue this stronger route. Nor does $\Delta_{\text{max}} \leq 0$ alone imply the downstream stage-diagonal scalar condition: with the same witness and stage budget 4, the displayed defect $4 - 3$ makes $\Delta_{\text{diag}} > 0$. Adding active max-pivoting, active/off-diagonal budget control, and compact-product smallness with budget $B = 1/16$ still does not rescue this scalar-stage shortcut. The active-pivot row-max-defect counterexample rules out deriving nonpositive row-max defect from active max-pivoting plus active/off-diagonal budget bounds alone. A separate compact-product counterexample rules out deriving the same nonpositive row-max defect merely from upper-triangular nonsingularity plus a satisfiable product compact-smallness inequality, and the product/active-budget strengthening shows that adding the active-pivot budget surface still does not rescue this row-max defect route. The same product-smallness surface also fails to imply the scalar stage-diagonal lower-bound condition for the fixed stage budget $B_k = 2$, and adding the active-pivot budget surface still does not rescue that scalar stage-diagonal route. The active-only implication is also refuted as a consequence: if active max-pivoting and active/off-diagonal budgets alone proved the scalar stage-diagonal condition, the product-plus-active implication would follow. Even adding product compact-smallness to the active-pivot budget surface does not rescue the comparison $B_k \leq R_{k,i}^{\text{max}}$. The active-pivot oversized-budget counterexample separately rules out deriving that comparison from the same active surface. The probability-level equation (8) wrapper now exposes these row-max fields directly at the sampled least-squares objective theorem, rather than hiding them behind a diagonal lower-bound family, but it does not prove them. Its horizon-clamped sibling applies the same row-max bridge and then reuses the active-pivot horizon theorem, so the visible row-max probability surface no longer exposes samplewise global budget-monotonicity. The local source-control and solver actual-unit siblings likewise remove only the abstract `gammaValid` fields from the row-max active-pivot local route, deriving them from $(m : \mathbb{R})u < 1$. Its actual-unit-roundoff sibling removes the abstract sampled `gammaValid` fields by deriving them from $(s : \mathbb{R})u < 1$; this is only a validity-surface reduction. The actual-unit horizon sibling combines that scalar guard with the row-max horizon theorem, removing both the sampled validity fields and samplewise global budget-monotonicity from this row-max probability surface. The scalar-comparison explicit-validity horizon sibling first extracts the displayed row-max comparison from `storedQRStageRowMaxComparisonDefectBudget` ≤ 0 , then reuses that horizon row-max wrapper, so the two-scalar probability surface no longer needs samplewise global budget-monotonicity. Its actual-unit horizon sibling derives the sampled validity fields from $(s : \mathbb{R})u < 1$ and calls that explicit-validity wrapper, while still keeping both scalar defects visible. The exact no-pivot shortcut for this scalar condition is now ruled out: the existing two-stage exact Householder counterexample has positive `storedQRStageDiagLowerDefectBudget` for the constant stage budget 2. The exact no-pivot comparison-scalar shortcut is separately ruled out as well: the same exact witness has positive `storedQRStageRowMaxComparisonDefectBudget` for the nonnegative constant stage budget 3, because the displayed strict-upper row maximum is 2. Thus exact recurrence plus signed squared-norm identities, nonzero denominators, and nonnegative stage budgets still do not imply the comparison scalar. The scalar conditions themselves remain open for the concrete pivoted/sorted/off-diagonal-controlled loop. The stored-recurrence active-pivot shortcut is also ruled out: the same exact-with- $u = 0$ stored-panel witness satisfies the stored recurrence, signed-alpha equation, nonzero source denominators, and $(2 : \mathbb{R})u < 1$, but its displayed first pivot is not the finite active max-pivot selector because the second column has larger active trailing norm. Thus the active-pivot equation must be proved by a concrete pivoted/sorted loop invariant or kept visible. The stored-recurrence scalar

row-budget shortcut is also ruled out: the same exact-with- $u = 0$ stored-panel witness satisfies the stored recurrence, signed-alpha equation, nonzero source denominators, and $(2 : \mathbb{R})u < 1$, while constant stage budgets 2 and 3 make the scalar stage-diagonal and stage-budget/row-max comparison defects positive. Thus those scalar fields must also come from a concrete pivoted/sorted off-diagonal-control invariant or remain visible. The diagonal-dominant stored-surface shortcut is ruled out as well: the exact stored sequence from $\begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}$ remains locally diagonally dominant and satisfies the stored, signed-alpha, source-denominator, actual-unit, and nonnegative-stage-budget fields, but stage budgets 4 and 3 still make the scalar diagonal-lower and comparison defects positive. Row sorting itself now preserves the rectangular least-squares residual objective and normal equations, by the row-permutation lemmas `lsObjective\_permuteRows`, `rectLSGram\_permuteRows`, and `rectLSRhs\_permuteRows`. Column pivoting now also has the matching coefficient-relabeling foundation: `rectMatMulVec\_permuteCols`, `RectLSNormalEquations.of\_permuteCols`, and `IsLeastSquaresMinimizer.of\_permuteCols`. The combined wrappers `rectLSGram\_permuteRowsCols`, `rectLSRhs\_permuteRowsCols`, `RectLSNormalEquations.of\_permuteRowsCols`, `lsObjective\_permuteRowsCols`, and `IsLeastSquaresMinimizer.of\_permuteRowsCols` now package the row-sorted, column-pivoted least-squares pullback in one step. On the unpivoted source-controlled branch, the stored-QR final triangular block and back-substitution solution are now connected directly to the equation (8) high-probability rounded sampled-row objective theorem. The remaining source dependency is therefore to supply the determinant/lower-zero, global compact-step budget, and row-budget/diagonal-lower-bound fields for the chosen concrete rounded loop, or to replace that branch by a pivoted/sorted theorem family with its own proved growth invariant;

- distribution-specific random-projection/FJLT/Gaussian/Rademacher uniformization guarantees for Algorithm 3. The weaker coordinate-Hoeffding leverage-probability event and the induced one-step uniform-row Loewner bound are now formalized. The deterministic-after-preconditioning iid uniform sample-average concentration theorem is also formalized in tail-budget finite-Loewner form, and the signed-Hadamard preprocessing randomness is now composed with the iid uniform sampling randomness on the product law for the scoped coordinate-Hoeffding route. The corresponding floating-point uniform-sketch transfer is also formalized with an explicit sample-dependent FP perturbation budget. A deterministic-radius refinement is also formalized: any scalar τ dominating that budget over all joint outcomes gives the same high-probability event with radius $\varepsilon + \tau$, and a sampled-row norm cap gives a closed-form fixed budget. The source-directed expected Euclidean row-norm step, deterministic row-norm convexity and Lipschitz inputs from Tropp’s Lemma 3.3 proof, the affine scaling that turns Ledoux’s $[0, 1]^m$ concentration constant into Tropp’s Rademacher constant, the finite MGF/Herbst/Laplace algebraic substrate, the corrected-log-MGF quotient monotonicity step, the right-limit at zero and final finite Herbst extraction to a log-Laplace bound, the entropy-bound-to-tail wrapper, the finite product-law entropy chain rule, the unbiased Bernoulli coordinate expectation and entropy formulas, the fair-Bernoulli coordinate log-Sobolev inequality, the first one-coordinate product peel-off for $P \otimes \nu$, the finite L^2 section-norm bridge, the abstract Bernoulli-product induction lift, the concrete `RademacherTrace` m cube entropy-gradient theorem, the conditional exponential-tilt reduction from that theorem, the conditional finite-cube Chernoff wrapper, scalar half-tilt exponential inequalities, uniform Rademacher flip invariance, non-sharp pointwise half-difference and coordinate-absolute-

difference bridges into the tilt flip-gradient hypothesis, the concrete signed-Hadamard row-norm positive-flip self-bounding estimate $\sum_k \max(F_i(\omega) - F_i(\omega^k), 0)^2 \leq 4/m$, the positive-drop exponential-tilt bridge, the source-sharp one-row and all-row SRHT row-norm tails, the resulting leverage-probability cap, the logarithmic $t = \sqrt{8 \log(m/\delta)/m}$ preprocessing-budget corollary, and the exact source-sharp SRHT preprocessing plus iid uniform-sampling composition theorem, and the matching source-sharp rounded sample-Gram floating-point transfer are formalized in explicit t -parameter, logarithmic-preprocessing, tail-budget, and fixed FP-budget forms. The remaining Algorithm 3 distribution-specific claims concern non-SRHT Gaussian/FJLT/input-sparsity preprocessing;

- the low-rank structural condition of equation (9). The exact rank-factorization vocabulary, the left/right basis-product rank-at-most certificates, the column-sketch projector rank bridge, and the explicit residual-certificate and coefficient-handoff surfaces are now formalized. Rectangular SVD, pseudoinverse, full-rank $V_k^T Z$, best rank- k , unitarily invariant norm, and computed SVD/projector routine certificates are still open;
- the matrix-completion claims around equations (10)–(11);
- the effective-resistance sparsification and Laplacian-solver claims described later in the paper.

The deterministic transfer and sketch-objective theorems above are real Lean theorems, but they should not be read as proofs of those still-open randomized or structural results. Consequently, the current full-paper formalization gate is intentionally marked as failed until the open paper-level claims in the ledger have matching Lean theorems.

12 File Map

Algorithm 1 update and trace stability. `RandNLA/ElementwiseSampling.lean`

Algorithm 1 concentration. `RandNLA/HitCountConcentration.lean`

Algorithm 1 one-step trace-MGF product-law adapters. `RandNLA/ElementwiseTraceMGF.lean`

Algorithm 1 equation (2) spectral transfer. `RandNLA/ElementwiseSpectral.lean`

Finite symmetric matrix spectral bridge. `Analysis/MatrixSpectral.lean`

Finite-real to complex C*-matrix bridge. `Analysis/CStarMatrixBridge.lean`

Operator-log/exponential bridge. `Analysis/OperatorLog.lean`

Complex C*-matrix expectation/order/Jensen bridge. `Analysis/CStarMatrixExpectation.lean`

Lieb trace and relative-entropy route vocabulary. `Analysis/LiebTrace.lean`

Trace-exponential probability bridge. `Analysis/MatrixConcentration.lean`

Algorithm 2 sampled rows. `RandNLA/RowSampling.lean`

Algorithm 2 Gram analysis and FP equation (5). `RandNLA/RowSamplingGram.lean`

Algorithm 2 leverage scores and equation (7). `RandNLA/RowSamplingLeverage.lean`

Algorithm 2 row-trace and leverage log-CGF prerequisites. `RandNLA/RowSamplingTraceMGF.lean`

`RandNLA/RowSamplingLeverageMGF.lean`

Algorithm 3 preconditioning. `RandNLA/UniformRowSampling.lean`

`RandNLA/UniformRowSamplingMGF.lean`

`RandNLA/UniformRowSamplingComposition.lean`

`RandNLA/UniformRowSamplingFP.lean`

`RandNLA/Preconditioning.lean`

Equation (8) sketched least squares. `RandNLA/LeastSquaresSketch.lean`

Equation (9) low-rank foundations. `RandNLA/LowRankApprox.lean`

QR and Householder stability substrate. `Algorithms/InverseBounds.lean`

`Algorithms/QR/HouseholderSpec.lean`

`Algorithms/QR/HouseholderApply.lean`

`Algorithms/QR/HouseholderQR.lean`

`Algorithms/LeastSquares/LSQRSolve.lean`

Full theorem ledger. `docs/RANDNLA_CACM_THEOREM_LEDGER.md`

Open full-paper backlog. `docs/RANDNLA_CACM_NOT_PROVED_LEDGER.md`